Harmonic analysis

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1 Haar Measure

1.1 LCH Groups

Definition. A topological space X is **locally compact** if every point of X admits a compact neighborhood. An **LCH group** is a topological group whose underlying topology is locally compact Hausdorff.

1. A subset of a space X is **relatively compact** if its closure is compact in X.

2. A subset of a space X is **σ-compact** if it is a countable union of compact subsets.

Proposition 1.1. Let G be an LCH group.

1. For a closed subgroup $H \leqslant G$, the quotient space G/H is an LCH space.

2. G admits a σ-compact open subgroup.

3. The countable union of σ-compact open subgroups of G generates a σ-compact open subgroup.

Proof.

1. Since $G \to G/H$ is open and surjective, G/H is also locally compact. To show G/H is Hausdorff, let $xH \neq yH \in G/H$. Let $U \subseteq G$ be a relatively compact open neighborhood of x with $\overline{U} \cap yH = \varnothing$ (U exists since $yH \subseteq G$ is closed). $\overline{U}$ being compact and H being closed, the set $\overline{U}H$ is also closed in G, so there exists a relatively compact open neighborhood V of y such that $V \cap UH = \varnothing$. Hence $VH \cap UH = \varnothing$.

2. Let V be a symmetric relatively compact open unit-neighborhood. For $n \in \mathbb{N}$, one has
$$\overline{V}^n \subseteq \overline{V^n} \subseteq V.V^n = V^{n+1}$$
so that $H := \bigcup_n \overline{V}^n = \bigcup_n V^{n+1}$, which is σ-compact, and is an open subgroup.

3. Let (L_n) be a sequence of σ-compact open subgroups of G. Put $L := \bigcup_n L_n$; then the subgroup generated by the L_n is $\bigcup_m L^m$, which is clearly open, and being a countable union of countable unions of σ-compact sets, it is also σ-compact.

□

1.2 Haar Measure on LCH Groups

1.2.1 Outer Radon measure

Definition. For a topological space X, put
$$C_c(X) = \{f : X \to \mathbb{C} \mid f \text{ is continuous with compact support}\}$$
where the **support** of a function f is defined as $\operatorname{supp} f := \overline{\{x \in X \mid f(x) \neq 0\}}$.

Definition. Let X be a topological space. The **Borel σ-algebra** $\mathcal{B}$ is the σ-algebra that generated by all open sets of X. Every element in $\mathcal{B}$ is called a **Borel set**.

1. Any measure $\mu : \overline{\mathcal{B}} \to [0, \infty]$ is called a **Borel measure**.
2. A Borel measure is **locally finite** if every point $x \in X$ admits a neighborhood U with $\mu(U) < \infty$.
3. A locally finite Borel measure μ on $\mathcal{B}$ is called a **outer Radon measure** if
   (a) (outer regular)
   $\mu(A) = \inf \left\{ \mu(U) \mid X \underset{\text{open}}{\supseteq} U \supseteq A \right\}$ for every $A \in \mathcal{B}$;
   (b) (weakly inner regular)
   $\mu(U) = \sup \left\{ \mu(K) \mid K \subseteq U,\ K \underset{\text{cpt}}{\subseteq} X \right\}$ for every open set U of X.

- We will use the term Radon measure for outer Radon measure for simplicity.
- For $A \in \mathcal{B}$ with finite measure, we have $\mu(A) = \sup \left\{ \mu(K) \mid K \subseteq A,\ K \underset{\text{cpt}}{\subseteq} X \right\}$.
- We can state (b) in terms of continuous functions with compact support. For each compact $K \subseteq U$, by Urysohn's Lemma we can find $f \in C_c^+(G)$ with $f|_K \equiv 1$, $0 \leqslant f \leqslant 1$ and $f|_{X \setminus U} \equiv 0$. Then

$$\mu(U) \geqslant \int_U f d\mu = \int_X f d\mu \geqslant \int_K f d\mu = \mu(K)$$

  Hence

$$\mu(U) = \sup_{\substack{f \in C_c^+(G) \\ f \leqslant \mathbf{1}_U}} \int_X f d\mu$$

Proposition 1.2. Let μ be an outer Radon measure on a LCH space X. Then $C_c(X)$ with compact supports is dense in $L^p(\mu)$ for every $1 \leqslant p < \infty$.

Proof. Fix $1 \leqslant p < \infty$ and let V be the closure of $C_c(X)$ in $L^p(\mu)$. We must show $V = L^p := L^p(\mu)$. It suffices to show $\mathbf{1}_A \in V$ for all measurable sets A with finite measure. By outer regularity, we can find open sets $(U_n)_n$ containing A such that $\mathbf{1}_{U_n} \to \mathbf{1}_A$ in L^p, so we can assume A is open. Again by weakly inner regularity we reduce to the case A is compact.

For each $\varepsilon > 0$, pick open $U \supseteq A$ with $\mu(U \setminus A) < \varepsilon$. By Urysohn's Lemma, there exists $g \in C_c(X)$ with $g|_A \equiv 1$, $g|_{X \setminus U} \equiv 0$ and $0 \leqslant g \leqslant 1$. Then

$$\|\mathbf{1}_A - g\|_p^p = \int_{U \setminus A} |g(x)|^p d\mu(x) \leqslant \mu(U \setminus A) < \varepsilon$$

□

Definition. Let G be a LCH group.

1. A measure μ on G is called **left-invariant** if $\mu(xA) = \mu(A)$ for all $x \in G$ and measurable A.

2. A *non-zero* left-invariant outer Radon measure on G is called a **Haar measure** on G.

Theorem 1.3. Let G be a LCH group. Then there exists a Haar measure on G which is uniquely determined up to positive scalars.

Corollary 1.3.1. Let μ be a Haar measure on the LCH group G.

1. Every nonempty open set U has strictly positive measure.

2. Every compact set has finite measure.

3. Every continuous positive function $f \geqslant 0$ with $\displaystyle\int_G f(x)d\mu(x) = 0$ vanishes identically.

4. The support of a measurable function f on G that is μ-integrable is contained is in a σ-compact open subgroup of G.

Proof.

1. Suppose there exists a nonempty open set U with zero measure. Then for every compact set K in G, K can be covered by finitely many left-translations xU of U, so that K is also of measure zero. Being weakly inner regular, μ vanishes on every open set. Hence μ is identically zero, contradicting to the non-triviality assumption imposed in the definition of a Haar measure.

2. Recall that a Radon measure is locally finite. Hence every compact set can be covered by finitely many open sets with finite measure.

3. It implies that $\mu(f^{-1}(0, \infty)) = 0$, so by 1. it is empty.

4. It suffices to show the set $A = \{x \in X \mid f(x) \neq 0\}$ is contained in a σ-compact open subgroup L. We make the following reductions.

   - $A = \bigcup_n A_n$, where $A_n = \{x \in X \mid |f(x)| \geqslant 1/n\}$ is of finite measure. Hence it suffices to show a set with finite measure is contained in a σ-compact open subgroup.
   - By outer regularity, we can find open $U \supseteq A$ with $\mu(U) < \infty$, so it suffices to show an open set with finite measure is contained in a σ-compact open subgroup.

   Let $H \leqslant G$ be any σ-compact open subgroup. Then G is a disjoint union of the xH with $x \in G$. Since $\mu(U) < \infty$, U can only meet countably many cosets xH, for either $xH \cap U = \varnothing$ or $\mu(xU \cap U) > 0$ by 1. Let L be the group generated by H and those cosets xH that meet U. Then $L \supseteq U \supseteq A$ and L is σ-compact and open.

$\square$

1.2.2 Uniform continuity

Definition. A map $f : G \to X$ to a metric space (X, d) is **uniformly continuous** if for all $\varepsilon > 0$ there exists a unit-neighborhood U such that for $x^{-1}y \in U$ or $yx^{-1} \in U$ one has $d(f(x), f(y)) < \varepsilon$.

Lemma 1.4. Any function $f \in C_c(G)$ is uniformly continuous.

Proof. Let $K = \operatorname{supp}(f)$. Fix $\varepsilon > 0$ and a compact unit-neighborhood V. Since f is continuous, for every $x \in G$ there exists a unit-neighborhood $V_x \subseteq V$ such that if $y \in xV_x$, then $d(f(y), f(x)) < \varepsilon/2$. Let U_x be the symmetric open unit-neighborhood with $U_x^2 \subseteq V_x$. Then the sets xU_x $(x \in KV)$ form an open cover of the compact set KV, so there exists $x_1, \ldots, x_n \in KV$ such that $KV \subseteq x_1U_{x_1} \cup \cdots \cup x_nU_{x_n}$. Put $U_i = U_{x_i}$ and $U = U_1 \cap \cdots \cap U_n$; note that U is symmetric.

Let $x, y \in G$ with $x^{-1}y \in U$.

- If $x \notin KV$, then $y \notin K$, for $x \in yU^{-1} = yU \subseteq yV$. Then $f(x) = f(y) = 0$.
- If $x \in KV$, then there exists j with $x \in x_jU_j$, so that $y \in x_jU_jU \subseteq x_jV_{x_j}$. Hence
$$d(f(x), f(y)) \leqslant d(f(x), f(x_j)) + d(f(x_j), f(j)) < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$$

So far we have found a unit-neighborhood U such that if $x^{-1}y \in U$, then $d(f(x), f(y)) < \varepsilon$. It goes verbatim that we can find V with the condition replaced by $yx^{-1} \in V$ and having the same result. Then the uniform continuity is satisfied with the neighborhood $U \cap V$. □

1.2.3 Existence

To prove Theorem, we invoke Rieze's representation theorem. It suffices to show that up to positive scalars there exists a nontrivial positive linear functional $I : C_c(G) \to \mathbb{C}$ that is invariant in the sense $I(L_xf) = I(f)$ for every $x \in G$ and $f \in C_c(F)$, where $L_xf(y) := f(x^{-1}y)$ is the left translation. Likewise define $R_xf(y) := f(xy)$. The "inverse" in the definition of L_x is made so that $L_{xy} = L_xL_y$.

Lemma 1.5. Let G be an LCH group and $\Lambda : C_c(G) \to \mathbb{C}$ be a positive linear functional. Then Λ is left-invariant if and only if its induced outer Radon measure on G is left-invariant.

Proof. The if part is obvious. For the only if part, let μ be an induced outer Radon measure on G. We need to show $\mu(xA) = \mu(A)$ for all measurable A and $x \in G$. By outer regularity, we may assume $A = U$ is open, and by weakly inner regularity we have
$$\mu(U) = \sup_{\substack{f \in C_c^+(G) \\ f \leqslant \mathbf{1}_U}} \int_X f d\mu = \sup_{\substack{f \in C_c^+(G) \\ f \leqslant \mathbf{1}_U}} \Lambda(f)$$
Thus for all $g \in G$,
$$\mu(gU) = \sup_{\substack{f \in C_c^+(G) \\ f \leqslant \mathbf{1}_{gU}}} \Lambda(f) = \sup_{\substack{f \in C_c^+(G) \\ L_{g^{-1}}f \leqslant \mathbf{1}_U}} \Lambda(f) = \sup_{\substack{f \in C_c^+(G) \\ f \leqslant \mathbf{1}_U}} \Lambda(L_gf) = \sup_{\substack{f \in C_c^+(G) \\ f \leqslant \mathbf{1}_U}} \Lambda(f) = \mu(U)$$
□

Definition. A function $f : G \to \mathbb{C}$ is **positive** if $f(G) \subseteq \mathbb{R}_{\geqslant 0}$; we write $f \geqslant 0$ in this case.

- Put $C_c^+(G)$ for the set of all positive continuous functions with compact support.
- For any two $f, g \in C_c^+(G)$ with $g \neq 0$, there exist finitely many $s_j \in G$ and positive numbers c_j such that for all $x \in G$, $f(x) \leqslant \sum_{j=1}^{n} c_j g(s_j^{-1}x)$, or simply $f \leqslant \sum_{j=1}^{n} c_j L_{s_j} g$.

  Indeed, let U be a compact neighborhood such that g does not vanishes on U and say $s_1U, \ldots, s_nU$ cover the support of f. Put $c_j = \dfrac{\max\{f(x) \mid x \in s_jU\}}{\min\{g(x) \mid x \in U\}}$. Then for $x \in s_jU$, we have $f(x) \leqslant c_j g(s_j^{-1}x)$. Summing over $j = 1, \ldots, n$ gives the desired inequality.

For $f, g \in C_c^+(G)$ with $g \neq 0$, define the **index**

$$(f : g) = \inf \left\{ \sum_{j=1}^{n} c_j \mid \text{there exists } s_j \in G \text{ such that } f \leqslant \sum_{j=1}^{n} c_j L_{s_j} g \right\}$$

Lemma 1.6. For $f, f_1, f_2, g, h \in C_c^+(G)$ with $g, h \neq 0$ and $c > 0$, one has

1. $(L_y f : g) = (f : g)$ for every $y \in G$.
2. $(f_1 + f_2 : g) \leqslant (f_1 : g) + (f_2 : g)$.
3. $(cf : g) = c(f : g)$.
4. $f_1 \leqslant f_2 \Rightarrow (f_1 : g) \leqslant (f_2 : g)$.
5. $(f : h) \leqslant (f : g)(g : h)$.
6. $(f : g) \geqslant \dfrac{\max f}{\max g}$, where $\max f := \max\{f(x) \mid x \in G\}$.

In the following we fix a nonzero function $f_0 \in C_c^+(G)$. For $f, \phi \in C_c^+(G)$ with $\phi \neq 0$, define

$$J(f, \phi) = J_{f_0}(f, \phi) := \frac{(f : \phi)}{(f_0 : \phi)}$$

Lemma 1.7. For $f, g, \phi \in C_c^+(G)$ with $f, \phi \neq 0$ and $c > 0$, one has

1. $\dfrac{1}{(f_0 : f)} \leqslant J(f, \phi) \leqslant (f : f_0)$.
2. $J(L_s f, \phi) = J(f, \phi)$ for every $s \in G$.
3. $J(f + g, \phi) \leqslant J(f, \phi) + J(g, \phi)$.
4. $J(cf, \phi) = cJ(f, \phi)$.

Lemma 1.8. Let $f_1, f_2 \in C_c^+(G)$ and $\varepsilon > 0$. There exists a unit-neighborhood V in G such that

$$J(f_1, \phi) + J(f_2, \phi) \leqslant J(f_1 + f_2, \phi) + \varepsilon$$

holds for every $\phi \in C_c^+(G)\backslash\{0\}$ with support in V.

Proof. Pick an $f' \in C_c^+(G)$ such that $f \equiv 1$ on the support of $f_1 + f_2$. Choose $\delta > 0$ such that

$$\delta(f_1 + f_2 : f_0) + (\delta + \delta^2)(f' : f_0) < \varepsilon$$

and set

$$f = f_1 + f_2 + \delta f', \qquad h_1 = \frac{f_1}{f}, \qquad h_2 = \frac{f_2}{f}$$

where we set $h_j(x) = 0$ if $f(x) = 0$. Then $h_j \in C_c^+(G)$ for $j = 1, 2$. For if $f(x) = 0$, then $f_j(x) = f'(x) = 0$ so that $x \notin \operatorname{supp}(f_1 + f_2)$. Hence there exists a unit-neighborhood U such that if $y \in xU$, then $f_1(y) + f_2(y) = 0$, implying $f_1(y) = f_2(y) = 0$. Hence $h_j \equiv 0$ in xU, demonstrating the continuity of h_j at x.

Being compactly supported, the h_j are uniformly continuous, so there exists a unit-neighborhood V such that for $x, y \in G$ with $x^{-1}y \in V$ and $j = 1, 2$, one has $|h_j(x) - h_j(y)| < \delta/2$. Let $\phi \in C_c^+(G)\backslash\{0\}$ with support in V, and choose finitely many $s_k \in G$, $c_k > 0$ with $f \leqslant \sum_k c_k L_{s_k}\phi$. Then $\phi(s_k^{-1}x) \neq 0$ (so that $s_k^{-1}x \in V$) implies $|h_j(x) - h_j(s_k)| < \delta/2$, and for all x, one has

$$\begin{aligned} f_j(x) = f(x)h_j(x) &\leqslant \sum_k c_k \phi(s_k^{-1}x) h_j(x) \\ &\leqslant \sum_k c_k \phi(s_k^{-1}x) \left(h_j(s_k) + \frac{\delta}{2} \right) \end{aligned}$$

so that $(f_j : \phi) \leqslant \sum_k c_k(h_j(s_k) + \delta/2)$, implying

$$(f_1 : \phi) + (f_2 : \phi) \leqslant \sum_k c_k(1 + \delta)$$

which yields

$$\begin{aligned} J(f_1, \phi) + J(f_2, \phi) &\leqslant J(f, \phi)(1 + \delta) \\ &\leqslant (J(f_1 + f_2, \phi) + \delta J(f', \phi))(1 + \delta) \\ &= J(f_1 + f_2, \phi) + \delta J(f_1 + f_2, \phi) + (\delta + \delta^2) J(f', \phi) \\ &\overset{1.7.1}{\leqslant} J(f_1 + f_2, \phi) + \delta(f_1 + f_2 : f_0) + (\delta + \delta^2)(f' : f_0) \\ &< J(f_1 + f_2, \phi) + \varepsilon \end{aligned}$$

□

Lemma 1.6.(5) together with $(f : f) = 1$ gives $\dfrac{1}{(f_0 : f)} \leqslant (f : f_0)$. For $f \in C_c^+(G)\backslash\{0\}$, let S_f be the compact interval

$$S_f := \left[\frac{1}{(f_0 : f)}, (f : f_0) \right]$$

The product $S := \prod_{f \in C_c^+(G)\backslash\{0\}} S_f$ is compact by Tychonov's theorem. For every $\phi \in C_c^+(G)\backslash\{0\}$, we get an element $J(f,\phi) \in S_f$ from Lemma 1.7.(1), hence an element $(J(f,\phi))_f \in S$.

For a unit-neighborhood V let

$$L_V := \overline{\left\{ (J(f,\phi))_f \in S \mid \phi \in C_c^+(G)\backslash\{0\} \text{ with } \operatorname{supp}(\phi) \subseteq V \right\}} \underset{\text{closed}}{\subseteq} S$$

Since S is compact, the intersection

$$\bigcap \{L_V \mid V : \text{ unit neighborhood in } G\}$$

is nonempty. Choose an element $(I_{f_0}(f))_f$ in this intersection.

Lemma 1.9. $I := I_{f_0} : C_c^+(G) \to \mathbb{C}$ is a left-invariant positive homogeneous additive map.

Proof. By definition and Lemma 1.7.2. and 4., I is positive, left-invariant and homogeneous. Let $f, g \in C_c^+(G)\backslash\{0\}$. We need to show $I(f+g) = I(f) + I(g)$. For each $\varepsilon > 0$, by Lemma 1.8 we can find unit-neighborhood V such that

$$J(f,\phi) + J(g,\phi) \leqslant J(f+g,\phi) + \varepsilon$$

holds for every $\phi \in C_c^+(G)\backslash\{0\}$ with support in V. In particular, this forces $I(f) + I(g) \leqslant I(f+g) + \varepsilon$. Letting $\varepsilon \to 0^+$, and since $(I(f))_f \in \bigcap_V L_V$, we must have $I(f) + I(g) \leqslant I(f+g)$. The reversed inequality follows from Lemma 1.7.3. □

Extending I by linearity we obtain a well-defined positive invariant linear functional $I : C_c(G) \to \mathbb{C}$. This proves the existence of a Haar measure on G.

1.2.4 Uniqueness

Lemma 1.10. Let ν be a Haar measure on G. Then for every $f \in C_c(G)$ the map

$$s \mapsto \int_G f(xs)d\nu(x)$$

is continuous on G.

Proof. We must show for each $s \in G$ there exists a neighborhood U of s such that for all $t \in U$ one has

$$\left| \int_G (f(xs) - f(xt))d\nu(x) \right| < \varepsilon$$

Replacing f by $R_s f$, we reduce to the case $s = e$. Let $K = \operatorname{supp}(f)$, and V a compact symmetric unit-neighborhood. For $t \in V$, one has $\operatorname{supp}(R_t f) \subseteq KV$. Since f is uniformly continuous, there exists

a symmetric unit-neighborhood W such that for $t \in W$ one has $|f(xt) - f(x)| < \dfrac{\varepsilon}{\nu(KV)}$. Finally, for $t \in W \cap V$, we see

$$\begin{aligned}\left|\int_G (f(xt) - f(x))d\nu(x)\right| &\leqslant \int_{KV} |f(xt) - f(x)|d\nu(x) \\ &< \frac{\varepsilon}{\nu(KV)}\nu(KV) = \varepsilon\end{aligned}$$

□

Suppose note μ, ν are two Haar measures. We aim to show that $\nu = c\mu$ for some $c > 0$. For $f \in C_c(G)$ with $\displaystyle\int_G f(t)d\mu(t) =: I_\mu(f) \neq 0$, set

$$D_f(s) := \frac{1}{I_\mu(f)} \int_G f(ts)d\nu(t)$$

The function D_f is continuous by Lemma 1.10. Let $g \in C_c(G)$. By Fubini's theorem and the invariance of the measures μ, ν, we obtain

$$\begin{aligned}I_\mu(f)I_\nu(g) &= \int_G \int_G f(s)g(t)d\nu(t)d\mu(s) \overset{\text{inv}}{=} \int_G \int_G f(s)g(s^{-1}t)d\nu(t)d\mu(s) \\ &\overset{\text{Fubini}}{=} \int_G \int_G f(s)g(s^{-1}t)d\mu(s)d\nu(t) \overset{\text{inv}}{=} \int_G \int_G f(ts)g(s^{-1})d\mu(s)d\nu(t) \\ &\overset{\text{Fubini}}{=} \int_G \int_G f(ts)g(s^{-1})d\nu(t)d\mu(s) \\ &= \int_G \left(\int_G f(ts)d\nu(t)\right) g(s^{-1})d\mu(s) \\ &= I_\mu(f) \int_G D_f(s)g(s^{-1})d\mu(s)\end{aligned}$$

Since $I_\mu(f) \neq 0$, we conclude

$$I_\nu(g) = \int_G D_f(s)g(s^{-1})d\mu(s)$$

Pick another $f' \in C_c(G)$ with $I_\mu(f') \neq 0$. It follows that

$$\int_G (D_f(s) - D_{f'}(s))g(s^{-1})d\mu(s) = 0$$

for every $g \in C_c(G)$. Replacing g with the function $\tilde{g}$ given by

$$\tilde{g}(s) = |g(s)|^2\overline{(D_f(s^{-1}) - D_{f'}(s^{-1}))}$$

one obtains

$$\int_G |(D_f(s) - D_{f'}(s))g(s^{-1})|^2 d\mu(s) = 0$$

It follows that $(D_f(s) - D_{f'}(s))g(s^{-1}) = 0$. Since g is arbitrary, $D_f = D_{f'}$; call this function D. Now for each $f \in C_c(G)$ with $I_\mu(f) \neq 0$, one has

$$\int_G f(t)d\mu(t)D(e) = \int_G f(t)d\nu(t)$$

By linearity, it follows that this equality holds everywhere in $C_c(G)$, and hence proving the uniqueness.

1.3 Modular Characters

From now on, unless otherwise mentioned, for an LCH group G we always fix a Haar measure, and for the integral we will write $\int_G f(x)dx$. For the measure of a set $A \subseteq G$ we write $\mathrm{vol}(A)$. If G is compact, we always normalize the Haar measure so that $\mathrm{vol}(G) = 1$. For $p \geqslant 1$, put $L^p(G)$ for the L^p-space of G with respect to a Haar measure.

Definition. Let G be an LCH group and μ a Haar measure on G.

1. To each $x \in G$ we assign a Haar measure μ_x by $\mu_x(A) := \mu(Ax)$.

By the uniqueness of Haar measures, there exists $\Delta(x) > 0$ with $\mu_x = \Delta(x)\mu$.

2. The map $\Delta : G \to \mathbb{R}_{>0}$ is called the **modular function/modular character** of G.

3. G is called **unimodular** if $\Delta \equiv 1$.

For X a set and and a function $f : X \to \mathbb{C}$, define the **sup-norm** of f by

$$\|f\|_X := \sup_{x \in X} |f(x)|$$

Theorem 1.11.

1. The modular function $\Delta : G \to \mathbb{R}_{>0}$ is a continuous group homomorphism.

2. If we let $[G, G]$ to be the closure of the commutator subgroup of G, then $\Delta \equiv 1$ if $G/[G, G]$ is compact. In particular, G is unimodular if G is abelian or compact.

3. For $y \in G$ and $f \in L^1(G)$, one has $R_y f \in L^1(G)$ and

$$\int_G R_y f(x)dx = \int_G f(xy)dx = \Delta(y^{-1}) \int_G f(x)dx$$

   In a concise form, $d(xy) = \Delta(y)dx$.

4. For $f \in L^1(G)$,

$$\int_G f(x^{-1})\Delta(x^{-1})dx = \int_G f(x)dx$$

   In a concise form, $d(x^{-1}) = \Delta(x^{-1})dx$.

In fact, 3. says that $\Delta(x^{-1})dx$ is a right invariant Haar measure, and 4. implies that $x \mapsto x^{-1}$ takes dx to $\Delta(x^{-1})dx$.

Proof.

3. It is clear for the case $f = \mathbf{1}_A$ with A measurable and $\mathrm{vol}(A) < \infty$. The general case follows as usual.

1.
   - Group homomorphism. For measurable A and $x, y \in G$, we have

     $$\Delta(xy)\mu(A) = \mu_{xy}(A) = \mu(Axy) = \mu_y(Ax) = \Delta(y)\mu(Ax) = \Delta(y)\mu_x(A) = \Delta(y)\Delta(x)\mu(A)$$

     Pick A such that $\mu(A) \neq 0$; then it follows that $\Delta(xy) = \Delta(x)\Delta(y)$.
   - Continuity. Let $f \in C_c(G)$ with $c = \int_G f(x)dx$. By 3. one has

     $$\Delta(y) = \frac{1}{c}\int_G f(xy^{-1})dx = \frac{1}{c}\int_G R_{y^{-1}}f(x)dx$$

2. Since $\mathbb{R}_{>0}$ is abelian, we have $\Delta(aba^{-1}b^{-1}) = 1$ for all $a, b \in G$. Since Δ is continuous, we have $\Delta([G, G]) = 1$. Thus Δ factors through $G/[G, G]$. The only compact subgroup of $\mathbb{R}_{>0}$ is $\{1\}$, so $\Delta \equiv 1$.

4. Let $f \in C_c(G)$ and $I(f) = \int_G f(x^{-1})\Delta(x^{-1})dx$. By 3.

   $$\begin{aligned} I(L_z f) &= \int_G f(z^{-1}x^{-1})\Delta(x^{-1})dx = \int_G f((xz)^{-1})\Delta(x^{-1})dx \\ &= \Delta(z^{-1})\int_G f(x^{-1})\Delta((xz^{-1})^{-1})dx = \int_G f(x^{-1})\Delta(x^{-1})dx \\ &= I(f) \end{aligned}$$

   Hence I is an invariant positive linear functional on $C_c(G)$, so by Lemma 1.5 and the uniqueness of Haar measures there exists $c > 0$ such that $I(f) = c\int_G f(x)dx$.

   It remains to show $c = 1$. Let $\varepsilon > 0$ and choose a symmetric unit-neighborhood V with $|1 - \Delta(s)| < \varepsilon$ for every $s \in V$, and take a nonzero symmetric function $f \in C_c^+(V)$. Then

   $$\begin{aligned} |1 - c|\int_G f(x)dx &= \left|\int_G f(x)dx - I(f)\right| \leqslant \int_G |f(x) - f(x^{-1})\Delta(x^{-1})|dx \\ &= \int_V f(x)|1 - \Delta(x^{-1})|dx < \varepsilon\int_G f(x)dx \end{aligned}$$

   so that $|1 - c| < \varepsilon$. Since ε is arbitrary, $c = 1$.

$\square$

Remark. Denote by $\mathrm{inv} : G \to G$ the inversion on G and consider the pushforward measure $\nu := \mathrm{inv}_*\mu$ of the left-invariant Haar measure μ on G. It is a Radon measure on G, and satisfies the integration formula, valid for all $f \in C_c(G)$

$$\int_G f(x)d\nu := \int_G f(x^{-1})d\mu$$

Now

$$\int_G f(xy)d\nu := \int_G f(y^{-1}x^{-1})d\mu = \int_G f(x^{-1})d\mu = \int_G f(x)d\nu$$

so ν is a right Haar measure. From 2. if we write μ as dx, then $\Delta(x^{-1})dx$ is also a right Haar measure, so the uniqueness says $dx^{-1} = c\Delta(x^{-1})dx$ for some unique scalar $c > 0$. This is exactly the same as the first part of 4.

Proposition 1.12. Let G be a LCH group. TFAE:

1. There exists $x \in G$ such that $\mathrm{vol}(\{x\}) > 0$.
2. $\mathrm{vol}\{e\} > 0$.
3. The Haar measure is a multiple of the counting measure.
4. G is a discrete group.

In particular, a discrete group is unimodular.

Proof. That 1. $\Leftrightarrow$ 2. follows from the left-invariant of vol. Assume 2.. Then for any finite set $E \subseteq G$, $\mathrm{vol}(E) = \mathrm{vol}(\{e\})\#E$. The general case follows from monotonicity. For 3. $\Rightarrow$ 4., note that this implies every open set with finite measure is a finite set. By Hausdorff axiom one see every point in U is open. Hence G is discrete. Finally, if G is discrete, every singleton is open so that each of them has a positive measure. $\square$

Proposition 1.13. Let G be a LCH group. Then G has finite measure if and only if G is compact.

Proof. The if part is clear. Now assume $\mathrm{vol}(G) < \infty$. Let U be a compact unit-neighborhood. Then

$$\max\{n \in \mathbb{N} \mid \text{there exists } (x_i)_{1 \leqslant i \leqslant n} \subseteq G \text{ such that } x_iU \cap x_jU = \varnothing \text{ if } i \neq j\} < \infty$$

Let $z_1U, \ldots, z_nU$ be such pairwise disjoint translation, and set K to be their disjoint union. Then K is compact, and for every $x \in G$, $xK \cap K \neq \varnothing$, so that $x \in KK^{-1}$. Hence $G = KK^{-1}$ is compact. $\square$

1.3.1 Modulus

Let G be an LCH group with a Haar measure μ. For a topological group automorphism $\sigma : G \to G$, we consider the measure μ_σ on G defined by $\mu_\sigma(A) := \mu(\sigma(A))$. Since σ is a homeomorphism, μ_σ is again a Radon measure, and for each $g \in G$,

$$\mu_\sigma(gA) = \mu(\sigma^{-1}(g)\sigma^{-1}(A)) = \mu(\sigma^{-1}(A)) = \mu_\sigma(A)$$

so μ_σ is also a Haar measure. The uniqueness of Haar measure assures the existence of the positive constant $\mathrm{mod}_G(\sigma)$, called the **modulus of the automorphism** σ, such that $\mu_\sigma = \mathrm{mod}_G(\sigma)\mu$. For instance, we have $\mathrm{mod}_G(g \mapsto gx) = \Delta_G(x)$. For $f \in L^1(G)$, we have the formula

$$\int_G f(\sigma^{-1}(x))d\mu(x) = \mathrm{mod}_G(\sigma)\int_G f(x)d\mu(x)$$

Symbolically, we write $d(\sigma(x)) = \mathrm{mod}_G(\sigma)dx$, where dx denotes $d\mu$.

For $g \in G$, we can consider the inner automorphism $\mathrm{Inn}_g : G \to G$ defined by $\mathrm{Inn}_g(x) := gxg^{-1}$. We put $\mathrm{mod}_G(g) := \mathrm{mod}_G(\mathrm{Inn}_g)$. Then we have the integral formula

$$\int_G f(g^{-1}xg)d\mu(x) = \mathrm{mod}_G(g)\int_G f(x)d\mu(x)$$

Comparing to the formula (1.11.3), we see $\mathrm{mod}_G = \Delta_G^{-1}$.

1.4 Quotient Integral Formula

1.4.1 Quotient measure

Definition. Let G be a LCH group and $H \leqslant G$ a closed subgroup. Then G/H is a LCH space. For each $f \in C_c(G)$, put

$$f^H(x) := \int_H f(xh)dh = \int_H L_{x^{-1}}f(h)dh$$

Note that the support of $L_{x^{-1}}f$ is compact, so that the integral exists.

- For $y = xk$ for some $k \in H$, we have

  $$f^H(y) = \int_H f(yh)dh = \int_H f(xkh)dh = \int_H L_{x^{-1}}f(kh)dh = \int_H L_{x^{-1}}f(h)dh = f^H(x)$$

  so that f^H descends to a map $f^H : G/H \to \mathbb{C}$.

- $f^H : G \to \mathbb{C}$ is continuous. Indeed, put $K = \mathrm{supp}(f)$, and let $x \in G$ and U a compact neighborhood of x. Then $\mathrm{supp}\, L_{x^{-1}}f \subseteq U^{-1}K \cap H$. Put $d = \mu_H(U^{-1}K \cap H)$, where μ_H denotes the given Haar measure on H. Given $\varepsilon > 0$, by uniform continuity of f there is a neighborhood $V \subseteq U$ of x such that $|f(yh) - f(xh)| < \dfrac{\varepsilon}{d}$ for all $y \in V$. It follows that

  $$|f^H(y) - f^H(x)| \leqslant \int_{U^{-1}K\cap H} |f(yh) - f(xh)|dh < \varepsilon$$

  for $y \in V$, so that f^H is continuous at x.

- The support of $f^H : G/H \to \mathbb{C}$ is contained in KH/H, where $K = \mathrm{supp}(f)$. Since KH/H is compact in G/H, f^H has compact support.

- $f^H : G \to \mathbb{C}$ descends to a continuous map $f^H : G/H \to \mathbb{C}$ with compact support.

Lemma 1.14. Let G be a LCH group and $H \leqslant G$ a closed subgroup. For every compact $C \subseteq G/H$, there exists compact $K \subseteq G$ such that $\pi(K) = C$.

Proof. Put $\pi : G \to G/H$ to be the canonical projection. For each $c \in C$, pick $y_c \in \pi^{-1}(C)$ and an open relatively compact neighborhood U_c of y_c. Since π is open, $\{\pi(U_c)\}_{c \in C}$ forms an open cover of C, and thus we can find $c_1, \ldots, c_n \in C$ such that $C \subseteq K'$ with $K' = \overline{U_{c_1}} \cup \cdots \cup \overline{U_{c_n}} \underset{\text{cpt}}{\subseteq} G$. It suffices to take $K = K' \cap \pi^{-1}(C)$. □

Lemma 1.15. The map

$$\begin{aligned} C_c(G) &\longrightarrow C_c(G/H) \\ f &\longmapsto f^H \end{aligned}$$

is surjective. In addition, the fibre of $h \in C_c^+(G/H)$ intersects with $C_c^+(G)$

Proof. Take $g \in C_c(G/H) \backslash \{0\}$. Apply Lemma with $C = \operatorname{supp}(g)$ and denote K be the compact set constructed. Use Urysohn's Lemma to obtain $\phi \in C_c(G)$ such that $\phi \geqslant 0$ and $\phi|_K \equiv 1$. Set $f = g\phi/\phi^H$; this definition makes sense for $\phi^H > 0$ on the support of g. Then $f^H = g\phi^H/\phi^H = g$. The last assertion is clear from the above construction. □

Definition. Let X be a G-space. A measure ν on the Borel σ-algebra of X is called an **invariant measure** if $\nu(xA) = \nu(A)$ for every $x \in G$ and every measurable $A \subseteq X$.

We need a generalized version of Lemma 1.5, though the proof is exactly the same.

Lemma 1.16. Let X be an LCH G-space, $\Lambda : C_c(X) \to \mathbb{C}$ a positive linear functional and $\chi : G \to \mathbb{R}_{>0}$ a continuous group homomorphism. For each $g \in G$,

- For $f \in C_c(X)$, define $g.f \in C_c(X)$ by $g.f(x) := f(g^{-1}.x)$.
- Define $g.\Lambda : C_c(X) \to \mathbb{C}$ by $g.\Lambda(f) := \Lambda(g^{-1}.f)$

Then Λ is a χ^{-1}-eigendistribution in the sense that $g.\Lambda = \chi^{-1}(g)\Lambda$ for all $g \in G$ if and only if the associated outer Radon measure μ is a χ-eigenmeasure, in the sense that $\mu(xA) = \chi(x)\mu(A)$.

Proof. Again, the if part is clear. For the only if part, let μ be an outer Radon measure on X induced by Λ. As in Lemma 1.5, it suffices to say $\mu(xU) = \chi(x)\mu(U)$ for open U and $x \in G$, and we have

$$\mu(U) = \sup_{\substack{f \in C_c^+(X) \\ f \leqslant \mathbf{1}_U}} \int_X f d\mu = \sup_{\substack{f \in C_c^+(X) \\ f \leqslant \mathbf{1}_U}} \Lambda(f)$$

Thus for all $x \in G$,

$$\mu(xU) = \sup_{\substack{f \in C_c^+(X) \\ f \leqslant \mathbf{1}_{xU}}} \Lambda(f) = \sup_{\substack{f \in C_c^+(X) \\ x^{-1}.f \leqslant \mathbf{1}_U}} \Lambda(f) = \sup_{\substack{f \in C_c^+(X) \\ f \leqslant \mathbf{1}_U}} \Lambda(x.f) = \sup_{\substack{f \in C_c^+(X) \\ f \leqslant \mathbf{1}_U}} \chi(x)\Lambda(f) = \chi(x)\mu(U)$$

□

Theorem 1.17. Let G be an LCH group and $H \leqslant G$ a closed subgroup.

(i) There exists a nonzero invariant Radon measure ν on the quotient G/H if and only if $\Delta_G|_H = \Delta_H$.

(ii) The nonzero invariant Radon measure ν on G/H is, if exists, unique up to a positive scalar.

(iii) Given Haar measures on G and H, there is a unique choice for ν, such that for every $f \in C_c(G)$ one has the quotient integral formula

$$\int_G f(x)dx = \int_{G/H} \int_H f(xh)dhd\nu(x)$$

This quotient integral formula is even valid for every $f \in L^1(G)$. We will always assume this normalization and call the ensuing measure on G/H the **quotient measure**.

Proof.

(i) Assume first that there exists a nonzero invariant Radon measure ν on G/H. Define a linear functional $I : C_c(G) \to \mathbb{C}$ by

$$I(f) := \int_{G/H} f^H(x)d\nu(x)$$

It's a nonzero invariant positive linear functional on $C_c(G)$, so it is given by a Haar measure. Let us write

$$I(f) = \int_G f(x)dx$$

For $h_0 \in H$, one gets

$$\begin{aligned}
\Delta_G(h_0) \int_G f(x)dx &\overset{1.11.3}{=} \int_G f(xh_0^{-1})dx = \int_G R_{h_0^{-1}} f(x)dx \\
&= \int_{G/H} \int_H R_{h_0^{-1}} f(xh)dhd\nu(x) \\
&\overset{1.11.3}{=} \Delta_H(h_0) \int_{G/H} \int_H f(xh)dhd\nu(x) \\
&= \Delta_H(h_0) \int_G f(x)dx
\end{aligned}$$

Choose $f \in C_c(G)$ such that $\int_G f(x)dx \neq 0$; then $\Delta_G|_H = \Delta_H$. Conversely, assume $\Delta_G|_H = \Delta_H$. Define a linear functional $I : C_c(G/H) \to \mathbb{C}$ by

$$I(g) = \int_G f(x)dx$$

where $f \in C_c(G)$ is such that $f^H = g$. To show this is well-defined, we must show if $f \in C_c(G)$ with $f^H = 0$, then $\int_G f(x)dx = 0$. For this, let $\phi \in C_c(G)$. Then

$$\begin{aligned}
0 &= \int_G \int_H f(xh)\phi(x)dhdx \overset{\text{Fubini}}{=} \int_H \int_G \phi(x)f(xh)dxdh \\
&\overset{1.11.3}{=} \int_H \left(\Delta_G(h^{-1}) \int_G \phi(xh^{-1})f(x)dx \right) dh \\
&\overset{\substack{\text{Fubini\&} \\ \text{assumption}}}{=} \int_G \left(\int_H \Delta_H(h^{-1})\phi(xh^{-1})dh \right) f(x)dx \\
&\overset{1.11.4}{=} \int_G \left(\int_H \phi(xh)dh \right) f(x)dx \\
&= \int_G \phi^H(x)f(x)dx
\end{aligned}$$

We can find ϕ with $\phi^H \equiv 1$ on the support of f, and thus $\int_G f(x)dx = 0$. Hence $I : C_c(G/H) \to \mathbb{C}$ is well-defined, and is clear that it is nonzero and positive. I is invariant for $L_x g = (L_x f)^H$ for each $x \in G$. Hence it follows from Lemma 1.16 (with trivial character χ) that I comes from an invariant Radon measure. In particular, the quotient integral formula is valid for $f \in C_c(G)$.

(ii) Having a nonzero invariant Radon measure ν on G/H is the same as having a nonzero positive invariant linear functional $I : C_c(G/H) \to \mathbb{C}$. The quotient integral formula implies we can normalize I so that the following diagram commutes

$$\begin{tikzcd}
C_c(G) \arrow[r] \arrow[dr, "\int_G dx"'] & C_c(G/H) \arrow[d, "I"] \\
& \mathbb{C}
\end{tikzcd}$$

where the upper horizontal arrow is the same as in Lemma 1.15. The same lemma says it is surjective, so the value of I is completely determined by the Haar integral on G; this implies (ii).

(iii) The first part results from the end of the proof of (i) and (ii). It remains to show the quotient integral formula for the integrable function f on G. We may assume $f \geqslant 0$. By monotone convergence theorem we can assume f is a step function, and by linearity we can even reduce to the case $f = \mathbf{1}_A$ where A is measurable of finite Haar measure. We have to show

- $\mathbf{1}_A^H$ is measurable on G/H;
- $\int_{G/H} \mathbf{1}_A^H d\nu(x) = \int_G \mathbf{1}_A dx$.

We start with the case $A = U$ being open. By Lemma U.23, the function

$$g(xH) := \sup_{\substack{\phi \in C_c(G) \\ 0 \leqslant \phi \leqslant \mathbf{1}_U}} \int_H \phi(xh)dh$$

is measurable on G/H and coincides with $\mathbf{1}_U^H$. A repeated use of Urysohn's Lemma and Lemma U.23 shows

$$\begin{aligned}
\int_{G/H} \int_H \mathbf{1}_U(xh)dhd\nu(x) &= \int_{G/H} \int_H \sup_{0 \leqslant \phi \leqslant \mathbf{1}_U} \phi(xh)dhd\nu(x) \\
&= \sup_{0 \leqslant \phi \leqslant \mathbf{1}_U} \int_{G/H} \int_H \phi(xh)dhd\nu(x) \\
&= \sup_{0 \leqslant \phi \leqslant \mathbf{1}_U} \int_G \phi(x)dx = \int_G \sup_{0 \leqslant \phi \leqslant \mathbf{1}_U} \phi(x)dx \\
&= \int_G \mathbf{1}_U(x)dx
\end{aligned}$$

so that the case $A = U$ is proved. If $A = K$ is a compact set, let V be a relative compact open neighborhood of K. Then $\mathbf{1}_K = \mathbf{1}_V - \mathbf{1}_{V \setminus K}$ so that this case is done. For general A of finite measure and given $n \in \mathbb{N}$, by regularity and Lemma U.1, there are compact K_n and open U_n with $K_n \subseteq A \subseteq U_n$ and $\mu(U_n \setminus K_n) < \frac{1}{n}$. We can assume the K_n are increasing and the U_n are decreasing. Let $g = \lim_{n \to \infty} \mathbf{1}_{K_n}^H$ and $h = \lim_{n \to \infty} \mathbf{1}_{U_n}^H$; then

- g, h are integrable on G/H;
- $0 \leqslant g \leqslant \mathbf{1}_A^H \leqslant h$;
- $h - g \geqslant 0$ has integral zero.

Hence

$$\int_{G/H} \mathbf{1}_A^H dx = \int_{G/H} g(x)dx = \lim_{n \to \infty} \int_{G/H} \mathbf{1}_{K_n}^H(x)dx = \lim_{n \to \infty} \int_G \mathbf{1}_{K_n}(x)dx = \int_G \mathbf{1}_A(x)dx$$

□

Corollary 1.17.1. Suppose $H \leqslant G$ is a closed subgroup such that there exists a nonzero invariant Radon measure on G/H. Let $f : G \to \mathbb{C}$ be a measurable function such that $A = \{x \in G \mid f(x) \neq 0\}$ is σ-finite. If the iterated integral

$$\int_{G/H} \int_H |f(xh)|dhdx < \infty$$

exists, then $f \in L^1(G)$.

Proof. It suffices to show $|f| \in L^1(G)$. Choose an increasing sequence of measurable sets $(A_n)_{n \in \mathbb{N}}$ with finite measure whose union is A, and define $f_n : G \to \mathbb{C}$ by $f_n = \min\{|f| \cdot \mathbf{1}_{A_n}, n\}$. Then $(f_n)_{n \in \mathbb{N}}$ is an increasing sequence of integrable functions that converges pointwise to $|f|$. Then by Theorem

$$\int_G f_n(x)dx = \int_{G/H} \int_H f_n(xh)dhdx \leqslant \int_{G/H} \int_H |f_n(xh)|dhdx < \infty$$

The result follows from monotone convergence theorem. $\square$

Corollary 1.17.2.

1. If $H \trianglelefteq G$ is a normal closed subgroup, then $\Delta_G|_H = \Delta_H$.

2. Let $H = \ker(\Delta_G : G \to \mathbb{R}_{>0})$ be the kernel. Then H is unimodular.

Proposition 1.18. Let G be an LCH group, $K \leqslant G$ a compact subgroup and $H \leqslant G$ a closed subgroup such that $G = HK$. Then one can arrange the Haar measure on $G,$, H, K in a way that for every $f \in L^1(G)$ one has

$$\int_G f(x)dx = \int_H \int_K f(hk)dkdh$$

Proof. The group $H \times K$ acts on G by $(h,k).g = hgk^{-1}$. This operation is transitive so that we have the identification $G = \dfrac{H \times K}{H \cap K}$, where we embed $H \cap K$ into $H \times K$ diagonally. Now

- the group $H \cap K$ is compact, so its modular character is trivial.

- The modular character of $H \times K$ is trivial on $H \cap K$.

By Theorem, there is a unique $H \times K$-invariant Radon measure on G up to scaling. We claim the Haar measure on G is also $H \times K$-invariant. Indeed, the Haar measure on G is H-invariant for H acts as left-translation. Since K is compact, we have $\Delta_G|_K \equiv 1$ so that

$$\int_G f(xk)dx = \Delta_G(k^{-1}) \int_G f(x)dx = \int_G f(x)dx$$

for all $k \in K$, $f \in C_c(G)$ by Theorem 1.11.3.

Now by uniqueness of the Haar measure, the Haar measure on G can be normalized in the way that the quotient integral formula

$$\int_H \int_K f(hk)dkdh = \int_{H \times K} f(hk)dh \otimes dk = \int_G \int_{H \cap K} f(x\ell)d\ell dx$$

holds for $f \in L^1(G)$. Finally, we see

$$\int_G \int_{H \cap K} f(x\ell)d\ell dx = \int_{H \cap K} \int_G f(x\ell)dxd\ell = \int_{H \cap K} \left(\Delta_G(\ell^{-1}) \int_G f(x)dx \right) d\ell = \int_G f(x)dx$$

$\square$

Example 1.19. Let $\mathbb{H}$ denote the (open) upper half complex plane. The action of $\mathrm{SL}_2(\mathbb{R})$ on $\mathbb{C}$ by Möbius transformation stabilizes $\mathbb{H}$, and induces a homeomorphism (in fact, a diffeomorphism) $\mathrm{SL}_2(\mathbb{R})/\mathrm{SO}(2) \cong \mathbb{H}$; here think of $\mathrm{SO}(2)$ as the stabilizer of i.

If we write $g = \begin{pmatrix} x & y \\ z & w \end{pmatrix} \in \mathrm{SL}_2(\mathbb{R})$, a Haar measure dg on $\mathrm{SL}_2(\mathbb{R})$ is given by $dg = dxdydzdw$, where the latter four measures are the Lebesgue measures on $\mathbb{R}$. Also, every element in $\mathrm{SO}(2)$ has the form

$$k_\theta = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix} \qquad (\theta \in \mathbb{R}).$$

In particular, the Haar measure $d\theta$ on $\mathrm{SO}(2)$ can be given as follows: for $f \in C(\mathrm{SO}(2))$, the integration against $d\theta$ is

$$\int_0^{2\pi} f(k_\theta) d\theta.$$

We have specified the Haar measures on $\mathrm{SL}_2(\mathbb{R})$ and $\mathrm{SO}(2)$. Since both groups are unimodular, there exists a unique $\mathrm{SL}_2(\mathbb{R})$-measure dh on $\phi : \mathrm{SL}_2(\mathbb{R})/\mathrm{SO}(2) \cong \mathbb{H}$ determined by the integration formula

$$\int_{\mathrm{SL}_2(\mathbb{R})} f(g) dg = \int_{\mathbb{H}} \int_0^{2\pi} f(\phi^{-1}(h) k_\theta) d\theta dh,$$

valid for all $f \in C_c(\mathrm{SL}_2(\mathbb{R}))$. We aim to find an explicit expression of dh in terms of known measures.

In fact, it is well-known that the hyperbolic measure $\dfrac{dxdy}{y^2}$ on $\mathbb{H}$ is $\mathrm{SL}_2(\mathbb{R})$-invariant.

1.4.2 Eigenmeasure

Let G be an LCH group. Instead of considering left invariant Radon measures on G, for a continuous group homomorphism $\chi : G \to \mathbb{R}_{>0}$, let us consider the "$\chi$-eigenmeasure" μ, in the sense that $\mu(xA) = \chi(x)\mu(A)$ for all $x \in G$ and measurable $A \subseteq G$.

Lemma 1.20. For each a continuous group homomorphism $\chi : G \to \mathbb{R}_{>0}$, there exists a unique χ-eigen Radon measure, up to a positive scalar, on G.

Proof. For a function f on G, denote by $f\chi$ the function on G defined by $f\chi(x) := f(x)\chi(x)$. Then we have a linear isomorphism

$$\begin{aligned} \mathrm{Hom}_{\mathbb{C}}(C_c(G), \mathbb{C})^G &\longrightarrow \mathrm{Hom}_{\mathbb{C}}(C_c(G), \mathbb{C})^{\chi^{-1}} \\ \Lambda &\longmapsto \Lambda_\chi : f \mapsto \Lambda(f\chi) \end{aligned}$$

that sends positive to positive. The lemma now follows from Lemma 1.16 and the uniqueness of Haar measures. □

Theorem 1.21. Let $H \leqslant G$ be a closed subgroup and $\chi : G \to \mathbb{R}_{>0}$ a continuous group homomorphism. There exists a χ-eigen Radon measure on G/H if and only if $\chi(h)\Delta_G(h) = \Delta_H(h)$ for all $h \in H$.

Proof. Suppose μ is a χ-eigen Radon measure on G/H. For $f \in C_c(G)$, define $I : C_c(G) \to \mathbb{C}$ by

$$I(f) = \int_{G/H} (f\chi^{-1})^H d\mu.$$

This is a positive linear function on $C_c(G)$, and for $g \in G$,

$$g.I(f) = \int_{G/H} \int_H f(gxh)\chi^{-1}(xh)dhd\mu(x) = \chi(g) \int_{G/H} g^{-1}.(f\chi)^H d\mu = \chi(g)\chi^{-1}(g)I(f) = I(f).$$

Thus $I(f)$ is given by a Haar measure ν on G, i.e.,

$$\int_{G/H} \int_H f(xh)\chi^{-1}(xh)dhd\mu(x) = \int_G f(x)d\nu(x)$$

for all $f \in C_c(G)$. Now for $h_0 \in H$, one gets

$$\begin{aligned}\Delta_G(h_0) \int_G f(x)d\nu(x) &\overset{1.11.3}{=} \int_G f(xh_0^{-1})d\nu(x) = \int_G R_{h_0^{-1}} f(x)dx = \int_{G/H} \int_H R_{h_0^{-1}} f(xh)\chi^{-1}(xh)dhd\mu(x) \\ &\overset{1.11.3}{=} \chi^{-1}(h_0)\Delta_H(h_0) \int_{G/H} \int_H f(xh)dhd\mu(x) = \chi^{-1}(h_0)\Delta_H(h_0) \int_G f(x)d\nu(x)\end{aligned}$$

This shows $\chi(h_0)\Delta_G(h_0) = \Delta_H(h_0)$. The converse is proved in the same way as the proof of Theorem 1.17.(i) with slight modification. □

1.4.3 Quasi-invariant measures

Let G be an LCH group and $H \leqslant G$ a closed subgroup. Let μ be a Radon measure on the homogeneous space G/H. For $g \in G$ define the translate μ_g of μ by the formula $\mu_g(A) := \mu(gA)$.

Definition. Let the notations be as in above.

1. μ is called **quasi-invariant** if all the measures μ_x have the σ-ideal of zero sets; equivalently, μ is quasi-invariant if the μ_x are mutually absolutely continuous.

2. μ is called **strongly quasi-invariant** if there exists a continuous map $\lambda : G \times G/H \to \mathbb{R}_{>0}$ such that $d\mu_x(p) = \lambda(x,p)d\mu(p)$ for all $x \in G$ and $p \in G/H$; equivalently, μ is strongly quasi-invariant if $\mu$is quasi-invariant and the Radon-Nikodym derivatives $(d\mu_x/d\mu)(p)$ is jointly continuous in (x,p).

Lemma 1.22. If μ is a nonzero quasi-invariant measure on G/H, then $\mu(U) > 0$ for every nonempty open set U.

Proof. Suppose otherwise. Say $\mu(U) = 0$ for some open set U. Then $\mu(gU) = 0$ for all $g \in G$ as μ_g and μ are mutually absolutely convergent. By weakly inner regularity it suffices to show μ vanishes on every compact set. But this is clear. □

Lemma 1.23. Let V be a relatively compact symmetric open unit-neighborhood of G. Then there exists a set $A \subseteq G$ such that

(i) for every $g \in G$ there exists $a \in A$ with $gH \cap Va \neq \varnothing$, and

(ii) if $K \subseteq G$ is compact, $\#\{a \in A \mid KH \cap \overline{V}a \neq \varnothing\} < \infty$.

Proof. By Zorn's lemma there exists a maximal set $A \subseteq G$ such that if $a, b \in A$ then $a \neq VbH$ (note that this condition is symmetric in a and b). For any $g \in G$, gH intersects some Va, for otherwise $g \notin VaH$ for all $a \in A$, contradicting the maximality. Also, if $K \subseteq G$ is compact and $KH \cap \overline{V}a \neq \varnothing$ for infinitely many a, then we can find $(a_n)_n$ with each $a_n \in A$ distinct to each other and $(h_n)_n$ in H such that $a_n h_n \in \overline{V}K$ for all n. Since $\overline{V}K$ is compact, by passing to subsequence we can assume $a_n h_n$ converges to, say, $z \in \overline{V}K$. Pick a symmetric unit-neighborhood W in G such that $W^2 \subseteq V$. Then there are integers $n \neq m$ such that $a_n h_n$, $a_m h_m \in Wz$, and thus $a_n h_n \in V a_m h_m$. But then $a_n \in V a_m H$, a contradiction to the definition of A. □

Lemma 1.24. There exists a continuous function $f : G \to [0, \infty)$ satisfying

(i) $\{y \in G \mid f(y) > 0\} \cap gH \neq \varnothing$ for all $g \in G$, and

(ii) $\operatorname{supp} f \cap KH$ is compact for every compact $K \subseteq G$.

Proof. Pick symmetric $h \in C_c^+(G)$ with $g(1) > 0$, let $V = h^{-1}(0, \infty)$, choose $A \subseteq G$ as in Lemma 1.23 for this V and set

$$f(g) = \sum_{a \in A} h(ga^{-1}).$$

By Lemma 1.23.(ii), this sum is finite when g is in a fixed compact subset of G, so f defines a continuous map on G. Since

$$\operatorname{supp} f = \overline{\bigcup_{a \in A} Va} \subseteq \bigcup_{a \in A} \overline{V}a$$

for any compact $K \subseteq G$, $\operatorname{supp} f \cap KH$ is contained in a finite union of the $\overline{V}a$ by Lemma 1.23.(ii) again, so $\operatorname{supp} f \cap KH$ is compact. Finally, by Lemma 1.23.(i), $\{y \in G \mid f(y) > 0\} = \bigcup_{a \in A} Va$ intersects every coset gH ($g \in G$). □

Definition. A **rho-function** for the pair (G, H) is a continuous map $\rho : G \to \mathbb{R}_{>0}$ such that

$$\rho(gh) = \frac{\Delta_H(h)}{\Delta_G(h)} \rho(g)$$

for all $g \in G$, $h \in H$.

Proposition 1.25. For any LCH group G and any closed subgroup $H \leqslant G$, (G, H) admits a rho-function.

Proof. Let f be as in Lemma 1.24 and set

$$\rho(g) := \int_H \frac{\Delta_G(h)}{\Delta_H(h)} f(gh)dh.$$

By Lemma 1.24.(ii), this integral converges for each $g \in G$ and defines a continuous function on G, and by (i) it is positive. Moreover,

$$\rho(gh_0) = \int_H \frac{\Delta_G(h)}{\Delta_H(h)} f(gh_0h)dh = \int_H \frac{\Delta_G(h_0^{-1}h)}{\Delta_H(h_0^{-1}h)} f(gh)dh = \frac{\Delta_H(h_0)}{\Delta_G(h_0)}\rho(g).$$

□

Lemma 1.26. If $f \in C_c(G)$ and $f^H = 0$, then the integral $\int_G f(g)\rho(g)dg = 0$ for every rho-function ρ.

Proof. By Lemma 1.15, we can find $\phi \in C_c^+(G)$ with $\phi^H = 1$ on $\pi(\operatorname{supp} f)$. Then

$$\begin{aligned} 0 &= \int_G \int_H \rho(g)\phi(g)f(gh^{-1})\Delta_H(h^{-1})dhdg \overset{1.11.3}{=} \int_H \int_G \rho(gh)\phi(gh)f(g)\Delta_H(h^{-1})\Delta_G(h^{-1})dgdh \\ &= \int_G \int_H \rho(g)\phi(gh)f(g)dhdg = \int_G f(g)\rho(g)\phi^H(g)dg = \int_G f(g)\rho(g)dg. \end{aligned}$$

Here ϕ is to make the integrals absolutely convergent. □

Theorem 1.27. Every strongly quasi-invariant measure on G/H arises from a rho-function constructed in Proposition 1.25, and all such measures are strongly equivalent.

Proof. Suppose μ is strongly equivalent, so that $(d\mu_x/d\mu)(p) = \lambda(x,p)$ is a positive continuous function on $G \times G/H$. For $x, y \in G$, since $\mu_{xy} = (\mu_x)_y$, the "chain rule" for the Radon Nikodym derivatives implies

$$\lambda(xy,p) = \lambda(x,yp)\lambda(y,p)$$

μ-a.e. in p. Since both sides are continuous functions, this identity must hold for every p by Lemma 1.22.

If $f \in C_c(G)$ and $y \in G$, we have

$$\begin{aligned} \int_{G/H}\int_H f(y^{-1}xh)\lambda(xh,H)^{-1}dhd\mu(xH) &= \int_{G/H}\int_H f(xh)\lambda(yxh,H)^{-1}\lambda(y,xH)dhd\mu(xH) \\ &= \int_{G/H}\int_H f(xh)\lambda(xh,H)^{-1}dhd\mu(xH) \end{aligned}$$

by the above identity. Hence

$$f \mapsto \int_{G/H}\int_H f(xh)\lambda(xh,H)^{-1}dhd\mu(xH)$$

defines a left-invariant positive linear function on $C_c(G)$, so there exists $c > 0$ such that

$$\int_{G/H}\int_H f(xh)\lambda(xh,H)^{-1}dhd\mu(xH) = c\int_G f(x)dx$$

Define $\rho(x) = c\lambda(x, H)$. Replacing f by $f\lambda(\cdot, H)$, we obtain

$$\int_{G/H}\int_H f(xh)dhd\mu(xH) = \int_G f(x)\rho(x)dx.$$

We show ρ is a rho-function. Indeed, for $h_0 \in H$,

$$\begin{aligned}\int_G f(x)\rho(xh_0)dx &= \Delta_G(h_0^{-1})\int_G R_{h_0^{-1}}f(x)\rho(x)dx \\ &= \Delta_G(h_0^{-1})\int_{G/H}\int_H f(xhh_0^{-1})dhd\mu(xH) \\ &= \Delta_G(h_0^{-1})\Delta_H(h_0)\int_{G/H}\int_H f(xh)dhd\mu(xH) = \Delta_G(h_0^{-1})\Delta_H(h_0)\int_G f(x)\rho(x)dx\end{aligned}$$

holds for every $f \in C_c(G)$, so

$$\rho(xh) = \frac{\Delta_H(h)}{\Delta_G(h)}\rho(x)$$

holds for every $x \in G$, $h \in H$. Since ρ is continuous and positive, this shows ρ is a rho-function.

It remains to show the last assertion. Suppose μ and μ' are strongly equivalent measures with associated rho-function ρ and ρ'. The functional equation for rho-function implies the ratio ρ'/ρ defines a positive continuous function ϕ on G/H. For $f \in C_c(G)$, we have $(f\rho'/\rho)^H = f^H\phi$, so

$$\int_{G/H} f^H d\mu' = \int_G f\rho' = \int_G f(\rho'/\rho)\rho = \int_{G/H} f^H\phi d\mu.$$

This proves $d\mu' = \phi d\mu$. □

1.4.4 Homomorphisms and measures

Lemma 1.28. Let X, Y be LCH spaces and $f : X \to Y$ a proper continuous map. If μ is a Radon measure on X, then the pushforward measure $f_*\mu$ is a Radon measure on Y.

Proof. Local finiteness is clear (the properness is used here). For U open in Y,

$$f_*\mu(U) = \mu(f^{-1}(U)) = \sup_{K \underset{\text{cpt}}{\subseteq} f^{-1}(U)} \mu(K) \leqslant \sup_{K \underset{\text{cpt}}{\subseteq} U} \mu(f^{-1}(K)) \leqslant \mu(f^{-1}(U)) = f_*\mu(U).$$

so $f_*\mu$ is weakly inner regular (the properness is not used here). Finally, for B Borel in Y, by Corollary T.7.1 we have

$$f_*\mu(B) = \mu(f^{-1}(B)) = \inf_{f^{-1}(B)\subseteq U \underset{\text{open}}{\subseteq} X} \mu(U) \geqslant \inf_{B\subseteq V \underset{\text{open}}{\subseteq} U} \mu(f^{-1}(V)) \geqslant \mu(f^{-1}(B)) = f_*\mu(B)$$

so $f_*\mu$ is outer regular. □

Proposition 1.29. Let G and H be two LCH groups and $\varphi : G \to H$ a surjective open continuous homomorphism with $\ker\varphi$ compact. If A, B are two subset of H with finite measure, then

$$\frac{\mathrm{vol}(A)}{\mathrm{vol}(B)} = \frac{\mathrm{vol}(\varphi^{-1}(A))}{\mathrm{vol}(\varphi^{-1}(B))}$$

Proof. We show that φ is a proper map. For this, under our assumption we have $G/\ker\varphi \cong H$ as topological groups, so it suffices to show $\pi : G \to G/\ker\varphi$ is proper. Let $C \subseteq G/\ker\varphi$ be a compact subset. By Lemma 1.15 we can find compact $K \subseteq G$ with $\pi(K) = C$. We claim $\pi^{-1}(C) = K\ker\varphi$.

- As $\pi(K\ker\varphi) = \pi(K) = C$, we have $K\ker\varphi \subseteq \pi^{-1}(C)$.
- Let $x \in \pi^{-1}(C)$. Then $\pi(x) \in C$, so we can find $k \in K$ such that $x\ker\varphi = k\ker\varphi$, or $k^{-1}x \in \ker\varphi$. Thus $x \in k\ker\varphi \subseteq K\ker\varphi$, so that $\pi^{-1}(C) \subseteq K\ker\varphi$.

Since $\ker\varphi$ is compact, being a product of two compact sets, $\pi^{-1}(C)$ is compact. Thus φ is a proper map.

Let μ be a Haar measure on G and consider the pushforward measure $\varphi_*\mu$ on H. The previous lemma shows $\varphi_*\mu$ is a Radon measure, and it is nontrivial (perhaps by surjectivity). We show this is left invariant. Let $A \subseteq H$ be a measurable subset and $h \in H$. Let g be in the fibre of h. Then

$$\varphi_*\mu(hA) = \mu(\varphi^{-1}(hA)) = \mu(g\varphi^{-1}(A)) = \mu(\varphi^{-1}(A)) = \varphi_*\mu(A)$$

Thus $\varphi_*\mu$ is a Haar measure on H, and the proposition follows. □

Remark 1.30. If G is σ-compact, then a continuous surjective homomorphism is automatically open. See Theorem 5.13.

1.5 Convolution

Definition. An **algebra over** $\mathbb{C}$ is a complex vector space $\mathcal{A}$ together with a map $\mathcal{A} \times \mathcal{A} \to \mathcal{A}$, called the product of multiplication and written $(a, b) \mapsto ab$, which is bilinear and associative. $\mathcal{A}$ is called a **commutative algebra** if in addition the multiplication is commutative.

Definition. Let G be an LCH group. For two measurable $f, g : G \to \mathbb{C}$ define the **convolution product** as

$$f * g(x) := \int_G f(y)g(y^{-1}x)dy$$

whenever the integral exists.

Theorem 1.31. Let $f, g \in L^1(G)$. Then

(i) The integral $f * g$ exists almost everywhere in x and defines a function in $L^1(G)$.

(ii) The L^1-norm satisfies $\|f * g\|_1 \leqslant \|f\|_1 \|g\|_1$.

The convolution product endows $L^1(G)$ with the structure of an algebra.

Proof. Note that f, g are measurable in the sense that the preimages of Borel sets under f, g are in the completed Borel σ-algebra. Define the function

$$\begin{aligned} \psi : G \times G &\longrightarrow \mathbb{C} \\ (y, x) &\longmapsto f(y)g(y^{-1}x) \end{aligned}$$

We can write ψ as

$$\begin{aligned} \psi : G \times G &\xrightarrow{\alpha} G \times G \xrightarrow{f \times g} \mathbb{C} \times \mathbb{C} \xrightarrow{\mu} \mathbb{C} \\ (y, x) &\longmapsto (y, y^{-1}x) \end{aligned}$$

where $\mu : \mathbb{C} \times \mathbb{C} \to \mathbb{C}$ is multiplication. Since $f \times g$ and μ are measurable, to show ψ is measurable it suffices to show that α is. Since α is continuous, it is Borel-measurable, and we must show the preimage under α of a null set is again null (completeness). But this follows from the formula, valid for each $\phi \in C_c(G \times G)$.

$$\int_{G \times G} \phi(y, x) dx \otimes dy \overset{\text{Fubini}}{=} \int_G \int_G \phi(y, x) dx dy \overset{\text{inv}}{=} \int_G \int_G \phi(y, y^{-1}x) dx dy$$

Let $S(f)$ and $S(g)$ be the supports and f and g, respectively. By Corollary 1.3.1.4, $S(f)$ and $S(g)$ are σ-compact. The support of ψ is contained in $S(f) \times S(f)S(g)$, so it's also σ-compact. Hence by Fubini's Theorem,

$$\begin{aligned} \|f * g\|_1 &\leqslant \int_G \int_G |f(y)g(y^{-1}x)| dy dx = \int_G \int_G |f(y)g(y^{-1}x)| dx dy \\ &= \int_G \int_G |f(y)g(x)| dx dy \\ &= \|f\|_1 \|g\|_1 < \infty \end{aligned}$$

and Fubini's theorem again shows that $\psi(\cdot, x)$ is integrable a.e. in x and that $f * g \in L^1(G)$.

It remains to show the associativity and bilinearity. The bilinearity is clear. For associativity, let $f, g, h \in L^1(G)$. Then

$$\begin{aligned} (f * g) * h(x) &= \int_G (f * g)(y) h(y^{-1}x) dy = \int_G \int_G f(z) g(z^{-1}y) h(y^{-1}x) dz dy \\ &= \int_G \int_G f(z) g(z^{-1}y) h(y^{-1}x) dy dz = \int_G \int_G f(z) g(y) h(y^{-1}z^{-1}x) dy dz \\ &= \int_G f(z)(g * h)(z^{-1}x) dz = f * (g * h)(x) \end{aligned}$$

Using Fubini's Theorem to compute the integral is valid, which is justified in the preceding paragraphs. $\square$

Proposition 1.32. Let G be unimodular. Let $1 \leqslant p, q, r \leqslant \infty$ with $\frac{1}{p} + \frac{1}{q} = \frac{1}{r} + 1$. Then for $f \in L^p(G)$ and $g \in L^q(G)$, one has $f * g \in L^r(G)$ and $\|f * g\|_r \leqslant \|f\|_p \|g\|_q$

Proof.

$$\begin{aligned}|f*g(x)| &\leqslant \int_G |f(y)g(y^{-1}x)|dy \\ &= \int_G |f(y)|^{1-\frac{p}{r}}|g(y^{-1}x)|^{1-\frac{q}{r}}|f(y)^{\frac{p}{r}}g(y^{-1}x)^{\frac{q}{r}}|dy \\ &= \int_G (|f(y)|^p)^{\frac{r-p}{rp}}(|g(y^{-1}x)|^q)^{\frac{r-q}{rq}}|f(y)^p g(y^{-1}x)^q|^{\frac{1}{r}}dy\end{aligned}$$

We invoke Hölder's inequality. One checks that $r \geqslant p, q$ and

$$\frac{r-p}{rp}+\frac{r-q}{rq}+\frac{1}{r}=\frac{1}{r}\left(\frac{r}{p}-1+\frac{r}{q}-1+1\right)=\frac{1}{r}(1+r-1)=1$$

Hence

$$|f*g(x)| \leqslant \left(\int_G |f(y)|^p dy\right)^{\frac{r-p}{rp}}\left(\int_G |g(y^{-1}x)|^q dy\right)^{\frac{r-q}{rq}}\left(\int_G |f(y)^p g(y^{-1}x)^q|dy\right)^{\frac{1}{r}}$$

Taking r-power both sides, we see

$$|f*g(x)|^r \leqslant \|f\|_p^{r-p}\|g\|_q^{r-q}\int_G |f(y)^p g(y^{-1}x)^q|dy$$

Note that f^p, $g^q \in L^1(G)$, so by Theorem 1.31 the right hand side is integrable, so that

$$\|f*g\|_r^r \leqslant \|f\|_p^{r-p}\|g\|_q^{r-q}\|f\|_p^p\|g\|_p^q = \|f\|_p^r\|g\|_q^r$$

and thus $\|f*g\|_r \leqslant \|f\|_p\|g\|_q$. The function ψ defined in the previous theorem then lie in $L^r(G)$, and by Fubini's theorem that $f*g \in L^r(G)$. (Fubini's theorem is available, as explained in Theorem 1.31.) □

Lemma 1.33. For $f, g \in L^1(G)$ and $y \in G$, one has

$$R_y(f*g) = f*(R_y g) \qquad L_y(f*g) = (L_y f)*g$$

Proof.

$$R_y(f*g)(x) = (f*g)(xy) = \int_G f(z)g(z^{-1}xy)dz = \int_G f(z)R_y g(z^{-1}x)dx = f*(R_y g)(x)$$

Likewise for L. □

Theorem 1.34. The algebra $L^1(G)$ is commutative if and only if G is abelian.

Proof. Suppose G is abelian. Then

$$(f*g)(x) = \int_G f(y)g(y^{-1}x)dy \overset{\text{inv}}{=} \int_G f(xy)g(y^{-1})dy \overset{1.11.4}{=} \int_G \Delta_G(y^{-1})f(xy^{-1})g(y)dy = (g*f)(x)$$

for $\Delta_G \equiv 1$. Suppose $L^1(G)$ is commutative and let $f, g \in L^1(G)$. For $x \in G$ we have

$$\begin{aligned} 0 = f * g(x) - g * f(x) &= \int_G \big(f(y)g(y^{-1}x) - g(y)f(y^{-1}x)\big)\, dy \\ &\overset{\text{inv}}{=} \int_G \big(f(xy)g(y^{-1}) - g(y)f(y^{-1}x)\big)\, dy \\ &\overset{1.11.4}{=} \int_G g(y)\left(\Delta_G(y^{-1})f(xy^{-1}) - f(y^{-1}x)\right) dy \end{aligned}$$

This holds for all $g \in L^1(G)$, so that $\Delta_G(y^{-1})f(xy^{-1}) - f(y^{-1}x) = 0$ for all $f \in C_c(G)$. Taking $x = 1$ gives $\Delta_G \equiv 1$, and thus G is unimodular and $f(xy^{-1}) = f(y^{-1}x)$ for every $f \in C_c(G)$, $x, y \in G$. This shows G is abelian. (If $xy^{-1} \neq y^{-1}x$ for some x, y, apply Urysohn's Lemma to find an f that make the equality fails.) □

1.5.1 Dirac Net

Definition. A **Dirac function** is a function $\phi \in C_c(G)$ such that

(i) $\phi \geqslant 0$,

(ii) $\displaystyle\int_G \phi(x)dx = 1$ and

(iii) $\phi(x^{-1}) = \phi(x)$ for all $x \in G$.

A **Dirac family** is a family $(\phi_U)_U$ of Dirac functions indexed by the set $\mathcal{U}$ of all unit-neighborhoods U such that $\operatorname{supp}\phi_U \subseteq U$.

Lemma 1.35.

1. The convolution product of two Dirac functions is a Dirac function.

2. To every unit-neighborhood U there exists a Dirac function ϕ_U such that ϕ_U and $\phi_U * \phi_U$ have support inside U.

Proof.

1. Let ϕ, ψ be Dirac functions. That $\phi * \psi \geqslant 0$ is clear.

$$\int_G \phi*\psi(x)dx = \int_G\int_G \phi(y)\psi(y^{-1}x)dydx \overset{\text{Fubini}}{=} \int_G\int_G \phi(y)\psi(y^{-1}x)dxdy \overset{\text{inv}}{=} \left(\int_G \phi(y)dy\right)\left(\int_G \psi(x)dx\right) = 1$$

$$\phi * \psi(x^{-1}) = \int_G \phi(y)\psi(y^{-1}x^{-1})dy \overset{\text{inv}}{=} \int_G \phi(xy)\psi(y^{-1})dy = \int_G \psi(y^{-1}x^{-1})\psi(y)dy = \phi * \psi(x)$$

2. Let U be a given unit-neighborhood and let $W \subseteq U$ be any symmetric unit-neighborhood such that $W^2 \subseteq U$. By Urysohn's Lemma, there is an $h \in C_c(G)$ with $0 \neq h \geqslant 0$ and $\operatorname{supp}(h) \subseteq W$. Set $\phi_U(x) = h(x) + h(x^{-1})$ and normalize this function so that ϕ_U has integral 1. Then $\operatorname{supp}(\phi_U * \phi_U) \subseteq \operatorname{supp}(\phi_U)^2 \subseteq W^2 \subseteq U$.

□

1.5.2 Regular Representations

Lemma 1.36. For given $1 \leqslant p < \infty$ and $g \in L^p(G)$, the maps $y \mapsto L_y g$ and $y \mapsto R_y g$ are continuous maps from G to $L^p(G)$. Moreover, $y \mapsto L_y g$ is uniformly continuous, and if G is unimodular, so is $y \mapsto R_y g$.

Proof. By the invariance of Haar integrals,

$$\|L_y g - L_x g\|_p = \|L_{x^{-1}y} g - g\|_p$$

so the uniform continuity will follow once the continuity at e is established. Likewise,

$$\|R_y g - R_x g\|_p = \Delta(x^{-1})^{1/p} \|R_{x^{-1}y} g - g\|_p$$

so the last assertion will follows for the same reason.

Now we prove the continuity at the unit element e. First consider the case $g \in C_c(G)$. Choose $\varepsilon > 0$ and put $K = \operatorname{supp}(g)$; then $\operatorname{supp}(L_y g) = yK$. Let U_0 be a compact symmetric unit-neighborhood. Then for $y \in U_0$, one has $\operatorname{supp}(L_y g) \subseteq U_0 K$. By uniform continuity there exists a unit-neighborhood $U \subseteq U_0$ such that for $y \in U$, $\|L_y g - g\|_G < \dfrac{\varepsilon}{\operatorname{vol}(U_0 K)^{\frac{1}{p}}}$. Then for $y \in U$,

$$\|L_y g - g\|_p = \left(\int_G |g(y^{-1}x) - g(x)|^p dx \right)^{\frac{1}{p}} < \varepsilon$$

For general g, pick $f \in C_c(G)$ with $\|f - g\|_p < \varepsilon/3$. Choose a unit-neighborhood U with $\|f - L_y f\|_p < \varepsilon/3$ for every $y \in U$. Then for $y \in U$,

$$\|g - L_y g\|_p \leqslant \|g - f\|_p + \|f - L_y f\|_p + \|L_y f - L_y g\|_p < \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon$$

Note we use the fact $\|L_y f - L_y g\|_p = \|f - g\|_p$. For case for the right translation is similar, except in the last step we use $\|R_y f - R_y g\|_p = \Delta(y^{-1})^{1/p} \|f - g\|_p$ instead. □

Lemma 1.37. Let $H \leqslant G$ be a closed subgroup of an LCH group G such that G/H admits a nonzero G-invariant Radon measure; fix such a measure. For given $1 \leqslant p < \infty$ and $g \in L^p(G/H)$, the map $y \mapsto L_y g$ is a uniformly continuous map from G to $L^p(G/H)$.

Proof. This is a generalization of Lemma 1.36, and their proofs are really similar. □

Lemma 1.38. Let $\varepsilon > 0$. For every $f \in L^1(G)$ there exists a unit-neighborhood U such that for every Dirac function ϕ_U with support in U one has

$$\|f * \phi_U - f\|_1 < \varepsilon, \qquad \|\phi_U * f - f\|_1 < \varepsilon$$

For every $f \in C(G)$ and every compact $K \subseteq G$ there exists a unit-neighborhood U such that for every Dirac function ϕ_U with support in U one has

$$\|f * \phi_U - f\|_K < \varepsilon, \qquad \|\phi_U * f - f\|_K < \varepsilon$$

where $\|g\|_K = \sup_{x \in K} |g(x)|$.

In other words this means that the net $(\phi_U * f)_U$ indexed by the set of all unit-neighborhoods, converges to f in the L^1 sense if $f \in L^1(G)$, and compactly if $f \in C(G)$.

Proof.

$$\begin{aligned}
\|f * \phi_U - f\|_1 &= \int_G \left| \int_G f(y)\phi_U(y^{-1}x)dy - f(x) \right| dx = \int_G \left| \int_G (f(xy) - f(x))\phi_U(y)dy \right| dx \\
&\leqslant \int_G \int_G |(f(xy) - f(x))|\phi_U(y)dydx \\
&= \int_G \|R_y f - f\| \, \phi_U(y)dy
\end{aligned}$$

Then by Lemma 1.36, it suffices to pick U small.

For the second, let $f \in C(G)$ and $K \subseteq G$ compact. Since $f|_K$ is uniformly continuous, for every $\varepsilon > 0$ there exists a unit-neighborhood U such that for all $x, y \in K$ with $y^{-1}x \in U$ one has $|f(y) - f(x)| < \varepsilon$. Now let ϕ_U be a Dirac function with support in U. Then

$$|f * \phi_U(x) - f(x)| \leqslant \int_G |f(xy) - f(x)|\phi_U(y)dy = \int_G |f(y) - f(x)|\phi_U(x^{-1}y)dy < \varepsilon$$

□

2 Banach algebras

2.1 Banach Algebras

Definition.

1. A **Banach algebra** is an algebra $\mathcal{A}$ over $\mathbb{C}$ together with a norm $\|\cdot\|$ such that $(\mathcal{A}, \|\cdot\|)$ is a Banach space and that the norm is submultiplicative, i.e.

$$\|a \cdot b\| \leqslant \|a\| \, \|b\|$$

holds for all $a, b \in \mathcal{A}$.

2. An algebra $\mathcal{A}$ is **unital** if the multiplication has an identity element, which we denote by $1_{\mathcal{A}}$.

3. If $\mathcal{A}$ is a unital Banach algebra, denote by $\mathcal{A}^\times$ the group of invertible elements in $\mathcal{A}$.

- In particular, the inequality implies the multiplication on a Banach algebra $\mathcal{A}$ is continuous, so that $\mathcal{A}$ is a topological ring.

- As usual, for $a \in \mathcal{A}$, $r > 0$, we use $B_r(a) = \{b \in \mathcal{A} \mid \|b - a\| < r\}$ to denote the open ball of radius r centered at a.

Definition. Let $\mathcal{A}$, $\mathcal{B}$ be Banach algebras.

1. A **homomorphism** of Banach algebras $\phi : \mathcal{A} \to \mathcal{B}$ is a continuous algebra homomorphism.

2. A **topological isomorphism** of Banach algebras is a homomorphism with continuous inverse.

3. An **(isometric) isomorphism** of Banach algebras is an topological isomorphism which is also an isometry.

Example 2.1.

1. For a compact space X, the $\mathbb{C}$-vector space $C(X) := C(X, \mathbb{R})$ of complex-valued continuous functions is a commutative Banach algebra with the sup-norm $\|f\|_X := \sup\limits_{x \in X} |f(x)|$.

2. If G is an LCH group, then $(L^1(G), *)$ with $\|\cdot\|_1$ is a Banach algebra, and $L^1(G)$ is commutative iff G is abelian.

3. Let V be a Banach space. For $T \in \mathrm{End}_{\mathbb{C}}(V)$, define the **operator norm** by

$$\|T\|_{\mathrm{op}} := \sup_{v \neq 0} \frac{\|Tv\|}{\|v\|}$$

The operator T is **bounded** if $\|T\|_{\mathrm{op}} < \infty$, which is equivalent to saying that T is continuous. The set $\mathcal{B}(V)$ of all bounded operators on V is a Banach algebra with $\|\cdot\|_{\mathrm{op}}$.

Lemma 2.2. Let $\mathcal{A}$ be a unital Banach algebra and write $1 = 1_{\mathcal{A}}$. Then $\|1\| \geqslant 1$, and there exists an equivalent norm $\|\cdot\|'$ such that $(\mathcal{A}, \|\cdot\|')$ is again a Banach algebra with $\|1\|' = 1$.

Proof. Since $\|1\| \neq 0$ and $\|1\|\,\|1\| \geqslant \|1\|$, $\|1\| \geqslant 1$. For the second statement, define $\|\cdot\|'$ by

$$\|a\|' := \sup_{v \neq 0} \frac{\|av\|}{\|v\|}$$

Since $\|av\| \leqslant \|a\|\,\|v\|$, we have $\|a\|' \leqslant \|a\|$. Conversely,

$$\|a\|' = \sup_{v \neq 0} \frac{\|av\|}{\|v\|} \geqslant \frac{\|a\|}{\|1\|}$$

This shows $\|\cdot\|$ and $\|\cdot\|'$ are equivalent. Trivially, $\|1\|' = 1$. □

Remark 2.3. With the Lemma in mind, we always assume the identity of a unital Banach algebra has norm 1.

Proposition 2.4. Let G be an LCH group. The algebra $L^1(G)$ is unital if and only if G is discrete.

Proof. If G is discrete, then every singleton has positive measure. The identity element of $L^1(G)$ is $\mathbf{1}_{\{e\}}$, where e is the identity element of G. Conversely, assume $L^1(G)$ is unital; say $\phi \in L^1(G)$ is the identity. Suppose G is not discrete. Then any unit-neighborhood U contains at least two points. By Urysohn's lemma there exists two Dirac functions ϕ_U and ψ_U with support in U such that $\operatorname{supp} \phi_U \cap \operatorname{supp} \psi_U = \varnothing$.

- Indeed, let $x \neq y \in U$ and neighborhood $x \in U_x$, $y \in U_y$ such that $U_x \cap U_y = \varnothing$ and $U_x^{-1} \cap U_y = \varnothing = U_x \cap U_y^{-1}$. Use Urysohn's lemma to find f_x, $f_y \in C_c(G)$ with $f_x, f_y \geqslant 0$ and $\operatorname{supp} f_x \subseteq U_x$, $\operatorname{supp} f_y \subseteq U_y$. Define $F_x(z) = f_x(z) + f_x(z^{-1})$ and $F_y(z)$ similarly. Then F_x and F_y are positive symmetric. One can normalize them so that they have integral 1.

In particular, $\|\phi_U - \psi_U\|_1 = 2$ for every U. Now by Lemma 1.38 (or its proof) we can find U such that $\|\phi_U * \phi - \phi\|_1 < 1$ and $\|\psi_U * \phi - \phi\|_1 < 1$. Hence

$$2 = \|\phi_U - \psi_U\|_1 \leqslant \|\phi_U * \phi - \phi\|_1 + \|\psi_U * \phi - \phi\|_1 < 2$$

a contradiction. Hence G is discrete. □

Lemma 2.5. Let $\mathcal{A}$ be a unital Banach algebra and let $a \in \mathcal{A}$ with $\|a\| < 1$. Then $1 - a \in \mathcal{A}^\times$ with inverse

$$(1-a)^{-1} = \sum_{n=0}^{\infty} a^n$$

The unit group $\mathcal{A}^\times$ is open in $\mathcal{A}$, and under the subspace topology, $\mathcal{A}^\times$ is a topological group.

Proof. Since $\|a\| < 1$, the series $\sum_{n=0}^{\infty} a^n$ converges in $\mathcal{A}$, and

$$(1-a)\sum_{n=0}^{N} a^n = 1 - a^N \to 1 \text{ as } N \to \infty$$

Hence $\sum_{n=0}^{\infty} a^n = (1-a)^{-1}$. For $x \in \mathcal{A}^\times$ and $y \in \mathcal{A}$ with $\|y-x\| < \|x^{-1}\|^{-1}$, we have

$$\left\|yx^{-1} - 1\right\| \leqslant \|y-x\| \left\|x^{-1}\right\| < 1$$

so $yx^{-1} \in \mathcal{A}^\times$, giving $y \in \mathcal{A}^\times$. This shows $\mathcal{A}^\times$ is open.

It remains to show that inversion is continuous in $\mathcal{A}^\times$. But for $x, y \in \mathcal{A}^\times$,

$$x^{-1} - y^{-1} = x^{-1}(y-x)y^{-1} = x^{-1}(y-x)(y^{-1} - x^{-1}) + x^{-1}(y-x)x^{-1}$$

Hence

$$(1 - x^{-1}(y-x))(x^{-1} - y^{-1}) = x^{-1}(y-x)x^{-1}$$

Fix x and let $\|x^{-1}(y-x)\| < \frac{1}{2}$ so that $1 - x^{-1}(y-x) \in \mathcal{A}^\times$. Then $x^{-1} - y^{-1} = \dfrac{x^{-1}(y-x)x^{-1}}{1 - x^{-1}(y-x)}$ and thus

$$\left\|x^{-1} - y^{-1}\right\| \frac{\|x^{-1}\|^2 \|y-x\|}{\|1 - x^{-1}(y-x)\|} \leqslant 2\left\|x^{-1}\right\|^2 \|y-x\|$$

Alternatively, for $\|a\| < 1$, we compute

$$\left\|(1-a)^{-1}\right\| = \left\|\sum_{n=0}^{\infty} a^n\right\| \leqslant \sum_{n=0}^{\infty} \|a\|^n = \frac{1}{1 - \|a\|}$$

so that $a \mapsto (1-a)^{-1}$ is a continuous map at on $B_1(0)$. This implies the inversion $a \mapsto a^{-1} = (1-(1-a))^{-1}$ is continuous on $B_1(1)$, and hence on $xB_1(1)$ for all $x \in \mathcal{A}^\times$. □

2.1.1 The Spectrum $\sigma_{\mathcal{A}}(a)$

Let $\mathcal{A}$ be a unital Banach algebra. For $a \in \mathcal{A}$ we denote by

$$\mathrm{Res}(a) := \{\lambda \in \mathbb{C} \mid \lambda 1 - a \in \mathcal{A}^\times\}$$

the **resolvent set** of $a \in \mathcal{A}$. This is an open set in $\mathbb{C}$ for $\lambda \mapsto \lambda 1 - a$ is a continuous map and $\mathcal{A}^\times \subseteq \mathcal{A}$ is open. Its complement

$$\sigma_{\mathcal{A}}(a) := \mathbb{C} \backslash \mathrm{Res}(a)$$

is called the **spectrum** of a, and it is a closed set.

Example 2.6.

(a) $\mathcal{A} = M_n(\mathbb{C}) = \mathrm{End}(\mathbb{C}^n)$. Then the spectrum of $M \in \mathcal{A}$ consists of the eigenvalues of M in $\mathbb{C}$.

(b) Let X be a compact space and $\mathcal{A} = C(X)$. Then $\sigma(f) = \mathrm{Im}\, f = f(X)$ for any $f \in \mathcal{A}$.

(c) For a unital Banach algebra $\mathcal{A}$, we have $\sigma(ab)\backslash\{0\} = \sigma(ba)\backslash\{0\}$ for every $a, b \in \mathcal{A}$. For if $1 - ab$ is invertible with inverse c, then $1 - ba$ is invertible with inverse $1 + bca$:

$$(1 - ba)(1 + bca) = 1 + bca - ba - babca = 1 + b(c - 1 - abc)a = 1 + b(c(1 - ab) - 1)a = 1$$

and $(1 + bca)(1 - ba) = 1 - ba + bca - bcaba = 1 - b(1 - c(1 - ab))a = 1$.

Lemma 2.7. Let $\mathcal{A}$ be a unital Banach algebra. Then for every $a \in \mathcal{A}$ we have $\sigma_{\mathcal{A}}(a) \subseteq \{z \in \mathbb{C} \mid |z| \leqslant \|a\|\}$. In particular, $\sigma_{\mathcal{A}}(a)$ is compact.

Proof. If $|\lambda| > \|a\|$, then $\left\|\frac{a}{\lambda}\right\| < 1$ so that $\lambda 1 - a$ is invertible. □

Definition. Let $D \subseteq \mathbb{C}$ be an open set, and let $f : D \to V$ be a map, where V is a Banach space. We say f is **holomorphic** if for every $z \in D$ the limit

$$f'(z) := \lim_{h \to 0} \frac{f(z + h) - f(z)}{h}$$

exists in V.

- If f is holomorphic and $\alpha : V \to \mathbb{C}$ is continuous and $\mathbb{C}$-linear, then $\alpha \circ f : D \to \mathbb{C}$ is holomorphic.
- A holomorphic function is continuous.

Lemma 2.8. Let $a \in \mathcal{A}$. Then the map

$$\begin{aligned} f : \mathrm{Res}(a) &\longrightarrow \mathcal{A}^\times \\ \lambda &\longmapsto (\lambda 1 - a)^{-1} \end{aligned}$$

is holomorphic.

Proof. For $\lambda, \rho \in \mathrm{res}(a)$, we have

$$f(\lambda) - f(\rho) = (\lambda 1 - a)^{-1}((\rho 1 - a) - (\lambda 1 - a))(\rho 1 - a)^{-1} = (\rho - \lambda) f(\lambda) f(\rho)$$

Put $\lambda = \rho + h$ with $h \in \mathbb{C}^\times$; then $\frac{1}{h}(f(\rho + h) - f(\rho)) = -f(\rho + h) f(\rho)$. Since f is continuous, letting $h \to 0$ gives $f'(\rho) = -f(\rho)^2$. □

Theorem 2.9. Let $\mathcal{A}$ be a unital Banach algebra, and let $a \in \mathcal{A}$. Then $\sigma_{\mathcal{A}}(a) \neq \varnothing$.

Proof. Suppose otherwise; then $a \neq 0$, $\mathrm{Res}(a) = \mathbb{C}$ and the map $f : \lambda \mapsto (\lambda 1 - a)^{-1}$ in the above lemma is entire. Hence for every continuous linear map $\alpha : \mathcal{A} \to \mathbb{C}$, the map $\alpha \circ f$ is entire in the usual sense. Break $\mathbb{C}$ (domain of f) into two parts $A := \{\lambda \in \mathbb{C} \mid |\lambda| \leqslant 2\|a\|\}$ and $B := \mathbb{C} \backslash A$; A is compact, and so is $(\alpha \circ f)(A)$. For $\lambda \in B$, we have $|\lambda| > 2\|a\|$ and

$$\left\|(\lambda 1 - a)^{-1}\right\| = |\lambda^{-1}| \left\|(1 - \lambda^{-1} a)^{-1}\right\| \leqslant |\lambda^{-1}| \sum_{n=0}^{\infty} \left\|\lambda^{-1} a\right\|^n \leqslant 2|\lambda|^{-1} < \|a\|^{-1}$$

so that $f(B)$ is bounded. Since α is bounded, $\alpha(f(B)) \subseteq \mathbb{C}$ is bounded as well. It follows by Liouville's theorem that $\alpha \circ f$ is constant. To determine the constant, from the inequality above we see $(\lambda 1 - a)^{-1} \to 0$ as $\lambda \to \infty$, so $\lim_{\lambda \to \infty} (\alpha \circ f)(\lambda) = 0$; hence $\alpha \circ f \equiv 0$ for each α; by Hahn-Banach theorem, we obtain $f \equiv 0$, a contradiction. □

Corollary 2.9.1 (Gelfand-Mazur)**.** Let $\mathcal{A}$ be a unital Banach algebra that is also a division ring. Then $\mathcal{A} = \mathbb{C}1$.

Proof. For $a \in \mathcal{A}$, if $a \notin \mathbb{C}1$, then $\sigma_{\mathcal{A}}(a)$ is empty, which is absurd by Theorem. Hence $a \in \mathbb{C}1$. □

Definition. For an element a of a unital Banach algebra $\mathcal{A}$ we define the spectral radius $r(a)$ of a by

$$r(a) := \sup\{|\lambda| \mid \lambda \in \sigma_{\mathcal{A}}(a)\} \leqslant \|a\|$$

Theorem 2.10 (Spectral radius formula)**.** Let $\mathcal{A}$ be a unital Banach algebra. Then

$$r(a) = \lim_{n \to \infty} \|a^n\|^{\frac{1}{n}}$$

Proof. If $\lambda 1 - a$ is not invertible, then neither is $\lambda^n 1 - a^n$ for each $n \in \mathbb{N}$. Hence $\lambda^n \in \sigma(a^n)$ and $|\lambda| \leqslant \|a^n\|^{\frac{1}{n}}$ so that $r(a) \leqslant \|a^n\|^{\frac{1}{n}}$. For the converse, we show $\limsup_{n \to \infty} \|a^n\|^{\frac{1}{n}} \leqslant r(a)$. Recall

$$(\lambda 1 - a)^{-1} = \lambda^{-1}(1 - \lambda^{-1} a)^{-1} = \sum_{n=0}^{\infty} \frac{a^n}{\lambda^{n+1}} \tag{$\diamond$}$$

holds for all $|\lambda| > \|a\|$. From the Cauchy estimate in complex analysis (c.f. Corollary U.40.1) we see expansion is in fact valid when $|\lambda| > r(a)$ (if $|\lambda| > r(a)$, then $(\lambda 1 - a)^{-1}$ is defined). It follows that $\limsup_{n \to \infty} \|a^n\|^{\frac{1}{n}} < |\lambda|$ (view $(\diamond)$ as a power series in λ^{-1}) for all $|\lambda| > r(a)$, and hence $\limsup_{n \to \infty} \|a^n\|^{\frac{1}{n}} \leqslant r(a)$. □

Lemma 2.11. Suppose $\mathcal{A}$ is a closed subalgebra of the unital Banach algebra $\mathcal{B}$ such that $1 \in \mathcal{A}$. Then

$$\partial \sigma_{\mathcal{A}}(a) \subseteq \sigma_{\mathcal{B}}(a) \subseteq \sigma_{\mathcal{A}}(a)$$

for all $a \in \mathcal{A}$, where $\partial \sigma_{\mathcal{A}}(a)$ denotes the boundary of $\sigma_{\mathcal{A}}(a)$ in $\mathcal{C}$.

Proof. If $\lambda 1 - a$ is not invertible in $\mathcal{B}$, it cannot be invertible in $\mathcal{A}$ either; this gives $\sigma_{\mathcal{B}}(a) \subseteq \sigma_{\mathcal{A}}(a)$. Suppose $\lambda \in \partial\sigma_{\mathcal{A}}(a)$ and let $\lambda_n \notin \sigma_{\mathcal{A}}(a)$ be a sequence that converges to λ. Then $\lambda_n 1 - a$ converges to $\lambda 1 - a$. If $\lambda \notin \sigma_{\mathcal{B}}(a)$, then $(\lambda_n 1 - a)^{-1} \to (\lambda 1 - a)^{-1} \in \mathcal{A}$ since $\mathcal{A}$ is closed, and thus $\lambda \notin \sigma_{\mathcal{A}}(a)$, a contradiction since $\partial\sigma_{\mathcal{A}}(a) \subseteq \sigma_{\mathcal{A}}(a)$. □

Example 2.12 (Disc-algebra)**.** Let $\mathbb{D} \subseteq \mathbb{C}$ be the open unit disc in $\mathbb{C}$. The **disc algebra** $\mathcal{A}$ is by definition the subalgebra of $C(\overline{\mathbb{D}})$ consisting of all functions that are holomorphic on $\mathbb{D}$. $\mathcal{A}$ is closed since a uniform limit of holomorphic functions is again holomorphic.

Let $\mathbb{T} = \partial\mathbb{D}$ be the circle group. By maximum principal, the restriction $\mathcal{A} \to C(\mathbb{T})$ is an isometry so that $\mathcal{A}$ can be viewed as a Banach subalgebra of $C(\mathbb{T})$. We know $\sigma_{\mathcal{A}}(f) = f(\overline{\mathbb{D}})$ and $\sigma_{C(\mathbb{T})}(f) = f(\mathbb{T})$ for all $f \in \mathbb{A}$.

Proposition 2.13 (Spectral mapping theorem for polynomials)**.** Let $\mathcal{A}$ is a unital algebra and $a \in \mathcal{A}$ with $\sigma_{\mathcal{A}}(a) \neq \varnothing$. If $p \in \mathbb{C}[z]$, then $\sigma_{\mathcal{A}}(p(a)) = p(\sigma_{\mathbb{A}}(a))$.

Proof. If p is a constant, this is clear. Suppose otherwise. For $\mu \in \mathbb{C}$, we write

$$p(z) - \mu = \lambda_0(\lambda_1 - z)\cdots(\lambda_n - z)$$

for some $n \in \mathbb{N}$, $\lambda_i \in \mathbb{C}$ with $\lambda_0 \neq 0$.

If $\mu \notin \sigma_{\mathcal{A}}(p(a))$, then $p(a) - \mu 1_{\mathcal{A}}$ is invertible, and hence so is each $\lambda_i 1_{\mathcal{A}} - a$. The converse is true obviously. Hence we have $\mu \in \sigma_{\mathcal{A}}(p(a))$ if and only if $\lambda_{\in}\sigma_{\mathbb{A}}(a)$ for some $1 \leqslant i \leqslant n$, and thus $\sigma_{\mathcal{A}}(p(a)) \subseteq p(\sigma_{\mathcal{A}}(a)$. For the reversed inclusion, if $\lambda \in \sigma_{\mathcal{A}}(a)$, then $p(a) - p(\lambda) = (\lambda 1_{\mathcal{A}} - a)b$ for some $b \in \mathcal{A}$, and hence $p(\lambda) \in \sigma_{\mathcal{A}}(p(a))$. □

2.1.2 Adjoining a Unit

Definition. For any Banach algebra $\mathcal{A}$, the **unitization** of $\mathcal{A}$ is the Banach algebra

$$\mathcal{A}^e := \mathcal{A} \times \mathbb{C}$$

with obvious vector space structure, multiplication defined by $(a, \lambda)(b, \mu) = (ab + \mu a + \lambda b, \lambda\mu)$ and norm $\|(a, \lambda)\| := \|a\| + |\lambda|$.

- We view $\mathcal{A}$ as $\mathcal{A} \times \{0\} \subseteq \mathcal{A}^e$, and this is an embedding of Banach algebras.
- If $\mathcal{A}$ has unit 1, then we have an isomorphism of algebras

$$\begin{aligned} \mathcal{A}^e &\longrightarrow \mathcal{A} \oplus \mathbb{C} \\ (a, \lambda) &\longmapsto (a + \lambda 1, \lambda) \end{aligned}$$

  where the multiplication on $\mathcal{A} \oplus \mathbb{C}$ is defined componentwise.

- The unitization defines a functor from the category of Banach algebras to the category of unital Banach algebras: for a homomorphism $\phi : \mathcal{A} \to \mathcal{B}$, define $\phi^e : \mathcal{A}^e \to \mathcal{B}^e$ by $\phi^e(a, \lambda) = (\phi(a), \lambda)$.

If $\mathcal{A}$ is a Banach space without unit, for each $a \in \mathcal{A}$ we define the **spectrum** of a to be

$$\sigma_{\mathcal{A}}(a) := \sigma_{\mathcal{A}^e}(a)$$

where we use the identification of $\mathcal{A}$ mentioned above.

Definition. Let X be an LCH space.

1. A function $f : X \to \mathbb{C}$ is said to **vanish at infinity** if for every $\varepsilon > 0$ there exists a compact set $K \subseteq X$ such that $|f(x)| < \varepsilon$ for all $x \in X \backslash K$.

2. Denote by $C_0(X)$ the subalgebra of $C(X)$ consisting of all continuous functions on X that vanish at infinity.

Then $C_0(X)$ is a Banach algebra with the sup-norm $\|f\|_X = \sup_{x \in X} |f(x)|$. Note that $C_0(X)$ is unital if and only if X is compact, and in this case $C_0(X) = C(X)$.

Proof. We must show $C_0(X)$ is complete with respect to $\|\cdot\|_X$. Let $(f_n)_n$ be a Cauchy sequence in $C_0(X)$. About each $x \in X$ pick a compact neighborhood K_x. Then $(f_n|_{K_x})_n$ is uniformly Cauchy, so that we can define $f(x) := \lim_{n \to \infty} f_n(x)$; it is clear f is well-defined and is continuous. Now fix $\varepsilon > 0$ and take $N \in \mathbb{N}$ such that $\|f_n - f_N\|_X < \varepsilon/3$ whenever $n \geqslant N$. Let $K \underset{\text{cpt}}{\subseteq} X$ such that $|f_N(x)| < \varepsilon/3$ for all $x \in X \backslash K$. For $x \in X - K$ there exists $m \geqslant N$ such that $|f(x) - f_m(x)| < \varepsilon/3$, and thus $|f(x)| \leqslant |f(x) - f_m(x)| + |f_m(x) - f_N(x)| + |f_N(x)| < \varepsilon$. This proves f vanishes at infinity. $\square$

Example 2.14. Let X be an LCH space and put $X^\infty = X \cup \{\infty\}$ to be the one point compactification of X. Every open neighborhood of ∞ in X^∞ is of the form $X \backslash K$ with $K \underset{\text{cpt}}{\subseteq} X$. With this in mind, we can identify $C_0(X)$ as a subspace of $C(X^\infty)$ consisting of all continuous functions f such that $f(\infty) = 0$. This justifies the notion "vanishing at infinity".

Lemma 2.15. There is a canonical topological isomorphism of Banach algebras $C(X^\infty) \cong C_0(X)^e$.

Proof. Define

$$\begin{aligned} \Phi : C(X^\infty) &\longrightarrow C_0(X)^e \\ f &\longmapsto (f - f(\infty), f(\infty)) \end{aligned}$$

It is clear that Φ is bijective.

- Homomorphism:

$$\begin{aligned} \Phi(f)\Phi(g) &= ((f - f(\infty))(g - g(\infty)) + g(\infty)(f - f(\infty)) + f(\infty)(g - g(\infty)), f(\infty)g(\infty)) \\ &= (fg - f(\infty)g(\infty), f(\infty)g(\infty)) = \Phi(fg) \end{aligned}$$

- Continuity:

$$\|\Phi(f) - \Phi(g)\| = \|f - f(\infty) - g + g(\infty)\|_X + |f(\infty) - g(\infty)| \leqslant 3\,\|f - g\|_{X^\infty}$$

- Continuous inverse: Let (f, λ), $(g, \rho) \in C_0(X)^e$.

$$\|(f + \lambda) - (g + \rho)\|_{X^\infty} \leqslant \|f - g\|_{X^\infty} + |\lambda - \rho| = \|f - g\|_X + |\lambda - \rho| = \|(f, \lambda) - (g - \rho)\|$$

□

2.1.3 The Gelfand Map

Definition. Let $\mathcal{A}$ be a commutative Banach algebra. The **structure space** $\Delta_{\mathcal{A}}$ is the set of all nonzero continuous algebra homomorphisms $\mathcal{A} \to \mathbb{C}$.

- If $\mathcal{A}$ is unital, then for each $m \in \Delta_{\mathcal{A}}$, $m(1) = m(1)^2$ and the assumption $m \neq 0$ imply $m(1) = 1$.
- For each $m \in \Delta_{\mathcal{A}}$, there exists precisely one extension $m^e : \mathcal{A}^e \to \mathbb{C}$ of m defined by

$$m^e(a, \lambda) = m(a) + \lambda$$

  Indeed, we must have $m^e(a, 0) = m(a)$ and $m^e(1_{\mathcal{A}^e}) = m^e(0, 1) = 1$.

- If $m' \in \Delta_{\mathcal{A}^e}$ is not extended from $\mathcal{A}$, then it must vanish on $\mathcal{A}$ by the uniqueness of extension. Hence we must have $m' = m_\infty$ is the **augmentation functional**, where $m_\infty \in \Delta_{\mathcal{A}^e}$ is defined by

$$m_\infty(a, \lambda) = \lambda$$

  In other words, $\Delta_{A^e} = \{m^e \mid m \in \Delta_{\mathcal{A}}\} \cup \{m_\infty\}$.

Definition. For each Banach space V, denote by $V^\vee$ the **dual space** consisting of all continuous linear functionals $\alpha : V \to \mathbb{C}$, which is also a Banach space with the operator norm $\|\alpha\| := \sup\limits_{v \in V - \{0\}} \dfrac{|\alpha(v)|}{\|v\|}$.

Lemma 2.16. Let $\mathcal{A}$ be a commutative Banach algebra and $m \in \Delta_{\mathcal{A}} \subseteq \mathcal{A}^\vee$. Then $\|m\| \leqslant 1$. If $\mathcal{A}$ is unital, $\|m\| = 1$.

Proof. Suppose $\mathcal{A}$ is unital. Then

$$\|m\| = \sup_{\|a\|=1} |m(a)| \geqslant |m(1)| = 1$$

On the other hands, $m(a - m(a)1) = m(a) - m(a) = 0$ so $m(a) \in \sigma_{\mathcal{A}}(a)$; this gives $|m(a)| \leqslant \|a\|$, so that $\|m\| \leqslant 1$. If $\mathcal{A}$ is not unital, then we still have

$$1 = \|m^e\| = \sup_{\|(a,\lambda)\|=1} |m(a) + \lambda| \geqslant \sup_{\|(a,\lambda)\|=1, \lambda=0} |m(a) + \lambda| = \|m\|$$

or just $1 = \|m^e\| \geqslant \|m^e|_{\mathcal{A}}\|$. □

Definition. For each normed space V, the **weak-*-topology** on $V^\vee$ is the initial topology induced by the set of functions $\{\delta_v \mid v \in V\}$, where $\delta_v : V^\vee \to \mathbb{C}$ is the evaluation at $v \in V$, namely

$$\delta_v(\alpha) := \alpha(v) \in \mathbb{C}$$

- The weak-*-topology is the same as the topology of pointwise convergence, i.e. the subspace topology given by the product space $\mathbb{C}^V = \prod_{v \in V} \mathbb{C}$.

  Proof. By definition a subbasis for the weak-*-topology consists of subsets of the form

  $$L(v, U) := \{\alpha \in V^\vee \mid \alpha(v) \in U\}$$

  where $U \underset{\text{open}}{\subseteq} \mathbb{C}$ and $v \in V$. Then by definition $\bigcap_{i=1}^{n} L(v_i, U_i)$ is the basis for the product space $\mathbb{C}^V$. □

- A net $(\alpha_j)_j$ in $V^\vee$ converges to $\alpha \in V^\vee$ in the weak-*-topology if and only if $\alpha_j(v) \to \alpha(v)$ for all $v \in V$.

For any commutative Banach algebra $\mathcal{A}$, we equip $\Delta_{\mathcal{A}}$ with the subspace topology induced from the weak-*-topology on the dual space $\mathcal{A}^\vee$.

Theorem 2.17 (Banach-Alaoglu)**.** Let V be a complex normed space. Then the closed unit ball

$$\overline{B}' := \{f \in V^\vee \mid \|f\| \leqslant 1\} \subseteq V^\vee$$

is a compact Hausdorff space under the weak-*-topology.

Proof. Since $\mathbb{C}^V$ is Hausdorff, so is $\overline{B}'$. For the compactness, consider the product $K := \prod_{v \in V} \overline{B_1(0)} \subseteq \mathbb{C}^V$ equipped with subspace topology; this is compact by Tychonov's theorem. Since $\overline{B}' \subseteq K$ and $\overline{B}' = V^\vee \cap K$, it suffices to show $V^\vee$ is closed in $\mathbb{C}^V$.

- Take $(x_v)_v \in \mathbb{C}^V$ such that $x_{v+w} \neq x_v + x_w$ for some $v, w \in V$. Then take respective neighborhoods U_v, U_w, U_{v+w} of x_v, x_w, x_{v+w} such that $(U_v + U_w) \cap U_{v+w} = \varnothing$. Then $U_v \times U_w \times U_{v+w} \times \prod \mathbb{C}$ is disjoint from $V^\vee$.

- If $(x_v)_v \in \mathbb{C}^V$ is such that $\lambda x_v \neq x_{\lambda v}$ for some $\lambda \in \mathbb{C}$ and $v \in V$, then take U, V such that $\lambda U \cap V = \varnothing$ with $x_v \in V$, $x_{\lambda v} \in U$. Then $U \times V \times \prod \mathbb{C}$ again disjoint from $V^\vee$.

□

Lemma 2.18. Let $\mathcal{A}$ be a commutative Banach algebra. Then the inclusion

$$\begin{aligned} \Phi : \Delta_{\mathcal{A}} &\longrightarrow \Delta_{\mathcal{A}^e} \\ m &\longmapsto m^e \end{aligned}$$

is a homeomorphism onto its image.

Proof. Φ is clearly injective for $m^e|_{\mathcal{A}} = m$. To show it is a topological embedding, we have the following equivalence

$$\begin{aligned} m_j \to m \text{ in } \Delta_{\mathcal{A}} &\Leftrightarrow m_j(a) \to m(a) \text{ in } \mathbb{C} \text{ for each } a \in \mathcal{A} \\ &\Leftrightarrow m_j(a) + \lambda \to m(a) + \lambda \text{ in } \mathbb{C} \text{ for each } a \in \mathcal{A} \\ &\Leftrightarrow m_j^e(a) \to m^e(a) \text{ in } \mathbb{C} \text{ for each } a \in \mathcal{A}^e \\ &\Leftrightarrow m_j^e \to m^e \text{ in } \Delta_{\mathcal{A}}^e \end{aligned}$$

□

Definition. For each $a \in \mathcal{A}$ define

$$\begin{aligned} \hat{a} : \Delta_{\mathcal{A}} &\longrightarrow \mathbb{C} \\ m &\longmapsto \hat{a}(m) := m(a) \end{aligned}$$

- $\hat{a}$ is continuous. Indeed, for each open U in $\mathbb{C}$,

$$\hat{a}^{-1}(U) = \{m \in \Delta_{\mathcal{A}} \mid m(a) \in U\} = \left(U \times \prod \mathbb{C}\right) \cap \mathbb{C}^{\mathcal{A}} \underset{\text{open}}{\subseteq} \mathbb{C}^{\mathcal{A}}$$

The association $\mathcal{A} \ni a \mapsto \hat{a} \in C(\Delta_{\mathcal{A}})$ is an algebra homomorphism, called the **Gelfand transform**.

Theorem 2.19. Let $\mathcal{A}$ be a commutative Banach algebra.

1. $\Delta_{\mathcal{A}}$ is an LCH space.
2. If $\mathcal{A}$ is unital, $\Delta_{\mathcal{A}}$ is compact.
3. For every $a \in \mathcal{A}$, $\hat{a} \in C_0(\Delta_{\mathcal{A}})$. The Gelfand transform

$$\begin{aligned} \mathcal{A} &\longrightarrow C_0(\Delta_{\mathcal{A}}) \\ a &\longmapsto \hat{a} \end{aligned}$$

   is an algebra homomorphism.
4. For every $a \in \mathcal{A}$ one has $\|\hat{a}\|_{\Delta_{\mathcal{A}}} \leqslant \|a\|$ so that the Gelfand transform is continuous.

Proof. Note that by a previous lemma, we have the containment $\Delta_{\mathcal{A}} \subseteq \overline{B}' \subseteq \mathcal{A}^\vee$ where $\overline{B}'$ is defined in the statement of Banach-Alaoglu. This means the closure $\overline{\Delta_{\mathcal{A}}}$ is thus compact in $\mathcal{A}^\vee$. We contend that

- $\overline{\Delta_{\mathcal{A}}} = \Delta_{\mathcal{A}}$ if $\mathcal{A}$ is unital. (This proves 2.)
- $\overline{\Delta_{\mathcal{A}}} = \Delta_{\mathcal{A}}$ or $\Delta_{\mathcal{A}} \cup \{0\}$ if $\mathcal{A}$ is not unital, where 0 is the zero map. (This proves 1.)

We first show that every element of $\overline{\Delta_{\mathcal{A}}}$ is an algebra homomorphism. The set of continuous algebra homomorphism is described by

$$S = \{(x_a)_a \in \mathcal{A}^\vee \mid x_{ab} = x_a x_b \text{ for all } a, b \in \mathcal{A}\}$$

Suppose $(x_a)_a \in \mathcal{A}^\vee$ is such that $x_{ab} \neq x_a x_b$ for some $a, b \in \mathcal{A}$. Then take respective neighborhoods U_a, U_b, U_{ab} of x_a, x_b, x_{ab} such that the product $U_a U_b$ is disjoint from U_{ab}. Then $(U_a \times U_b \times U_{ab} \times \prod \mathbb{C}) \cap \mathcal{A}^\vee$ is a neighborhood of $(x_a)_a$ that does not meet S. Hence S is closed, which implies that $\overline{\Delta}_{\mathcal{A}} \subseteq S$. Note that by definition $S \backslash \Delta_{\mathcal{A}} \subseteq \{0\}$. It shows our contention except the case $\mathcal{A}$ is unital. In the unital case, we have $\Delta_{\mathcal{A}} \subseteq (\{1\} \times \prod \mathbb{C}) \cap S$; the latter is a closed set in S so that $0 \notin \overline{\Delta}_{\mathcal{A}}$. Hence $\overline{\Delta}_{\mathcal{A}} = S = \Delta_{\mathcal{A}}$.

Now we prove 3. If $\Delta_{\mathcal{A}}$ is compact, the result ensues automatically. Otherwise, what we have proved above shows $\overline{\Delta_{\mathcal{A}}} = \Delta_{\mathcal{A}} \cup \{0\}$ coincides with the one point compactification of $\Delta_{\mathcal{A}}$. Taking a small neighborhood of 0 in $\mathcal{C}$ not touching 1 proves the vanishing at infinity of $\hat{a}$. Finally for 4., we have

$$\|\hat{a}\| = \sup_{m \in \Delta_{\mathcal{A}}} |\hat{a}(m)| = \sup_{m \in \Delta_{\mathcal{A}}} |m(a)| \leqslant \|m\| \, \|a\| = \|a\|$$

□

Example 2.20. Consider the case $\mathcal{A} = C_0(X)$ for an LCH space X. For each $x \in X$ let $m_x : \mathcal{A} \to \mathbb{C}$ be the evaluation map at x. Then $x \mapsto m_x$ is a homeomorphism from X to the structure space $\Delta_{\mathcal{A}}$. If X is not compact, consider the diagram

$$\begin{tikzcd} X^\infty \arrow[r] & \Delta_{C(X^\infty)} \\ X \arrow[r] \arrow[u, hook] & \Delta_{C_0(X)} \arrow[u, hook] \end{tikzcd}$$

where the vertical maps are canonical inclusions (Lemma 2.18 and 2.15) and the horizontal maps are $x \mapsto m_x$. If the case when the space is compact is shown, then the upper horizontal map is a homeomorphism, implying that the lower horizontal one is also a homeomorphism. Hence it suffices to deal with compact X.

Since both X and $\Delta_{\mathcal{A}}$ are compact Hausdorff, it suffices to show $x \mapsto m_x$ is a continuous bijection. The continuity is clear. The injectivity follows from Urysohn's lemma. For the surjectivity, let $m \in \Delta_{\mathcal{A}}$. Consider the vanishing set

$$V = \{x \in X \mid f(x) = 0 \text{ for all } f \in \ker m\}$$

If V is not empty, take $x \in V$. Then $\ker m = \ker m_x$. For each $f \in \Delta_{\mathcal{A}}$, $m_x(f - f(x)1) = 0$ so that $0 = m(f - f(x)1)$, namely, $m(f) = m(f(x)1) = m_x(f)$, showing the surjectivity. If V is empty, then for each $x \in X$ there exists $f_x \in \ker m$ such that $f_x(x) \neq 0$, and by continuity $f_x(y) \neq 0$ for $y \in U_x$ and some neighborhood U_x of x. By compactness we can choose $x_1, \ldots, x_n$ such that $U_{x_1} \cup \cdots \cup U_{x_n} = X$. Then $f_{x_1}^2 + \cdots + f_{x_n}^2$ vanishes nowhere on X, so it is unit in $C(X)$, contradicting to $f_{x_1}^2 + \cdots + f_{x_n}^2 \in \ker m$.

Lemma 2.21. Suppose that $\phi : \mathcal{A} \to \mathcal{B}$ is an algebra homomorphism between commutative Banach algebras such that $m \circ \phi \neq 0$ for every $m \in \Delta_{\mathcal{B}}$. Then the pullback map

$$\begin{aligned} \phi^* : \Delta_{\mathcal{B}} &\longrightarrow \Delta_{\mathcal{A}} \\ m &\longmapsto m \circ \phi \end{aligned}$$

is continuous. Moreover, it is a homeomorphism if it is bijective.

Proof. Let $U_1, \ldots, U_n$ be open sets in $\mathcal{C}$. We must show that for any $a_1, \ldots, a_n \in \mathcal{A}$, the set

$$S := \{m \in \Delta_{\mathcal{B}} \mid (m \circ \phi)(a_i) \in U_i \text{ for } i = 1, \ldots, n\}$$

is open in $\Delta_{\mathcal{B}}$. Indeed, if we put $b_i = \phi(a_i)$, then $\{m \in \Delta_{\mathcal{B}} \mid m(b_i) \in U_i \text{ for } i = 1, \ldots, n\}$ is contained in S, and the former set is open in $\Delta_{\mathcal{B}}$ by the very definition. Now consider the diagram

$$\begin{tikzcd} \Delta_{\mathcal{B}^e} \arrow[r, "\phi^{e*}"] & \Delta_{\mathcal{A}^e} \\ \Delta_{\mathcal{B}} \arrow[u, hook] \arrow[r, "\phi^*"] & \Delta_{\mathcal{A}} \arrow[u, hook] \end{tikzcd}$$

where $\phi^e : \mathcal{A}^e \to \mathcal{B}^e$ is the map defined by $\phi^e(a, \lambda) = (\phi(a), \lambda)$. ϕ^{e*} is well-defined for ϕ^* is. We have

$$(\phi^*(m))^e(a, \lambda) = \phi^*(m)(a) + \lambda = m(\phi(a)) + \lambda = m^e(\phi(a), \lambda) = m^e(\phi^e(a, \lambda)) = \phi^{e*}(m^e)(a, \lambda)$$

so the diagram commutes. If ϕ^* is bijective, so is ϕ^{e*}. Since Δ_e and $\Delta_{\mathcal{A}^e}$ are compact, ϕ^{e*} is a homeomorphism. By restricting to $\Delta_{\mathcal{B}}$ we see ϕ^* is also a homeomorphism if ϕ is bijective. $\square$

2.1.4 Maximal Ideals

Definition. Let $\mathcal{A}$ be an algebra. An linear subspace I of $\mathcal{A}$ is called an **ideal** if I is stable under the action of $\mathcal{A}$ on two sides.

1. I is **proper** if $I \subsetneq \mathcal{A}$.
2. I is **maximal** if I is proper and is maximal among all proper ideal with respect to the inclusion.

- If $\mathcal{A}$ is unital, then any proper ideal contains no unit in $\mathcal{A}$.
- In the case $\mathcal{A}$ is a unital Banach algebra, since $\mathcal{A}^\times$ is open, the closure of any proper ideal is still a proper ideal. In particular, a maximal ideal is closed.
- By a Zorn's lemma argument we see if $\mathcal{A}$ is a unital Banach algebra, then every proper ideal is contained in a maximal ideal.
- Let $\mathcal{A}$ be as above. If $\mathcal{A}$ is commutative, then the set of non unit elements forms a proper ideal.

Definition. Let $\mathcal{A}$ be a Banach algebra and $I \trianglelefteq \mathcal{A}$ a closed ideal. The quotient algebra $\mathcal{A}/I$ equipped with the quotient norm defined in Lemma V.5 is again a Banach algebra.

Proof. We still need to show $\|ab + I\| \leqslant \|a + I\| \, \|b + I\|$ for $a, b \in \mathcal{A}$.

$$\|a + I\| \, \|b + I\| \geqslant \inf_{x,y \in I} \|(a + x)(b + y)\| = \inf_{x,y \in I} \|ab + (ay + xb + xy)\| \geqslant \inf_{z \in I} \|ab + z\| = \|ab + I\|$$

□

Theorem 2.22. Let $\mathcal{A}$ be a commutative unital Banach algebra.

1. The map $m \mapsto \ker m$ is a bijection between $\Delta_{\mathcal{A}}$ and the maximal spectrum of $\mathcal{A}$.
2. $a \in \mathcal{A}^\times$ if and only if $m(a) \neq 0$ for every $m \in \Delta_{\mathcal{A}}$.
3. For $a \in \mathcal{A}$ one has $\sigma(a) = \operatorname{Im} \hat{a}$.

Proof. We first derive 2 and 3 from 1. The only if part of 2. is obvious. Conversely, if a is not invertible, then it is contained in some maximal ideal of $\mathcal{A}$, and by 1. this means $m(a) = 0$ for some $m \in \Delta_{\mathcal{A}}$. For 3. we have the following equivalences

$$\begin{aligned} \lambda \in \sigma(a) &\Leftrightarrow a - \lambda 1 \notin \mathcal{A}^\times \\ &\Leftrightarrow m(a - \lambda 1) = 0 \text{ for some } m \in \Delta_{\mathcal{A}} \\ &\Leftrightarrow m(a) = \lambda \text{ for some } m \in \Delta_{\mathcal{A}} \\ &\Leftrightarrow \lambda \in \operatorname{Im} \hat{a} \end{aligned}$$

It remains to show 1. Suppose $\ker m = \ker n$ for some $m, n \in \Delta_{\mathcal{A}}$. Then for each $a \in \mathcal{A}$, $m(a - m(a)) = 0 = n(a - m(a))$, i.e., $m(a) = n(a)$. Hence $m = n$. Now given an maximal ideal $\mathfrak{m}$ of $\mathcal{A}$. Then $\mathcal{A}/\mathfrak{m}$ is a unital Banach algebra that is also a field. By Gelfand-Mazur this implies $\mathcal{A}/\mathfrak{m} \cong \mathbb{C}$. This gives a nontrivial algebra homomorphism $m : \mathcal{A} \to \mathcal{A}/\mathfrak{m} \cong \mathbb{C}$ with kernel $\mathfrak{m}$. □

Corollary 2.22.1. Let $\mathcal{A}$ be a commutative non-unital Banach algebra. Then for all $a \in \mathcal{A}$, $\sigma(a) = \operatorname{Im} \hat{a} \cup \{0\}$.

Proof. By Theorem, $\sigma(a) = \operatorname{Im} \widehat{(a, 0)}$, where $(a, 0) \in \mathcal{A}^e$. Recall that $\Delta_{\mathcal{A}^e} = \{m^e \mid \in \mathcal{A}\} \cup \{m_\infty\}$. Hence

$$\operatorname{Im} \widehat{(a, 0)} = \{m^e(a, 0) \mid m \in \mathcal{A}\} \cup \{m_\infty(a, 0)\} = \operatorname{Im} \hat{a} \cup \{0\}$$

□

2.2 The Gelfand-Naimark Theorem

2.2.1 C^*-algebra

Definition. Let $\mathcal{A}$ be an algebra over $\mathbb{C}$.

1. An **involution** $^* : \mathcal{A} \to \mathcal{A}$ is an abelian group homomorphism such that for each $\lambda \in \mathbb{C}$ and $a, b \in \mathcal{A}$
   - $(\lambda a)^* = \overline{\lambda} a^*$.
   - $(ab)^* = b^* a^*$.
   - $(a^*)^* = a$.
2. $\mathcal{A}$ is called a **Banach *-algebra** if it is a Banach algebra equipped with an involution * such that $\|a^*\| = \|a\|$ for each $a \in \mathcal{A}$.
3. A Banach *-algebra $\mathcal{A}$ is called a **C^*-algebra** if $\|a^* a\| = \|a\|^2$ for each $a \in \mathcal{A}$.
4. An element a of a Banach *-algebra is called **self-adjoint** if $a = a^*$.

Example 2.23.

1. Let H be a Hilbert space and $\mathcal{B}(H)$ the Banach algebra of all bounded linear operators on H. Then the map $T \mapsto T^*$ is an involution, where T^* is the adjoint of T. $\mathcal{B}(H)$ is a C^*-algebra

   Proof. For each $v, w \in H$ one has $\langle Tv, w\rangle = \langle v, T^* w\rangle$. For each $v \in V$ by Cauchy-Schwarz
   $$\|Tv\|^2 = \langle T^* Tv, v\rangle \leqslant \|T^* Tv\| \, \|v\|$$
   so that $\dfrac{\|Tv\|^2}{\|v\|} \leqslant \dfrac{\|T^* Tv\|}{\|v\|}$. Hence $\|T\|^2 \leqslant \|T^* T\| \leqslant \|T^*\| \, \|T\|$, and thus $\|T\| \leqslant \|T^*\|$. By symmetry one has the equality, which forces $\|T\|^2 = \|T^* T\|$. □

2. Let X be an LCH space. Then $C_0(X)$ is a C^*-algebra where the involution is defined by the complex conjugation.

3. The disc algebra $\mathcal{A}$ (in Example 2.12) with involution defined by $f^*(z) := \overline{f(\overline{z})}$ is a Banach *-algebra but not a C^*-algebra.

   Proof. It is clear it is an Banach *-algebra. Let $f \in \mathcal{A}$.
   $$\|f^* f\| = \sup_{z \in \overline{\mathbb{D}}} |\overline{f(\overline{z})} f(z)| = \sup_{z \in \overline{\mathbb{D}}} |f(\overline{z}) f(z)|$$
   Consider the function $f(z) = e^{iz}$. Then
   $$\sup_{z \in \overline{\mathbb{D}}} |f(\overline{z}) f(z)| = \sup_{z \in \overline{\mathbb{D}}} |e^{i(\overline{z} + z)}| = 1$$
   On the other hand $\|f\|^2 = \sup |e^{iz}|^2 = \sup |e^{-\operatorname{Im}(z)}| = e^2$. □

Proposition 2.24. Let G be an LCH group. The Banach algebra $L^1(G)$ with the involution

$$f^*(x) := \Delta_G(x^{-1})\overline{f(x^{-1})}$$

is a Banach *-algebra but not a C^*-algebra unless G is trivial, in which case $L^1(G) = \mathbb{C}$.

Proof.

$$(f^*)^*(x) = \Delta_G(x^{-1})\overline{f^*(x^{-1})} = \Delta_G(x^{-1})\overline{\Delta_G(x)\overline{f(x)}} = f(x)$$

so it is an involution (the other conditions are obvious). Also

$$\|f^*\|_1 = \int_G |\Delta_G(x^{-1})\overline{f(x^{-1})}|dx \overset{1.11.4}{=} \int_G |f(x)|dx = \|f\|_1$$

For the last assertion, we need a lemma.

Lemma 2.25.

(a) Let X be an LCH space and let $x_1, \ldots, x_n \in X$ be distinct points. Let $\lambda_1, \ldots, \lambda_n \in \mathbb{C}$ be any given numbers. Then there exists $f \in C_c(G)$ with $f(x_j) = \lambda_j$ for each j.

(b) Let G be an LCH group and $g \in C_c(G)$ be with the property $\left|\int_G g(y)dy\right| = \int_G |g(y)|dy$. Then there exists $\theta \in \mathbb{T}$ such that $g(x) \in \theta[0, \infty)$ for every $x \in G$.

Proof.

(a) Let $U_1, \ldots, U_n$ be pairwise disjoint open neighborhoods of $x_1, \ldots, x_n$ respective. By Urysohn's lemma we can find $f_i \in C_c(G)$ with $\operatorname{supp} f_i \subseteq U_i$ and $f_i(x_i) = 1$. Take $f(x) = \sum_{i=1}^{n} \lambda_i f_i(x)$.

(b) If $\int_G g(y)dy = 0$, then $\int_G |g(y)|dy = 0$ so that $g \equiv 0$. Let us suppose $\int_G g(y)dy \neq 0$. Replacing g by λg for some $\lambda \in \mathbb{T}$ we may assume $\int_G g(y)dy > 0$. Then

$$\begin{aligned} 0 &= \int_G |g(y)|dy - \int_G g(y)dy = \int_G |g(y)|dy - \operatorname{Re}\left(\int_G g(y)dy\right) \\ &= \int_G |g(y)|dy - \int_G \operatorname{Re}(g(y))dy \\ &= \int_G (|g(y)| - \operatorname{Re}(g(y)))dy \end{aligned}$$

so that $|g(y)| = \operatorname{Re}(g(y))$ for each $y \in G$. This means $g(y) \geqslant 0$.

□

Return to the proof of Proposition. We assume $L^1(G)$ is a C^*-algebra. Then for all $f \in C_c(G)$, one has $\|f * f^*\|_1 = \|f\|_1^2$, or

$$\int_G \left| \int_G \Delta_G(x^{-1}y) f(y) \overline{f(x^{-1}y)} dy \right| dx = \int_G \int_G |f(x)f(y)| dy dx \overset{1.11.4}{=} \int_G \int_G \Delta_G(x^{-1}y) |f(y)\overline{f(x^{-1}y)}| dy dx$$

It follows $\left| \int_G \Delta_G(x^{-1}y) f(y) \overline{f(x^{-1}y)} dy \right| = \int_G \Delta_G(x^{-1}y) |f(y)\overline{f(x^{-1}y)}| dy$, so by Lemma (b) for each $x \in G$ we can find θ_x such that $\Delta_G(x^{-1}y) f(y)\overline{f(x^{-1}y)} \in \theta_x[0, \infty)$ for all $y \in G$; since $\Delta_G \geqslant 0$, in fact $f(y)\overline{f(x^{-1}y)} \in \theta_x[0, \infty)$.

Suppose G is non-nontrivial. Pick $x_0 \neq 1$ in G. By Lemma (a) we can find $f \in C_G(x)$ with $f(x_0) = f(x_0^{-1}) = i$ and $f(1) = 1$. Now for $x = x_0$, we see

$$y = 1 \Rightarrow f(y)\overline{f(x^{-1}y)} = -i$$
$$y = x_0 \Rightarrow f(y)\overline{f(x^{-1}y)} = i$$

a contradiction. Hence G is trivial. □

Lemma 2.26. Suppose $\mathcal{A}$ is a C^*-algebra and that $a \in \mathcal{A}$ is self-adjoint, i.e., $a = a^*$. Then $r(a) = \|a\|$, where $r(a)$ is the spectral radius of a.

Proof. We already have $r(a) \leqslant \|a\|$. By C^* property have $\|a^2\| = \|a^*a\| = \|a\|^2$; it follows that $\left\|a^{2^n}\right\| = \|a\|^{2^n}$ for each n. By spectral radius formula, we obtain

$$r(a) = \lim_{n\to\infty} \left\|a^{2^n}\right\|^{\frac{1}{2^n}} = \|a\|$$

(If $\mathcal{A}$ is not unital, the spectrum of a is computed in the unitization of $\mathcal{A}$.) □

Corollary 2.26.1. If $\mathcal{A}$ is a C^*-algebra, then $\mathcal{A}$ has a unique C^*-norm.

Proof. Suppose that $\|\cdot\|$ is a norm on $\mathcal{A}$ such that $\mathcal{A}$ becomes a C^*-algebra. By Lemma for each $a \in \mathcal{A}$ we see $\|a\|^2 = \|a^*a\| = r(a^*a)$. The result now follows from the nature that the spectrum does nothing with the norm. □

Lemma 2.27. Let $\mathcal{A}$ be a C^*-algebra. Then $\mathcal{A}^e$ can be made into a C^*-algebra. Moreover, the embedding $\mathcal{A} \subseteq \mathcal{A}^e$ is isometric.

Proof. For $(a, \lambda) \in \mathcal{A}^e$, define $(a, \lambda)^* = (a^*, \overline{\lambda})$; it is clearly an involution on $\mathcal{A}^e$ extending that of on $\mathcal{A}$. For the norm, we first consider the case $\mathcal{A}$ being unital. Then there is an algebra homomorphism $\mathcal{A}^e \cong \mathcal{A} \oplus \mathbb{C}$, where the latter is equipped with componentwise multiplication. It is also a *-homomorphism if for each $(a, \lambda) \in \mathcal{A} \oplus \mathbb{C}$ we put $(a, \lambda)^* = (a^*, \overline{\lambda})$. Now defining $\|(a, \lambda)\| := \max\{\|a\|, |\lambda|\}$ clearly makes $\mathcal{A} \oplus \mathbb{C}$ a C^*-algebra and the embedding $a \mapsto (a, 0)$ is clearly isometric.

Now let us assume $\mathcal{A}$ is non-unital. Consider the homomorphism

$$\begin{aligned} L : \mathcal{A}^e &\longrightarrow \mathcal{B}(\mathcal{A}) \\ (a, \lambda) &\longmapsto L_{(a,\lambda)} : b \mapsto ab + \lambda b \end{aligned}$$

and define $\|(a, \lambda)\| := \left\|L_{(a,\lambda)}\right\|_{\mathrm{op}}$.

1° $\|\cdot\|$ is a Banach algebra norm. It suffices to show L is injective. Suppose $ab + \lambda b = 0$ for all $b \in \mathcal{A}$.

- $\lambda = 0$. Then $ab = 0$ for all $b \in \mathcal{A}$. Letting $b = a^*$ and taking norm, by C^* property we have $0 = \|aa^*\| = \|a\|^2$ so $a = 0$.
- $\lambda \neq 0$. Then $b = \frac{a}{-\lambda} b$; we may simply assume $b = ab$. In particular, $a^* = aa^*$, implying a is self-adjoint. Hence $b^* = (ab)^* = b^* a$ for all $b \in \mathcal{A}$. It follows that a is the unit in $\mathcal{A}$, a contradiction.

It follows $(a, \lambda) = (0, 0)$ so that L is injective.

2° The inclusion $a \mapsto (a, 0)$ is isometric. On the one hand $\|ab\| \leqslant \|a\| \|b\|$ gives $\left\|L_{(a,0)}\right\| \leqslant \|a\|$. On the other hand $\|aa^*\| = \|a\| \|a^*\|$ tells $\left\|L_{(a,0)}\right\| \geqslant \|a\|$.

3° $\mathcal{A}^e$ is complete. Since $\mathcal{A}$ is a complete subspace of $\mathcal{A}^e$ of finite codimension, it follows from Lemma V.6 that $\mathcal{A}^e$ is also complete.

4° $\mathcal{A}^e$ is a C^*-algebra. Let $\varepsilon > 0$. By definition there exists $b \in \mathcal{A}$ with norm 1 such that $\|ab + \lambda b\| \geqslant \|(a, \lambda)\| (1 - \varepsilon)$. This implies

$$\begin{aligned} (1-\varepsilon)^2 \|(a,\lambda)\|^2 \leqslant \|ab + \lambda b\|^2 &= \|(ab + \lambda b)^* ab + \lambda b\| \\ &= \left\|(b^*, 0)(a^*, \overline{\lambda})(a, \lambda)(b, 0)\right\| \\ &\leqslant \|(b^*, 0)\| \left\|(a^*, \overline{\lambda})(a, \lambda)\right\| \|(b, 0)\| \\ &= \left\|(a^*, \overline{\lambda})(a, \lambda)\right\| \end{aligned}$$

Since $\varepsilon > 0$ is arbitrary, it follows

$$\left\|(a, \lambda)^2\right\| \leqslant \left\|(a^*, \overline{\lambda})(a, \lambda)\right\| \leqslant \left\|(a^*, \overline{\lambda})\right\| \|(a, \lambda)\|$$

In particular we have $\|(a, \lambda)\| \leqslant \left\|(a^*, \overline{\lambda})\right\|$; by symmetry we have equality and hence we have equality everywhere.

□

Definition. If $\mathcal{A}$ is a Banach *-algebra and $a \in \mathcal{A}$, we define the **real part** and the **imaginary part** of a as

$$\operatorname{Re} a = \frac{1}{2}(a + a^*) \text{ and } \operatorname{Im} a = \frac{1}{2i}(a - a^*)$$

Then $\operatorname{Re} a$ and $\operatorname{Im} a$ are self-adjoint with $a = \operatorname{Re} a + i \operatorname{Im} a$.

2.2.2 Gelfand-Naimark

Lemma 2.28. If $\mathcal{A}$ is a commutative C^*-algebra, then $m(a^*) = \overline{m(a)}$ for every $a \in \mathcal{A}$ and every $m \in \Delta_{\mathcal{A}}$.

Proof. By passing to $\mathcal{A}^e$ if necessary, we may assume $\mathcal{A}$ is unital. If the statement holds for self-adjoint elements, then for general $a \in \mathcal{A}$,

$$m(a^*) = m(\operatorname{Re} a - i \operatorname{Im} a) = m(\operatorname{Re} a) - im(\operatorname{Im} a) = \overline{m(\operatorname{Re} a)} - i\overline{m(\operatorname{Im} a)} = \overline{m(\operatorname{Re} a) + im(\operatorname{Im} a)} = \overline{m(a)}$$

Hence we may assume a is self-adjoint; in this case we must show $m(a) \in \mathbb{R}$. Write $m(a) = x + iy$ with $x, y \in \mathbb{R}$ and put $a_t = a + it$ for all $t \in \mathbb{R}$. Then

$$a_t^* a_t = (a - it)(a + it) = a^2 + t^2$$

and hence

$$x^2 + (y + t)^2 = |m(a_t)|^2 \leqslant \|a_t\|^2 = \|a_t^* a_t\| = \left\|a^2 + t^2\right\| \leqslant \|a\|^2 + t^2$$

so that $x^2 + y^2 + 2yt \leqslant \|a\|^2$, which forces $y = 0$. Thus $m(a) \in \mathbb{R}$ as we wished. □

Theorem 2.29 (Gelfand-Naimark)**.** If $\mathcal{A}$ is a commutative C^*-algebra, then the Gelfand transform

$$\begin{aligned} \mathcal{A} &\longrightarrow C_0(\Delta_{\mathcal{A}}) \\ a &\longmapsto \hat{a} \end{aligned}$$

is an isometric *-isomorphism. Furthermore, $\Delta_{\mathcal{A}}$ is compact if and only if $\mathcal{A}$ is unital.

Proof. We show $\mathcal{A} \to C_0(\Delta_{\mathcal{A}})$ is an isometric *-homomorphism with dense image $\hat{\mathcal{A}} = \{\hat{a} \mid a \in \mathcal{A}\}$. In particular, this shows $\hat{\mathcal{A}}$ is complete (thanks to isometry), and hence closed. Being dense, it follows that $\hat{\mathcal{A}} = C_0(\Delta_{\mathcal{A}})$. We first show it is an isometry. Indeed, by 2.22.3 and 2.26, for any $a \in \mathcal{A}$ one has

$$\|\hat{a}\|^2 = \left\|\overline{\hat{a}}\hat{a}\right\|^2 = \left\|\widehat{a^*a}\right\|^2 = r(a^*a) = \|a^*a\| = \|a\|^2$$

To show $\hat{\mathcal{A}}$ is dense in $C_0(\Delta_{\mathcal{A}})$, we use the Stone-Weierstrass, which is valid for $\Delta_{\mathcal{A}}$ is LCH by 2.19.

- $\hat{\mathcal{A}}$ separates points. Indeed, if $m, n \in \Delta_{\mathcal{A}}$ such that $\hat{a}(m) = \hat{a}(n)$ for each $a \in \mathcal{A}$, then $m = n$ since they agree everywhere.
- $\hat{\mathcal{A}}$ vanishes nowhere. This follows from our definition that $m \in \Delta_{\mathcal{A}}$ is nontrivial; hence we can find $a \in \mathcal{A}$ with $0 \neq m(a) = \hat{a}(m)$.
- $\hat{\mathcal{A}}$ is stable under complex conjugation. By 2.28, one has $\overline{\hat{a}(m)} = \overline{m(a)} = m(a^*) = \hat{a^*}(m)$ so that $\overline{\hat{a}} \in \hat{\mathcal{A}}$.

Hence $\hat{\mathcal{A}}$ is dense in $C_0(\Delta_{\mathcal{A}})$ with respect to the sup-norm. Finally, if $\Delta_{\mathcal{A}}$ is compact, then $\mathcal{A} \cong C_0(\Delta_{\mathcal{A}})$ is unital. The converse is 2.19. □

Let $\mathcal{A}$ and $\mathcal{B}$ be two commutative C^*-algebras. A morphism from $\mathcal{A}$ to $\mathcal{B}$ is a *-homomorphism $\Phi : \mathcal{A} \to \mathcal{B}$ which defines a map $\Phi^* : \Delta_{\mathcal{B}} \to \Delta_{\mathcal{A}}$. This defines a category, and denote it by C^***-Alg**. Let **LCH** be a full subcategory of **Top** whose objects are locally compact Hausdorff spaces. In this section we met several (contravariant) functors:

$$\begin{aligned} F : \mathbf{LCH} &\longrightarrow C^*\text{-}\mathbf{Alg} & G : C^*\text{-}\mathbf{Alg} &\longrightarrow \mathbf{LCH} \\ X \xrightarrow{f} Y &\longmapsto C_0(X) \xleftarrow{f^*} C_0(Y) & \mathcal{A} \xrightarrow{\Phi} \mathcal{B} &\longmapsto \Delta_{\mathcal{A}} \xleftarrow{\Phi^*} \Delta_{\mathcal{B}} \end{aligned}$$

Gelfand-Naimark says that the Gelfand map defines a natural isomorphism from the identity functor to $F \circ G$: for $\Phi : \mathcal{A} \to \mathcal{B}$, one has a commutative diagram

$$\begin{tikzcd} \mathcal{A} \arrow[r, "\sim"] \arrow[d, "\Phi"'] & C_0(\Delta_{\mathcal{A}}) \arrow[d, "(\Phi^*)^*"] \\ \mathcal{B} \arrow[r, "\sim"] & C_0(\Delta_{\mathcal{B}}) \end{tikzcd}$$

On the other hand, the map in Example 2.20 defines a natural isomorphism from the identity functor of $G \circ F$: for $f : X \to Y$, one has a commutative diagram

$$\begin{tikzcd} X \arrow[r, "\sim"] \arrow[d, "f"'] & \Delta_{C_0(X)} \arrow[d, "(f^*)^*"] \\ Y \arrow[r] & \Delta_{C_0(Y)} \end{tikzcd}$$

(It is not quite clear from definition that F defines a functor, but the above commutative diagram shows that F is really a functor.) Hence, F and G define an equivalence of categories between **LCH** and C^***-Alg**.

There are typical subcategories of **LCH** and C^***-Alg**, namely the full subcategory of **LCH** whose objects are compact Hausdorff spaces, and the subcategory of C^***-Alg** whose objects are commutative unital C^*-algebras and morphisms are unital *-homomorphisms. Then F and G restrict to an equivalence of categories between them.

2.3 The Continuous Functional Calculus

Definition. Let $\mathcal{A}$ be a commutative unital C^*-algebra and $a \in \mathcal{A}$. The assignment

$$\begin{aligned} C(\sigma_{\mathcal{A}}(a)) &\longrightarrow C(\Delta_{\mathcal{A}}) \cong \mathcal{A} \\ f &\longmapsto f \circ \hat{a} =: f(a) \end{aligned}$$

is called the **(continuous) functional calculus** for a. (The last isomorphism is given by Gelfand-Naimark.)

- The assignment is well-defined for $\sigma_{\mathcal{A}}(a) = \operatorname{Im} \hat{a}$ by 2.22.3.

Definition. Let $\mathcal{A}$ be a C^*-algebra.

1. An element $a \in \mathcal{A}$ is **normal** if $aa^* = a^*a$.
2. For each $a \in \mathcal{A}$, denote by $C^*(a)$ the **smallest C^*-subalgebra of $\mathcal{A}$ containing a.**
3. If $\mathcal{A}$ is unital, $C^*(a, 1)$ is called the **unital C^*-algebra generated by** a.

- One sees that $C^*(a)$ is the closure of the subalgebra generated by a and a^*. Indeed, denote by X the latter space. X is a C^*-subalgebra of $\mathcal{A}$ for * is continuous. Since $C^*(a)$ is a closed *-algebra containing a and a^*, $X \subseteq C^*(a)$. By minimality $X = C^*(a)$.
- Similarly, $C^*(a, 1)$ is the closure of the subalgebra generated by 1, a and a^*.
- $a \in \mathcal{A}$ is normal if and only if $C^*(a)$ is commutative.

Lemma 2.30. Let $\Phi : \mathcal{A} \to \mathcal{B}$ be a *-homomorphism with $\mathcal{A}$ Banach * and $\mathcal{B}$ C^*. Then $\|\Phi(a)\| \leqslant \|a\|$ for every $a \in \mathcal{A}$. In particular, Φ is continuous.

Proof. To show Φ is continuous, we only need to show $\Phi^e : \mathcal{A}^e \to \mathcal{B}^e$ is continuous. If this is shown, then $\Phi = \Phi^e|_{\mathcal{A}}^{\mathcal{B}}$ is also continuous. Thus we can assume $\mathcal{A}$ and $\mathcal{B}$ are unital and Φ is unital. We can assume a is self-adjoint. Indeed, if the self-adjoint case is proved, then

$$\|\Phi(a)\|^2 = \|\Phi(a)\Phi(a)^*\| = \|\Phi(aa^*)\| \leqslant \|aa^*\| \leqslant \|a\| \, \|a^*\| = \|a\|^2$$

We claim $\sigma_{\mathcal{B}}(\Phi(a)) \subseteq \sigma_{\mathcal{A}}(a)$. To see this, suppose $\Phi(a) - \lambda = \Phi(a - \lambda)$ is not invertible in $\mathcal{B}$. Then $a - \lambda$ is not invertible, either. It follows that

$$\|\Phi(a)\| \overset{2.26}{=} r(\Phi(a)) \leqslant r(a) \leqslant \|a\|$$

□

Lemma 2.31. Let $\mathcal{A} \subseteq \mathcal{B}$ be unital C^*-algebra. Then $\mathcal{B}^\times \cap \mathcal{A} = \mathcal{A}^\times$. In particular, for $a \in \mathcal{A}$, we have $\sigma_{\mathcal{A}}(a) = \sigma_{\mathcal{B}}(a)$.

Proof. The inclusion $\supseteq$ is clear, so we suppose $a \in \mathcal{B}^\times \cap \mathcal{A}$. First assume $\mathcal{A}$ and $\mathcal{B}$ are commutative. The restriction res : $\Delta_{\mathcal{B}} \to \Delta_{\mathcal{A}}$ defines a continuous homomorphism, whence a map res* : $C(\Delta_{\mathcal{A}}) \to C(\Delta_{\mathcal{B}})$. Then the diagram

$$\begin{tikzcd} C(\Delta_{\mathcal{A}}) \arrow[r, "\sim"] \arrow[d, "\mathrm{res}^*"] & \mathcal{A} \arrow[d] \\ C(\Delta_{\mathcal{B}}) \arrow[r, "\sim"] & \mathcal{B} \end{tikzcd}$$

where the horizontal maps are Gelfand transforms, is commutative. Since a is invertible in $\mathcal{B}$, its image $\widehat{a} \in C(\Delta_{\mathcal{B}})$ has no zero, whence $\widehat{a} \in C(\Delta_{\mathcal{A}})$ has no zero as well. This proves $a \in \mathcal{A}^\times$.

Now assume $\mathcal{A}$ is commutative, while $\mathcal{B}$ possibly not. We can find $b \in \mathcal{B}$ such that $ab = ba = 1$. Then $a^*b = baa^*b = ba^*ab = ba^*$, i.e., b commutes with a^*. Similarly, $b = a^{-1}$ commutes with b^*. This means a is invertible in the commutative C^*-algebra $C^*(1, a, b)$ generated by $1, a, b$. Then $a \in \mathcal{A}^\times$ by the first paragraph.

Finally, drop commutative condition on $\mathcal{A}$. First consider the case a being normal. Then the C^*-algebra $C^*(1, a)$ is commutative, and the second paragraph shows $a \in C^*(1, a)^\times \subseteq \mathcal{A}^\times$. Last, suppose $a \in \mathcal{A}$ is arbitrary. Note that a is invertible if and only if aa^* and a^*a are invertible (purely algebraic). Since aa^* and a^*a are normal, the last paragraph implies aa^* and a^*a are invertible in $\mathcal{A}$, so a is invertible in $\mathcal{A}$ as well. $\square$

Theorem 2.32. Let $\mathcal{A}$ be a unital C^*-algebra and let $a \in \mathcal{A}$ be a normal element. Then there exists a unique unital *-homomorphism

$$\Phi_a : C(\sigma(a)) \to \mathcal{A}$$

with the property that $\Phi_a(\mathrm{id}) = a$. Write $f(a) := \Phi_a(f)$. Then

(i) Φ_a is isometric with image $C^*(a, 1) \subseteq \mathcal{A}$.

(ii) If $\mathcal{A}$ is commutative, the $\widehat{\Phi_a(f)} = f \circ \hat{a}$.

(iii) If $\mathcal{A} = C^*(a, 1)$, then $\hat{a} : \Delta_{\mathcal{A}} \to \sigma(a)$ is a homeomorphism.

(iv) If $f : \sigma(a) \to \mathbb{C}$ has a power series expansion $f(z) = \sum_{n=0}^{\infty} a_n(z - z_0)^n$ which converges uniformly on $\sigma(a)$, then $f(a) = \sum_{n=0}^{\infty} a_n(a - z_0 1)^n$, and the series converges in norm.

Proof. The uniqueness follows from Stone-Weierstrass and Lemma 2.30. Let us prove the existence. Consider $a \in C^*(1, a) \subseteq \mathcal{A}$ and its Gelfand transform

$$\hat{a} : \Delta_{C^*(1,a)} \longrightarrow \operatorname{Im} \hat{a} = \sigma_{C^*(1,a)}(a) = \sigma_{\mathcal{A}}(a)$$

This is injective, for if $\hat{a}(m_1) = \hat{a}(m_2)$, then $m_1(a) = m_2(a)$; together with Lemma 2.28, we see m_1 and m_2 agree on the linear span of elements of the form $a^\ell(a^*)^k$ which is dense in $C^*(1, a)$, and thus $m_1 = m_2$. Since $C^*(1, a)$ is unital, $\Delta_{C^*(1,a)}$ is compact, and thus $\hat{a}$ is a homeomorphism. Thus we get a unital isometric *-isomorphism

$$\begin{aligned} \Psi : C(\sigma(a)) &\longrightarrow C(\Delta_{C^*(1,a)}) \\ f &\longmapsto f \circ \hat{a} \end{aligned}$$

Composing this with the inverse of the Gelfand transform $C^*(1, a) \to C(\Delta_{C^*(1,a)})$ gives us a unital isometric *-isomorphism

$$\Phi_a : C(\sigma(a)) \longrightarrow C^*(1, a)$$

Now compute $\Phi(1) = 1$ and $\Psi(\mathrm{id}) = \mathrm{id} \circ \hat{a} = \hat{a}$, so that $\Phi_a(\mathrm{id}) = a$. For the last assertion (iv), for each N put $f_N(z) = \sum_{n=0}^{N} a_n(z - z_0)^n$. Then

$$\Psi_a(f_N) = \sum_{n=0}^{N} a_n(a - z_0 1)^n$$

and since Ψ_a is an isometry,

$$\|\Psi_a(f_N) - \Psi_a(f)\|_{\mathcal{A}} = \|\Psi_a(f_N - f)\|_{\mathcal{A}} = \|f_N - f\|_\infty \to 0 \text{ as } N \to \infty$$

□

Example 2.33. Let X be a compact Hausdorff space. Then for each $f \in C(X)$, the functional calculus sends each function $g \in C(\sigma(f)) = C(f(X))$ to the function $g \circ f \in C(X)$. This follows quickly from the uniqueness part of Theorem and the fact that $g \mapsto g \circ f$ satisfies $\mathrm{id} \circ f = f$.

Corollary 2.31.1.

(a) If $a \in \mathcal{A}$ is normal, then a is self-adjoint if and only $\sigma_{\mathcal{A}}(a) \subseteq \mathbb{R}$.

(b) Let $\Psi : \mathcal{A} \to \mathcal{B}$ be a unital *-homomorphism between C^*-algebras and let $a \in \mathcal{A}$ be a normal element. Then $\Psi(a)$ is normal and $\sigma(\Psi(a)) \subseteq \sigma(a)$. Then diagram

$$\begin{tikzcd} C(\sigma(a)) \arrow[r, "\Phi_a"] \arrow[d, "\mathrm{res}"] & \mathcal{A} \arrow[d, "\Psi"] \\ C(\sigma(\Psi(a))) \arrow[r, "\Phi_{\Psi(a)}"] & \mathcal{B} \end{tikzcd}$$

commutes. In particular, one has $f(\Psi(a)) = \Psi(f(a))$ for every $f \in C(\sigma(a))$.

(c) Suppose $a \in \mathcal{A}$ is a normal element in the unital C^*-algebra $\mathcal{A}$ and let $f \in C(\sigma_{\mathcal{A}}(a))$. Then

(i) $f(a) \in \mathcal{A}$ is normal,

(ii) (Spectral mapping theorem) $\sigma_{\mathcal{A}}(f(a)) = f(\sigma_{\mathcal{A}}(a))$, and

(iii) $g(f(a)) = (g \circ f)(a)$ for all $g \in C(\sigma_{C(\sigma_{\mathcal{A}}(a))}(f)) = C(f(\sigma_{\mathcal{A}}(a))) \overset{\text{(ii)}}{=} C(\sigma_{\mathcal{A}}(f(a)))$.

Proof.

(a) $\Phi_a(\mathrm{id}) = a$, so $a^* = \Phi_a(\mathrm{id})^* = \Phi_a(\overline{\mathrm{id}})$. Since Φ_a is isometric, it is injective, so $a = a^*$ if and only if $\mathrm{id} = \overline{\mathrm{id}}$, if and only if $\sigma_{\mathcal{A}}(a) \subseteq \mathbb{R}$.

(b) The first two statements are clear. For the commutativity we have

$$\Psi\Phi_a(\mathrm{id}_{\sigma(a)}) = \Psi(a) = \Phi_{\Psi(a)}(\mathrm{id}_{\sigma(\Psi(a))}) = \Phi_{\Psi(a)}(\mathrm{res}\,\mathrm{id}_{\sigma(a)})$$

Since all maps are *-homomorphism, it follows from Stone-Weierstrass that the diagram commutes.

(c) (i) is clear since $f(a) \in C^*(1, a)$. For (iii), consider the commutative diagram

$$\begin{tikzcd} C(\sigma(f)) \arrow[r, "\Phi_f"] \arrow[d, "\mathrm{res}"] & C(\sigma_{\mathcal{A}}(a)) \arrow[d, "\Phi_a"] \\ C(\sigma(\Phi_a(f))) \arrow[r, "\Phi_{f(a)}"] & \mathcal{A} \end{tikzcd}$$

Note that by (ii) the left vertical arrow res is the identity map, so

$$g(f(a)) = \Phi_{f(a)}(g) = \Phi_{f(a)}(\operatorname{res} g) = \Phi_a \Phi_f(g) = \Phi_a(g \circ f) = (g \circ f)(a) = (g \circ f)(a)$$

It remains to show (ii). Trivially we have $\sigma_{\mathcal{A}}(f(a)) \subseteq \sigma_{C(\sigma_{\mathcal{A}}(a))}(f) = f(\sigma_{\mathcal{A}}(a))$. Note that the spectrum is preserved under $\mathbb{C}-$algebra isomorphisms; then

$$\sigma_{\mathcal{A}}(f(a)) = \sigma_{C^*(a,1)}(f(a)) = \sigma_{C(\sigma_{\mathcal{A}}(a))}(f) = f(\sigma_{\mathcal{A}}(a))$$

□

Corollary 2.31.2. Suppose $\Psi : \mathcal{A} \to \mathcal{B}$ is an injective *-homomorphism between C^*-algebras. Then Ψ is already isometric.

Proof. By Lemma 2.30 we have $\|\Psi(a)\| \leqslant \|a\|$ for all $a \in \mathcal{A}$. As in the proof of Lemma 2.30, we may assume everything is unital. Assume there exists $a \in \mathcal{A}$ with $\|\Psi(a)\| < \|a\|$; we may assume $\|a\| = 1$. Then

$$\alpha := \|\Psi(a^*a)\| = \|\Psi(a)\|^2 < \|a\|^2 = \|a^*a\| = 1$$

Put $c = a^*a$; then c is self-adjoint with $\sigma(c) \subseteq [-1, 1]$ and $\sigma(\Psi(c)) \subseteq [-\alpha, \alpha]$. By Lemma 2.26 $\sigma(c)$ contains either 1 or -1 (note that $\sigma(c)$ is compact, so $r(a) = |\lambda|$ for some $\lambda \in \sigma(c)$), so we can find a function $0 \neq f \in C(\sigma(c))$ with $f \equiv 0$ on $\sigma(\Psi(c))$. Injectivity of Ψ shows that $0 \neq \Psi(f(c)) = f(\Psi(c)) = 0$, a contradiction. □

Corollary 2.31.3 (Fuglede's)**.** Let $\mathcal{A}$ be a C^*-algebra and $a \in \mathcal{A}$ be normal. If $b \in \mathcal{A}$ is such that $ab = ba$, then $a^*b = ba^*$.

Proof. We may assume $\mathcal{A}$ is unital. Define $f : \mathbb{C} \to \mathcal{A}$ by

$$f(\lambda) = e^{i\lambda a^*} b e^{-i\lambda a^*}.$$

Clearly f is a holomorphic function. Write $\lambda a^* = c_1(\lambda) + ic_2(\lambda)$ with $c_i(\lambda)$ self-adjoint. Since a is normal, $c_1(\lambda)$ commutes with $c_2(\lambda)$ and

$$e^{i\lambda a^*} = e^{ic_1(\lambda)} e^{-c_2(\lambda)} = e^{2ic_1(\lambda)} e^{-(c_2(\lambda) + ic_1(\lambda))} = e^{2ic_1(\lambda)} e^{-i\bar{\lambda} a}.$$

Since $ab = ba$, we have $e^{-i\bar{\lambda}a} b = b e^{-i\bar{\lambda}a}$, so

$$f(\lambda) = e^{2ic_1(\lambda)} \left(e^{-i\bar{\lambda}a} b e^{i\bar{\lambda}a} \right) e^{-2ic_1(\lambda)} = e^{2ic_1(\lambda)} b e^{-2ic_1(\lambda)}.$$

As $c_1(\lambda)$ is self-adjoint, we have $\sigma(c_1(\lambda)) \subseteq \mathbb{R}$, so $e^{2ic_1(\lambda)}$ is bounded, which implies that $f(\lambda)$ is a bounded entire function. By Liouville's theorem, it follows that f is a constant. But

$$0 = f'(0) = i(a^*b - ba^*)$$

this shows $a^*b = ba^*$. $\square$

We now extend Theorem 2.32 to the non-unital case. Recall for a non-unital C^*-algebra $\mathcal{A}$, the spectrum of an element is computed in the unitization; this implies $0 \in \sigma(a)$ for all $a \in \mathcal{A}$. For each element $a \in \mathcal{A}$, let us defined

$$C_0(\sigma(a)) := \{f \in C(\sigma(a)) \mid f(0) = 0\}$$

This notation may cause confusion. However, observe that $\sigma(a)$ is the one-point compactification of $\sigma(a)\backslash\{0\}$, so $\{f \in C(\sigma(a)) \mid f(0) = 0\}$ is exactly the space $C_0(\sigma(a)\backslash\{0\}) \subseteq C(\sigma(a))$. This kind of justifies our confusing notation.

Corollary 2.31.4. Let $\mathcal{A}$ be a *non-unital* C^*-algebra and $a \in \mathcal{A}$ be a normal element. There exists a unique *-homomorphism

$$\Phi_a : C_0(\sigma(a)) \to \mathcal{A}$$

with the property that $\Phi_a(\mathrm{id}) = a$. Moreover, it is isometric with image $C^*(a) \subseteq \mathcal{A}$.

Proof. The uniqueness follows as in Theorem 2.32. For the existence, we apply Theorem 2.32 to the unitization $\mathcal{A}^e$ of $\mathcal{A}$. We then obtain a unique unital *-homomorphism $\Phi_a : C(\sigma(a)) \to \mathcal{A}^e$ with the following properties:

(i) $\Phi_a(\mathrm{id}) = a$, where $\mathrm{id} = \mathrm{id}_{\sigma(a)}$ and (by definition) $\sigma(a) = \sigma_{\mathcal{A}^e}((a, 0))$.

(ii) Φ_a is isometric with image $C^*(a, 1) \subseteq \mathcal{A}^e$.

Let us study the image of $C_0(\sigma(a))$ under Φ_a. Since Φ_a is isometric and $C_0(\sigma(a))$ is closed in $C(\sigma(a))$, $\Phi_a(C_0(\sigma(a)))$ is also closed. Clearly $C_0(\sigma(a))$ contains all polynomials in z and $\overline{z}$ without constant term, so $\Phi_a(C_0(\sigma(a)))$ contains a dense subset of $C^*(a)$; being closed, we have $C^*(a) \subseteq \Phi_a(C_0(\sigma(a))) \subsetneq C^*(a, 1)$.

Lemma 2.32. $\dim_{\mathbb{C}} C^*(a, 1)/C^*(a) = 1$.

Proof. Let $x \in C^*(a, 1)\backslash C^*(a)$. Since $C^*(a)$ is closed in $C^*(a, 1)$, we can find a sequence of polynomials with nonzero constant term $p_n(X, Y) \in \mathbb{C}[X, Y]$ such that $p_n(a, a^*) \to x$. Let $c_n = p_n(0, 0)$; then $c_n \to c$ for some $c \in \mathbb{C}$. Put $q_n(X, Y) := p_n(X, Y) - c_n$; then $q_n(a, a^*) \to x - c1$ (recall that $1 = 1_{\mathcal{A}^e} = (0, 1) \in \mathcal{A} \times \mathbb{C}$). But $C^*(a)$ is closed and $q_n(a, a^*) \in C^*(a)$, it follows that $x - c1 \in C^*(a)$. $\square$

By the lemma, since $\Phi_a(C_0(\sigma(a)))$ is proper in $C^*(a, 1)$, it forces that $\Phi_a(C_0(\sigma(a))) = C^*(a)$, as desired. $\square$

3 More on C^*-algebras

We collect some terminology in C^*-algebra.

Definition. Let $\mathcal{A}$ be a C^*-algebra and $a \in \mathcal{A}$.

1. a is **normal** if $a^*a = aa^*$.
2. a is **self-adjoint** if $a = a^*$. Let $\mathcal{A}_{sa}$ denote the set of self-adjoint elements in $\mathcal{A}$.
3. a is **positive** if a is self-adjoint and $\sigma_{\mathcal{A}}(a) \subseteq [0, \infty)$. Let $\mathcal{A}_+$ denote the set of positive elements in $\mathcal{A}$. (Recall for the non-unital case, the spectrum is computed in the unitization.)
4. a is a **projection** if a is self-adjoint and $a^2 = a$.
5. $a, b \in \mathcal{A}$ are called **orthogonal** if $ab = ba = 0$.
6. If $\mathcal{A}$ is unital, a is called **unitary** if a is normal and $aa^* = 1$.

We can define an partial order $\leqslant$ on $\mathcal{A}_{sa}$ as follows. For $a, b \in \mathcal{A}_{sa}$, say $a \leqslant b$ if $b - a \in \mathcal{A}_+$. This is really a partial order, for if $0 \leqslant a \geqslant 0$, i.e., $\sigma(a) = \{0\}$, then by Lemma 2.26, $\|a\| = r(a) = 0$, so that $a = 0$.

Lemma 3.1. Let X be an LCH space. Consider the C^*-algebra $C_0(X)$; the norm is sup-norm, and the * is complex conjugation.

(i) For a function $f \in C_0(X)$, $f \geqslant 0$ is positive if and only if $f(x) \geqslant 0$ for all $x \in X$.

(ii) If $f \in C_0(X)$ is positive, then for all $n \in \mathbb{N}$, there exists a unique positive function $g \in C_0(X)$ such that $f = g^n$.

(iii) For self-adjoint $f \in C_0(X)$, there exist unique positive functions f^+, $f^- \in C_0(X)$ such that $f = f^+ - f^-$ with $f^+f^- = f^-f^+ = 0$.

Proof.

(i) Note that $\sigma(f) = f(X)$. Thus $f \geqslant 0 \Leftrightarrow \sigma(f) \subseteq [0, \infty) \Leftrightarrow f(X) \subseteq [0, \infty)$.

(ii) For the existence, just take $g \in C_0(X)$ defined by $g(x) := f(x)^{\frac{1}{n}}$. The uniqueness is clear for we are in $\mathbb{R}_{\geqslant 0}$.

(iii) Since f is self-adjoint, $f(X) \subseteq \mathbb{R}$. Take $f^+ := \max\{0, f\}$ and $f^- = \max\{0, -f\}$

□

Proposition 3.2. Let $\mathcal{A}$ be a C^*-algebra and $a \in \mathcal{A}_+$ be positive. For each $n \in \mathbb{N}$, there exists a unique positive element $b \in \mathcal{A}_+$ with $b^n = a$.

Proof. If $\mathcal{A}$ is unital, we use Theorem 2.32; otherwise, we use Corollary 2.31.4 instead. Since $\sigma(a) \subseteq [0, \infty)$, we can find $f(x) \in C_0(\sigma(a))$ with $f \geqslant 0$ and $f(x)^n = x$ for $x \in \sigma(a)$. Define $b := f(a) \in \mathcal{A}$; b is positive by Spectral mapping theorem, and $b^n = a$. This shows the existence. For the uniqueness, let $b' \in \mathcal{A}_+$ be another such element. Then $a \in C^*(b')$, and then

$$b \in C^*(a) \subseteq C^*(b') \cong C_0(\sigma(b'))$$

The uniqueness now follows from Lemma 3.1.(ii). □

Proposition 3.3. Let $\mathcal{A}$ be a C^*-algebra.

(i) Every element is a linear combination of two self-adjoint elements.

(ii) Every self-adjoint element can be written uniquely as $x - y$ with x, y positive and orthogonal.

(iii) Every element can be written as a linear combination of four positive elements.

(iv) If $\mathcal{A}$ is unital, every element is a linear combination of four unitary elements.

Proof. We already see that for $a \in \mathcal{A}$, we can write $a = \operatorname{Re} a + i \operatorname{Im} a$ with

$$\operatorname{Re} a = \frac{1}{2}(a + a^*) \text{ and } \operatorname{Im} a = \frac{1}{2i}(a - a^*)$$

The existence part of (ii) follows directly from Lemma 3.1(ii), and if $a = x - y$ with x, y positive and $xy = 0$, then $(a^2)^{\frac{1}{2}} = x + y$, so $x = \frac{1}{2}(a + (a^2)^{\frac{1}{2}})$ is uniquely determined by a. (iii) follows from (i) and (ii). For (iv), by (i) it suffices to show every self-adjoint element is a linear combination of two unitary elements. Let $a \in \mathcal{A}_{sa}$; scaling if necessarily, we may assume $\|a\| \leqslant 1$. Then $1 - a^2 \geqslant 0$; indeed, it is self-adjoint, and if we put $f(t) = 1 - t^2$, then by Spectral mapping theorem we see $\sigma(1 - a^2) = f(\sigma(a)) \subseteq [0, 1]$. Let $u_1 = a - i\sqrt{1 - a^2}$ and $u_2 = a + i\sqrt{1 - a^2}$; then the u_i are unitary, and $a = \frac{1}{2}(u_1 + u_2)$, as wanted. □

3.1 Positive elements

Proposition 3.4. Let $\mathcal{A}$ be a C^*-algebra. Then $\mathcal{A}_+$ is a closed convex cone.

Proof. WLOG we assume $\mathcal{A}$ is unital. For $a \in \mathcal{A}_+$ and $\lambda \geqslant 0$, it is easy to check $\lambda a \in \mathcal{A}_+$, so we only need to show that the sum of two positive elements remains positive.

We first show that for $X \subseteq \mathbb{R}$ compact and $f \in C(X)$, f is positive if there exists $r \in \mathbb{R}_{\geqslant 0}$ such that $\|f - r\| \leqslant r$. If $r = 0$, then $f = 0$ so it is positive trivially. Suppose $r > 0$ but f is not positive; we then can find $t \in X$ with $f(t) < 0$. Then $|f(t) - r| > r$ so that $\|f - r\| = \sup_{t \in X} |f(t) - r| > r$, a contradiction. This proves our claim. Secondly, note that if $f \in \mathcal{A}$ is positive, then $\|f - r\| \leqslant \|f\|$ if $0 \leqslant r \leqslant \|f\|$. This can be seen easily by Lemma 2.26.

Now let $a, b \in \mathcal{A}_+$. Since $a+b$ is self-adjoint, we may identify $C^*(a+b, 1_{\mathcal{A}}) \cong C(\sigma_{\mathcal{A}}(a+b))$ and $a+b$ with $f(t) = t$. Thus it suffices to find $r \geqslant 0$ such that $\|a+b-r\| \leqslant r$. Take $r := \|a\| + \|b\|$; then

$$\|a+b-r\| = \|a+b-\|a\|-\|b\|\| \leqslant \|a-\|a\|\| + \|b-\|b\|\| \leqslant \|a\| + \|b\| = r$$

The last inequality follows from what mentioned in the end of the second paragraph. This shows $a+b$ is positive.

It remains to show $\mathcal{A}_+$ is closed. From the second paragraph we have

$$B := \{a \in \mathcal{A}_+ \mid \|a\| \leqslant 1\} = \{a \in \mathcal{A} \mid \|a - 1_{\mathcal{A}}\| \leqslant 1\} \cap \mathcal{A}_{sa}$$

The last two sets are closed, so B is closed as well. Thus $\mathcal{A}_+ = \mathbb{R}_{\geqslant 0}B$ is closed: if a_n is a sequence in $\mathbb{R}_{\geqslant 0}B$ with $a_n \to a \in \mathcal{A}$ and $a \neq 0$, then $\dfrac{a_n}{\|a_n\|} \to \dfrac{a}{\|a\|}$, and since the $\dfrac{a_n}{\|a_n\|} \in B$, it forces $\dfrac{a}{\|a\|} \in B$, so $a \in \mathbb{R}_{\geqslant 0}B$. □

Corollary 3.4.1. The poset $(\mathcal{A}_{sa}, \geqslant)$ is upward directed, i.e., any two element in $\mathcal{A}_{sa}$ has a common upper bound in $\mathcal{A}_{sa}$.

Proposition 3.5. For $a \in \mathcal{A}$, the element a^*a is positive. In particular, we have $\mathcal{A}_+ = \{a^*a \mid a \in \mathcal{A}\}$.

Proof. Every a^*a is self-adjoint. Suppose $-a^*a$ is positive. Since $\sigma_{\mathcal{A}}(-a^*a)\backslash\{0\} = \sigma_{\mathcal{A}}(-aa^*)\backslash\{0\}$, $-aa^*$ is positive as well. Write $a = b + ic$ for $b, c \in \mathcal{A}_{sa}$. Then

$$\begin{aligned} aa^* &= b^2 + ibc - icb + c^2 \\ a^*a &= b^2 - ibc + icd + c^2 \end{aligned}$$

so $a^*a = 2b^2 + 2c^2 + (-aa^*)$ is a sum of positive elements, implying a^*a is also positive. Hence $0 = \|a^*a\| = \|a\|^2$, and therefore $\|a\| = 0$.

Write $a^*a = b - c$ with $b, c \in \mathcal{A}_+$ and $bc = cb = 0$. We claim $c = 0$. We have

$$-(ac)^*(ac) = -ca^*ac = -c(b-c)c = c^3 \in \mathcal{A}_+$$

By the first paragraph, it follows $ac = 0$, and hence

$$0 = a^*ac = (b-c)c = -c^2 = -c^*c$$

so that $0 = \|c^*c\| = \|c\|^2$, i.e., $c = 0$. Hence $a^*a = b$ is positive.

For the last equality, note that if $a \geqslant 0$, then there exists a positive square root $a^{\frac{1}{2}}$ of a by Proposition 3.2, so $a = (a^{\frac{1}{2}})^* a^{\frac{1}{2}}$. □

Proposition 3.6. Let $\mathcal{A}$ be a C^*-algebra and $a, b \in \mathcal{A}_{sa}$ with $a \leqslant b$. Then

(i) for any $c \in \mathcal{A}$, we have $c^*ac \leqslant c^*bc$;

(ii) $\|a\| \leqslant \|b\|$;

(iii) if $\mathcal{A}$ is unital and $a, b \in \mathcal{A}^\times \cap \mathcal{A}_+$, then $b^{-1} \leqslant a^{-1}$

Proof.

(i) Since $b - a \geqslant 0$, it has a positive square root $(b-a)^{\frac{1}{2}}$ by Proposition 3.2. Thus

$$c^*bc - c^*ac = c^*(b-a)c = ((b-a)^{\frac{1}{2}}c)^*((b-a)^{\frac{1}{2}}c) \geqslant 0$$

by Proposition 3.5.

(ii) WLOG we may assume $\mathcal{A}$ is unital. By continuous functional calculus, we have $b \leqslant \|b\|$; to see this, we can prove it in $C^*(b, 1_{\mathcal{A}}) \cong C(\sigma_{\mathcal{A}}(b))$, and this becomes an obvious identity $t \leqslant \max_{t \in \sigma_{\mathcal{A}}(b)} |t|$. Hence $a \leqslant b \leqslant \|b\|$. Finally, the same reasoning in $C^*(a, 1_{\mathcal{A}}) \cong C(\sigma_{\mathcal{A}}(a))$ gives $\|a\| \leqslant \|b\|$.

(iii) Note that for $c \in \mathcal{A}^\times$, we have $\sigma_{\mathcal{A}}(c^{-1}) = (\sigma_{\mathcal{A}}(c))^{-1}$, so $b^{-1}, a^{-1} \in \mathcal{A}_+$ as well. By (i) we have

$$b^{-\frac{1}{2}}ab^{-\frac{1}{2}} \leqslant b^{-\frac{1}{2}}bb^{-\frac{1}{2}} = 1_{\mathcal{A}}$$

so (ii) implies $\left\|b^{-\frac{1}{2}}a^{\frac{1}{2}}\right\|^2 \leqslant 1$. (Here we use the convention that the identity in a unital Banach algebra has norm 1.) Thus

$$a^{\frac{1}{2}}b^{-1}a^{\frac{1}{2}} \leqslant \left\|a^{\frac{1}{2}}b^{-1}a^{\frac{1}{2}}\right\| \leqslant 1 = 1_{\mathcal{A}}$$

By (i) we have

$$b^{-1} = a^{-\frac{1}{2}}\left(a^{\frac{1}{2}}b^{-1}a^{\frac{1}{2}}\right)a^{-\frac{1}{2}} \leqslant a^{-\frac{1}{2}}1_{\mathcal{A}}a^{-\frac{1}{2}} = a^{-1}$$

□

Proposition 3.7. Let $\mathcal{A}$ be a C^*-algebra.

(i) Every projection is positive. More generally, if $p \in \mathcal{A}$ is idempotent, i.e. $p^2 = p$, then $\sigma_{\mathcal{A}}(p) \subseteq \{0, 1\}$.

(ii) Let $p, q \in \mathcal{A}$ be projections. Then $p \leqslant q$ if and only if $pq = qp = p$.

Proof. (i) follows from Spectral mapping theorem for polynomials: if we take $f(z) = z - z^2$, then

$$f(\sigma_{\mathcal{A}}(p)) = \sigma_{\mathcal{A}}(f(p)) = \sigma_{\mathcal{A}}(0) = \{0\}.$$

so we must have $\sigma_{\mathcal{A}}(p) \subseteq \{0, 1\}$. For (ii), first assume $p \leqslant q$. Then $0 \leqslant q - p \leqslant 1_{\mathcal{A}^e} - p$, so by Proposition 3.6.(i)

$$0 = p0p \leqslant p(q-p)p \leqslant p(1_{\mathcal{A}^e} - p)p = p - p = 0$$

and thus $0 = p(q-p)p = pqp - p$, whence

$$\|qp - p\|^2 = \|(qp-p)^*(qp-p)\| = \|pqp - pqp - pqp + p\| = \|p - pqp\| = 0$$

i.e., $qp = p$. Also, $pq - p = (qp - p)^* = 0$. Conversely, suppose $pq = qp = p$. Then

$$(q-p)(q-p) = q^2 - qp - pq + p^2 = q - p - p + p = q - p$$

and $(q-p)^* = q - p$, so $q - p$ is a projection. By (i), $q - p \geqslant 0$, or $q \geqslant p$. □

3.2 Approximate identity

Definition. Let $\mathcal{B}$ be a Banach algebra. A net $(e_\lambda)_{\lambda\in\Lambda}$ is called a **left approximate identity** if for all $a \in \mathcal{B}$ we have

$$a = \lim_{\lambda\in\Lambda} e_\lambda a, \qquad \text{or} \qquad \lim_{\lambda\in\Lambda} \|a - e_\lambda a\| = 0$$

Similarly we define **right approximate identity**. A net in $\mathcal{B}$ is called an **approximate identity** if it is simultaneously a left and right approximate identity.

Lemma 3.8. Let $\mathcal{A}$ be a C^*-algebra and let $\mathcal{A}_+^1 := \{a \in \mathcal{A}_+ \mid \|a\| < 1\}$. Then the poset $(\mathcal{A}_+^1, \leqslant)$ is upward directed.

Proof. Put $\mathcal{A}^e$ to be its unitization if $\mathcal{A}$ is not unital, and $\mathcal{A}^e = \mathcal{A}$ if it is unital. First consider $a, b \in \mathcal{A}_+$. Then $1_{\mathcal{A}^e} + a$, $1_{\mathcal{A}^e} + b \in (\mathcal{A}^e)^\times \cap \mathcal{A}_+^e$. Suppose $a \leqslant b$. Then $1_{\mathcal{A}^e} + a \leqslant 1_{\mathcal{A}^e} + b$, so $(1_{\mathcal{A}^e} + b)^{-1} \leqslant (1_{\mathcal{A}^e} + a)^{-1}$ by Proposition 3.6.(iii) and

$$a(1_{\mathcal{A}^e} + a)^{-1} = 1_{\mathcal{A}^e} - (1_{\mathcal{A}^e} + a)^{-1} \leqslant 1_{\mathcal{A}^e} - (1_{\mathcal{A}^e} + b)^{-1} = b(1_{\mathcal{A}^e} + b)^{-1}$$

Note that $a(1_{\mathcal{A}^e} + a)^{-1}$ and $b(1_{\mathcal{A}^e} + b)^{-1}$ are both in $\mathcal{A}_+^1$. To see this, expanding in power series we can see $a(1_{\mathcal{A}^e} + a)^{-1} \in \mathcal{A}$. Since $\sigma_{\mathcal{A}^e}(1_{\mathcal{A}^e} + a) \subseteq [1, \|a\| + 1]$, we have

$$\left\|a(1_{\mathcal{A}^e} + a)^{-1}\right\| \leqslant \frac{\|a\|}{\|1_{\mathcal{A}^e} + a\|} \leqslant \frac{\|a\|}{1 + \|a\|} < 1$$

Now suppose $a, b \in \mathcal{A}_+^1$. Let $x = a(1_{\mathcal{A}^e} - a)^{-1}$ and $y = b(1_{\mathcal{A}^e} - b)^{-1}$ and $c = (x + y)(1_{\mathcal{A}^e} + x + y)^{-1}$. Since $x \leqslant x + y$, by the first paragraph we have

$$a = x(x + 1_{\mathcal{A}^e})^{-1} \leqslant (x + y)(1_{\mathcal{A}^e} + x + y)^{-1} = c$$

Similarly, $b \leqslant c$. □

Theorem 3.9. Every C^*-algebra $\mathcal{A}$ has an increasing approximate identity consisting of positive elements with norm less than 1. If $\mathcal{A}$ is separable, the constructed approximate identity can be countable.

Proof. Let $\Lambda = (\mathcal{A}_+^1, \leqslant)$ and $u_\lambda = \lambda$ for all $\lambda \in \Lambda$. We claim (u_λ) is an approximate identity of $\mathcal{A}$. We must show that

$$a = \lim_{\lambda\in\Lambda} u_\lambda a = \lim_{\lambda\in\Lambda} a u_\lambda$$

for all $a \in \mathcal{A}$. By linearity it suffices to show this holds for $a \in \mathcal{A}_+$. Let $\Gamma : C^*(a) \to C_0(\sigma_{\mathcal{A}}(a)) = C_0(X)$ be the inverse of the continuous functional calculus, where $X = \sigma_{\mathcal{A}}(a)\backslash\{0\}$. Put $f = \Gamma(a)(= \mathrm{id})$ Let $0 < \varepsilon < 1$ and $K := \{x \in X \mid |f(x)| \geqslant \varepsilon\}$.

Let $\delta > 0$ such that $\delta < 1$ and $(1 - \delta)\|f\| < \varepsilon$. Define $g_\delta \in C_0(X)$ by setting $g_\delta(x) = \delta$ for $x \in K$ and $g \equiv 0$ outside a open neighborhood of K; this is possible since $K \subseteq X$ is compact. Then $g_\delta \in C_0(X)_+^1$

and $\|g_\delta f - f\| < \varepsilon$: for $x \in K$, $|\delta f(x) - f(x)| \leqslant (1-\delta)\|f\| < \varepsilon$, and for $x \notin K$, $|g_\delta(x)f(x) - f(x)| \leqslant \varepsilon(1 - g_\delta(x)) < \varepsilon$. Since Γ is isometric, it preserves norm, so that $\Gamma^{-1}(g_\delta) = \mu$ for some $\mu \in \Lambda$, and

$$\|u_\mu a - a\| = \|au_\mu - a\| = \|g_\delta f - f\| < \varepsilon$$

Suppose $\lambda \in \Lambda$ such that $\mu \leqslant \lambda$. Then $1_{\mathcal{A}^e} - u_\lambda \leqslant 1_{\mathcal{A}^e} - u_\mu$, so

$$a^{\frac{1}{2}}(1_{\mathcal{A}^e} - u_\lambda)a^{\frac{1}{2}} \leqslant a^{\frac{1}{2}}(1_{\mathcal{A}^e} - u_\mu)a^{\frac{1}{2}}$$

and hence

$$\|a - au_\lambda\| = \left\|a^{\frac{1}{2}}(1_{\mathcal{A}^e} - u_\lambda)a^{\frac{1}{2}}\right\| \leqslant \left\|a^{\frac{1}{2}}(1_{\mathcal{A}^e} - u_\mu)a^{\frac{1}{2}}\right\| = \|a - au_\mu\| < \varepsilon$$

Therefore, $a = \lim\limits_{\lambda\in\Lambda} u_\lambda a = \lim\limits_{\lambda\in\Lambda} au_\lambda$.

Now assume $\mathcal{A}$ is separable and let $\{a_n\}_{n\in\mathbb{N}}$ be a countable dense subset of $\mathcal{A}$. Let $\lambda_1 \in \Lambda$ be such that $\|a_1u_\lambda - a_1\|, \|u_\lambda a_1 - a_1\| < \dfrac{1}{2}$ for $\lambda \geqslant \lambda_1$. Then choose $\lambda_2 \geqslant \lambda_1$ such that $\|a_ju_\lambda - a_j\|, \|u_\lambda a_j - a_j\| < \dfrac{1}{4}$ for $j = 1, 2$ and $\lambda \geqslant \lambda_2$. Continuing inductively, we obtain $\{\lambda_n\}_{n\in\mathbb{N}} \subseteq \Lambda$ such that for $j \in \mathbb{N}$

$$\|a_ju_{\lambda_n} - a_j\|, \|u_{\lambda_n}a_j - a_j\| \to 0 \text{ as } n \to \infty$$

Now let $\varepsilon > 0$ and for $a \in \mathcal{A}$, let $j \in \mathbb{N}$ such that $\|a - a_j\| < \dfrac{\varepsilon}{3}$. Take $n \gg 0$ such that $\|a_ju_{\lambda_m} - a_j\| < \dfrac{\varepsilon}{3}$ for $m \geqslant n$. Then

$$\|au_{\lambda_m} - a\| \leqslant \|(a - a_j)u_{\lambda_m}\| + \|a_ju_{\lambda_m} - a_j\| + \|a_j - a\| < \varepsilon$$

for $\|u_{\lambda_m}\| \leqslant 1$. Similarly $\|u_{\lambda_m}a - a\| < \varepsilon$. This finishes the proof. □

Remark 3.10. In the following when we talk about an approximate identity of a C^*-algebra, we always assume it is an increasing net consisting of positive elements with norm less than 1.

Definition. A C^*-algebra is called **σ-unital** if it possesses a sequential approximate identity.

- The above theorem then shows that every separable C^*-algebra is σ-unital. Moreover, it admits an increasing sequential approximate identity consisting of positive element with norm less than 1.

3.3 Hereditary Subalgebras and Ideals

Definition. Let $\mathcal{A}$ be a C^*-algebra.

1. A C^*-subalgebra $\mathcal{B}$ of $\mathcal{A}$ is called **hereditary** if for $a, b \in \mathcal{A}$ with $a \leqslant b$, $b \in \mathcal{B}$ implies $a \in \mathcal{B}$.

2. For a projection $p \in \mathcal{A}$, the C^*-subalgebra $p\mathcal{A}p$ is called a **corner**.

Lemma 3.11. Let $I \subseteq \mathcal{A}$ be a closed left ideal. Then I has a right approximate identity.

Proof. Observe that $I \cap I^*$ is a C^*-subalgebra of $\mathcal{A}$, so it admits an approximate identity $(u_\lambda)_{\lambda \in \Lambda}$. Let $a \in I$; then $a^*a \in I \cap I^*$, and

$$\begin{aligned} \|a - au_\lambda\|^2 &= \|(a - au_\lambda)^*(a - au_\lambda)\| \\ &\leqslant \|a^*a - u_\lambda a^*a\| + \|u_\lambda(a^*a - u_\lambda a^*a)\| \leqslant 2\,\|a^*a - a^*au_\lambda\| \to 0 \end{aligned}$$

□

Theorem 3.12. Let $\mathcal{A}$ be a C^*-algebra and $I \trianglelefteq \mathcal{A}$ a closed (two-sided) ideal.

1. $I = I \cap I^*$; in particular, I is a *-ideal of $\mathcal{A}$.
2. $\mathcal{A}/I$ is a C^*-algebra with quotient norm (c.f. Lemma V.5) and the induced *-operation from $\mathcal{A}$.

Proof. Let (u_λ) be the right approximate identity constructed in the lemma. Then for $a \in I$,

$$a^* = (\lim_\lambda au_\lambda)^* = \lim_\lambda u_\lambda a^* \in I$$

since I is a closed ideal. Now for $a \in \mathcal{A}$, we have

$$\|a + I\| := \inf_{b \in I} \|a + b\| = \lim_\lambda \|a - u_\lambda a\| = \lim_\lambda \|a - au_\lambda\|$$

This is because $\|a + b\| \leqslant \|a - au_\lambda\| + \|au_\lambda + b\|$ and $au_\lambda \in I$. Hence

$$\|a^*a + I\| = \lim_\lambda \|a^*a(1 - u_\lambda)\| \geqslant \lim_\lambda \|(1 - u_\lambda)a^*a(1 - u_\lambda)\| = \lim_\lambda \|a(1 - u_\lambda)\|^2 = \|a + I\|^2$$

The second equality results from $\|1 - u_\lambda\| \leqslant 1$. For the reverse inequality, note that

$$\|a^* + I\| = \lim_\lambda \|(a - au_\lambda)^*\| = \lim_\lambda \|a - au_\lambda\| = \|a + I\|$$

so that $\|aa^* + I\| \leqslant \|a + I\|\,\|a^* + I\| = \|a + I\|^2$. □

Corollary 3.12.1. Let $\Phi : \mathcal{A} \to \mathcal{B}$ be a homomorphism of C^*-algebras. Then

(i) Φ is norm-decreasing;

(ii) $\Phi(\mathcal{A})$ is a C^*-subalgebra of $\mathcal{B}$;

(iii) Φ is injective if and only if Φ is isometric.

Proof. (i) is Lemma 2.30, and (iii) is Corollary 2.31.2. For (ii), since $\mathcal{A}/\ker\Phi \to \mathcal{B}$ is injective, it is isometric by (iii), implying the image $\Phi(\mathcal{A})$ is closed in $\mathcal{B}$, as wanted.

□

Corollary 3.12.2. Let $\mathcal{A}$ be a C^*-algebra. If $I \trianglelefteq \mathcal{A}$ is a closed ideal, and $J \trianglelefteq I$ is a closed ideal of I, then J is also a closed ideal of $\mathcal{A}$.

Proof. Let (u_λ) be an approximate identity of I. Then for $a \in \mathcal{A}$ and $x \in J$, we have $ax = a\lim_\lambda u_\lambda x = \lim_\lambda (au_\lambda)x \in J$ since $au_\lambda \in I$. □

3.4 Positive Linear Functionals

Definition.

(i) A $\mathbb{C}$-linear map $\phi : \mathcal{A} \to \mathcal{B}$ between C^*-algebras is called **positive** if $\phi(\mathcal{A}_+) \subseteq \mathcal{B}_+$.

(ii) A positive linear map $\phi : \mathcal{A} \to \mathbb{C}$ is called a **positive linear functional**.

(iii) A positive linear functional ϕ is called a **state** if $\|\phi\| = 1$.

(iv) A positive linear functional ϕ is called a **trace** if $\phi(ab) = \phi(ba)$ for all $a, b \in \mathcal{A}$

(v) A **tracial state** is a state that is also a trace.

(vi) A state ϕ is **faithful** if $\phi(a^*a) = 0$ implies $a = 0$.

The set of all states (reps. tracial states) on $\mathcal{A}$ is denoted by $S(\mathcal{A})$ (resp. $T(\mathcal{A})$).

Proposition 3.13. Let $\mathcal{A}$ be a C^*-algebra and $\phi : \mathcal{A} \to \mathbb{C}$ a linear functional. Then the pairing $(\,,\,)_\phi : \mathcal{A} \times \mathcal{A} \to \mathbb{C}$ defined by

$$(a, b)_\phi := \phi(ab^*)$$

is a nonnegative Hermitian form on $\mathcal{A}$ if and only if ϕ is positive.

Proof. Let ϕ be a linear functional. To say $(\,,\,)_\phi$ is nonnegative Hermitian, it is equivalent to the following conditions.

(i) $(ra + b, c)_\phi = r(a, c)_\phi + (b, c)_\phi$ for all $a, b, c \in \mathcal{A}$ and $r \in \mathbb{C}$.

(ii) $\overline{(a, b)_\phi} = (b, a)_\phi$ for all $a, b \in \mathcal{A}$.

(iii) $(a, a)_\phi \geqslant 0$ for all $a \in \mathcal{A}$.

Since ϕ is linear, (i) always holds. By Proposition 3.5, (iii) holds if and only if ϕ is positive. When ϕ is positive, (ii) holds automatically by Proposition 3.3. $\square$

Corollary 3.13.1. Let $\mathcal{A}$ be a C^*-algebra and $\phi : \mathcal{A} \to \mathbb{C}$ a positive linear functional. Then

$$|\phi(ab^*)|^2 \leqslant \phi(aa^*)\phi(bb^*)$$

for all $a, b \in \mathcal{A}$.

Proof. By the previous proposition, the statement is reduced to a linear algebra lemma.

Lemma 3.14. Let V be a complex vector space and $f : V \times V \to \mathbb{C}$ a nonnegative Hermitian form. Then for $v, w \in V$, we have $|f(v, w)|^2 \leqslant f(v, v)f(w, w)$.

Proof. We may assume $f(v,w) \neq 0$. Since f is nonnegative, for all $\lambda \in \mathbb{C}$

$$0 \leqslant f(\lambda v + w, \lambda v + w) = |\lambda|^2 f(v,v) + 2\operatorname{Re}(\lambda f(v,w)) + f(w,w)$$

In particular, this holds for $\lambda = t\dfrac{|f(v,w)|}{f(v,w)}$ for any $t \in \mathbb{R}$, so that (note that $|\lambda| = |t|$)

$$t^2 f(v,v) + 2t|f(v,w)| + f(w,w) \geqslant 0$$

If $f(v,v) = 0$, then $2t|f(v,w)| + f(w,w) \geqslant 0$ for all $t \in \mathbb{R}$, which is impossible since we are assuming $f(v,w) \neq 0$. Therefore $f(v,v) \neq 0$, whence

$$4|f(v,w)|^2 - 4f(v,v)f(w,w) \leqslant 0$$

as wanted. □

It suffices to take $f = (\,,\,)_\phi$ in the lemma. □

Proposition 3.15. Any positive linear function on a C^*-algebra $\mathcal{A}$ is bounded.

Proof. Let $\phi : \mathcal{A} \to \mathbb{C}$ by a positive linear functional. First note that if $M := \sup\limits_{a \in \mathcal{A}_+,\, \|a\|=1} \phi(a) < \infty$, then ϕ is bounded. Indeed, for an element $x \in \mathcal{A}$, write $x = \operatorname{Re} x + i \operatorname{Im} x$; then $\operatorname{Re}\phi(x) = \phi(\operatorname{Re} x)$ and $\operatorname{Im}\phi(x) = \phi(\operatorname{Im} x)$ since $\phi(\mathcal{A}_{sa}) \subseteq \mathbb{R}$. So to show ϕ is bounded it suffices to show ϕ is bounded on $\mathcal{A}_{sa}$. But for $x \in \mathcal{A}_{sa}$, we can write $x = x^+ - x^-$ with $x^\pm$ positive and $\|x^\pm\| \leqslant \|x\|$; the inequality holds since continuous functional calculus is isometric. Then $\|\phi(x)\| = \|\phi(x^+) - \phi(x^-)\| \leqslant M\,\|x^+\| + M\,\|X^-\| \leqslant 2M\,\|x\|$.

Now suppose for contradiction that ϕ is unbounded. The previous paragraph implies $\sup\limits_{a \in \mathcal{A}_+,\, \|a\|=1} |\phi(a)| = \infty$. In particular, we can find a sequence $(a_n)_{n\in\mathbb{N}}$ with $a_n \in \mathcal{A}_+$ and $\|a_n\| = 1$ such that $\phi(a_n) \geqslant 2^n$. But if we put $a := \sum\limits_{n=1}^{\infty} 2^{-n} a_n \in \mathcal{A}_+$ (Proposition 3.4), then for each $N \in \mathbb{N}$ we have $a - \sum\limits_{n=1}^{N-1} 2^{-n} a_n \geqslant 0$, so

$$\phi(a) \geqslant \phi\left(\sum_{n=1}^{N-1} 2^{-n} a_n\right) = \sum_{n=1}^{N-1} 2^{-n}\phi(a_n) \geqslant N - 1$$

a contradiction. □

Proposition 3.16. Let $\mathcal{A}$ be a C^*-algebra and let $(u_\lambda)_\lambda$ be an approximate unit. Let ϕ be a linear functional on $\mathcal{A}$. Then ϕ is positive if and only if $\lim_\lambda \phi(u_\lambda) = \|\phi\|$. In particular, if $\mathcal{A}$ is unital, ϕ is positive and only if $\phi(1_{\mathcal{A}}) = \|\phi\|$.

Proof. □

3.5 The Gelfand-Naimark-Segal construction

Definition. Let $\mathcal{A}$ be a C^*-algebra. A **$*$-representation** of $\mathcal{A}$ is *-homomorphism $\pi : \mathcal{A} \to \mathcal{B}(H)$ of $\mathcal{A}$ into the C^*-algebra of bounded operators on a Hilbert space H.

- We say π is **faithful** if $\pi : \mathcal{A} \to \mathcal{B}(H)$ is injective. In this case, by Corollary 2.31.2 we see $\mathcal{A}$ is $*$-isomorphic to a closed self-adjoint subalgebra of $\mathcal{B}(H)$, or, simply put, a concrete C^*-algebra.

We now prove every C^*-algebra has many representations. Let $\mathcal{A}$ be a C^*-algebra. Given a positive linear functional ϕ, put

$$N_\phi := \{a \in \mathcal{A} \mid \phi(a^*a) = 0\}$$

which is a closed left ideal in $\mathcal{A}$. Indeed, closedness is clear, and if $x \in N_\phi$ and $a \in \mathcal{A}$, then by Cauchy-Schwarz, we have

$$\phi((ax)^*(ax))^2 = \phi(x^*(a^*ax))^2 \leqslant \phi(x^*x)\phi(x^*a^*aa^*ax) = 0$$

Now consider the vector space quotient $\mathcal{A}/N_\phi$. Define a pairing on the quotient:

$$\begin{aligned} \langle , \rangle_\phi : \mathcal{A}/N_\phi \times \mathcal{A}/N_\phi &\longrightarrow \mathbb{C} \\ (a + N_\phi, b + N_\phi) &\longmapsto \phi(b^*a) \end{aligned}$$

By Proposition 3.13, we see this pairing defines an inner product on $\mathcal{A}/N_\phi$. Let H_ϕ denote the completion of $\mathcal{A}/N_\phi$ with respect to this inner product.

For $a \in \mathcal{A}$, define a linear map $\pi_\phi(a) \in \operatorname{End}_{\mathbb{C}} \mathcal{A}/N_\phi$ by

$$\pi_\phi(a)(b + N_\phi) = ab + N_\phi$$

(this is well-defined for N_ϕ is a left ideal). Let $b \in N_\phi \in \mathcal{A}/N_\phi$ be such that $\phi(b^*b) \leqslant 1$. Then

$$\|\pi_\phi(a)b\|^2 = \langle ab + N_\phi, ab + N_\phi\rangle_\phi = \phi(b^*a^*ab) \leqslant \phi(b^* \|a\|^2 b) \leqslant \|a\|^2$$

by Proposition 3.6.(i) (note that $a^*a \leqslant \|a^*a\| = \|a\|^2$). This shows $\pi_\phi(a)$ is bounded, whence extends to an operator on H_ϕ. Finally, $\pi_\phi : \mathcal{A} \to \mathcal{B}(H_\phi)$ is a $*$-homomorphism. Indeed,

$$\langle b + N_\phi, \pi_\phi(a^*)(c + N_\phi)\rangle_\phi = \phi((a^*c)^*b) = \phi(c^*ab) = \langle \pi_\phi(a)(b + N_\phi), c + N_\phi\rangle_\phi.$$

In particular, π_ϕ is a *-representation of $\mathcal{A}$. The *-representation (π_ϕ, H_ϕ) is called the **Gelfand-Naimark-Segal representation**, or simply the **GNS representation**, associated to ϕ.

Now given a family $(\pi_\lambda, H_\lambda)_{\lambda\in\Lambda}$ of *-representation of a C^*-algebra $\mathcal{A}$, consider the homomorphism

$$\begin{aligned} \oplus_\lambda \pi_\lambda : \mathcal{A} &\longrightarrow B\left(\bigoplus_{\lambda\in\Lambda} H_\lambda\right) \\ a &\longmapsto \oplus_\lambda \pi_\lambda(a) \end{aligned}$$

This again gives a *-representation of $\mathcal{A}$ on the Hilbert space direct sum $\bigoplus_{\lambda\in\Lambda} H_\lambda$.

Recall that $S(\mathcal{A})$ consists of all states of $\mathcal{A}$, i.e., positive linear functional with norm 1. Note that for a positive linear functional ϕ and $\alpha > 0$, the GNS constructions π_ϕ and $\pi_{\alpha\phi}$ are equivalent. So we may confine ourselves to those associated to states. The representation

$$\left(\bigoplus_{\phi\in S(\mathcal{A})} \pi_\phi, \bigoplus_{\phi\in S(\mathcal{A})} H_\phi\right)$$

is called the **universal representation** of $\mathcal{A}$.

Theorem 3.17 (Gelfand-Naimark)**.** The universal representation is faithful.

Proof. It suffices to show for each $a \in \mathcal{A}\backslash\{0\}$, we can find some state ϕ such that $\pi_\phi(a)$ is a nontrivial operator on H_ϕ, i.e., we can find $b \in \mathcal{A}$ such that $\phi((ab)^*ab) \neq 0$.

□

4 Duality for Abelian Groups

4.1 The Dual Group

Definition.

1. A locally compact Hausdorff abelian group will be called an **LCA group** for short.
2. A **(unitary) character** of an LCA group A is a continuous group homomorphism $\chi : A \to S^1$.
3. For an LCA group A, denote by $\widehat{A}$ the set of all characters on A, which is a group under pointwise multiplication, called the **Pontryagin dual**, or the **dual group** of A.

Example 4.1.

1. Consider the additive group $\mathbb{R}$. For each $t \in \mathbb{R}$, define a character

$$\begin{aligned} \chi_t : \mathbb{R} &\longrightarrow S^1 \\ x &\longmapsto e^{2\pi i x t} \end{aligned}$$

The association $t \mapsto \chi_t$ is a group isomorphism $\mathbb{R} \cong \widehat{\mathbb{R}}$.

Proof. Let $f : \mathbb{R} \to S^1$ be a continuous group homomorphism. This means f is a continuous map with a functional equation $f(x+y) = f(x)f(y)$ for each $x, y \in \mathbb{R}$, and $f(0) = 1$. Pick a $\delta \in \mathbb{R}$ such that $\alpha := \int_0^\delta f(t)dt \neq 0$. Then the functional equation implies

$$\alpha f(x) = \int_0^\delta f(t+x)dt = \int_x^{x+\delta} f(t)dt$$

Since f is continuous, the right hand side is differentiable and thus f is itself differentiable. Differentiating the functional equation with respect to x, we obtain $f'(x+y) = f'(x)f(y)$; if we put $A = f(1)$ and let $x = 0$, we derive a differential equation $f'(y) = Af(y)$ of f. It has a unique solution $f(y) = Ce^{sy}$ for some (unique) $s \in \mathbb{C}$ and $C \in \mathbb{C}$.

- $f(0) = 1$, so that $C = 1$.
- f is bounded, so $\operatorname{Re} t = 0$. Let us write $s = 2\pi i t$ for a unique $t \in \mathbb{R}$.

Then $f(y) = e^{2\pi i t y} = \chi_t(y)$. □

2. $\widehat{\mathbb{Z}} \cong \mathbb{R}/\mathbb{Z}$, and $\widehat{\mathbb{R}/\mathbb{Z}} \cong \mathbb{Z}$. The isomorphisms are the same as the case of $\mathbb{R}$.

Definition. Let X and Y be spaces. The compact-open topology on Y^X is the topology generated by the subbasis consisting of the sets

$$L(K,U) := \{f : X \to Y \mid f(K) \subseteq U\}$$

where $K \underset{\text{cpt}}{\subseteq} X$ and $U \underset{\text{open}}{\subseteq} Y$.

- The compact-open topology is finer than the product topology on Y^X.
- If Y is Hausdorff, then Y^X is Hausdorff. Indeed, for $f \neq g$ pick x such that $f(x) \neq g(x)$. Being Hausdorff, we can find disjoint open sets U and V with $f(x) \in U$ and $g(x) \in V$. Then $L(\{x\}, U)$ and $L(\{x\}, V)$ are disjoint open neighborhood of f and g, respectively.
- For each $x \in X$, the evaluation map $\delta_x : Y^X \to Y$ is continuous. Indeed, for each $U \underset{\text{open}}{\subseteq} Y$, $\delta_x^{-1}(U) = L(\{x\}, U)$ is open in Y^X.

Lemma 4.2. Let $\mathcal{K}$ be a collection of compact subsets of X containing a neighborhood basis at each point of X (such a collection exists when, for example, X is LCH), and let $\mathcal{B}$ be a subbasis for the open sets of Y. Then the sets $L(K,U)$ with $K \in \mathcal{K}$ and $U \in \mathcal{B}$ form a subbasis for the compact-open topology.

Proof. Since $L(K,U) \cap L(K,V) = L(K, U \cap V)$, we may assume $\mathcal{B}$ is a basis for the topology of Y. We must show for each $K \underset{\text{cpt}}{\subseteq} X$, $U \underset{\text{open}}{\subseteq} Y$ and $f \in L(K,U)$, we can find $K_1, \ldots, K_n \in \mathcal{K}$ and $U_1, \ldots, U_n \in \mathcal{B}$ such that $f \in \bigcap_{i=1}^{n} L(K_i, U_i) \subseteq L(K,U)$.

For each $x \in K$, take an open neighborhood $f(x) \in U_x \in \mathcal{B}$ in U and a $K_x \in \mathcal{K}$ such that $x \in f(K_x) \subseteq U_x$. Since K is compact, there exist $x_1, \ldots, x_n$ such that $K \subseteq K_{x_1} \cup \cdots \cup K_{x_n}$. Then $f \in \bigcap_{i=1}^{n} L(K_{x_i}, U_{x_i}) \subseteq L(K,U)$. $\square$

Lemma 4.3. Let Y be a metric space. Then for each $f \in C(X,Y)$, the sets of the form

$$B_{K,\varepsilon}(f) = \{g \in C(X,Y) \mid d_K(f,g) := \sup_{x \in K} d_Y(f(x), g(x)) < \varepsilon\}$$

with $\varepsilon > 0$ and $K \underset{\text{cpt}}{\subseteq} X$ forms a neighborhood basis of f.

Proof. We show that for each $L(K,U)$ and $f \in L(K,U)$ there exists a ball $B_{K,\varepsilon}(f) \subseteq L(K,U)$. Since $f(K)$ is compact, there is a distance $\varepsilon > 0$ from $f(K)$ to the complement of U; thus if $g \in B_{\varepsilon/2}(f)$, then $g(K) \subseteq U$ so that $g \in L(K,U)$.

It remains to show $B_{K,\varepsilon}(f)$ is open in compact-open topology. Let $g \in B_{K,\varepsilon}(f)$ and put $\delta = \varepsilon - d_K(f,g)$. Since $g(K)$ is compact, it is covered by $B_{\delta/3}(g(x_i))$ for some $x_1, \ldots, x_n \in K$. Put $K_i = K \cap \overline{g^{-1}(B_{\delta/3}(g(x_i)))}$; then

$$g(K_i) \subseteq g(\overline{g^{-1}(B_{\delta/3}(g(x_i)))}) \subseteq \overline{g(g^{-1}(B_{\delta/3}(g(x_i))))} \subseteq \overline{B_{\delta/3}(g(x_i))} \subseteq B_{\delta/2}(g(x_i)) =: U_i$$

Then for $h \in \bigcap_{i=1}^{n_g} L(K_i, U_i)$ and $x \in K_i$, one has

$$d_Y(f(x), h(x)) \leqslant d_Y(f(x), g(x)) + d_Y(g(x), g(x_i)) + d_Y(g(x_i), h(x)) \leqslant d_K(f, g) + \frac{\delta}{2} + \frac{\delta}{2} < \varepsilon$$

so that $h \in B_{K,\varepsilon}(f)$. Therefore, $g \in \bigcap_{i=1}^{n_g} L(K_i, U_i) \subseteq B_{K,\varepsilon}(f)$, i.e. $B_{K,\varepsilon}(f)$ is open. □

Corollary 4.3.1. If X is compact and Y is a metric space, then the compact open topology on the space of continuous functions $C(X, Y)$ is the same as the topology induced by the sup-norm.

Definition. Let A be an LCA group. We equip $\widehat{A}$ with the subspace topology of the compact-open topology on $\mathbb{C}^A$.

- $\widehat{A}$ is a closed subspace of the space of continuous functions $C(A)$. Indeed, write

$$\widehat{A} = \bigcap_{x,y \in A} \{\chi : A \to S^1 \mid \chi(xy) = \chi(x)\chi(y)\}$$

  It suffices to show each set in the intersection is closed. Let $x, y \in A$. Suppose $f \in C(A)$ is such that $f(xy) \neq f(x)f(y)$. Take open neighborhoods U, V, W of $f(x)$, $f(y)$, $f(xy)$, respectively, such that W is disjoint from $UV = \{uv \mid (u, v) \in U \times V\}$. Since A is LCH, we can find compact $x \in K_U \subseteq f^{-1}(U), y \in K_V \subseteq f^{-1}(V)$, $xy \in K_W \subseteq f^{-1}(W)$. Then all functions g in $L(K_U, U) \cap L(K_V, V) \cap L(K_W, W)$ satisfy $g(xy) \neq g(x)g(y)$.

Example 4.4.

1. The compact open topology on $\widehat{\mathbb{R}} \cong \mathbb{R}$ is the usual topology on $\mathbb{R}$.

   Proof. We must show the group isomorphism

$$\begin{aligned} \Phi : \mathbb{R} &\longrightarrow \widehat{\mathbb{R}} \\ t &\longmapsto (\chi_t : x \mapsto e^{2\pi i x t}) \end{aligned}$$

   is a homeomorphism. Suppose $(t_n) \to t$ in $\mathbb{R}$. We must show $\chi_{t_n} \to \chi_t$ in compact-open topology. By Lemma 4.2, we only need to show for each $a \leqslant b \in \mathbb{R}$ and open $U \subseteq S^1$ with $\chi_t \in L([a, b], U)$, $\chi_{t_n} \in L([a, b], U)$ if $n \gg 1$.

   - Let $\varepsilon > 0$ be the distance between $\chi_t([a, b])$ and the complement of U in S^1.
   - Let $\delta > 0$ be such that $|e^{2\pi i y} - 1| < \varepsilon/2$ for each $|y| < \delta$.

   Let $N \in \mathbb{N}$ be such that $[|t_n - t|a, |t_n - t|b] \subseteq [-\delta/2, \delta/2]$ for all $n \geqslant N$. Then for each $x \in [a, b]$

$$|\chi_{t_n}(x) - \chi(x)| = |e^{2\pi i t_n x} - e^{2\pi i t x}| = |e^{2\pi i (t_n - t)x} - 1| < \frac{\varepsilon}{2}$$

   Hence $\chi_{t_n}([a, b]) \subseteq U$ for $n \geqslant N$, i.e. $\chi_{t_n} \in L([a, b], U)$ if $n \geqslant N$. Conversely, let χ_t □

Proposition 4.5. With the compact-open topology $\widehat{A}$ is a Hausdorff topological group.

Proof. We have to show the map

$$\begin{aligned} \Phi : \widehat{A} \times \widehat{A} &\longrightarrow \widehat{A} \\ (\chi, \psi) &\longmapsto \chi\psi^{-1} \end{aligned}$$

is continuous. For (χ, ψ), (χ', ψ') and $x \in A$ one has

$$\begin{aligned} |\chi(x)\psi^{-1}(x) - \chi'(x)\psi'^{-1}(x)| &\leqslant |\chi(x)\psi^{-1}(x) - \chi(x)\psi'^{-1}(x)| + |\chi(x)\psi'^{-1}(x) - \chi'(x)\psi'^{-1}(x)| \\ &= |\psi^{-1}(x) - \psi'^{-1}(x)| + |\chi(x) - \chi'(x)| \end{aligned}$$

Let $K \subseteq A$ be compact and let $\varepsilon > 0$. Then

$$B_{K,\varepsilon}(\chi\psi^{-1}) = \{\gamma \in \widehat{A} \mid \left\|\gamma - \chi\psi^{-1}\right\|_K < \varepsilon\}$$

is an open neighborhood of $\chi\psi^{-1}$, and sets of this form are a neighborhood basis (Lemma 4.3). The estimate above shows that $B_{K,\varepsilon/2}(\chi) \times B_{K,\varepsilon/2}(\psi) \subseteq \Phi^{-1}(B_{K,\varepsilon}(\chi\psi^{-1}))$, so Φ is continuous. □

Proposition 4.6. Let A be an LCA group.

1. If A is compact, then $\widehat{A}$ is discrete.
2. If A is discrete, then $\widehat{A}$ is compact.

Proof.

1. Suppose A is compact. Put $U = \{z \in S^1 \mid \operatorname{Re} z > 0\}$. Since A is compact, the set $L(A, U)$ is open by definition. But U contains no nontrivial subgroup of S^1, so if $\chi \in L(A, U)$, then $\chi(A) = \{1\}$; hence $L(A, U)$ contains only trivial character.
2. Suppose A is discrete. Then the compact subspaces of A are precisely the finite subsets of A. Hence the compact open topology on $\widehat{A}$ coincides with the topology of pointwise convergence, and $\widehat{A} \subseteq (S^1)^A$ is a closed subspace; by Tychonov's theorem, the latter space is compact, and hence so is $\widehat{A}$.

□

4.2 The Fourier Transform

Definition. Let A be an LCA group. For $f \in L^1(A)$ we define the **Fourier transform** of f to be the map $\hat{f} : \widehat{A} \to \mathbb{C}$ defined by

$$\hat{f}(\chi) := \int_A f(x)\overline{\chi(x)}dx$$

This integral exists since χ is bounded.

Lemma 4.7. For $f, g \in L^1(A)$ and $\chi \in \widehat{A}$, one has $|\hat{f}(\chi)| \leqslant \|f\|$ and $\widehat{f * g} = \hat{f}\hat{g}$.

Proof. The first assertion is clear. The second follows from a direct computation:

$$\begin{aligned}
\widehat{f * g}(\chi) &= \int_G (f * g)(x)\overline{\chi(x)}dx \\
&= \int_G \left(\int_G f(y)g(y^{-1}x)dy \right) \overline{\chi(x)}dx \\
&\overset{\text{Fubini}}{=} \int_G \int_G f(y)\overline{\chi(y)}g(y^{-1}x)\overline{\chi(y^{-1}x)}dxdy \\
&\overset{\text{inv}}{=} \int_G f(y)\overline{\chi(y)}dy \cdot \int_G g(x)\overline{\chi(x)}dx = \hat{f}\hat{g}
\end{aligned}$$

□

Lemma 4.8. For $f \in L^1(A)$ we have $\hat{f}^* = \overline{\hat{f}}$ so that the Fourier transform $\hat{\cdot} : L^1(A) \to C(\widehat{A})$ is *-invariant.

Proof. Let $\chi \in \widehat{A}$. Then

$$\hat{f}^*(\chi) = \int_A \Delta_A(y^{-1})\overline{f(y^{-1})}\chi(y)dy = \overline{\int_A f(y^{-1})\chi(y^{-1})dy} \overset{1.11.4}{=} \overline{\int_A f(y)\chi(y)dy} = \overline{\hat{f}}(\chi)$$

□

Theorem 4.9. Let A be an LCA group. The map

$$\begin{aligned}
d : \widehat{A} &\longrightarrow \Delta_{L^1(A)} \\
\chi &\longmapsto d_\chi
\end{aligned}$$

where $d_\chi(f) := \int_A f(x)\overline{\chi(x)}dx$, is a homeomorphism. In particular,

1. $\widehat{A}$ is locally compact Hausdorff (Theorem 2.19) so that $\widehat{A}$ is an LCA group.
2. For every $f \in L^1(A)$, the Fourier transform $\hat{f} \in C_0(\widehat{A})$. (Theorem 2.19) Precisely, the triangle commutes

$$\begin{tikzcd}
L^1(A) \arrow[rr, "\text{Fourier}"] \arrow[dr, "\text{Gelfand}"'] & & C_0(\widehat{A}) \\
& C_0(\Delta_{L^1(A)}) \arrow[ur, "d^*"'] &
\end{tikzcd}$$

Proof. It follows from the above lemma that $d_\chi \in \Delta_{L^1(A)}$.

- Injectivity. If $d_\chi = d_\psi$, then $\int_A f(x)\overline{(\chi(x) - \psi(x))}dx = 0$ for all $f \in L^1(G)$; in particular, letting $f = \chi - \psi$ we see $\int_A |\chi(x) - \psi(x)|^2 dx = 0$ so that $\chi = \psi$.

- Surjectivity. Let $m \in \Delta_{L^1(A)}$. Since $C_c(A)$ is dense in $L^1(A)$, we can find $g \in C_c(A)$ such that $m(g) \neq 0$. For each $x \in A$ define

$$\chi(x) = \frac{\overline{m(L_x g)}}{\overline{m(g)}}$$

That m is continuous implies χ is continuous, and for $x, y \in A$, one compute

$$m(L_x g)m(L_y g) = m(L_x g * L_y g) = m(L_{xy} g * g) = m(L_{xy} g)m(g)$$

so that $\chi(xy) = \chi(x)\chi(y)$, i.e. $\chi : A \to \mathbb{C}^\times$ is a group homomorphism.

Let $f \in C_c(A)$. Then by Lemma U.39 we can view $f * g$ as $\int_A f(x) L_x g dx$. Since m is continuous, by Proposition U.36 one obtains

$$\begin{aligned}\int_A f(x)\overline{\chi(x)}dx &= \frac{1}{m(g)}\int_A f(x)m(L_x g)dx = \frac{1}{m(g)} m\left(\int_A f(x)L_x g dx\right) \\ &= \frac{1}{m(g)} m(f * g) = \frac{m(f)m(g)}{m(g)} = m(f)\end{aligned}$$

Let (ϕ_U) be a Dirac net in $C_c(A)$. Then $\phi_U * \overline{\chi}$ converges pointwise to $\overline{\chi}$, so for $x \in A$ and $\varepsilon > 0$ there exists a unit-neighborhood U such that

$$\begin{aligned}|\chi(x)| \leqslant |\phi_U * \overline{\chi}(x)| + \varepsilon &= \left|\int_A L_x \phi_U(y)\overline{\chi(y)}dy\right| + \varepsilon \\ &= |m(L_x \phi_U)| + \varepsilon \leqslant \lim_U \|L_x \phi_U\|_1 + \varepsilon = 1 + \varepsilon\end{aligned}$$

Since $\varepsilon > 0$ is arbitrary, we get $|\chi(x)| \leqslant 1$ for every $x \in A$; but $\chi(x^{-1}) = \chi(x)^{-1}$ so it is equality. In conclusion $\chi \in \widehat{A}$ and $m = d_\chi$.

- Continuity. Let $f \in L^1(A)$ and $U \subseteq \mathbb{C}$ be open. We must show the inverse image under d the set $M(f, U) := \{m \in \Delta_{L^1(A)} \mid m(f) \in U\}$ is open in $\widehat{A}$, i.e.

$$d^{-1}M(f, U) = \{\chi \in \widehat{A} \mid d_\chi(f) \in U\} = \{\chi \in \widehat{A} \mid \hat{f}(\chi) \in U\} \underset{\text{open}}{\subseteq} \widehat{A}$$

Let $\chi \in d^{-1}M(f, U)$ and $\varepsilon > 0$ such that $B_\varepsilon(\hat{f}(\chi)) \subseteq U$. Let $g \in C_c(A)$ with $\|f - g\|_1 < \varepsilon/3$. Put $K = \operatorname{supp}(g)$ and $V := B_{K,\varepsilon/3\|g\|_1}(\chi)$ (Lemma 4.3). Then for all $\psi \in V$ we have

$$\begin{aligned}|\hat{f}(\chi) - \hat{f}(\psi)| &\leqslant |\hat{f}(\chi) - \hat{g}(\chi)| + |\hat{g}(\chi) - \hat{g}(\psi)| + |\hat{g}(\psi) - \hat{f}(\psi)| \\ &< \|f - g\|_1 + \|g\|_1 \frac{\varepsilon}{3\|g\|_1} + \|f - g\|_1 \\ &< \varepsilon\end{aligned}$$

so that $\hat{f}(\psi) \in U$. Hence $\chi \in V \subseteq d^{-1}M(f, U)$, proving that $d^{-1}M(f, U)$ is open.

- Continuous inverse. We prove the following lemma.

Lemma 4.10. Let $\chi_0 \in \widehat{A}$, $K \subseteq A$ compact and $\varepsilon > 0$. Then there exist $\ell \in \mathbb{N}$, $f_0, \ldots, f_\ell \in L^1(A)$, and $\delta > 0$ such that for $\chi \in \widehat{A}$ the condition $|\hat{f}_j(\chi) - \hat{f}_j(\chi_0)| < \delta$ for all $j = 0, \ldots, \ell$ implies $|\chi(x) - \chi_0(x)| < \varepsilon$ for all $x \in K$; in other words,

$$d_{\chi_0} \in \bigcap_{j=0}^{\ell} M(f_j, B_\delta(\hat{f}_j(\chi_0))) \subseteq dB_{K,\varepsilon}(\chi_0)$$

Proof. We first make a reduction. For $f \in L^1(A)$, one has

$$\hat{f}(\chi) - \hat{f}(\chi_0) = \widehat{f\overline{\chi_0}}(\chi\overline{\chi_0}) - \widehat{f\overline{\chi_0}}(1)$$

so that we may assume $\chi_0 = 1$. Let $f \in L^1(A)$ with $\widehat{f}(1) = \int_A f(x)dx = 1$. Then there exists a unit neighborhood U of A such that $\|L_u f - f\|_1 < \frac{\varepsilon}{3}$ for all $u \in U$. Since K is compact, there are finitely many $x_1, \ldots, x_\ell \in A$ such that $K \subseteq x_1 U \cup \cdots \cup x_\ell U$. Set $f_j := L_{x_j} f$ and $f_0 = f$, and let $\delta = \frac{\varepsilon}{3}$. Let $\chi \in \widehat{A}$ with $|\hat{f}_j(\chi) - 1| < \frac{\varepsilon}{3}$ for every $j = 0, \ldots, \ell$. Now let $x \in K$. Then there exists $1 \leqslant j \leqslant \ell$ and $u \in U$ such that $x = x_j u \in x_j U$. One has

$$\begin{aligned}
|\chi(x) - 1| &= |\overline{\chi(x)} - 1| \\
&\leqslant |\overline{\chi(x)} - \overline{\chi(x)}\hat{f}(\chi)| + |\overline{\chi(x)}\hat{f}(\chi) - \hat{f}_j(\chi)| + |\hat{f}_j(\chi) - 1| \\
&= |1 - \hat{f}(\chi)| + |\widehat{L_x f}(\chi) - \widehat{L_{x_j} f}(\chi)| + |\hat{f}_j(\chi) - 1| \\
&< \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon
\end{aligned}$$

where the last inequality uses

$$|\widehat{L_x f}(\chi) - \widehat{L_{x_j} f}(\chi)| \leqslant \left\|L_x f - L_{x_j} f\right\|_1 = \|L_u f - f\|_1 < \frac{\varepsilon}{3}$$

□

□

4.3 Group C^*-Algebra of an LCA Group

Let $f \in L^1(A)$ and $\phi, \psi \in L^2(A)$. For every $y \in A$, one has

$$|\langle L_y \phi, \psi \rangle| \leqslant \|L_y \phi\|_2 \|\psi\|_2 = \|\phi\|_2 \|\psi\|_2$$

This implies that the integral $\int_A f(y)\langle L_y \phi, \psi \rangle dy$ exists, and one has the estimate

$$\left| \int_A f(y)\langle L_y \phi, \psi \rangle \right| \leqslant \|f\|_1 \|\phi\|_2 \|\psi\|_2$$

In other word, the anti-linear map

$$\begin{aligned} L^2(A) &\longrightarrow \mathbb{C} \\ \psi &\longmapsto \int_A f(y)\langle L_y\phi, \psi\rangle dy \end{aligned}$$

is bounded. By Rieze's representation theorem, there exists a unique vector, denoted by $L(f)\phi$, in $L^2(A)$ such that

$$\langle L(f)\phi, \psi\rangle = \int_A f(y)\langle L_y\phi, \psi\rangle dy$$

for all $\psi \in L^2(A)$. The above estimate gives $|\langle L(f)\phi, \psi\rangle| \leqslant \|f\|_1 \|\phi\|_2 \|\psi\|_2$; in particular, for $\psi = L(f)\phi$ one concludes

$$\|L(f)\phi\|_2^2 \leqslant \|f\|_1 \|\phi\|_2 \|L(f)\phi\|_2$$

so that $\|L(f)\phi\|_2 \leqslant \|f\|_1 \|\phi\|_2$, which implies the map

$$\begin{aligned} L(f) : L^2(A) &\longrightarrow L^2(A) \\ \phi &\longmapsto L(f)\phi \end{aligned}$$

is bounded, and hence continuous.

Lemma 4.11. If $f \in L^1(A)$ and $\phi \in L^1(A) \cap L^2(A)$, then $L(f)\phi = f * \phi = \phi * f$.

Proof. By Proposition 1.32, $f * \phi \in L^1(A) \cap L^2(A)$. Let $\psi \in C_c(A)$. Then

$$\begin{aligned} \langle f * \phi, \psi\rangle &= \int_A \int_A f(y)\phi(y^{-1}x)\overline{\psi(x)} dy dx \\ &\overset{\text{Fubini}}{=} \int_A f(y) \left(\int_A L_y\phi(x)\overline{\psi(x)} dx \right) dy \\ &= \int_A f(y)\langle L_y\phi, \psi\rangle dy \\ &= \langle L(f)\phi, \psi\rangle \end{aligned}$$

Since $C_c(G)$ is dense in $L^2(A)$, this implies $L(f)\phi = f * \phi$. □

Lemma 4.12. The map

$$\begin{aligned} L : L^1(A) &\longrightarrow \mathcal{B}(L^2(A)) \\ f &\longmapsto L(f) \end{aligned}$$

is an injective continuous homomorphism of Banach *-algebras.

Proof.

- Continuity. Since L is clearly linear and $\|L(f)\|_{\text{op}} \leqslant \|f\|_1$, so L is bounded and thus continuous.

- Homomorphism. Let $f, g \in L^1(A)$ and $\phi \in C_c(A)$. One computes

$$L(f * g)(\phi) \overset{4.11}{=} (f * g) * \phi = f * (g * \phi) \overset{4.11}{=} f * L(g)\phi \overset{4.11}{=} L(f)L(g)\phi$$

  Lemma 4.11 is valid for $L(g)\phi = g * \phi \in L^1(A) \cap L^2(G)$ by Proposition 1.32. Since $C_c(A)$ is dense in $L^2(G)$, this implies $L(f * g) = L(f)L(g)$.

- Injectivity. Suppose $f \in L^1(A)$ with $L(f)\phi = 0$ for all $\phi \in L^2(A)$. In particular, $f * \psi = 0$ for all $\psi \in C_c(G)$. Letting ψ run over a Dirac net of A, we see $f = 0$.

- *-equivariant. Recall that since A is abelian

$$f^*(x) = \Delta_A(x^{-1})\overline{f(x^{-1})} = \overline{f(x^{-1})}$$

  For $\phi, \psi \in C_c(A)$, one has

$$\begin{aligned}
\langle L(f)\phi, \psi\rangle = \langle f * \phi, \psi\rangle &= \int_A \int_A f(y)\phi(y^{-1}x)\overline{\psi(x)}dxdy \\
&\overset{1.11}{=} \int_A \int_A f(y^{-1})\phi(yx)\overline{\psi(x)}dxdy \\
&\overset{\text{inv}}{=} \int_A \int_A \phi(x)f(y^{-1})\overline{\psi(y^{-1}x)}dxdy \\
&= \int_A \int_A \phi(x)\overline{\overline{f(y^{-1})}\psi(y^{-1}x)}dxdy \\
&= \langle \phi, f^* * \psi\rangle
\end{aligned}$$

  so that $L(f^*) = L(f)^*$.

□

Definition. For an LCA group A, the **group C^*-algebra** $C^*(A)$ is defined to be the closure of $L(L^1(A))$ in the C^*-algebra $\mathcal{B}(L^2(A))$.

- Since $L^1(A)$ is a commutative Banach algebra, $C^*(A)$ is a commutative C^*-algebra.

Theorem 4.13. The pullback map $L^* : \Delta_{C^*(A)} \to \Delta_{L^1(A)}$ is a homeomorphism.

Proof. Since the image of L is dense in $C^*(A)$, it follows that $m \circ L \neq 0$ for all $m \in \Delta_{C^*(A)}$ and that L^* is injective. By Lemma 2.21, it suffices to show L^* is surjective.

To show this, let $m \in \Delta_{L^1(A)}$ and $\chi \in \widehat{A}$ such that $m = d_\chi$, i.e., $m(f) = \hat{f}(\chi)$ for all $f \in L^1(A)$. We have to show m is continuous in C^*-norm, for then it has a continuous extension to $C^*(A)$. To this end, let $\mu_0 \in \Delta_{C^*(A)}$ be fixed. Then there exists $\chi_0 \in \hat{A}$ such that $\mu_0(L(f)) = \hat{f}(\chi_0)$ for all $f \in L^1(A)$. Then for $f \in L^1(A)$,

$$m(f) = \int_A f(x)\overline{\chi(x)}dx = \int_A f(x)\overline{\chi(x)}\chi_0(x)\overline{\chi_0(x)}dx = \widehat{f\overline{\chi}\chi_0}(\chi_0) = \mu_0(L(f\overline{\chi}\chi_0))$$

It follows that $|m(f)| = |\mu_0(L(f\overline{\chi}\chi_0))| \leqslant \|L(f\overline{\chi}\chi_0)\|_{C^*(A)}$. So it suffices to show that for $f \in L^1(A)$, $\|L(f)\|_{C^*(A)} = \|L(f\eta)\|_{C^*(A)}$ for all $\eta \in \widehat{A}$. As C^*-norm is the operator norm in $\mathcal{B}(L^2(A))$, we consider $\phi, \psi \in L^2(A)$ and compute

$$\begin{aligned}
\langle L(\eta f)\phi, \psi\rangle &= \int_A \eta(x)f(x)\langle L_x\phi, \psi\rangle dx \\
&= \int_A \eta(x)f(x)\int_A \phi(x^{-1}y)\overline{\psi(y)}dydx \\
&= \int_A \eta(x)f(x)\int_A (\overline{\eta\phi})(x^{-1}y)\overline{(\overline{\eta}\psi)(y)}dydx \\
&= \langle L(\eta f)\overline{\eta}\phi, \overline{\eta}\psi\rangle
\end{aligned}$$

Putting $\psi = L(\eta f)\phi$, we get

$$\|L(\eta f)\phi\|_2^2 = \langle L(f)(\overline{\eta}\phi), \overline{\eta}L(\eta f)\phi\rangle \leqslant \|L(f)(\overline{\eta}\phi)\|_2 \|\overline{\eta}L(\eta f)\phi\|_2$$

Since $\|L(\eta f)\phi\|_2 = \|\overline{\eta}L(\eta f)\phi\|_2$, it follows that $\|L(\eta f)\phi\|_2 \leqslant \|L(f)(\overline{\eta}\phi)\|_2$, and hence $\|L(\eta f)\|_{C^*(A)} \leqslant \|L(f)\|_{C^*(A)}$; by symmetry we get equality. □

Corollary 4.13.1. Let A be an LCA group. Then the Fourier transform $L^1(A) \to C_0(\widehat{A})$ is injective.

Proof. By functoriality, from $L^1(A) \to C^*(A)$ we obtain a commutative square

$$\begin{tikzcd}
L^1(A) \arrow[r, "\text{Gelfand}_1"] \arrow[d, "L"] & C_0(\Delta_{L^1(A)}) \arrow[d, "(L^*)^*"] \\
C^*(A) \arrow[r, "\text{Gelfand}_2"] & C_0(\Delta_{C^*(A)})
\end{tikzcd}$$

The vertical right arrow is bijective by Theorem 4.13. In Theorem 4.9 we have a commutative triangle

$$\begin{tikzcd}
L^1(A) \arrow[rr, "\text{Fourier}"] \arrow[dr, "\text{Gelfand}_1"'] & & C_0(\widehat{A}) \\
& C_0(\Delta_{L^1(A)}) \arrow[ur, "d^*"'] &
\end{tikzcd}$$

Hence we see the Fourier transform is the composition of injective maps

$$L^1(A) \xrightarrow{L} C^*(A) \xrightarrow{((L^*)^*)^{-1}\circ \text{Gelfand}_2} C_0(\Delta_{L^1(A)}) \xrightarrow{d^*} C_0(\widehat{A})$$

(In fact, the injectivity simply results from the commutative triangle in Theorem 4.9.) □

4.4 The Plancherel Theorem

The Plancherel Theorem says that the Fourier transform extends to a unitary equivalence $L^2(A) \cong L^2(\widehat{A})$. The proof of the theorem is postponed to the next subsection. In this subsection we do some preparations.

Lemma 4.14. Let $\phi, \psi \in L^2(A)$. Then the convolution product $\phi * \psi(x) := \int_A \phi(y)\psi(y^{-1}x)dy$ exists for every $x \in A$ and defines a continuous function in x. The convolution $\phi * \psi$ lies in $C_0(A)$ and its sup-norm satisfies $\|\phi * \psi\|_A \leqslant \|\phi\|_2 \|\psi\|_2$. Finally one has $\phi * \phi^*(1) = \|\phi\|_2^2$.

Proof. Since A is abelian (and hence unimodular), we have

$$\phi * \psi(x) = \int_A \phi(y)\psi(y^{-1}x)dy = \int_A \phi(y)\overline{L_x\psi^*(y)}dy = \langle \phi, L_x\psi^* \rangle$$

Since ϕ and ψ are L^2, $\langle \phi, L_x\psi^* \rangle$ exists for every x by Hölder's inequality. The continuity follows from Lemma 1.36. Next, by Cauchy-Schwarz, we obtain

$$\|\phi * \psi\|_A = \sup_{x \in A} |\langle \phi, L_x\psi^* \rangle| \leqslant \|\phi\|_2 \|\psi\|_2$$

By taking $(\phi_n)_n$ and $(\psi_n)_n$ in $C_c(A)$ such that $\|\phi_n - \phi\|_2 \to 0$ and $\|\psi_n - \psi\|_2 \to 0$, we see from the above inequality that $\phi_n * \psi_n \to \phi * \psi$ uniformly. Since $C_c(A) \subseteq C_0(A)$ and $C_0(A)$ is complete, $\phi * \psi \in C_0(A)$. Finally, $\phi * \phi^*(1) = \langle \phi, \phi \rangle = \|\phi\|_2^2$. □

The space $C_0(A) \times C_0(\widehat{A})$ is a Banach space with the norm

$$\|(f, \eta)\|_0^* := \max\{\|f\|_A, \|\eta\|_{\widehat{A}}\}$$

We embed $C_0(A) \cap L^1(A)$ into this product space by

$$\begin{aligned} C_0(A) \cap L^1(A) &\longrightarrow C_0(A) \times C_0(\widehat{A}) \\ f &\longmapsto (f, \hat{f}) \end{aligned}$$

and denote its closure in the product space by $C_0^*(A)$; it is a Banach space with norm $\|\cdot\|_0^*$. We have two natural projections

$$C_0(A) \xleftarrow{\pi_0} C_0(A) \times C_0(\widehat{A}) \xrightarrow{\pi_*} C_0(\widehat{A})$$

Lemma 4.15. Both restrictions to $C_0^*(A)$ of π_0 and π_* are injective. Hence we can view $C_0^*(A)$ as a subspace of $C_0(A)$ as well as of $C_0(\widehat{A})$.

Proof. Let $f \in C_0^*(A)$. We must show if one of $\pi_0(f)$ and $\pi_*(f)$ is zero, then so is the other. Let $(f_n)_n \subseteq C_0(A) \cap L^1(A)$ such that $f_n \to f$; then $\pi_*(f_n) \to \pi_*(f)$ in $C_0(\widehat{A})$ and $\pi_0(f_n) \to \pi_0(f)$ in $C_0(A)$.

- Recall the isomorphism $C^*(A) \cong C_0(\widehat{A})$. Then $\pi_*(f_n) \to \pi_*(f)$ in $C_0(\widehat{A})$ if and only if $L(f_n) \to \pi_*(f)$ (view $\pi_*(f)$ as an element in $C^*(A)$), if and only if $f_n * \psi \to \pi_*(f)(\psi)$ in $L^2(A)$ for all $\psi \in C_c(A)$.
- Also, $f_n * \psi \to \pi_0(f) * \psi$ in $C_0(A)$.

Hence for $\phi \in C_c(A)$, $\langle f_n * \psi, \phi \rangle$ converges to $\langle \pi_*(f)(\psi), \phi \rangle$, and also to $\langle \pi_0(f) * \psi, \phi \rangle$, and therefore,

$$\langle \pi_*(f)(\psi), \phi \rangle = \langle \pi_0(f) * \psi, \phi \rangle$$

holds for all $\psi, \phi \in C_c(A)$. Then $\pi_*(f) = 0$ if and only if $\pi_0(f) = 0$. □

In the sequel, given an element $f \in C_0^*(A)$ we will freely view it as an element of $C_0(A)$ or of $C^*(A) \cong C_0(\widehat{A})$. If we want to emphasize the distinction, we write f for the former and write $\widehat{f}$ for the latter.

Lemma 4.16. $f \in C_0^*(A)$. If the Fourier transform $\widehat{f}$ is real-valued, then $f(1) \in \mathbb{R}$. If $\widehat{f} \geqslant 0$, then $f(1) \geqslant 0$. Here $1 \in A$ is the identity element.

Proof. Suppose $\widehat{f}$ is real-valued. Then $\widehat{f} = \overline{\hat{f}} = \hat{f}^*$, and since $\hat{\cdot}$ is injective, $f = f^*$ and therefore $f(1) = f^*(1) = \overline{f(1)}$. Now suppose $\hat{f} \geqslant 0$. Then there exists $g \in C_0(\widehat{A}) \cong C^*(A)$ with $g \geqslant 0$ sand $\hat{f} = g^2$. Let $\phi = \phi^* \in C_c(A)$. Then $L(g)\phi \in L^2(A)$, so

$$(L(g)\phi) * (L(g)\phi)^*(1) = \|L(g)\phi\|_2^2 \geqslant 0$$

Take $(g_n)_n \subseteq L^1(A)$ such that $L(g_n) \to g$ in $C^*(A)$. Then

$$\begin{aligned}
(L(g)\phi) * (L(g)\phi)^* &= \lim_n (L(g_n)\phi) * (L(g_n)\phi)^* \\
&= \lim_n (g_n * \phi) * (g_n * \phi)^* \\
&= \lim_n g_n * \phi * \phi * g_n^* \\
&= \lim_n g_n * g_n^* * \phi * \phi \\
&= \lim_n L(g_n * g_n^*)(\phi * \phi) = L(f)(\phi * \phi) = f * \phi * \phi
\end{aligned}$$

where the second last equality is due to $g^2 = \hat{f}$. $\square$

Lemma 4.17.

(a) The space $L^1(A) * C_c(A)$ is a subspace of $C_0(A)$.

(b) Let $f \in C^*(A)$, and let $\phi, \psi \in C_c(A)$. Then $L(f)(\phi * \psi) \in C_0^*(A) \cap L^2(A)$, viewed as a subspace of $C_0(A)$. One has $\widehat{L(f)(\phi * \psi)} = \widehat{f}\widehat{\phi}\widehat{\psi}$.

Proof.

(a) For $f \in L^1(A)$, let $f_n \in C_c(A)$ such that $f_n \to f$ in L^1. Then for $\phi \in C_c(A)$,

$$|(f_n - f_m) * \phi(x)| = \left| \int_A (f_n - f_m)(y)\phi(y^{-1}x)dy \right| \leqslant \|f_n - f_m\|_1 \|\phi\|_\infty$$

Thus $f_n * \phi$ is Cauchy in $C_0(A)$, and since $C_0(A)$ is complete, their limit $f * \phi$ lies in $C_0(A)$.

(b) For $f \in C^*(A)$, we can find $f_n \in L^1(A)$ such that $L(f_n) \to f$ in $C^*(A)$. We have $L(f_n)(\phi * \psi) = f_n * \phi * \psi \in C_0(A) \cap L^1(A)$ by Lemma 4.11 and (a). We show that $f_n * \phi * \psi$ is Cauchy in $C_0^*(A)$, i.e. it is Cauchy in $C_0(A)$ and its Fourier transform is Cauchy is $C_0(\widehat{A})$.

Since $\widehat{f_n * \phi * \psi} = \widehat{f_n}\widehat{\phi}\widehat{\psi}$ and $\widehat{f_n}$ converges uniformly on $\widehat{A}$ (recall $C^*(A) \cong C_0(\widehat{A})$), we see $\widehat{f_n}\widehat{\phi}\widehat{\psi} \to \widehat{f}\widehat{\phi}\widehat{\psi}$ uniformly on $\widehat{A}$. On the other hand, by Lemma 4.14,

$$\|(f_m - f_n) * \phi * \psi\|_A \leqslant \|(f_m - f_n) * \phi\|_2 \|\psi\|_2 \leqslant \|f_m - f_n\|_1 \|\phi\|_2 \|\psi\|_2$$

so $f_n * \phi * \psi$ is Cauchy in $C_0(A)$ as well.

□

Lemma 4.18. Let (ϕ_U) be a Dirac net in $C_c(A)$. Then

(a) $(f * \phi_U)$ converges to f in $C^*(A)$ for all $f \in C^*(A)$.

(b) $(f * \phi_U)$ converges uniformly to f for every $f \in C_0(A)$.

(c) $(f * \phi_U)$ converges to f in $C_0^*(A)$ for every $f \in C_0^*(A)$.

(d) $(\widehat{\phi_U})$ converges locally uniformly to 1 on $\widehat{A}$.

Proof.

1. It holds when $f \in L^1(A)$ by Lemma 1.38. Since $L^1(A) \subseteq C^*(A)$ is dense, the result follows.

2. Similar to (a), with $L^1(A)$ replaced by $C_c(A)$, which is dense in $C_0(A)$.

3. This follows from (a) and (b).

4. Let $C \subseteq \widehat{A}$ be compact and pick positive $\psi \in C_c(\widehat{A})$ such that $\psi \equiv 1$ on C. Let $f \in C^*(A)$ such that $\hat{f} = \psi$ (see the last composition in Corollary 4.13.1). Then

$$\left\|\hat{\phi_U}\psi - \psi\right\|_{\widehat{A}} = \|\phi_U * f - f\|_{\mathrm{op}} \to 0$$

by (a), and the result follows. (Recall that the Gelfand transform is isometric by Gelfand-Naimark so that map $C^*(A) \cong C_0(\widehat{A})$ is isometric; see Corollary 4.13.1 as well.)

□

Lemma 4.19. Let $\eta \in C_c(\widehat{A})$ be real-valued and let $\varepsilon > 0$. Then there are $f_1, f_2 \in C_0^*(A) \cap L^2(A)$, considered as subspaces of $C_0(A)$, such that

(i) the Fourier transforms $\hat{f}_1$, $\hat{f}_2$ lie in $C_c(\widehat{A})$,

(ii) they satisfy $\hat{f}_1 \leqslant \eta \leqslant \hat{f}_2$, further $\left\|\hat{f}_1 - \hat{f}_2\right\|_{\widehat{A}} < \varepsilon$, and $\operatorname{supp} \hat{f}_i \subseteq \operatorname{supp} \eta$ for $i = 1, 2$,

(iii) as well as $0 \leqslant f_2(1) - f_1(1) < \varepsilon$

In particular, every $\eta \in C_c(\widehat{A})$ is the uniform limit of functions of the form $\hat{f}$ with $f \in C_0^*(A)$ of support contained in $\operatorname{supp} \eta$.

Proof. For any Dirac function $\phi \in C_c(A)$ one has $\hat{\phi} \in C_0(\widehat{A})$ by Theorem 4.9.2 and Lemma 4.18 says that the ensuing function $\hat{\phi}$ can be chosen to approximate the constant 1 arbitrarily close on any compact set.

Note that the Fourier transform $\widehat{h * h^*} = \hat{h}\hat{h}^* = \hat{h}\overline{\hat{h}} \geqslant 0$. Let $K := \operatorname{supp} \eta \subseteq \widehat{A}$. Since $C_c(A)$ contains Dirac functions of arbitrary small support, we conclude that for every $\delta > 0$ there exists a function $\phi_\delta \in C_c^+(A)$ such that the function $\psi_\delta := \phi_\delta * \phi_\delta^*$ satisfies

$$1 - \delta \leqslant \hat{\psi}_\delta(\chi) \leqslant 1 + \delta \qquad \text{for every } \chi \in K$$

Fix $\phi \in C_c^*(A)$ such that $\psi := \phi * \psi^*$ satisfies $\hat{\psi}(\chi) \geqslant 1$ for all $\chi \in K$. Let $f \in C^*(A)$ with $\hat{f} = \eta$ and set

$$f_1 := f * (\psi_\delta - \delta\psi), \qquad f_2 := f * (\psi_\delta + \delta\psi)$$

By Lemma 4.17, $f_1, f_2 \in C_0^*(A) \cap L^2(A)$. For every $\chi \in \widehat{A}$, we have

$$\hat{f}_1(\chi) = \hat{f}(\chi)(\hat{\psi}_\delta(\chi) - \delta\hat{\psi}(\chi)) \leqslant \eta(\chi) \leqslant \hat{f}_2(\chi)$$

Further, as $\hat{f}(\chi) = \eta(\chi)$, one has $\operatorname{supp}(\hat{f}_i) \subseteq \operatorname{supp} \eta$. The other properties follow by choosing δ small enough and Lemma 4.16. □

Proposition 4.20. Let $\psi \in C_c(\widehat{A})$ be real-valued. Then

$$\sup\{f(1) \mid f \in C_0^*(A),\ \hat{f} \leqslant \psi\} = \inf\{f(1) \mid f \in C_0^*(A),\ \hat{f} \geqslant \psi\}$$

Denote this common value by $I(\psi)$. Extending linearly, we obtain a positive linear functional $I : C_c(\widehat{A}) \to \mathbb{C}$. Then I is a Haar integral. We write this integral as

$$I(\psi) = \int_{\widehat{A}} \psi(\chi) d\chi$$

Proof. By Lemma 4.16, if we pick $f \in$ LHS and $g \in$ RHS, then $\widehat{g - f} \geqslant 0$ so that $g(1) \geqslant f(1)$; this implies

$$\sup\{f(1) \mid f \in C_0^*(\widehat{A}),\ \hat{f} \leqslant \psi\} \leqslant \inf\{f(1) \mid f \in C_0^*(A),\ \hat{f} \geqslant \psi\}$$

The equality holds, for their difference is arbitrarily small by Lemma 4.19 (particularly (ii) and (iii)).

- I linear. Clear.
- I positive. If $\psi \geqslant 0$, then $f(1) \geqslant 0$ for all $f \in C_0^*(A)$ with $\hat{f} \geqslant \psi$ by Lemma 4.16. Thus $I(\psi) \geqslant 0$.
- Invariance. Let $\psi \in C_c(\widehat{A})$ and $f \in C_0^*(A)$ with $\hat{f} \leqslant \psi$. For $\chi \in \widehat{A}$ we have $L_\chi \hat{f} \leqslant L_\chi \psi$. Further,

  $$L_\chi \hat{f}(\rho) = \hat{f}(\chi^{-1}\rho) = \int_A f(y)\overline{\chi^{-1}\rho(y)}dy = \int_A f\chi(y)\overline{\rho(y)}dy = \widehat{f\chi}(\rho)$$

  and $\chi f(1) = f(1)$. This shows the invariance.

□

The following will be proved in the next subsection as a consequence of the Pontryagin duality theorem.

Theorem 4.21 (Plancherel theorem)**.** For a given Haar measure on A there exists a uniquely determined Haar measure on $\widehat{A}$, called the **Plancherel measure**, such that for $f \in L^1(A) \cap L^2(A)$ one has

$$\|f\|_2 = \left\|\hat{f}\right\|_2$$

This implies that the Fourier transform extends to an isometry from $L^2(A)$ to $L^2(\widehat{A})$. Indeed, it is also surjective, so the Fourier transform extends to a canonical unitary equivalence $L^2(A) \cong L^2(\widehat{A})$.

Corollary 4.21.1. Let A be a compact LCA group. Then the elements of the dual group $\widehat{A}$ form an orthonormal basis of $L^2(A)$.

Proof. We normalized the Haar measure on A so that the volume of A is 1. A being compact, every continuous function on A is L^2; particularly, every character of A is L^2. We show the characters of A form an orthonormal system, i.e. that for $\chi, \eta \in \widehat{A}$ we have

$$\langle \chi, \eta \rangle = \delta_{\chi,\eta} = \begin{cases} 1 & , \chi = \eta \\ 0 & , \chi \neq \eta \end{cases}$$

This is standard, and can be proved as in the finite group case. It follows that the Fourier transform of a character χ is the map δ_χ with $\delta_\chi(\eta) := \delta_{\chi,\eta}$.

Since A is compact, $\widehat{A}$ is discrete so that the Plancherel measure is just a multiple of the counting measure; say the constant is $c > 0$. Let S be any finite subset of $\widehat{A}$. For each $f \in L^2(\widehat{A})$, one has

$$\left\| f - \sum_{\chi \in S} f(\chi)\delta_\chi \right\|_2^2 = c \left(\sum_{\chi \in \widehat{A}} |f(\chi)|^2 - \sum_{\chi \in S} |f(\chi)|^2 \right)$$

It follows from the definition of the integral (associated to the counting measure) that we can pick S large enough that the difference is arbitrarily small; hence $\{\delta_\chi\}_{\chi \in \widehat{A}}$ forms an orthonormal basis of $L^2(\widehat{A})$. By Plancherel theorem, the Fourier transform is a unitary equivalence, the characters form an orthonormal basis of $L^2(A)$. □

4.5 Pontryagin Duality

For an LCA group A, we see in Theorem 4.9 that $\widehat{A}$ is again an LCA group, so we can consider the dual group $\widehat{\widehat{A}}$ of $\widehat{A}$, and it is again an LCA group. There is a canonical association, called the **Pontryagin map**, δ which assigns each $x \in A$ to a group homomorphism $\delta_x : \widehat{A} \to S^1$ given by $\delta_x(\chi) = \chi(x)$.

Lemma 4.22. For each $x \in A$, the homomorphism $\delta_x : \widehat{A} \to S^1$ is continuous, and thus $\delta_x \in \widehat{\widehat{A}}$.

Proof. For each $\varepsilon > 0$, $\delta_x^{-1}(B_\varepsilon(1)) = \{\chi \in \widehat{A} \mid |\chi(x) - 1| < \varepsilon\}$ is open in $\widehat{A}$ by the definition of the compact-open topology (note that every singleton is compact). □

Proposition 4.23. The Pontryagin map $\delta : A \to \widehat{\widehat{A}}$ is an injective continuous group homomorphism. In particular, if $1 \neq x \in A$, there exists some $\chi \in \widehat{A}$ such that $\chi(x) \neq 1$.

Proof.

- Group homomorphism. Clear.
- Continuity. We show the Pontryagin map is continuous at the identity element. Let $V \subseteq \widehat{\widehat{A}}$ be a unit-neighborhood. By Lemma 4.3 we can find compact $K' \subseteq \widehat{A}$ and $\varepsilon > 0$ such that

$$B_{K',\varepsilon} = B_{K',\varepsilon}(1) = \{\alpha \in \widehat{\widehat{A}} \mid |\alpha(\chi) - 1| < \varepsilon \text{ for all } \chi \in K'\} \subseteq V$$

  Let $L \subseteq A$ be a compact unit-neighborhood. Since K' is compact, we can find $\chi_1, \ldots, \chi_n \in K'$ such that $K' \subseteq B_{L,\varepsilon/2}(\chi_1) \cup \cdots \cup B_{L,\varepsilon/2}(\chi_n)$, where

$$B_{L,\varepsilon/2}(\chi) := \{\psi \in \widehat{A} \mid |\psi(x) - \chi(x)| < \varepsilon/2 \text{ for all } x \in L\}$$

  For $j = 1, \ldots, n$, let $U_j = \{x \in A \mid |\chi_j(x) - 1| < \varepsilon/2\}$. Finally, let

$$U = \operatorname{int} L \cap U_1 \cap \cdots \cap U_n$$

  Then for $x \in U$ and $\chi \in K'$ (say $\chi \in B_{L,\varepsilon/2}(\chi_i)$), we have

$$|\delta_x(\chi) - 1| = |\chi(x) - 1| \leqslant |\chi(x) - \chi_i(x)| + |\chi_i(x) - 1| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$$

  so that $\delta(U) \subseteq B_{K',\varepsilon} \subseteq V$.
- Injectivity. Assume $1 \neq x \in A$ with $\delta_x = \mathbf{1}_{\widehat{A}}$. Then $\chi(x) = 1$ for every $\chi \in \widehat{A}$. Choose $g \in C_c(A)$ with $g(1) = 1$ and $g(x^{-1}) = 0$. Then $L_x(g) \neq g$, but

$$\widehat{L_x(g)}(\chi) = \overline{\chi}(x)\widehat{g}(\chi) = \widehat{g}(\chi)$$

  for all $\chi \in \widehat{A}$, a contradiction to injectivity of Fourier transform.

□

Lemma 4.24. Let $f \in C_0^*(A)$ be such that it Fourier transform lies in $C_c(\widehat{A})$. Then for every $x \in A$ one has $f(x) = \widehat{\widehat{f}}(\delta_{x^{-1}})$.

Proof. One has for $x \in A$,

$$f(x) = L_{x^{-1}} f(1) = \int_{\widehat{A}} \widehat{L_{x^{-1}} f}(\chi) d\chi = \int_{\widehat{A}} \widehat{f}(\chi)\delta_x(\chi) d\chi = \widehat{\widehat{f}}(\delta_{x^{-1}})$$

The second equality holds for $\hat{f} \in C_c(\widehat{A})$ and by Proposition 4.20. □

Lemma 4.25. For an LCA group A, the following hold.

(i) $C_c(A)$ is dense in $C_0^*(A)$.

(ii) $C_c(\widehat{A}) \cap \{\hat{f} \mid f \in C_0^*(A) \cap L^2(A)\}$ is dense in $C_0^*(\widehat{A})$.

(iii) $C_c(\widehat{A}) \cap \{\hat{f} \mid f \in C_0^*(A) \cap L^2(A)\}$ is dense in $L^2(\widehat{A})$.

Proof.

(i) By definition, $C_0(A) \cap L^1(A)$ is dense $C_0^*(A)$. It suffices to show that for a given $f \in C_0(A) \cap L^1(A)$, there exists a sequence in $C_c(A)$ converging in $\|\cdot\|_A$ and $\|\cdot\|_1$. For each $n \geqslant 1$, let $K_n := \{x \in A \mid |f(x)| \geqslant \frac{1}{n}\}$; this is compact since $f \in C_0(A)$. Choose $\chi_n \in C_c(A)$ with $\chi_n|_{K_n} \equiv 1$ and set $f_n = \chi_n f$. Then $f_n \to f$ in both norms.

(ii) and (iii) follow from (i) and Lemma 4.19.

□

Theorem 4.26 (Pontryagin Duality)**.** The Pontryagin map $\delta : A \to \widehat{\widehat{A}}$ is an isomorphism of LCA groups.

Proof. We have seen that δ is an injective group homomorphism. It remains to show δ is surjective with a continuous inverse. We will show that δ is a closed map with dense image.

- <u>δ has a dense image.</u> Suppose otherwise. Then there exists open $U \subseteq \widehat{\widehat{A}}$ disjoint from $\delta(A)$. By Lemma 4.19 (with $\widehat{A}$ in place of A) together with Urysohn's lemma, we can find $\psi \in C_0^*(\widehat{A})$ which is *nonzero* and $\widehat{\psi}$ is supported in U, i.e., $\widehat{\psi}(\delta(A)) = 0$. By Lemma 4.25, we can find $(f_n) \subseteq C_0^*(A)$ such that $\psi_n := \widehat{f_n}$ lies in $C_c(\widehat{A})$ with $\psi_n \to \psi$ in $C_0^*(\widehat{A})$. By inversion formula, we have

$$f_n(x) = \widehat{\psi_n}(\delta_{x^{-1}}) \text{ for all } x \in A.$$

  This implies (for $\widehat{\psi}(\delta(A)) = 0$) that $f_n \to 0$ uniformly on A. On the other hand, $\widehat{f_n} =: \psi_n \to \psi$ uniformly on $\widehat{A}$. These two imply that (f_n) is a Cauchy sequence in $C_0^*(A)$, so it converges in this space. Since the limit is unique, it follows from Lemma 4.15 that $\psi = 0$, a contradiction. Hence the image of δ must be dense in $\widehat{\widehat{A}}$.

- <u>δ is a proper map.</u> It suffices to show $\breve{\delta}(x) := \delta(x^{-1})$. Let $K \subseteq \widehat{\widehat{A}}$ be compact. By Lemma 4.19 (with Urysohn's lemma) there exists $\psi \in C_0^*(\widehat{A})$ such that $\hat{\psi}$ has compact support, $\geqslant 0$ on $\widehat{\widehat{A}}$, and $\geqslant 1$ on K. As above, there is a sequence $(f_n) \subseteq C_0^*(A)$ such that $\psi_n := \hat{f}_n \geqslant 0$ lies in $C_c(\widehat{A})$ and converges to ψ in $C_0^*(\widehat{A})$. Fix n with $\left\|\hat{\psi} - \hat{\psi}_n\right\|_{\widehat{\widehat{A}}} < \dfrac{1}{2}$. We have $f_n(x) = \hat{\psi}_n(\delta_{x^{-1}})$ for all $x \in A$, and since $f_n \in C_0(A)$, we can find compact $C \subseteq A$ such that $|f_n| < \dfrac{1}{2}$ outside C. As $\hat{\psi}_n|_K \geqslant \dfrac{1}{2}$, it follows that $\breve{\delta}^{-1}(K) \subseteq C$.

- δ is a closed map. This is Proposition T.7.

□

Proposition 4.27. The Fourier transform induces an isometric isomorphism of Banach spaces

$$\mathcal{F} : C_0^*(A) \to C_0^*(\widehat{A})$$

with inverse map given by the dual Fourier transform

$$\begin{aligned} \widehat{\mathcal{F}} : C_0^*(\widehat{A}) &\longrightarrow C_0^*(A) \\ \psi &\longmapsto [x \mapsto \hat{\psi}(\delta_{x^{-1}})] \end{aligned}$$

Proof. Let $B = \{f \in C_0^*(A) \mid \widehat{f} \in C_c(\widehat{A})\}$. For $f \in B$ we have $\widehat{\mathcal{F}} \circ \mathcal{F}(f) = f$ by Lemma 4.24. Further, one has

$$\begin{aligned} \|f\|_0^* := \max\{\left\|\hat{f}\right\|_{\widehat{A}}, \|f\|_A\} &= \max\{\left\|\hat{f}\right\|_{\widehat{A}}, \left\|\widehat{\mathcal{F}} \circ \mathcal{F}(f)\right\|_A\} \\ &= \max\{\left\|\hat{f}\right\|_{\widehat{A}}, \left\|\hat{\hat{f}}\right\|_{\hat{\widehat{A}}}\} \\ &= \|\mathcal{F}(f)\|_0^* \end{aligned}$$

By Lemma 4.25 $\mathcal{F}(B)$ is dense in $C_0^*(\widehat{A})$, so the Fourier transform defines a surjective isometry from the closure of B to $C_0^*(\widehat{A})$, and thus $\widehat{F}$ is an isometry from $C_0^*(\widehat{A})$ to $C_0^*(A)$. Since $\widehat{\mathcal{F}} = \mathcal{F}_{\widehat{A}} \circ \delta^{-1}$ (in an appropriate sense), where $\mathcal{F}_{\widehat{A}}$ denotes the Fourier transform on $\widehat{A}$, and since $\mathcal{F}_{\widehat{A}}(C_0^*(\widehat{A}))$ contains a subset of $C_c(\hat{\widehat{A}})$ that is dense in $C_0^*(\hat{\widehat{A}})$ by Lemma 4.25, it follows from Pontryagin duality that $\widehat{F}(C_0^*(\widehat{A}))$ is dense in $C_*(A)$. Since $\widehat{F}$ is isometric, it is then an isomorphism of Banach spaces. □

Theorem 4.28 (Inversion Formula)**.** Let $f \in L^1(A)$ be such that $\widehat{f} \in L^1(\widehat{A})$. Then f is continuous, and for $x \in A$ one has

$$f(x) = \hat{\hat{f}}(\delta_{x^{-1}})$$

Proof. Let $f \in L^1(A)$ with $\widehat{f} \in L^1(\widehat{A})$. Then $\widehat{f} \in C_0(\widehat{A}) \cap L^1(\widehat{A}) \subseteq C_0^*(\widehat{A})$ by Theorem 4.9.2. and definition of $C_0^*(\widehat{A})$. By Proposition 4.27 we can find $g \in C_0^*(A)$ with $\widehat{g} = \widehat{f}$ and $g(x) = \hat{\hat{f}}(\delta_{x^{-1}})$ for every $x \in A$. Since the Fourier transform $C^*(A) \to C_0(\widehat{A})$ is injective, we call $f = g$ in $C^*(A)$. □

Proof. (of Plancherel theorem) Let $f \in L^1(A) \cap L^2(A)$. By Lemma 4.14, we have $f * f^* \in L^1(A) \cap C_0(A)$. The continuous function $h = \widehat{f * f^*} = |\widehat{f}|^2 \in C_0(\widehat{A})$ is positive. Let $\phi \in C_c(\widehat{A})$ satisfy $0 \leqslant \phi \leqslant h$. Then by Proposition 4.20 and Lemma 4.14, we have

$$\int_{\widehat{A}} \phi(\chi) d\chi \leqslant f * f^*(1) = \|f\|_2^2 < \infty$$

Thus h is integrable, so $\widehat{f * f^*} \in L^1(A)$. By inversion formula we have

$$\|f\|_2^2 = f * f^*(1) = \widehat{\widehat{f * f^*}}(1) = \widehat{|\hat{f}|^2}(1) = \left\|\hat{f}\right\|_2^2$$

As $L^1(A) \cap L^2(A)$ is dense in $L^2(A)$, the Fourier transform extends uniquely to an isometric linear map $L^2(A) \to L^2(\widehat{A})$. By Lemma 4.19 the image in $L^2(A)$ is dense, whence the map is surjective. $\square$

Proposition 4.29. Let $\phi, \psi \in L^1(A) \cap L^2(A)$ and let $f = \phi * \psi$. Then $f \in L^1(A)$ and $\hat{f} \in L^1(\widehat{A})$, so the inversion formula applies to f.

Proof. We have $\hat{f} = \widehat{\phi * \psi} = \hat{\phi}\hat{\psi} \in L^1(\widehat{A})$ by Hölder's inequality. $\square$

4.6 Poisson Summation Formula

Let A be an LCA group. Recall we have a pairing

$$\begin{aligned} \langle\, , \rangle : A \times \widehat{A} &\longrightarrow \mathbb{C} \\ (a, \chi) &\longmapsto \chi(a) = \delta_a(\chi) \end{aligned}$$

From this we introduce some notation. For $E \subseteq A$, let

$$E^\perp := \{\chi \in \widehat{A} \mid \langle E, \chi \rangle = 1\} = \{\chi \in \widehat{A} \mid \delta_a(\chi) = 1 \text{ for all } a \in E\}$$

and for $L \subseteq \widehat{A}$, let

$$L^\perp = \{a \in A \mid \langle a, L \rangle = 1\} = \{a \in A \mid \chi(a) = 1 \text{ for all } \chi \in L\}$$

By definition (and perhaps Proposition 4.23), $E^\perp$ is closed in $\widehat{A}$, and $L^\perp$ is closed in A.

Proposition 4.30. Let A be an LCA group and $B \leqslant A$ be closed. Then

(i) $\begin{aligned} B^\perp &\longrightarrow \widehat{A/B} \\ \chi &\longmapsto \overline{\chi} : xB \mapsto \chi(x) \end{aligned}$ is an isomorphism.

(ii) $(B^\perp)^\perp = B$.

(iii) $\begin{aligned} \widehat{A}/B^\perp &\longrightarrow \widehat{B} \\ \chi B^\perp &\longmapsto \chi|_B \end{aligned}$ is an isomorphism.

Proof.

(i) Since $A \to A/B$ is surjective, we have an injection $\widehat{A/B} \to \widehat{A}$. Its image is clearly $B^\perp$.

(ii) From definition follows $B \subseteq (B^\perp)^\perp$. Similarly, if $L \subseteq \widehat{\mathbb{A}}$, we have $L \subseteq (L^\perp)^\perp$. It follows that

$$B^\perp \supseteq ((B^\perp)^\perp)^\perp \supseteq B^\perp$$

so $B^\perp = ((B^\perp)^\perp)^\perp$.

(iii) $(\widehat{A}/B^\perp)^\wedge = (B^\perp)^\perp = B$.

□

Corollary 4.30.1. Let A be an LCA group and $B \leqslant A$ be closed. Then we have an short exact sequence

$$1 \longrightarrow \widehat{A/B} \longrightarrow \widehat{A} \xrightarrow{(\cdot)|_B} \widehat{B} \longrightarrow 1$$

In other words, the Pontryagin dual $\widehat{\cdot}$ is an exact functor in the category of LCA groups.

Theorem 4.31 (Poisson's Summation Formula)**.** Let A be an LCA group and $B \leqslant A$ be closed. For $f \in L^1(A)$, define $f^B \in L^1(A/B)$ by

$$f^B(xB) := \int_B f(xb)db$$

If we identify $\widehat{A/B}$ with $B^\perp$, we get $\widehat{f^B} = \widehat{f}|_B^\perp$. If, in addition, $\widehat{f}|_{B^\perp} \in L^1(B^\perp)$, we get

$$\int_B f(xb)db = \int_{B^\perp} \widehat{f}(\chi)\chi(x)d\chi$$

for almost all $x \in A$. Here the Haar measure on $B^\perp \cong \widehat{A/B}$ is the Plancherel measure with respect to the chosen Haar measure on A/B such that the quotient integral formula holds. Finally, if f^B is everywhere defined and continuous, the above equation holds for all $x \in A$.

Proof. By the quotient integral formula, for $\chi \in B^\perp$

$$\widehat{f^B}(\chi) = \int_{A/B} f^B(xB)\overline{\chi}(xB)dxB = \int_{A/B}\int_B f(xb)\chi(xb)dbdxB = \int_A f(x)\chi(x)dx = \widehat{f}(\chi)$$

Moreover, if $\widehat{f}|_{B^\perp} \in L^1(B^\perp) = L^1(\widehat{A/B})$, then the inversion formula implies

$$\int_B f(xb)db = f^B(xB) = \widehat{\widehat{f^B}} \quad (\delta_{x^{-1}B}) = \widehat{\widehat{f}|_{B^\perp}} \quad (\delta_{x^{-1}B}) = \int_{B^\perp} \widehat{f}(\chi)\overline{\chi(x)}d\chi$$

holds almost everywhere. If f^B exists everywhere and is continuous, then this formula holds everywhere.

□

Example 4.32. Let A be the additive group $\mathbb{R}$ with euclidean topology. Then A is self-dual, where the isomorphism $A \cong \widehat{A}$ is given by $y \mapsto [\chi_y : x \mapsto e^{2\pi ixy}]$. Let B be the discrete subgroup $\mathbb{Z}$. Then $y \mapsto \chi_y$ maps B bijectively to $B^\perp$.

For $f \in L^1(\mathbb{R})$ such that $\widehat{f}|_{\mathbb{Z}} \in L^1(\mathbb{Z})$, the equality

$$\sum_{k\in\mathbb{Z}} f(x+k) = \sum_{k\in\mathbb{Z}} \widehat{f}(k)e^{2\pi ikx} \tag{$\spadesuit$}$$

holds for almost all x, where $\widehat{f}(x) = \displaystyle\int_{\mathbb{R}} f(y)e^{-2\pi ixy}dy$.

Define the **Schwartz space** $\mathcal{S}(\mathbb{R})$ as the space of all smooth functions $f : \mathbb{R} \to \mathbb{C}$ such that for all $m, n \in \mathbb{Z}$ the function $x^n f^{(m)}(x)$ is bounded. Then the Fourier transform is a bijection on $\mathcal{S}(\mathbb{R})$. For $f \in \mathcal{S}(\mathbb{R})$, both sums in ($\spadesuit$) converge uniformly and define continuous functions, so by taking $x = 0$ we obtain

$$\sum_{k\in\mathbb{Z}} f(k) = \sum_{k\in\mathbb{Z}} \widehat{f}(k)$$

5 The Structure of LCA-Groups

5.1 Connectedness

Recall that for a topological space X with base point x_0, the set $\pi_0(X, x_0)$ is defined to be the collection of all connected components with a mark point being the one containing x_0.

Definition. Let G be a topological group. The connected component of G containing e is called the **identity component**, and is denoted by G°.

Proposition 5.1. Let G be a topological group.

(i) The identity component G° is a closed normal subgroup of G.

(ii) For any $x \in G$, the coset xG° is the connected component containing x. In particular, the natural map $G \to \pi_0(G, e)$ induces a bijection $G/G^\circ \cong \pi_0(G, e)$.

(iii) The quotient group G/G° is totally disconnected.

Proof. To show G/G° is totally disconnected, we prove that every subset $A \subseteq G/G^\circ$ with $\#A \geqslant 2$ is not connected. Let $\pi : G \to G/G^\circ$ be the projection and put $B = \pi^{-1}(A)$. Then B contains at least two distinct cosets of G°, so B is not connected. Thus we can find open $W_1, W_2 \subseteq G$ with $B \cap W_i \neq \varnothing$, $i = 1, 2$, $B \cap W_1 \cap W_2 = \varnothing$ and $B \subseteq W_1 \cup W_2$. Then $xG^\circ \cap W_1 \cap W_2 = \varnothing$ and $xG^\circ \subseteq W_1 \cup W_2$ for all $x \in B$; since xG° is connected, we have $xG^0 \subseteq W_i$ for some $i = 1, 2$. Hence, $V_i := \pi(W_i)$, $i = 1, 2$ are nonempty disjoint open sets in G/G° separating A. □

Proposition 5.2. Let G be a topological group and let U be an open compact unit-neighborhood of G. Then U contains an open compact subgroup of G.

Proof. By compactness of U we can find a symmetric unit-neighborhood V of G such that $UV = U = VU$. The subgroup $K := \bigcup_{n \in \mathbb{N}} V^n$ generated by V is then an open, hence closed, subgroup contained in U. Since U is compact, K is also compact. □

Proposition 5.3. Every totally disconnected LCH space X has a basis consisting open and compact subsets of X.

Proof. It suffices to show that for each $x \in X$ and each compact neighborhood U of x, there exists an open and closed subset V of X such that $x \in V \subseteq U$. For this, let

$$M := \{y \in U \mid \text{there exists a relatively open and closed subset } C_y \subseteq U \text{ with } y \in C_y,\ x \notin C_y\}$$

By definition M is the union of all such C_y, showing that M is relatively open in U. Let $A = U \backslash M$; then $x \in A$ and A is closed. We claim

- the set A is connected.

Assuming this, since X is totally disconnected, we have $A = \{x\}$, so the compact boundary $\partial U := U \backslash \operatorname{int} U$ lies in M, hence lies in the union of finitely many sets $C_{y_1}, \ldots, C_{y_n}$. Then $V := U \backslash \bigcup_{i=1}^{n} C_{y_i}$ contains x, is relatively open in U and is closed in X. Since $V \subseteq \operatorname{int} U$, V is also open in X.

It remains to show A is connected. Let B_1, B_2 be closed, hence compact in U, subsets of A such that $B_1 \cap B_2 = \varnothing$ and $A = B_1 \cup B_2$. Assuming $x \in B_1$, we must show B_2 is empty. Since B_1 and B_2 are compact, we can find $W \subseteq U$ open in U such that $B_2 \subseteq W$ and $\overline{W} \cap B_1 = \varnothing$. Since $\overline{W} \backslash W \subseteq M$ is compact in U, it is covered by finitely many of the open sets C_y; say $C \subseteq M$ is their (finite) union. Then $\overline{W} \backslash W \subseteq C$ and C is open and closed in U. Set $\tilde{W} := W \cup C$; since $W \cup C = \overline{W} \cup C$, $\tilde{W}$ is open and closed in U. Also, $\tilde{W} \cap A = W \cap A = B_2$, so $x \notin \tilde{W}$, and $\tilde{W} \subseteq M$, which implies $B_2 = \varnothing$. $\square$

Theorem 5.4. A totally disconnected LCH group admits an open neighborhood basis of identity consisting of open compact subgroups.

5.1.1 Profinite groups

Definition. A **profinite group** is a topological group isomorphic to a projective limit of finite groups.

Theorem 5.5. For a topological group G, TFAE:

(i) G is profinite.

(ii) G is compact, Hausdorff and totally disconnected.

(iii) G is compact, Hausdorff and has an open neighborhood basis of the identity consisting of normal subgroups.

Proof. (i)$\Rightarrow$(ii) is clear. Assume (ii). By Theorem 5.4 G has an open neighborhood basis of identity consisting of open subgroups. Let U be an open subgroup of G; since G is compact, the index $[G : U]$ is finite. Consider the natural homomorphism $G \to S_{G/U}$; the fibre of g is of the form $g \bigcup_{h \in G/U} h^{-1}Uh$, an open neighborhood of g. Hence the homomorphism is continuous, so its kernel N is an open normal subgroup of G contained in U. Hence (iii).

Now assume (iii). We have a natural map $\Phi : G \to H := \varprojlim_N G/N$, where N runs over all open normal subgroups of G. Since G is compact, each G/N is a finite group. We have to check it is an isomorphism of topological groups.

- Injectivity. The kernel is the intersection of all open normal subgroups. Since they form a neighborhood basis of the identity e of G and G is Hausdorff, it follows that $\ker \Phi = \{e\}$.

- Continuity. Let S be a finite collection of open normal subgroups of G. Then the inverse image of $H_S := \left(\prod_{N \in S} \{eN\} \times \prod_{N \notin S} G/N \right) \cap \varprojlim_N G/N$ under Φ is $\bigcap_{N \in S} N$, which is open in G.

- Dense image. Let S and H_S be as in previous item. Let $g \in G$ and write $\Phi(g) = \prod_N g_N \in H$. Then $\Phi(g)H_S$ is a neighborhood of $\Phi(g)$, and if we take $g' \in G$ that is mapped under $G \to G/\bigcap_{N \in S} N$ to $g_{\bigcap_{N \in S} N}$, then $\Phi(g') \in \Phi(g)H_S$.
- Closed mapping. For G is compact.

$\square$

Example 5.6 (Infinite Galois theory)**.** Let Ω/k be a (finite or infinite) Galois extension, i.e, a normal separable extension. The Galois group $G := \mathrm{Gal}(\Omega/k)$ is given a topology, called the **Krull topology**, defined as follows. For $\sigma \in G$, take the cosets $\sigma \,\mathrm{Gal}(\Omega/K)$ as an open neighborhood basis of $\sigma \in G$, where K/k runs over all finite Galois subextensions of Ω/k.

- The multiplication $G \times G \to G$ is continuous, since for $(\sigma, \tau) \in G \times G$, the preimage of $\sigma\tau \,\mathrm{Gal}(\Omega/K)$ contains a neighborhood $\sigma \,\mathrm{Gal}(\Omega/\tau(K)) \times \tau \,\mathrm{Gal}(\Omega/K)$ of (σ, τ).
- The inversion $G \to G$ is continuous, for the preimage of $\sigma^{-1} \,\mathrm{Gal}(\Omega/K)$ is

$$\mathrm{Gal}(\Omega/K)\sigma = \sigma\sigma^{-1} \,\mathrm{Gal}(\Omega/K)\sigma = \sigma \,\mathrm{Gal}(\Omega/\sigma^{-1}(K))$$

In this way $G = \mathrm{Gal}(\Omega/k)$ becomes a topological group. In fact, one can easily see that this topology is just the subspace topology inherited from the pointwise convergence topology on Ω^Ω, where Ω is thought of as a discrete space. In particular, this implies G is Hausdorff. Now consider the map

$$\begin{aligned} \Phi : G &\longrightarrow H := \prod_K \mathrm{Gal}(K/k) \\ \sigma &\longmapsto (\sigma|_K)_K \end{aligned}$$

where K/k runs over all finite Galois subextensions of Ω/k. By Tychonov's theorem, H is compact, and to show G is compact it suffices to show Φ is a closed embedding.

- Injective. Clear.
- Continuity. Let K_0/k be a finite Galois subextension of Ω/k. The inverse image of the set $\prod_{K \neq K_0} \mathrm{Gal}(K/k) \times \{\sigma_0\}$ with $\sigma_0 \in \mathrm{Gal}(K_0/k)$ is $\sigma \,\mathrm{Gal}(\Omega/K_0)$, where $\sigma \in G$ is any extension of σ_0 to Ω.
- Open mapping. Let the notation be as in the previous item. Then the image of $\sigma \,\mathrm{Gal}(\Omega/K_0)$ under Φ is $\Phi(G) \cap \left(\prod_{K \neq K_0} \mathrm{Gal}(K/k) \times \{\sigma_0\}\right)$, which is open in $\Phi(G)$.
- $\Phi(G)$ is closed in H. Indeed, we have

$$\Phi(G) = \bigcap_{L \subseteq L'} \{(\sigma_K)_K \in H \mid \sigma_{L'}|_L = \sigma_L\}$$

where $L \subseteq L'$ runs over all finite subextensions of Ω/k. For such a pair $L \subseteq L'$, if we write $\mathrm{Gal}(L/k) = \{\sigma_i\}_{i=1}^n$ and $S_i \subseteq \mathrm{Gal}(L'/k)$ such that $S_i|_L = \{\sigma_i\}$, then

$$\{(\sigma_K)_K \in H \mid \sigma_{L'}|_L = \sigma_L\} = \bigcup_{i=1}^n \left(\prod_{K \neq L, L'} \mathrm{Gal}(K/k) \times S_i \times \{\sigma_i\} \right)$$

is a closed set in H, and hence $\Phi(G)$ is closed in H.

This shows G is a compact Hausdorff topological space. Moreover, since $G/\mathrm{Gal}(\Omega/K) \cong \mathrm{Gal}(K/k)$ is a group, $\mathrm{Gal}(\Omega/K)$ is normal, i.e. G admits an open neighborhood basis of the identity consisting of normal subgroups. Therefore G is profinite by Theorem 5.5; in particular, we have an isomorphism of topological groups

$$\mathrm{Gal}(\Omega/k) \cong \varprojlim_K \mathrm{Gal}(K/k)$$

The infinite Galois theory states that the association $K \mapsto \mathrm{Gal}(\Omega/K)$ gives an 1-1 correspondence between the subextensions K/k of Ω/k and the closed subgroups of $\mathrm{Gal}(\Omega/k)$, under which the open subgroups of $\mathrm{Gal}(\Omega/k)$ correspond to the finite subextensions of Ω/k.

- Each open subgroup is closed.
- If K/k is a finite subextension of Ω/k, take N/k be its Galois closure in Ω. Then for $\mathrm{Gal}(\Omega/N)$ is an open unit-neighborhood of $\mathrm{Gal}(\Omega/K)$, so $\mathrm{Gal}(\Omega/K)$ is open in G.
- If K/k is any subextension of Ω/k, then $\mathrm{Gal}(\Omega/K) = \bigcap_{K'} \mathrm{Gal}(\Omega/K')$, where K' runs over all finite subextensions of K/k, so $\mathrm{Gal}(\Omega/K)$ is closed.
- Injectivity. For a subextension K/k of Ω/k, the fixed field of $\mathrm{Gal}(\Omega/K)$ in Ω is K. Clearly, $K \subseteq \Omega^{\mathrm{Gal}(\Omega/K)}$. If $x \in \Omega^{\mathrm{Gal}(\Omega/K)}$, let $\sigma \in \mathrm{Gal}(K(x)/K)$ and extend it to an element $\sigma' \in \mathrm{Gal}(\Omega/K) \subseteq G$. Then $\sigma(x) = \sigma'(x) = x$, so $\mathrm{Gal}(K(x)/K) = 1$, i.e., $x \in K$.
- Surjectivity. Let $H \leqslant G$ be a closed subgroup. Put $K = \Omega^H$. We claim $H = \mathrm{Gal}(\Omega/K)$. Clearly, $H \subseteq \mathrm{Gal}(\Omega/K)$. Conversely, let $\sigma \in \mathrm{Gal}(\Omega/K)$. If L/K be a finite Galois subextension of Ω/K, then $\sigma\,\mathrm{Gal}(\Omega/L)$ is an open neighborhood of σ in $\mathrm{Gal}(\Omega/K)$. The restriction to H of $\mathrm{Gal}(\Omega/K) \to \mathrm{Gal}(L/K)$ is surjective, since K is the fixed field of the image $H|_L$ of H, so $H|_L = \mathrm{Gal}(L/K)$ by the finite Galois theory. Now choose $\tau \in H$ such that $\tau|_L = \sigma|_L$. Then $\tau \in H \cap \sigma\,\mathrm{Gal}(\Omega/L)$, showing that σ lies in the closure of H. Since H is closed, $\sigma \in H$.
- If H is an open subgroup of G, since it is closed, it is of the form $H = \mathrm{Gal}(\Omega/K)$. Since $G/H = \mathrm{Gal}(\Omega/k)/\mathrm{Gal}(\Omega/K) \cong \mathrm{Gal}(K/k)$ is a finite group by compactness of G, K/k has finite degree.

Example 5.7. If A is a discrete abelian torsion group, then the dual group $\widehat{A} = \mathrm{Hom}(A, S^1)$ is profinite. By Proposition 4.6, $\widehat{A}$ is compact. Consider the canonical map

$$\Phi : \widehat{A} \longrightarrow \varprojlim_{\alpha} \mathrm{Hom}(A_\alpha, S^1)$$

where A_α runs over all finite subgroups of A. Since A is torsion, $A = \bigcup A_\alpha$, so Φ is a continuous bijection. Since $\widehat{A}$ is compact, this implies Φ is a homeomorphism.

Example 5.8. The rings $\mathbb{Z}/n\mathbb{Z}$, $n \in \mathbb{N}$ together with the projection $\mathbb{Z}/n\mathbb{Z} \to \mathbb{Z}/m\mathbb{Z}$, $m \mid n$ form a projective system. The projective limit

$$\widehat{\mathbb{Z}} := \varprojlim_{n \in \mathbb{N}} \mathbb{Z}/n\mathbb{Z}$$

is **the profinite completion of** $\mathbb{Z}$. For each $n \in \mathbb{N}$ by Chinese Remainder theorem we have $\mathbb{Z}/n\mathbb{Z} \cong \prod_p \mathbb{Z}/p^{\mathrm{ord}_p n}\mathbb{Z}$. Passing to inverse limit, we have

$$\widehat{\mathbb{Z}} = \varprojlim_{n \in \mathbb{N}} \mathbb{Z}/n\mathbb{Z} \cong \varprojlim_{n \in \mathbb{N}} \prod_p \mathbb{Z}/p^{\mathrm{ord}_p n}\mathbb{Z} = \prod_p \varprojlim_{n \in \mathbb{N}} \mathbb{Z}/p^{\mathrm{ord}_p n}\mathbb{Z} \cong \prod_p \mathbb{Z}_p$$

The third equality is canonical, and the last equality is given by the projection.

A profinite group G is **procyclic** if there exists $g \in G$ such that $G = \overline{\langle g \rangle}$. Such an element g is called a **topological generator** of G. Let G be a procyclic group. For instance, $\mathbb{Z}_p$ and $\widehat{\mathbb{Z}}$ are procyclic, and 1 is a topological generator of them.

- G is abelian. More generally, if G is a Hausdorff topological group and A an abelian subgroup, then $\overline{A}$ is abelian. To show that, consider the map $c : G \times G \to G$ defined by $(x, y) \mapsto xyx^{-1}y^{-1}$. Since A is abelian, $A \times A \subseteq \ker c$. Since G is Hausdorff, $\ker c = c^{-1}(e)$ is closed. Then $\overline{A} \times \overline{A} \subseteq \overline{A \times A} \subseteq \ker c$, so that $\overline{A}$ is abelian.

- The open subgroups of G have the forms nG, $n \in \mathbb{N}$. Since nG is a continuous image of G, nG is closed. The quotient G/nG is then Hausdorff, and is finite since it contains a dense finite subgroup $\{kg \mod nG \mid 0 \leqslant k < n\}$. Hence nG is open.

  Conversely, let H be an open subgroup of index n. Then $nG \subseteq H \subseteq G$, and $n = [G : H] = [G : nG] \leqslant n$, so that $H = nG$.

- G is a quotient of $\widehat{\mathbb{Z}}$. For each $n \in \mathbb{N}$, we have a surjective map $\mathbb{Z}/n\mathbb{Z} \to G/nG$. Passing to limits, we obtain $\widehat{\mathbb{Z}} \to G$.

Lemma 5.9.

1. A connected topological group G having a path-connected neighborhood of the identity is path connected.

2. If G is a connected LCA group, then the dual $\widehat{G}$ is torsion-free.

3. If G is a compact LCA group with $\widehat{G}$ torsion-free, then G is connected.

Proof.

1. Let U be a path connected unit-neighborhood of G. Since G is connected, the subgroup generated by U is the whole G, namely, $G = \bigcup_{n=1}^{\infty} U^n$. Since each U^n is path-connected and they share a common point e, G is also path-connected.

2. We prove the contrapositive. Let $\chi \in \widehat{G}$ be nontrivial with finite order, say n. Then $\chi(G) \subseteq \mu_n(\mathbb{C})$, so $\chi : G \to \mu_n(\mathbb{C})$ and hence $\ker \chi$ is a proper open and closed subgroup of G. Thus G is not connected.

3. Suppose G is not connected and let U be a proper nontrivial closed and open unit-neighborhood. Since G is compact, so is U, so we can find a symmetric open unit-neighborhood V of G such that $VU = U = UV$. Let H be the (open) subgroup generated by V; then $HU = U$, and $H \subseteq U$ particularly. Then G/H is a nontrivial finite group, and any character of G/H gives a torsion character of G, so $\widehat{G}$ is not torsion free.

□

Lemma 5.10. Let $\Lambda \leqslant G$ be a discrete subgroup of a topological group G and let $\pi : G \to G/\Lambda$ be the quotient map. Suppose $\sigma : [0,1] \to G/\Lambda$ is a path starting from 0. Then there exists a path $\tilde{\sigma} : [0,1] \to G$ starting from 0 such that $\sigma = \pi \circ \tilde{\sigma}$.

Proof. The discreteness of Λ makes the projection $\pi : G \to G/\Lambda$ a covering map, and the statement now follows from the path lifting property of the covering map.

□

Theorem 5.11. Let K be a second countable path-connected compact Hausdorff group. Then K is isomorphic to a product of countably many circle groups S^1.

Proof. Let $D = \widehat{K}$ be its dual group. By Proposition 4.6 D is discrete, and by Lemma 5.9, D is torsion-free. Also, by Corollary 4.21.1, the D forms an orthonormal basis for the Hilbert space $L^2(K)$. Since $L^2(K)$ has countable orthonormal basis by Lemma U.28, it follows that D is countable by Proposition V.15.

For an abelian group A, the rank of A is defined as $\dim_{\mathbb{Q}} A \otimes_{\mathbb{Z}} \mathbb{Q}$. We show that every finite rank subgroup of D is finitely generated. Let $F \subseteq D$ be a finite rank subgroup of rank r; since D is with discrete topology, the inclusion $F \subseteq D$ is continuous. Dualizing gives a surjection $K \to \widehat{F} =: L$, so L is compact and path-connected. Now since F is of finite rank, $F \cong \varinjlim_j F_j$, where F_j runs over all finitely generated subgroups of F with full rank r. We claim the limit stops, i.e., $F_j = F$ for some j.

Dualizing gives (since each F_j is discrete) $L \cong \varprojlim_j T_j$, where $T_j = \widehat{F_j}$ and each projection in the projective system is surjective. Since $F_j \cong \mathbb{Z}^r$, $T_j \cong \mathbb{R}^r/\mathbb{Z}^r$. Fix an index ν and an isomorphism $T_\nu \cong \mathbb{R}^r/\mathbb{Z}^r$. Then

by some isomorphism theorem for each $j \geqslant \nu$ there is a subgroup $\Lambda_j \subseteq \mathbb{Z}^r$ of full rank and an isomorphism $\psi_j : T_j \to \mathbb{R}^r/\Lambda_j$ such that diagram

$$\begin{tikzcd}
T_j \arrow[r, two heads, "\pi_\nu^j"] \arrow[d, "\psi_j"] & T_\nu \arrow[d, "\wr"] \\
\mathbb{R}^r/\Lambda_j \arrow[r, "p_j"] & \mathbb{R}^r/\mathbb{Z}^r
\end{tikzcd}$$

where p_j is the natural projection. Let $\Lambda = \bigcap_{j \geqslant \nu} \Lambda_j \subseteq \mathbb{R}^r$. The group $G := \mathbb{R}^r/\Lambda$ then injects into the $\varprojlim_j T_j = L$, and we claim that it is an isomorphism. For this let $x \in L$ and let p be a path connecting the identity element of L to x. Lemma 5.10, for each projection $\pi_j : L \to T_j \cong$ with $j \geqslant \nu$, the path $\pi_j \circ p$ lifts to a path in G, as depicted below (the commutativity of the leftmost trapezoid is what we need)

$$\begin{tikzcd}
 & & \mathbb{R}^r \arrow[d, two heads] \arrow[ddr] & \\
 & & G \arrow[d] \arrow[dl, hook] & \\
{[0,1]} \arrow[r, "p"] \arrow[uurr] & L \arrow[r, "\pi_j"] & T_j \arrow[r, equal] & \mathbb{R}^r/\Lambda_j
\end{tikzcd}$$

so the path p actually lies in G, whence $x \in G$. Thus $L \cong \mathbb{R}^r/\Lambda$, and since L is compact, Λ has full rank. So the limit stops and hence every finite rank subgroup of D is finitely generated.

Dualizing the theorem, our assertion is equivalent to saying that D is isomorphic to a direct sum of countably many cyclic groups $\mathbb{Z}$. Hence the theorem is reduced to the following purely algebraic property.

Lemma 5.12. Let D be a countable torsion free abelian group such that every finite rank subgroup of D is finitely generated. Then D is a direct sum of cyclic groups.

Proof. The $\mathbb{Q}$-module $D_\mathbb{Q} = D \otimes_\mathbb{Z} \mathbb{Q}$ is generated by D, hence it contains a basis of consisting of elements of D. Let $v_1, v_2, \ldots$ be a basis for $D_\mathbb{Q}$ with $v_j \in D$ for each j. We shall construct inductively a basis $w_1, w_2, \ldots$ of $D_\mathbb{Q}$ such that for each $n \in \mathbb{N}$

$$\mathbb{Q}w_1 \oplus \cdots \oplus \mathbb{Q}w_n = \mathbb{Q}v_1 \oplus \cdots \oplus \mathbb{Q}v_n$$

and

$$(\mathbb{Q}w_1 \oplus \cdots \oplus \mathbb{Q}w_n) \cap D = \mathbb{Z}w_1 \oplus \cdots \oplus \mathbb{Z}w_n$$

To start, let $F = \mathbb{Q}v_1 \cap D$. Then F has rank 1, so it is finitely generated, say $F = \mathbb{Z}w_1$ for some $w_1 \in D$. Now assume $w_1, \ldots, w_n$ have been constructed. The group $G = (\mathbb{Q}w_1 \oplus \cdots \oplus \mathbb{Q}w_n \oplus \mathbb{Q}v_{n+1}) \cap D$ has

rank $n+1$. The fundamental theorem for finitely generated abelian groups guarantees the existence of $u_1, \ldots, u_{n+1} \in D$ and $a_1, \ldots, a_n \in \mathbb{Z}$ such that $G = \mathbb{Z}u_1 \oplus \cdots \oplus \mathbb{Z}u_{n+1}$ and

$$\mathbb{Z}w_1 \oplus \cdots \oplus \mathbb{Z}w_n = \mathbb{Z}a_1u_1 \oplus \cdots \oplus \mathbb{Z}a_nu_n \subseteq G$$

This equality implies that for $1 \leqslant i \leqslant n$, we have

$$u_i \in (\mathbb{Q}w_1 \oplus \cdots \oplus \mathbb{Q}w_n) \cap D = \mathbb{Z}w_1 \oplus \cdots \oplus \mathbb{Z}w_n = \mathbb{Z}a_1u_1 \oplus \cdots \oplus \mathbb{Z}a_nu_n$$

and hence $a_i = \pm 1$. Thus in effect we have $\mathbb{Z}w_1 \oplus \cdots \oplus \mathbb{Z}w_n = \mathbb{Z}u_1 \oplus \cdots \oplus \mathbb{Z}u_n$, so we may pick $w_{n+1} = u_{n+1}$. This finishes the construction, and thus

$$D = \mathbb{Z}w_1 \oplus \mathbb{Z}w_2 \oplus \cdots = \bigoplus_{n \in \mathbb{N}} \mathbb{Z}w_n$$

is a direct sum of cyclic groups $\mathbb{Z}$. □

□

5.2 The Structure Theorems

Theorem 5.13 (Open mapping theorem)**.** Suppose G and H are LCH groups and G is σ-compact. Then a continuous surjective homomorphism $\phi : G \to H$ is automatically an open map.

Proof. Let 1_G and 1_H denote the identity elements in G and H, respectively. It suffices to show $\phi(U)$ is a neighborhood of 1_H whenever U is a neighborhood of 1_G. To see this, choose a compact symmetric neighborhood of 1_G such that $V^2 \subseteq U$. As G is σ-compact, write $G = \bigcup_{n \geqslant 1} K_n$ with each K_n compact in G. For each $n \in \mathbb{N}$ find a finite subset $F_n \subseteq K_n$ such that $K_n \subseteq \bigcup_{x \in F_n} xV$. Put $F = \bigcup_{n \geqslant 1} F_n$; then F is a countable subset of G such that $G = \bigcup_{x \in F} xV$. Since ϕ is surjective, it follows that $H = \bigcup_{x \in F} \phi(xV)$. Each $\phi(xV)$ is compact, so it is closed. Since H is a Baire space by Theorem T.5, it follows that there exists $x \in F$ such that $\phi(xV)$ has nonempty interior. By translation it then implies $\phi(V)$ has nonempty interior. Finally, if choose a nonempty open set W in H with $W \subseteq \phi(V)$, then $W^{-1}W$ is open in H and

$$1_H \in W^{-1}W \subseteq \phi(V)^{-1}\phi(V) \subseteq \phi(V^{-1}V) = \phi(V^2) \subseteq \phi(U)$$

□

6 Operators on Hilbert Spaces

The space $\mathcal{B}(H)$ of all bounded linear operators on a Hilbert space H is a Banach space with the operator norm, and is even a C^*-algebra as seen in Example 2.23.1. We will write

$$\sigma(T) := \sigma_{\mathcal{B}(H)}(T)$$

for the spectrum of T with respect to the C^*-algebra $\mathcal{B}(H)$, and simply call it **the spectrum of the operator T.**

6.1 Functional Calculus

Let H be a Hilbert space and T a bounded **normal operator** on H, i.e., T commutes with its adjoint T^*, or, equivalently, T is normal as an element of the C^*-algebra $\mathcal{B}(H)$. We then can apply the results of Section 2.3, which for any continuous function $f : \sigma(T) \to \mathbb{C}$ gives a unique element $f(T) \in \mathcal{B}(H)$ that commutes with T and satisfies

$$\widehat{f(T)} = f \circ \widehat{T}$$

where the has means the Gelfand transform with respect to the unital C^*-algebra generated by T. Recall by Lemma 2.31 the spectrum of a normal operator T does not depend on the C^*-algebra. The map

$$\begin{aligned} C(\sigma(T)) &\longrightarrow \mathcal{B}(H) \\ f &\longmapsto f(T) \end{aligned}$$

is the **continuous functional calculus**. We summarize some important properties in the next proposition.

Proposition 6.1. Let T be a bounded normal operator on the Hilbert space H and let $\mathcal{A} = C^*(T, 1)$ be the unital C^*-algebra generated by T.

(a) The map $f \mapsto f(T)$ is a unital isometric C^*-isomorphism from $C(\sigma(T))$ to $\mathcal{A}$ which sends the identity map $\mathrm{id}_{\sigma(T)}$ to T.

(b) Let $V \subseteq H$ be a closed subspace stable under T and T^*. Then V is stable under $\mathcal{A}$, and $f(T)|_V = f(T|_V)$.

(c) Let $V = \ker f(T)$. Then V is stable under T and T^*, and the spectrum of $T|_V$ is contained in the zero locus of f.

(d) If $f(z) = \sum_{n=0}^{\infty} a_n z^n$ is a power series that converges for $z = \|T\|$, then $f(T) = \sum_{n=0}^{\infty} a_n T^n$.

Proof.

(a) This follows from Theorem 2.32.(i).

(b) V is then stable under $\mathcal{A}$ for the linear combinations of operators of the form $(T)^k(T^*)^\ell$ are dense in $\mathcal{A}$. We then obtain a well-defined *-homomorphism

$$\begin{aligned} \Psi : \mathcal{A} &\longrightarrow \mathcal{B}(V) \\ S &\longmapsto S|_V \end{aligned}$$

The result now follows from Corollary 2.31.1.(b).

(c) The first assertion is clear, and the second is obtained by Corollary 2.31.1

$$f(\sigma(T|_V)) = \sigma(f(T|_V)) = \sigma(f(T)|_V) = 0$$

(d) This follows from Theorem 2.32.(iv).

□

There are two important classes of normal operators.

(i) **Self-adjoint operators**, i.e., $T = T^*$. By Corollary 2.31.1.(a), $\sigma(T) \subseteq \mathbb{R}$.

(ii) **Unitary operators**, i.e., $UU^* = U^*U = 1$. Note that a normal operator $U \in \mathcal{B}(H)$ is unitary if and only if $\sigma(U) \subseteq \mathbb{T}$. This follows from Theorem 2.32, for $U^*U = 1$ if and only if $\overline{\mathrm{id}_{\sigma(U)}}\,\mathrm{id}_{\sigma(U)} = 1$, i.e., $\sigma(U) \subseteq \mathbb{T}$.

Recall the **Schwartz space** $\mathcal{S}(\mathbb{R})$ consists of all smooth functions $f : \mathbb{R} \to \mathbb{C}$ such that for any two integers $m, n \geqslant 0$ the function $x^n f^{(m)}(x)$ is bounded, i.e., those smooth functions f with all its derivatives **rapidly decreasing**. An element in $\mathcal{S}(\mathbb{R})$ is called an **Schwartz function**.

For $f \in \mathcal{S}(\mathbb{R})$, the **Fourier inversion formula** says that

$$f(x) = \int_{\mathbb{R}} \hat{f}(y) e^{2\pi i x y} dy$$

where $\hat{f}(y) = \displaystyle\int_{\mathbb{R}} f(x) e^{-2\pi i x y} dx$ is the **Fourier transform**.

Proposition 6.2. Let H be a Hilbert space and $T \in \mathcal{B}(H)$ self-adjoint. Then for every $f \in \mathcal{S}(\mathbb{R})$,

$$f(T) = \int_{\mathbb{R}} \hat{f}(y) e^{2\pi i y T} dy$$

where $e^{2\pi i y T}$ is a unitary operator defined by the continuous functional calculus, and the integral should be viewed as a Bochner integral with value in $\mathcal{B}(H)$.

Proof. Put $g(x) = e^{2\pi i y x}$ and let

$$\begin{aligned} \Phi : C(\sigma(T)) &\longrightarrow \mathcal{B}(H) \\ f &\longmapsto f(T) \end{aligned}$$

be the continuous functional calculus of T. Then $e^{2\pi i y T} = g(T) = \Phi(g)$. To see $e^{2\pi i y T}$ is unitary, we compute

$$(e^{2\pi i y T})^* = \Phi(g)^* = \Phi(\overline{g}) = \Phi(g^{-1}) = \Phi(g)^{-1} = (e^{2\pi i y T})^{-1}$$

The second equality holds for Φ is *-invariant, by Theorem 2.32, and the third equality holds for $g(x) \in \mathbb{T}$. The function

$$\begin{aligned} \mathbb{R} &\longrightarrow \mathcal{B}(H) \\ y &\longmapsto \hat{f}(y) e^{2\pi i y T} \end{aligned}$$

then satisfies the condition Proposition U.38.1.2., so the integral $\int_{\mathbb{R}} \hat{f}(y) e^{2\pi i y T} dy$ exists. To see the equality, note that the Fourier inversion formula implies that

$$f|_{\sigma(T)} = \int_{\mathbb{R}} \hat{f}(y) e^{2\pi i y \operatorname{id}_{\sigma(T)}} dy$$

so by the continuity of Φ (and the very definition of Lebesgue integral) we have

$$f(T) = \Phi(f|_{\sigma(T)}) = \Phi\left(\int_{\mathbb{R}} \hat{f}(y) e^{2\pi i y \operatorname{id}_{\sigma(T)}} dy\right) = \int_{\mathbb{R}} \hat{f}(y) \Phi(e^{2\pi i y \operatorname{id}_{\sigma(T)}}) dy = \int_{\mathbb{R}} \hat{f}(y) e^{2\pi i y T} dy$$

□

6.1.1 Positive Operators

Definition. A self-adjoint operator $T \in \mathcal{B}(H)$ is called **positive** (or positive semi-definite), if

$$\langle Tv, v \rangle \geqslant 0$$

for all $v \in H$.

Theorem 6.3. Let $T \in \mathcal{B}(H)$ be self-adjoint. TFAE:

(a) T is positive;

(b) The spectrum $\sigma(T) \subseteq [0, \infty)$, i.e., T is positive in the sense of C^*-algebras;

(c) There exists an operator $R \in \mathcal{B}(H)$ such that $T = R^* R$;

(d) There exists a unique positive operator S with $T = S^2$. In this case we write $S = \sqrt{T}$.

Proof. That (d) $\Rightarrow$ (c) $\Rightarrow$ (a) is trivial, and (b) $\Rightarrow$ (d) follows from Proposition 3.2, so it remains to show (a) $\Rightarrow$ (b).

For (a) $\Rightarrow$ (b), WLOG we assume $\|T\| = 1$. Since T is self-adjoint, $\sigma(T) \subseteq \mathbb{R} \cap \overline{B_1(0)} = [-1, 1]$. We contend that $T_\mu := T + \mu 1$ is invertible for every $\mu > 0$, which implies that T has no negative spectral value. By assumption, we have

$$\|T_\mu v\| \, \|v\| \geqslant \langle T_\mu v, v\rangle = \langle Tv, v\rangle + \mu\langle v, v\rangle \geqslant \mu \, \|v\|^2$$

which implies that $\|T_\mu n\| \geqslant \mu \, \|v\|$ for every $v \in H$. It follows that T_μ is injective. (Note that the proof ends here if H is finite dimensional.) Since T_μ is self-adjoint, we also get

$$(T_\mu(H))^\perp = \ker T_\mu = 0$$

for $w \in (T_\mu(H))^\perp \Leftrightarrow 0 = \langle T_\mu v, w\rangle = \langle v, T_\mu w\rangle$ for every $v \in H \Leftrightarrow T_\mu w = 0$. By Lemma V.11 we then see $\overline{T_\mu(H)} = H$. Thus for each $w \in H$ we can find $(v_n)_n \subseteq H$ with $Tv_n \to w$. Since

$$\|v_n - v_m\| \leqslant \frac{1}{\mu} \, \|T_\mu v_n - T_\mu v_m\|$$

for $m, n \in \mathbb{N}$, it follows that $(v_n)_n$ is a Cauchy sequence in H and hence converges to, say, $v \in H$. Thus $T_\mu v = w$, showing that T_μ is surjective. By Open Mapping Theorem, T_μ^{-1} is continuous, so T_μ is invertible in $\mathcal{B}(H)$.

□

Definition. For $T \in \mathcal{B}(H)$, define the operator $|T| := \sqrt{T^*T}$. This is well-defined by the Theorem above.

Proposition 6.4 (Polar decomposition)**.** Let $T \in \mathcal{B}(H)$.

(i) The norm of $|T|v$ coincides with $\|Tv\|$ for every $v \in H$.

(ii) There is an isometry $U : \overline{\operatorname{Im}|T|} \to \overline{\operatorname{Im} T}$ such that $T = U|T|$.

(iii) The decomposition in (ii) is unique, in the sense that if $T = U'P$ with P positive and $U' : \overline{\operatorname{Im} P} \to H$ isometric, then $U' = U$ and $P = |T|$.

Proof. For $v \in H$,

$$\||T|v\|^2 = \langle |T|v, |T|v\rangle = \langle |T|^2 v, v\rangle = \langle T^*Tv, v\rangle = \langle Tv, Tv\rangle = \|Tv\|^2$$

so (i) holds. For $v \in H$, define

$$\begin{aligned} U : \operatorname{Im}|T| &\longrightarrow \operatorname{Im} T \\ |T|v &\longmapsto Tv \end{aligned}$$

This is well-defined, for if $|T|v = |T|w$, then $0 = \||T|(v - w)\| = \|T(v - w)\|$ so that $Tv = Tw$. By (i) this is an isometry. Also, U extends to a bounded operator $U : \overline{\operatorname{Im}|T|} \to \overline{\operatorname{Im} T}$.

- Suppose $(v_n)_n \subseteq H$ is such that $|T|v_n$ is Cauchy. Then $U(|T|v_n) = Tv_n$ is also Cauchy by (i), so $Tv_n \to v$ for some $v \in \overline{\operatorname{Im} T}$. For $w = \lim_{n\to\infty} |T|v_n \in \overline{\operatorname{Im}|T|}$, define $Uw := v$. This is again well-define by (i), and $\|w\| = \|v\|$ by continuity of norm.

This proves (ii). For (iii), extend U to a bounded operator on H by setting $U \equiv 0$ on $\overline{\operatorname{Im}|T|}^{\perp}$ and do likewise for U'. Then U^*U is the orthogonal projection to $\overline{\operatorname{Im}|T|}$ and U'^*U' is the orthogonal projection to $\overline{\operatorname{Im} P}$. Note

$$|T| = \sqrt{T^*T} = \sqrt{(U'P)^*U'P} = \sqrt{P^*U'^*U'P} = \sqrt{P^*P} = \sqrt{P^2} = P$$

where the last equality is the uniqueness part of Theorem 6.3.(d). This also proves $U = U'$. □

6.1.2 Schur's Lemma

Lemma 6.5. Let H be a Hilbert space and $T \in \mathcal{B}(H)$ be normal. If $\sigma(T) = \{\lambda\}$ is a singleton, then $T = \lambda \operatorname{id}_H$.

Proof. $\sigma(T) = \{\lambda\} \Leftrightarrow \operatorname{id}_{\sigma(T)} = \lambda 1|_{\sigma(T)}$, and therefore $T = \operatorname{id}_{\sigma(T)}(T) = \lambda 1|_{\sigma(T)}(T) = \lambda \operatorname{id}_H$ ($1|_{\sigma(T)}$ being the identity element in $C(\sigma(T))$).

If T is self-adjoint, this can be proved by Lemma 2.26, for then we have $\|T - \lambda \operatorname{id}_H\| = r(T - \lambda \operatorname{id}_H) = 0$. □

Theorem 6.6 (Schur)**.** Suppose $A \subseteq \mathcal{B}(H)$ is a self-adjoint set of bounded operators (namely, $S \in A$ implies $S^* \in A$). TFAE:

(a) A is **topological irreducible**, i.e., if $0 \neq L \subseteq H$ is any A-invariant *closed* subspace, then $L = H$.

(b) If $T \in \mathcal{B}(H)$ commutes with all $S \in A$, then $T = \mu \operatorname{id}$ for some $\mu \in \mathbb{C}$.

Proof. Assume (b) holds and that $0 \neq L \subseteq H$ is any A-invariant closed subspace. Then the orthogonal complement $L^{\perp}$ is also A-invariant, for with $v \in L$, $u \in L^{\perp}$ and $S \in A$, we have

$$\langle v, Su\rangle = \langle \underbrace{S^*v}_{\in L}, u\rangle = 0$$

So the orthogonal projection $P_L : H \to L$ commutes with A, and thus P_L is multiple of the identity. But $P_L|_L = \operatorname{id}_L$, this implies $P_L = \operatorname{id}_H$ and $L = H$.

Assume (a) holds, and let $T \in \mathcal{B}(H)$ commute with A. Then also T^* commutes with A since A is self-adjoint. By writing $T = \frac{1}{2}(T + T^*) + \frac{1}{2i}(T - T^*)$, we may assume $T \neq 0$ is self-adjoint. We will show that $\sigma(T)$ is a singleton. Note that every operator commuting with T also commutes with $f(T)$ for every $f \in C(\sigma(T))$. Now suppose there exist $x \neq y \in \sigma(T)$. Then there are two functions $f, g \in C(\sigma(T))$ with $f(x) \neq 0 \neq g(y)$ and $fg = 0$. Then $f(T) \neq 0 \neq g(T)$ and $f(T)g(T) = fg(T) = 0$. Since $g(T)$ commutes with A, the space $L := \overline{g(T)H}$ is a nonzero A-invariant subspace of H. By (a) we get $L = H$, but then $0 \neq f(T)H = f(T)\overline{g(T)H} \subseteq \overline{f(T)g(T)H} = 0$, a contradiction. □

6.2 Compact Operators

Definition. Let $T \in \mathcal{B}(H)$.

1. T is a **finite rank operator** if $\dim_{\mathbb{C}} \operatorname{Im} T < \infty$.
2. T is a **compact operator** if T sends bounded subsets to relatively compact subsets.

- It follows from definition that if T is compact and S is bounded, then both TS and ST is compact.
- T is compact if and only if for a given bounded sequence $v_j \in H$, the sequence Tv_j has a convergent subsequence.

Proposition 6.7. For $T \in \mathcal{B}(H)$, TFAE:

(a) T is compact.

(b) For every orthonormal sequence e_j, the sequence Te_j has a convergent subsequence.

(c) For every orthonormal sequence e_j, the sequence $Te_j \to 0$.

(d) There exists a sequence F_n of finite rank operators such that $\|T - F_n\|_{\mathrm{op}} \to 0$ as $n \to \infty$.

Proof. (a) $\Rightarrow$ (b) is trivial. (d) $\Rightarrow$ (a) follows from a diagonal argument. For (b) $\Rightarrow$ (c), we only need to show every subsequence of Te_j admits a subsequence converging to 0. Passing to subsequence, we assume $Te_j \to v \in H$. As said, we must show $v = 0$.

Lemma 6.8. Every orthonormal sequence in an inner product space converges weakly to 0.

Proof. Let v_n be an orthonormal sequence. For $y \in H$, by Bessel's inequality one has $\sum_{n=1}^{\infty} \langle v_n, y\rangle^2 \leqslant \|y\|^2 < \infty$. In particular, $\langle v_n, y\rangle \to 0 = \langle 0, y\rangle$ as $n \to \infty$, as claimed. $\square$

By this lemma, we then have $\langle T^*v, e_j\rangle \to 0$ as $j \to \infty$. Now for $\varepsilon > 0$, we can find $N \gg 0$ such that $\|T^*Te_j - T^*v\| < \varepsilon$ and $|\langle T^*v, e_j\rangle| < \varepsilon$ whenever $j \geqslant N$. Then $\|Te_j\|^2 = \langle T^*Te_j, e_j\rangle \leqslant 2\varepsilon$.

For (c) $\Rightarrow$ (d), we will construct an adequate orthonormal sequence e_n, as follows. Choose $e_1 \in H$ such that $\|Te_1\| \geqslant \frac{1}{2}\|T\|$. Assume that $e_1, \ldots, e_n$ $(n \geqslant 2)$ are constructed, and let $U_n := \operatorname{span}_{\mathbb{C}}\{e_1, \ldots, e_n\}$. Let $P_n : H \to U_n$ be the orthogonal projection. If $\|T - TP_n\|_{\mathrm{op}} = 0$, then $T = TP_n$ is of finite rank. Otherwise, choose

$$e_{n+1} \in U_n^{\perp} \cap \{x \in H \mid \|x\| = 1\}$$

with $\|Te_{n+1}\| \geqslant \frac{1}{2}\|T(\mathrm{id} - P_n)\|_{\mathrm{op}}$. This is possible. For $\|T - TP_n\|_{\mathrm{op}} > \frac{1}{2}\|T - TP_n\|$, so we can find $\|v\| = 1$ such that $\|(T - TP_n)v\| \geqslant \frac{1}{2}\|T - TP_n\|_{\mathrm{op}} > 0$. If we put $w' = (\mathrm{id} - P_n)v \in U_n^{\perp}$, then

$$\|(T - TP_n)v\| = \|T(\mathrm{id} - P_n)v\| = \|Tw'\| \leqslant \|T(w'/\|w'\|)\|\,\|w'\| \leqslant \|T(w'/\|w'\|)\|$$

Thus $e_{n+1} := \dfrac{w'}{\|w'\|}$ works. Now, by (c), we have

$$\|T - TP_n\| = \|T(\mathrm{id} - P_n)\| \leqslant 2\,\|Te_{n+1}\| \to 0 \text{ as } n \to \infty$$

so that $\|T - TP_n\|_{\mathrm{op}} \to 0$ as $n \to \infty$.

□

Corollary 6.8.1. For a bounded operator T, TFAE:

(a) T is compact.

(b) T^*T is compact.

(c) TT^* is compact.

(d) T^* is compact.

Proof. Suppose (a) and let v_n is a bounded sequence. Then we can find a convergent subsequence of Tv_n, and by continuity of T^*, T^*Tv_n also has a convergent subsequence. Also, by continuity of T^* we know T^*v_n is again bounded, so by compactness of T we can find a convergent subsequence of TT^*v_n; this shows (b) and (c). Assume (b) and let e_n be an orthonormal sequence in H. Then $T^*Te_n \to 0$ by Proposition 6.7.(c), and by Cauchy-Schwarz, we have

$$\|Te_n\|^2 = \langle T^*Te_n, e_n\rangle \leqslant \|T^*Te_n\|\,\|e_n\| = \|T^*Te_n\| \to 0$$

so $Te_n \to 0$. By Proposition 6.7.(c) again we deduce T is compact.

By symmetry we see (d) $\Rightarrow$ (b),(c) and (c) $\Rightarrow$(d), and hence the proof is finished. □

Proposition 6.9. A bounded operator T on a Hilbert space H is compact if and only if the image of the closed unit ball is compact.

Proof. Suppose $T(\overline{B_1(0)})$ is compact. Then for each $r > 0$, $T(\overline{B_r(0)}) = T(r\overline{B_1(0)}) = rT(\overline{B_1(0)})$ is still compact, so T is compact by its very definition. Now suppose T is compact. By definition we have $T(\overline{B_1(0)})$ is relatively compact, so it is enough to show it is closed. Let $(v_n)_n$ be a sequence in $\overline{B_1(0)}$ with $Tv_n \to w$ for some $w \in H$. By Banach Alaoglu, we can find $\alpha \in H^\vee$ with $\|\alpha\| \leqslant 1$ such that, by passing to subsequence, $\langle v_n, x\rangle \to \alpha(x)$ as $n \to \infty$ for all $x \in H$. By Riesz's Representation theorem, we can find $v \in H$ such that $\alpha = \langle v, \cdot\rangle$ with $\|v\| = \|\alpha\| \leqslant 1$. Finally, take $\varepsilon > 0$ and pick $n \gg 0$ so that $\|Tv_n - w\| < \varepsilon$ and $|\langle v - v_n, T^*(Tv - w)\rangle| < \varepsilon$. Then

$$\begin{aligned}\|Tv - w\|^2 &= \langle v, T^*(Tv - w)\rangle - \langle v_n, T^*(Tv - w)\rangle + \langle Tv_n, Tv - w\rangle - \langle w, Tv - w\rangle \\ &\quad \langle v - v_n, T^*(Tv - w)\rangle + \langle Tv_n - w, Tv - w\rangle \\ &\quad < \varepsilon + \|Tv_n - w\|\,\|Tv - w\| < \varepsilon(1 + \|Tv - w\|)\end{aligned}$$

Since ε is arbitrary, we see $\|Tv - w\| = 0$, i.e., $Tv = w$. □

6.2.1 Spectral Theorem for Compact Normal Operators

Theorem 6.10. Let T be a compact normal operator on a Hilbert space. Then there exists a sequence λ_n of non-zero complex numbers, which is either finite or tends to zero, such that one has an orthogonal decomposition

$$H = \ker T \oplus \overline{\bigoplus_n \operatorname{Eig}(T, \lambda_n)}$$

Each eigenspace $\operatorname{Eig}(T, \lambda_n) := \{v \in H \mid Tv = \lambda_n v\}$ is finite dimensional, and the eigenspaces are pairwise orthogonal.

Proof. The proof will go as follows.

1° Show a compact normal operator $T \neq 0$ has a nonzero eigenvalue by reducing to the case T being self-adjoint.

2° Show H is a direct sum of eigenspaces of T.

3° Show every eigenspace corresponding to a nonzero eigenvalue is finite dimensional, and the eigenvalues do not accumulate away from zero.

We first show that we can reduce to the case T being self-adjoint. Write

$$T = \frac{1}{2}(T + T^*) - \frac{i}{2}(iT - iT^*) := T_1 + iT_2$$

as a linear combination of commuting compact self-adjoint operators. If $T_2 = 0$, we are done. Otherwise, T_2 has a nonzero real eigenvalue ν. The corresponding eigenspace is T_1-invariant (for the T_i commute), so the restriction of T_1 on it also compact self-adjoint, and hence has an eigenvalue $\mu \in \mathbb{R}$. Then $\lambda := \mu + i\nu$ is a nonzero eigenvalue of T.

Now we have to show that a compact self-adjoint operator $T \neq 0$ has a nonzero eigenvalue.

Lemma 6.11. For a bounded self-adjoint operator T on a Hilbert space H, we have

$$\|T\| = \sup\{|\langle Tv, v\rangle| \mid \|v\| = 1\}$$

Proof. Let C be the RHS. By Cauchy-Schwarz, we have $C \leqslant \|T\|$. On the other hand, for $v, w \in H$ with $\|v\|, \|w\| \leqslant 1$, one has

$$\begin{aligned}
C &\geqslant \frac{1}{2}C(\|v\|^2 + \|w\|^2) = \frac{1}{4}C(\|v+w\|^2 + \|v-w\|^2) \\
&\geqslant \frac{1}{4}|\langle T(v+w), v+w\rangle - \langle T(v-w), v-w\rangle| \\
&= \frac{1}{2}|\langle Tv, w\rangle + \langle Tw, v\rangle| = \frac{1}{2}|\langle Tv, w\rangle + \langle w, Tv\rangle| \\
&= |\operatorname{Re}\langle Tv, w\rangle|
\end{aligned}$$

Replacing v with θv for some $\theta \in S^1$ we get $C \geqslant |\langle Tv, w\rangle|$ for all $\|v\|, \|w\| \leqslant 1$, and so $\|T\| \leqslant C$. $\square$

We proceed to show a compact self-adjoint operator $T \neq 0$ has a nonzero eigenvalue. In fact, we will show either $\|T\|$ or $-\|T\|$ is an eigenvalue. By the lemma, there is a sequence $v_n \in H$ with $\|v_n\| = 1$ such that $\langle Tv_n, v_n\rangle \to \pm\|T\|$. Passing to a subsequence and replacing T with $-T$ if necessary, we assume $\langle Tv_n, v_n\rangle \to \|T\|$. Since T is compact, passing to subsequence again we can assume $Tv_n \to u \in H$. Then $\|u\| \leqslant \|T\|$, and we get

$$0 \leqslant \|Tv_n, \|T\|\, v_n\|^2 = \|Tv_n\|^2 - 2\,\|T\|\,\langle Tv_n, v_n\rangle + \|T\|^2\,\|v_n\|^2 \to \|u\|^2 - \|T\|^2 \leqslant 0$$

which implies $\|Tv_n - \|T\|\, v_n\| \to 0$. Thus $v := \lim\limits_{n\to\infty} v_n = \dfrac{1}{\|T\|}u$ exists and $Tv = \lim_n Tv_n = u = \|T\|\, v$. This finishes 1°.

Let $0 \neq U \subseteq V$ be the closure of the (direct) sum of all eigenspaces of T corresponding to nonzero eigenvalues. Since every eigenvector for T is also an eigenvector for its adjoint T^*, U is stable under T and T^*, and hence the orthogonal complement $U^\perp$ is stable under T and T^* as well. The operator T restricts to a compact normal operator on $U^\perp$. Since $T|_{U^\perp}$ cannot have nonzero eigenvalue, $T|_{U^\perp} \equiv 0$ and $U^\perp = \ker T$. This shows 2°, and it is clear that the eigenspaces are pairwise orthogonal.

Need more explanation!! It remains to show 3°. For $\lambda \in \mathbb{C}^\times$, put $E_\lambda := \mathrm{Eig}(T, \lambda) = \ker(T - \lambda\,\mathrm{id})$. We show $\dim_{\mathbb{C}} E_\lambda < \infty$. Suppose otherwise; then there exists an orthonormal sequence f_n in E_λ. Then $\|f_n - f_m\| = \sqrt{2}$ for $n \neq m$, and $\|Tf_n - Tf_m\| = \sqrt{2}|\lambda| \neq 0$; in particular, this shows no subsequence of Tf_n is Cauchy, a contradiction to the compactness of T. Let $\mu > 0$ and consider the space $V_\mu := \sum\{E_\lambda \mid |\lambda| \geqslant \mu,\ E_\lambda \neq 0\}$. We claim $\dim_{\mathbb{C}} V_\mu < \infty$, and this will imply no spectral values of T can accumulate away from zero. Suppose otherwise. Then there exists an infinite sequence λ_n of distinct eigenvalues of T; let v_n be a corresponding eigenvector with norm 1. Then v_n is an orthonormal sequence, and

$$\|Tv_n - Tv_m\|^2 = \|\lambda_n v_n - \lambda_m v_m\|^2 = |\lambda_n|^2 + |\lambda_m|^2 \geqslant 2\mu^2 > 0$$

again a contradiction to the compactness of T.

□

6.2.2 Singular Values

Lemma 6.12. For a compact operator T, $\sqrt{T^*T}$ is also compact.

Proof. By Corollary 6.8.1.(b) we know T^*T is compact. To see $\sqrt{T^*T}$ is compact, let e_n be an orthonormal sequence. Then

$$\left\|\sqrt{T^*T}e_n\right\|^2 = \langle T^*Te_n, e_n\rangle \leqslant \|T^*Te_n\|\,\|e_n\| = \|T^*Te_n\| \to 0$$

by Proposition 6.7.(c). □

Definition. Let T be a compact operator. Then by the lemma we see $|T|$ is a positive compact operator. Let $s_j(T)$ be the family of nonzero eigenvalues of $|T|$ repeated with multiplicities and such that $s_{j+1}(T) \leqslant s_j(T)$ for all j. The $s_j(T)$ are called the **singular values** of T.

Proposition 6.13. Let T be a compact operator.

1. We have $s_1(T) = \|T\|$ and

$$s_{j+1}(T) = \inf_{v_1,\dots,v_j \in H} \sup\{\|Tw\| \mid w \perp v_1, \dots, v_j,\ \|w\| = 1\}$$

   where the vectors $v_1, \dots, v_j$ are unit eigenvectors for the eigenvalues $s_1(T), \dots, s_j(T)$, respectively.

2. For any bounded operator S on H one has $s_j(ST) \leqslant \|S\|\, s_j(T)$.

Proof. To be filled

□

6.3 Hilbert-Schmidt and Trace Class

6.3.1 Hilbert-Schmidt Operators

Let $T \in \mathcal{B}(H)$ and let (e_j) be an orthonormal basis of H. The **Hilbert-Schmidt norm** $\|T\|_{\mathrm{HS}}$ is defined by

$$\|T\|_{\mathrm{HS}}^2 := \sum_j \langle Te_j, Te_j \rangle = \sum_j \|Te_j\|^2 \in [0, +\infty]$$

Lemma 6.14. $\|T\|_{\mathrm{HS}}$ does not depend on the choice of the orthonormal basis of H, and $\|T\|_{\mathrm{HS}} = \|T^*\|_{\mathrm{HS}}$.

Proof. Let ϕ_α be another orthonormal basis of H. Then by Parseval's identity, we have

$$\begin{aligned}\sum_j \|Te_j\|^2 &= \sum_j \sum_\alpha \langle Te_j, \phi_\alpha \rangle \langle \phi_\alpha, Te_j \rangle = \sum_j \sum_\alpha \langle e_j, T^*\phi_\alpha \rangle \langle T^*\phi_\alpha, e_j \rangle \\ &= \sum_\alpha \sum_j \langle e_j, T^*\phi_\alpha \rangle \langle T^*\phi_\alpha, e_j \rangle = \sum_\alpha \|T^*\phi_\alpha\|^2\end{aligned}$$

The change of order of summation is justified since the general terms are non-negative. Applying this equality to $\{e_j\} = \{\phi_\alpha\}$ first and then to T^* instead of T, we obtain

$$\sum_j \|Te_j\|^2 = \sum_j \|T^*e_j\|^2 = \sum_\alpha \|T\phi_\alpha\|^2$$

□

Definition. A bounded operator T is called a **Hilbert-Schmidt operator** if $\|T\|_{\mathrm{HS}} < \infty$.

Lemma 6.15. Let $S, T \in \mathcal{B}(H)$. Then

(a) $\|ST\|_{\mathrm{HS}} \leqslant \|S\|_{\mathrm{op}}\, \|T\|_{\mathrm{HS}}$.

(b) $\|ST\|_{\mathrm{HS}} \leqslant \|S\|_{\mathrm{HS}}\, \|T\|_{\mathrm{op}}$.

(c) $\|T\|_{\mathrm{op}} \leqslant \|T\|_{\mathrm{HS}}$.

(d) For every unitary operator U, $\|UT\|_{\mathrm{HS}} = \|TU\|_{\mathrm{HS}} = \|T\|_{\mathrm{HS}}$.

Proof. Let $(e_j)_j$ be an orthonormal basis.

(a) $\|ST\|_{\mathrm{HS}}^2 = \sum_j \|STe_n\|^2 \leqslant \sum_j \|S\|_{\mathrm{op}}^2 \|Te_j\|^2 = \|S\|_{\mathrm{op}}^2 \|T\|_{\mathrm{HS}}^2$.

(b) $\|ST\|_{\mathrm{HS}} = \|(ST)^*\|_{\mathrm{HS}} = \|T^*S^*\|_{\mathrm{HS}} \leqslant \|T^*\|_{\mathrm{op}} \|S^*\|_{\mathrm{HS}} = \|T\|_{\mathrm{op}} \|S\|_{\mathrm{HS}}$.

(c) Let $v \in H$ with $\|v\| = 1$. Then we can find an orthonormal basis $(\phi_i)_i$ with $\phi_1 = v$. Then

$$\|Tv\|^2 = \|T\phi_1\|^2 \leqslant \sum_i \|T\phi_i\|^2 = \|T\|_{\mathrm{HS}}^2$$

(d) This is clear for $(Ue_j)_j$ is also an orthonormal basis when U is unitary.

□

Proposition 6.16. For an orthonormal basis e_j and Hilbert Schmidt operators B, C, the sum

$$(B, C) := \sum_j \langle Be_j, Ce_j \rangle$$

is (absolutely) convergent. Further, (B, C) is independent of the choice of orthonormal basis e_j.

Proof. The first is proved by applying Cauchy-Schwarz twice:

$$\sum_j |\langle Be_j, Ce_j \rangle| \leqslant \sum_j \|Be_j\| \|Ce_j\| \leqslant \left(\sum_j \|Be_j\|^2 \right)^{\frac{1}{2}} \left(\sum_j \|Ce_j\|^2 \right)^{\frac{1}{2}} = \|B\|_{\mathrm{HS}} \|C\|_{\mathrm{HS}} < \infty$$

The second assertion follows from the independence result for $\|\cdot\|_{\mathrm{HS}}$ and the polarization: $(i = \sqrt{-1})$

$$(B, C) = \frac{1}{4} \sum_{k=1}^{4} i^k \left\| B + i^k C \right\|_{\mathrm{HS}}^2$$

□

Theorem 6.17. Let H be a Hilbert space and let $\mathcal{HS}$ denote the space of Hilbert-Schmidt operators on H.

(i) $\mathcal{HS}$ with the pairing defined in Proposition 6.16 is a Hilbert space.

(ii) The map

$$\begin{aligned} H \hat{\otimes} H^\vee &\longrightarrow \mathcal{HS} \\ v \otimes \alpha &\longmapsto [T_{v,\alpha} : w \mapsto \alpha(w)v] \end{aligned}$$

is well-defined and is a Hilbert space isomorphism.

Proof. Let $(e_j)_{j\in I}$ be an orthonormal basis of H.

(i) The pairing is clear linear in the first argument, and sesquilinear in the second argument. It is positive definite, for $(B,B) = \sum_j \|Be_j\|^2 \geqslant 0$, and $(B,B) = 0$ if and only if $\|Be_j\|^2 = 0$ for all j, if and only if $B = 0$.

It remains to show $\mathcal{HS}$ is complete with respect to the Hilbert-Schmidt norm. Consider the map

$$\begin{aligned} \mathcal{HS} &\longrightarrow \ell^2(I\times I) \\ T &\longmapsto M_T : (i,j) \mapsto \langle Te_i, e_j\rangle \end{aligned}$$

The map is well-defined and is norm-preserving, for

$$\|T\|_{\mathrm{HS}}^2 = \sum_i \|Te_i\|^2 = \sum_{i,j} |\langle Te_i, e_j\rangle|^2 = \sum_{i,j} |M_T(i,j)|^2$$

This map admits an inverse, i.e., for $f \in \ell^2(I\times I)$, define $T_f : H \to H$ by setting $T_f(e_i) = \sum_j f(i,j)e_j$ and extending linearity. This is well-defined for $f(i,j)$ is square-summable. It is clear that $f \mapsto T_f$ is inverse to $T \mapsto M_T$, and thus $\mathcal{HS}$ is isomorphic to $\ell^2(I\times I)$ as normed spaces. Since $\ell^2(I\times I)$ is known to be complete, $\mathcal{HS}$ is also complete.

(ii) By Rieze's representation theorem, we can find $v_\alpha \in H$ such that $\alpha(w) = \langle w, v_\alpha\rangle$ for all $w \in H$. Then

$$\|T_{v,\alpha}\|_{\mathrm{HS}}^2 = \sum_j \|\alpha_j(e_j)v\|^2 = \|v\|^2 \sum_j |\langle e_j, v_\alpha\rangle|^2 = \|v\|^2 \|v_\alpha\|^2 = \|v\|^2 \|\alpha\|^2 = \|v\otimes\alpha\|^2 < \infty$$

By polarization we see the described map is a Hilbert space homomorphism. For the surjectivity, let $T \in \mathcal{HS}$. For each $w \in H$, write $w = \sum_i \langle w, e_i\rangle e_i = \sum_i \hat{e}_i(w)e_i$. Then $T(w) = \sum_i \hat{e}_i(w)T(e_i)$ so that T is mapped by $\sum_i T(e_i)\otimes\hat{e}_i \in H\widehat{\otimes}H^\vee$.

□

For a measure space $(X,\mathcal{A},\mu)$, consider the Hilbert space $L^2(X)$. We suppose either μ is σ-finite or X is LCH and μ is outer Radon so that the Fubini's theorem holds with respect to the product measure $\mu\otimes\mu$ on $L^2(X\times X)$. An function k in $L^2(X\times X)$ is called an **L^2-kernel** on X.

Proposition 6.18. Suppose $k(x,y)$ is an L^2-kernel on X. For $\phi \in L^2(X)$ define

$$K\phi(x) := \int_X k(x,y)\phi(y)d\mu(y)$$

Then this integral exists a.e. in x, and $K\phi \in L^2(X)$. Moreover, K extends to a Hilbert-Schmidt operator $K : L^2(X) \to L^2(X)$ with

$$\|K\|_{\mathrm{HS}}^2 = \int_X\int_X |k(x,y)|^2 d\mu(x)d\mu(y)$$

Proof. To see the integral exists a.e. in $x \in X$, let $\psi \in L^2(X)$ be any element. Then $[(x,y) \mapsto \psi(x)\phi(y)] \in L^2(X \times X)$, and therefore the function $[(x,y) \mapsto k(x,y)\psi(x)\phi(y)] \in L^1(X \times X)$. By Fubini, it follows that

$$\int_X k(x,y)\psi(x)\phi(y)d\mu(y) = \psi(x)\int_X k(x,y)\phi(y)d\mu(y)$$

exists a.e. in $x \in X$.

Lemma 6.19. Let $k \in L^2(X \times X)$. Then there exists a σ-finite subset A of X such that for all $x \notin A$, we have $k(x,y) = 0$ a.e. in y.

Proof. Use Fubini to write $\infty > \int_X \left(\int_Y |k(x,y)|^2 d\mu(y)\right) d\mu(x)$. For $m \geqslant 1$ consider

$$A_m = \left\{x \in X \mid \int_Y |k(x,y)|^2 d\mu(y) \geqslant \frac{1}{m}\right\}.$$

Each A_m has finite measure, and if we put $A = \bigcup_{m \geqslant 1} A_m$, then for $x \notin A$, we have $\int_Y |k(x,y)|^2 d\mu(y) = 0$, so $k(x,y) = 0$ a.e. in y. □

Let A be the σ-finite set as in the lemma, and write $A = \bigcup_{m \geqslant 1} A_m$ with A_m increasing. Take $\psi_m = \mathbf{1}_{A_m}$. Then as $m \to \infty$

$$\psi_m(x)\int_X k(x,y)\phi(y)d\mu(y) \to \mathbf{1}_A(x)\int_X k(x,y)\phi(y)d\mu(y) = \int_X k(x,y)\phi(y)dy$$

proving that the last integral exists a.e. in x.

Now use Cauchy-Schwarz to estimate

$$\begin{aligned}
\|K\phi\|^2 &= \int_X |K\phi(x)|^2 dx = \int_X \left|\int_X k(x,y)\phi(y)d\mu(y)\right|^2 d\mu(x) \\
&\leqslant \int_X \left(\int_X |k(x,y)|^2 dy\right)\left(\int_X |\phi(y)|^2 dy\right) dx \\
&= \int_X \int_X |k(x,y)|^2 dx dy \, \|\phi\|^2
\end{aligned}$$

showing that $K : \phi \mapsto K\phi$ is a bounded operator on $L^2(X)$. Last, let $(e_j)_j$ be an orthonormal basis of $L^2(X)$. Then

$$\begin{aligned}
\|K\|_{\mathrm{HS}}^2 &= \sum_j \|Ke_j\|^2 = \sum_j \int_X Ke_j(x)\overline{Ke_j(x)}dx \\
&= \sum_j \int_X \left(\int_X k(x,y)e_j(y)dy \overline{\int_X k(x,y)e_j(y)dy}\right) dx \\
&= \sum_j \int_X \langle k(x,\cdot), e_j\rangle\langle e_j, k(x,\cdot)\rangle dx = \int_X \sum_j \langle k(x,\cdot), e_j\rangle\langle e_j, k(x,\cdot)\rangle dx \\
&= \int_X \langle k(x,\cdot), k(x,\cdot)\rangle dx = \int_X \int_X |k(x,y)|^2 dx dy
\end{aligned}$$

where the second last equality is Parseval's identity. □

Proposition 6.20. An operator T is Hilbert-Schmidt if and only if it is compact and its singular values satisfy $\sum_n s_n(T)^2 < \infty$. In this case, one has $\|T\|_{\mathrm{HS}}^2 = \sum_n s_n(T)^2$.

Proof. For a bounded operator T, we have

$$\|T\|_{\mathrm{HS}}^2 = \sum_j \langle Te_j, Te_j\rangle = \sum_j \langle T^*Te_j, e_j\rangle = \sum_j \langle |T|^2 e_j, e_j\rangle = \sum_j \langle |T|e_j, |T|e_j\rangle = \||T|\|_{\mathrm{HS}}^2$$

so that T is Hilbert-Schmidt if and only if $|T|$ is.

Let T be Hilbert-Schmidt. To see T is compact, we use Proposition 6.7.(c). Let $\{e_j\}_{j=1}^{\infty}$ be an orthonormal sequence and let A be an orthonormal basis for T containing the e_j. Then

$$\infty > \|T\|_{\mathrm{HS}}^2 = \sum_{h\in H} \|Th\|^2 \geqslant \sum_{j=1}^{\infty} \|Te_j\|^2$$

implies that $\lim_{j\to\infty} Te_j = 0$.

Now assume T is compact; in particular, $|T|$ is compact. By using an orthonormal basis consisting eigenvectors of $|T|$, which exists by Spectral Theory, to compute $\||T|\|_{\mathrm{HS}}$, we can see $|T|$ is Hilbert-Schmidt if and only if $\sum_j s_j(T)^2 < \infty$, and the sum equals $\||T|\|_{\mathrm{HS}}^2$. The last assertion follows from the computation in the first paragraph. □

6.3.2 Trace Class Operators

Definition. A bounded operator T is called a **trace class operator** if it is compact and the **trace norm**

$$\|T\|_{\mathrm{tr}} := \sum_j s_j(T)$$

is finite, where the $s_j(T)$ are singular values of T.

- By Cauchy-Schwarz and Proposition 6.20, every trace class operator is Hilbert-Schmidt.

Lemma 6.21.

1. Let T be a trace class operator and S a bounded operator. Then $\|TS\|_{\mathrm{tr}}, \|ST\|_{\mathrm{tr}} \leqslant \|S\|_{\mathrm{op}} \|T\|_{\mathrm{tr}}$.

2. If T is a compact operator, then

$$\|T\|_{\mathrm{tr}} = \sup_{(e_i),(h_i)} \sum_i |\langle Te_i, h_i\rangle|$$

where (e_i) and (h_i) run over all orthonormal bases.

Proof.

(i) This follows from Proposition 6.13 and the easy facts that $\|T^*\| = \|T\|$ and $\|T\|_{\mathrm{tr}} = \|T^*\|_{\mathrm{tr}}$.

(ii) For simplicity, write $s_j = s_j(T)$. Let f_j be an orthonormal sequence consisting of eigenvectors of $|T|$. Then for each $v \in H$, we have

$$|T|v = \sum_j s_j \langle v, f_j \rangle f_j$$

Use polar decomposition to obtain isometric $U : \operatorname{Im}|T| \to \operatorname{Im} T$ such that $T = U|T|$. If we put $g_j := Uf_j$, then for $v \in H$,

$$Tv = U\left(\sum_j s_j \langle v, f_j, \rangle f_j\right) = \sum_j s_j \langle v, f_j \rangle g_j$$

Now for any two orthonormal bases e_i, h_j, by Cauchy-Schwarz and Parseval's identity, we have

$$\begin{aligned}
\sum_i |\langle Te_i, h_i \rangle| &= \sum_i \left| \sum_j s_j \langle e_i, f_j \rangle \langle g_j, h_i \rangle \right| \\
&\leqslant \sum_j s_j \sum_i |\langle e_i, f_j \rangle \langle g_j, h_i \rangle| \\
&\leqslant \sum_j s_j \left(\sum_i |\langle e_i, f_j \rangle|^2 \right)^{\frac{1}{2}} \left(\sum_i |\langle g_j, h_i \rangle|^2 \right)^{\frac{1}{2}} \\
&= \sum_j s_j \, \|f_j\| \, \|g_j\| = \sum_j s_j
\end{aligned}$$

For the reversed inequality, take e_j to be any orthonormal basis that prolongs the orthonormal sequence f_j and h_j to be any orthonormal basis prolonging g_j. Then

$$\sum_i |\langle Te_i, h_i \rangle| = \sum_i \left| \sum_j s_j \langle e_i, f_j \rangle \langle g_j, h_i \rangle \right| = \sum_i \left| \sum_j s_j \delta_{ij} \right| = \sum_i \sum_j s_j \delta_{ij} = \sum_j s_j$$

□

Theorem 6.22. For a trace class operator T, the **trace**

$$\operatorname{tr} T := \sum_j \langle Te_j, e_j \rangle$$

does not depend on the choice of an orthonormal basis (e_j). If T is trace class and normal, we have

$$\operatorname{tr} T = \sum_n \lambda_n \dim \operatorname{Eig}(T, \lambda_n)$$

where the sum runs over the sequence of non-zero eigenvalues (λ_n) of T, and the sum converges absolutely.

Proof. Write $T = U|T|$ for the polar decomposition. By Spectral Theory we see that the image of $S_2 := \sqrt{|T|}$ equals that of $|T|$, and therefore we can define $S_1 := U\sqrt{|T|}$. It is clear that S_1, S_2, S_1^* are Hilbert-Schmidt, and $T = S_1 S_2$. Therefore

$$\sum_i \langle Te_i, e_i \rangle = \sum_i \langle S_2 e_i, S_1^* e_i \rangle \qquad ((\spadesuit))$$

The sum converges (absolutely) and is independent of choice of orthonormal basis e_j by Proposition 6.16. Now applying this identity first to $(\phi_\alpha) = (e_j)$ and then to $(S_2, S_1^*) \mapsto (S_1, S_2^*)$, we see

$$\sum_i \langle S_2 e_i, S_1^* e_i \rangle = \sum_i \langle S_1 e_i, S_2^* e_i \rangle = \sum_\alpha \langle S_2 \phi_\alpha, S_1^* \phi_\alpha \rangle$$

When T is normal, use a basis consisting of eigenvectors to compute trace. □

Theorem 6.23. Let H be a Hilbert space. We consider the following spaces.

- $\mathcal{F}$: the space of bounded operators of finite rank.
- $\mathcal{T}$: the space of trace class operators.
- $\mathcal{HS}$: the space of Hilbert-Schmidt operators.
- $\mathcal{K}$: the space of compact operators.

Further, we write $\mathcal{HS}^2 := \operatorname{span}\{ST \mid S, T \in \mathcal{HS}\}$. Then

(a) $\mathcal{F}$, $\mathcal{T}$, $\mathcal{HS}$, $\mathcal{K}$ are all *-ideals of the C^*-algebra $\mathcal{B}(H)$.

(b) $\mathcal{K}$ is the norm closure of $\mathcal{F}$.

(c) One has the inclusions

$$\mathcal{F} \subseteq \mathcal{T} = \mathcal{HS}^2 \subseteq \mathcal{HS} \subseteq \mathcal{K}$$

where the inclusions are strict if $\dim H = \infty$.

Proof.

1. These follow from Lemma 6.21, Lemma 6.15 and Corollary 6.8.1.
2. This is Proposition 6.7.
3. The first inclusion is clear. $\mathcal{T} \subseteq \mathcal{HS}^2$ is shown in the proof of Theorem 6.22. If $S \in \mathcal{HS}$, then $S^*S \in \mathcal{T}$ by Proposition 6.20 and the fact $s_j(S^*S) = s_j(S)^2$. For general, let $S, T \in \mathcal{HS}$. Write $S = S_1 + iS_2$ and $T = T_1 + iT_2$ with S_i, T_j self-adjoint; being linear combinations of S and T, the S_i and T_j are Hilbert-Schmidt as well. So we may assume S, T are self-adjoint. But $ST = \frac{1}{2}((S+T)^*(S+T) - S^*S - T^*T)$, so $ST \in \mathcal{T}$ follows. The third inclusion is Lemme 6.15.(a), and the fourth is Proposition 6.20.

   It remains to show the inclusions are strict when $\dim H = \infty$. Let $\{e_n\}_{n\in\mathbb{N}}$ be an orthonormal sequence in H and define $T_i : H \to H$ $(i = 1, 2)$ by setting $T_i(e_n) = \frac{1}{n^{i/2}} e_n$, $T_i|_{\overline{\operatorname{span}_{\mathbb{C}}\{e_n\}_n}^{\perp}} \equiv 0$ and extending by linearity. Then the T_i are bounded, compact and self-adjoint. Then one can easily see that $T_2 \in \mathcal{HS} \backslash \mathcal{T}$ and $T_1 \in \mathcal{K} \backslash \mathcal{HS}$.

□

Proposition 6.24.

(a) For a trace class operator T, we have $\operatorname{tr} T^* = \overline{\operatorname{tr} T}$.

(b) For Hilbert-Schmidt S, T, we have $\operatorname{tr}(ST) = \operatorname{tr}(TS)$.

Proof.

(a) $\operatorname{tr} T^* = \sum_j \langle T^* e_j, e_j \rangle = \sum_j \overline{\langle T e_j, e_j \rangle} = \overline{\operatorname{tr} T}$.

(b) We use the notation in Proposition 6.16. If T is unitary, then

$$\operatorname{tr}(ST) = \operatorname{tr}(T^*(TS)T) = (TST, T) = (TS, \mathrm{id}) = \operatorname{tr} TS$$

for the pairing is independent of the choice of orthonormal basis. For general follows at once, since the desired identity is linear in T, and every bounded operator is a linear combination of unitary operators by Proposition 3.3.(iv).

□

6.4 Spectral Theorem for Normal Operators

6.4.1 Resolution of the Identity

Definition. Let $(X, \mathcal{A})$ be a measurable space and H a Hilbert space. A **resolution of the identity** on $\mathcal{A}$ is a map $E : \mathcal{A} \to \mathcal{B}(H)$ with the following properties:

(a) $E(\varnothing) = 0$, $E(\Omega) = \mathrm{id}_H$.

(b) Each $E(\omega)$ is an orthogonal projection.

(c) $E(\omega' \cap \omega'') = E(\omega')E(\omega'')$.

(d) If $\omega' \cap \omega'' = \varnothing$, then $E(\omega' \cup \omega'') = E(\omega') + E(\omega'')$.

(e) For every $v, w \in H$, the set function $E_{v,w} : \mathcal{A} \to \mathbb{C}$ defined by

$$E_{v,w}(\omega) = \langle E(\omega)v, w \rangle$$

is a complex measure on $(X, \mathcal{A})$.

When $(X, \mathcal{A})$ is the Borel σ-algebra on a locally compact Hausdorff space, we further require each $E_{v,w}$ to be regular.

We derive some formal properties of a resolution of the identity.

(i)

(ii) Since each $E(\omega)$ is an orthogonal projection, for each $v \in H$, we have

$$E_{v,v}(\omega) = \langle E(\omega)v, v\rangle = \|E(\omega)v\|^2 .$$

Thus each $E_{v,v}$ is a positive measure on $(X, \mathcal{A})$ with total variation $\|E_{v,v}\| = E_{v,v}(X) = \|v\|^2$.

(iii) The complex measure $E_{v,w}$ has total variation less than $\|v\|\,\|w\|$. Indeed, let $(\omega_n)_{n=1}^N$ be a finite measurable partition of X. Write $|E_{v,w}(\omega_n)| = \alpha_n E_{v,w}(\omega_n)$ for some $|\alpha_n| = 1$. Then

$$\sum_{n=1}^N |E_{v,w}(\omega_n)| = \sum_{n=1}^N \alpha_n E_{v,w}(\omega_n) = \left\langle \sum_{n=1}^N \alpha_n E(\omega_n)v, w \right\rangle \leqslant \left\| \sum_{n=1}^N \alpha_n E(\omega_n)v \right\| \|w\|$$

The $E(\omega_n)v$ are pairwise orthogonal, so

$$\left\| \sum_{n=1}^N E(\omega_n)\alpha_n v \right\|^2 = \sum_{n=1}^N \|E(\omega_n)v\|^2 = \left\| \sum_{n=1}^N E(\omega_n)v \right\|^2 = \|E(X)v\|^2 = \|v\|^2$$

This proves the claim.

(iv) If $(\omega_n)_n$ is a sequence such that $E(\omega_n) = 0$ for all $n \geqslant 1$, then $E\left(\bigcup_{n\geqslant 1} \omega_n\right) = 0$. Indeed, if we put ω to be the union, then for each $x \in H$, we have

$$E_{x,x}(\omega) = \sum_{n=1}^\infty E_{x,x}(\omega_n) = \sum_{n=1}^\infty \langle E(\omega_n)x, x\rangle = 0.$$

Since $\|E(\omega)x\|^2 = E_{x,x}(\omega)$, we deduce $E(\omega) = 0$.

(v) By (c), any two of the projections $E(\omega)$ commute with each other.

(vi) By (a), (c), if $\omega' \cap \omega''$, then the ranges of $E(\omega')$ and $E(\omega'')$ are orthogonal.

(vii) By (b), we have $\overline{E_{v,w}(\omega)} = E_{w,v}(\omega)$.

(viii) By (d), the set function E is finitely additive.

It is natural to ask whether E is countably additive in the norm topology. Let $(\omega_n)_n$ be a sequence of disjoint sets in $\mathcal{A}$. Since $\|E(\omega_n)\|$ is either 0 or 1, partial sums of the series $\sum_{n=1}^\infty E(\omega_n)$ cannot form a Cauchy sequence unless $E(\omega_n) = 0$ for all but finitely many n.

However, E is in fact countably additive in strong operator topology. Let $(\omega_n)_n$ be as above and $v \in H$. By (e), we have

$$\sum_{n=1}^\infty \langle E(\omega_n)v, w\rangle = \langle E(\omega)v, w\rangle$$

for all $w \in H$, where ω denotes the union of the ω_n. Note that the vectors $E(\omega_n)v$ are pairwise orthogonal by (vi), so Proposition V.16 implies

$$\sum_{n=1}^{\infty} E(\omega_n)v = E(\omega)v$$

In other words, $\sum_{n=1}^{\infty} E(\omega_n) = E(\omega)$ in strong operator topology. In a fancier term, for each $v \in H$, the map $\omega \mapsto E(\omega)v$ is a countably additive H-valued measure on $(X, \mathcal{A})$.

Recall that $B = B(X, \mathcal{A}, \mathbb{C})$ be the set of all bounded $\mathcal{A}$-measurable functions on X; this is a unital commutative C^*-algebra with respect to the sup norm $\|\cdot\|_{\sup}$. Let $E : \mathcal{A} \to \mathcal{B}(H)$ be a resolution of the identity. Let $\{D_n\}_n$ be a countable collection of open discs which forms a basis for the topology of $\mathbb{C}$. For $f \in B$, let $V = V_f$ be the union of those D_n with $E(f^{-1}(D_n)) = 0$; then $E(f^{-1}(V)) = 0$ (by (iv)) and V is the largest open subspace of $\mathbb{C}$ with this property (use (c)).

For $f \in B$, the **essential image** $\operatorname{ess.im}(f)$ of f is the complement of V; this is the smallest closed subset of $\mathbb{C}$ that contains $f(p)$ for almost every $p \in X$ (with respect to the H-valued measure E). Explicitly,

$$\operatorname{ess.im}(f) = \{z \in \mathbb{C} \mid E(\{x \in X \mid |f(x) - z| < \varepsilon\}) > 0 \text{ for all } \varepsilon > 0\}$$

We say f is **essentially bounded** if its essential image is bounded in $\mathbb{C}$ (hence compact). In this case, we define the **essential supremum**

$$\|f\|_\infty := \sup_{\lambda \in \operatorname{ess.im}(f)} |\lambda| = \inf_{E(N)=0} \sup_{x \notin N} |f(x)|$$

Let $N := \{f \in B \mid \|f\|_\infty = 0\}$; then N is a *-ideal, and a similar argument to the proof of Theorem U.27 and (iv) shows N is closed. Then B/N is a Banach algebra, and we denote it by $L^\infty(E)$.

Lemma 6.25.

1. Every $f \in B$ is a uniform limit of simple functions.
2. The quotient norm for a coset $f + N$ is $\|f\|_\infty$.
3. The spectrum $\sigma_{L^\infty(E)}(f + N)$ is the essential image of f.

Proof.

1. This is due to boundedness.
2. Let $f_0 \in N$ and $N = f_0^{-1}(V_{f_0})$. Then $\|f + f_0\|_{\sup} \geqslant \sup_{x \notin N} |f(x) + f_0(x)|$. Since $f_0(x)$ lies in the essential image of f_0 for $x \notin N$, $\|f_0\|_\infty = 0$ implies $f_0(x) = 0$ for all $x \notin N$, whence,

$$\|f + f_0\|_{\sup} \geqslant \sup_{x \notin N} |f(x)| \geqslant \|f\|_\infty$$

The reverse estimate is given by

$$\|f\|_\infty = \inf_{E(N)=0} \sup_{x \notin N} |f(x)| = \inf_{E(N)=0} \left\| f + \|f\|_{X \setminus N} \mathbf{1}_N \right\|_{\sup} \geqslant \inf_{f_0 \in N} \|f + f_0\|_{\sup} = \|f + N\|_{\sup}.$$

3.

$\square$

The use of this lemma will be implicit. Say a subalgebra A of $\mathcal{B}(H)$ **normal** if it is commutative and self-adjoint.

Theorem 6.26. If E is a resolution of the identity as above, then there exists an isometric *-homomorphism

$$\Psi : L^\infty(E) \longrightarrow \mathcal{B}(H)$$

onto a closed unital normal subalgebra A of $\mathcal{B}(H)$ given by the formula

$$\langle \Psi(f)v, w\rangle = \int_X f dE_{v,w}$$

for all $v, w \in H$ and $f \in B$. Symbolically we will write

$$\Psi(f) = \int_X f dE.$$

Moreover,

1. $\|\Psi(f)v\|^2 = \displaystyle\int_X |f|^2 dE_{v,v}$.
2. $C_{\mathcal{B}(H)}(\{E(\omega) \mid \omega \in \mathcal{A}\}) = C_{\mathcal{B}(H)}(\{\Psi(f) \mid f \in L^\infty(E)\})$.

In particular, we have

$$\|f\|_\infty^2 = \sup_{\|v\|\leqslant 1} \int_X |f|^2 dE_{v,v}$$

Proof. For v, w and $f \in B$, define $B_f(v, w) := \displaystyle\int_X f dE_{v,w}$ (integration against a complex measure). Then by (iii) $|B_f(v, w)| \leqslant \|f\|_\infty \|E_{v,w}\| \leqslant \|f\|_\infty \|v\| \|w\|$, so by Riesz's Representation theorem we find a unique operator $\Psi(f) \in \mathcal{B}(H)$ such that

$$\langle \Psi(f)v, w\rangle = \int_X f dE_{v,w}$$

for all $v, w \in H$. The estimate and uniqueness at once show that $\Psi : B \to \mathcal{B}(H)$ is a continuous linear map.

If s is a simple function, say $s = \sum_{i=1}^n \alpha_i \mathbf{1}_{\omega_i}$, where $\{\omega_1, \ldots, \omega_n\}$ is a measurable partition of X, then

$$B_s(v, w) = \sum_{i=1}^n \alpha_i E_{v,w}(\omega_i) = \left\langle \sum_{i=1}^n \alpha_i E(\omega_i)v, w \right\rangle$$

The uniqueness shows $\Psi(s) = \sum_{i=1}^{n} \alpha_i E(\omega_i)$. Now

$$\Psi(s)^* = \sum_{i=1}^{n} \overline{\alpha_i} E(\omega_i) = \Psi(\overline{s}).$$

Since every function in B is a uniform limit of simple functions, this shows $\Psi(f)^* = \Psi(\overline{f})$. If $t = \sum_{j=1}^{m} \beta_j \mathbf{1}_{\omega'_j}$ is another simple function, where $\{\omega'_j\}$ is a measure partition of X, then

$$st = \sum_{i,j} \alpha_i \beta_j \mathbf{1}_{\omega_i \cap \omega'_j}$$

On the other hand, by (c)

$$\Psi(s)\Psi(t) = \sum_{i,j} \alpha_i \beta_j E(\omega_i) E(\omega'_j) = \sum_{i,j} \alpha_i \beta_j E(\omega_i \cap \omega'_j).$$

This show $\Psi(st) = \Psi(s)\Psi(t)$, thus $\Psi(fg) = \Psi(f)\Psi(g)$ for all $f, g \in B$. This proves Ψ is a continuous *-homomorphism. Also,

$$\|\Psi(s)v\|^2 = \langle \Psi(s)^* \Psi(s)v, v\rangle = \langle \Psi(|s|^2)v, v\rangle = \int_X |s|^2 dE_{v,v},$$

so 1. holds and $\|\Psi(s)v\| \leqslant \|s\|_\infty \|v\|$ by (ii). On the other hand, if v lies in the image of $E(\omega_i)$ for some i, then by (vi)

$$\Psi(s)v = \alpha_i E(\omega_i)v = \alpha_i v.$$

If we choose i so that $|\alpha_i| = \|s\|_\infty$, then together with a previous estimate we obtain $\|s\|_\infty = \|\Psi(s)\|$. This shows Ψ is isometric. Finally, Q commutes with every $E(\omega)$ if and only Q commutes with every $\Psi(s)$ with s simple. A limit process shows 2. holds. □

6.4.2 Spectral theorem

In the following let H be a Hilbert space. Recall from Lemma 2.31 that if A is a closed unital *-subalgebra of $\mathcal{B}(H)$, then $\sigma_A(T) = \sigma_{\mathcal{B}(H)}(T)$ for all $T \in A$.

Theorem 6.27. Let A be closed unital normal subalgebra of $\mathcal{B}(H)$.

(I) There exists a unique resolution of the identity E on the Borel subsets of the structure space $\Delta = \Delta_A$ which satisfies

$$T = \int_\Delta \widehat{T} dE$$

for every $T \in A$, where $\widehat{T}$ is the Gelfand transform.

(II) The inverse of the Gelfand transform extends to an isometric *-homomorphism $\Phi : L^\infty(E) \to \mathcal{B}(H)$ onto the closed subalgebra B of $\mathcal{B}(H)$ containing A, given by

$$\Phi f = \int_\Delta f dE.$$

(III) B is the norm closure of the set of all finite linear combinations of the projections $E(\omega)$.

(IV) If $\omega \subseteq \Delta$ is nonempty and open, then $E(\omega) \neq 0$.

(V) $C_{\mathcal{B}(H)}A = C_{\mathcal{B}(H)}(\{E(\omega) \mid \omega \subseteq \Delta : \text{Borel}\})$

Proof. Recall by Gelfand-Naimark Theorem, Δ is compact and the Gelfand transform

$$\begin{aligned} A &\longrightarrow C(\Delta) \\ T &\longmapsto \widehat{T} \end{aligned}$$

is an isometric *-isomorphism. We first prove the uniqueness part in 1. Suppose E is a resolution of identity on Δ satisfying

$$\langle Tv, w\rangle = \int_\Delta \widehat{T} dE_{v,w}$$

for all $v, w \in H$, $T \in A$. Since $\widehat{T}$ runs over $C(\Delta)$ and $E_{v,w}$ is complex regular, so the uniqueness part of Riesz's representation theorem shows that each $E_{v,w}$ is uniquely determined by the above identity. Since $E_{v,w}(\omega) = \langle E(\omega)v, w\rangle$, each projection $E(\omega)$ is also uniquely determined.

We now consider the existence. For any $v, w \in H$, by Gelfand-Naimark Theorem, the functional

$$\begin{aligned} C(\Delta) &\longrightarrow \mathbb{C} \\ \widehat{T} &\longmapsto \langle Tv, w\rangle \end{aligned}$$

is a bounded linear function on $C(\Delta)$ of norm $\leqslant \|v\| \, \|w\|$. So by Riesz's representation theorem we can find a unique regular complex Borel measure $\mu_{v,w}$ on Δ such that

$$\langle Tv, w\rangle = \int_\Delta \widehat{T} d\mu_{v,w}$$

holds for all $v, w \in H$ and $T \in A$. The right hand side still defines a bounded sesquilinear functional on H if $\widehat{T}$ is replaced by bounded Borel functions f. To each such f it corresponds to a unique operator $\Phi f \in \mathcal{B}(H)$ (by Riesz's representation theorem) such that

$$\langle \Phi f v, w\rangle = \int_\Delta f d\mu_{v,w}.$$

Comparing, we see $\Phi\widehat{T} = T$, so it extends the inverse of Gelfand transform. The uniqueness part shows that Φ is linear. Note that T is self-adjoint if and only if $\widehat{T}$ is real-valued, and for such T, we have

$$\int_\Delta \widehat{T} d\mu_{v,w} = \langle Tv, w\rangle = \langle v, Tw\rangle = \overline{\langle Tw, v\rangle} = \overline{\int_\Delta \widehat{T} d\mu_{w,v}}$$

implying $\mu_{v,w} = \overline{\mu_{w,v}}$, thus

$$\langle \Phi\overline{f}v, w\rangle = \int_\Delta \overline{f} d\mu_{v,w} = \overline{\int_\Delta f d\mu_{w,v}} = \overline{\langle \Phi f w, v\rangle} = \langle v, \Phi f w\rangle$$

for all $v, w \in H$. This shows $\Phi\overline{f} = (\Phi f)^*$. Next we prove $\Phi(fg) = (\Phi f)(\Phi g)$ for all bounded Borel f, g on Δ. For $S, T \in A$, we have $\widehat{ST} = \widehat{S}\widehat{T}$, so

$$\int_\Delta \widehat{S}\widehat{T} d\mu_{v,w} = \langle STv, w\rangle = \int_\Delta \widehat{S} d\mu_{Tv,w}.$$

This holds for all $\widehat{S} \in C(\Delta)$, so it remains true if $\widehat{S}$ is replaced by any bounded Borel f. Thus

$$\int_\Delta f\widehat{T} d\mu_{v,w} = \int_\Delta f d\mu_{Tv,w} = \langle \Phi f(Tv), w\rangle = \langle Tv, u\rangle = \int_\Delta \widehat{T} d_{v,u}$$

where we put $u = (\Phi f)^* w$. Again, this remains true if $\widehat{T}$ is replaced by any g, giving

$$\langle \Phi(fg)v, w\rangle = \int_\Delta fg d\mu_{v,w} = \int_\Delta g d\mu_{v,u} = \langle \Phi g v, u\rangle = \langle \Phi g v, (\Phi f)^* w\rangle = \langle \Phi(f)\Phi(g)v, w\rangle.$$

Now we define E: for $\omega \subseteq \Delta$ Borel, define $E(\omega) := \Phi(\mathbf{1}_\omega)$.

(c) Since Φ is multiplicative, this shows $E(\omega' \cap \omega'') = E(\omega')E(\omega'')$.

(b) Since Φf is self-adjoint when f is real, this shows $E(\omega)$ is self-adjoint, and by (c) (with $\omega' = \omega'' = \omega$) we see $E(\omega)$ is a projection.

(a) $E(\varnothing) = \Phi(0) = 0$, and $E(\Delta) = \Phi(\mathrm{id}_\Delta) = \mathrm{id}_H$.

(d) The integral representation $\langle \Phi f v, w\rangle = \int_\Delta f d\mu_{x,y}$ show that E is finitely additive.

(e) For $v, w \in H$, we have

$$E_{v,w}(\omega) = \alpha E(\omega)v, w\rangle = \int_\Delta \mathbf{1}_\omega d\mu_{v,w} = \mu_{v,w}(\omega)$$

so $E_{v,w}$ is a complex measure, and thus $\Phi f = \int_\Delta f dE$.

To finish (II) we must show Φ is an isometry, but this follows from Theorem 6.26. (III) is clear for every function in $L^\infty(E)$ is a uniform limit of simple functions. For (IV), let ω be open with $E(\omega) = 0$. Let $T \in A$ with $\operatorname{supp} \widehat{T} \subseteq \omega$. Then by (I)

$$T = \int_\Delta \widehat{T} dE = \int_\omega \widehat{T} dE = 0$$

whence $\widehat{T} = 0$. From Urysohn's Lemma we see $\omega = \varnothing$. This proves (IV). To prove (V), choose $S \in \mathcal{B}(H)$, $v, w \in H$, and put $u = S^* w$. For any $T \in A$ and Borel $\omega \subseteq \Delta$, we have

$$\begin{aligned}
\langle STv, w\rangle = \langle Tv, u\rangle &= \int_\Delta \widehat{T} dE_{v,u} \\
\langle Tsv, w\rangle &= \int_\Delta \widehat{T} dE_{Sv,w} \\
\langle SE(\omega)v, w\rangle = \langle E(\omega)v, u\rangle &= E_{v,u}(\omega) \\
\langle E(\omega)Sv, w\rangle &= E_{Sv,w}(\omega).
\end{aligned}$$

Thus $S \in C_{\mathcal{B}(H)}(A)$ if and only if $E_{Sv,w} = E_{v,u}$, if and only if $SE(\omega) = E(\omega)S$ for all Borel ω. □

Now we specialize to a single normal operator T.

Theorem 6.28. If $T \in \mathcal{B}(H)$ is normal, then there exists a unique resolution of the identity E on the Borel subsets of $\sigma(T) = \sigma_{\mathcal{B}(H)}(T)$ which satisfies

$$T = \int_{\sigma(T)} \lambda dE(\lambda).$$

Furthermore, every projection $E(\omega)$ commutes with every $S \in \mathcal{B}(H)$ that commutes with T.

We refer to this E as the **spectral decomposition of** T. Sometimes it is convenient to extend E to all Borel sets in $\mathbb{C}$, by setting $E(\omega) = 0$ if $\omega \cap \sigma(T) = \varnothing$.

Proof. Let $A = C^*(\mathrm{id}_H, T, T^*) \subseteq \mathcal{B}(H)$ be the smallest unital commutative C^*-algebra generated by T in $\mathcal{B}(H)$. Now Theorem 6.27 applies. In the proof of continuous functional calculus we see Δ_A can be identified with $\sigma(T)$ via the Gelfand transform $\widehat{T}$; explicitly, we have $\widehat{T}(\lambda) = \lambda$ for all $\lambda \in \sigma(T)$. The proves the existence of E.

On the other hand, if E exists so that $T = \int_{\sigma(T)} \lambda dE(\lambda)$ holds, then Theorem 6.26 implies

$$p(T, T^*) = \int_{\sigma(T)} p(\lambda, \overline{\lambda}) dE(\lambda)$$

holds for all $p \in \mathbb{C}[X, Y]$. The uniqueness then follows from Stone-Weierstrass (and argue as in the first paragraph of the proof of Theorem 6.26).

For the last statement, if $ST = TS$, then $ST^* = T^*S$ by Corollary 2.31.3; hence S commutes with A. By Theorem 6.27.(V), this shows $SE(\omega) = E(\omega)S$ for all Borel $\omega \subseteq \sigma(T)$. □

This theorem is clearly an extension of the continuous functional calculus of normal operators. Let T be a normal operator and A is closed normal subalgebra generated by T. Let E be the spectral decomposition of T. Then Theorem 6.27 gives an isometric *-homomorphism

$$\begin{aligned} \Psi : L^\infty(E) &\longrightarrow \mathcal{B}(H) \\ f &\longmapsto \Psi f \end{aligned}$$

with

$$\Psi f = \int_{\sigma(T)} f dE.$$

Let us write $f(T) := \Psi f$ for any bounded Borel function f. Then $f \mapsto f(T)$ satisfies

1. $\overline{f}(T) = f(T)^*$,
2. $\|f(T)\| = \|f\|_\infty \leqslant \sup\limits_{\lambda \in \sigma(T)} |f(\lambda)|$ with equality when f is continuous.

Thus $f \mapsto f(T)$ gives an isometric *-homomorphism from $C(\sigma(T))$ to $\mathcal{B}(H)$ (with image A) satisfying

$$\|f(T)v\|^2 = \int_{\sigma(T)} |f|^2 dE_{x,x}.$$

The image of Borel functional calculus is the closure of finite linear combinations of projections $E(\omega)$ with $\omega \subseteq \sigma(T)$ Borel.

7 Representation

Let V be a Banach space. Put

$$\mathrm{GL}_{\mathrm{cts}}(V) := \{T \in \mathrm{Aut}_{\mathbb{C}}(V) \mid T \text{ is continuous}\}$$

It follows from the Open Mapping theorem that $\mathrm{GL}_{\mathrm{cts}}(V)$ is a group.

Definition. Let G be a topological group.

1. A **representation** of G on a Banach space V is a group homomorphism $\pi : G \to \mathrm{GL}_{\mathrm{cts}}(V)$ such that the action map

$$\begin{aligned} G \times V &\longrightarrow V \\ (g, v) &\longmapsto \pi(g)v \end{aligned}$$

   is continuous.

2. For two representations (π, V_π), (η, V_η) of G, a continuous linear operator $T : V_\pi \to V_\eta$ is a **G-homomorphism**, or is called **G-equivariant/intertwining**, if $T\pi(g) = \eta(g)T$ for all $g \in G$. The set of all G-intertwining operators from V_π to V_η is denoted by $\mathrm{Hom}_G(V_\pi, V_\eta)$, or $\mathrm{Hom}_G(\pi, \eta)$.

Lemma 7.1. Let G be a topological group, V be a Banach space and $\pi : G \to \mathrm{GL}_{\mathrm{cts}}(V)$ be a group homomorphism. Then π is a representation if and only if

(a) the map $g \mapsto \pi(g)v$ is continuous at $g = 1$ for every $v \in V$, and

(b) the map $g \mapsto \|\pi(g)\|_{\mathrm{op}}$ is bounded in a neighborhood of the unit of G.

Proof. Suppose π is a representation. Then (a) is obvious. For (b), let $R > 0$ be any real number. Then by continuity, we can find $r > 0$ and a unit neighborhood U of G such that $\pi(U)B_r(0) \subseteq B_R(0)$. Then for $g \in U$,

$$\|\pi(g)\|_{\mathrm{op}} = \sup_{\|v\|=1} \|\pi(g)v\| = \frac{2}{r} \sup_{\|v\|=1} \left\|\pi(g)\frac{rv}{2}\right\| \leqslant \frac{2R}{r}$$

Conversely, we write

$$\begin{aligned} \|\pi(g)v - \pi(h)w\| &= \left\|\pi(h)(\pi(h^{-1}g)v - w)\right\| \leqslant \|\pi(h)\| \\ &\leqslant \|\pi(h)\| \left\|\pi(h^{-1}g)v - \pi(h^{-1}g)w + \pi(h^{-1}g)w - w\right\| \\ &\leqslant \|\pi(h)\| \left(\left\|\pi(h^{-1}g)\right\| \|v - w\| + \left\|\pi(h^{-1}g)w - w\right\|\right) \end{aligned}$$

By assumption, as $(g, v) \to (h, w)$, the last term tends to 0, hence the continuity. $\square$

Definition. A representation (π, V_π) is called a **subrepresentation** of a representation (η, V_η) if

- V_π is a closed subspace of V_π, and

- $\eta|_{V_\pi} = \pi$.

In other words, a subrepresentation of (η, V_η) is a closed subspace invariant under η.

Definition. Let H be a Hilbert space. A representation π of G on H is called a **unitary representation** if

$$\langle \pi(g)v, \pi(g)w \rangle = \langle v, w \rangle$$

for all $g \in G$ and $v, w \in H$; in other words, all $\pi(g)$, $g \in G$ are unitary operators on H.

- For a Hilbert space H, denote by $\mathcal{U}(H) \subseteq \mathrm{GL}_{\mathrm{cts}}(H)$ the subgroup of unitary operators.
- By Lemma 7.1, a homomorphism $\pi : G \to \mathcal{U}(H)$ is a representation of G if and only if $g \mapsto \pi(g)v$ is continuous at $g = 1$ for all $v \in V$.

Lemma 7.2. A representation π of the group G on a Hilbert space V is unitary if and only if $\pi(g^{-1}) = \pi(g)^*$ holds for every $g \in G$.

Proof. For $g \in G$, $\pi(g)$ is unitary if and only if $\pi(g)$ is invertible and $\pi(g)^* = \pi(g^{-1})$. □

Example 7.3.

1. For a continuous group homomorphism $\chi : G \to \mathbb{C}^\times$, we can define a representation π_χ of G on $V = \mathbb{C}$ by $\pi_\chi(g)v = \chi(g)v$, and π_χ is unitary if and only if χ maps into the unit circle $\mathbb{T} = S^1$.
2. Let G be an LCH group. Then one can consider the **left regular representation**

$$\begin{aligned} \pi_{\mathrm{reg}} : G &\longrightarrow \mathrm{GL}_{\mathrm{cts}}(L^2(G)) \\ x &\longmapsto L_x \end{aligned}$$

This is really a representation by Lemma 7.1, Lemma 1.36 and

$$\begin{aligned} \langle L_x\phi, L_x\psi \rangle &= \int_G L_x\phi(y)\overline{L_x\psi(y)}dy \\ &= \int_G \phi(x^{-1}y)\overline{\psi(x^{-1}y)}dy \\ &= \int_G \phi(y)\overline{\psi(y)}dy = \langle \phi, \psi \rangle \end{aligned}$$

so $\|L_x\| = 1$ $(x \in G)$ and π_{reg} is unitary (essentially by the left-invariant of Haar measure).

7.1 Construction

In this subsection let G be a topological group.

Definition. Let (π_i, H_i) be unitary representations of G on Hilbert spaces H_i $(i \in I)$. We define the **direct sum representation** $\bigoplus_{i\in I} \pi_i : G \to \mathrm{GL}_{\mathrm{cts}}(\widehat{\bigoplus_{i\in I}} H_i)$ by

$$\left(\bigoplus_{i\in I} \pi_i\right)(g) \sum_{i\in I} v_i := \sum_{i\in I} \pi_i(g) v_i$$

Example 7.4. Let $G = \mathbb{R}/\mathbb{Z}$ and $V = L^2(\mathbb{R}/\mathbb{Z})$. Let π be the left regular representation of $\mathbb{R}/\mathbb{Z}$. By the Plancherel theorem, $\widehat{G} = \{e_k(x) := e^{2\pi i k x} \mid k \in \mathbb{Z}\}$ forms a orthonormal basis of $L^2(\mathbb{R}/\mathbb{Z})$, so π admits a direct sum decomposition

$$V = \widehat{\bigoplus_{k\in\mathbb{Z}}} \mathbb{C} e_k$$

and G acts on $\mathbb{C}e_k$ through the character e_k.

Definition. Let G_1, G_2 be two topological group, and (π_i, H_i) be a representation of G_i on the Hilbert space H_i. Define the **tensor product representation** $\pi_1 \otimes \pi_2 : G_1 \times G_2 \to \mathrm{GL}_{\mathrm{cts}}(H_1 \widehat{\otimes} H_2)$ by

$$(\pi_1 \otimes \pi_2)(g_1, g_2)(v_1 \otimes v_2) = \pi_1(v_1) \otimes \pi_2(v_2)$$

Definition. Let (π_i, V_i) be two representations of G on Banach spaces. Then $\mathrm{Hom}_G(V_1, V_2)$ is a closed subspace of the Banach space $\mathrm{Hom}_{\mathrm{cts}}(V_1, V_2)$. Define a representation $\pi : G \to \mathrm{GL}_{\mathrm{cts}}(\mathrm{Hom}_G(V_1, V_2))$ of G by

$$(\pi_L(g)T)(v) = \pi_2(g)T(v) = T(\pi_1(g)v)$$

7.2 Schur's Lemma

Definition. A representation (π, V_π) is called a **subrepresentation** of a representation (η, V_η) if

- V_π is a closed subspace of V_π, and
- $\eta|_{V_\pi} = \pi$.

In other words, a subrepresentation of (η, V_η) is a closed subspace invariant under η.

Definition. Let (π, V) be a representation on a group G.

1. (π, V) is called **irreducible** if it does not contain any proper nontrivial subrepresentation.
2. A vector $v \in V$ is called a **cyclic vector** if the linear span of the G-orbit $Gv := \{\pi(g)v \mid g \in G\}$ of v is dense in V.

- A representation is irreducible if and only if every nonzero vector is cyclic.

Example 7.5.

1. Every one-dimensional representation is irreducible.

2. Consider the unitary group $\mathrm{U}(n)$ and its natural action on $\mathbb{C}^n$. Then this representation is irreducible. The case $n = 1$ is trivial. When $n \geqslant 2$, we show for any two vectors $v, u \in \mathbb{C}^n$ with $\|v\| = \|u\| = 1$, we can find $A \in \mathrm{U}(n)$ such that $Av = u$. Consider the plane spanned by v, u; up to a (unitary) rotation, we can assume $v = (1, 0, 0, \ldots, 0)$ and $u = (a, b, 0, \ldots, 0)$ with $|a|^2 + |b|^2 = 1$. Then $A := \begin{pmatrix} a & -\overline{b} \\ b & \overline{a} \end{pmatrix} \oplus I_{n-2}$ does the job. Hence a nontrivial $\mathrm{U}(n)$-invariant subspace of $\mathbb{C}^n$ contains all directions, and hence it is the whole space.

Lemma 7.6 (Schur)**.** Let (π, H) be a unitary representation of the topological group G. TFAE

(a) (π, H) is irreducible.

(b) $\mathrm{Hom}_G(H, H) = \mathbb{C}\,\mathrm{id}_H$.

Proof. Since $\pi(g^{-1}) = \pi(g)^*$, the set $\{\pi(g) | g \in G\} \subseteq \mathcal{B}(H)$ is self-adjoint. Then the result follows from Theorem 6.6. □

Definition. For two unitary representations (π, V_π), (η, V_η), they are called **unitarily equivalent** if there exists a unitary G-intertwining operator $T : V_\pi \to V_\eta$.

Example 7.7. Let $G = \mathbb{R}$ and let $V_\pi = V_\eta = L^2(\mathbb{R})$. The representation π is given by $\pi(x)\phi(y) = \phi(x+y)$, and η is given by $\eta(x)\phi(y) = e^{2\pi i x y}\phi(y)$. The Fourier transformation is then unitary equivalence $V_\pi \to V_\eta$. This follows from the Plancherel theorem and a direct computation.

Corollary 7.6.1. Let (π, V_π) and (η, V_η) be two irreducible unitary representations. Then a G-homomorphism T from V_π to V_η is either zero or invertible with continuous inverse. In the latter case there exists $c > 0$ such that cT is unitary. Hence the space $\mathrm{Hom}_G(\pi, \eta)$ is zero unless π and η is unitary equivalent, in which case the space is of dimension 1.

Proof. Let $T : V_\pi \to V_\eta$ be a G-homomorphism. Then its adjoint $T^* : V_\eta \to V_\pi$ is also a G-homomorphism:

$$\begin{aligned} \langle v, T^*\eta(g)w\rangle &= \langle Tv, \eta(g)w\rangle = \langle \eta(g^{-1})Tv, w\rangle \\ &= \langle T\pi(g^{-1})v, w\rangle = \langle v, \pi(g)T^*w\rangle \end{aligned}$$

Thus $T^*T \in \mathrm{End}_G(V_\pi)$, so by Schur's lemma $T^*T = \lambda\,\mathrm{id}_{V_\pi}$ for some $\lambda \in \mathbb{C}$. If $T \neq 0$, then T^*T is nonzero and positive, so $\lambda > 0$. If we let $c = \sqrt{\lambda^{-1}} > 0$, then $(cT)^*(cT) = \mathrm{id}_{V_\pi}$; this shows cT is unitary. A similar argument shows TT^* is bijective, which implies cT is bijective, whence T is a homeomorphism by Open Mapping theorem. □

Definition. For an LCH group G, we denote by $\widehat{G}$ the set of all equivalence classes the irreducible unitary representations of G. We call $\widehat{G}$ the **unitary dual** of G.

- We need to explain why $\widehat{G}$ forms a set. First, we show that there exists a cardinality α depending on G such that every irreducible representation (π, V_π) of G satisfies $\dim V_\pi \leqslant \alpha$. Indeed, every irreducible representation has a cyclic vector, so we can take $\alpha = \#G$. Secondly, a representation of G is just a vector space V together with an appropriate function $V \times V \to \mathbb{C}$ and a map $G \to \mathrm{GL}(V)$, and for each cardinality $\beta \leqslant \alpha$ there exists a unique vector space V_β of dimension β. Hence every equivalence class of irreducible representations of G has a representation in the set $\bigcup_{\beta \leqslant \alpha} \mathbb{C}^{V_\beta \times V_\beta} \times \mathrm{GL}(V_\beta)^G$.

Example 7.7. If G is LCA, then every irreducible representation is one-dimensional, so the unitary dual of G coincides with the Pontryagin dual of G. To see this, let (π, V_π) be any nonzero irreducible unitary representation of G. Then $\pi(x)\pi(y) = \pi(xy) = \pi(yx) = \pi(y)\pi(x)$ for all $x, y \in G$, so that $\pi(x) \in \mathrm{Hom}_G(V_\pi, V_\pi)$. By Schur's lemma, for all $x \in G$ we have $\pi(x) = \lambda(x)\,\mathrm{id}_{V_\pi}$ for some $\lambda(x) \in \mathbb{C}$. But this means every closed subspace of V_π is G-invariant, so $\dim_{\mathbb{C}} V_\pi = 1$ particularly. By Lemma 7.2 we have $\lambda(x) \in S^1$ for all $x \in G$, so λ is a unitary character of G. The converse is clear.

7.3 Representation of $L^1(G)$

Proposition 7.8. Let (π, V_π) be a unitary representation of an LCH group G. For each $f \in L^1(G)$ there exists a unique bounded operator $\pi(f)$ on V_π such that

$$\langle \pi(f)v, w\rangle = \int_G f(x)\langle \pi(x)v, w\rangle dx$$

holds for all $v, w \in V_\pi$. The induced map $\pi : L^1(G) \to \mathcal{B}(V_\pi)$ is a continuous homomorphism of Banach *-algebras.

Proof. We invoke Riesz's Representation theorem. Taking complex conjugation, it suffices to show

$$\langle w, \pi(f)v\rangle = \int_G \overline{f(x)}\langle w, \pi(w)v\rangle dx$$

Now the map $w \mapsto \int_G \overline{f(x)}\langle w, \pi(w)v\rangle dx$ is $\mathbb{C}$-linear and bounded, since

$$\begin{aligned}\left|\int_G \overline{f(x)}\langle w, \pi(w)v\rangle dx\right| &\leqslant \int_G |f(x)\langle w, \pi(w)v\rangle|\, dx \\ &\leqslant \int_G |f(x)|\, \|w\|\, \|\pi(x)v\|\, dx \\ &= \|f\|_1\, \|w\|\, \|v\|\end{aligned}$$

Here we use π is unitary. Therefore there exists a unique vector, denoted by $\pi(f)v \in V_\pi$, such that the equality holds. The same theorem shows $v \mapsto \pi(f)v$ is linear. To show it is bounded, note that the above computation gives

$$\|\pi(f)v\|^2 = \langle \pi(f)v, \pi(f)v\rangle \leqslant \|f\|_1 \|\pi(f)v\| \|v\|$$

so that $\|\pi(f)v\| \leqslant \|f\|_1 \|v\|$. It remains to show $\pi(f * g) = \pi(f)\pi(g)$ and $\pi(f)^* = \pi(f^*)$. For the former,

$$\begin{aligned}
\langle \pi(f * g)v, w\rangle &= \int_G \left(\int_G f(y)g(y^{-1}x)dy\right) \langle \pi(x)v, w\rangle dx \\
&\overset{\text{Fubini}}{=} \int_G \int_G f(y)g(y^{-1}x)\langle \pi(x)v, w\rangle dxdy \\
(x \mapsto yx) &= \int_G \int_G f(y)g(x)\langle \pi(yx)v, w\rangle dxdy \\
&= \int_G \int_G f(y)g(x)\langle \pi(x)v, \pi(y^{-1})w\rangle dxdy \\
&= \int_G f(y)\langle \pi(g)v, \pi(y^{-1})w\rangle dy \\
&= \int_G f(y)\langle \pi(y)\pi(g)v, w\rangle dy = \langle \pi(f)\pi(g)v, w\rangle
\end{aligned}$$

and for the latter,

$$\begin{aligned}
\langle \pi(f^*)v, w\rangle &= \int_G f^*(x)\langle \pi(x)v, w\rangle dx \\
&= \int_G \Delta_G(x^{-1})\overline{f(x^{-1})}\langle v, \pi(x^{-1})w\rangle dx \\
&= \overline{\int_G \Delta_G(x^{-1})f(x^{-1})\langle \pi(x^{-1})w, v\rangle dx} \\
&= \overline{\int_G f(x)\langle \pi(x)w, v\rangle dx} \\
&= \overline{\langle \pi(f)w, v\rangle} = \langle v, \pi(f)w\rangle
\end{aligned}$$

□

Remark 7.9. Alternatively, one can define $\pi(f)$ as the Bochner integral

$$\pi(f) := \int_G f(x)\pi(x)dx$$

in the Banach space $\mathcal{B}(V_\pi)$. To see why this integral exists, we use Proposition U.38.2. By linearity, we may assume $f \in L^1(G)$ is nonnegative. Consider the finite positive measure $d\mu(x) = f(x)dx$ on G. Then $\pi(f) = \int_G \pi(x)d\mu(x)$. By Corollary 1.3.1.4 the measure $d\mu$ is supported on a σ-compact open subset K of G, so the integral $\int_G \pi(x)d\mu(x)$ really takes place on a σ-compact set. Since π is continuous, by Lemma U.37 π is separable when restricting to K, and thus $\int_G \pi(x)d\mu(x)$ exists by Proposition U.38.2.

Lemma 7.10. Let (π, V_π) be a representation of G. Then for every $v \in V_\pi$ and every $\varepsilon > 0$ there exists a unit-neighborhood U such that for every Dirac function ϕ_U with support in U one has $\|\pi(\phi_U)v - v\| < \varepsilon$. In particular, for every Dirac net $(\phi_U)_U$, the net $(\pi(\phi_U)v)_U$ converges to v in norm topology.

Proof. For any open set U and ϕ_U a Dirac function supported in U, we have

$$\|\pi(\phi_U)v - v\| = \left\| \int_G \phi_U(x)(\pi(x)v - v)dx \right\| \leqslant \int_G \phi_U(x) \|\pi(x)v - v\| \, dx$$

The first equality is clear, and the second is Proposition U.36.(b). For given $\varepsilon > 0$ there exists a unit-neighborhood U_0 such that $\|\pi(x)v - v\| < \varepsilon$ whenever $x \in U_0$. Then $U = U_0$ does the job. □

Definition. A *-representation $\pi : L^1(G) \to \mathcal{B}(V)$ of $L^1(G)$ on a Hilbert space V is **non-degenerate** if the subspace

$$\pi(L^1(G))V := \operatorname{span}\{\pi(f)v \mid f \in L^1(G),\, v \in V\}$$

is dense in V.

- It follows from Lemma 7.10 that every representation of $L^1(G)$ constructed from a unitary representation (π, V_π) as in Proposition 7.8 is non-degenerate.

Proposition 7.11. Let $\pi : L^1(G) \to \mathcal{B}(V)$ be a non-degenerate *-representation of $L^1(G)$ on a Hilbert space V. Then there exists a unique unitary representation $(\tilde{\pi}, V)$ of G such that

$$\langle \pi(f)v, w \rangle = \int_G f(x) \langle \tilde{\pi}(x)v, w \rangle dx$$

holds for all $f \in L^1(G)$ and all $v, w \in V$.

Proof. Note that π is continuous by Lemma 2.30. Let $x \in G$. We first define an operator $\tilde{\pi}(x)$ on the dense subspace $\pi(L^1(G))V$. Each element in $\pi(L^1(G))V$ has the form $\sum_{i=1}^n \pi(f_i)v_i$, $f_i \in L^1(G)$, $v_i \in V$. Define

$$\tilde{\pi}(x) \sum_{i=1}^n \pi(f_i)v_i := \sum_{i=1}^n \pi(L_x f_i)v_i$$

We must show this is well-defined, which amounts to show that $\sum_{i=1}^n \pi(f_i)v_i = 0$ implies $\sum_{i=1}^n \pi(L_x f_i)v_i = 0$. For $f, g \in L^1(G)$, we have $g^* * L_x f = (L_{x^{-1}} g)^* * f$. Indeed,

$$\begin{aligned}
g^* * L_x f(y) &= \int_G \Delta_G(z^{-1}) \overline{g(z^{-1})} f(x^{-1}z^{-1}y) dz \\
&\overset{1.11.4.}{=} \int_G \overline{g(z)} f(x^{-1}zy) dz \\
(z \mapsto xz) &= \int_G \overline{g(xz)} f(zy) dz \\
&\overset{1.11.4.}{=} \int_G \Delta_G(z^{-1}) \overline{g(xz^{-1})} f(z^{-1}y) dz = (L_{x^{-1}} g)^* * f(y)
\end{aligned}$$

Then for $g, f_1, \ldots, f_n \in L^1(G)$, $v_1, \ldots, v_n, w \in V$, $x \in G$

$$\begin{aligned}\left\langle \sum_{i=1}^n \pi(L_x f_i)v_i, \pi(g)w \right\rangle &= \sum_{i=1}^n \langle \pi(g^* * L_x f_i)v_i, w\rangle \\ &= \sum_{i=1}^n \langle \pi((L_{x^{-1}}g)^* * f_i)v_i, w\rangle = \left\langle \sum_{i=1}^n \pi(f_i)v_i, \pi(L_{x^{-1}}g)w \right\rangle\end{aligned}$$

Now assume $\sum_{i=1}^n \pi(f_i)v_i = 0$. Then the above shows that $\sum_{i=1}^n \pi(L_x f_i)v_i$ is orthogonal to $\pi(L^1(G))V$, which is dense in V, so $\sum_{i=1}^n \pi(L_x f_i)v_i = 0$. Thus $\tilde{\pi}(x)$ is a well-defined operator on $\pi(L^1(G))V$, and the above computation also shows $\tilde{\pi}(x)$ is unitary on $\pi(L^1(G))V$. Therefore $\tilde{\pi}(x)$ extends to a unique unitary operator on V with inverse $\tilde{\pi}(x^{-1})$, and clearly $\tilde{\pi}(xy) = \tilde{\pi}(x)\tilde{\pi}(y)$ for all $x, y \in G$. By Lemma 1.36 for each $f \in L^1(G)$ the map $x \mapsto L_x f$ is continuous, so $x \mapsto \tilde{\pi}(x)v$ is continuous for every $v \in V$. Thus $(\tilde{\pi}, V)$ is a unitary representation of G by Lemma 7.1.

It remains to show $\pi(f) = \tilde{\pi}(f)$ for all $f \in L^1(G)$. By continuity we may assume $f \in C_c(G)$ and it suffices to show $\langle \tilde{\pi}(f)\pi(g)v, w\rangle = \langle \pi(f)\pi(g)v, w\rangle$ for all $g \in C_c(G)$, $v, w \in V$.

$$\begin{aligned}\langle \tilde{\pi}(f)\pi(g)v, w\rangle &= \int_G f(x)\langle \tilde{\pi}(x)\pi(g)v, w\rangle dx \\ &= \int_G f(x)\langle \pi(L_x g)v, w\rangle dx \\ &= \int_G \langle \pi(f(x)L_x g)v, w\rangle dx \\ &\overset{U.36.(c)}{=} \left\langle \pi\left(\int_G f(x)L_x g dx\right)v, w \right\rangle \\ &\overset{U.39}{=} \langle \pi(f * g)v, w\rangle = \langle \pi(f)\pi(g)v, w\rangle\end{aligned}$$

□

Example 7.12. Let G be an LCH group, and consider the left regular representation $L : G \to \mathcal{U}(L^2(G))$. By Proposition 7.8 we then have a continuous *-homomorphism $L : L^1(G) \to \mathcal{B}(L^2(G))$ defined by

$$L(f)g = f * g \in L^2(G).$$

By Lemma 1.38, $L : L^1(G) \to \mathcal{B}(L^2(G))$ is injective.

In particular, the case G being LCA recovers Lemma 4.12. Also, using the Plancherel isomorphism $L^2(G) \cong L^2(\widehat{G})$ we obtain a map $L^1(G) \to \mathcal{B}(L^2(\widehat{G}))$ given by $f \mapsto [g \mapsto \widehat{f} \cdot g]$. Hence

$$\|L(f)\| = \left\| g \mapsto \widehat{f}g \right\| = \left\| \widehat{f} \right\|_{\sup} = \sup_{\chi \in \widehat{G}} |\widehat{f}(\chi)|.$$

7.4 Square Integrable Representations

Definition. Let G be a topological group and (π, H) be a representation on the Hilbert space H.

(i) For $v, w, \in H$, the function $c_{v,w} : G \to C(G)$ defined as

$$c_{v,w}(g) := \langle \pi(g)v, w \rangle$$

is called a **matrix coefficient of** π.

(ii) The representation π is called **square integrable** if (π, H) is irreducible, unitary, and it has a nonzero square integrable matrix coefficient.

Lemma 7.13. Let (π, H) be a unitary representation of an LCH group G. For $v \in H$, define

$$D_v := \{w \in V \mid c_{w,v} \in C(G) \cap L^2(G)\}.$$

Then the linear map $T : D_v \to L^2(G)$ defined by $Tw = c_{w,v}$ is a **closed operator**, i.e., the graph

$$\Gamma_T := \{(w, Tw) \mid w \in D_v\} \subseteq D_v \times L^2(G)$$

is a closed subset.

Proof. Let $(w_n)_n$ be a sequence in D_v such that $w_n \to w \in H$ and $Tw_n \to \phi \in L^2(G)$. Passing to a subsequence (U.27), we may assume $Tw_n \to \phi$ pointwise almost everywhere. But then for $g \in G$,

$$|\phi(g) - Tw_n(g)| = |Tw(g) - Tw_n(g)| = |\langle \pi(g)(w - w_n), v \rangle| \leqslant \|w - w_n\| \, \|v\| \to 0$$

as $n \to \infty$. This implies $\phi = Tw$ almost everywhere. □

Lemma 7.14. Let G be an LCH group, and let (π, H) be a square integrable representation.

(i) All matrix coefficients of π is square integrable.

(ii) There exists a unitary operator $T \in \mathrm{Hom}_G(H, L^2(G))$ with closed range and $T(H) \leqslant C(G) \cap L^2(G)$.

Proof. Take $v', w' \in H$ such that $c_{v',w'} : G \to C(G)$ is nonzero and square integrable; we may assume v', w' have norm 1. Define

$$D = \{v \in H \mid c_{v,w'} \in L^2(G)\}$$

□

8 Compact Groups

8.1 Finite Dimensional Representations

In this subsection, let K be a compact (Hausdorff topological) group and (τ, V_τ) a finite dimensional (complex) representation of K.

Lemma 8.1. On V_τ there exists an inner product making τ a unitary representation. If τ is irreducible, this inner product is determined up to a positive scalar.

Proof. Let $(\,,\,)$ be any inner product on V_τ, and define a map $\langle\,,\rangle : V_\tau \times V_\tau \to \mathbb{C}$ by the formula

$$\langle v, w\rangle := \int_K (\tau(k)v, \tau(k)w)dk$$

where dk is the normalized Haar measure on K so that $\mathrm{vol}(K, dk) = 1$. This clearly defines an inner product on V_τ, and τ becomes a unitary representation because dk is unimodular.

Now assume τ is irreducible and let $\langle\,,\rangle_1, \langle\,,\rangle_2$ be two inner products making τ unitary. For $i = 1, 2$, denote by (τ_i, V_i) the representation (τ, V_τ) when equipped with the inner product $\langle\,,\rangle_i$. Since V_τ is finite dimensional, $\mathrm{id} : V_1 \to V_2$ is bounded nonzero intertwining operator from τ_1 to τ_2. Since V_τ is irreducible, by Corollary 7.6.1 (in which Schur's lemma can be replaced by the usual Schur's lemma) there exists $c > 0$ such that $c \cdot \mathrm{id}$ is unitary. But this implies that $c^2\langle v, w\rangle_2 = \langle v, w\rangle_1$ for all $v, w \in V_\tau$. □

Remark 8.2. In fact, the first paragraph of the proof does not use the finite dimensionality of V_τ: it holds for any Hilbert space representation of K. The second part of this lemma does not holds in general, but when V_τ is a Hilbert space, we can show that the constructed unitary representation defines an equivalent norm to the original one.

To prove this, let V_τ be a Hilbert space with inner product $(\,,\,)$. For any $g \in K$, define $T_g : V_\tau \to V_\tau$ by $T_g(v) = \pi(g)v$. For any $v \in V$, the map $K \to \mathbb{R}$, $g \mapsto \|\pi(g)v\|$ is continuous, so it attains a maximum by compactness of K. It follows by uniform boundedness principal that there exists $C > 0$ such that $\|T_g(v)\| \leqslant C\,\|v\|$ for any $g \in K$, $v \in V_\tau$. In particular, we have $(v, v) \leqslant C^2(\pi(g)v, \pi(g)v)$ for any $g \in K$, $v \in V_\tau$. The inner product $\langle,\rangle : V_\tau \times V_\tau \to \mathbb{C}$ defined by

$$\langle v, w\rangle := \int_K (\pi(g)v, \pi(g)w)dg$$

then satisfies $C^{-2}(v, v) \leqslant \langle v, v\rangle \leqslant C^2\langle v, v\rangle$ (as $\mathrm{Vol}(K) = 1$). This shows the equivalence (as Banach spaces).

Proposition 8.3. A finite dimensional representation of a compact group is a direct sum of irreducible representations.

Proof. We prove this by induction on the dimension of representations; the cases $\dim = 0, 1$ are evident. Let (τ, V) be a representation with $\dim_{\mathbb{C}} V \geqslant 2$. If τ is irreducible, we are done. Otherwise, V admits a

nonzero proper invariant subspace U. By the previous lemma, we equip V with an inner product so that τ is unitary. Let W be the orthogonal complement of U. If we can show W is also invariant, the result will follow from the induction. Now let $w \in W$ and $k \in K$. Then for all $u \in U$

$$\langle \tau(k)w, u \rangle \overset{\tau \text{ unitary}}{=} \langle w, \underbrace{\tau(k^{-1})u}_{\in U} \rangle = 0$$

so that $\tau(k)w \in W$. This proves W is invariant under K. $\square$

Definition. Let (τ, V) be a finite dimensional complex representation of a compact group K. The dual space $V^* = \mathrm{Hom}_{\mathbb{C}}(V, \mathbb{C})$ carries a natural representation of K, called the **contragredient representation** τ^*, defined by

$$\tau^*(k)\alpha(v) = \alpha(\tau(k^{-1})v)$$

for all $k \in K$, $\alpha \in V^*$, $v \in V$.

Suppose V is equipped with an inner product so that τ is unitary. By Riesz's Representation theorem for every $\alpha \in V^*$ there exists a unique vector v_α such that

$$\alpha(w) = \langle w, v_\alpha \rangle$$

for all $w \in V$. One installs an inner product on V^* by

$$\langle \alpha, \beta \rangle := \langle v_\beta, v_\alpha \rangle$$

Lemma 8.4. If (τ, V_τ) is irreducible (resp. unitary), then so is the contragredient τ^*. Install V_τ with the inner product making τ unitary. Then for $x \in K$ and $\alpha \in V_\tau^*$, one has

$$v_{\tau^*(x)\alpha} = \tau(x)v_\alpha$$

so the map $\alpha \mapsto v_\alpha$ is an antilinear K-intertwining operator from V_τ^* to V_τ.

Proof. Let $U \subseteq V_\tau^*$ be a K-invariant subspace; then $U^\perp := \{v \in V_\tau \mid \alpha(v) = 0 \text{ for all } \alpha \in U\}$ is a K-invariant subspace of V. Thus if V_τ is irreducible, then $U^\perp$ is either 0 or V_τ, whence $U = 0$ or V_τ^*.

Next we show the intertwining relation. For $w \in V_\tau$,

$$\langle w, v_{\tau^*(x)\alpha} \rangle = \tau^*(x)\alpha(w) = \alpha(\tau(x^{-1})w) = \langle \tau(x^{-1})w, v_\alpha \rangle = \langle w, \tau(x)v_\alpha \rangle$$

here we use τ is unitary. Hence $v_{\tau^*(x)\alpha} = \tau(x)v_\alpha$ holds for all $v \in V_\tau$. Finally,

$$\langle \tau^*(x)\alpha, \tau^*(x)\beta \rangle = \langle v_{\tau^*(x)\beta}, v_{\tau^*(x)\alpha} \rangle = \langle \tau(x)v_\beta, \tau(x)v_\alpha \rangle = \langle v_\beta, v_\alpha \rangle = \langle \alpha, \beta \rangle$$

so that τ^* is unitary as well. $\square$

8.2 The Peter-Weyl Theorem

Let K be a compact group and let $\widehat{K}$ denote its unitary dual. Let $\widehat{K}_{\text{fin}}$ be the subset of $\widehat{K}$ consisting of all finite dimensional irreducible representations. We are going to show $\widehat{K} = \widehat{K}_{\text{fin}}$.

Definition. For a unitary representation (τ, V_τ) of K, a **matrix coefficient** is a function on K of the form $k \mapsto \langle \tau(k)v, w\rangle$ for some $v, w \in V_\tau$.

- A matrix coefficient is continuous, so it lies in the Hilbert space $L^2(K)$.
- The set of matrix coefficients of all finite dimensional representations is closed under complex conjugation. To see this, let $v, v' \in V_\tau$ and $\alpha, \beta \in V_\tau^*$ be their Riesz's dual, i.e., $v = v_\alpha$, $v' = v_\beta$. Then by Lemma 8.4,

$$\overline{\langle \tau(x)v, v'\rangle} = \langle v_\beta, \tau(x)v_\alpha\rangle = \langle v_\beta, v_{\tau^*(x)\alpha}\rangle = \langle \tau^*(x)\alpha, \beta\rangle$$

For every class in $\widehat{K}_{\text{fin}}$ choose a representation (τ, V_τ). Choose an orthonormal basis $e_1, \ldots, e_n$ of V_τ and write $\tau_{ij}(k) = \langle \tau(j)e_i, e_j\rangle$ for the corresponding matrix coefficients. Clearly, for every $v, w \in V_\tau$, the function $k \mapsto \langle \tau(k)v, w\rangle$ is a linear combination of the τ_{ij}, $1 \leqslant i, j, \leqslant \dim_{\mathbb{C}} V_\tau$. In what follows we shall write $\dim(\tau)$ for $\dim_{\mathbb{C}} V_\tau$.

Theorem 8.5 (Peter-Weyl)**.**

(a) For $\tau \neq \gamma$ in $\widehat{K}_{\text{fin}}$, one has

$$\langle \tau_{ij}, \gamma_{rs}\rangle = \int_K \tau_{ij}(k)\overline{\gamma_{rs}}(k)dk = 0$$

so the matrix coefficients of non-equivalent representations are orthogonal in $L^2(K)$.

(b) For $\tau \in \widehat{K}_{\text{fin}}$, one has

$$\langle \tau_{ij}, \tau_{rs}\rangle = \frac{1}{\dim(\tau)}\delta_{ir}\delta_{js}$$

In other words, the family $(\sqrt{\dim(\tau)}\tau_{ij})_{\tau,i,j}$ is an orthogonal set in $L^2(K)$.

(c) $(\sqrt{\dim(\tau)}\tau_{ij})_{\tau,i,j}$ is an orthogonal basis of $L^2(K)$.

(d) The regular representations $(L, L^2(K))$ and $(R, L^2(K))$ decompose into direct sums of finite-dimensional irreducible representations. More precisely, the space of matrix coefficients (of all finite dimensional representations of K) is dense in $L^2(K)$.

Proof.

(a) Let $\tau \neq \gamma$ in $\widehat{K}_{\text{fin}}$. Let $T : V_\tau \to V_\gamma$ be a (bounded) linear operator and set

$$S = S_T := \int_K \gamma(k^{-1}) T \tau(k) dk \in \mathcal{B}(V_\tau)$$

Note that this integral takes place in a finite dimensional vector space, and it exists since the integrand is continuous and K is compact. Then one has $S\tau(k) = \gamma(k)S$, hence $S = 0$ by Schur's lemma.

Let (e_j) and (f_s) be orthonormal bases (chosen above) for V_τ and V_γ, respectively, and choose $T_{js} : V_\tau \to V_\gamma$ given by $T_{js}(v) = \langle v, e_j \rangle f_s$. Let $S_{js} = S_{T_{js}}$ as above. Then

$$\begin{aligned}
0 = \langle S_{js} e_i, f_r \rangle &= \int_K \langle \gamma(k^{-1}) T_{js} \tau(k) e_i, f_r \rangle dk \\
&= \int_K \langle \gamma(k^{-1}) \langle \tau(k) e_i, e_j \rangle f_s, f_r \rangle dk \\
&= \int_K \langle \tau(k) e_i, e_j \rangle \langle \gamma(k^{-1}) f_s, f_r \rangle dk \\
&= \int_K \langle \tau(k) e_i, e_j \rangle \overline{\langle \gamma(k) f_s, f_r \rangle} dk = \langle \tau_{ij}, \gamma_{rs} \rangle
\end{aligned}$$

(b) Let the notation be as in (a). The same computation gives

$$\langle S_{js} e_i, e_r \rangle = \langle \tau_{ij}, \tau_{rs} \rangle$$

By Schur's lemma, $S_{js} = \lambda \,\mathrm{id}$ for some $\lambda \in \mathbb{C}$, so

$$\langle \tau_{ij}, \tau_{rs} \rangle = \lambda \delta_{ir}$$

We compute λ:

$$\begin{aligned}
\lambda \dim V_\tau = \operatorname{tr}(S) &= \operatorname{tr}\left(\int_K \tau(k^{-1}) T_{js} \tau(k) dk \right) \\
&= \int_K \operatorname{tr}(\tau(k^{-1}) T_{js} \tau(k)) dk = \int_K \operatorname{tr}(T_{js}) dk = \operatorname{tr}(T_{js}) = \delta_{js}
\end{aligned}$$

Therefore,

$$\langle \tau_{ij}, \tau_{rs} \rangle = \frac{1}{\dim(\tau)} \delta_{js} \delta_{ir}$$

(c) Let $\tau \in \widehat{K}_{\text{fin}}$, and let M_τ be the subspace of $L^2(K)$ spanned by all matrix coefficients of the representation τ. If we put $h : K \to \mathbb{C}$ given by $h(k) = \langle \tau(k) v, w \rangle$ for some $v, w \in V_\tau$, then

$$\begin{aligned}
h^*(k) := \overline{h(k^{-1})} &= \langle \tau(k) w, v \rangle \in M_\tau \\
L_{k_0} h(k) = h(k_0^{-1} k) &= \langle \tau(k) v, \tau(k_0) w \rangle \in M_\tau \\
R_{k_0} h(k) = h(k k_0) &= \langle \tau(k) \tau(k_0) v, w \rangle \in M_\tau
\end{aligned}$$

This means M_τ is closed under adjoints, and left and right translations. Let M be the closure in $L^2(K)$ of the span of all M_τ, $\tau \in \widehat{K}_{\text{fin}}$. We want to show $M = L^2(K)$, or equivalently $M^\perp = 0$.

Assume for contradiction that $M^\perp \neq 0$. Let $H \neq 0 \in M^\perp$, and let $(\phi_U)_U$ be a Dirac net. Then $\phi * H \to H$ in $L^2(K)$. Since $M^\perp$ is closed under translation, it follows that $\phi_U * H \in M^\perp$ for all U. For the detail, we have

$$\langle \phi_U * H, f\rangle \overset{U.39}{=} \left\langle \int_K \phi_U(k) L_k H dk, f \right\rangle = \int_K \phi_U(k) \langle L_k H, f\rangle dk = 0$$

for all $f \in M$. Therefore, there must exist U such that $\phi_U * H \neq 0$. This shows $M^\perp$ contains a nonzero continuous function. Next, let $F_1 \neq 0 \in M^\perp$ be continuous; we may assume $F_1(e) > 0$. Set

$$F_2(x) := \int_K f_1(y^{-1}xy)dy$$

Since $M^\perp$ is closed under left and right translation, $F_2 \in M^\perp$, and $F_2(e) > 0$ and it is invariant under conjugation. Finally put $F(x) := F_2(x) + \overline{F_2(x^{-1})}$. Then the function $F : K \to \mathbb{C}$ is (i) continuous, (ii) lies in $M^\perp$, (iii) $F(e) > 0$ and (iv) $F = F^*$.

Consider the operator $T : L^2(K) \to L^2(K)$ defined by

$$T(f) := f * F = R(F)f \overset{\text{(ii)}}{\in} M^\perp$$

This is nonzero by (iii). Since $R : L^1(K) \to \mathcal{B}(L^2(K))$ is a *-representation by Proposition 7.8, $T^* = R(F)^* = R(F^*) \overset{\text{(iv)}}{=} R(F) = T$, i.e., T is self-adjoint. Further,

$$Tf(x) = f * F(x) = \int_K f(y)F(y^{-1}x)dy$$

T is an integral operator with kernel $k(x, y) := F(y^{-1}x)$. By Proposition 6.18 $T = T^* \neq 0$ is a Hilbert-Schmidt operator, hence compact; it follows that T has a real eigenvalue $\lambda \neq 0$ with finite dimensional eigenspace V_λ. We claim V_λ is stable under left-translation. For this, let $f \in V_\lambda$, so $f * F = T(f) = \lambda f$. Then for $k \in K$, one has

$$L_k f * F \overset{1.33}{=} L_k(f * F) = \lambda L_k f$$

as desired. This implies (L, V_λ) is a finite dimensional unitary representation of K, hence it contains an irreducible subrepresentation $0 \neq W \subseteq V_\lambda \subseteq M^\perp$. Let $f, g \in W$, and let $h(k) = \langle L_k f, g\rangle$ be the corresponding matrix coefficient. One has

$$h(k) = \int_K f(k^{-1}x)\overline{g(x)}dx = \overline{\int_K g(x)f^*(x^{-1}k)dx} = \overline{g * f^*}(k)$$

so $h \in M^\perp$. On the other hand, $h \in M$, so $\langle h, h\rangle = 0$, a contradiction to (b). Hence $M^\perp = 0$, i.e., $M = L^2(K)$.

(d) Since $L^2(K) = M$, where M is as in (c), we have

$$L^2(K) = \overline{\bigoplus_{\tau \in \widehat{K}_{\text{fin}}} M_\tau}$$

Each M_τ has finite dimension, and is invariant under left and right translations. By Proposition 8.3, this proves (d).

□

If (π, V_π) is a representation of an LCH group G, for convenience, by an **irreducible subspace** we mean a closed subspace $U \subseteq V_\pi$ which is stable under $\pi(G)$ such that the representation (π, U) obtained by restricting each $\pi(k)$ to U is irreducible.

Theorem 8.6. Let K be a compact group.

(a) $\widehat{K} = \widehat{K}_{\text{fin}}$, i.e., every irreducible unitary representation a compact group is finite dimensional.

(b) Every unitary representation of K is an orthogonal direct sum of irreducible representations.

Proof. Let (π, V_π) be a unitary representation of K. We show that V_π can be written as a direct sum

$$V_\pi = \overline{\bigoplus_{i \in I} V_i}$$

with each V_i a finite dimensional irreducible subspace of V_π. This proves (b), and if we apply this to a given irreducible unitary representation, it also implies (a).

So let (π, V_π) be a unitary representation of K. Consider the collection S of all families $(V_i)_{i \in I}$, where each V_i is a finite dimensional irreducible subspace of V_π and $V_i \perp V_j$ whenever $i \neq j \in I$. Introduce a partial order $\leqslant$ on S by

$$(V_i)_{i \in I} \leqslant (W_\alpha)_{\alpha \in A} \Leftrightarrow I \subseteq A \text{ and } V_i = W_i \text{ for all } i \in I$$

By Zorn's lemma, S admits a maximal element, denoted by $(V_i)_{i \in I}$. We contend that the orthogonal (algebraic) direct sum $\bigoplus_{i \in I} V_i$ is dense in V, which is equivalent to the orthogonal space $W := \left(\bigoplus_{i \in I} V_i\right)^\perp$ being the zero space.

Assume for contradiction that $W \neq 0$. By maximality, it suffices to show the following.

Lemma 8.7. Every nonzero unitary representation (η, W_η) contains a finite dimensional irreducible subspace.

Proof. Let $v, w \in W_\eta$ and let $\psi_{v,w}(x) := \langle \eta(x)v, w \rangle$ be the corresponding matrix coefficient; then $\psi_{v,w} \in C(K) \subseteq L^2(K)$ and

$$\psi_{\eta(y)v,w}(x) = \langle \eta(xy)v, w \rangle = \psi_{v,w}(xy) = R_y \psi_{v,w}(x)$$

In other words, for each fixed w, the map $v \mapsto \psi_{v,w} \in \mathrm{Hom}_K(\eta, (R, L^2(K)))$. Let us assume $\langle v, w\rangle \neq 0$ so that this map is nonzero. By (d) of Peter-Weyl theorem, we know $(R, L^2(K))$ is a direct sum of finite dimensional irreducible representations, so there exists an orthogonal projection $P : L^2(K) \to F$ to a finite dimensional irreducible subrepresentation such that $P(\psi_{v,w}) \neq 0$. Consequently there exists a nonzero K-intertwining operator $T : W_\eta \to F$ which is surjective by irreducibility of F, and hence it induces an isomorphism from $(\ker T)^\perp \subseteq W_\eta$ to F. Therefore, $(\ker T)^\perp$ is the desired finite dimensional irreducible subspace. □

□

Remark 8.8. There is a much shorter proof of (a). Let (π, V_π) be an irreducible unitary representation of a compact group K. Pick any $v \in V_\pi$ of norm 1 and define $T : V_\pi \to V_\pi$ by

$$T(w) = \int_K \pi(g)v\langle \pi(g)v, w\rangle dg \in V_\pi.$$

A simple computation shows that T is G-equivariant, $T = \lambda\,\mathrm{id}_{\mathbb{C}}$ by Schur's lemma. Applying $\langle \cdot, v\rangle$, we obtain

$$\overline{\lambda} = \langle v, Tv\rangle = \int_K \langle \pi(g)v, v\rangle\langle \pi(g)v, v\rangle dg = \int_K |\langle \pi(g)v, v\rangle|^2 dg > 0.$$

Now let $W \subseteq V$ be any finite dimensional subspace and let $E : V \to W$ be the orthogonal projection, which exists by Theorem V.10. Then

$$\lambda E = ETE = \int_K E\pi(g)\langle \pi(g)v, E\cdot\rangle dg.$$

Taking trace (which makes sense as the image lies in W) on both sides gives

$$\lambda \dim W = \int_K \mathrm{tr}\left(E\pi(g)\langle \pi(g)v, E\cdot\rangle\right) dg = \int_K \|E\pi(g)v\|^2\, dg \leqslant \mathrm{Vol}(K) < \infty.$$

Here Parseval's identity is used. Hence $\dim W$ has a uniform bound, implying $\dim V < \infty$ in turn.

Remark 8.9. There is alternative proof of lemma in Theorem that is independent of the Peter-Weyl theorem. Let (π, V_π) be a unitary representation of a compact group K. Let $T_0 \in \mathrm{Hom}_{\mathbb{C}}(V_\pi, V_\pi)$ be the orthogonal projection onto a nonzero finite dimensional subspace W of V. Define $T \in \mathrm{Hom}_{\mathbb{C}}(V_\pi, V_\pi)$ by

$$Tv = \int_K \pi(g)^{-1}T_0\pi(g)v dg.$$

The integrand lands in a finite dimensional subspace W, so T is of finite rank. Moreover, T is positive as

$$\langle Tv, w\rangle = \int_K \langle T_0\pi(g)v, \pi(g)w\rangle dg = \int_K \langle \pi(g)v, T_0\pi(g)w\rangle dg = \langle v, Tw\rangle$$

and $\langle Tv, v\rangle \geqslant 0$. Now by Spectral theorem for compact normal operators, there exists $\lambda \in \mathbb{C}^\times$ such that $0 < \dim\ker(T - \lambda\,\mathrm{id}) < \infty$. In other words, we find a nonzero finite dimensional subrepresentation of V_π.

8.2.1 Characters

Definition. Let π be a finite dimensional representation of a compact group K. The function $\chi_\pi : K \to \mathbb{C}$ defined by

$$\chi_\pi(k) := \operatorname{tr} \pi(k)$$

is called the **character** of the representation π.

- We can write χ_π as a sum of matrix coefficients. Let (e_i) be the chosen orthonormal basis for V_π. Then

$$\operatorname{tr} \pi(k) = \sum_{i=1}^{\dim(\pi)} \langle \pi(k)e_i, e_i \rangle = \sum_{i=1}^{\dim(\pi)} \pi_{ii}(k)$$

- If π and τ are unitarily equivalent, then $\chi_\pi = \chi_\tau$. To see this, say $\alpha : \pi \to \tau$ is a unitary K-isomorphism. If e_i is an orthonormal basis for π, then for all $k \in K$

$$\operatorname{tr} \tau(k) = \sum \langle \tau(k)\alpha e_i, \alpha e_i \rangle = \sum \langle \alpha\pi(k)e_i, \alpha e_i \rangle = \sum \langle \pi(k)e_i, e_i \rangle = \operatorname{tr} \pi(k)$$

Theorem 8.10. Let π, η be two finite dimensional irreducible unitary representations of the compact group K. For their characters, we have

$$\langle \chi_\pi, \chi_\eta \rangle = \delta_{\pi\eta} = \begin{cases} 1 & \text{, if } \pi = \tau \text{ in } \widehat{K} \\ 0 & \text{, else} \end{cases}$$

Here the inner product is the one of $L^2(K)$.

Proof. By (a) and (b) of Peter-Weyl theorem,

$$\langle \chi_\pi, \chi_\eta \rangle = \sum_{i=1}^{\dim(\pi)} \sum_{j=1}^{\dim(\tau)} \langle \pi_{ii}, \tau_{kk} \rangle = \sum_{i=1}^{\dim(\pi)} \sum_{j=1}^{\dim(\tau)} \delta_{\pi\tau}\delta_{ij} \frac{1}{\dim(\pi)} = \delta_{\pi\tau}$$

□

Let π be any finite dimensional representation of K. By Proposition 8.3 we can write π as a direct sum of K-subrepresentations. This shows χ_π is a finite sum of characters of representations appearing in π. In particular,

Corollary 8.10.1. Let π be a finite dimensional representation of K. Then π is irreducible if and only if $\langle \chi_\pi, \chi_\pi \rangle = 1$.

8.2.2 Fourier Transform

We now give a reformulation of the Peter-Weyl theorem. We have a unitary representation η of the group $K \times K$ on the Hilbert space $L^2(K)$ given by

$$\eta(k_1, k_2)f(x) := f(k_1^{-1}xk_2)$$

On the other hand, for $(\tau, V_\tau) \in \widehat{K}$, the group $K \times K$ acts on the finite dimensional vector space $\operatorname{End} V_\tau = \operatorname{Hom}_K(V_\tau, V_\tau)$ by

$$\eta_\tau(k_1, k_2)T = \tau(k_1)T\tau(k_2^{-1})$$

On $\operatorname{End} V_\tau$, we have a natural inner product

$$\langle S, T\rangle = \dim_{\mathbb{C}}(V_\tau)\operatorname{tr}(ST^*)$$

making the representation of $K \times K$ unitary.

Theorem 8.11 (Peter-Weyl)**.** There is a natural unitary isomorphism

$$L^2(K) \cong \widehat{\bigoplus_{\tau\in\widehat{K}}} \operatorname{End} V_\tau$$

which intertwines the conjugation representation η of $K \times K$ on $L^2(K)$ with $\bigoplus_{\tau\in\widehat{K}} \eta_\tau$.

This isomorphism maps a given $f \in L^2(K) \subseteq L^1(K)$ to $\sum_{\tau\in\widehat{K}} \tau(f)$, where

$$\tau(f) := \int_K f(x)\tau(x)dx$$

In particular, if for a given $f \in L^2(K)$, we define the **Fourier transform**

$$\begin{aligned} \widehat{f} : \widehat{K} &\longrightarrow \widehat{\bigoplus}_{\tau\in\widehat{K}} \operatorname{End} V_\tau \\ \tau &\longmapsto \tau(f) \end{aligned}$$

then

(i) $\widehat{f * g}(\tau) = \tau(f) \circ \tau(f)$ for $f, g \in L^2(G)$ and $\tau \in \widehat{K}$,

(ii) $\tau(f)^* = \tau(f^*)$ for $\tau \in \widehat{K}$, $f \in L^2(G)$ (where $f^*(x) := \overline{f(x^{-1})}$, c.f. Proposition 2.24),

(iii) $\|f\| = \left\|\widehat{f}\right\|$ for every $f \in L^2(K)$.

In this way the Peter-Weyl theorem presents itself as a generalization of the Plancherel formula to the compact groups.

For $\sum_{\tau\in\widehat{K}} g(\tau) \in \widehat{\bigoplus_{\tau\in\widehat{K}}} \operatorname{End} V_\tau$, define the **inverse transform** $\check{g} \in L^2(K)$ by

$$\check{g}(x) = \sum_{\tau\in\widehat{K}} \dim(\tau)\operatorname{tr}(g(\tau)\tau(x^{-1}))$$

This is the inverse of the isomorphism $f \mapsto \widehat{f}$.

Proof. Since $\tau \mapsto \tau^*$ is a bijection on $\widehat{K}$, (c) of Peter-Weyl theorem yields the orthonormal basis $\sqrt{\dim(\tau)}\tau^*_{k\ell}$ for $L^2(K)$. For $f \in L^2(K)$ and indices i, j, one has

$$\langle \tau(f)e_i, e_j\rangle = \int_K f(x)\tau_{ij}(x)dx = \langle f, \overline{\tau_{ij}}\rangle$$

where the e_i are the orthonormal basis for V_τ. Applying this formula to $f = \sigma^*_{k\ell} = \overline{\sigma_{k\ell}}$ for some $\sigma \in \widehat{K}$, we see $\widehat{\sigma^*_{k\ell}}(\tau) = \tau(\overline{\sigma_{k\ell}}) = 0$ for $\sigma \neq \tau$, and

$$\langle \widehat{\tau^*_{k\ell}}(\tau)e_i, e_j\rangle = \frac{1}{\dim(\tau)}\delta_{ki}\delta_{\ell j}$$

It follows that $\tau^*_{k\ell}$ is mapped to the operator $\frac{1}{\dim(\tau)}E^\tau_{k\ell} \in \operatorname{End} V_\tau$, where $E^\tau_{k\ell}$ is given by $e_i \mapsto \delta_{ik}e_\ell$. Hence the basis element $\sqrt{\dim(\tau)}\tau^*_{k\ell}$ is mapped to $\dim(\tau)^{-\frac{1}{2}}E^\tau_{k\ell}$. We verify the maps $\sqrt{\dim(\tau)}E^\tau_{k\ell}$ form an orthonormal basis for $\operatorname{End} V_\tau$, which will conclude the proof.

$$\langle \dim(\tau)^{-\frac{1}{2}}E^\tau_{ij}, \dim(\tau)^{-\frac{1}{2}}E^\tau_{k\ell}\rangle = \operatorname{tr}(E^\tau_{ij}(E^\tau_{k\ell})^*) = \sum_{s=1}^{\dim(\tau)} \langle (E^\tau_{k\ell})^*e_s, (E^\tau_{ij})^*e_s\rangle = \sum_{s=1}^{\dim(\tau)} \delta_{s\ell}\delta_{sj}\delta_{ik} = \delta_{\ell j}\delta_{ki}$$

For the inverse map, we compute

$$\operatorname{tr}(E^\tau_{k\ell}\tau(x^{-1})) = \sum_{i=1}^{\dim(\tau)} \langle E^\tau_{k\ell}(x^{-1})e_i, e_i\rangle = \sum_{i=1}^{\dim(\tau)} \langle \tau(x^{-1})e_i, \delta_{\ell i}e_k\rangle = \tau^*_{k\ell}(x)$$

so $\dim(\tau)^{-\frac{1}{2}}E^\tau_{k\ell}$ is mapped, under the inverse transform, back to $\sqrt{\dim(\tau)}\tau^*_{k\ell}$. □

For a compact group K, a function $f : K \to \mathbb{C}$ is called a **class function** if $f(k^{-1}xk) = f(x)$ for all $x, k \in K$. In other words, f descends to a function on the conjugacy class of K. Denote by $L^2(K/\text{conj})$ the space of all square-integrable class functions of K.

Corollary 8.11.1. $L^2(K/\text{conj})$ is a closed subspace of $L^2(K)$, and $(\chi_\pi)_{\pi\in\check{K}}$ forms an orthonormal basis for $L^2(K/\text{conj})$.

Proof. $L^2(K/\text{conj}) = \bigcap_{k\in K} \ker(R_k L_k - \text{id})$, so it is a closed subspace of $L^2(K)$. For the second assertion, since each trace χ_π is a class function, it suffices to show if $f \in L^2(K/\text{conj})$ such that $\langle f, \chi_\pi\rangle = 0$ for all $\pi \in \widehat{K}$, then $f = 0$. By Peter-Weyl theorem we must show the Fourier transform $\widehat{f}$ vanishes, i.e., $\pi(f) = 0$ for all $\pi \in \widehat{K}$.

We first show $\pi(f) \in \operatorname{End} V_\pi$ is K-intertwining. For all $k \in K$, we compute

$$\pi(k)\pi(f) = \int_K f(x)\pi(kx)dx = \int_K f(k^{-1}x)\pi(x)dx = \int_K f(xk^{-1})\pi(x)dx = \int_K f(x)\pi(xk)dx = \pi(f)\pi(k)$$

By Schur's lemma, there exists $\lambda \in \mathbb{C}$ such that $\pi(f) = \lambda \operatorname{id}_{V_\pi}$. Now taking trace, one sees

$$\lambda \dim(\pi) = \operatorname{tr} \pi(f) = \int_K f(x) \operatorname{tr} \pi(x)dx = \langle f(x), \chi_{\pi^*}\rangle = 0$$

so that $\lambda = 0$, i.e., $\pi(f) = 0$. □

Corollary 8.11.2. For $f \in \operatorname{span}(L^2(K) * L^2(K)) \subseteq C(K)$ (c.f. Lemma 4.14), we have

$$f(k) = \sum_{\pi\in\widehat{K}} \dim(\pi)\widehat{R_k f}(\pi).$$

Proof. It suffices to consider $f = f_1 * f_2$ with $f_i \in L^2(K)$. For $k \in K$, by Lemma 1.33 we have $R_k f = f_1 * R_k f_2$. Put $f_3(g) = \overline{(R_k f_2)(g^{-1})} = (R_k f_2)^*(g)$ Then

$$\begin{aligned} f(k) = R_k f(e) &= \int_K f_1(g)(R_k f_2)(g^{-1})dg = \int_K f_1(g)\overline{f_3(g)}dg = \langle f_1, f_3\rangle_{L^2(K)} = \langle \widehat{f_1}, \widehat{f_3}\rangle \\ &= \sum_{\tau\in\widehat{K}} \dim(\tau) \operatorname{tr}(\tau(f_1) \circ \tau(f_3)^*) = \sum_{\tau\in\widehat{K}} \dim(\tau) \operatorname{tr}(\tau(f_1 * f_3^*)) = \sum_{\tau\in\widehat{K}} \dim(\tau) \operatorname{tr}(\tau(f_1 * R_k f_2)) \\ &= \sum_{\tau\in\widehat{K}} \dim(\tau) \operatorname{tr}(\tau(R_k(f_1 * f_2))) = \sum_{\tau\in\widehat{K}} \dim(\tau)\widehat{R_k f}(\tau) \end{aligned}$$

□

Definition. Let (τ, V_τ) and (γ, V_γ) be finite dimensional representations of the compact group K. There is a natural representation of the group $K \times K$ on the tensor product space $V_\tau \otimes V_\gamma$ given by

$$(\tau \otimes \gamma)(k_1, k_2) = \tau(k_1) \otimes \gamma(k_2) \in \operatorname{GL}(V_\tau \otimes V_\gamma)$$

Lemma 8.12. For a given $\tau \in \widehat{K}$, there is a natural unitary isomorphism

$$\psi : V_\tau \otimes V_{\tau^*} \to \operatorname{End} V_\tau \in \operatorname{Isom}_{K\times K}(\tau \otimes \tau^*, \eta_\tau)$$

Proof. The map ψ given by

$$\psi(v \otimes \alpha) = [w \mapsto \alpha(w)v] \in \operatorname{End} V_\tau$$

is linear and sends the simple tensors onto to the subspace of operators of rank one, so ψ is surjective since $\operatorname{End} V_\tau$ is generated by the operator of ranks one. Since $\dim V_\tau \otimes V_{\tau*} = \dim \operatorname{End} V_\tau$, ψ is bijective. For $k, \ell \in K$, one has

$$\begin{aligned}
\psi(\tau \otimes \tau^*(k,\ell)(v \otimes \alpha))(w) &= \psi(\tau(k)v \otimes \tau^*(\ell)\alpha)(w) \\
&= \tau^*(\ell)\alpha(w)\tau(k)v = \alpha(\tau(\ell^{-1})w)\tau(k)v \\
&= \tau(k)(\alpha(\tau(\ell^{-1})w)v) \\
&= \tau(k)\psi(v \otimes \alpha)(\tau(\ell^{-1})w) \\
&= \eta_\tau(k,\ell)\psi(v \otimes \alpha)(w)
\end{aligned}$$

Let $e_1, \ldots, e_n$ be an orthonormal basis for V_τ. We compute the adjoint $\psi(v \otimes \alpha)^*$; we have

$$\psi(v \otimes \alpha)e_i = \alpha(e_i)v = \sum_{j=1}^n \alpha(e_i)\langle v, e_j\rangle e_j$$

so the adjoint is $\psi(v \otimes \alpha)^* e_i = \sum_{j=1}^n \overline{\alpha(e_j)\langle v, e_i\rangle} e_j$. Then

$$\begin{aligned}
\langle \psi(v \otimes \alpha), \psi(v \otimes \alpha)\rangle &= \dim(\tau)\operatorname{tr}(\psi(v \otimes \alpha)\psi(v \otimes \alpha)^*) \\
&= \dim(\tau)\sum_{i,j=1}^n \alpha(e_i)\langle v, e_j\rangle\overline{\alpha(e_i)\langle v, e_j\rangle} \\
&= \dim(\tau)\,\|\alpha\|^2\,\|v\|^2 = \langle v \otimes \alpha, v \otimes \alpha\rangle
\end{aligned}$$

Then $\Psi := \dim(\tau)^{-\frac{1}{2}}\psi$ is a unitary intertwining isomorphism. □

Corollary 8.12.1. There is a natural unitary isomorphism

$$L^2(K) \cong \widehat{\bigoplus_{\tau \in \widehat{K}}} V_\tau \otimes V_{\tau*}$$

that intertwines the conjugation representation η with $\bigoplus_{\tau \in \widehat{K}} \tau \otimes \tau^*$. In particular, we have

$$L \cong \widehat{\bigoplus_{\tau \in \widehat{K}}} 1_{V_\tau} \otimes \tau^* \qquad \text{and} \qquad R \cong \widehat{\bigoplus_{\tau \in \widehat{K}}} \tau \otimes 1_{V_{\tau*}}$$

for the left and right regular representations of K. Here 1 means the trivial representation.

Proof. The first isomorphism results from Peter-Weyl and the previous lemma. The second assertion follows from restricting to one fact of the group $K \times K$. □

8.2.3 K-finite vectors and matrix coefficients

Definition. Let G be a group and V a representation of G. A vector $v \in V$ is called **G-finite** if its orbit Gv is contained in some finite dimensional subspace of V. Denote by V^{fin} the subspace of all G-finite vectors.

- It is clear that V^{fin} is invariant under G action.

Let K be a compact group. Consider the subspace $C(K)^{\text{fin}}$ of finite vectors in $C(K)$. We let K acts on $C(K)^{\text{fin}}$ from the left. If $\pi \in \widehat{K}$, then the matrix coefficient $k \mapsto \langle \pi(k)v, w\rangle$ is clearly K-finite, as it is contained in M_π (c.f. the notation in the proof of (c) of Peter-Weyl), which is a finite dimensional K-subrepresentation of $C(K)$. In fact,

Lemma 8.13. $C(K)^{\text{fin}} = \bigoplus_{\pi \in \widehat{K}} M_\pi$ consists exactly of matrix coefficients of K.

Proof. Let $f \in C(K)^{\text{fin}}$ and V a finite dimensional K-subrepresentation of $C(K)$ containing f. By Riesz's Representation theorem, let $\alpha \in \overline{V}$ be the unique element such that $\langle h, \alpha\rangle = h(e)$ for all $h \in \overline{V}$. Then

$$\overline{f}(k) = (\ell_{g^{-1}}\overline{f})(e) = \langle \ell_{k^{-1}}\overline{f}, \alpha\rangle = \langle \overline{f}, \ell_g \alpha\rangle$$

so that $f(k) = \langle \ell_g \alpha, \overline{f}\alpha\rangle$ is a matrix coefficient of $\overline{V}$. $\square$

The same argument shows that $\bigoplus_{\pi \in \widehat{K}} M_\pi$ consists of all K-finite vectors when the action of K acts from the right. In particular, this shows

Corollary 8.13.1. View $C(K)$ as an $K \times K$-representation. Then the matrix coefficients are exactly those $K \times K$-finite vectors in $C(K)$. Using various interpretations, we have

$$C(K)^{\text{fin}} = \bigoplus_{\pi \in \widehat{K}} M_\pi \cong \bigoplus_{\tau \in \widehat{K}} V_\tau \otimes V_\tau^* \cong \bigoplus_{\tau \in \widehat{K}} \operatorname{End} V_\tau$$

In the rest of this subsubsection, we change our notation: let G be a compact group, and put

$$R(G, \mathbb{C}) = C(G)^{\text{fin}} = \bigoplus_{\pi \in \widehat{G}} M_\pi$$

to be the $\mathbb{C}$-subalgebra of $C(G)$ consisting of all matrix coefficients. Similarly, put

$$R(G, \mathbb{R}) = R(G, \mathbb{C}) \cap C(G, \mathbb{R})$$

to be the $\mathbb{R}$-subalgebra of $R(G, \mathbb{C})$ consisting of real-valued functions. Since the complex conjugation leaves $R(G, \mathbb{C})$ invariant, we have

$$R(G, \mathbb{C}) = R(G, \mathbb{R}) \otimes_{\mathbb{R}} \mathbb{C}$$

under the natural identification $C(G) = C(G, \mathbb{R}) \otimes_{\mathbb{R}} \mathbb{C}$.

8.3 Isotypes

Let (π, V_π) be a unitary representation of the compact group K. For $(\tau, V_\tau) \in \widehat{K}$, define the **isotype** of τ, or the **isotypical component** of τ in π as the subspace

$$V_\pi[\tau] := \sum_{\substack{U \subseteq V_\pi \\ U \cong V_\tau}} U$$

This is the sum of all invariant subspaces U which are K-isomorphic to V_τ. Alternatively one can describe $V_\pi[\tau]$ as follows. There is a canonical map

$$\begin{aligned} T_\tau : \mathrm{Hom}_K(V_\tau, V_\pi) \widehat{\otimes} V_\tau &\longrightarrow V_\pi \\ \alpha \otimes v &\longmapsto \alpha(v) \end{aligned}$$

By definition we have $T_\tau \in \mathrm{Hom}_K(1 \otimes \tau, \pi)$. We claim the image of T_τ in V_π is the whole $V_\pi[\tau]$.

- Let $0 \neq \alpha \in \mathrm{Hom}_K(V_\tau, V_\pi)$; since V_τ is irreducible, α is injective. Therefore $T_\tau|_{\alpha \otimes V_\tau} = \alpha$ is a K-homomorphism mapping onto a subspace of V_π that is K-isomorphic to V_τ.
- Conversely, let $U \subseteq V_\pi$ be a subspace with $\pi|_U \cong \tau$ via the map $\alpha : V_\tau \to U$. Then $U = T_\tau(\alpha \otimes V_\tau)$.

Note also that if (τ, V_τ) and (σ, V_σ) are non-equivalent irreducible representations of K, then $V_\pi[\tau] \perp V_\pi[\sigma]$. This follows from the fact that if $U, U' \subseteq V_\pi$ are subspaces with $U \cong V_\tau$, $U' \cong V_\sigma$, then the orthogonal projection $P : V_\pi \to U'$ restricts to a K-homomorphism $P|_U : U \to U'$, which must be zero.

Lemma 8.14. On the vector space $\mathrm{Hom}_K(V_\tau, V_\pi)$ there exists an inner product making it a Hilbert space such that T_τ is an isometry.

Proof. Let $v_0 \in V_\tau$ be a norm-one vector. For $\alpha, \beta \in \mathrm{Hom}_K(V_\tau, V_\pi)$, define

$$\langle \alpha, \beta \rangle := \langle \alpha(v_0), \beta(v_0) \rangle$$

Since an element in $\mathrm{Hom}_K(V_\tau, V_\pi)$ is either zero or injective, it follows at once that $\langle\, , \rangle$ defines an inner product on $\mathrm{Hom}_K(V_\tau, V_\pi)$. We claim with this inner product, the space $\mathrm{Hom}_K(V_\tau, V_\pi)$ is complete. For this let $(\alpha_n)_n$ be a Cauchy sequence in $\mathrm{Hom}_K(V_\tau, V_\pi)$. By definition $(\alpha_n(v_0))_n$ is a Cauchy sequence in V_π; say $\alpha_n(v_0) \to w_0 \in V_\pi$. Then for all $k \in K$, the sequence $\alpha_n(\tau(k)v_0) = \pi(k)\alpha_n(v_0) \to \pi(k)w_0$. Likewise, for each $f \in L^1(K)$, we have

$$\alpha_n(\tau(f)v_0) = \int_K f(k)\alpha_n(\tau(k)v_0)dk = \int_K f(k)\pi(k)\alpha_n(v_0)dk = \pi(f)\alpha_n(v_0) \to \pi(f)w_0 \qquad (\spadesuit)$$

Consider the annihilator $\operatorname{ann} v_0 = \operatorname{ann}_{L^1(K)}(v_0) \subseteq L^1(K)$ of v_0, i.e,

$$\operatorname{ann}_{L^1(K)}(v_0) := \{f \in L^1(K) \mid \tau(f)v_0 = 0\}$$

Since V_τ is irreducible and $\dim_{\mathbb{C}} V_\tau < \infty$, the map $\begin{array}{ccc} L^1(K) & \longrightarrow & V_\tau \\ f & \longmapsto & \tau(f)v_0 \end{array}$ induces a K-isomorphism $V_\tau \cong L^1(K)/\operatorname{ann} v_0$. By (♠) we have $\operatorname{ann} v_0 \subseteq \operatorname{ann} w_0$, and therefore the map

$$\begin{array}{ccc} \alpha : V_\tau \cong L^1(K)/\operatorname{ann} v_0 & \longrightarrow & V_\pi \\ \tau(f)v_0 & \longmapsto & \pi(f)w_0 \end{array}$$

is well-defined and is K-intertwining. It follows that $\alpha_n \to \alpha$ in $\operatorname{Hom}_K(V_\tau, V_\pi)$.

We now show T_τ is an isometry. For a fixed $\alpha \in \operatorname{Hom}_K(V_\tau, V_\pi)$, the inner product on V_τ defined by $(v, w) := \langle \alpha(v), \alpha(w) \rangle$ is K-invariant, so by Schur's lemma there exists $c(\alpha) > 0$ such that $(v, w) = c(\alpha)\langle v, w \rangle$ for all $v, w \in V_\tau$. Thus

$$\langle T_\tau(\alpha \otimes v), T_\tau(\alpha \otimes v) \rangle = (v, v) = c(\alpha)\langle v, v \rangle$$

Setting $v = v_0$ we conclude that $c(\alpha) = \langle \alpha, \alpha \rangle$, which proves T_τ is an isometry. □

It follows from the above lemma that $V_\pi[\tau]$ is isometrically isomorphic to Hilbert space tensor product $\operatorname{Hom}_K(V_\tau, V_\pi) \widehat{\otimes} V_\tau$, and that $\pi|_{V_\pi(\tau)}$ is unitarily equivalent to the representation $1 \otimes \tau$ on the tensor product. If we choose an orthonormal basis $(\alpha_i)_{i \in I}$ of $\operatorname{Hom}_K(V_\tau, V_\pi)$, then we get an isomorphism

$$\begin{array}{ccc} \operatorname{Hom}_K(V_\tau, V_\pi) \widehat{\otimes} V_\tau & \xrightarrow{\sim} & \widehat{\bigoplus_{i \in I}} V_\tau \\ \alpha \otimes v & \longmapsto & \sum_{i \in I} \langle \alpha, \alpha_i \rangle v \end{array}$$

Thus we see that $V_\pi(\tau)$ is unitarily equivalent to a direct sum of V_τ's with multiplicity $\#I = \dim_{\mathbb{C}} \operatorname{Hom}_K(V_\tau, V_\pi)$.

Theorem 8.15. Let K be a compact group.

(a) $V_\pi[\tau]$ is a closed invariant subspace of V_π.

(b) $V_\pi[\tau]$ is K-isomorphic to a Hilbert space direct sum of copies of V_τ.

(c) V_π is the Hilbert space direct sum of the isotypes $V_\pi[\tau]$, where τ runs over $\widehat{K}$.

Proof. Since $V_\pi[\tau]$ is an isometric image of a complete space, it is itself complete, whence closed in V_π. Since $V_\pi[\tau]$ is a sum of invariant subspaces, it is invariant as well. Now let $V_\pi = \widehat{\bigoplus}_i V_i$ be any decomposition into irreducibles. Set

$$\tilde{V}_\pi(\tau) := \widehat{\bigoplus_{i : V_i \cong V_\tau}} V_i \subseteq V_\pi$$

It follows that $\tilde{V}_\pi(\tau) \subseteq V_\pi[\tau]$ since the latter contains the direct sum and is closed. Now clearly V_π is the Hilbert space direct sum of all spaces $\tilde{V}_\pi(\tau)$, and hence it is also that of all spaces $V_\pi[\tau]$, as the latter are pairwise orthogonal. This implies (c), and a fortiori $\tilde{V}_\pi(\tau) = V_\pi[\tau]$, and thus (b). □

By the lemma, for a unitary representation (π, V_π) of a compact groups K, there are K-equivariant isomorphisms

$$V_\pi \cong \widehat{\bigoplus_{(\tau,V_\tau)\in\widehat{K}}} V_\pi[\tau] \cong \widehat{\bigoplus_{(\tau,V_\tau)\in\widehat{K}}} \mathrm{Hom}_K(V_\tau, V_\pi) \widehat{\otimes} V_\tau$$

The next proposition gives an explicit formula for the first isomorphism.

Proposition 8.16. Let (π, V_π) be a unitary representation of the compact group K. For $\tau \in \widehat{K}$, the orthogonal projection $P : V_\pi \to V_\pi[\tau]$ is given by the formula

$$P(v) = \dim(\tau) \int_K \overline{\chi_\tau}(x)\pi(x)v dx$$

Proof. We need to show for each $v, w \in V_\pi$ we have

$$\langle Pv, w\rangle = \dim(\tau) \int_K \overline{\chi_\tau}(x)\langle \pi(x)v, w\rangle\, dx$$

Denote the right hand side by (v, w). Write $v = v_0 + v_1$ where $v_0 \in V_\pi[\tau]$, $v_1 \in V_\pi[\tau]^\perp$. Decompose $w = w_0 + w_1$ likewise. Then $\langle Pv, w\rangle = \langle v_0, w_0\rangle$. We claim

$$\begin{aligned}(v_0, w_0) &= \langle v_0, w_0\rangle \\ (v_i, w_j) &= 0 \text{ for } (i, j) \neq (0, 0)\end{aligned}$$

Then since $(\ ,\ )$ is bilinear, it follows $(v, w) = (v_0, w_0) = \langle v_0, w_0\rangle = \langle Pv, w\rangle$, as wanted.

To show the claim, decompose $V_\pi[\tau]$ into irreducibles, each equivalent to V_τ. Since the sum is orthogonal, we may assume v_0, w_0 lie in the same component. If we let (e_i) be an orthonormal basis for V_τ, then by (b) of Peter-Weyl theorem,

$$\begin{aligned}(v_0, w_0) &= \dim(\tau) \int_K \overline{\chi_\tau(x)}\langle \pi(x)v_0, w_0\rangle dx \\ &= \dim(\tau) \sum_{i,j} \langle v_0, e_i\rangle \overline{\langle w_0, e_j\rangle} \int_K \sum_k \overline{\tau_{kk}(x)}(\pi|_{V_\tau})_{ij}(x) dx \\ &= \sum_i \langle v_0, e_i\rangle \overline{\langle w_0, e_i\rangle} = \langle v_0, w_0\rangle\end{aligned}$$

For the remaining equalities, since $V_\pi[\tau]$ and $V_\pi[\tau]^\perp$ are invariant under K, we have $(v_1, w_0) = 0 = (v_0, w_1)$, and since $V_\pi[\tau]^\perp$ is an orthogonal direct sum of isotypes other than τ, (a) of Peter-Weyl theorem (and a similar computation to the above) shows $(v_1, w_1) = 0$. □

Example 8.17. Let us consider the regular representation of a compact group K. From the proof of Peter-Weyl theorem, we see there is a direct sum decomposition

$$L^2(K) = \overline{\bigoplus_{\tau\in\widehat{K}} M_\tau}$$

where for each $\tau \in \widehat{K}$, the finite dimensional subspace M_τ is the linear span of all matrix coefficients of τ and is invariant under left and right translations. It is then easy to see that $L^2(K)_R[\tau] = M_\tau$ and $L^2(K)_L[\tau] = M_{\tau*}$. Alternatively, one can verify this by the last proposition. For instance, for $f \in L^2(K)$, we compute

$$\int_K \overline{\chi_\tau}(x) L_x f(k) dx = \int_K \overline{\chi_\tau}(kx^{-1}) f(x) dx = \sum_{i,j=1}^{\dim(\tau)} \overline{\tau_{ji}}(k) \left(\int_K \overline{\tau_{ij}}(x^{-1}) f(x) dx \right) = \sum_{i,j=1}^{\dim(\tau)} \langle f, \overline{\tau_{ij}} \rangle \overline{\tau_{ij}}(k)$$

so $L^2(K)_L[\tau]$ is spanned by the $\overline{\tau_{ij}}$.

Example 8.18 (Irreducible representations of product groups)**.** Let G_1, G_2 be compact groups. Let W be a unitary representation of $G_1 \times G_2$. If we think it of a representation of $\{1\} \times G_2 \cong G_2$, there is a decomposition

$$W = \widehat{\bigoplus_{(\tau, V_\tau) \in \widehat{G_2}}} \mathrm{Hom}_{G_2}(V_\tau, W) \widehat{\otimes} V_\tau$$

For $T \in \mathrm{Hom}_{G_2}(V_\tau, W)$ and $g \in G_1$, define $g.T \in \mathrm{Hom}_{G_2}(V_\tau, W)$ by $(g.T)(x) = (g, 1)T(x)$; this is indeed G_2-equivariant, as for $g' \in G_2$ and $v \in V_\tau$,

$$(g.T)(g'v) = (g, 1)T(g'.v) = (g, 1)(1, g')T(v) = (1, g')(g.1)T(v) = (1, g')(g.T)(v).$$

If we equip each hom set with such G_1-actions, this decomposition is in fact $G_1 \times G_2$-equivariant. Indeed, for $\alpha \otimes v \in \mathrm{Hom}_{G_2}(V_\tau, W) \widehat{\otimes} V_\tau$,

$$(g_1.\alpha)(g_2.v) = (g_1, 1)\alpha(g_2.v) = (g_1, 1)(1, g_2)\alpha(v) = (g_1, g_2)\alpha(v).$$

In sum, we obtain a $G_1 \times G_2$-equivariant isomorphism

$$W = \widehat{\bigoplus_{(\tau, V_\tau) \in \widehat{G_2}}} \mathrm{Hom}_{G_2}(V_\tau, W) \widehat{\otimes} V_\tau.$$

Recall on $\mathrm{Hom}_{G_2}(V_\tau, W)$ we defined a Hilbert space structure in the proof of Lemma 8.14: if $v_0 \in V_\tau$ is a norm-one vector, for $\alpha, \beta \in \mathrm{Hom}_{G_2}(V_\tau, W)$, we put

$$\langle \alpha, \beta \rangle := \langle \alpha(v_0), \beta(v_0) \rangle.$$

For $g \in G_1$, since W is unitary

$$\langle g.\alpha, g.\beta \rangle := \langle (g, 1)\alpha(v_0), (g, 1)\beta(v_0) \rangle = \langle \alpha(v_0), \beta(v_0) \rangle = \langle \alpha, \beta \rangle.$$

This means the just defined G_1-action on $\mathrm{Hom}_{G_2}(V_\tau, W)$ is unitary. Hence we have a G_1-equivariant decomposition

$$\mathrm{Hom}_{G_2}(V_\tau, W) = \widehat{\bigoplus_{(\sigma, V_\sigma) \in \widehat{G_1}}} \mathrm{Hom}_{G_1}(V_\sigma, \mathrm{Hom}_{G_2}(V_\tau, W)) \widehat{\otimes} V_\sigma$$

The usual adjunction gives a G_1-equivariant isomorphism

$$\mathrm{Hom}_{G_1}(V_\sigma, \mathrm{Hom}_{G_2}(V_\tau, W)) \cong \mathrm{Hom}_{G_1\times G_2}(V_\sigma \otimes V_\tau, W).$$

Putting things together yields

$$W = \widehat{\bigoplus_{(\sigma,\tau)\in\widehat{G_1}\times\widehat{G_2}}} \mathrm{Hom}_{G_1\times G_2}(V_\sigma \otimes V_\tau, W) \widehat{\otimes} (V_\sigma \otimes V_\tau)$$

It is easy to see that $V_\sigma \otimes V_\tau$ is irreducible as a $G_1 \times G_2$ representation: indeed, $\chi_{V_\sigma\otimes V_\tau} = \chi_\sigma \otimes \chi_\tau$, so

$$\int_{G_1\times G_2} |\chi_{V_\sigma\otimes V_\tau}(g_1, g_2)|^2 dg_1 \otimes dg_2 = \int_{G_1} |\chi_\sigma(g_1)|^2 dg_1 \cdot \int_{G_2} |\chi_\tau(g_2)|^2 dg_2 = 1 \cdot 1 = 1.$$

The irreducibility then follows from Corollary 8.10.1. Putting ourselves in the case $W = L^2(G_1 \times G_2)$, part (d) of Peter-Weyl shows that $V_\sigma \otimes V_\tau$ exhausts $\widehat{G_1 \times G_2}$, demonstrating the bijection

$$\widehat{G_1} \times \widehat{G_2} \cong \widehat{G_1 \times G_2}$$

By this, the natural inclusion $C(G_1, \mathbb{R}) \otimes_{\mathbb{R}} C(G_2, \mathbb{R}) \to C(G_1 \times G_2, \mathbb{R})$ restricts to an algebra isomorphism

$$R(G_1, \mathbb{R}) \otimes_{\mathbb{R}} R(G_2, \mathbb{R}) \cong R(G_1 \times G_2, \mathbb{R}).$$

Clearly the above remains valid with all $\mathbb{R}$ replaced by $\mathbb{C}$.

8.4 Induced Representations

Let K be a compact group and $M \leqslant K$ a closed subgroup. Let (σ, V_σ) be a unitary representation of M. We are going to define the **induced representation** $\pi_\sigma = \mathrm{Ind}_M^K \sigma$. First, define the space

$$L^2(K, V_\sigma) := \left\{ f : K \to V_\sigma \mid f \text{ is measurable with } \int_K \|f(x)\|_\sigma^2 \, dk < \infty \right\} / \sim$$

where $f \sim g$ if and only if $f - g$ is a null function, where $\|\cdot\|_\sigma$ is the norm on V_σ. On $L^2(K, V_\sigma)$ there is an inner product

$$\langle f, g\rangle := \int_K \langle f(k), g(k)\rangle_\sigma dk$$

where $\langle\, ,\rangle_\sigma$ is the inner product on V_σ. By taking an orthonormal basis (e_i) of V_σ, we see there is an isometric isomorphism $L^2(K, V_\sigma) \cong L^2(K) \widehat{\otimes} V_\sigma$; in particular, this shows $L^2(K, V_\sigma)$ is a Hilbert space.

The representation space of π_σ is the subspace

$$\mathrm{Ind}_M^K V_\sigma := \left\{ f \in L^2(K, V_\sigma) \mid \text{for all } m \in M,\ f(mk) = \sigma(m) f(k) \text{ a.e. in } k \in K \right\}$$

This is a closed subspace of $L^2(K, V_\sigma)$, as if for each $m \in M$ we define $T_m \in \mathcal{B}(L^2(K, V_\sigma))$ by $T_m(f) = L_{m^{-1}} f - \sigma(m) f$, then

$$\operatorname{Ind}_M^K V_\sigma = \bigcap_{m \in M} \ker T_m$$

The representation π_σ is defined as the right translation:

$$\pi_\sigma(y) f(x) = f(xy)$$

Since K is unimodular, π_σ is clearly unitary. The following theorem allows us to compute the irreducible components in the induced representation $\operatorname{Ind}_M^K V_\sigma$. We begin with an easy lemma.

Lemma 8.19. The subspace $C(K, V_\sigma)$ of all continuous functions $K \to V_\sigma$ is dense in $L^2(K, V_\sigma)$.

Proof. Obviously we have $C(K, V_\sigma) \cong C(K) \otimes V_\sigma$, so the assertion follows. Explicitly, since the action π_σ on $L^2(K, V_\sigma)$ is isomorphic to the tensor $R \otimes 1$ on $L^2(K) \otimes V_\sigma$, for $\phi \in C(K)$, $f \in L^2(K)$, $v \in V_\sigma$, we have

$$\pi_\sigma(\phi)(f \otimes v) = \int_K \phi(k) \pi_\sigma(k)(f \otimes v) dk = \left(\int_K \phi(k) R_k f dk \right) \otimes v = (f * \phi) \otimes v \in C(K, V_\sigma)$$

Since $f * \phi \to f$ in $L^2(K)$ as ϕ runs over all Dirac functions, we have $\pi_\sigma(\phi)(f \otimes v) \to f \otimes v$ in $L^2(K, V_\sigma)$. $\square$

Theorem 8.20 (Frobenius reciprocity)**.** If $\sigma \in \widehat{M}$ is irreducible, for every irreducible representation $(\tau, U) \in \widehat{K}$, there is a canonical isomorphism

$$\operatorname{Hom}_K(U, \operatorname{Ind}_M^K V_\sigma) \xrightarrow{\sim} \operatorname{Hom}_M(U|_M, V_\sigma)$$

where $U|_M$ is the representation $\pi|_M$ of M.

Proof. Before writing down the isomorphism, we make some preparation. Put $V^c := C(K, V_\sigma) \cap \operatorname{Ind}_M^K V_\sigma$. The space V^c is K-invariant, and is dense in $\operatorname{Ind}_M^K V_\sigma$ by the previous lemma. Let $\alpha \in \operatorname{Hom}_K(U, \operatorname{Ind}_M^K V_\sigma)$. We show that the image of α lies in V^c. For this recall $L^2(K)$ decomposes into a direct sum of isotypes $L^2(K)[\gamma]$ for $\gamma \in \widehat{K}$ (here we consider the right regular representation) with each $L^2(K)[\gamma]$ finite dimensional consisting of continuous functions (matrix coefficients). Consider the composition

$$\alpha : U \xrightarrow{\alpha} \operatorname{Ind}_M^K V_\sigma \hookrightarrow L^2(K, V_\sigma) \xrightarrow{\sim} L^2(K) \otimes V_\sigma$$

The map α (by abuse of notation) intertwines τ and $R \otimes 1$, so $\alpha(U) \subseteq (L^2(K) \otimes V_\sigma)[\tau] \subseteq L^2(K)[\tau] \otimes V_\sigma$ consists of continuous functions, and this implies $\alpha(U) \subseteq V^c$.

Define

$$\begin{aligned} \operatorname{Hom}_K(U, \operatorname{Ind}_M^K V_\sigma) &\longrightarrow \operatorname{Hom}_M(U|_M, V_\sigma) \\ \alpha &\longmapsto [u \mapsto \alpha(u)(1)] \end{aligned}$$

and

$$\begin{aligned} \operatorname{Hom}_M(U|_M, V_\sigma) &\longrightarrow \operatorname{Hom}_K(U, \operatorname{Ind}_M^K V_\sigma) \\ \beta &\longmapsto [(u, k) \mapsto \beta(\tau(k) u)] \end{aligned}$$

It is direct to check these two maps are mutually inverse and intertwine. $\square$

Example 8.21. Let K be a compact group and $M \leqslant K$ a closed subgroup. Then K/M carries a unique Radon measure μ that is invariant under left translation of K and normalized so that $\mu(K/M) = 1$. The group K acts on the Hilbert space $L^2(K/M, \mu)$ by left translation, and this is a unitary representation. This representation is K-isomorphic to the representation $\operatorname{Ind}_M^K \mathbb{C}$ induced from the trivial representation via the map

$$\begin{aligned} L^2(K/M, \mu) &\longrightarrow \operatorname{Ind}_M^K \mathbb{C} \\ \psi &\longmapsto [k \mapsto \psi(k^{-1}M)] \end{aligned}$$

By Frobenius reciprocity for any $\tau \in \widehat{K}$, the multiplicity in $L^2(K/M, \mu)$ is

$$\operatorname{Hom}_K(V_\tau, L^2(K/M, \mu)) \cong \operatorname{Hom}_M(V_\tau|_M, \mathbb{C}) \cong V_\tau^M.$$

Hence, we obtain

$$L^2(K/M) \cong \widehat{\bigoplus_{\tau \in \widehat{K}}} V_\tau^M \otimes V_\tau.$$

An immediate consequence of this example is

Corollary 8.21.1. Let K be a compact group with $\#K \geqslant 2$ and M a proper closed subgroup. Then there exists a nontrivial irreducible representation of K whose restriction to M possesses a trivial representation of M.

Proof. Since M is proper, the quotient K/M has at least two points. Consider the decomposition in Example 8.21:

$$L^2(K/M) \cong \widehat{\bigoplus_{\tau \in \widehat{K}}} V_\tau^M \otimes V_\tau.$$

Since $\#K/M \geqslant 2$, LHS has dimension $\geqslant 2$. The trivial irreducible representation of K only contributes one dimension to RHS, so there must be some nontrivial $(\tau, V_\tau) \in \widehat{K}$ with $V_\tau^M \neq 0$. But then $V_\tau|_M$ contains a copy of the trivial representation. $\square$

Corollary 8.21.2. Retain the notation. There exists a nontrivial $(\pi, V_\pi) \in \widehat{K}$ and $v \in V_\pi$ such that the stabilizer subgroup $K_v = \{g \in K \mid \pi(g)v = v\}$ is a proper subgroup of K containing M.

Proof. Let (π, V_π) be the nontrivial irreducible representation found in the previous corollary. Pick any nonzero $v \in V_\pi^M$; then $M \subseteq K_v$. Since π is nontrivial, K_v cannot be the whole K, so that $K_v \subsetneq K$, as wanted.

$\square$

8.5 Compact Lie groups

8.5.1 Characterization among compact groups

Theorem 8.22. For any unit-neighborhood U of K, there exists a finite dimensional representation τ of K such that $\ker \tau \subseteq U$.

Proof. Let $k \in K$ with $k \neq e$. Since $C(K)$ separates points and the space of matrix coefficients is dense in $L^2(K)$ by (d) of Peter-Weyl theorem, it follows that $m(k) \neq m(e)$ for some matrix coefficient m of a finite dimensional (unitary) representation τ of K. In particular, this implies $\tau(k) \neq \tau(e)$. Hence the open sets $G \backslash \ker \tau$, where τ runs over all finite dimensional representations of K, form an open cover of $G \backslash \{e\}$, and hence an open cover of the compact set $G \backslash U$. We then can find some finite dimensional representations $\tau_1, \ldots, \tau_n$ of K with $G \backslash U \subseteq \bigcup_{i=1}^{n} G \backslash \ker \tau_i$. If we write τ for the direct sum of the τ_i, we see that $\ker \tau = \bigcap_{i=1}^{n} \ker \tau_i$ and thus $\ker \tau \subseteq U$. □

Corollary 8.22.1. Every compact group is isomorphic, as topological groups, to a projective limit of compact Lie groups.

Proof. Denote by F the set of finite dimensional (unitary) representations of a compact group K. If $(\tau, V) \in F$, then $\tau(K) \subseteq \mathrm{GL}(V)$. Since $\tau(K)$ is compact and $\mathrm{GL}(V)$ is a Lie group, it follows from Theorem W.49 that $\tau(K)$ is itself a compact Lie group. Since $K/\ker \tau \cong \tau(K)$, we see $K/\ker \tau$ is isomorphic to a compact Lie group, as topological groups.

The canonical projections $K \to K/\ker \tau \, (\tau \in F)$ give rises to a continuous homomorphism $\phi : K \to \varprojlim_{\tau \in F} G/\ker \tau$ with dense image. Since K is compact, ϕ is surjective. To show ϕ is a homeomorphism, it remains to show ϕ is injective. Note that $\ker \phi = \bigcap_{\tau \in F} \ker \tau$, so we have to show the intersection is trivial. From Theorem 8.22 that $\{\ker \tau \mid \tau \in F\}$ forms a neighborhood basis of identity of K. Now our claim follows as K is Hausdorff. □

Corollary 8.22.2. For a compact group K, TFAE:

(i) There exists a unit-neighborhood of K containing no nontrivial closed normal subgroup.

(ii) K is isomorphic to a closed subgroup of $\mathrm{GL}(V)$ for some finite dimensional complex vector space V.

(iii) K is a Lie group.

Proof. (i)⇒(ii) follows from Theorem 8.22, which implies that there exists a finite dimensional representation (τ, V) of K with $\ker \tau = \{e\}$, i.e., $\tau : K \to \mathrm{GL}(V)$ is injective. (ii)⇒(iii) follows from Theorem W.49. (iii)⇒(i) is Proposition W.40. □

Corollary 8.22.3. For every proper closed subgroup of a compact Lie group K, there exists a finite dimensional representation (π, V_π) of K and a vector $v \in V_\pi$ such that $M = K_v = \{g \in K \mid \pi(g)v = v\}$.

Proof. Let M be a closed subgroup of K.

□

8.5.2 Matrix coefficients

Now let K be a compact group with a faithful representation $\pi : K \to \mathrm{GL}(V)$ in a finite dimensional complex vector space V. Choose any basis $e_1, \ldots, e_n$ for V and define the matrix coefficients $\pi_{ij} : K \to \mathbb{C}$ by

$$\pi_{ij}(g) = \langle \pi(g)e_i, e_j \rangle.$$

These π_{ij} $(1 \leqslant i, j \leqslant n)$ vanish nowhere as $\pi(K) \subseteq \mathrm{GL}(V)$. Since π is injective, these π_{ij} $(1 \leqslant i, j \leqslant n)$ separate points. Hence, by Stone-Weierstrass the subalgebra of $C(K)$ generated by the π_{ij}, $\overline{\pi_{ij}}$ $(1 \leqslant i, j \leqslant n)$ is dense in $C(G)$.

In fact, by Lemma 8.1 we can choose a K-invariant inner product on V so that (π, V) is unitary. As mentioned before the statement of Theorem 8.5, the functions $\overline{\pi_{ij}}$ are the matrix coefficients of the contragredient representation (π^*, V^*) of π. Moreover, the "monimials" $\pi_{i_1 j_1} \cdots \pi_{i_n j_n} \overline{\pi_{k_1 l_1}} \cdots \overline{\pi_{k_m l_m}}$ are the matrix coefficients of the representation $V^{\otimes n} \otimes (V^*)^{\otimes m}$.

Lemma 8.23.

(i) Every irreducible unitary representation of K appears in some $V^{\otimes n} \otimes (V^*)^{\otimes m}$.

(ii) π_{ij}, $\overline{\pi_{ij}}$ $(1 \leqslant i, j \leqslant n)$ generate $C(K)^{\mathrm{fin}}$. In particular, the algebra $C(K)^{\mathrm{fin}}$ is of finite type over $\mathbb{C}$.

Proof. For a finite dimensional representation τ of K, denote by M_τ the space of matrix coefficients for τ. If we denote by A the unital subalgebra of $C(K)$ generated by the π_{ij}, $\overline{\pi_{ij}}$ $(1 \leqslant i, j \leqslant n)$, the discussion above shows that

$$A = \sum_{n,m \geqslant 0} M_{V^{\otimes n} \otimes (V^*)^{\otimes m}}.$$

Let S be the subset of $\widehat{K}$ consisting of those representations appearing in $V^{\otimes n} \otimes (V^*)^{\otimes m}$ $(n, m \geqslant 0)$; decomposing those tensor product representations into irreducibles shows

$$A = \bigoplus_{\tau \in S} M_\tau \subseteq \bigoplus_{\tau \in \widehat{K}} M_\tau = C(K)^{\mathrm{fin}}.$$

The last subspace is dense in $L^2(K)$ by Peter-Weyl, and the subalgebra A is dense as said before. Since the direct sum is orthogonal, A cannot be dense unless $S = \widehat{K}$. This shows (i) and (ii) simultaneously. □

8.5.3 Compact Lie groups as real algebraic groups

In this subsubsection let G be compact Lie group. Lemma 8.23 shows that $R(G,\mathbb{C})$ is of finite type over $\mathbb{C}$. In fact, if we view V as a real vector space in the above discussion, the same argument shows that $R(G,\mathbb{R})$ is of finite type over $\mathbb{R}$. The affine scheme

$$X = \operatorname{Spec} R(G,\mathbb{R})$$

is then a (possibly reducible) algebraic variety over $\mathbb{R}$. We are going to show this is an algebraic group whose $\mathbb{R}$-point $X(\mathbb{R})$ is isomorphic to the original compact Lie group G. Let k be either $\mathbb{R}$ or $\mathbb{C}$. To give it a group scheme structure it is the same as giving $R(G,k)$ a commutative Hopf algebra structure. Define

$$\begin{aligned} \Delta : R(G,k) &\longrightarrow R(G\times G,k) = R(G,k)\otimes R(G,k) \\ f &\longmapsto [(g,g')\mapsto f(gg')] \end{aligned}$$

where the last isomorphism is from Example 8.18. This is well-defined as matrix coefficients are $G\times G$-finite by Corollary 8.13.1. Define

$$\begin{aligned} \iota : R(G,k) &\longrightarrow R(G,k) \\ f &\longmapsto [g\mapsto f(g^{-1})] \end{aligned}$$

and

$$\begin{aligned} \varepsilon : R(G,k) &\longrightarrow k \\ f &\longmapsto f(e) \end{aligned}$$

Finally let $m : R(G,k)\otimes R(G,k)\to R(G,k)$ and $e : k\to R(G,k)$ be the multiplication and the inclusion of constant maps; they are the structure maps for the k-algebra $R(G,k)$. These just defined morphisms satisfy three commutative diagrams

$$\begin{tikzcd}
 & R(G,k)\otimes R(G,k) \arrow[r, "\mathrm{id}\otimes\Delta"] & R(G,k)\otimes R(G,k)\otimes R(G,k) \arrow[dd, equal] \\
R(G,k) \arrow[ur, "\Delta"] \arrow[dr, "\Delta"'] & & \\
 & R(G,k)\otimes R(G,k) \arrow[r, "\Delta\otimes\mathrm{id}"] & R(G,k)\otimes R(G,k)\otimes R(G,k)
\end{tikzcd}$$

$$\begin{tikzcd}
 & R(G,k)\otimes R(G,k) \arrow[r, "\varepsilon\otimes\mathrm{id}"] & k\otimes R(G,k) \arrow[d, "m\circ(e\otimes\mathrm{id})"] \\
R(G,k) \arrow[ur, "\Delta"] \arrow[dr, "\Delta"'] \arrow[rr, equal] & & R(G,k) \\
 & R(G,k)\otimes R(G,k) \arrow[r, "\mathrm{id}\otimes\varepsilon"] & R(G,k)\otimes k \arrow[u, "m\circ(\mathrm{id}\otimes e)"']
\end{tikzcd}$$

$$\begin{tikzcd}
 & R(G,k)\otimes R(G,k) \arrow[rr, "\mathrm{id}\otimes\iota"] & & R(G,k)\otimes R(G,k) \arrow[dr, "m"] & \\
R(G,k) \arrow[ur, "\Delta"] \arrow[rr, "\varepsilon"] \arrow[dr, "\Delta"'] & & k \arrow[rr, "e"] & & R(G,k) \\
 & R(G,k)\otimes R(G,k) \arrow[rr, "\iota\otimes\mathrm{id}"] & & R(G,k)\otimes R(G,k) \arrow[ur, "m"'] &
\end{tikzcd}$$

Taking spec shows that $X = \operatorname{Spec} R(G,\mathbb{R})$ is really an affine group scheme over $\mathbb{R}$. Each point $g \in G$ determines an evaluation map $\mathrm{ev}_g : R(G,\mathbb{R}) \to \mathbb{R}$, giving rise to a map

$$\begin{aligned} \Phi : G &\longrightarrow X(\mathbb{R}) = \mathrm{Hom}_{\mathbf{Alg}_{\mathbb{R}}}(R(G,\mathbb{R}),\mathbb{R}) \\ g &\longmapsto \mathrm{ev}_g \end{aligned}$$

This is injective as $R(G,\mathbb{R})$ separates points. Equip $X(k)$ with the initial topology with respect to the maps $X(k) = \mathrm{Hom}_{\mathbf{Alg}_k}(R(G,k),k) \ni p \mapsto p(f) \in k\,(f \in R(G,k))$, where k is with its usual euclidean topology. If $f_1,\dots,f_n$ is a generating set for $R(G,k)$, the topology just defined is the same as the subspace topology inherited from the injection

$$\begin{aligned} X(k) &\longrightarrow k^n \\ p &\longmapsto (p(f_1),\dots,p(f_n)) \end{aligned}$$

In particular, this shows $X(k)$ is Hausdorff under this topology.

Lemma 8.24.

(i) $X(k)$ is a topological group.

(ii) $\Phi : G \to X(\mathbb{R})$ is a continuous injective group homomorphism.

In particular, since G is compact and $X(\mathbb{R})$ is Hausdorff, via Φ we can view G as a closed subgroup of $X(\mathbb{R})$

Proof. The topology on $X(k)$ enjoys the universal property: if Y is another topological space with a set-theoretic map $\varphi : Y \to X(k)$, then φ is continuous if and only if $Y \xrightarrow{\varphi} X(k) \xrightarrow{p\mapsto p(f)} k$ is continuous for every $f \in R(G,k)$. It is then direct to see $X(k)$ is a topological group and Φ is continuous.

It remains to show Φ is a group homomorphism. For $f \in R(G,k)$, write

$$\Delta(f) = \sum f_i \otimes f_i'$$

for some $f_i, f_i' \in R(G,k)$. Then for $g,h \in G$,

$$(\mathrm{ev}_g \otimes \mathrm{ev}_h) \circ \Delta(f) = \sum f_i(g) f_i'(h) = f(gh) = \mathrm{ev}_{gh}(f)$$

where the third identity results from the definition that $\Delta(f)(g,h) = f(gh)$. Varying $f \in R(G,k)$ shows $(\mathrm{ev}_g \otimes \mathrm{ev}_h) \circ \Delta = \mathrm{ev}_{gh}$, which is exactly what we want. $\square$

Let $\pi : G \to \mathrm{GL}_n(k)$ be a representation of G. Then the matrix coefficients π_{ij} lie in $R(G,k)$. Consider the k-algebra homomorphism

$$\begin{aligned} k[x_{ij}] &\longrightarrow R(G,k) \\ x_{ij} &\longrightarrow \pi_{ij} \end{aligned}$$

Since $(\pi_{ij})_{ij} \in \mathrm{GL}_n(k)$, the determinant $\det(x_{ij})$ is invertible in $R(G,k)$. This induces a homomorphism $k[x_{ij}, \det(x_{ij})^{-1}] \to R(G,k)$ on the localization. Taking spec gives a k-morphism

$$X \to \mathrm{GL}_{n,k}$$

of varieties. It is straightforward to verify that $k[x_{ij}, \det(x_{ij})^{-1}] \to R(G,k)$ is in fact a Hopf algebra homomorphism, so $X \to \mathrm{GL}_{n,k}$ is an algebraic group homomorphism. Taking k-points yields a group homomorphism

$$\begin{aligned} \pi_k : X(k) &\longrightarrow \mathrm{GL}_n(k) \\ p &\longmapsto (p(\pi_{ij}))_{ij}. \end{aligned}$$

This is continuous if and only if each $p \mapsto p \circ \pi_{ij}$ is, which is the case by definition. If we choose π to be faithful (possible by Corollary 8.22.2), by Lemma 8.23.(ii) the homomorphism $k[x_{ij}, \det(x_{ij})^{-1}] \to R(G,k)$ is surjective, meaning that π_k is a closed embedding. It is worth-noting that, since $\Phi(g) = \mathrm{ev}_g$, we have a commuting triangle

$$\begin{tikzcd} X(\mathbb{R}) \arrow[r, "\pi_{\mathbb{R}}"] & \mathrm{O}(n) \\ G \arrow[u, "\Phi"] \arrow[ur, "\pi"] & \end{tikzcd}$$

A similar argument implies that if $k = \mathbb{R}$ and $\pi(G) \subseteq \mathrm{O}(n)$, then $\pi_{\mathbb{R}}(X(\mathbb{R})) \subseteq \mathrm{O}(n)$.

Theorem 8.25. $X(\mathbb{R})$ is a compact Lie group and $\Phi : G \to X(\mathbb{R})$ is a Lie group isomorphism.

Proof. Pick any faithful representation $\pi : G \to \mathrm{GL}_n(\mathbb{R})$ of π. By Lemma 8.1, the representation space $\mathbb{R}^n$ admits a G-invariant inner product, so up to a conjugation by an element in $\mathrm{GL}_n(\mathbb{R})$ we may assume $\pi(G) \subseteq \mathrm{O}(n)$. Now consider the homomorphism $\pi_{\mathbb{R}} : X(\mathbb{R}) \to \mathrm{GL}_n(\mathbb{R})$. It has image in $\mathrm{O}(n)$ as mentioned above, and is a closed embedding. By Theorem W.49, $X(\mathbb{R})$ is a compact Lie group.

It remains to show Φ is surjective. The inclusion $\Phi : G \to X(\mathbb{R})$ induces an algebra homomorphism $\Phi^* : R(X(\mathbb{R}), \mathbb{R}) \to R(G, \mathbb{R})$ on the algebras of matrix coefficients. On the other hands, if $\rho : G \to \mathrm{GL}_n(\mathbb{R})$ is a representation and f is its matrix coefficient, then the evaluation $\lambda_f : X(\mathbb{R}) \ni p \mapsto p(f) \in \mathbb{R}$ is a matrix coefficient for $\rho_{\mathbb{R}} : X(\mathbb{R}) \to \mathrm{GL}_n(\mathbb{R})$. This follows from the construction of $\rho_{\mathbb{R}}$ in the above. Hence we obtain an algebra homomorphism

$$\begin{aligned} \lambda : R(G, \mathbb{R}) &\longrightarrow R(X(\mathbb{R}), \mathbb{R}) \\ f &\longmapsto \lambda_f. \end{aligned}$$

Since π is faithful, so is $\pi_{\mathbb{R}}$, and by Lemma 8.23.(ii), the algebra $R(X(\mathbb{R}), \mathbb{R})$ is generated by the $(\pi_{\mathbb{R}})_{ij}$. By construction $\lambda(\pi_{ij}) = (\pi_{\mathbb{R}})_{ij}$ and $\Phi^*((\pi_{\mathbb{R}})_{ij}) = \pi_{ij}$, so λ and Φ^* are mutually inverses, whence isomorphisms. Consider the commutative square

$$\begin{tikzcd} R(X(\mathbb{R}), \mathbb{R}) \arrow[r, "\Phi^*"] \arrow[d, hook] & R(G, \mathbb{R}) \arrow[d, hook] \\ C(X(\mathbb{R}), \mathbb{R}) \arrow[r, "\Phi^*"] & C(G, \mathbb{R}) \end{tikzcd}$$

Since the upper horizontal arrow is an isomorphism and all vertical maps have dense images with sup-norm topology, the lower horizontal arrow is also an isomorphism. (Note that Φ^* is continuous as Φ is injective.) This forces $\Phi : G \to X(\mathbb{R})$ to be surjective by an Urysohn lemma argument. $\square$

We turn to the affine group scheme $X_{\mathbb{C}} = \operatorname{Spec} R(G, \mathbb{C})$ over $\mathbb{C}$. The argument that precedes Theorem 8.25 shows that $X_{\mathbb{C}}(\mathbb{C}) = X(\mathbb{C})$ is a closed subgroup of $\mathrm{GL}_n(\mathbb{C})$. A general fact in algebraic geometry says that $X(\mathbb{C})$ contains at least a smooth point, so by translation we see $X(\mathbb{C})$ is a smooth algebraic group over $\mathbb{C}$, and hence a complex Lie group.

8.6 Examples

8.6.1 SU(2)

Consider the n-dimensional sphere $S^n \subseteq \mathbb{R}^{n+1}$. We equip S^n with the subspace topology from $\mathbb{R}^{n+1}$ with euclidean topology.

Lemma 8.26. For a subset $A \subseteq S^n$, A is Borel-measurable in S^n if and only if $IA := \{ta \mid 0 \leqslant t \leqslant 1,\ a \in A\}$ is Borel-measurable in $\mathbb{R}^{n+1}$.

Proof. The Borel sets in S^n is the intersection of S^n and those of $\mathbb{R}^{n+1}$, so if IA is Borel, so is $A = IA \cap S^n$. For the converse, if A is closed, since the map $[0,1] \times S^n \to \mathbb{R}^{n+1}$ defined by $(t, a) \mapsto ta$ is closed (the domain being compact), IA is closed. If $A \neq \varnothing$ is open, then $IA = (\overline{B_1(0)} \backslash IA^c) \cup \{0\}$ is Borel. It remains to show the family $\mathcal{S} := \{A \subseteq S^n \mid IA \text{ is Borel}\}$ is a σ-algebra.

- $S^n \in \mathcal{S}$ and $\varnothing \in \mathcal{S}$.
- If $\varnothing \neq A \in \mathcal{S}$, then $IA^c = (\overline{B_1(0)} \backslash IA) \cup \{0\}$ is Borel.
- If $A_i \in \mathcal{S}$ with $\bigcup_i A_i \neq \varnothing$, then $I \bigcup_i A_i = \bigcup_i IA_i$ is Borel.
- If $A_i \in \mathcal{S}$ with $\bigcap_i A_i \neq \varnothing$, then $I \bigcap_i A_i = (\overline{B_1(0)} \backslash I \bigcup_i A_i^c) \cup \{0\}$ is Borel.

$\square$

Let λ be the Lebesgue measure on $\mathbb{R}^{n+1}$. For a measurable set $A \subseteq S^n$, define the normalized Lebesgue measure as

$$\mu(A) := \frac{\lambda(IA)}{\lambda(IS^n)} = \frac{\lambda(IA)}{\lambda(B_1(0))}$$

The Lebesgue measure λ is invariant under the action of the orthogonal group $\mathrm{O}(n+1)$, and it follows that μ is also invariant under $\mathrm{O}(n+1)$.

Lemma 8.27. Let $n \in \mathbb{N}$ and $e_1 = (1, 0, \ldots, 0)^t \in \mathbb{R}^{n+1}$. The matrix multiplication $g \mapsto ge_1$ gives an identification

$$S^n \cong \mathrm{O}(n+1)/\mathrm{O}(n) \cong \mathrm{SO}(n+1)/\mathrm{SO}(n)$$

This map is invariant under left translation, and the normalized Lebesgue measure on S^n is the unique normalized invariant measure on this quotient space.

Proof. Here we embed $\mathrm{O}(n)$ into $\mathrm{O}(n+1)$ by the map $g \mapsto \begin{pmatrix} 1 & 0 \\ 0 & g \end{pmatrix}$. Then $\mathrm{O}(n)$ is the stabilizer of e_1 under the action of $\mathrm{O}(n+1)$; this gives an injective continuous map $\mathrm{O}(n+1)/\mathrm{O}(n) \to S^n$. It is then easy to see this is a homeomorphism, and similarly $S^n \cong \mathrm{SO}(n+1)/\mathrm{SO}(n)$. The normalized Lebesgue is transferred to the quotient $\mathrm{O}(n+1)/\mathrm{O}(n)$, and as said above it is invariant under $\mathrm{O}(n+1)$. Now the uniqueness follows the uniqueness part in Theorem 1.17.

□

Recall that $\mathrm{SU}(2)$ consists of 2×2 unitary matrices with determinant 1. In other words,

$$\mathrm{SU}(2) = \left\{ \begin{pmatrix} a & -\overline{b} \\ b & \overline{a} \end{pmatrix} \mid \begin{pmatrix} a \\ b \end{pmatrix} \in S^3 \subseteq \mathbb{C}^2 \right\}$$

From this description and the fact that the normalized Lebesgue measure on S^3 is invariant under $\mathrm{O}(4)$ as well as the uniqueness of the invariant measures, we obtain the following lemma.

Lemma 8.28. The map $\mathrm{SU}(2) \to S^3$, mapping the matrix $g \in \mathrm{SU}(2)$ to its first column, is a homeomorphism. Via this homeomorphism, the normalized Lebesgue measure on S^3 coincides with the normalized Haar measure on $\mathrm{SU}(2)$.

Lemma 8.29. For measurable $A \subseteq S^n$, define $\sigma(A) = (n+1)\lambda(IA)$. Then for any Borel-measurable function $f : \mathbb{R}^{n+1} \to [0, \infty]$, we have

$$\int_{\mathbb{R}^{n+1}} f(x) d\lambda(x) = \int_0^\infty r^n \left(\int_{S^n} f(ru) d\sigma(u) \right) dr$$

Lemma 8.30. Let $f : S^n \to \mathbb{C}$ be integrable and for $n \in \mathbb{N}_0$ let $F_m : \mathbb{R}^{n+1} \to \mathbb{C}$ be defined by

$$F_m(rx) = r^m f(x)$$

for all $x \in S^n$ and $r \geqslant 0$. Then

$$\int_{S^n} f(x)d\mu(x) = c_{n,m} \int_{\mathbb{R}^{n+1}} F_m(x)e^{-\|x\|^2} d\lambda(x)$$

where $c_{n,m}$ will be specified in the proof.

Proof. By the previous lemma, we have

$$\begin{aligned}
\int_{\mathbb{R}^{n+1}} F_m(x)e^{-\|x\|^2} d\lambda(x) &= c \int_0^\infty r^n \int_{S^n} F_m(rx)e^{-r^2} d\mu(x)dr \\
&= c \int_0^\infty r^{n+m} e^{-r^2} dr \int_{S^n} f(x)d\mu(x) \\
(r \mapsto r^2) &= \frac{c}{2}\Gamma\left(\frac{n+m}{2} + 1\right) \int_{S^3} f(x)d\mu(x)
\end{aligned}$$

for some constant c. Let $f \equiv 1$ and $m = 0$ in the above identity; then

$$\frac{c}{2}\Gamma\left(\frac{n}{2} + 1\right) = \int_{\mathbb{R}^{n+1}} e^{-\|x\|^2} d\lambda(x) = \left(\int_{\mathbb{R}} e^{-x^2} dx\right)^{n+1} = \pi^{\frac{n+1}{2}}$$

so that

$$c_{n,m} := \left(\frac{c}{2}\Gamma\left(\frac{n+m}{2} + 1\right)\right)^{-1} = \pi^{-\frac{n+1}{2}} \frac{\Gamma\left(\frac{n}{2} + 1\right)}{\Gamma\left(\frac{n+m}{2} + 1\right)}$$

□

For $n \in \mathbb{N}_0$, denote by $\mathcal{P}_n$ the set of homogeneous polynomials on $\mathbb{C}^2$ of degree n. Then

$$\mathcal{P}_n = \bigoplus_{k=0}^{n} \mathbb{C} z_1^k z_2^{n-k}$$

For $p, q \in \mathcal{P}_n$, we define

$$\langle p, q\rangle_n := \langle p|_{S^3}, q|_{S^3}\rangle_{L^2(S^3)} = \int_{S^3} p(x)\overline{q(x)}d\mu(x)$$

It follows from the previous lemma that

$$\langle p, q\rangle_n = c_{3,2n} \int_{\mathbb{C}^2} p(z)\overline{q(z)}e^{-\|z\|^2} d\lambda(z)$$

We define a representation π_n of SU(2) on $\mathcal{P}_n$ by

$$\pi_n(g)p(z) := p(g^{-1}z)$$

As a preparation, we compute the inner product $\langle z_1^k z_2^{n-k}, z_1^r z_2^{n-r}\rangle_n$ for $0 \leqslant k, r \leqslant n$.

$$\int_{\mathbb{C}^2} (z_1^k z_2^{n-k})\overline{z_1^r z_2^{n-r}} e^{-\|z\|^2} d\lambda(z) = \left(\int_{\mathbb{C}} z^k \overline{z}^r e^{-|z|^2} dz\right)\left(\int_{\mathbb{C}} z^{n-k}\overline{z}^{n-r} e^{-|z|^2} dz\right)$$

Compute

$$\int_{\mathbb{C}} z^k \overline{z}^r dz = \int_0^{2\pi} y \int_0^{\infty} (\rho e^{i\theta})^k (\rho e^{-i\theta})^r e^{-\rho^2} \rho d\rho d\theta = \delta_{kr}\pi\Gamma(k+1) = \delta_{kr}\pi k!.$$

Hence

$$\langle z_1^k z_2^{n-k}, z_1^r z_2^{n-r}\rangle_n = c_{3,2n}\delta_{kr}\pi^2 k!(n-k)! = \delta_{kr}\frac{k!(n-k)!}{n!} = \delta_{kr}\binom{n}{k}^{-1}.$$

In view of the computation, we find that $\left\{\binom{n}{k}^{\frac{1}{2}} z_1^k z_2^{n-k}\right\}_{k=0}^{n}$ forms an orthonormal basis for $\mathcal{P}_n$.

Lemma 8.31. The representation $\pi_n : \mathrm{SU}(2) \to \mathrm{GL}(\mathcal{P}_n)$ is unitary and irreducible

Proof. By Lemma 8.28, we choose the measure μ on S^3 respecting the homeomorphism $\mathrm{SU}(2) \cong S^3$; the homeomorphism is explicitly defined by $g \mapsto g.e_1$, where $e_1 = (1,0,0,0) \in \mathbb{C}^2$. Then

$$\begin{aligned}\langle \pi_n(h)p, \pi_n(h)q\rangle_n &= \int_{S^3} p(h^{-1}x)\overline{q(h^{-1}x)} d\mu(x) \\ &= \int_{\mathrm{SU}(2)} p(h^{-1}g.e_1)\overline{q(h^{-1}g.e_1)} dg \overset{\text{inv}}{=} \int_{\mathrm{SU}(2)} p(g.e_1)\overline{q(g.e_1)} dg = \langle p, q\rangle_n\end{aligned}$$

for any $h \in \mathrm{SU}(2)$ and $p, q \in \mathcal{P}_m$.

To show irreducibility, □

Theorem 8.32. $\widehat{\mathrm{SU}(2)} = \{[\mathcal{P}_n, \pi_n] \mid n \in \mathbb{N}_0\}$, where $[\mathcal{P}_n, \pi_n]$ denotes the equivalence class of $(\mathcal{P}_n, \pi_n)$.

Proof. Regard S^1 as a subgroup of $\mathrm{SU}(2)$ by $e^{i\theta} \mapsto \begin{pmatrix} e^{i\theta} & \\ & e^{-i\theta}\end{pmatrix}$. Let $\chi_n = \operatorname{tr}\pi_n$ be the character of the representation π_n. We have

$$\chi_n(e^{i\theta}) = \sum_{k=0}^{n} \langle \pi(e^{i\theta})p_k, p_k\rangle_n = \sum_{k=0}^{n} e^{i\theta(n-2k)}$$

where the $p_k = \binom{n}{k}^{\frac{1}{2}} z_1^k z_2^{n-k}$ are the orthonormal bases for $\mathcal{P}_n$. The key observation is that

$$\operatorname{span}_{\mathbb{C}}\{\chi_n(e^{i\theta}) \mid n \in \mathbb{N}_0\} = \operatorname{span}_{\mathbb{C}}\{\cos n\theta \mid n \in \mathbb{N}_0\}$$

which is more-or-less obvious (as $e^{i\theta} + e^{-i\theta} = 2\cos\theta$). Since each element in $\mathrm{SU}(2)$ is conjugate to S^1, by classical Fourier analysis on S^1 we see the restriction

$$\begin{aligned} L^2(\mathrm{SU}(2)/\mathrm{conj}) \cap C(\mathrm{SU}(2)) &\longrightarrow C_{\mathrm{even}}(S^1) \\ f &\longmapsto f|_{S^1}\end{aligned}$$

is a linear isomorphism, where

$$C_{\text{even}}(S^1) = \{f \in C(S^1) \mid f(x) = f(x^{-1}) \text{ for all } x \in S^1\}$$

and it contains $\operatorname{span}_{\mathbb{C}}\{\cos n\theta \mid n \in \mathbb{N}_0\}$ as a dense subset. Moreover, it is (almost) norm-preserving:

$$\langle z^k \overline{z}^{n-k}, z^r \overline{z}^{n-r} \rangle_{L^2(S^1)} = \delta_{kr}.$$

Hence $\operatorname{span}_{\mathbb{C}}\{\chi_n(e^{i\theta}) \mid n \in \mathbb{N}_0\}$ is a dense subset of $L^2(\mathrm{SU}(2)/\mathrm{conj}) \cap C(\mathrm{SU}(2))$, and hence of $L^2(\mathrm{SU}(2)/\mathrm{conj})$. It follows from Corollary 8.11.1 that the χ_n are the only characters of $\mathrm{SU}(2)$. In view of Theorem 8.10, the proof is completed. □

Corollary 8.32.1. The $\mathrm{SU}(2)$ representation on $L^2(S^3)$ is isomorphic to the Hilbert space direct sum $\widehat{\bigoplus_{n \geqslant 0}} (n+1)\mathcal{P}_n$.

Proof. Since $S^3 \cong \mathrm{SU}(2)$ and $\dim \mathcal{P}_n = n+1$, this follows at once from Example 8.21.

□

8.6.2 SO(n)

Let $n, m \geqslant 0$, and put

$$V_{n,m} = \mathbb{C}[x_1, \ldots, x_n]_m = \bigoplus_{\substack{1 \leqslant k_1, \ldots, k_n \\ k_1 + \cdots + k_n = m}} \mathbb{C} x_1^{k_1} \cdots x_n^{k_n}$$

to be the space of n-variable homogeneous polynomials of degree m over $\mathbb{C}$. The space $\mathcal{P}_n$ that appeared in the previous subsubsection is now the space $V_{2,n}$. The general linear group $\mathrm{GL}_n(\mathbb{R})$ acts on $V_{n,m}$ naturally on the left:

$$g.p(x) := p(g^{-1}x)$$

In particular, the orthogonal group $\mathrm{O}(n)$ acts on $V_{n,m}$; we denote the representation it affords by $\pi_{n,m} : \mathrm{O}(n) \to \mathrm{GL}(V_{n,m})$. Let $\Delta \in \operatorname{Der} C^\infty(\mathbb{R}^n)$ be the Laplacian on $\mathbb{R}^n$:

$$\Delta = \sum_{i=1}^n \partial_i^2 = \frac{\partial^2}{\partial x_1^2} + \cdots + \frac{\partial^2}{\partial x_n^2}$$

Lemma 8.33. For $p \in \mathbb{C}[x_1, \ldots, x_n]$ and $g \in \mathrm{O}(n)$, one has $g.\Delta(p) = \Delta(g.p)$.

Proof. For $g \in \mathrm{GL}_n(\mathbb{C})$, by chain rules one computes

$$\Delta(g.p) = \sum_{i,j=1}^n ((gg^t)^{-1})_{ij} \cdot g. \left(\frac{\partial^2}{\partial x_i \partial x_j} p \right).$$

Hence, for $g \in \mathrm{O}(n)$, we have

$$\Delta(g.p) = \sum_{i,j=1}^n \delta_{ij} \cdot g. \left(\frac{\partial^2}{\partial x_i \partial x_j} p \right) = g.\Delta(p)$$

□

Write $H_{n,m} := \ker \Delta \cap V_{n,m}$ for the space of all **harmonic polynomial of degree m in n-variables**. The previous lemma then show that $g.H_{n,m} \subseteq H_{n,m}$ for $g \in \mathrm{O}(n)$, so $H_{n,m}$ is actually an $\mathrm{O}(n)$-representation.

As before, define an inner product on $V_{n,m}$ by

$$\langle p, q\rangle := \langle p|_{S^{n-1}}, q|_{S^{n-1}}\rangle_{L^2(S^{n-1})} = \int_{S^{n-1}} p(x)\overline{q(x)}d\mu(x) \overset{(8.30)}{=} c_{n-1,2m}\int_{\mathbb{R}^n} p(x)\overline{q(x)}e^{-\|x\|^2}d\lambda(x)$$

where the measure μ is chosen as in (and before) Lemma 8.27. Here we view complex polynomials as functions on $\mathbb{R}$. In particular, this inner product is $\mathrm{O}(n)$-invariant.

To proceed further, we introduce the Laplacian on S^{n-1}. First, we associate with $f : S^{n-1} \to \mathbb{C}$ a function $F : \mathbb{R}^n \to \mathbb{C}$ given by $F(x) = f(x/\|x\|)$. Then for $f \in C^\infty(S^{n-1})$, define

$$\Delta_{S^{n-1}} f := (\Delta F)|_{S^{n-1}}$$

Let $f \in V_{n,m}$. Recall the Euler identity

$$\sum_{i=1}^n x_i \frac{\partial}{\partial x_i} f = mf.$$

To see this, differentiate the identity $f(tx) = t^m f(x)$ with respect to t; we have

$$\sum_{i=1}^n \left(\frac{\partial}{\partial x_i} f\right)(tx) \cdot x_i = mt^{m-1} f(x).$$

Taking $t = 1$ yields the identity.

Let $f \in V_{n,m}$. Its extension $F : \mathbb{R}^n \to \mathbb{C}$ to $\mathbb{R}^n$ is $F(x) = f(x/\|x\|) = \|x\|^{-m} f(x)$. A tedious computation along with the Euler identity yields that

$$\Delta_{S^{n-1}} f = -m(m+n-2)f + (\Delta f)|_{S^{n-1}}$$

Lemma 8.34. For $f \in V_{n,m}$, f is harmonic if and only if f is an eigenfunction of $\Delta_{S^{n-1}}$, in which case f has eigenvalue $-m(m+n-2)$.

Proof. It is clear from the above identity when f is harmonic. Conversely if $\Delta_{S^{n-1}} f = \lambda f$ for some $\lambda \in \mathbb{C}$, then

$$\lambda f = -m(m+n-2)f + (\Delta f)|_{S^{n-1}},$$

or

$$(\lambda + m(m+n-2))\, f = (\Delta f)|_{S^{n-1}}.$$

LHS is homogeneous of degree m, while RHS is homogeneous of degree $m-2$. Hence $(\Delta f)|_{S^{n-1}} = 0$, and thus $\Delta f = 0$. $\square$

8.6.3 Spin(n)

Recall that $\mathrm{Spin}(n)$ is a double cover $\mathrm{SO}(n)$ and it fits into a short exact sequence

$$1 \longrightarrow \{\pm\} \longrightarrow \mathrm{Spin}(n) \longrightarrow \mathrm{SO}(n) \longrightarrow 1.$$

In particular, every representation of $\mathrm{SO}(n)$ results in a representation of $\mathrm{Spin}(n)$. Precisely, for every topological group G, there is a bijection

$$\{\pi \in \mathrm{Hom}_{\mathbf{TopGp}}(\mathrm{Spin}(n), G) \mid \pi(-1) = e \in G\} \longrightarrow \mathrm{Hom}_{\mathbf{TopGp}}(\mathrm{SO}(n), G)$$

Definition. A representation $\pi : \mathrm{Spin}(n) \to \mathrm{GL}(V)$ is called **spinorial** if $\pi(-1) \neq \mathrm{id}_V \in \mathrm{GL}(V)$.

In this subsubsection we are going to construct two spinorial representations, called the **spin** (resp. **half-spin**) **representation**.

Let $(\ ,\ )$ be the symmetric bilinear form on $\mathbb{C}^n$ given by the (real) inner product. Let $m = \left\lfloor \frac{n}{2} \right\rfloor$; then $n = 2m$ if $2 \mid n$, and $n = 2m+1$ if $2 \nmid n$. Take

$$\begin{aligned} W &= \{(z_1, \dots, z_m, iz_1, \dots, iz_m(,0)) \mid z_k \in \mathbb{C}\} \\ W' &= \{(z_1, \dots, z_m, -iz_1, \dots, -iz_m(,0)) \mid z_k \in \mathbb{C}\} \end{aligned}$$

to be maximal isotropic (complex) subspaces and $e_0 = (0, \dots, 0, 1)$. Then

$$\mathbb{C}^n = \begin{cases} W \oplus W' & , n \text{ even} \\ W \oplus W' \oplus \mathbb{C}e_0 & , n \text{ odd.} \end{cases}$$

Recall the Clifford algebra C_n associated to the real quadratic space $(\mathbb{R}^n, \|\cdot\|^2)$. Put $C_n(\mathbb{C}) := C_n \otimes_{\mathbb{R}} \mathbb{C}$; then by definition $C_n(\mathbb{C}) = T(\mathbb{C}^n)/\langle z \otimes z + (z,z) \mid z \in \mathbb{C}^n \rangle$. In fact,

Lemma 8.35. There are $\mathbb{C}$-algebra isomorphisms

$$C_n(\mathbb{C}) \cong \begin{cases} \mathrm{End}_{\mathbb{C}} \bigwedge W & , \text{if } n \text{ even} \\ \mathrm{End}_{\mathbb{C}} \bigwedge W \oplus \mathrm{End}_{\mathbb{C}} \bigwedge W & , \text{if } n \text{ odd.} \end{cases}$$

Proof. We begin with the case n being even. For $z = w + w' \in \mathbb{C}^n$ with $w \in W$, $w' \in W'$, define $\Phi : \mathbb{C}^n \to \mathrm{End}_{\mathbb{C}} \bigwedge W$ by

$$\Phi(z) = \epsilon(w) - 2\iota(w').$$

Here, $\epsilon(w) \in \mathrm{End}_{\mathbb{C}} \bigwedge W$ is given by $x \mapsto w \wedge x$, and $\iota(w') \in \mathrm{End}_{\mathbb{R}} \bigwedge W$ by

$$\iota(w')(x_1 \wedge \cdots \wedge x_\ell) = \sum_{i=1}^{\ell} (-1)^{i+1} (w', x_i) x_1 \wedge \cdots \wedge \widehat{x_I} \wedge \cdots \wedge x_\ell.$$

A basic property for ϵ and ι is that $\epsilon(w)^2 = 0 = \iota(w')^2$. We claim that $\Phi(z)^2 = -2(w, w')\,\mathrm{id} = -(z, z)\,\mathrm{id}$, i.e.,

$$\epsilon(w)\iota(w') + \iota(w')\epsilon(w) = (w, w').$$

This follows from an easy calculation. In particular, this show that Φ, extending multiplicatively, defines a map

$$\Phi : C_n(\mathbb{C}) \to \mathrm{End}_{\mathbb{C}} \bigwedge W.$$

Since $\dim_{\mathbb{R}} C_n = 2^n$ and $\dim_{\mathbb{C}} W = m = \dfrac{n}{2}$, both sides have complex dimension 2^n, so to show Φ is an isomorphism it suffices to show it is surjective. To this end, let $w_1, \ldots, w_m$ be a basis for W and let $w'_1, \ldots, w'_n$ be its dual basis with respect to $(\ ,\ )$. Then for $1 \leqslant i_1 < \cdots < i_k \leqslant m$, a moment's thought shows that the operator

$$\Phi(w_{i_1} \cdots w_{i_k} w'_{i_1} \cdots w'_{i_k}) : \bigwedge W \to \bigwedge W$$

kills $\bigwedge^p W$ for $p < k$, maps $\bigwedge^k W$ onto $\mathbb{C} w_{i_1} \wedge \cdots \wedge w_{i_k}$, and stabilizes $\bigwedge^p W$ for $p > k$; in fact, it kills every pure k-form except $w_{i_1} \wedge \cdots \wedge w_{i_k}$. Easy manipulation shows that the image of Φ contains all maps on $\bigwedge W$ that maps $w_{i_1} \wedge \cdots \wedge w_{i_k}$ to $w_{i_1} \wedge \cdots \wedge w_{i_k}$ and all the other pure forms to 0. For $1 \leqslant j_1 < \cdots < j_l \leqslant m$, the operator $\Phi(w_{i_1} \cdots w_{i_k} w'_{j_1} \cdots w'_{j_l})$ maps $w_{j_1} \wedge \cdots \wedge w_{j_l}$ to $\pm w_{i_1} \wedge \cdots \wedge w_{i_k}$, and any other pure form to the others. Composing shows that the image of Φ contains all maps sending $w_{j_1} \wedge \cdots \wedge w_{j_l}$ to $w_{i_1} \wedge \cdots \wedge w_{i_k}$ and sending other basis elements to 0. This shows the surjectivity.

Now we consider the odd n case. This times, for $z = w + w' + ce_0 \in \mathbb{C}^n$ with $w \in W$, $w' \in W'$, define $\Phi^{\pm} : \mathbb{C}^n \to \mathrm{End}_{\mathbb{C}} \bigwedge W$ by

$$\Phi^{\pm}(z) = \epsilon(w) - 2\iota(w') \pm (-1)^{\deg} ic\,\mathrm{id}\,.$$

The factor $(-1)^{\deg}$ is added so that $\Phi^{\pm}(z)^2 = -2(w, w') - c^2 = -(z, z)$. Again this implies that $\Phi^{\pm}$ defines an algebra homomorphism $\Phi^{\pm} : C_n(\mathbb{C}) \to \mathrm{End}_{\mathbb{C}} \bigwedge W$, and hence

$$\Phi : C_n(\mathbb{C}) \to \mathrm{End}_{\mathbb{C}} \bigwedge W \oplus \mathrm{End}_{\mathbb{C}} \bigwedge W$$

by $z \mapsto (\Phi^+ z, \Phi^- z)$. Both sides having the same dimension 2^n, we can prove Φ is an isomorphism by showing Φ is surjective. This is similar to the proof in the even case; to "separate" the action, one makes use of use $\Phi(ie_0) = ((-1)^{\deg}, -(-1)^{\deg})$. $\square$

Corollary 8.35.1. There are $\mathbb{C}$-algebra isomorphisms

$$C_n^0(\mathbb{C}) := C_n^0 \otimes_{\mathbb{R}} \mathbb{C} \cong \begin{cases} \mathrm{End}_{\mathbb{C}} \bigwedge^{\mathrm{even}} W \oplus \mathrm{End}_{\mathbb{C}} \bigwedge^{\mathrm{odd}} W & \text{, if } n \text{ even} \\ \mathrm{End}_{\mathbb{C}} \bigwedge W & \text{, if } n \text{ odd.} \end{cases}$$

Here $\mathrm{End}_{\mathbb{C}} \bigwedge^{\mathrm{even}} W, \mathrm{End}_{\mathbb{C}} \bigwedge^{\mathrm{odd}} W$ stand for their obvious meanings.

Proof. For n even, it is clear that $\Phi(C_n^0(\mathbb{C}))$ stabilizes $\bigwedge^{\text{even}} W$ and $\bigwedge^{\text{odd}} W$, so restricting to them yields an injective algebra homomorphism

$$C_n^0(\mathbb{C}) \longrightarrow \operatorname{End}_{\mathbb{C}} \bigwedge^{\text{even}} W \oplus \operatorname{End}_{\mathbb{C}} \bigwedge^{\text{odd}} W$$

Since both sides have dimension 2^{n-1}, this is an isomorphism.

For n odd, restricting Φ^+ to $C_n^0(\mathbb{C})$ yields

$$\Phi^+ : C_n^0(\mathbb{C}) \longrightarrow \operatorname{End}_{\mathbb{C}} \bigwedge W.$$

In the proof of surjectivity of Φ in the even case, note that the crucial operator $\Phi(w_{i_1} \cdots w_{i_k} w'_{i_1} \cdots w'_{i_k})$ lies in the image of $C_n^0(\mathbb{C})$, and the operator $\Phi(w_{i_1} \cdots w_{i_k} w'_{j_1} \cdots w'_{j_l})$ in following argument can be modified, by adding some $\Phi(e_0)$, so that it also lies in the image of $C_n^0(\mathbb{C})$. To sum up, this means Φ^+ remains surjective. To show it is an isomorphism it suffices to notice that both sides have dimension 2^{n-1}. We remark that we could also use Φ^+ to establish an isomorphism. $\square$

Now we can construct the spin and half-spin representation. The isomorphism Φ (resp. Φ^+ or Φ^-) from the last lemma allows as to define a representation of $\operatorname{Spin}(n) \subseteq C_n^0(\mathbb{C})$ on $\bigwedge W$:

$$\Phi \text{ (resp. } \Phi^{\pm}) : \operatorname{Spin}(n) \to \operatorname{GL}(\bigwedge W)$$

This is the desired **spin representation**. When n is even, we can further restrict Φ to $\bigwedge^{\text{even}} W$ or $\bigwedge^{\text{odd}} W$, yielding

$$\Phi : \operatorname{Spin}(n) \to \operatorname{GL}(\bigwedge^{\text{even}} W) \text{ (resp. } \operatorname{GL}(\bigwedge^{\text{odd}} W)).$$

They are called the **half-spin representation**. These constructed representations are clearly spinorial, as $\Phi(-1) = -\operatorname{id}$.

Lemma 8.36. For n even, the half-spin representations are irreducible. For n odd, the spin representation is irreducible.

Proof. Let $\{e_i\}_{i=1}^n$ be the standard basis for $\mathbb{C}^n$. In $C_n(\mathbb{C})$, compute

$$(e_j \pm i e_{j+m})(e_k \pm i e_{k+m}) = e_j e_k \pm i(e_j e_{k+m} + e_{j+m} e_k) - e_{j+m} e_{k+n}.$$

Note that each summand on the right lies in $\operatorname{Spin}(n)$. This implies the operators $\epsilon(e_j)\epsilon(e_k)$ and $\iota(e_j)\iota(e_k)$ lie in the span of $\Phi(\operatorname{Spin}(n))$ in $\operatorname{End}_{\mathbb{C}} \bigwedge W$. Hence a $\operatorname{Spin}(n)$-invariant subspace is also invariant under $\epsilon(e_j)\epsilon(e_k)$ and $\iota(e_j)\iota(e_k)$. It is then clear that when n is even, two half spin representations are irreducible.

For n odd, compute (note that $e_n = e_0$ in the previous notation)

$$(e_j \pm i e_{j+m}) e_n = e_j e_n \pm i e_{j+m} e_n$$

so that $\epsilon(e_j)(-1)^{\deg}$ and $\iota(e_j)(-1)^{\deg}$ lie in the span of $\Phi(\operatorname{Spin}(n))$ in $\operatorname{End}_{\mathbb{C}} \bigwedge W$. It is then clear that the spin representation is irreducible.

$\square$

9 Direct Integrals

9.1 Von Neumann Algebras

Let H be a Hilbert space. For a subset $M \subseteq \mathcal{B}(H)$, we define the **commutant**, or the centralizer, of M as

$$M' = C_{\mathcal{B}(H)}(M) := \{T \in \mathcal{B}(H) \mid Tm = mT \text{ for all } m \in M\}$$

Clearly, for $N \subseteq M \subseteq \mathcal{B}(H)$, we have $M' \subseteq N'$. The commutant of M' is called the **bi-commutant** or **double commutant** of M. Also, for a subset $M \subseteq \mathcal{B}(H)$, define the **adjoint set** $M^* = \{m^* \mid m \in M\}$. A subset M is called **self-adjoint** if $M = M^*$.

Definition. A **von Neumann algebra** is a *-subalgebra $\mathcal{A}$ of $\mathcal{B}(H)$ with $\mathcal{A}'' = \mathcal{A}$.

- A von Neumann algebra is necessarily norm closed, so it is itself a C^*-algebra. The converse need not hold.
- For a subset $M \subseteq \mathcal{B}(H)$, we have $M \subseteq M''$, so $M''' \subseteq M'$. On the other hand $M' \subseteq (M')'' = M'''$, so $M' = M'''$. Thus, if $M^* = M$, then M' is a von Neumann algebra.
- For self-adjoint $M \subseteq \mathcal{B}(H)$, its double commutant M'' is the smallest von Neumann algebra in $\mathcal{B}(H)$ containing M, and it is called the von Neumann algebra generated by M.

For a von Neumann algebra $\mathcal{A} \subseteq \mathcal{B}(H)$, $Z(\mathcal{A}) = A \cap A'$ is called the **center** of $\mathcal{A}$. A von Neumann algebra $\mathcal{A}$ is called a **factor** if its center is trivial.

Example 9.1.

1.

9.2 Weak and Strong Topologies

Let H be a Hilbert space. On $\mathcal{B}(H)$ we can define three topologies.

1. The topology induced by the operator norm. This is called the **norm topology**.
2. The topology induced by the family of seminorms $T \mapsto \|Tv\|$, where v runs over H. This is called the **strong operator topology (SOT)**.
3. The topology induced by the family of seminorms $T \mapsto |\langle Tv, w\rangle|$, where v, w run over H. This is called the **weak operator topology (WOT)**.

It is clear that for a subset $\mathcal{A} \subseteq \mathcal{B}(H)$, we have the inclusions

$$\mathcal{A} \subseteq \overline{\mathcal{A}}^{\text{NORM}} \subseteq \overline{\mathcal{A}}^{\text{SOT}} \subseteq \overline{\mathcal{A}}^{\text{WOT}} \subseteq \mathcal{A}''$$

Theorem 9.2 (von Neumann's Bi-commutant Theorem)**.** Let H be a Hilbert space, and let $\mathcal{A}$ be a unital *-subalgebra of $\mathcal{B}(H)$. Then $\overline{\mathcal{A}}^{\mathrm{SOT}} = \overline{\mathcal{A}}^{\mathrm{WOT}} = \mathcal{A}''$.

Proof. It suffices to show $A'' \subseteq \overline{\mathcal{A}}^{\mathrm{SOT}}$.

□

9.3 Representations

9.4 Hilbert Integrals

A family of vectors $(\xi_i)_{i\in I}$ in a Hilbert space H is called **quasi-orthonormal basis** if the nonzero members of this family form an orthonormal basis of H.

Let $(X, \mathcal{D})$ be a measurable space. A **Hilbert bundle** over X is a family of Hilbert space $(H_x)_{x\in X}$ together with a family of maps $\xi_i : X \to \bigsqcup_{x\in X} H_x$ $(i \in I)$ with $\xi_i(x) \in H_x$ such that

- for each $x \in X$, the family $(\xi_x(x))_{i\in I}$ is a quasi-orthonormal basis of H_x, and
- for each $i \in I$, $\{x \in X \mid \xi_i(x) = 0\} \in \mathcal{D}$.

A **section** is a map $s : X \to \bigsqcup_{x\in X} H_x$ with $s(x) \in H_x$ for each $x \in X$. A section s is called a **measurable section** if

- for every $i \in I$, the function $x \mapsto \langle s(x), \xi_i(x)\rangle$ is $\mathcal{D}$-measurable on X, and
- there exists a countable subset $I_s \subseteq I$ such that the function $x \mapsto \langle s(x), \xi_i(x)\rangle$ vanishes identically for every $i \notin I_s$.

Let μ be a measure on $\mathcal{D}$. A measurable section s is called a **null-section** if it vanishes outside a null set. The **direct Hilbert integral** is the vector space of all measurable sections s satisfying

$$\|s\|^2 := \int_X \|s(x)\|^2 \, d\mu(x) < \infty$$

modulo the space of null-sections. We denote the space by $H = \displaystyle\int_X^{\oplus} H_x d\mu(x)$.

Lemma 9.3. For $s, t \in H$, define

$$\langle s, t\rangle := \int_X \langle s(x), t(x)\rangle \, d\mu(x)$$

This defines an inner product on H, and H becomes a Hilbert space.

Proof. We must show the completeness. For $i \in I$, let $X_i := \{x \in X \mid \xi_i(x) \neq 0\}$ and $P_i : H \to L^2(X_i)$ be given by $P_i(s)(x) := \langle s(x), \xi_i(x)\rangle$. We claim this is surjective. For $f \in L^2(X_i)$, define $s_f : X \to \bigsqcup_{x\in X} H_x$ by $s_f(x) = f(x)\xi_i(x)$.

- For each $j \in I$ and $x \in X$, $\langle s_f(x), \xi_j(x)\rangle = f(x)\delta_{ij}$ is measurable.
- The function $x \mapsto \langle s_f(x), \xi_j(x)\rangle = f(x)\delta_{ij}$ vanishes if $j \neq i$.

The s_f defines a measurable section. Moreover,

$$\|s_f\|^2 = \int_X \|s_f(x)\|^2 \, d\mu(x) = \int_{X_i} |f(x)|^2 d\mu(x) = \|f\|_2^2 < \infty$$

so $s_f \in H$. Thus the P_i defines a map with dense image

$$H = \int_X^{\oplus} H_x \, d\mu(x) \longrightarrow \widehat{\bigoplus_{i\in I}} L^2(X_i)$$

Now it suffices to show this map is isometric. For $s \in H$, by Parseval's identity

$$\begin{aligned}
\|s\|^2 &= \int_X \|s(x)\|^2 \, d\mu(x) = \int_X \left(\sum_{i\in I_s} |\langle s(x), \xi_i(x)\rangle|^2 \right) d\mu(x) \\
&= \sum_{i\in I_s} \int_X |\langle s(x), \xi_i(x)\rangle|^2 d\mu(x) \\
&= \sum_{i\in I_s} \int_{X_i} |P_i(s)(x)|^2 d\mu(x) = \sum_{i\in I} \|P_i(s)\|_2^2
\end{aligned}$$

□

Example 9.4. Direct sums are special cases of direct integrals. Let $H = \bigoplus_{j\in I} H_j$ be a direct sum of separable Hilbert spaces. This space equals the direct integral $\int_X^{\oplus} H_x \, d\mu(x)$ with $X = I$ and μ the counting measure on X.

Let (H_x, ξ_j) be a Hilbert bundle and μ a measure on X. Let G be an LCH group, and for every $x \in X$, let η_x be a unitary representation of G on H_x, such that for every $g \in G$ and all $i, j \in I$, the map $x \mapsto \langle \eta_x(g)\xi_i(x), \xi_j(x)\rangle$ is measurable. Then

$$\eta(g)s(x) := \eta_x(g)s(x)$$

defines a unitary representation of G on the direct integral $H = \int_X^{\oplus} H_x \, d\mu(x)$.

Example 9.5. Let A be an LCA group with the dual group $\widehat{A}$ equipped with the Plancherel measure. Each character $\chi \in \widehat{A}$ determines a one-dimensional representation of A on $H_\chi = C$. Put $\eta_\chi(y) := \chi(y)$. Then the direct integral (with constant section $\xi_1(\chi) := 1 \in \mathbb{C} = H_\chi$) satisfies

$$\int_{\widehat{A}}^{\oplus} H_\chi d\chi \cong L^2(\widehat{A})$$

with $\eta(y)\xi(\chi) = \chi(y)\xi(\chi)$. It follows from the Plancherel theorem that $(\eta, L^2(\widehat{A}))$ is unitarily equivalent to the left regular representation $(L, L^2(A))$ of A via the Fourier transform.

9.5 The Plancherel Theorem

Theorem 9.6. Let G be a second countable, unimodular, locally compact group of type I. There is a unique measure μ on $\widehat{G}$ such that for $f \in L^1(G) \cap L^2(G)$ one has

$$\|f\|_2^2 = \int_{\widehat{G}} \|\pi(f)\|_{\mathrm{HS}}^2 \, d\mu(\pi)$$

The map $f \mapsto (\pi(f))_\pi$ extends to a unitary $G \times G$ equivariant map

$$L^2(G) \cong \int_{\widehat{G}}^{\oplus} \mathcal{H}(V_\pi) d\mu(\pi)$$

where the representation of η_π of $G \times G$ on the space of Hilbert-Schmidt operators $\mathcal{H}(V_\pi)$ is given by $\eta_\pi(x, y)T = \pi(x)T\pi(y^{-1})$ for each $\pi \in \widehat{G}$ and $x, y \in G$.

10 The Selberg Trace Formula

10.1 Cocompact Groups and Lattices

Definition. Let G be a topological group.

1. A subgroup $H \leqslant G$ is called **cocompact** if the quotient G/H is a compact space.

2. A subgroup $\Lambda \leqslant G$ is called **discrete** if the subspace topology on Λ is the discrete topology.

Proposition 10.1. Let G be an LCH group. If G admits a unimodular closed cocompact subgroup, then G is unimodular itself.

Proof. Let $H \leqslant G$ be a unimodular closed cocompact subgroup. Since $G/H \cong H\backslash G$ via $gH \mapsto Hg^{-1}$, $H\backslash G$ is compact as well and the map $C_c(G) \to C(H\backslash G)$ given by $g \mapsto {}^H g$ is surjective by Lemma 1.15, where ${}^H g(x) := \int_H g(hx)dh$. We install a Radon measure μ on the right coset space $H\backslash G$ by Riesz's representation theorem as follows. For $f \in C(H\backslash G)$, choose $g \in C_c(G)$ with ${}^H g = f$, and define

$$\int_{H\backslash G} f(x)d\mu(x) := \int_G g(x)dx$$

To show this is well-define, suppose ${}^H g = 0$. Note that since H is unimodular, $\int_H g(h^{-1}x)dh = \int_H g(hx)dh = 0$. Let $\phi \in C_c(G)$ with ${}^H\phi = 1$. One gets

$$\int_G g(x)dx = \int_G \int_H \phi(hx)g(x)dhdx = \int_H \int_G \phi(hx)g(x)dxdh = \int_H \int_G \phi(x)g(h^{-1}x)dxdh = 0$$

Now, for $f \in C_c(H\backslash G)$, if we take $g \in C_c(G)$ with ${}^H g = f$, then

$$\int_{H\backslash G} f(xy)d\mu(x) = \int_G g(xy)dx = \Delta(y^{-1}) \int_G g(x)dx = \Delta(y^{-1}) \int_{H\backslash G} f(x)d\mu(x)$$

In particular, for $f \equiv 1 \in C(H\backslash G)$, we have

$$0 < \int_{H\backslash G} f(x)d\mu(x) = \int_{H\backslash G} f(xy)d\mu(x) = \Delta(y^{-1}) \int_{H\backslash G} f(x)d\mu(x)$$

so that $\Delta(y^{-1}) = 1$ for all $y \in G$. $\square$

Lemma 10.2. Let G be a topological group.

1. A subgroup $\Gamma \leqslant G$ is discrete if and only if there exists a unit-neighborhood $U \subseteq G$ with $\Gamma \cap U = \{1\}$.

2. A discrete subgroup is closed in G if G is Hausdorff.

Proof.

1. Clear.

2. We prove the following generalization.

   Lemma 10.3. Let G be a Hausdorff topological group and $H \leqslant G$ be a locally compact subgroup. Then H is closed in G.

   Proof. Replacing G with the closure of H in G, we may assume H is dense in G. Let $x \in H$ and choose a neighborhood U of x in H with compact closure C. Write $U = V \cap H$ for some open $V \subseteq G$. Since C is compact and G is Hausdorff, C is closed in G, and thus $V \backslash C$ is open in G. But for $V \cap H = U \subseteq C$, it forces $(V \backslash C) \cap H = \varnothing$, and since H is dense in G, it must be the case $V \subseteq C$; in particular, $V \subseteq H$. This shows H is open in G, and since they are topological groups, H is closed in G. □

   A discrete space is automatically locally compact and Hausdorff, so the lemma applies.

□

Definition. Let G a LCH group. A discrete subgroup $\Gamma \leqslant G$ such that G/Γ carries an invariant Radon measure μ with $\mu(G/\Gamma) < \infty$ is called a **lattice** in G. A cocompact lattice is called a **uniform lattice**.

- The quotient G/Γ admits a invariant Radon measure, so by Theorem 1.17, $\Delta_G|_\Gamma = \Delta_\Gamma \equiv 1$.
- When we speak of the Haar measure on a lattice, we always measure the counting measure, and on the quotient G/Γ we always equip it with the measure so that the quotient measure formula holds.

Proposition 10.4. Let G be an LCH group. A discrete cocompact subgroup Γ is a uniform lattice.

Proof. By Proposition 1.12, Γ is unimodular, so Proposition 10.1 together with Lemma 10.2.2. implies G is unimodular. By Theorem 1.17 G/Γ admits an invariant Radon measure, and since G/Γ is compact, it has finite volume, so it is a lattice. □

Proposition 10.5. Let G be an LCH group. If G admits a lattice, then G is unimodular.

Proof. Let Δ be the modular function of G and let $H = \ker \Delta$; then H is unimodular by Corollary 1.17.2.2. Let $\Gamma \leqslant G$ be a lattice. As observed above, we have $\Delta_G|_\Gamma = 1$, so $\Gamma \leqslant H$. Since $H/\Gamma \leqslant G/\Gamma$, by Theorem 1.17 we have

$$\infty > \operatorname{vol}(G/\Gamma) = \int_{G/\Gamma} 1 dx = \int_{G/H} \int_{H/\Gamma} 1 dx dy = \operatorname{vol}(G/H) \operatorname{vol}(H/\Gamma)$$

In particular, $\operatorname{vol}(G/H) < \infty$, so G/H is compact by Proposition 1.13. Now it follows from Proposition 10.1 that G is unimodular. □

We conclude this subsection with some interesting facts about lattices.

Proposition 10.6 (Pigeonhole principal)**.** Let G be an LCH group and Γ a cocompact lattice. If $X \subseteq G$ is a measurable subset such that $\mathrm{vol}(X) > \mathrm{vol}(G/\Gamma)$, then there are two distinct elements $x, x' \in X$ satisfying $x^{-1}x' \in \Gamma$.

Proof. Let $\pi : G \to G/\Gamma$ be the canonical projection. We prove that for each measurable $X \subseteq G$, we have $\mathrm{vol}(X) \geqslant \mathrm{vol}(\pi(X))$ with equality if π is injective on X. The proposition follows by taking contrapositive.

Let $X \subseteq G$ be measurable. Then

$$\begin{aligned}\mathrm{vol}(X) &= \int_G \mathbf{1}_X(x)dx = \int_{G/\Gamma} \sum_{\gamma\in\Gamma} \mathbf{1}_X(x\gamma)dx \\ &= \int_{G/\Gamma} \#\{\gamma \in \Gamma \mid x\gamma \in X\}dx = \int_{G/\Gamma} \#\{y \in X \mid x^{-1}y \in \Gamma\}\end{aligned}$$

On the other hand, we have $x\Gamma \in \pi(X)$ if and only if $\#\{y \in X \mid x^{-1}y \in \Gamma\} \geqslant 1$, which implies

$$\mathrm{vol}(\pi(X)) = \int_{G/\Gamma} \mathbf{1}_{\pi(X)}(x)dx \leqslant \int_{G/\Gamma} \#\{y \in X \mid x^{-1}y\}dx = \mathrm{vol}(X)$$

When $\pi|_X$ is injective, $x\Gamma \in \pi(X)$ if and only $\#\{y \in X \mid x^{-1}y \in \Gamma\} = 1$, so the equality holds. □

Proposition 10.7. Let G be an LCH group and Γ a cocompact lattice. If Γ' is another lattice of G containing Γ, then $\mathrm{vol}(G/\Gamma) = [\Gamma' : \Gamma]\,\mathrm{vol}(G/\Gamma')$.

Proof. Since G/Γ is compact, by Lemma 1.15 we can find $f_0 \in C_c^+(G)$ such that

$$f_1(x) := f_0^\Gamma(x) = \sum_{\gamma\in\Gamma} f_0(x\gamma) = 1$$

for all $x \in G$. If we form $f_2(x) = f_0^{\Gamma'}(x)$, then $f_2(x) = [\Gamma' : \Gamma]$ for all $x \in G$. Now by quotient integral formula we have

$$\int_G f_0(x)dx = \int_{G/\Gamma} \sum_{\gamma\in\Gamma} f_0(x\gamma)dx = \int_{G/\Gamma} 1dx = \mathrm{vol}(G/\Gamma)$$

and

$$\int_G f_0(x)dx = \int_{G/\Gamma'} \sum_{\gamma\in\Gamma'} f_0(x\gamma)dx = \int_{G/\Gamma'} [\Gamma' : \Gamma]dx = [\Gamma' : \Gamma]\,\mathrm{vol}(G/\Gamma')$$

□

Proposition 10.8. Let G be an LCH group, Γ is cocompact lattice and $\sigma : G \to G$ an topological group automorphism. Then $\mathrm{vol}(G/\sigma(\Gamma)) = \mathrm{mod}_G(\sigma)\,\mathrm{vol}(G/\Gamma)$.

Proof. Let f_0 and f_1 be as in the proof of the previous proposition. Then

$$\int_G f_0(\sigma^{-1}(x))dx = \mathrm{mod}_G(\sigma)\int_G f_0(x)dx = \mathrm{mod}_G(\sigma)\int_{G/\Gamma} \sum_{\gamma\in\Gamma} f_0(x\gamma)dx = \mathrm{mod}_G(\sigma)\,\mathrm{vol}(G/\Gamma)$$

On the other hand,

$$\int_G f_0(\sigma^{-1}x)dx = \int_{G/\sigma(\Gamma)} \sum_{\gamma\in\sigma(\Gamma)} f_0(\sigma^{-1}(x\gamma))dx = \int_{G/\sigma(\Gamma)} \sum_{\gamma\in\Gamma} f_0(\sigma^{-1}(x)\gamma)dx = \mathrm{vol}(G/\sigma(\Gamma))$$

□

10.2 Discreteness of the Spectrum

Let G be an LCH group and $\Gamma \leqslant G$ a cocompact lattice. Then the right coset space $\Gamma\backslash G$ admits a right-invariant Radon measure μ with $\mu(\Gamma\backslash G) < \infty$. Therefore the action of G on $L^2(\Gamma\backslash G)$ by right translation R gives rise to a unitary representation of G.

Theorem 10.9. Let G be an LCH group and $\Gamma \leqslant G$ a uniform lattice. The representation R on $L^2(\Gamma\backslash G)$ decomposes as a direct sum of irreducible representations with finite multiplicities, i.e.,

$$L^2(\Gamma\backslash G) \cong \widehat{\bigoplus_{\pi\in\widehat{G}}} N_\Gamma(\pi)\pi$$

where $\widehat{G}$ is the unitary dual of G, and $N_\Gamma(\pi) \in \mathbb{N}_0$.

The proof of this theorem will occupy the rest of this subsection. First, we need a lemma to tell us that for $f \in C_c(G)$ the operator $R(f)$ is given a continuous integral kernel. For later use we will extend this to a greater class of functions f. Let U be a compact unit-neighborhood in G. For a continuous function f on G, let $f_U : G \to [0, \infty)$ be defined by

$$f_U(y) := \sup_{x,z\in U} |f(xyz)|$$

Lemma 10.10. The function f_U is continuous.

Proof. It suffices to show that for all $a \geqslant 0$, the sets $f^{-1}((a, \infty))$ and $f_U^{-1}([0, a))$ are open. For the former, assume $f_U(x) > a$. Then we can find $u_1, u_2 \in U$ with $|f(u_1xu_2)| > a$. As the function $y \mapsto f(u_1yu_2)$ is continuous, we can find a neighborhood V of x such that $|f(u_1vu_2)| > a$ for all $v \in V$. This implies $f_U(v) > a$ for every $v \in V$.

For the latter, let assume $a > f_U(y) \geqslant 0$ and let $\varepsilon > 0$ be given so that $a - \varepsilon > f_U(y)$. The function $(u, x, v) \mapsto |f(uxv)|$ is continuous, so for all $(u, v) \in U \times U$, we can find open neighborhood U_{uv} of u, v in U and open neighborhood V_{uv} of y in G such that $|f(u_1xu_2)| < a - \varepsilon$ for all $u_1, u_2 \in U_{uv}$ and $x \in V_{uv}$. The family $\{U_{uv} \times U_{uv}\}_{u,v\in U}$ of open sets covers $U \times U$, so there exist $u_i, v_j \in U$ $(1 \leqslant i, j \leqslant n < \infty)$ such that $U \times U = \bigcup_{i=1}^{n} U_{u_iv_i} \times U_{u_iv_i}$. Let $V = \bigcap_{i=1}^{n} V_{u_iv_i} \underset{\text{open}}{\subseteq} G$. Then for $x \in V$ and $u, v \in U$, we have $(u, v) \in U_{u_iv_i} \times U_{u_iv_i}$ for some $1 \leqslant i \leqslant n$, so $|f(uxv)| < a - \varepsilon$. Thus $f_U(x) \leqslant a - \varepsilon < a$ for all $x \in V$.

□

Definition. A continuous function $f : G \to \mathbb{C}$ is called **uniformly integrable** if there exists a compact unit-neighborhood U such that $f_U \in L^1(G)$. Denote by $C_{\text{unif}}(G)$ the set of all uniformly integrable continuous function on G.

- If $f \in C_{\text{unif}}(G)$, then $|f| \leqslant f_U$, so $f \in L^1(G)$.

Lemma 10.11. Let G be unimodular. Then $C_{\text{unif}}(G) \subseteq C_0(G)$ and it is an algebra under convolution.

Proof. Let $f \in C_{\text{unif}}(G)$ and let U be a compact symmetric unit-neighborhood of G such that $f_U \in L^1(G)$. If f does not vanish at infinity, then we can find $\varepsilon > 0$ such that for all compact set K of G there exists $x \in G \backslash K$ with $|f(x)| \geqslant \varepsilon$. Let $x_1 \in G$ such that $|f(x_1)| \geqslant \varepsilon$, and let $x_2 \notin x_1 U^2$ such that $|f(x_2)| \geqslant \varepsilon$. Since U is symmetric, $x_1 U \cap x_2 U = \varnothing$. Next, pick $x_3 \notin x_1 U \cup x_2 U$ with $|f(x_3)| \geqslant \varepsilon$. Continuing in this way, we construct a sequence $(x_n)_n$ in G such that $x_n U \cap x_m U = \varnothing$ whenever $n \neq m$, and $|f(x_n)| \geqslant \varepsilon$ for every n. But then $f_U \geqslant \varepsilon$ on $x_n U$, which contradicts the integrability of f_U. Note that in this paragraph the unimodularity of G is not used.

Since $L^1(G) \cap C_0(G) \subseteq L^2(G)$, we have $C_{\text{unif}}(G) \subseteq L^2(G)$. Let $f, g \in C_{\text{unif}}(G)$. We can write $f * g(x) = \langle f, L_x g^* \rangle$. Since $x \mapsto L_x g^*$ is continuous by Lemma 1.36 and the inner product is continuous, the convolution $f * g(x)$ is also continuous in x. Finally, take a compact unit-neighborhood U such that f_U, $g_U \in L^1(G)$. Then

$$\begin{aligned}
(f * g)_U(y) &= \sup_{x,z \in U} \left| \int_G f(\xi) g(\xi^{-1} x y z) d\xi \right| \\
&= \sup_{x,z \in U} \left| \int_G f(x\xi) g(\xi^{-1} y z) d\xi \right| \\
&\leqslant \sup_{x,z \in U} \int_G |f(x\xi) g(\xi^{-1} y z)| d\xi \\
&\leqslant \int_G f_U(\xi) g_U(\xi^{-1} y) d\xi = f_U * g_U(y)
\end{aligned}$$

This implies that $(f * g)_U \in L^1(G)$, so $f * g \in C_{\text{unif}}(G)$. □

Lemma 10.12. For $f \in C_{\text{unif}}(G)$ and $\phi \in L^2(\Gamma \backslash G)$ one has

$$R(f)\phi(x) = \int_{\Gamma \backslash G} k(x, y) \phi(y) dy$$

where $k(x, y) := \sum_{\gamma \in \Gamma} f(x^{-1} \gamma y)$. The kernel is continuous on $\Gamma \backslash G \times \Gamma \backslash G$.

Proof. Let $f \in L^1(G)$. For $\phi \in L^2(\Gamma \backslash G)$, by quotient integral formula one computes

$$\begin{aligned}
R(f)\phi(x) &= \int_G f(y) R(y) \phi(x) dy = \int_G f(y) \phi(xy) dy \\
&= \int_G f(x^{-1} y) \phi(y) dy \\
&= \int_{\Gamma \backslash G} \sum_{\gamma \in \Gamma} f(x^{-1} \gamma y) \phi(\gamma y) dy = \int_{\Gamma \backslash G} k(x, y) \phi(y) dy
\end{aligned}$$

The obtained formula is valid almost everywhere in $x \in G$. In particular, for $f \in C_{\text{unif}}(G)$ this argument works with f replaced by f_U for a suitable compact symmetric unit-neighborhood U to get a kernel k_U. Now choose $g \in C_c(G)$ with $g \geqslant 0$ and ${}^{\Gamma}g(x) := \sum_{\gamma \in \Gamma} g(\gamma x) = 1$ for all $x \in G$ (c.f. Lemma 1.15) and use

quotient integral formula to get

$$\begin{aligned}\int_{\Gamma\backslash G\times\Gamma\backslash G} k_U(x,y)dxdy &= \int_{\Gamma\backslash G}\int_{\Gamma\backslash G}\sum_{\gamma\in\Gamma} g(\gamma x)\sum_{\tau\in\Gamma} f_U(x^{-1}\gamma^{-1}\tau y)dxdy \\ &= \int_{\Gamma\backslash G}\int_G g(x)\sum_{\tau\in\Gamma} f_U(x^{-1}\tau y)dxdy \\ &= \int_G\int_G g(x)f_U(x^{-1}y)dxdy = \|g*f_U\|_1 < \infty\end{aligned}$$

Thus $\sum_{\gamma\in\Gamma} f(x^{-1}\gamma y)$ converges almost everywhere in (x,y), so it converges on a dense set of (x,y). Let (x_0,y_0) be such a point of convergence. We are going to show $k(x,y)$ is continuous on the subset $x_0U\times y_0U$. Let $\varepsilon>0$ be given and choose a finite subset $S\subseteq\Gamma$ so that $\sum_{\gamma\notin S} f_U(x_0^{-1}\gamma y_0) < \frac{\varepsilon}{2}$. This means for $(x,y)\in x_0U\times y_0U$, one has $\sum_{\gamma\notin S}|f(x^{-1}\gamma y)| < \frac{\varepsilon}{2}$, which implies that for $(x,y),(x',y')\in x_0U\times y_0U$

$$|k(x,y)-k(x',y')| \leqslant \sum_{\gamma\in S}|f(x^{-1}\gamma y)-f(x'^{-1}\gamma y')|+\varepsilon$$

Letting $(x',y')\to(x,y)$, the continuity of k follows from that of f. □

As $k\in C(\Gamma\backslash G\times\Gamma\backslash G)\subseteq L^2(\Gamma\backslash G\times\Gamma\backslash G)$, the operator $R(f)$ is Hilbert-Schmidt by Proposition 6.18, whence compact by Proposition 6.20. The theorem follows from the next lemma.

Lemma 10.13. Let A be a *-closed subspace of $C_c(G)$. Let (η,V_η) be a unitary representation of G such that for every $f\in A$, the operator $\eta(f)$ is compact and such that for every nonzero $v\in V_\eta$, the space $\eta(A)v$ is nonzero. Then η is a direct sum of irreducible representations with finite multiplicities.

Notice here $C_c(G)$ is viewed as a *-subalgebra of $L^1(G)$.

Proof. By a Zorn's lemma argument we can find a subspace $E\leqslant V$ maximal with the property that it decomposes as a sum of irreducibles. The assumption of the lemma also holds for the orthogonal complement $E^\perp$ of E in $V=V_\eta$. By maximality of E this orthogonal complement cannot contain any irreducible subspace. We must show $E^\perp=0$. Equivalently, we show that a representation η as in the assumption always contains an irreducible subspace.

The space A is generated by the self-adjoint elements. Let $f\in A$ be self-adjoint; then the operator $\eta(f)$ is self-adjoint as well. By assumption, $\eta(f)$ is nonzero and compact. Therefore by the Spectral Theorem $\eta(f)$ has a nonzero eigenvalue, say μ. Let V_μ be the corresponding eigenspace. Consider the collection

$$\{E\cap V_\mu \mid E\leqslant V \text{ is closed and invariant, } E\cap V_\mu\neq 0\}$$

This is nonempty. Indeed, if we choose $0\neq v\in V_\mu$, then $\overline{\eta(L^1(G))v}$ is closed and invariant with $0\neq\mu v\in\overline{\eta(L^1(G))v}\cap V_\mu$. Among this collection we pick W with minimal dimension. Let

$$E^1 := \bigcap\{E\leqslant V \mid E \text{ is closed and invariant, } E\cap V_\mu = W\}\supseteq W$$

Then $E^1 \leqslant V$ is closed and invariant. To conclude it suffices to show E^1 is irreducible. Let $F \leqslant E$ be a closed and invariant subspace. Note the invariance implies $\eta(f)E \subseteq E$ and $\eta(f)F \subseteq F$. Replacing F by its orthogonal complement in E, we may assume $F \cap V_\mu \neq 0$. But $0 \neq F \cap V_\mu \subseteq E \cap V_\mu = W$, the minimality forces $F \cap V_\mu = W$, and thus $E^1 \subseteq F$, or $F = E^1$. This shows the irreducibility.

It remains to show the multiplicities are finite. For this note first that if τ and σ are unitarily equivalent representations, then λ is an eigenvalue of $\tau(f)$ if and only if it is one for $\sigma(f)$. Thus any collection of orthogonal subspaces of V that are mutually unitarily equivalent must all have nontrivial intersection with the same eigenspaces of some $f \in A$. But each eigenspace is finite dimensional, the finiteness of multiplicity follows. □

We include a lemma similar to the above one.

Lemma 10.14. Let G be an LCH group and (π, V) a unitary representation of G. Suppose that for each unit-neighborhood U of G there exists an integrable function f on G such that

(i) f is nonnegative and symmetric with support in U,

(ii) $\displaystyle\int_G f(g)dg = 1$, and

(iii) $\pi(f) \in \mathcal{B}(V)$ is compact.

Then π is a direct sum of irreducibles with finite multiplicities.

Proof. Similar to the proof of Lemma 10.13, it suffices to show such a representation always admits an irreducible subspace. Let $v \in V$ be of norm one. By continuity, we may choose a unit-neighborhood U so small that $\|v - \pi(g)v\| < \dfrac{1}{2}$ for all $g \in U$. Pick f as in the statement of the theorem; then

$$\|\pi(f)v - v\| \leqslant \int_U f(g)\,\|\pi(g)v - v\|\,dg < \frac{1}{2}\int_U f(g)dg = \frac{1}{2}$$

Note that $f^*(x) := \Delta_G(x^{-1})\overline{f(x^{-1})}$ also satisfies those conditions in the theorem, with the same estimate $\|\pi(f^*)v - v\| \leqslant \dfrac{1}{2}$, so

$$\|\pi(f + f^*)v - 2v\| \leqslant \|\pi(f)v - v\| + \|\pi(f^*)v - v\| \leqslant 1$$

In particular, if we put $h = f + f^*$, then $\pi(h)v \neq 0$. In sum, $\pi(h)$ is a nonzero self-adjoint compact operator. The rest of the proof is the same as that of the previous lemma. □

10.3 The Trace Formula

Definition. Let X be an LCH space and μ a Radon measure on X. A continuous L^2-kernel k on X is called **admissible** if there exists a function $g \in C(X) \cap L^2(X)$ such that $|k(x, y)| \leqslant g(x)g(y)$ for all $x, y \in X$.

- If X is compact, then every continuous kernel is admissible.

An operator $S : L^2(X) \to L^2(X)$ is called an **admissible operator** if there exists an admissible kernel k such that

$$S\phi(x) = \int_X k(x,y)\phi(y)d\mu(y)$$

for all $\phi \in L^2(X)$, $x \in X$.

- Note that an integral operator on $L^2(X)$ with L^2-kernel is necessarily a bounded operator.

Proposition 10.15. Let X be an LCH space equipped with a Radon measure dx. Assume X is either first countable or compact. Let T be an integral operator with continuous L^2-kernel k on X. Assume that there exists admissible operators S_1, S_2 with $T = S_1S_2$. Then T is of trace class and

$$\mathrm{tr}(T) = \int_X k(x,x)dx$$

The last formula holds as long as T is of trace class when the diagonal $k(x,x)$ is defined in a correct way. See [Bri91] and also mathoverflow post.

Proof. Replacing S_2 with S_2^*, we may assume $T = S_1S_2^*$. By polarization, to show this proposition we may further assume $T = SS^*$ with S admissible. Since S is Hilbert-Schmidt, $T = SS^*$ is of trace class by Theorem 6.23.(c), and

$$\mathrm{tr}(T) = \sum_j \langle SS^*e_j, e_j\rangle = (S^*, S^*) = \|S^*\|_{\mathrm{HS}}^2 = \|S\|_{\mathrm{HS}}^2$$

where $(e_j)_j$ is an orthonormal basis for $L^2(X)$, and $(\,,\,)$ is the pairing defined in Proposition 6.16. The last equality is Lemma 6.14, and the first equality is Theorem 6.22. Let ℓ be the admissible kernel of S.

Lemma 10.16. If T is an integral operator on $L^2(X)$ with kernel k, then its adjoint T^* is also an integral operator with kernel $\overline{k}$.

Proof. For $\phi, \psi \in C_c(X)$, compute

$$\langle T^*\phi, \psi\rangle = \langle \phi, T\psi\rangle = \int_X \phi(x)\overline{T\psi}(x)dx = \int_X\int_X \phi(x)\overline{k(x,y)\psi(y)}dydx = \left\langle \int_X \phi(x)\overline{k(x,\cdot)}dx, \psi \right\rangle$$

Since $C_c(X)$ is dense in $L^2(X)$, the result follows. □

Thus for $\phi \in C_c(X)$ we have

$$SS^*\phi(x) = \int_X \ell(x,u)S^*\phi(u)du = \int_X\int_X \ell(x,u)\overline{\ell(y,u)}\phi(y)dydu = \int_X \left(\int_X \ell(x,u)\overline{\ell(y,u)}du\right)\phi(y)dy$$

So the operator SS^* has the kernel $\ell * \ell^*(x,y) := \displaystyle\int_X \ell(x,z)\overline{\ell(y,z)}dz$. We prove $\ell * \ell^*$ is continuous.

- If X is first countable, the continuity can be checked with sequences. Since $z \mapsto\mapsto \ell(x,z)\overline{\ell(y,z)}$ is integrable over X for each (x,y) with a common integrable upper bound by virtue of admissibility, the continuity then follows from DCT.

- If X is compact, then ℓ is uniformly continuous, in the sense that for all $x_0 \in X$ and $\varepsilon > 0$ there exists a neighborhood U of x_0 such that for all $(x,z) \in U \times X$ we have $|\ell(x,z) - \ell(x_0,z)| < \varepsilon$. Together with $\mathrm{vol}(X, dx) < \infty$, this implies $\ell * \ell^*$ is continuous.

Last, for ϕ, $\psi \in C_c(X)$, we have

$$\int_X \int_X \ell * \ell^*(x,y)\phi(y)\phi(x)dydx = \langle SS^*\phi, \overline{\psi}\rangle = \int_X \int_X k(x,y)\phi(y)\psi(x)dydx$$

Since $C_c(X)$ is dense in $L^2(X)$, we conclude that $\ell * \ell^* = k$. Therefore,

$$\int_X k(x,x)dx = \int_X \int_X |\ell(x,u)|^2 dudx = \|S\|_{\mathrm{HS}}^2 = \mathrm{tr}(T)$$

where the third equality is Proposition 6.18. □

Recall that for a uniform lattice Γ of G and $\pi \in \widehat{G}$, the number $N_\Gamma(\pi) \geqslant 0$ is the multiplicity of π as a subrepresentation of $(R, L^2(\Gamma\backslash G))$. Denote by $\widehat{G}_\Gamma$ the set of all $\pi \in \widehat{G}$ with positive multiplicity.

Write $C_{\mathrm{unif}}(G)^2 = C_{\mathrm{unif}}(G) * C_{\mathrm{unif}}(G)$ for the linear span of functions of the form $g*h$ with $g, h \in C_{\mathrm{unif}}(G)$.

If Γ is a lattice in G and $\gamma \in \Gamma$, denote by $[\gamma]$ the conjugacy class of γ in Γ, $G_\gamma = C_G(\gamma)$ the centralizer of γ in G, and $\Gamma_\gamma = G_\gamma \cap \Gamma$ the centralizer of γ in Γ. Note that the map $\Gamma \ni \nu \mapsto \nu^{-1}\gamma\nu \in [\gamma]$ induces a bijection $\Gamma_\gamma\backslash\Gamma \cong [\gamma]$.

Theorem 10.17 (Trace Formula)**.** Let G be an LCH group and $\Gamma \leqslant G$ a uniform lattice. Then for every $\pi \in \widehat{G}_\Gamma$ and $f \in C_{\mathrm{unif}}(G)^2$, the operator $\pi(f)$ is of trace class and

$$\sum_{\pi \in \widehat{G}_\Gamma} N_\Gamma(\pi)\,\mathrm{tr}\,\pi(f) = \sum_{[\gamma]} \mathrm{vol}(\Gamma_\gamma\backslash G_\gamma)\mathcal{O}_\gamma(f)$$

where the summation on the right runs over all conjugacy classes $[\gamma]$ in the group Γ, and $\mathcal{O}_\gamma(f)$ denotes the **orbital integral**

$$\mathcal{O}_\gamma(f) := \int_{G_\gamma\backslash G} f(x^{-1}\gamma x)dx$$

We shall see in the sequel that the centralizer G_γ is unimodular and that $\Gamma_\gamma\backslash G_\gamma$ is of finite measure for all $\gamma \in \Gamma$. The expression $\mathrm{vol}(\Gamma_\gamma\backslash G_\gamma)\mathcal{O}_\gamma(f)$ is therefore well-defined, in the sense that it is independent of the choice of Haar measure on G_γ.

The left hand side is called the **spectral side**, while the right hand side is called the **geometric side**.

Proof. The algebra $C_{\text{unif}}(G)^2$ consists of all finite linear combinations of functions of the form $g * h^*$ with $g, h \in C_{\text{unif}}(G)$, so it suffices to prove the trace formula for $f = g * h^*$. By Lemma 10.12, the operators $R(g)$ and $R(h)$ are integral operators with continuous kernels. Since $\Gamma\backslash G$ is compact, $R(g)$ and $R(h)$ are admissible operators. By Proposition 10.15, the operator $R(f) = R(g)R(h)^*$ is of trace class with trace $\operatorname{tr} R(f) = \int_{\Gamma\backslash G} k_f(x,x)dx$. By Theorem 10.9 and the definition of trace, all restriction of $R(f)$ to subrepresentations are of trace class and

$$\sum_{\pi \in \widehat{G}_\Gamma} N_\Gamma(\pi) \operatorname{tr} \pi(f) = \operatorname{tr} R(f) = \int_{\Gamma\backslash G} k_f(x,x)dx = \int_{\Gamma\backslash G} \sum_{\gamma\in\Gamma} f(x^{-1}\gamma x)dx$$

We order the sum in accordance with the conjugacy classes $[\gamma]$ in Γ, interchange integration and summation (valid as $\Gamma\backslash G$ is compact), and quotient integral formula to obtain

$$\begin{aligned}\operatorname{tr} R(f) &= \int_{\Gamma\backslash G} \sum_{[\gamma]} \sum_{\sigma\in\Gamma_\gamma\backslash\Gamma} f(x^{-1}\sigma^{-1}\gamma\sigma x)dx \\ &= \sum_{[\gamma]} \int_{\Gamma\backslash G} \sum_{\sigma\in\Gamma_\gamma\backslash\Gamma} f(x^{-1}\sigma^{-1}\gamma\sigma x)dx = \sum_{[\gamma]} \int_{\Gamma_\gamma\backslash G} f(x^{-1}\gamma x)dx\end{aligned}$$

Lemma 10.18. For every $\gamma \in \Gamma$, the centralizer G_γ is unimodular and $\Gamma_\gamma\backslash G_\gamma$ has finite invariant measure.

Proof. The above computation shows that for $f \in C_{\text{unif}}(G)^2$ with $f \geqslant 0$, one has $\int_{\Gamma_\gamma\backslash G} f(x^{-1}\gamma x)dx < \infty$ for all $\gamma \in \Gamma$. Since G is unimodular by Lemma 10.5, the space $\Gamma_\gamma\backslash G$ carries an invariant Radon measure ν. Consider the projection $p : \Gamma_\gamma\backslash G \to G_\gamma\backslash G$ and let $\mu = p_*\nu$ be the pushforward measure of ν via p; in other word, for $f \in C_c(G_\gamma\backslash G)$, we have

$$\int_{G_\gamma\backslash G} f d\mu = \int_{\Gamma_\gamma\backslash G} f \circ p \, d\nu$$

We show μ is a Radon measure, i.e., the integral above is finite. For this let $0 \leqslant f \in C_c(G_\gamma\backslash G)$ and let $\Phi : G_\gamma\backslash G \to G$ be given by $\Phi(G_\gamma x) = x^{-1}\gamma x$. Then the subset $\Phi(\operatorname{supp} f) \subseteq G$ is compact, so by Tietze's extension theorem we have find $0 \leqslant \tilde{f} \in C_c(G)$ such that $\tilde{f}(y^{-1}\gamma y) = f(G_\gamma y)$ for every $y \in G$ with $f(G_\gamma y) > 0$. Choose $0 \leqslant F \in C_c(G)^2$ such that $\tilde{f} \leqslant F$.

- To show the existence of F, let $g \in C_c^+(G)$ be such that $g > 0$ in a neighborhood of $\operatorname{supp} \tilde{f}$. There exists a unit-neighborhood U such that $(\phi_U * g)|_{\operatorname{supp} \tilde{f}} > 0$, where ϕ_U is a Dirac function supported in U. Set $F := c\phi_U * g$ for $c \gg 0$ so that $F \geqslant \tilde{f}$.

Since $F \in C_c(G)^2$, as mentioned in the beginning of the proof of this lemma, we have

$$\int_{G_\gamma\backslash G} f d\mu = \int_{\Gamma_\gamma\backslash G} f \circ p \, d\nu = \int_{\Gamma_\gamma\backslash G} \tilde{f}(y^{-1}\gamma y)d\nu(y) \leqslant \int_{\Gamma_\gamma\backslash G} F(y^{-1}\gamma y)d\nu(y) < \infty$$

This show μ is a Radon measure. μ is G-invariant since so is ν, and by Theorem 1.17 it follows that G_γ is unimodular. Finally, for $f \in C_c(G_\gamma \backslash G)$, we have

$$\begin{aligned}\infty > \int_{\Gamma_\gamma \backslash G} f(p(y))d\nu(y) &= \int_{G_\gamma \backslash G} \int_{\Gamma_\gamma \backslash G_\gamma} f(p(x\sigma))d\sigma d\mu(x) \\ &= \int_{G_\gamma \backslash G} \int_{\Gamma_\gamma \backslash G_\gamma} f(x)d\sigma d\mu(x) = \mathrm{vol}(\Gamma_\gamma \backslash G_\gamma) \int_{\Gamma_\gamma \backslash G} f(x)d\mu(x)\end{aligned}$$

so that $\mathrm{vol}(\Gamma_\gamma \backslash G_\gamma) < \infty$. □

To conclude the proof, we continue our computation with the help of the lemma.

$$\begin{aligned}\mathrm{tr}\, R(f) &= \sum_{[\gamma]} \int_{\Gamma_\gamma \backslash G} f(x^{-1}\gamma x)dx \\ &= \sum_{[\gamma]} \int_{G_\gamma \backslash G} \int_{\Gamma_\gamma \backslash G_\gamma} f((\sigma x)^{-1}\gamma(\sigma x))d\sigma dx = \sum_{[\gamma]} \mathrm{vol}(\Gamma_\gamma \backslash G_\gamma)\mathcal{O}_\gamma(f)\end{aligned}$$

□

Example 10.19. Consider the case $G = \mathbb{R}$ and $\Gamma = \mathbb{Z}$. By Plancherel theorem we have the decomposition

$$L^2(\mathbb{Z}\backslash\mathbb{R}) \cong \widehat{\bigoplus_{k\in\mathbb{Z}}} \mathbb{C}[t \mapsto e^{2\pi i t k}]$$

so if we identify $\widehat{G}$ with $\mathbb{R}$ via $x \mapsto [t \mapsto e^{2\pi i x}]$, we see $\widehat{G}_\Gamma$ is mapped onto $\mathbb{Z}$, and the multiplicities are all equal to one. Thus the spectral side equals $\sum_{k\in\mathbb{Z}} \widehat{f}(k)$, and the geometric side is $\sum_{k\in\mathbb{Z}} f(k)$. Thus in this case, the trace formula is the Poisson summation formula in disguise.

Lemma 10.20. Let G be an LCH group and A a *-subalgebra of $C_c(G)$ stable under left translation L_y $(y \in G)$ which contains a Dirac net $(\phi_U)_U$. Let (π, V) and (σ, W) be two unitary representations of G such that for each $f \in A$ the operators $\pi(f)$ and $\sigma(f)$ are of trace class. If

$$\mathrm{tr}\, \pi(f)^*\pi(f) \geqslant \mathrm{tr}\, \sigma(f)^*\sigma(f)$$

for all $f \in A$, then σ is a subrepresentation of π. If the equality holds for all $f \in A$, then σ is isomorphic to π.

Proof. By Lemma 7.10 the orbit $\pi(A)v$ is nonzero for each nonzero $v \in V$. Thus by Lemma 10.13 both π and σ decomposes into irreducibles with finite multiplicity. By a Zorn's lemma argument there exists a maximal subrepresentation of σ with the property of being isomorphic to a subrepresentation of π. Restricting to their orthogonal complements, respectively, we assume π and σ have no isomorphic subrepresentations. To show the first assertion, we must show $\sigma = 0$. If this is shown, the second assertion follows once we reverse the roles of π and σ.

Let $V = \bigoplus_{\alpha \in I} V_\alpha$ be a decomposition into pairwise orthogonal subrepresentations and let $(v_{\alpha,\mu})_\mu \in V_\alpha$ be such that

$$\sum_\alpha \sum_\mu \|\pi(f) v_{\alpha,\mu}\|^2 < \infty$$

for every $f \in A$. This is possible for each $\pi(f)$ is of trace class, and thus Hilbert Schmidt. We can simply take $(v_{\alpha,\mu})_{\alpha,\mu}$ to be an orthonormal basis. Choose any nonzero vector $w \in W$. We claim that for every $\varepsilon > 0$ we can find $f \in A$ such that

$$\sum_\alpha \sum_\mu \|\pi(f) v_{\alpha,\mu}\|^2 < \varepsilon \|\sigma(f) w\|^2$$

Suppose otherwise. For each μ let $V_{\alpha,\mu}$ be a copy of V_α, and define the space

$$L := \overline{\left\{ \sum_\alpha \sum_\mu \pi(f) v_{\alpha,\mu} \mid f \in A \right\}} \subseteq V' := \bigoplus_\alpha \bigoplus_\mu V_{\alpha,\mu}$$

Then for every $f \in A$ the map

$$\sum_\alpha \sum_\mu \pi(f) v_{\alpha,\mu} \mapsto \sigma(f) w$$

would define a bounded unitary G-equivariant operator T from L to W. It is nontrivial as said in the very beginning of the proof. Extend T to the whole V' by setting $T|_{L^\perp} = 0$. Consider the restriction of T to $\bigoplus_\mu V_{\alpha,\mu} \subseteq V'$. Since $V_{\alpha,\mu}$ does not occur in W, this restriction must be trivial. Hence $T = 0$, a contradiction.

Now assume $\sigma \neq 0$. As said in the very beginning of the proof, there exists $h \in A$ with $\sigma(h) \neq 0$. If we put $f = h * h^*$, then $\sigma(f)$ is of trace class and positive. Therefore $\sigma(f)$ possesses a largest eigenvalue and we can scale h in such a way that this eigenvalue is 1. Let $w \in W$ be such that $\|w\| = 1$ and $\sigma(f) w = w$. Let $\lambda > 0$ be the largest eigenvalue of $\pi(f)$. For every α let $(v_{\alpha,\mu})_\mu$ be an orthonormal basis of V_α consisting of eigenvectors of $\pi(f)$, and write $\pi(f) v_{\alpha,\mu} = \lambda_{\alpha,\mu} v_{\alpha,\mu}$. Then for every $g \in A$,

$$\sum_\alpha \sum_\mu \|\pi(g) v_{\alpha,\mu}\|^2 = \|\pi(g)\|_{\mathrm{HS}}^2 < \infty$$

and by the second paragraph we can find $g \in A$ with

$$\sum_\alpha \sum_\mu \|\pi(g) v_{\alpha,\mu}\|^2 < \frac{1}{\lambda^2} \|\sigma(g) w\|^2$$

Then

$$\begin{aligned} \operatorname{tr} \pi(g * f)^* \pi(g * f) &= \sum_{\alpha,\mu} \|\pi(g * f) v_{\alpha,\mu}\|^2 = \sum_{\alpha,\mu} \lambda_{\alpha,\mu}^2 \|\pi(g) v_{\alpha,\mu}\|^2 \leqslant \lambda^2 \sum_{\alpha,\mu} \|\pi(g) v_{\alpha,\mu}\|^2 < \|\sigma(g) w\|^2 \\ &= \|\sigma(g * f) w\|^2 = \operatorname{tr} \sigma(g * f)^* \sigma(g * f) \leqslant \operatorname{tr} \pi(g * f)^* \pi(g * f) \end{aligned}$$

a contradiction. This implies $\sigma = 0$. $\square$

10.4 Locally Constant Functions

Definition. Let X be a topological space.

1. A function $f : X \to \mathbb{C}$ is called **locally constant** if each point $x \in X$ admits an neighborhood U to which the restriction of f is constant.

Suppose, in addition, $X = G$ is a topological group.

2. A function $f : G \to \mathbb{C}$ is called **uniformly locally constant** if there exists a unit-neighborhood U such that f is constant on every set of the form UxU, $x \in G$.

- If f is locally constant with compact support, then f is uniformly locally constant.

Proposition 10.21. Let G be a totally disconnected LCH group and f a uniformly locally constant and integrable function on G. Then $f \in C_{\text{unif}}(G)^2$. In particular, the trace formula is valid for f.

Proof. Let U denote the open compact symmetric unit neighborhood of G so that f is constant on UxU for every $x \in G$. Then $f_U = |f|$, and therefore $f \in C_{\text{unif}}(G)$. The same holds for the function $e_U := \dfrac{1}{\text{vol}(U)}\mathbf{1}_U$. We have

$$e_U * f(g) = \frac{1}{\text{vol}(U)} \int_G \mathbf{1}_U(h) f(h^{-1}g) dh = \frac{1}{\text{vol}(U)} \int_U f(g) dh = f(g)$$

for every $g \in G$. Thus $f = f * e_U \in C_{\text{unif}}(G)^2$. □

10.5 Lie Groups

Theorem 10.22. Let G be a Lie group of dimension n and let $\Gamma \leqslant G$ be a cocompact lattice. Let $f \in C(G) \cap L^1(G)$ such that the kernel

$$k(x, y) := \sum_{\gamma \in \Gamma} f(x^{-1}\gamma y)$$

converges uniformly and is $2r$-times continuously differentiable in the first argument, where $r = \left\lceil \frac{n}{2} \right\rceil$. Then the trace formula is valid for f.
In particular, the trace formula holds for every $f \in C_c^{2r}(G)$.

To show this theorem we need a partition of unity with a smooth square root. The following lemma is proved in Appendix W.7.1.

Lemma 10.23. Let M be a smooth manifold and let $(U_i)_{i \in I}$ be an open covering of M. There there are smooth functions $u_i : M \to [0, 1]$, such that $\operatorname{supp} u_i \subseteq U_i$ and that

$$\sum_{i \in I} u_i \equiv 1$$

where the sum is locally finite. Moreover, one can choose the u_i in a way that for each $i \in I$ the function $\sqrt{u_i}$ is smooth as well.

A Borel measure ν on $\mathbb{R}^n$ is called a **smooth measure** if the Radon-Nikodym derivative of ν with respect to the Lebesgue measure dx is smooth and non-negative, i.e., there exists a smooth function $h : \mathbb{R}^n \to [0, \infty)$ such that

$$\nu(A) = \int_A h(x)dx$$

for all Borel A. A measure μ on a smooth manifold M is called a **smooth measure** if for every chart $\phi : U \to \mathbb{R}^n$, the pushforward measure $\phi_*\mu$ on $\mathbb{R}^n$ is smooth.

Proposition 10.24. Let M be a compact smooth manifold of dimension n with a smooth measure μ. Let $k : M \times M \to \mathbb{C}$ be continuous and $2r$-times continuously differentiable in the first argument, where $r = \left\lceil \frac{n}{2} \right\rceil$. Then the induced integral operator $T_k : L^2(M) \to L^2(M)$

$$T_k(\phi)(x) := \int_M k(x, x)\phi(y)d\mu(y)$$

is of trace class and

$$\operatorname{tr} T_k = \int_M k(x, x)dx.$$

Proof. We first prove the assertion in the case $M = \mathbb{R}^n/\mathbb{Z}^n$ and μ being the Lebesgue measure (the Haar measure with total volume 1). In this case we define

$$l(x, y) = \sum_{k \in \mathbb{Z}^n} \left(\frac{1}{1 + 4\pi^2 \|k\|^2} \right)^r e_k(x)\overline{e_k(y)}$$

where $e_k(x) = e^{2\pi i \langle x, k \rangle}$. By integral test we see this sum converges absolutely and uniformly, and the kernel $l(x, y)$ is therefore continuous. Let Δ be the Laplacian

$$\Delta = -\frac{\partial^2}{\partial x_1^2} - \cdots - \frac{\partial^2}{\partial x_n^2}.$$

Lemma 10.25. For $\phi \in C^{2r}(\mathbb{R}^n/\mathbb{Z}^n)$ one has

$$T_l(1 + \Delta)^r \phi = \phi.$$

Proof. Both sides being continuous functions, it suffices to show they are equal in L^2-sense. Since $\{e_k \mid k \in \mathbb{Z}^n\}$ forms an orthonormal basis for $L^2(\mathbb{R}^n/\mathbb{Z}^n)$, it is enough to show $T_l(1 + \Delta)^r e_k = e_k$ for each $k \in \mathbb{Z}^n$. This follows from the computations

$$(1 + \Delta)^r e_k = (1 + 4\pi^2 \|k\|^2)^r e_k$$

and

$$T_l e_k(x) = \sum_{\ell \in \mathbb{Z}^n} \left(\frac{1}{1 + 4\pi^2 \|\ell\|^2} \right)^r e_\ell(x) \int_M \overline{e_\ell(y)} e_k(y)dy = \left(\frac{1}{1 + 4\pi^2 \|k\|^2} \right)^r e_k(x)$$

□

Now let k be as in the theorem. Then for $\phi \in L^2(M)$, the function $T_k\phi$ is $C^{2r}(M)$ (for M is compact we can differentiate under the integral sign), so by the lemma

$$T_l(1+\Delta)^r T_k\phi = T_k\phi.$$

By the same reason one has $(1+\Delta)^r T_k = T_{k'}$, where

$$k'(x,y) := (1+\Delta_x)^r k(x,y)$$

so $T_k = T_l T_{k'}$ is a product of two Hilbert Schmidt operator, hence of trace class. As both T_l and $T_{k'}$ are admissible (as M is compact), the theorem follows from Proposition 10.15.

Next let M be an arbitrary smooth compact manifold of dimension n. Let $((U_i,\psi_i))_{i=1}^s$ be an open cover of M consisting of charts $\psi_i : U_i \to \mathbb{R}^n/\mathbb{Z}^n$; so for each i, the map ψ_i is a homeomorphism of U_i into some open set $V_i \subseteq \mathbb{R}^n/\mathbb{Z}^n$. We choose $V_1,\dots,V_s$ in a way that they are pairwise disjoint. Let $(u_i)_i$ be a smooth partition of unity with smooth square root subordinate to the cover $(U_i)_i$. For $1 \leqslant i,j \leqslant s$ let

$$k_{i,j}(x,y) = \sqrt{u_i(x)}k(x,y)\sqrt{u_j(y)} \in C(M\times M)$$

and define a continuous kernel on $\mathbb{R}^n/\mathbb{Z}^n \times \mathbb{R}^n/\mathbb{Z}^n$

$$\widetilde{k}_{i,j}(x,y) := \begin{cases} \sqrt{d_i(x)}k_{i,j}(\psi_i^{-1}(x),\psi_j^{-1}(y)\sqrt{d_j(y)} & \text{, if } (x,y)\in V_i\times V_j \\ 0 & \text{, otherwise} \end{cases}$$

Here d_j denotes the Radon-Nikodym derivative of $(\psi_j)_*\mu$ with respect to the Lebesgue measure. We define $\widetilde{k} := \sum_{i,j}\widetilde{k}_{i,j}$ and for $\phi \in L^2(M)$, we set

$$\phi_j(x) := \phi(x)\sqrt{u_j(x)}.$$

Define $\widetilde{\phi}_j \in L^2(\mathbb{R}^n/\mathbb{Z}^n)$ by

$$\widetilde{\phi}_j(x) := \begin{cases} \phi_j(\psi_j^{-1}(x))\sqrt{d_j(x)} & \text{, if } x\in V_j \\ 0 & \text{, otherwise} \end{cases}$$

Finally set $\widetilde{\phi} = \sum_j \widetilde{\phi}$.

Lemma 10.26. The map

$$\begin{aligned} \Psi : L^2(M) &\longrightarrow L^2(\mathbb{R}^n/\mathbb{Z}^n) \\ \phi &\longmapsto \widetilde{\phi} \end{aligned}$$

is a linear isometry, and

$$\Psi(T_k\phi) = T_{\widetilde{k}}\Psi(\phi)$$

for every $\phi \in L^2(M)$. The operator $T_{\widetilde{k}}$ equals $PT_{\widetilde{k}}P$, where P is the orthogonal projection $L^2(\mathbb{R}^n/\mathbb{Z}^n) \to \operatorname{Im}\Psi$. Finally, we have

$$\int_M k(x,x)d\mu(x) = \int_{\mathbb{R}^n/\mathbb{Z}^n} \widetilde{k}(x,x)dx$$

Proof. The map Ψ is linear. For $\phi \in L^2(M)$, we compute

$$\begin{aligned}\|\Psi(\phi)\|^2 &= \int_{\mathbb{R}^n/\mathbb{Z}^n} |\widetilde{\phi}(x)|^2 dx = \sum_j \int_{V_j} |\widetilde{\phi}_j(x)|^2 dx = \sum_j \int_{V_j} |\phi_j(\psi_j^{-1}(x))|^2 d_j(x) dx \\ &= \sum_j \int_{U_j} |\phi_j(x)|^2 d\mu(x) = \sum_j \int_M |\phi(x)|^2 u_j(x) d\mu(x) = \int_M |\phi(x)|^2 d\mu(x) = \|\phi\|^2\end{aligned}$$

so Ψ is an isometry. For the second, compute

$$\begin{aligned}T_{\widetilde{k}}\Psi(\phi)(x) &= T_{\widetilde{k}}\widetilde{\phi}(x) = \sum_j T_{\widetilde{k}}\widetilde{\phi}_j(x) = \sum_j \int_{V_j} \widetilde{k}(x,y)\widetilde{\phi}_j(y) dy = \sum_{i,j} \int_{V_j} \widetilde{k}_{i,j}(x,y)\widetilde{\phi}_j(y) dy \\ &= \sum_i \sqrt{d_i(x)}\sqrt{u_i(\psi_i^{-1}(x))} \int_M k(\psi_i^{-1}(x), y)\phi(y) d\mu(y) = \Psi T_{\widetilde{k}}(\phi)(x)\end{aligned}$$

To show $T_{\widetilde{k}} = PT_{\widetilde{k}}P$ one has to show for all $g \in (\operatorname{Im}\Psi)^\perp$,

$$T_{\widetilde{k}}(g) = 0 \text{ and } \langle T_{\widetilde{k}} h, g\rangle = 0$$

for all $h \in L^2(\mathbb{R}^n/\mathbb{Z}^n)$. These follow from the facts that $\operatorname{Im}(y \mapsto \widetilde{k}(x,y)) \subseteq \operatorname{Im}\Psi$ and $\operatorname{Im}(x \mapsto \widetilde{k}(x,y)) \subseteq \operatorname{Im}\Psi$ if the other argument is fixed. Finally,

$$\begin{aligned}\int_{\mathbb{R}^n/\mathbb{Z}^n} \widetilde{k}(x,x) dx &= \sum_i \int_{V_i} k_{i,i}(\psi_i^{-1}(x), \psi_i^{-1}(x)) d_i(x) dx = \sum_i \int_{U_i} k_{i,i}(x,x) d\mu(x) \\ &= \sum_i \int_M k(x,x) u_i(x) d\mu(x) = \int_M k(x,x) d\mu(x)\end{aligned}$$

□

We use this lemma to show T_k is compact, and next of trace class with trace $\operatorname{tr} T_k = \operatorname{tr} T_{\widetilde{k}}$. If these are shown, then the last assertion of the lemma implies

$$\operatorname{tr} T_k = \operatorname{tr} T_{\widetilde{k}} = \int_{\mathbb{R}^n/\mathbb{Z}^n} \widetilde{k}(x,x) dx = \int_M k(x,x) d\mu(x)$$

and the proposition follows.

For the compactness, we use Proposition 6.7.(c). Let (e_j) be an orthonormal sequence in $L^2(M)$. Then

$$\|T_k e_j\| = \|\Psi T_k e_j\| = \left\|T_{\widetilde{k}}\Psi e_j\right\| \to 0$$

as $T_{\widetilde{k}}$ is compact and $(\Psi e_j)_j$ is orthonormal. To show its trace norm is finite, we use Lemma 6.21. Let (e_i) and (h_j) be two orthonormal bases of $L^2(M)$. Then

$$\sum_i |\langle T_k e_i, h_i\rangle| = \sum_i |\langle \Psi T_k e_i, \Psi h_i\rangle| = \sum_i |\langle T_{\widetilde{k}}\Psi e_i, \Psi h_i\rangle| \leqslant \left\|T_{\widetilde{k}}\right\|_{\mathrm{tr}}$$

so that $\|T_k\|_{\mathrm{tr}}$ is finite. Finally, let e_i be any orthonormal basis of $L^2(M)$. Then

$$\operatorname{tr} T_k = \sum_i \langle T_k e_i, e_i \rangle = \sum_i \langle T_{\tilde{k}} \Psi e_i, \Psi e_i \rangle$$

On the other hand, $(\Psi e_i)_i$ forms an orthonormal basis of $\operatorname{Im} \Psi$. Let $(e'_j)_j$ be any orthonormal basis prolonging $(\Psi e_i)_i$; then

$$\operatorname{tr} T_{\tilde{k}} = \sum_j \langle T_{\tilde{k}} e'_j, e'_j \rangle = \sum_j \langle T_{\tilde{k}} P e'_j, P e'_j \rangle = \sum_i \langle T_{\tilde{k}} \Psi e_i, \Psi e_i \rangle = \operatorname{tr} T_k$$

This concludes the proof of the proposition. □

Now we deduce Theorem 10.22 from Proposition 10.24. The proof is the same as that of trace formula. Let ρ be the right translation by G on $L^2(\Gamma \backslash G)$. Let f be as in the theorem. Then $\rho(f)$ is of trace class, and

$$\operatorname{tr} \rho(f) = \sum_{\pi \in \hat{G}} N_\Gamma(\pi) \operatorname{tr} \pi(f)$$

where $N_\Gamma(\pi)$ is the multiplicity of π occurring in $L^2(\Gamma \backslash G)$. On the other hand, we have

$$\begin{aligned} \operatorname{tr} \rho(f) &= \int_{\Gamma \backslash G} k(x,x) dx = \int_{\Gamma \backslash G} \sum_{\gamma \in \Gamma} f(x^{-1} \gamma x) dx = \int_{\Gamma \backslash M} \sum_{[\gamma]} \sum_{\sigma \in \Gamma_\gamma \backslash \Gamma} f(x^{-1} \sigma^{-1} \gamma \sigma x) dx \\ &= \sum_{[\gamma]} \int_{\Gamma_\gamma \backslash G} f(x^{-1} \gamma x) dx = \sum_{[\gamma]} \int_{G_\gamma \backslash G} \int_{\Gamma_\gamma \backslash G_\gamma} f((\sigma x)^{-1} \gamma (\sigma x)) d\sigma dx = \sum_{[\gamma]} \operatorname{vol}(\Gamma_\gamma \backslash G_\gamma) \mathcal{O}_\gamma(f) \end{aligned}$$

where $[\gamma]$ runs over all conjugacy classes of Γ.

11 The Heisenberg Group

12 $\mathbf{SL}_2(\mathbb{R})$

12.1 The upper half plane

Denote by $\mathbb{H} = \{z \in \mathbb{C} \mid \operatorname{im} z > 0\}$ the upper half plane. The special linear group $G = \mathrm{SL}_2(\mathbb{R})$ acts on $\mathbb{H}$ by Möbius transformation, and the stabilizer of $i \in \mathbb{H}$ is $\mathrm{SO}_2(\mathbb{R})$, the special orthogonal group. This induces a continuous bijection

$$\begin{aligned} \mathrm{SL}_2(\mathbb{R})/\mathrm{SO}(2) &\xrightarrow{\quad\sim\quad} \mathbb{H} \\ g = \begin{pmatrix} a & b \\ c & d \end{pmatrix} &\longmapsto gi = \frac{ai+b}{ci+d} \end{aligned}$$

Theorem 12.1 (Iwasawa Decomposition)**.** Let $A = \left\{ \begin{pmatrix} a & 0 \\ 0 & a^{-1} \end{pmatrix} \mid a > 0 \right\}$, $N = \left\{ \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix} \mid x \in \mathbb{R} \right\}$ and $K = \mathrm{SO}(2)$. Then the map

$$\begin{aligned} \psi : A \times N \times K &\longrightarrow G \\ (a, n, k) &\longmapsto ank \end{aligned}$$

is a homeomorphism.

Proof. Let $g \in G$, and let $gi = x + yi$. If we put

$$a = \begin{pmatrix} y^{\frac{1}{2}} & \\ & y^{-\frac{1}{2}} \end{pmatrix}, \qquad n = \begin{pmatrix} 1 & xy^{-1} \\ & 1 \end{pmatrix},$$

then $gi = ani$, so that $g^{-1}an \in K$. This implies that $g = ank$ for some $k \in K$. Now define the inverse map

$$\begin{aligned} \phi : G &\longrightarrow A \times N \times K \\ g &\longrightarrow (\underline{a}(g), \underline{n}(g), \underline{k}(g)) \end{aligned}$$

where

$$\begin{aligned} \underline{a} \begin{pmatrix} a & b \\ c & d \end{pmatrix} &= \begin{pmatrix} \frac{1}{\sqrt{c^2+d^2}} & \\ & \sqrt{c^2+d^2} \end{pmatrix} \\ \underline{n} \begin{pmatrix} a & b \\ c & d \end{pmatrix} &= \begin{pmatrix} 1 & ac+bd \\ & 1 \end{pmatrix} \\ \underline{k} \begin{pmatrix} a & b \\ c & d \end{pmatrix} &= \frac{1}{\sqrt{c^2+d^2}} \begin{pmatrix} d & -c \\ c & d \end{pmatrix}. \end{aligned}$$

It is clear that $\phi \circ \psi = \mathrm{id}$ and $\psi \circ \phi = \mathrm{id}$. $\square$

We shall keep the notation $\underline{a}(g)$, $\underline{n}(g)$ and $\underline{k}(g)$ used in the above proof. Also, for x, t, $\theta \in \mathbb{R}$, put

$$a_t = \begin{pmatrix} e^t & \\ & e^{-t} \end{pmatrix} \in A \qquad n_x = \begin{pmatrix} 1 & x \\ & 1 \end{pmatrix} \in N \qquad k_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \in K$$

Say a function $f : G \to \mathbb{C}$ is a **smooth** if the map $\mathbb{R}^3 \to \mathbb{C}$ given by $(t, x, \theta) \mapsto f(a_t n_x k_\theta)$ is smooth. This is the same as saying $f : G \to \mathbb{C}$ is smooth when G is equipped with the usual manifold structure. We denote the space of smooth functions by $C^\infty(G)$, and $C_c^\infty(G) = C^\infty(G) \cap C_c(G)$.

Theorem 12.2. The group $G = \mathrm{SL}_2(\mathbb{R})$ is unimodular.

Proof. Let $\phi : G \to \mathbb{R}_{>0}$ be a continuous homomorphism. We claim $\phi \equiv 1$. Since K is compact, we have $\phi(K) = 1$. Since $\phi(a_{t+s}) = \phi(a_t a_s) = \phi(a_t)\phi(a_s)$ and ϕ is continuous, there exists $x \in \mathbb{R}$ such that $\phi(a_t) = e^{tx}$ for any $t \in \mathbb{R}$. Consider the element $w = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \in G$; we have $w a_t w^{-1} = a_{-t}$. Since $e^{-tx} = \phi(a_{-t}) = \phi(w a_t w^{-1}) = \phi(a_t) = e^{tx}$, and it follows that $x = 0$, i.e., $\phi(A) = 0$. Similarly, there exists $y \in \mathbb{R}$ such that $\phi(n_t) = e^{ty}$ for any $t \in \mathbb{R}$. As $a_r n_t a_r^{-1} = n_{e^{2r}t}$, it follows that $e^{ty} = e^{e^{2r}ty}$, which implies $t = 0$, i.e., $\phi(N) = 1$.

A quick proof: any homomorphism ϕ factors through the abelianization $\mathrm{SL}_2(\mathbb{R})^{\mathrm{ab}}$, which is trivial. Hence ϕ is trivial. $\square$

Theorem 12.3. For any given Haar measures on three of the four groups G, A, N K, there exists a unique Haar measure on the fourth such that for any $f \in L^1(G)$, the integration formula

$$\int_G f(x)dx = \int_A \int_N \int_K f(ank)dkdnda$$

holds. For $\phi \in L^1(K)$ and $x \in G$ one has

$$\int_K \phi(k)dk = \int_K \phi(\underline{k}(kx))e^{2\underline{t}(kx)}dk$$

where $\underline{t}(g)$ for the unique $t \in \mathbb{R}$ with $\underline{a}(g) = a_t$.

13 Wavelets

13.1 First Ideas

Let (X, μ) be a measure space. Roughly speaking, a wavelet is an element $\phi \in L^2(X, \mu)$ together with a family of transformations $t \mapsto \pi_t \phi$ indexed by the elements of some other measure space (T, ν). The corresponding **wavelet transform** is a map

$$\begin{aligned} \Phi : L^2(X, \mu) &\longrightarrow L^2(T, \nu) \\ \psi &\longmapsto t \mapsto \langle \psi, \pi_t \phi \rangle. \end{aligned}$$

One usually consider only cases where the wavelet transform is injective and allows some reconstruction formula for a given ψ out of the transformed data. In many interesting cases, the measure space T is a locally compact group equipped with Haar measure and the transformation $t \mapsto \pi_t$ is a unitary representation of T on $L^2(X, \mu)$, so it is evident that Harmonic Analysis is entering the picture. Although there are many important cases that do not fit directly into this setting (like most discretized versions), we shall restrict ourselves to this setting below. The basic example is given by the continuous wavelet transform of the real line:

Example 13.1. Recall the $ax + b$-group G consists of all matrices of the form $\begin{pmatrix} a & b \\ 0 & 1 \end{pmatrix}$ with $a \in \mathbb{R}^\times$ and $b \in \mathbb{R}$. It is the group of affine transformations $(a, b) \mapsto [T_{a,b} : x \mapsto ax + b]$. A short computation shows that a left Haar measure on G is given by

$$\int_G f(a, b) d(a, b) = \int_{\mathbb{R}^\times} \int_{\mathbb{R}} f(a, b) db \frac{da}{|a|^2}$$

and that $\Delta_G : (a, b) \mapsto \dfrac{1}{|a|}$ is the modular function on G.

The action of G on $\mathbb{R}$ induces a unitary representation $(a, b) \mapsto \pi(a, b)$ of the $ax + b$-group on $L^2(\mathbb{R})$ via

$$\pi(a, b)\phi(x) = \frac{1}{\sqrt{|a|}} \phi(T_{a,b}^{-1}(x)) = \frac{1}{\sqrt{|a|}} \phi\left(\frac{x - b}{a}\right)$$

Now fix an element $\phi \in L^2(\mathbb{R})$. Then for each $\psi \in L^2(\mathbb{R})$ we obtain a bounded continuous function $W_\phi(\psi) : G \to \mathbb{C}$ given by

$$(a, b) \mapsto W_\phi(\psi)(a, b) = \langle \psi, \pi(a, b)\phi \rangle.$$

In order to get useful transformations in this way, we have to find conditions on ϕ, which guarantee that the transform $\psi \mapsto W_\phi(\psi)$ is injective with image in $L^2(G)$ and allows for a suitable inversion formula.

Of course, the optimal solution to the above stated problem would be to find functions $\phi \in L^2(\mathbb{R})$ such that the resulting transform $\psi \mapsto W_\phi(\psi)$ is an isometry from $L^2(\mathbb{R})$ into $L^2(G)$. In that case the reconstruction formula would be given by the adjoint operator $W_\phi^* : L^2(G) \to L^2(\mathbb{R})$, since for every $\psi, \eta \in L^2(\mathbb{R})$ we would have

$$\langle \psi, \eta \rangle = \langle W_\phi(\psi), W_\phi(\eta) \rangle = \langle W_\phi^* W_\phi(\psi), \eta \rangle$$

and hence $\psi = W_\phi^* W_\phi(\psi)$ as elements in $L^2(\mathbb{R})$.

We put the problem into a more general framework. Suppose that G is an arbitrary LCH group and that (π, V_π) is a unitary representation of G on some Hilbert space V_π. Given any fixed vector $\xi \in V_\pi$, we obtain a corresponding transformation

$$\begin{aligned} W_\xi : V_\pi &\longrightarrow C(G) \\ \eta &\longmapsto x \mapsto \langle \eta, \pi(x)\xi \rangle. \end{aligned}$$

We call this the (generalized) **continuous wavelet transform** corresponding to π and ξ. Note that this transformation is injective if and only if ξ is a cyclic vector for π.

If W_ξ mapped V_π in to the Hilbert space $L^2(G)$, we would be able to use Hilbert space methods. But in general the function $x \mapsto \langle \eta, \pi(x)\xi \rangle$ may not be an L^2-function. So, in order to be able to use inner products, we have to restrict W_ξ to the linear subspace

$$D_\xi := \left\{ \eta \in W_\pi \mid W_\xi(\eta) \in L^2(G) \right\}.$$

We then get a linear operator $W_\xi : D_\xi \to L^2(G)$. Up to now we have always considered operators from Banach spaces to Banach spaces. The current example shows that sometimes linear operators occur naturally, which are only defined on linear subspaces of Banach spaces.

Let V, W be Banach spaces, and let $D \subseteq V$ be a linear subspace. A linear map $T : D \to W$ will loosely be called a **linear operator from V to W with domain D**.

- T is called **closed operator** if its graph

  $$\Gamma_T := \{(\eta, T\eta) \mid \eta \in D\} \subseteq V \times W$$

  is a closed subset of $V \times W$.

- T is called a **bounded operator** if its norm

  $$\|T\| := \sup_{v \in D,\, \|v\|=1} \|Tv\|$$

  is finite.

- T is called **densely defined** if D is dense in V.

Note that if T is densely defined and bounded, it extends uniquely to a bounded operator $T : V \to W$.

Lemma 13.2. The operator $W_\xi : D_\xi \to L^2(G)$ is closed.

Proof. Let $(\eta_n)_n$ be a sequence in D_ξ such that $\eta_n \to \eta \in V_\pi$ and $W_\xi \eta_n \to w \in L^2(G)$. Passing to subsequence, we may assume $W_\xi \eta_n \to w$ pointwise almost everywhere. But then for $x \in G$, we have

$$|W_\xi \eta(x) - W_\xi \eta_n(x)| = |\langle \eta - \eta_n, \pi(x)\xi \rangle| \leqslant \|\eta - \eta_n\| \, \|\xi\| \to 0 \text{ as } n \to \infty.$$

This implies $\phi = W_\xi(\eta)$ almost everywhere. □

Definition. A vector $\xi \in V_\pi$ is called **square integrable** if $D_\xi = V_\pi$, i.e., $W_\xi(\eta) : x \mapsto \langle \eta, \pi(x)\xi \rangle$ is square integrable on G for every $\eta \in V_\pi$.

- We put

$$D_\pi := \{\xi \in V_\pi \mid \xi \text{ is square integrable}\}.$$

  Then D_π is a G-invariant subspace of V_π, and it follows from Lemma 13.2 and Closed Graph theorem that $W_\xi : V_\pi \to L^2(G)$ is bounded for all $\xi \in D_\pi$.

The representation (V, π) is called a **square integrable representation** if D_π is dense in V_π. A square integrable vector is called an **admissible vector** if $W_\xi : V_\pi \to L^2(G)$ is isometric.

Having fixed notation as above, the general problems to be solved are now as follows:

1. Under what conditions does a unitary representation (π, V_π) have any nontrivial square integrable vectors, or when is π a square integrable representation?

2. Suppose the unitary representation (π, V_π) is square integrable. Characterize all admissible vectors in V_π (if they exist).

3. Given an admissible vector ξ for the representation (π, V_π). Find an explicit reconstruction formula for the corresponding continuous wavelet transform $W_\xi : V_\pi \to L^2(G)$.

As explained previously, Problem 3 reduces to the problem of finding an explicit formula for W_ξ^* the continuous wavelet transform. In the next section we shall give a general answer to the first two problem in the case π being irreducible.

Lemma 13.3. Suppose $\xi \in V_\pi$ is a cyclic square integrable vector of the unitary representation (π, V_π). Then π is a square integrable representation.

Proof. Because D_π contains a dense subspace $\pi(G)\xi$. □

Lemma 13.4. Suppose $\xi \in V_\pi$ is a square integrable vector for the unitary representation (π, V_π). Then the adjoint operator $W_\xi^* : L^2(G) \to V_\pi$ of the wavelet transform W_ξ is given by the formula

$$W_\xi^*(\phi) = \int_G \phi(x)\pi(x)\xi dx$$

Proof. This integral is understood as the unique vector in V_π such that

$$\langle \int_G \pi(x)\xi dx, \eta\rangle = \int_G \phi(x)\langle \pi(x)\xi, \eta\rangle dx$$

for every $\eta \in V_\pi$. The square integrability of ξ implies the integral on the right exists. For $\eta \in V_\pi$ and $\phi \in L^2(G)$, we have

$$\langle W_\xi^*\phi, \eta\rangle = \langle \phi, W_\xi\eta\rangle = \int_G \phi(x)\overline{W_\xi\eta(x)}dx = \int_G \phi(x)\langle \pi(x)\xi, \eta\rangle dx$$

□

Proposition 13.5. Suppose that $\xi \in V_\pi$ is an admissible vector for the unitary representation (π, V_π). Then for all $\eta \in V_\pi$ we have

$$\eta = W_\xi^* W_\xi \eta = \int_G \langle \eta, \pi(x)\xi\rangle \pi(x)\xi dx$$

Moreover, the image $W_\xi(V_\pi) \subseteq L^2(G)$ of the wavelet transform is a closed subspace of $L^2(G)$ with orthogonal projection $P_\xi : L^2(G) \to W_\xi(V_\pi)$ given by the formula

$$P_\xi\phi(y) = W_\xi W_\xi^*\phi(y) = \int_G \phi(x)\langle \pi(x)\xi, \pi(y)\xi\rangle dx.$$

Proof. This follows from the previous lemma and the fact for isometries $S : V \to W$ between Hilbert spaces that $S^*S = \mathrm{id}_V$ and $SS^* : W \to \operatorname{Im} S$ is the projection onto the closed range $\operatorname{Im} S \subseteq W$. □

Corollary 13.5.1. Suppose that $\xi \in V_\pi$ is an admissible vector for the unitary representation (π, V_π). Then the orthogonal projection $P_\xi : L^2(G) \to W_\xi(V_\pi)$ is given by the right convolution with $W_\xi(\xi) \in L^2(G)$, i.e.,

$$P_\xi(\phi) = \phi * W_\xi(\xi)$$

for every $\phi \in L^2(G)$.

Proof.

$$\phi * W_\xi\xi(y) = \int_G \phi(x)\langle \xi, \pi(x^{-1}y)\xi\rangle dx = \int_G \phi(x)\langle \pi(x)\xi, \pi(y)\xi\rangle dx.$$

□

We close this subsection by observing that every group has at least one square integrable representation.

Example 13.6. Consider the left regular representation of G. We claim that every $g \in C_c(G) \subseteq L^2(G)$ is square integrable, whence $(L, L^2(G))$ is square integrable.

14 p-Adic Numbers and Adeles

Appendix T: Topology

T.1 Compactness

Definition. Let X be a topological space.

1. A sequence $(a_n)_n$ in X is called **convergent** if there exists a point $a \in X$ such that for any open neighborhood U of a there exists $N \geqslant 0$ such that $a_n \in U$ for $n \geqslant N$.

2. For a subset S of X, a point $x \in X$ is called a **limit point** of S if for every neighborhood U of x in X, $(U \cap S) \backslash \{x\}$ is nonempty.

Definition. Let X be a topological space.

1. X is called **limit point compact** if every infinite subset of X has a limit point in X.

2. X is called **sequentially compact** if every sequence in X has a convergent subsequence in X.

3. X is called **(covering) compact** if every open cover of X admits a finite subcover.

4. X is called **Lindelöf** if every open cover of X admits a countable subcover.

5. X is called **countably compact** if every countable open cover of X admits a finite subcover.

Theorem T.1. Let X be a topological space.

(i) If X is compact, then it is limit point compact.

(ii) If X is sequentially compact, then X is countably compact.

(iii) If X is first countable and compact, then X is sequentially compact.

(iv) If X is first countable Hausdorff and limit point compact, then X is sequentially compact.

(v) If X is Lindelöf and sequentially compact, then X is compact.

Proof.

(v) Suppose for contrary that $\mathcal{U}$ is an open cover of X that does not admit a finite subcover.. Since X is Lindelöf, $\mathcal{U}$ admits a countable subcover, say $\mathcal{V} = \{U_n\}_{n=1}^{\infty}$. By assumption, we can find a sequence $(p_n)_n$ such that $p_n \in X \backslash \bigcup_{i=1}^{n} U_i$. Since X is sequentially compact, $(p_n)_n$ admits a convergent subsequence $(p_{n_k})_k$ converging to $p \in X$. Since $\mathcal{V}$ covers X, we can find $m \geqslant 1$ such that $p \in U_m$, and by convergence we can find $N \geqslant 0$ such that $p_{n_k} \in U_m$ whenever $k \geqslant N$. But for $\ell \geqslant m + N$, we have

$$p_{n_\ell} \in U_m \subseteq \bigcup_{i=1}^{m} U_i \subseteq \bigcup_{i=1}^{\ell} U_i \subseteq \bigcup_{i=1}^{n_\ell} U_i$$

a contradiction to our choice of $(p_n)_n$.

$\square$

T.2 Metric spaces

Definition. Let X be a set. A function $d : X \times X \to \mathbb{R}_{\geqslant 0}$ satisfies

(i) $d(x, y) = d(y, x)$ for any $x, y \in X$,

(ii) $d(x, y) \leqslant d(x, z) + d(z, y)$ for any $x, y, z \in X$, and

(iii) $d(x, y) = 0$ if and only if $x = y$

is called a metric on X. A pair (X, d) with d a metric on X is called a **metric space**.

For $x \in X$ and $r > 0$, the set

$$B_r(x) := \{y \in X \mid d(x, y) < r\}$$

is called an **open ball** centered at x with radius r. It is easy to see that the collection $\{B_r(x) \mid x \in X,\ r > 0\}$ defines a topology on (X, d), making X a topological space.

Lemma T.2. Let (X, d) be a metric space. TFAE:

1. X is second countable.
2. X is separable.
3. X is Lindelöf.

Lemma T.3. Let (X, d) be a metric space.

(i) If X is compact, then X is second countable.

(ii) If X is sequentially compact, then X is second countable.

Corollary T.3.1. A metric space (X, d) is compact if and only if it is sequentially compact.

Proof. This follows from Lemma T.2, Lemma T.3 and Theorem T.1. □

Lemma T.4. Let $A \subseteq X$ be a subset. The closure $\overline{A}$ is compact if and only if every sequence in A admits a convergent subsequence.

Corollary T.4.1. Let $(a_n)_n \subseteq X$ be a sequence. Then $(a_n)_n$ has compact closure if and only if every subsequence of $(a_n)_n$ admits a convergent subsequence.

Proof. Let $A = \{a_n \mid n \in \mathbb{Z}_{\geqslant 1}\}$. A subsequence of $(a_n)_n$ is a sequence in A, so the only if part follows directly from Lemma T.4.

For the if part, assume that every subsequence of $(a_n)_n$ admits a convergent subsequence. By Lemma T.4, it is enough to show every sequence in A admits a convergent subsequence. Let $(b_n)_n$ be a sequence

in A. We construct a subsequence of $(b_n)_n$ that is also a subsequence on $(a_n)_n$. Define $\psi, \phi : \mathbb{N} \to \mathbb{N}$ as follows. Let $\psi(1) = 1$ and $\phi(1)$ be any number such that $b_1 = a_{\phi(1)}$. For $n \geqslant 2$, define inductively that

$$\psi(n) = \min\{k \in \mathbb{Z} \mid k > \psi(n-1),\, b_k = a_m \text{ for some } m > \phi(n-1)\}$$

If the set on the right is empty, then for $k > \psi(n-1)$, we have $b_k = a_m$ implies $m \leqslant \phi(n-1)$, so that $(b_k)_{k>\psi(n-1)} \subseteq \{a_1, \ldots, a_{\phi(n-1)}\}$ a finite set. In this case it is possible to construct a subsequence of $(b_n)_n$ converging to some a_i, where $1 \leqslant i \leqslant \phi(n-1)$. Hence we may assume the set of the right is nonempty (for every n), so that $\psi(n) < \infty$ is well-defined. Define $\phi(n)$ to be any index $m > \phi(n-1)$ such that $b_{\psi(n)} = a_{\phi(n)}$. By construction ψ and ϕ are strictly increasing, so $(b_{\psi(n)})_n = (a_{\phi(n)})_n$ is a subsequence, and hence $(b_{\psi(n)})_n$ admits a convergent subsequence by assumption. □

T.3 Baire's Category

Definition. A topological space X is called a **Baire space** if every countable intersection of open and dense subspaces of X is again dense. Equivalently, if X is a countable union of closed sets in X, then at least one of the closed set has nonempty interior.

Theorem T.5. Every LCH space and every complete metric space is a Baire space.

Proof. Let X be either an LCH space or a complete metric space. Let U be an open subset of X and suppose $A_i \subseteq X$ is nowhere dense for $i = 0, 1, \ldots$. Let $V_0 = U$ and define recursively that $\overline{V}_{i+1} \subseteq V_i \backslash \overline{A_i}$.

- In the LCH case, we may find an open set V_{i+1} such that $V_{i+1} \subseteq \overline{V}_{i+1} \subseteq V_i \backslash \overline{A}_i$ with $\overline{V}_{i+1}$ compact. Note that the $\overline{V}_i$ satisfy the finite intersection property, so $\varnothing \neq \bigcap_i \overline{V}_i \subseteq U \backslash \bigcup_i \overline{A}_i$.
- In the complete metric case, we take $V_{i+1} = B_{\frac{1}{i+1}}(x)$ for some $x \in V_i \backslash \overline{A}_i$ such that $B_{\frac{2}{i+1}}(x) \subseteq V_i \backslash \overline{A}_i$. Thus a sequence $x_i \in V_i$ is Cauchy, and the completeness shows that the limit exists; moreover, $x = \lim_n x_n \in \overline{V}_i$ for all i, i.e, $x \in \bigcap_i \overline{V}_i \subseteq U \backslash \bigcup_i \overline{A}_i$.

In both cases we've shown $U \nsubseteq \bigcup_i \overline{A}_i$. Since U is arbitrary open set, we conclude $\operatorname{int} \bigcup_i \overline{A}_i = \varnothing$. □

T.4 Proper Map

Definition. Let X, Y be spaces. A map $f : X \to Y$ is called **proper** if for every compact $K \subseteq Y$, the preimage $f^{-1}(K)$ is compact.

Proposition T.6. If $f : X \to Y$ is a continuous closed map with compact fibre, then f is proper.

Proof. Let K be a compact set in Y and $\{U_\alpha\}_{\alpha \in I}$ an open cover of $f^{-1}(K)$. For each $y \in K$, let $I_y \subseteq I$ be a finite index set such that $\{U_\alpha\}_{\alpha \in I_y}$ covers $f^{-1}(y)$ and put $V_y := \bigcup_{\alpha \in I_y} U_\alpha$. Since f is closed, $W_y := Y \backslash f(X \backslash V_y)$ is an open neighborhood of y. By compactness we can find a finite subset $F \subseteq K$ such that $K \subseteq \bigcup_{y \in F} W_y$.

Then $f^{-1}(K) \subseteq \bigcup_{y \in F} V_y = \bigcup_{y \in F,\, \alpha \in I_y} U_\alpha$. Indeed, for $x \in X$ with $f(x) \in K$, take $y \in F$ such that $f(x) \in W_y$; this means $x \in V_y$. $\square$

Proposition T.7. Let Y be LCH and $f : X \to Y$ a continuous proper map. Then f is closed.

Proof. Let $C \subseteq X$ be closed and $L \subseteq Y$ be compact. We first show $f(C) \cap L$ is closed. As $C \cap f^{-1}(L)$ is compact,

$$f(C \cap f^{-1}(L)) = f(C) \cap L \subseteq Y$$

is compact, and hence closed in Y. Now for $y \in \overline{f(C)}$, we can find a compact neighborhood $L_y \subseteq Y$ of y. Since all neighborhood of y meet $L_y \cap f(C)$, we have

$$y \in \overline{L_y \cap f(C)} = L_y \cap f(C) \subseteq f(C)$$

$\square$

Corollary T.7.1. Let the setting be as above. For every $B \subseteq Y$ and every open neighborhood U of $f^{-1}(B)$ in X, there exists an open neighborhood V of B in Y such that $f^{-1}(V) \subseteq U$.

Proof. $V = Y \backslash f(X \backslash U)$ does the job.

$\square$

T.5 Urysohn's Lemma

Definition. Let X be a topological space.

1. X is called **regular** if for every point $x \in X$ and a closed subspace $C \subseteq X$ not containing x there are open neighborhoods U, V of x and C, respectively such that $U \cap V = \varnothing$.

2. X is called **normal** if for any two disjoint closed subspaces C_1, $C_2 \subseteq X$ there are open neighborhoods U, V of C_1, C_2, respectively such that $U \cap V = \varnothing$.

3. X is called **completely regular** if for each point $x \in X$ and a closed subspace $C \subseteq X$ not containing x, there exists a continuous map $f : X \to [0, 1]$ such that $f(x) = 0$ and $f \equiv 1$ on C.

Lemma T.8. Let X be an LCH space. If K is compact and U is open with $K \subseteq U$, then we can find a relatively compact open set V with $K \subseteq V \subseteq \overline{V} \subseteq U$.

Proof. Cover K by compact neighborhoods of points in K, and by compactness finitely many of them suffices to do so; let G be their union. If $U = X$, then $V = X$ does the job. If $U \subsetneq X$, let C be the complement of U. For each $p \in C$, let W_p be an open set such that $p \notin \overline{W_p}$ and $K \subseteq W_p$; such a W_p exists by compactness of K. Then $\{C \cap \overline{G} \cap \overline{W_p}\}_{p \in C}$ is a collection of compact sets with empty intersection. The finite intersection property assures that there are points $p_1, \ldots, p_n \in C$ such that

$$C \cap \overline{G} \cap \overline{W_{p_1}} \cap \cdots \cap \overline{W_{p_n}} = \varnothing$$

Then $V = G \cap W_{p_1} \cap \cdots \cap W_{p_n}$ does the job. $\square$

Lemma T.9. Let X be a topological space. Let $(r_n)_{n\geqslant 0}$ be an enumeration of $[0,1]\cap\mathbb{Q}$ with $r_0=0$ and $r_1=1$. Suppose $(V_n)_{n\geqslant 0}$ is a family of open sets with the property that $r_n\leqslant r_m$ if and only if $\overline{V_m}\subseteq V_n$. Then the function

$$f(x):=\begin{cases} 0 & \text{, if } x\notin V_0 \\ \sup\{r_n \mid x\in\overline{V_n}\} & \text{, if } x\in V_0 \end{cases}$$

is continuous.

Proof. For $r>s$, one has

$$f^{-1}(s,r)=\bigcup_{s<s'<s''<r} V_{s'}\backslash\overline{V_{s''}}$$

which is open. Similarly, $f^{-1}[0,s)$ and $f^{-1}(r,1]$ are open. Since such intervals generate the topology of $[0,1]$, this shows f is continuous. □

Lemma T.10. Let X be space and U an open set. Suppose either

- X is LCH and $K\subseteq U$ is compact, or
- X is normal and $K\subseteq U$ is closed in X.

Then we have the following.

(i) There is a continuous map $f:X\to[0,1]$ with $f|_K\equiv 1$ and $\operatorname{supp} f\subseteq U$. In the case X is LCH, f can be constructed so that $\operatorname{supp} f$ is compact.

(ii) Let $A\subseteq X$ be closed and $h\in C^+(A)$. If X is LCH we assume h vanishes at infinity. Suppose $h\geqslant 1$ on $K\cap A$. Then there exists $f\in C(X)$ as in (i) with additional property that $f(a)\leqslant h(a)$ for every $a\in A$.

Proof.

(i) Put $r_1=0$, $r_2=1$ and $\{r_n\}_{n\geqslant 3}$ an enumeration of $(0,1)\cap\mathbb{Q}$. Pick V_0,V_1 be open such that

$$K\subseteq V_1\subseteq\overline{V_1}\subseteq V_0\subseteq\overline{V_0}\subseteq U$$

This is possible by normality if X normal, and by Lemma T.8 if X is LCH.

Suppose $n\geqslant 2$ and $V_1,\dots,V_n$ are constructed in a way that $r_i<r_j$ implies $\overline{V_j}\subseteq V_i$. Now we construct V_{n+1}. Say $r_i<r_{n+1}<r_j$ with i maximal and j minimal among $\{1,\dots,n\}$. By normality or Lemma T.8 we can find V_n such that

$$\overline{V_j}\subseteq V_{n+1}\subseteq\overline{V_{n+1}}\subseteq V_i$$

Doing this indefinitely, we obtain a collection $\{V_n\}_{n\geqslant 0}$ of open sets such that

(a) $K \subseteq V_1 \subseteq \overline{V_0} \subseteq U$ with $\overline{V_0}$ compact if X is LCH,

(b) $i > j$ implies $\overline{V_i} \subseteq V_j$.

Define f as in Lemma T.9 with the $(V_n)_{n \geqslant 0}$ constructed above. Then f is continuous and $f|_K \equiv 1$.

(ii) At each stage of constructing V_{n+1} in (ii) (including $n = 1$), replace V_i by $V_i \cap \{a \in A \mid h(a) > r_{n+1}\}$.

□

Corollary T.10.1. normality ⇒ completely regularity; LCH ⇒ completely regularity.

Theorem T.11 (Tietze's)**.**

T.6 Paracompact Space

We begin with a series of definitions.

Definition. Let X be a topological space.

1. An open cover $\mathcal{V}$ of X is a **refinement** of another open cover $\mathcal{U}$ of X if each element of $\mathcal{V}$ is a subset of some element of $\mathcal{U}$.

2. A collection $\mathcal{U} \subseteq 2^X$ is called **locally finite** if each point $x \in X$ admits a neighborhood N which meets only a finite number of the members of $\mathcal{U}$.

3. The space X is called **paracompact** if X is Hausdorff and every open cover of X has a locally finite open refinement.

Theorem T.12. A metric space (X, d) is paracompact.

Proof. (Due to M.E. Rudin) Given an open cover $U_\alpha\,(\alpha \in A)$ of X, we aim to find a locally finite open refinement $V_{\alpha,n}\,((\alpha, n) \in A \times \mathbb{N})$ of U_α. By AC, A has a well-ordering, say $\leqslant$. For each $(\alpha, n) \in A \times \mathbb{N}$, define

$$V_{\alpha,n} = \bigcup_x B_{2^{-n}}(x)$$

where the union is taken over all $x \in X$ such that

(i) α is the smallest index with $x \in U_\alpha$. (such α exists since $\leqslant$ is a well-order)

(ii) $x \notin V_{\beta,j}$ for $j < n$ and $\beta \in A$.

(iii) $B_{3 \cdot 2^{-n}}(x) \subseteq U_\alpha$.

We claim the $V_{\alpha,n}$ is an locally finite open refinement of the U_α.

First, we show the $V_{\alpha,n}$ cover X. For each $x \in X$, let $\alpha \in A$ be the smallest index such that $x \in U_\alpha$ and pick n so large that $B_{3\cdot 2^{-n}}(x) \subseteq U_\alpha$; thus (i) and (iii) holds. If (ii) holds for x, then $x \in V_{\alpha,n}$; otherwise, $x \in V_{\alpha,j}$ for some $j < n$. In a nutshell, $x \in V_{\alpha,n}$ for $j \leqslant n$.

Second, we show the $V_{\alpha,n}$ is locally finite. Pick $x \in X$ and let $\alpha \in A$ be the smallest such that $x \in V_{\alpha,n}$ for some $n \in \mathbb{N}$ (such pair exists since we've shown the $V_{\alpha,n}$ cover X), and choose j so large that $B_{2^{-j}}(x) \subseteq V_{\alpha,n}$.

1. If $i \geqslant n+j$, then $B_{2^{-n-j}}(x) \cap D_{\beta,i} = \varnothing$: Since $i > n$, every ball $B_{2^{-i}}(y)$ that constitutes $V_{\beta,i}$ has its center y outside $V_{\alpha,n}$ (for otherwise that $y \in V_{\alpha,n} \cap V_{\beta,i}$ and $i > n$ would be in breach of (ii)). Also, since $B_{2^{-j}}(x) \subseteq V_{\alpha,n}$ (by our choice), $d_X(x,y) \geqslant 2^{-j}$. But $i \geqslant j+1$ and $n+j \geqslant j+1$,

$$2^{-n-j} + 2^{-i} \leqslant 2^{-j-1} + 2^{-j-1} = 2^{-j}$$

   so that $B_{2^{-n-j}}(x) \cap B_{2^{-i}}(y) = \varnothing$.

2. If $i < n+j$, then $\#\{\beta \in A \mid B_{2^{-n-j}}(x) \cap V_{\beta,i} \neq \varnothing\} \leqslant 1$: Suppose $p \in V_{\beta,i}$, $q \in V_{\gamma,i}$ and $\beta < \gamma$; we want to show $d(p,q) > 2^{-n-j+1}$ (for if it holds and $p \in B(x, 2^{-n-j})$, then

$$d_X(q,x) \geqslant |d(q,p) - d(p,x)| = 2^{-n-j+1} - 2^{-n-j} = 2^{-n-j}$$

   so that $q \notin B_{2^{-n-j}}(x)$). By definition there are points y, z such that $p \in B_{2^{-i}}(y)$ and $q \in B_{2^{-i}}(z)$. By (iii), $B_{3\cdot 2^{-i}}(y) \subseteq U_\beta$, and by (i), $z \notin U_\beta$. Hence $d_X(y,z) \geqslant 3 \cdot 2^{-i}$ and thus

$$d_X(p,q) \geqslant d(y,z) - d(y,p) - d(z,q) > 2^{-i} \geqslant 2^{-n-j+1}.$$

□

Remark. Why $n+j$? By definition, if $B_r(x) \cap V_{\beta,i} \neq \varnothing$, there exists y such that $B_r(x) \cap B_{2^{-i}}(y) \neq \varnothing$, i.e, $y \in B_{r+2^{-i}}(x)$. If r is picked so small that $r + 2^{-i} \leqslant 2^{-j}$, then

$$y \in B(x, r+2^{-i}) \subseteq B_{2^{-j}}(x) \subseteq V_{\alpha,n}$$

(ii) then forces $n \geqslant i$.

Hence, if $n < i$ and $j < i$, we can choose an $r \in (0, 2^{-j} - 2^{-i}]$ such that $B_r(x) \cap V_{\beta,i} = \varnothing$. Therefore for all $i \geqslant n+j+1$, we can choose r such that $B_r(x) \cap V_{\beta,i} = \varnothing$. Nevertheless, it's possible to choose other lower bound of i, $\max\{n,j\}+1$ for instance.

We list some properties of a paracompact space.

Proposition T.13. Let X be a paracompact space.

1. A closed subspace of X is paracompact.

2. X is normal.

Another notion that is closely related to the paracompactness is the following.

Definition. Let $(U_i)_{i\in I}$ be an open cover of the space X. A **partition of unity** subordinate to this cover is a collection continuous map $\{f_j : X \to [0,1] \mid j \in J\}$ such that

1. the collection $\{\operatorname{supp} f_j\}_j$ is a locally finite closed refinement of $(U_i)_i$, and
2. $\sum_j f_j(x) = 1$ for each $x \in X$.

Theorem T.14. A space is paracompact if and only if for every open cover there exists a partition of unity subordinate to it.

Theorem T.15. An LCH space X is paracompact if and only if it is a disjoint union of open σ-compact subspace.

Proof. We first prove the if part. It is clear that a disjoint union of paracompact spaces is paracompact, so we may assume X itself is σ-compact. Suppose $X = \bigcup_{n\geqslant 1} K_n$ with each K_n compact. By some modification we may assume $K_n \subseteq \operatorname{int} K_{n+1}$. Define compact sets $L_1 = K_1$ and $L_n = K_n \backslash \operatorname{int} K_{n-1}$ for $n \geqslant 2$. Then L_i can only intersect with L_{i-1} and L_{i+1} nontrivially. By Lemma T.8 we can find compact sets M_i whose interiors contain L_i and only intersect with M_{i-1} and M_{i+1}.

Now let an open cover of X be given. Consider the induced cover on L_i, and pick a finite open refinement such that each cover set is contained in M_i. Then these finite covers taken together provide a locally finite open refinement of the original cover. □

Theorem T.16. Let X be an second countable LCH space. Then the one-point compactification X^+ is second countable. In particular, X is σ-compact, and is paracompact by Theorem T.15.

Proof. Let $\mathcal{B}$ be a countable basis for X and let $K \subseteq X$ be compact. Each point x in K has a compact neighborhood N, and there exists $U_x \in \mathcal{B}$ with $x \in U_x \subseteq N$. By compactness there exists a finite subset I of K such that $K \subseteq \bigcup_{x\in I} U_x$. Put $V = X \backslash \bigcup_{x\in I} \overline{U_x}$. Then $V \cup \{\infty\}$ is a neighborhood of ∞ in X^+ contained in the arbitrary neighborhood $X^+ \backslash K$. This shows X^+ is second countable, and particularly X is σ-compact. □

T.7 The Stone-Weierstrass Theorem

Theorem T.17. Let X be LCH and suppose $A \subseteq C_0(X)$ is a $\mathbb{C}$-subalgebra of $C_0(X)$ such that

(i) A separates points, i.e., for $x \neq y \in X$ there exists $f \in A$ with $f(x) \neq f(y)$,

(ii) A is nowhere vanishing, i.e., for $x \in X$ there exists $f \in A$ with $f(x) \neq 0$, and

(iii) A is closed under complex conjugation.

Then A is dense in $C_0(X)$ in sup norm.

T.8 Isometry of locally compact metric spaces

In this subsection, fix a metric space (X, d).

Definition. A map $f : X \to X$ is called an **isometry** if it is bijective and $d(f(x), f(y)) = d(x, y)$ for any $x, y \in X$.

Denote by $I(X) = I(X, d)$ the set of all isometries on (X, d). It follows from the definition that $I(X) \subseteq \mathrm{Isom}_{\mathbf{Top}}(X, X)$, and $I(X)$ is a subgroup under function composition. We topologize $I(X)$ using the subspace topology inherited from the compact-open topology on X^X (c.f. Section 4.1). Hence it has a subbasis consisting of sets of the form

$$L(K, U) := \{f \in I(X) \mid f(K) \subseteq U\}$$

where K (resp. U) runs over all compact (resp. open) subsets of X.

In the following we further assume that (X, d) is connected locally compact.

Lemma T.18. (X, d) is separable.

Proof. By Theorem T.12 and Theorem T.15, X is σ-compact. But a compact metric space is already separable, being a countable union of separable spaces, X is itself a separable space. □

Lemma T.19. $I(X)$ is a second countable Hausdorff topological space.

Proof. That $I(X)$ is Hausdorff follows from the fact that X is Hausdorff, and that $I(X)$ is second countable follows from Lemma 4.2 and Lemma T.18. □

Lemma T.20. Let $(f_n)_n$ be a sequence in $I(X)$. If $(f_n)_n$ is pointwise Cauchy on a subset $A \subseteq X$, then $(f_n)_n$ also converges pointwise on the closure $\overline{A}$ of A.

Proof. Let $p \in \overline{A}$ and pick $r > 0$ such that the open ball $B_r(p)$ has compact closure. Let $0 < \varepsilon < r$ be given, pick a point $p' \in B_{\varepsilon/3}(p) \cap A$ and $N \gg 0$ such that $d(f_n(p'), f_m(p')) < \dfrac{\varepsilon}{3}$ whenever $n, m \geqslant N$. Then for $n, m \geqslant N$

$$\begin{aligned} d(f_n(p), f_m(p)) &\leqslant d(f_n(p), f_n(p')) + d(f_n(p'), f_m(p')) + d(f_m(p'), f_m(p)) \\ &< d(p, p') + \frac{\varepsilon}{3} + d(p', p) \\ &< \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon. \end{aligned}$$

Hence $(f_n(p))_{n \geqslant N} \subseteq B_\varepsilon(f_N(p))$. Since f_N is an isometry, $B_\varepsilon(f_N(p))$ has compact closure as $B_r(p)$ does. It follows that $(f_n(p))_{n \geqslant N}$ has a convergent subsequence in $B_\varepsilon(f_N(p))$ converging to a point $p^* \in \overline{B_\varepsilon(f_N(p))}$. The above estimate then shows that $f_n(p) \to p^*$ as well. □

Lemma T.21. Let $(f_n)_n$ be a sequence in $I(X)$ such that there exists a point $x \in X$ such that $(f_n(x))_n \subseteq X$ is convergent. Then for any $p \in X$, the sequence $(f_n(p))_n \subseteq X$ has compact closure.

Proof. Put

$$S = \{p \in X \mid (f_n(p))_n \subseteq X \text{ has compact closure.}\} \subseteq X$$

Since $(f_n(x))_n$ is convergent and X is LCH, it is easy to see $x \in S$. We are going to show S is closed and open, so that $S = X$ by connectedness of X.

- S is closed. Let $(p_n)_n \subseteq X$ be a sequence. By a diagonal argument, we may find a subsequence $(f_{n_k})_k$ of $(f_n)_n$ such that $(f_{n_k}(p_n))_k$ is convergent for any n. By Lemma T.20, $(f_{n_k})_k$ also converges pointwise on any limit point on $(p_n)_n$.

- S is open. Let $p \in S$ and pick $r > 0$ such that $\overline{B_r(p)}$ is compact. We claim $B_{r/4}(p) \subseteq S$. Let $q \in B_{r/4}(p)$, and we must show $(f_n(q))_n$ has compact closure.

  By Corollary T.4.1, it is enough to show that every subsequence of $(f_n(q))_n$ admits a convergent subsequence. Let $(\varphi_n(q))_n$ be any subsequence of $(f_n(q))_n$. Since $(\varphi_n(p))_n$ has compact closure, we can find a convergent subsequence $(\varphi_{n_k}(p))_k$ converging to, say p^*; by passing to a subsequence, we may assume $(\varphi_{n_k}(p))_k \subseteq B_{r/4}(p^*)$. Then

  $$d(\varphi_{n_k}(q), p^*) \leqslant d(\varphi_{n_k}(q), \varphi_{n_k}(p)) + d(\varphi_{n_k}(p), p^*) < d(q,p) + \frac{r}{4} < \frac{r}{4} + \frac{r}{4} = \frac{r}{2}$$

  so that $(\varphi_{n_k}(q))_k \subseteq B_{r/2}(p^*)$. The latter set has compact closure, as $\varphi_{n_k}(p) \in B_{r/4}(p^*)$ implies $B_{r/2}(p^*) \subseteq B_r(\varphi_{n_k}(p)) = \varphi_{n_k}(B_r(p))$, which has compact closure as $B_r(p)$ does. Hence $(\varphi_{n_k}(q))_k$, and thus $(\varphi_n(q))_n$, has a convergent subsequence.

□

Lemma T.22. Let $(f_n)_n$ be a sequence in $I(X)$ pointwise converging to a function $f : X \to X$. Then $f \in I(X)$ and $f_n \to f$ in the compact-open topology of $I(X)$.

Proof. Since the metric $d : X \times X \to \mathbb{R}$ is continuous, for $x, y \in X$, we have

$$d(f(x), f(y)) = \lim_{n\to\infty} d(f_n(x), f_n(y)) = \lim_{n\to\infty} d(x,y) = d(x,y).$$

This shows f preserves metric. Let K be a compact subset of X and $\varepsilon > 0$. Pick a finite subset $P \subseteq K$ such that $K \subseteq \bigcup_{p\in P} B_{\varepsilon/3}(p)$. Let $x \in K$ and pick $p \in F$ such that $d(x,p) < \dfrac{\varepsilon}{3}$. Then

$$d(f_n(p), f(p)) \leqslant d(f_n(p), f_n(x)) + d(f_n(x), f(x)) + d(f(x), f(p)) = d(p,x) + d(f_n(x), f(x)) + d(x,p) < \varepsilon$$

for $n \geqslant N$, where N is chosen so that $d(f_n(x), f(x)) < \dfrac{\varepsilon}{3}$ whenever $n \geqslant N$. By Lemma 4.3, this shows $f_n \to f$ in $\mathrm{Hom}_{\mathbf{Top}}(X,X)$. It remains to show $f \in I(X)$, i.e., $f(X) = X$.

Let $p \in X$ and $q = f(p) \in X$. Then

$$0 = d(q, f(p)) = \lim_{n\to\infty} d(q, f_n(p)) = \lim_{n\to\infty} d(f_n^{-1}(q), p)$$

implying that $(f_n^{-1}(q))_n$ converges to $p \in X$. By Lemma T.21, the sequence $(f_n^{-1}(p))_n$ has compact closure for any $p \in X$. Since X is separable, by a diagonal argument we may find a subsequence $(f_{n_k}^{-1})_k$ converging pointwise on a countable dense subset of X, whence on the whole X by Lemma T.20. Let $p' = \lim_{k\to\infty} f_{n_k}^{-1}(p)$; then

$$\lim_{k\to\infty} d(f_{n_k}(p'), p) = \lim_{k\to\infty} d(p', f_{n_k}^{-1}(p)) = d(p', p') = 0$$

so that $f(p') = \lim_{k\to\infty} f_{n_k}(p') = p$. Since p is arbitrary, this shows $f(X) = X$. □

Lemma T.23. Let $(f_n)_n$ be a sequence in $I(X)$ such that there exists a point $x \in X$ such that $(f_n(x))_n \subseteq X$ is convergent. Then $(f_n)_n$ admits a convergent subsequence in the compact-open topology of $I(X)$.

Proof. By Lemma T.21 and the fact that X is separable, we may use a diagonal argument to obtain a subsequence $(f_{n_k})_k$ of $(f_n)_n$ that converges pointwise on a countable dense subset of X, whence on the whole X by Lemma T.20. But then Lemma T.22 implies that $(f_{n_k})_k$ is convergent in the compact-open topology of $I(X)$. □

Theorem T.24. Let (X, d) be a connected locally compact metric space.

(i) The group of isometries $I(X)$ on X, equipped with the compact-open topology, is an LCH group, and its action on X is continuous.

(ii) For any $x \in X$, the isotropy subgroup of x is a compact subgroup of $I(X)$.

Proof. Let $(f_n)_n$ and $(g_n)_n$ be two sequences in $I(X)$ converging to f and g in compact-open topology. For any $x \in X$,

$$\begin{aligned} d(f_n(g_n(x)), f(g(x))) &\leqslant d(f_n(g_n(x)), f_n(g(x))) + d(f_n(g(x)), f(g(x))) \\ &= d(g_n(x), g(x)) + d(f_n(g(x)), f(g(x))) \to 0 \end{aligned}$$

as $n \to \infty$. Hence $f_n \circ g_n \to f \circ g$ pointwise, and thus in compact-open topology by Lemma T.22. This show the multiplication is continuous. Also,

$$d(f_n^{-1}(x), f^{-1}(x)) = d(x, f_n(f^{-1}(x))) = d(f(f^{-1}(x)), f_n(f^{-1}(x))) \to 0$$

as $n \to \infty$, so $f_n^{-1} \to f$ in compact-open topology again by Lemma T.22. This shows the inversion is continuous. In conclusion, $I(X)$ is a topological group.

Next, we show the action map $I(X) \times X \to X$ is continuous. Let $(f_n)_n$ be a sequence in $I(X)$ converging to $f \in I(X)$ and $(p_n)_n$ a sequence to X converging to $p \in X$. Then

$$d(f_n(p_n), f(p)) \leqslant d(f_n(p_n), f_n(p)) + d(f_n(p), f(p)) = d(p_n, p) + d(f_n(p), f(p)) \qquad (\spadesuit)$$

Let $\varepsilon > 0$ be given. Pick $N \gg 0$ such that $d(p_n, p) < \dfrac{\varepsilon}{2}$ for $n \geqslant N$. Put $K = \{p_n, p \mid n \geqslant N\}$, which is a compact set by Corollary T.4.1. Consider the subbase element $L(K, B_{\varepsilon/2}(f(p)))$ of $I(X)$; then

$f \in L(K, B_{\varepsilon/2})(f(p))$, as $d(f(p_n), f(p)) = d(p_n, p) < \dfrac{\varepsilon}{2}$. Take $N' \gg 0$ such that $f_n \in L(K, B_{\varepsilon/2}(f(p)))$ for $n \geqslant N'$. From the estimate (♠), we see $d(f_n(p_n), f(p)) < \varepsilon$ for $n \geqslant N + N'$. Hence $f_n(p_n) \to f(p)$, proving the continuity.

It remains to show $I(X)$ is locally compact. Let $x \in X$ and take an relatively compact open neighborhood U of x. By Theorem T.1.(v), Lemma T.19 and Lemma T.23, the open set $L(\{x\}, U)$ has compact closure. Finally, since the stabilizer $\{f \in I(X) \mid f(x) = x\}$ is contained in $L(\{p\}, U)$, to show it is compact it suffices to show it is closed. If $f \in I(X)$ is such that $f(x) \neq x$, pick any open neighborhood U of $f(x)$ not containing x and any compact neighborhood K of x contained in $f^{-1}(U)$. Then $f \in L(K, U)$, and for any $g \in L(K, U)$, we have $g(x) \subseteq U \notin x$, so $g(x) \neq x$. $\square$

Appendix U: Measure and Integration

U.1 Riesz's Representation Theorem

Lemma U.1. For an outer Radon measure μ on a topological space X and every measurable $A \subseteq X$ with $\mu(A) < \infty$, one has

$$\mu(A) = \sup_{\substack{K \subseteq A \\ K \,:\, \text{compact}}} \mu(K)$$

In particular, if $\mu(X) < \infty$, then μ is regular.

Theorem U.2 (Riesz's Representation Theorem)**.** Let X be an LCH space and let $\Lambda : C_c(X) \to \mathbb{C}$ be a positive linear functional. Then there exists a unique outer Radon measure

$$\Lambda f = \int_X f d\mu$$

for all $f \in C_c(X)$.

Theorem U.3. Let X be an LCH space. Then every bounded linear functional Φ on $C_0(X)$ is represented by a unique regular complex Borel measure μ, in the sense that

$$\Phi f = \int_X f d\mu$$

for every $f \in C_0(X)$. Moreover, the norm of Φ is the total variation of μ.

Proof. For a regular complex Borel measure μ on X, define $\Phi_\mu : C_0(X) \to \mathbb{C}$ by the formula

$$\Phi_\mu(f) := \int_X f d\mu$$

where $f \in C_0(X)$. Clearly, we have $|\Phi_\mu(f)| \leqslant \|f\|_\infty \|\mu\|$ for all $f \in C_0(X)$, so Φ_μ is a bounded operator. Thus we obtain a well-defined association

$$\begin{aligned} M_r(X, \mathbb{C}) &\longrightarrow C_0(X)^\vee \\ \mu &\longmapsto \Phi_\mu \end{aligned}$$

The above estimate show $\|\Phi_\mu\| \leqslant \|\mu\|$, and to show this is norm-preserving, we must show the reversed inequality. Let $\varepsilon > 0$ and suppose $\|\mu\| \neq 0$. By definition of $\|\mu\|$ we can find a finite Borel-measurable partition $\{A_n\}_{n=1}^N$ of X such that $\sum_{n=1}^N |\mu(A_n)| > \|\mu\| - \varepsilon$. By regularity of μ we can find compact $K_n \subseteq A_n$ such that $|\mu(A_n) - \mu(K_n)| \leqslant N^{-1}\varepsilon$, so

$$\|\mu\| - \varepsilon < \sum_{n=1}^N |\mu(A_n)| \leqslant \sum_{n=1}^N |\mu(K_n)| + \varepsilon \leqslant \sum_{n=1}^N |\mu|(K_n) + \varepsilon$$

or, $\|\mu\| - 2\varepsilon < \sum_{n=1}^{N} |\mu|(K_n)$; we may assume $\mu(K_n) \neq 0$ for each n. Now choose $f \in C_c(X)$ such that $\|f\|_\infty \leqslant 1$ and $f(x) = \dfrac{\overline{\mu(K_n)}}{|\mu(K_n)|}$ for each n (this is possible for there are only finitely many K_n and since X is LCH, we have Urysohn Lemma in hand). If we put $K = \bigcup_{n=1}^{N} K_n$, then

$$\int_K f d\mu = \sum_{n=1}^{N} |\mu(K_n)| > \|\mu\| - 2\varepsilon \text{ and } \left|\int_{K^c} f d\mu\right| \leqslant |\mu|(K^c) < 2\varepsilon$$

It follows that

$$\|\Phi_\mu\| \geqslant |\Phi_\mu(f)| = \left|\int_X f d\mu\right| > \|\mu\| - 4\varepsilon$$

Since ε is arbitrary, this proves $\|\Phi_\mu\| \geqslant \|\mu\|$. Being norm-preserving, this also show $\mu \mapsto \Phi_\mu$ is injective.

What is left to do is show that $\mu \mapsto \Phi_\mu$ is surjective. For this, we first the real case:

$$\begin{aligned} M_r(X, \mathbb{R}) &\longrightarrow C_0(X, \mathbb{R})^\vee \\ \mu &\longmapsto \Phi_\mu \end{aligned}$$

Let $\Phi : C_0(X) \to \mathbb{R}$ be a positive linear functional. The restriction of Φ to $C_c(X)$ is positive, so by Theorem U.2 there exists an outer Radon measure μ such that $\Phi(f) = \int_X f d\mu$ for all $f \in C_c(X)$. By weak inner regularity we have $\mu(X) = \sup_{\substack{f \in C_c(X) \\ 0 \leqslant f \leqslant 1}} \Phi(f) \leqslant \|\Phi\| < \infty$, so μ is finite. Now since Φ and Φ_μ are continuous and $C_c(X)$ is dense in $C_0(X)$, we have $\Phi = \Phi_\mu$ in $C_0(X, \mathbb{R})^\vee$. The proof of real case is completed by virtue of the following lemma.

Lemma U.4. Let X be an LCH space. Then for all $\Lambda \in C_0(X, \mathbb{R})^\vee$ we have the decomposition $\Lambda = \Lambda^+ - \Lambda^-$ with $\Lambda^\pm \in C_0(X, \mathbb{R})^\vee$ and positive.

The complex case follows from expressing a functional by the sum of its real and imaginary part, and the proof is finished. □

U.2 Signed and Complex Measures

U.2.1 Signed Measures

Definition. Let $(X, \mathcal{A})$ be a measurable space and $\mu : \mathcal{A} \to [-\infty, \infty]$ be a function.

1. μ is called **finitely additive** if for a finite disjoint family $\{A_i\}_{i=1}^n \subseteq \mathcal{A}$, we have

$$\mu\left(\bigcup_{i=1}^{n} A_i\right) = \sum_{i=1}^{n} \mu(A_i)$$

2. μ is called **countably additive** if for a sequence of disjoint measurable sets $\{A_i\}_{i=1}^{\infty}$, we have

$$\mu\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mu(A_i)$$

3. μ is called a **signed measure** if μ is countably additive and $\mu(\varnothing) = 0$.

4. A signed measure is called **finite** if it takes values in $\mathbb{R}$.

- Suppose μ is a signed measure on $(X, \mathcal{A})$. Then for $A \in \mathcal{A}$, the sum $\mu(A) + \mu(A^c)$ is defined, and is equal to $\mu(X)$. Hence, if $\mu(A) = \infty$ for some $A \in \mathcal{A}$, then $\mu(X) = \infty$; if $\mu(A) = -\infty$ for some $A \in \mathcal{A}$, then $\mu(X) = -\infty$. Hence a signed measure can only attain one of the values $\pm\infty$.

  In the same way we see if $\mu(A)$ is finite, then $\mu(B)$ is finite for all measurable subsets B of A.

- If no possible confusion occurs, we will omit the σ-algebra $\mathcal{A}$ and simply say X is a measurable space and an element in $\mathcal{A}$ a measurable set. Moreover, when we say "μ is a signed measure on X", we implicitly mean that X is a measurable space.

Lemma U.5. Let X be a measurable space and μ a signed measure on X. If $\{A_i\}$ is an increasing sequences of measurable sets, then

$$\mu\left(\bigcup_{i=1}^{n} A_i\right) = \lim_{i\to\infty} \mu(A_i)$$

If $\{A_i\}$ is a decreasing sequence of measurable sets with $\mu(A_n)$ finite for some n, then

$$\mu\left(\bigcap_{i=1}^{n} A_i\right) = \lim_{i\to\infty} \mu(A_i)$$

Definition. Let μ be a signed measure on a measurable space X.

1. A subset $A \subseteq X$ is called a **positive set** if it is measurable and for all measurable subset E of A, $\mu(E) \geqslant 0$.

2. A subset $A \subseteq X$ is called a **negative set** if it is measurable and for all measurable subset E of A, $\mu(E) \leqslant 0$.

Lemma U.6. Let μ be a signed measure on X and let $A \subseteq X$ be a measurable set such that $-\infty < \mu(A) < 0$. Then there exists a negative set B contained in A such that $\mu(B) \leqslant \mu(A)$.

Proof. We delete some subset with positive measure from A and take B to be the remaining part. Let

$$\delta_1 := \sup\{\mu(E) \mid E \subseteq A,\ E \text{ measurable}\} \geqslant 0 \qquad (\text{for } \mu(\varnothing) = 0)$$

and choose a measurable subset A_1 of A such that $\mu(A_1) \geqslant \min\left\{\frac{1}{2}\delta_1, 1\right\}$. Construct inductively the sequences $\{\delta_n\}$ and $\{A_n\}$ such that

$$\delta_n := \sup\left\{\mu(E) \mid E \text{ measurable}, E \subseteq A \backslash \bigcup_{i=1}^{n-1} A_i\right\}$$

and A_n is a measurable subset of $A \backslash \bigcup_{i=1}^{n-1} A_i$ such that $\mu(A_n) \geqslant \min\left\{\frac{1}{2}\delta_n, 1\right\}$. Now define $A_\infty := \bigcup_{n=1}^{\infty} A_n$ and $B = A \backslash A_\infty$. Since each A_n is disjoint with non-negative measure, $\mu(A_\infty) \geqslant 0$ by countable additivity, and hence

$$\mu(A) = \mu(A_\infty) + \mu(B) \geqslant \mu(B)$$

It remains to show B is a negative set. Since $\mu(A)$ is finite, so is $\mu(A_\infty) = \sum_{n=1}^{\infty} \mu(A_n)$, implying $\delta_n \to 0$. For any measurable subset E of B, we have $\mu(E) \leqslant \delta_n$ for all $n \in \mathbb{N}$, so $\mu(E) \leqslant 0$. □

Theorem U.7 (Hahn Decomposition Theorem)**.** Let X be a measurable space and let μ be a signed measure on X. Then we can find two disjoint subsets P and N of X such that P is a positive set, N is a negative set and $X = P \sqcup N$.

The pair (P, N) is called a **Hahn decomposition** for the signed measure μ.

Proof. Since μ cannot take $\pm\infty$ at the same time, for definiteness we assume μ does not take $-\infty$ as its values. Let

$$L := \inf\{\mu(A) \mid A \text{ is a negative set for } \mu\}$$

The set on the right is nonempty since $\varnothing$ is a negative set. Choose a sequence $\{A_n\}$ of negative sets for which $L = \lim_{n\to\infty} \mu(A_n)$, and put $N = \bigcup_{n=1}^{\infty} A_n$. Every measurable subset of N can be decomposed as a disjoint union of measurable subsets, each of which contained in some A_n, so N is negative as well. Hence $L \leqslant \mu(N) \leqslant \mu(A_n)$ for all $n \in \mathbb{N}$, and therefore $L = \mu(N)$. Since μ does not attain $-\infty$, $\mu(N)$ is finite.

Let $P = N^c$. We claim P is positive. If P contained some measurable set A with $\mu(A) < 0$, then A would contain some negative set B with $\mu(B) < 0$ by Lemma U.6, and $N \sqcup B$ would be a negative set such that

$$\mu(N \sqcup B) = \mu(N) + \mu(B) < \mu(N) = L$$

a contradiction to definition of L. Hence P must be positive for μ. □

Corollary U.7.1 (Jordan Decomposition Theorem)**.** Every signed measure is the difference of two positive measures, at least one of which is finite.

Proof. Let μ be a signed measure on X, and (P, N) a Hahn decomposition for μ. Define μ^+ and μ^- on X by

$$\mu^+(A) := \mu(A \cap P), \qquad \mu^-(A) := -\mu(A \cap N)$$

It follows from definition that μ^+ and μ^- are positive measure, and since μ cannot attain $\pm\infty$ simultaneously, at least one of the values $\mu(P)$ and $\mu(N)$ is finite, and hence at least one of the measures μ^+ and μ^- is finite. $\square$

Let (P, N) be a Hahn decomposition for the signed measure μ on X, and let μ^+ and μ^- be the positive measures constructed in the last proof. For a measurable set A and any of its measurable subset B, we have

$$\mu(B) = \mu^+(B) - \mu^-(B) \leqslant \mu^+(B) \leqslant \mu^+(A)$$

Since $\mu^+(A) = \mu(A \cap P)$, this implies

$$\mu^+(A) = \sup\{\mu(B) \mid B \text{ measurable}, B \subseteq A\}$$

Likewise

$$\mu^-(A) = \sup\{-\mu(B) \mid B \text{ measurable}, B \subseteq A\}$$

This says that μ^+ and μ^- are independent of the choice of Hahn decomposition (P, N) for μ. The measures μ^+ and μ^- are called the positive part and the negative part of μ, and the expression $\mu = \mu^+ - \mu^-$ is called the **Jordan decomposition** of μ. Moreover, they are minimal in the sense that if $\mu = \lambda_1 - \lambda_2$ with each λ_i positive, then

$$\begin{aligned} \mu^+(A) &= \mu(A \cap P) \leqslant \lambda_1(A \cap P) \leqslant \lambda_1(A) \\ \mu^-(A) &= \mu(A \cap N) \leqslant \lambda_2(A \cap N) \leqslant \lambda_2(A) \end{aligned}$$

Definition. The **variation** of a signed measure μ on X is the positive measure $|\mu|$ on X defined by $|\mu| := \mu^+ + \mu^-$. The **total variation** $\|\mu\|$ is defined by $\|\mu\| := |\mu|(X)$.

- It is easy to see $|\mu(A)| \leqslant |\mu|(A)$ for all measurable A. In fact, $|\mu|$ is the smallest positive measure possessing this property. Let ν be another positive measure such that $|\mu(A)| \leqslant \nu(A)$ for all measurable A. Let (P, N) be a Hahn decomposition for μ. Then for measurable A, we have $\mu^+(A) = |\mu(A \cap P)| \leqslant \nu(A \cap P)$ and $\mu^-(A) \leqslant \nu(A \cap N)$, so

$$|\mu|(A) = \mu^+(A) + \mu^-(A) \leqslant \nu(A \cap P) + \nu(A \cap N) = \nu(A)$$

U.2.2 Complex Measures

Definition. Let $(X, \mathcal{A})$ be a measurable space. A **complex measure** on $(X, \mathcal{A})$ is a function $\mu : \mathcal{A} \to \mathbb{C}$ that is countably additive, namely, for a sequence of disjoint measurable sets $\{A_i\}_{i=1}^{\infty}$, we have

$$\mu\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mu(A_i)$$

For a complex measure μ, define its **variation** $|\mu|$ by

$$|\mu|(A) := \sup\left\{\sum_{n=1}^{\infty} |\mu(A_n)| \,\middle|\, \{A_n\} \text{ is a measurable partition of } A\right\}$$

and the **total variation** $\|\mu\|$ is defined by $\|\mu\| := |\mu|(X)$.

- For a complex measure μ, we must have $\mu(\varnothing) = 0$ by countable additivity, and thus $|\mu|(\varnothing) = 0$.
- We can write $\mu = \mu' + i\mu''$, where μ', μ'' are finite signed measures on X, so by Jordan decomposition we have

  $$\mu = \mu_1 - \mu_2 + i\mu_3 - i\mu_4$$

  where the μ_i are the finite positive measures on X. This is called the **Jordan decomposition** of μ if $\mu_1 - \mu_2$ and $\mu_3 - \mu_4$ are the Jordan decomposition of the real and imaginary parts of μ.

  Since each μ_i is finite, this implies $|\mu|$ only takes finite values. In particular, $\|\mu\| < \infty$.
- It can be shown that in the definition of $|\mu|(A)$, it is enough to go through all *finite* measurable partitions of A. To see this, denote

  $$|\mu|_f(A) := \sup\left\{\sum_{n=1}^{N} |\mu(A_n)| \,\middle|\, \{A_n\}_{n=1}^{N} \text{ is a finite measurable partition of } A\right\}$$

  Then clearly $|\mu(A)| \leqslant |\mu|_f(A) \leqslant |\mu|(A)$ for all measurable A. Using the same argument in the next proposition, we can prove $|\mu|_f$ is a measure. But then for any measurable partition $\{A_n\}$ of A

  $$\sum_{n=1}^{\infty} |\mu(A_n)| \leqslant \sum_{n=1}^{\infty} |\mu|_f(A_n) = |\mu|_f(A)$$

  so by definition, $|\mu|(A) \leqslant |\mu|_f(A)$. These proves $|\mu|_f = |\mu|$.

Proposition U.8. Let μ be a complex measure on X. Then the variation $|\mu|$ of μ is a finite positive measure on X.

Proof. It remains to show $|\mu|$ is countably additive. Let E be a measurable set and $\{E_n\}$ be any measurable partition of E. If $\{A_n\}$ is any other measurable partition of E, then

$$\sum_n |\mu(A_n)| = \sum_n \left|\sum_m \mu(A_n \cap E_m)\right| \leqslant \sum_n \sum_m |\mu(A_n \cap E_m)| = \sum_m \sum_n |\mu(A_n \cap E_m)| \leqslant \sum_m |\mu|(E_m)$$

Conversely, let $t_n \in \mathbb{R}$ such that $t_n < |\mu|(E_n)$ and $\{A_{nm}\}_m$ a measurable partition of E_n such that $\sum_m |\mu(A_{nm})| > t_n$ for each n. Then

$$\sum_n t_n \leqslant \sum_{n,m} |\mu(A_{nm})| \leqslant |\mu|(E)$$

so that $\sum_n |\mu|(E_n) \leqslant |\mu|(E)$. This shows $|\mu|$ is countably additive. □

Let $(X, \mathcal{A})$ be a measurable space. Denote by $M(X, \mathcal{A}, \mathbb{R})$ the collection of all finite signed measures on $(X, \mathcal{A})$, and $M(X, \mathcal{A}, \mathbb{C})$ the collection of all complex measures. It is clear that $M(X, \mathcal{A}, \mathbb{R})$ and $M(X, \mathcal{A}, \mathbb{C})$ are vector spaces over $\mathbb{R}$ and $\mathbb{C}$, respectively, and the total variation defines a norm on each of them. For convenience, let call a finite signed measure a **real measure**.

Lemma U.9. Let μ be a finite signed measure. We can also think of μ as a complex measure. Then two variation coincide.

Proof. By $|\mu|$ we mean the variation defined by the Jordan decomposition. Let A be measurable and $\{A_n\}$ a measurable partition of A. Then

$$|\mu(A)| = \left|\sum_{n=1}^{\infty} \mu(A_n)\right| \leqslant \sum_{n=1}^{\infty} |\mu(A_n)| \leqslant \sum_{n=1}^{\infty} |\mu|(A_n) = |\mu|(A)$$

so that

$$|\mu(A)| \leqslant \sup\left\{\sum_{n=1}^{\infty} |\mu(A_n)| \,\middle|\, \{A_n\} \text{ is a measurable partition of } A\right\} \leqslant |\mu|(A)$$

We know the middle term defines a positive measure, and since $|\mu|$ is the smallest positive measure ν satisfying $|\mu(A)| \leqslant \nu(A)$ for all measurable A, this forces the second inequality to be an equality. □

Proposition U.10. Let $(X, \mathcal{A})$ be a measurable space. Then the space $M(X, \mathcal{A}, \mathbb{R})$ and $M(X, \mathcal{A}, \mathbb{C})$ are complete under the total variation norm.

Proof. It suffices to show every absolutely convergent series converges in $M(X, \mathcal{A}, \mathbb{R})$ or $M(X, \mathcal{A}, \mathbb{C})$. Let $\{\mu_n\}$ be a sequence of real or complex measures such that $\sum_{n=1}^{\infty} \|\mu_n\| < \infty$.

For a complex measure μ, we can write $\mu = \mu' + i\mu''$ with μ' and μ'' finite real, and we have $|\mu(A)| \geqslant |\mu'(A)|$ and $|\mu(A)| \geqslant |\mu''(A)|$, so by minimality we have $\|\mu\| \geqslant \|\mu'\|$ and $\|\mu\| \geqslant \|\mu''\|$. Hence it suffices to

deal with the real case. Still, write the Jordan decomposition $\mu = \mu^+ - \mu^-$, so that $|\mu| = \mu^+ + \mu^-$. This implies $\|\mu\| \geqslant \|\mu^\pm\|$, so we can further reduce to the positive measure case.

Now assume each μ_n is a finite positive measure such that $\sum_{n=1}^{\infty} \mu_n(X) < \infty$.

Lemma U.11. Let $\{\mu_n\}$ be a sequence of positive measures on X. Then the function μ defined by $\mu(A) := \sum_{n=1}^{\infty} \mu_n(A)$ is a positive measures on X.

Proof. We need to show μ is countably additive. Let $\{A_n\}$ be a disjoint sequence of measurable sets on X. What we must prove is

$$\sum_{n=1}^{\infty} \sum_{m=1}^{\infty} \mu_n(A_m) = \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} \mu_n(A_m)$$

Since each summand is non-negative, this clearly holds. □

By this lemma, we see the formula $\mu(A) := \sum_{n=1}^{\infty} \mu_n(A)$ defines a positive measure on X, and since $\mu(X) < \infty$, it is a finite measure. □

Finally we discuss the integration with respect to a real or complex measure. Let $(X, \mathcal{A})$ be a measurable space, and denote by $B(X, \mathcal{A}, \mathbb{R})$ and $B(X, \mathcal{A}, \mathbb{C})$ the space of bounded real-valued and complex-valued, respectively, $\mathcal{A}$-measurable functions on X. If μ is a real measure on $(X, \mathcal{A})$ and $\mu = \mu^+ - \mu^-$ is the Jordan decomposition of μ, then the integral for $f \in B(X, \mathcal{A}, \mathbb{R})$ with respect to μ is defined by

$$\int f d\mu := \int f d\mu^+ - \int f d\mu^-$$

and this defines a linear functional on $B(X, \mathcal{A}, \mathbb{R})$. If $\mu = \nu_1 - \nu_2$ for some finite positive ν_i, then $\nu_1 + \mu^- = \nu_2 + \mu^+$, so

$$\int_X f d\nu_1 + \int_X f d\mu^- = \int_X f d\nu_2 + \int_X f d\mu^+$$

or

$$\int f d\mu = \int f d\mu^+ - \int f d\mu^- = \int f d\nu_1 - \int f d\nu_2$$

If $A \in \mathcal{A}$ and $\mu \in M(X, \mathcal{A}, \mathbb{R})$, then $\int \mathbf{1}_A d\mu = \mu(A)$ holds, so the formula

$$\mu \mapsto \int f d\mu$$

defines a linear functional on $M(X, \mathcal{A}, \mathbb{R})$ when f is a step function. By DCT it follows that this holds for every $f \in B(X, \mathcal{A}, \mathbb{R})$. In sum, we have a bilinear pairing

$$\begin{aligned} B(X, \mathcal{A}, \mathbb{R}) \times M(X, \mathcal{A}, \mathbb{R}) &\longrightarrow \mathbb{R} \\ (f, \mu) &\longmapsto \int_X f d\mu \end{aligned}$$

Similarly if μ is a complex measure on $(X, \mathcal{A})$, we can write the Jordan decomposition of μ to define its integral, and the above statements remain valid in the complex case. Now we define a norm on $B(X, \mathcal{A}, \mathbb{R})$ and $B(X, \mathcal{A}, \mathbb{C})$ by

$$\|f\|_\infty := \sup\{|f(x)| \mid x \in X\}$$

If μ is a real or complex measure on $(X, \mathcal{A})$ and $f = \sum a_i \mathbf{1}_{A_i}$ is a simple function with each A_i disjoint, then

$$\left|\int_X f d\mu\right| = \left|\sum a_i \mu(A_i)\right| \leqslant \sum |a_i||\mu(A_i)| \leqslant \sum |a_i||\mu|(A_i) = \int_X |f| \, d|\mu|$$

Since each function in $B(X, \mathcal{A}, \mathbb{R})$ and $B(X, \mathcal{A}, \mathbb{C})$ is the uniform limit of a sequence of simple functions, it follows that the inequality holds for every bounded measurable function f. In particular, we have

$$\left|\int_X f d\mu\right| \leqslant \|f\|_\infty \|\mu\|$$

U.2.3 *-algebra of complex measures

Let $(X, \mathcal{A})$ and $(Y, \mathcal{B})$ be two measurable spaces and let μ and ν be real measures on $(X, \mathcal{A})$ and $(Y, \mathcal{B})$, respectively. Write the Jordan decomposition of $\mu = \mu^+ - \mu^-$, $\nu = \nu^+ - \nu^-$. We then define the product measure $\mu \otimes \nu$ by the formula

$$\mu \otimes \nu = \mu^+ \otimes \nu^+ - \mu^+ \otimes \nu^- - \mu^- \otimes \nu^+ + \mu^- \otimes \nu^-$$

Then for $A \in \mathcal{A}$ and $B \in \mathcal{B}$, the identity $(\mu \otimes \nu)(A \times B) = \mu(A)\nu(B)$ holds, and for $E \in \mathcal{A} \otimes \mathcal{B}$, we have the following integration formula

$$\mu \otimes \nu(E) = \int_X \nu(E_x) d\mu(x) = \int_Y \mu(E^y) d\nu(y)$$

All of these follow from Theorem U.21. To see the integration formulas above, for example, we have

$$\begin{aligned} \mu \otimes \nu(E) &= \mu^+ \otimes \nu^+(E) - \mu^+ \otimes \nu^-(E) - \mu^- \otimes \nu^+(E) + \mu^- \otimes \nu^-(E) \\ &= \int_X \nu^+(E_x) d\mu^+(x) - \int_X \nu^-(E_x) d\mu^+(x) - \int_X \nu^+(E_x) d\mu^-(x) + \int_X \nu^-(E_x) d\mu^-(x) \\ &= \int_X \nu^+(E_x) d\mu(x) - \int_X \nu^-(E_x) d\mu(x) \\ &= \int_X \nu(E_x) d\mu(x) \end{aligned}$$

If μ and μ' are two real measures, then

$$(\mu+\mu')\otimes\nu(E)=\int_X \nu(E_x)d(\mu+\mu')(x)=\int_X \nu(E_x)d\mu(x)+\int_X \nu(E_x)d\mu'(x)$$

so $(\mu+\mu')\otimes\nu=\mu\otimes\nu+\mu'\otimes\nu$. A similar formula holds for the second argument. In sum, the map

$$\begin{aligned} M(X,\mathcal{A},\mathbb{R})\times M(Y,\mathcal{B},\mathbb{R}) &\longrightarrow M(X\times Y,\mathcal{A}\otimes\mathcal{B},\mathbb{R}) \\ (\mu,\nu) &\longmapsto \mu\otimes\nu \end{aligned}$$

is a bilinear map. If μ and ν are complex measures, we define similarly $\mu\otimes\nu\in M(X\times Y,\mathcal{A}\otimes\mathcal{B},\mathbb{C})$ by using Jordan decomposition for complex measures and linearity, and all the above results hold as well.

Let G be a topological group and consider the Banach space of (finite) complex Borel measures $M(G,\mathbb{C})$ on G. For $\mu,\nu\in M(G,\mathbb{C})$, we define their convolution

$$\mu*\nu(A):=\int_{G\times G}\mathbf{1}_A(xy)d(\mu\otimes\nu)(x,y)$$

In other word, if we write mult : $G\times G\to G$ to be the multiplication of G, then $\mu*\nu$ is the pushforward measure of the product measure $\mu\otimes\nu$ by mult. It follows from DCT that for all bounded Borel-measurable function f we have

$$\int_G f d(\mu*\nu):=\int_{G\times G} f(xy)d(\mu\otimes\nu)(x,y)$$

When μ and ν are finite positive, by Fubini's theorem, we have

$$\begin{aligned}\int_{G\times G}\mathbf{1}_A(xy)d(\mu\otimes\nu)(x,y)&=\int_G\left(\int_G\mathbf{1}_A(xy)d\nu(y)\right)d\mu(x)=\int_G\nu(x^{-1}A)d\mu(x)\\&=\int_G\mu(Ay^{-1})d\nu(y)\end{aligned}$$

By linearity this holds for arbitrary complex measures μ and ν, and this implies $*$ defines a bilinear map on $M(G,\mathbb{C})$ particularly. Similarly, we have

$$\int_G f d(\mu*\nu)=\int_G\left(\int_G f(xy)d\mu(x)\right)d\nu(y)=\int_G\left(\int_G f(xy)d\nu(y)\right)d\mu(x)$$

for all bounded Borel-measurable functions f. For $\nu_i\in M(G)$, $i=1,2,3$, we have

$$\begin{aligned}\nu_1*(\nu_2*\nu_3)(A)&=\int_G(\nu_2*\nu_3)(x^{-1}A)d\nu_1(x)=\int_G\int_G\nu_3(y^{-1}x^{-1}A)d\nu_2(y)d\nu_1(x)\\(\nu_1*\nu_2)*\nu_3(A)&=\int_G\nu_3(u^{-1}A)d(\nu_1*\nu_2)(u)=\int_G\int_G\nu_3((xy)^{-1}A)d\nu_1(x)d\nu_2(y)\end{aligned}$$

It follows that $*$ is associative. Let μ_e be the Dirac measure at the identity $e \in G$; that is, for all Borel sets A of G, define $\mu_e(A) := \begin{cases} 1 & \text{, if } e \in A \\ 0 & \text{, if } e \notin A \end{cases}$. Then μ_e is the identity element of the operation $*$. Indeed, for $\mu \in M(G)$,

$$\mu * \mu_e(A) = \int_G \mu(Ay^{-1}) d\mu_e(y) = \mu(Ay^{-1})|_{y=e} = \mu(A)$$

and similarly $\mu_e * \mu(A) = \mu(A)$. Finally, if $\{A_i\}$ is a measurable partition of X, then

$$\begin{aligned} \sum_{i=1}^{\infty} |\mu * \nu(A_i)| &= \sum_{i=1}^{\infty} \left| \int_X \nu(x^{-1}A_i) d\mu(x) \right| \\ &\leqslant \sum_{i=1}^{\infty} \int_X |\nu(x^{-1}A_i)| d|\mu|(x) = \int_X \left(\sum_{i=1}^{\infty} |\nu(x^{-1}A_i)| \right) d|\mu|(x) \\ &\leqslant \int_X \|\nu\| \, d|\mu|(x) = \|\nu\| \, \|\mu\| \end{aligned}$$

Letting $\{A_i\}$ run over all possible measurable partitions of X, we obtain $\|\mu * \nu\| \leqslant \|\mu\| \, \|\nu\|$. In sum,

Proposition U.12. $(M(G, \mathbb{C}), *)$ is a unital Banach algebra.

Let $\mathrm{inv} : G \to G$ be the inversion on G. For $\mu \in M(G, \mathbb{C})$, define a new measure $\mu^* \in M(G, \mathbb{C})$

$$\mu^*(A) = \overline{(\mathrm{inv}_* \mu)(A)} = \overline{\mu(A^{-1})}$$

where $\bar{\cdot}$ is the complex conjugation; this is well-defined since inv is a homeomorphism. It is clear that $\mu \mapsto \mu^*$ defines an involution on $M(G, \mathbb{C})$, and by definition of total variation we have $\|\mu\| = \|\mu^*\|$; this makes $M(G, \mathbb{C})$ a unital Banach *-algebra.

Now assume G is a locally compact Hausdorff group. Choose a left Haar measure dx once and for all. There is a map

$$\begin{aligned} L^1(G) &\longrightarrow M(G, \mathbb{C}) \\ f &\longmapsto d\mu_f(x) := f(x)dx \end{aligned}$$

In fact, this is an algebra *-homomorphism. Indeed, if $f, g \in L^1(G)$, then

$$\begin{aligned} \mu_{f*g}(A) &\overset{\text{Fubini}}{=} \int_G \int_G \mathbf{1}_A(x) f(y) g(y^{-1}x) dx dy \\ (x \mapsto yx) &= \int_G \left(\int_G \mathbf{1}_A(yx) g(x) dx \right) f(y) dy \\ &= \int_G \mu_g(y^{-1}A) d\mu_f(y) \\ &= \mu_f * \mu_g(A) \end{aligned}$$

and

$$\mu_{f*}(A) = \int_G \mathbf{1}_A(x)\Delta_G(x^{-1})\overline{f(x^{-1})}dx \overset{1.11.4}{=} \int_G \mathbf{1}_A(x^{-1})\overline{f(x)}dx = \overline{\int_G \mathbf{1}_{A^{-1}}(x)d\mu_f(x)} = \mu_f^*(A)$$

It is also norm-preserving, for by Corollary U.34.1 we have $|\mu_f| = \mu_{|f|}$, so

$$\|\mu_f\| = \int_X d|\mu_f| = \int_X |f|dx = \|f\|_1$$

U.2.4 Regular measure

In what follows let X denote a locally compact Hausdorff space and $\mathcal{B}$ be the σ-algebra of Borel sets in X. For simplicity, we put $M(X, \mathbb{F}) = M(X, \mathcal{B}, \mathbb{F})$ where $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$.

Definition. For a topological space, a complex Borel measure μ on it is called **regular** if its total variation $|\mu|$ is a **regular** measure, in the sense that it is finite on compact sets and every measurable set can be approximated by open sets from above, and every open set can be approximated by compact sets from below.

- In our previous term, a positive measure is regular if and only if it is finite on compact sets, outer regular, and weakly inner regular. If X is LCH, then a positive regular measure is the same as a outer Radon measure.
- If μ is regular, for Borel A and $\varepsilon > 0$ we can find a compact set $K \subseteq A$ such that for all Borel B with $K \subseteq B \subseteq A$ we have $|\mu(A) - \mu(B)| < \varepsilon$. Indeed, since $|\mu|$ is regular, we can find compact K such that $|\mu|(A\backslash K) < \varepsilon$. Then for any such B, we have

$$|\mu(A) - \mu(B)| = |\mu(A\backslash B)| \leqslant |\mu|(A\backslash B) \leqslant |\mu|(A\backslash K) < \varepsilon$$

Proposition U.13. Let μ be a complex Borel measure on X. TFAE:

(i) μ is regular;

(ii) the positive measures appearing in the Jordan decomposition are regular;

(iii) μ is a linear combination of finite positive regular Borel measures.

Proof. Let μ be complex regular and let μ' be any positive finite measure appearing in the Jordan decomposition of μ. Then $\mu' \leqslant |\mu|$. Let $\varepsilon > 0$ be given.

- If A is Borel and $U \supseteq A$ is open such that $|\mu|(A) < |\mu|(U) + \varepsilon$, then $\mu'(U\backslash A) \leqslant |\mu|(U\backslash A) < \varepsilon$ so that

$$\mu'(U) = \mu'(A) + \mu'(U\backslash A) < \mu'(A) + \varepsilon$$

- If A is Borel and $K \subseteq A$ is compact such that $|\mu|(A) - \varepsilon < |\mu|(K)$, then $\mu'(A \backslash K) \leqslant |\mu|(A \backslash K) < \varepsilon$ so that

$$\mu'(K) = \mu'(A) - \mu'(A \backslash K) > \mu'(A) - \varepsilon$$

This shows μ' is regular, so (ii) holds. Clearly (ii) implies (iii). For (iii)$\Rightarrow$(i), the proof is similar to that of (i)$\Rightarrow$(ii) and use the fact that if $\mu = \alpha_1\mu_1 + \cdots + \alpha_n\mu_n$ where $\alpha_i \in \mathbb{C}$ and the μ_i are positive, then $|\mu| \leqslant |\alpha_1|\mu_1 + \cdots + |\alpha_n|\mu_n$. □

Let us denote by $M_r(X, \mathbb{F}) \subseteq M(X, \mathbb{F})$ the set of all $\mathbb{F}$-valued regular Borel measures, where $\mathbb{F} = \mathbb{R}, \mathbb{C}$.

Lemma U.14. $M_r(X, \mathbb{F})$ is a closed subspace of $M(X, \mathbb{F})$.

Proof. By the previous proposition we see $M_r(X, \mathbb{F})$ is a linear subspace. It remains to show $M_r(X, \mathbb{F})$ is closed. First note that if $\mu \in M_r(X, \mathbb{F})$ and $\nu \in M(X, \mathbb{F})$ with $\|\mu - \nu\| < \varepsilon$ for some $\varepsilon > 0$, then for any Borel A and open $U \supseteq A$ with $|\mu|(U \backslash A) < \varepsilon$, we have

$$|\nu|(U \backslash A) \leqslant \|\nu - \mu\| + |\mu|(U \backslash A) < 2\varepsilon$$

This shows the limit of sequences of outer regular measures is again outer regular. Similarly for weakly inner regular. This concludes the proof. □

Proposition U.15. Let G be an LCH group. The subspace $M_r(G, \mathbb{F})$ is *-closed. Hence $M_r(G, \mathbb{F})$ is a (non-unital) Banach *-algebra.

Proposition U.16. Let μ be a regular positive Borel measure on X and $f \in L^1(\mu)$. Then the complex measure μ_f is regular.

Proposition U.17. Let μ be a regular positive Borel measure on X and $\nu \in M_r(X, \mathbb{C})$. TFAE:

(i) There exists $f \in L^1(\mu)$ such that $\nu = \mu_f$.

(ii) $\nu \ll \mu$.

(iii) Each compact subset K of X that satisfies $\mu(K) = 0$ also satisfies $\nu(K) = 0$.

Proof. By regularity (ii)$\Leftrightarrow$(iii), and clearly (i)$\Rightarrow$(ii). It remains to show (ii)$\Rightarrow$(i). Since $|\nu|$ is finite regular, we can find an increasing sequence of compact subsets K_n in X such that $|\nu|(K_n) \to |\nu|(X)$. Locally finiteness of μ implies $\mu(K_n)$ is finite, so the measure μ_0 defined y $\mu_0(A) := \mu\left(A \cap \bigcup_{n \geqslant 1} K_n\right)$ is σ-finite. Now $\nu \ll \mu$ implies $\nu \ll \mu_0$, so by Radon-Nikodym theorem there exists $f \in L^1(\mu_0)$ such that $\nu = (\mu_0)_f$. By redefining f so that it vanishing outside $\bigcup_{n \geqslant 1} K_n$, we have $\nu = \mu_f$. □

Corollary U.17.1. Let X be an LCH space and μ a regular positive Borel measure on X. Then the map

$$\begin{aligned} L^1(\mu) &\longrightarrow M(X, \mathbb{C}) \\ f &\longmapsto d\mu_f(x) := f(x)dx \end{aligned}$$

induces a linear isometry onto the subspace $M_r(X, \mathbb{C})$ consisting of those ν with $\nu \ll \mu$.

U.3 Fubini's Theorem

U.3.1 Product Measure

Let $(X, \mathcal{A})$ and $(Y, \mathcal{B})$ be measurable spaces. Define $\mathcal{A} \otimes \mathcal{B}$ to be the smallest σ-algebra on $X \times Y$ which contains $\{A \times B \mid (A, B) \in \mathcal{A} \times \mathcal{B}\}$. For $E \subseteq X \times Y$, $x \in X$, $y \in Y$, define

$$E_x := \{y \in Y \mid (x, y) \in E\} \qquad E^y := \{x \in X \mid (x, y) \in E\}$$

These are called the **x-section** and **y-section** of E, respectively.

Lemma U.18. If $E \in \mathcal{A} \otimes \mathcal{B}$, then $E_x \in \mathcal{B}$ and $E^y \in \mathcal{A}$ for all $x \in X$ and $y \in Y$.

Similarly, for $f \in X \times Y \to \mathbb{C}$, $x \in X$, $y \in Y$, define

$$\begin{aligned} f_x : Y &\longrightarrow \mathbb{C} & \qquad f^y : X &\longrightarrow \mathbb{C} \\ y &\longmapsto f(x, y) & \qquad x &\longmapsto f(x, y) \end{aligned}$$

Lemma U.19. Let f be a $\mathcal{A} \otimes \mathcal{B}$-measurable function on $X \times Y$. Then

1. f_x is $\mathcal{B}$-measurable for each $x \in X$.
2. f^y is $\mathcal{A}$-measurable for each $y \in Y$.

Now comes to measure. Let $(X, \mathcal{A}, \mu)$ and $(Y, \mathcal{B}, \nu)$ be σ-finite measure spaces.

Lemma U.20. For $E \in \mathcal{A} \otimes \mathcal{B}$, $[x \mapsto \nu(E_x)]$ is $\mathcal{A}$-measurable and $[y \mapsto \mu(E^y)]$ is $\mathcal{B}$-measurable.

Theorem U.21. Let X, Y be as above. Then there exists a unique measure $\mu \otimes \nu$ on the σ-algebra $\mathcal{A} \otimes \mathcal{B}$ such that

$$(\mu \otimes \nu)(A \times B) = \mu(A)\nu(B)$$

for all $(A, B) \in \mathcal{A} \times \mathcal{B}$. Further, for $E \in \mathcal{A} \otimes \mathcal{B}$,

$$(\mu \otimes \nu)(E) = \int_X \nu(E_x) d\mu(x) = \int_Y \mu(E^y) d\nu(y)$$

U.3.2 Fubini's in σ-finite case

Theorem U.22 (Fubini's Theorem)**.** Let (X, μ) and (Y, ν) be two σ-finite measure spaces, and let f be a measurable function on $X \times Y$.

(a) If $f \geqslant 0$, then the two partial integrals $\int_X f(x, y) d\mu(x)$ and $\int_Y f(x, y) d\nu(y)$ define measurable functions such that Fubini's formula holds:

$$\int_{X \times Y} f(x, y) d(\mu \otimes \nu)(x, y) = \int_X \int_Y f(x, y) d\nu(y) d\mu(x) = \int_Y \int_X f(x, y) d\mu(x) d\nu(y)$$

(b) If f is complex valued and one of the iterated integrals

$$\int_X \int_Y |f(x, y)| d\nu(y) d\mu(x) \text{ or } \int_Y \int_X |f(x, y)| d\mu(x) d\nu(y)$$

is finite, then f is integrable and the Fubini's formula holds.

U.3.3 Fubini's in LCH cases

Lemma U.23. Let μ be an outer Radon measure on X, and let $\mathcal{F}$ be any subset of $C_c(X)$ consisting of functions $\phi \geqslant 0$ such that for $\phi, \psi \in \mathcal{F}$ there exists a function $\eta \in \mathcal{F}$ with $\eta \geqslant \phi$, ψ. Then $x \mapsto \sup_{\phi \in \mathcal{F}} \phi(x)$ is measurable and

$$\sup_{\phi \in \mathcal{F}} \int_X \phi(x) dx = \int_X \sup_{\phi \in \mathcal{F}} \phi(x) dx$$

Proof. Put $g(x) := \sup_{\phi \in \mathcal{F}} \phi(x)$; since each ϕ is measurable, so is g. For the integral formula, note that the estimate $\leqslant$ is obvious. Let $s = \sum_{i=1}^{m} a_i \mathbf{1}_{A_i}$ be a step function with $s \leqslant g$; by definition $\mu(A_i) < \infty$ for each i. By weakly regularity for a given $\varepsilon > 0$ we can find compact $K_i \subseteq A_i$ such that

$$\int_X s d\mu < \sum_{i=1}^{m} a_i \mu(K_i) + \varepsilon$$

Let $K = \bigcup_{i=1}^{m} K_i$ and write $s_0 = \sum_{i=1}^{m} a_i \mathbf{1}_{K_i}$. For given $0 < \delta < 1$ one has $(1-\delta)s_0(x) < g(x)$ for all $x \in K$. In particular, for each $x \in K$ we can find $\phi_x \in \mathcal{F}$ such that $(1-\delta)s_0(x) < \phi_x(x)$. Consider the open sets

$$U_x := \{y \in X \mid (1-\delta)s_0(y) < \phi_x(y)\}$$

These form an open cover of K, so by compactness we can find $x_1, \ldots, x_n \in K$ such that $K \subseteq U_{x_1} \cup \cdots \cup U_{x_n}$. By assumption there exists $\phi \in \mathcal{F}$ such that $\phi \geqslant \phi_{x_1}, \ldots, \phi_{x_n}$. Then $\phi > (1-\delta)s_0$, so that

$$\int_X s d\mu < \int_X s_0 d\mu + \varepsilon < \frac{1}{1-\delta} \int_X \phi d\mu + \varepsilon$$

Varying $\phi \in \mathcal{F}$ first, then $s \leqslant g$, we obtain

$$\int_X g d\mu \leqslant \frac{1}{1-\delta} \sup_{\phi \in \mathcal{F}} \int_X \phi d\mu + \varepsilon$$

Letting $\delta, \varepsilon \to 0^+$ concludes the proof. $\square$

Theorem U.24 (Fubini's Theorem for Radon measures)**.** Let μ and ν be outer Radon measures on the Borel sets of the LCH spaces X and Y, respectively. Then there exists a unique outer Radon measure $\mu \otimes \nu$ on $X \times Y$ such that

(a) If $f : X \times Y \to \mathbb{C}$ is $\mu \otimes \nu$-integrable, then the partial integrals $\int_X f(x,y) d\mu(x)$ and $\int_Y f(x,y) d\nu(y)$ define integrable functions such that Fubini's formula holds:

$$\int_{X \times Y} f(x,y) d(\mu \otimes \nu)(x,y) = \int_X \int_Y f(x,y) d\nu(y) d\mu(x) = \int_Y \int_X f(x,y) d\mu(x) d\nu(y)$$

(b) If $f : X \times Y \to \mathbb{C}$ is measurable such that $A = \{(x, y) \in X \times Y \mid f(x, y) \neq 0\}$ is σ-finite, then if one of the iterated integrals

$$\int_X \int_Y |f(x, y)| d\nu(y) d\mu(x) \text{ or } \int_Y \int_X |f(x, y)| d\mu(x) d\nu(y)$$

is finite, then f is integrable and the Fubini's formula holds.

Proof. The uniqueness follows from the Rieze's Representation Theorem, since the Fubini's formula determines the values of the integral on $C_c(X \times Y)$.

For the existence, observe that on each compact set $K = K_1 \times K_2$ there is a unique product measure $(\mu \otimes \nu)_K$ on K such that integration with respect to $(\mu \otimes \nu)_K$ is given by Fubini's formula, by the classical Fubini's. Since $(\mu \otimes \nu)_L$ restricts to $(\mu \otimes \nu)_K$ whenever $K \subseteq L$ with $L = L_1 \times L_2$ compact, these measures $(\mu \otimes \nu)_K$ determines a well-defined positive functional on $C_c(X \times Y)$. Let $\mu \otimes \nu$ denote the resulting outer Radon measure from the Riesz's Representation Theorem.

To show (a), a standard argument reduces us to the case $f = \mathbf{1}_A$ for some measurable $A \subseteq X \times Y$ with finite measure. If $A = U$ is open, a repeated use of Lemma U.23 shows

$$\begin{aligned}
\int_X \int_Y \mathbf{1}_U(x, y) dy dx &= \int_X \int_Y \sup_{0 \leqslant \phi \leqslant \mathbf{1}_U} \phi(x, y) dy dx = \sup_{0 \leqslant \phi \leqslant \mathbf{1}_U} \int_X \int_Y \phi(x, y) dy dx \\
&= \sup_{0 \leqslant \phi \leqslant \mathbf{1}_U} \int_{X \times Y} \phi(x, y) d(\mu \otimes \nu)(x, y) \\
&= \int_{X \times Y} \sup_{0 \leqslant \phi \leqslant \mathbf{1}_U} \phi(x, y) d(\mu \otimes \nu)(x, y) \\
&= \int_{X \times Y} \mathbf{1}_U(x, y) d(\mu \otimes \nu)(x, y)
\end{aligned}$$

If $A = K$ is compact, let V be a relatively compact open neighborhood of K; then $\mathbf{1}_K = \mathbf{1}_V - \mathbf{1}_{V \setminus K}$, so (a) holds in this case as well. It follows from MCT that (a) holds for A being σ-compact. But by weakly inner regularity a measurable set with finite measure is a disjoint union of σ-compact set and a null set, it remains to consider the case $A = N$ being null. Let $\varepsilon > 0$ and U an open set containing N with measure less than ε. Then

$$\int_X \int_Y \mathbf{1}_N(x, y) dy dx \leqslant \int_X \int_Y \mathbf{1}_U(x, y) dy dx = \int_{X \times Y} \mathbf{1}_U < \varepsilon$$

Letting $\varepsilon \to 0$ gives (a).

For (b), it suffices to show $|f|$ is integrable. Write $A = \bigcup_{n=1}^{\infty} A_n$ with $(A_n)_n$ increasing and each A_n being of finite measure. Define $f_n : X \times Y \to \mathbb{C}$ by $f_n = \min\{|f| \mathbf{1}_{A_n}, n\}$. Then $(f_n)_n$ is an increasing sequence of integral functions converging pointwise to $|f|$. By (a), we have

$$\int_{X \times Y} f_n(x, y) d(\mu \otimes \nu)(x, y) = \int_Y \int_X f_n(x, y) dx dy \leqslant \int_Y \int_X |f(x, y)| dx dy$$

for all $n \in \mathbb{N}$. Now (b) follows from MCT. $\square$

U.4 L^p-space and the Riesz-Fischer Theorem

Definition. Let $(X, \mathcal{A}, \mu)$ be a measure space.

1. For $1 \leqslant p < \infty$, let $\mathcal{L}^p(X)$ be the set of all measurable functions $f : X \to \mathbb{C}$ such that

$$\|f\|_p := \left(\int_X |f|^p d\mu\right)^{\frac{1}{p}} < \infty$$

2. Let $\mathcal{L}^\infty(X)$ be the set of measurable functions $f : X \to \mathbb{C}$ such that f is bounded outside a measure zero set. For such f, put

$$\begin{aligned}\|f\|_\infty &= \inf\{c > 0 \mid \mu\{|f(x)| > c\} = 0\} \\ &= \inf_{\mu(N)=0} \sup_{x \notin N} |f(x)|\end{aligned}$$

In other words, $\mathcal{L}^\infty(X)$ consists of measurable functions $f : X \to \mathbb{C}$ with finite $\|f\|_\infty < \infty$.

- We show two definitions for $\|\cdot\|_\infty$ coincide. Let $c > 0$ be such that $N_c := \{x \in X \mid |f(x)| > c\}$ is null. Then for $x \notin N_c$, we have $|f(x)| \leqslant c$, so that

$$\inf_{\mu(N)=0} \sup_{x \notin N} |f(x)| \leqslant c.$$

On the other hand, let N be a null set such that f is bounded outside N and put $c_N := \sup_{x \notin N} |f(x)|$. Then $\{x \in X \mid |f(x)| > c_N\} \subseteq N$ is null, and

$$\inf\{c > 0 \mid \mu\{|f(x)| > c\} = 0\} \leqslant c_N$$

- Let $f \in \mathcal{L}^\infty(X)$. The set $\{x \in X \mid |f(x)| \geqslant \|f\|_\infty\}$ is μ-null, since

$$\{x \in X \mid |f(x)| > \|f\|_\infty\} = \bigcap_{n \geqslant 1} \left\{x \in X \mid |f(x)| > \|f\| + \frac{1}{n}\right\}$$

and each set involved is null by definition.

Proposition U.25 (Hölder's inequality)**.** Let $1 \leqslant p_1, \ldots, p_n \leqslant \infty$ be such that $\sum_{i=1}^n \frac{1}{p_i} = 1$ and $f_i \in \mathcal{L}^{p_i}$, $i = 1, \ldots, n$. Then $\prod_{i=1}^n f_i \in \mathcal{L}^1$ and

$$\left\|\prod_{i=1}^n f_i\right\|_1 \leqslant \prod_{i=1}^n \|f_i\|_{p_i}$$

Proof. First assume $1 \leqslant p_1, \ldots, p_n < \infty$. By normalizing we may assume $\|f_i\|_{p_i} = 1$; then we must show $\left\|\prod_{i=1}^n f_i\right\| \leqslant 1$. To this end, we invoke the Jensen's inequality: for all $a_i > 0$, one has

$$\sum_{i=1}^n \frac{1}{p_i} \log a_i \leqslant \log \sum_{i=1}^n \frac{1}{p_i} a_i$$

Apply this inequality with $a_i = |f_i|^{p_i}$; we obtain

$$\log |f_1 \cdots f_n| \leqslant \log \left(\frac{|f_1|^{p_1}}{p_1} + \cdots + \frac{|f_n|^{p_n}}{p_n} \right)$$

or $|f_1 \cdots f_n| \leqslant \dfrac{|f_1|^{p_1}}{p_1} + \cdots + \dfrac{|f_n|^{p_n}}{p_n}$. This shows $f_1 \cdots f_n$ is integrable. Integrating both sides we see

$$\|f_1 \cdots f_n\|_1 = \int_X |f_1 \cdots f_n| d\mu \leqslant \int_X \left(\frac{|f_1|^{p_1}}{p_1} + \cdots + \frac{|f_n|^{p_n}}{p_n} \right) d\mu = \frac{1}{p_1} + \cdots + \frac{1}{p_n} = 1$$

Let us deal with the case $p_1 = \cdots = p_s = \infty$. Then $p_{s+1}^{-1} + \cdots + p_n^{-1} = 1$ and the result above shows $\|f_{s+1} \cdots f_n\|_1 \leqslant \|f_{s+1}\|_{p_{s+1}} \cdots \|f_n\|_{p_n}$. By induction we are reduced to the situation that if $\|f\|_1 < \infty$ and $\|g\|_\infty < \infty$, then $\|fg\|_1 < \infty$. This is trivial, for $|fg| \leqslant |f| \, \|g\|_\infty$ a.e., and thus

$$\int_X |fg| d\mu \leqslant \int_X |f| d\mu \, \|g\|_\infty = \|f\|_1 \, \|g\|_\infty$$

□

Proposition U.26 (Minkowski's inequality)**.** Let $1 \leqslant p \leqslant \infty$. Then for $f, g \in \mathcal{L}^p$, we have $f + g \in \mathcal{L}^p$ with

$$\|f + g\|_p \leqslant \|f\|_p + \|g\|_p$$

Thus $\|\cdot\|_p$ is a semi-norm on $\mathcal{L}^p$ for every $1 \leqslant p \leqslant \infty$.

Let $1 \leqslant p \leqslant \infty$. Since $\|\cdot\|_p$ is a semi-norm in $\mathcal{L}^p$, it follows that the zero set $N_p := \|\cdot\|_p^{-1}(0)$ of the semi-norm forms a $\mathbb{C}$-vector subspace of the $\mathcal{L}^p$. Define

$$L^p := \mathcal{L}^p / N_p$$

Note that $\|f\|_p = 0$ if and only if f is a **null function**, i.e., $f|_{X \setminus N} \equiv 0$ for some set N of measure zero. Then $\|\cdot\|_p$ is a norm on L^p. We ask whether L^p is a Banach space.

Theorem U.27 (Riesz-Fischer)**.** Let $1 \leqslant p \leqslant \infty$ and suppose (f_n) is a sequence in $\mathcal{L}^p(X)$ that is Cauchy with respect to $\|\cdot\|_p$. Then there exists $f \in \mathcal{L}^p(X)$ and a subsequence $(f_{n_k})_k$ of (f_n) such that $f_{n_k}(x) \to f(x)$ for every x outside a set of measure zero.

Proof. We distinguish the cases $p = \infty$ and $1 \leqslant p < \infty$. First consider the easier case, namely when $p = \infty$. Suppose $(f_n)_n$ is a Cauchy sequence in $\mathcal{L}^\infty$. Put

$$A_n = \{x \in X \mid |f_n(x)| > \|f_n\|_\infty\} \qquad (n \in \mathbb{N})$$
$$B_{n,m} = \{x \in X \mid |f_n(x) - f_m(x)| > \|f_n - f_m\|_\infty\} \qquad (n, m \in \mathbb{N})$$

All A_n and $B_{n,m}$ are null, so if we put E to be their (countable) union, then E is null, and on the complement of E, the sequence $(f_n)_n$ converges uniformly to a bounded function f on $X \backslash E$. Extend f across E by zero; then $f \in \mathcal{L}^\infty$ and

$$\|f_n - f\|_\infty \leqslant \sup_{x \notin E} |f(x) - f_n(x)| \to 0 \text{ as } n \to \infty.$$

□

Lemma U.28.

1. A Hilbert space is separable if and only if it has a countable orthonormal basis.

2. If X is a second countable Hausdorff space with a finite Radon measure μ, then the Hilbert space $L^2(X, \mu)$ is separable.

Proof.

1.

2. Let C be a countable basis of X. Any element in C has finite measure. By adding every finite intersection of sets in C to itself, we may assume C is closed under finite intersection and C is still countable. Consider the subspace H spanned by all $\mathbf{1}_U$ with $U \in C$. By exclusion-inclusion principal, H contains all characteristic functions of finite unions of sets in C.

   Let U be any open subset of X and say $U = \bigcup_{n \in \mathbb{N}} U_n$ for $U_n \in C$. Then $\mathbf{1}_{U_1 \cup \cdots \cup U_n} \to \mathbf{1}_U$ in L^2. It follows that $\overline{H} \subseteq L^2(X, \mu)$ contains all characteristic function of open sets. Since μ is Radon, $\overline{H}$ contains all simple functions. Since the space of simple functions are dense in $L^2(X, \mu)$, it follows that $\overline{H} = L^2(X, \mu)$.

□

Let μ be a positive measure on X, and suppose $1 \leqslant p \leqslant \infty$ and let q be the exponent conjugate of p, i.e., $\dfrac{1}{p} + \dfrac{1}{q} = 1$. By Hölder's inequality, if $g \in L^q(\mu)$, then the map $\Phi_g : L^p(\mu) \to \mathbb{C}$ defined by

$$\Phi_g(f) := \int_X fg d\mu$$

is a bounded linear functional with norm at most $\|g\|_q$. In sum, the map

$$\begin{aligned} \Phi : L^q(\mu) &\longrightarrow L^p(\mu)^\vee \\ g &\longmapsto \Phi_g \end{aligned}$$

is a norm-decreasing linear functional.

Proposition U.29. For $1 \leqslant p < \infty$, the map Φ is norm-preserving.

Proof. □

U.5 Radon-Nikodym Theorem

In this subsection we fix a measurable space $(X, \mathcal{A})$. By a (unadorned) measure on $(X, \mathcal{A})$ we mean either the positive or the complex measure.

Definition. Let μ be a positive measure on $(X, \mathcal{A})$. A measure λ is **absolutely continuous** with respect to μ if $\lambda(E) = 0$ for $E \in \mathcal{A}$ whenever $\mu(E) = 0$. In this case we write $\lambda \ll \mu$.

Definition.

1. Let λ be a measure on $(X, \mathcal{A})$. If there exists $A \in \mathcal{A}$ such that $\lambda(E) = \lambda(E \cap A)$ for all $E \in \mathcal{A}$, we say λ is **concentrated on A.**

2. Two measures λ_1 and λ_2 on $(X, \mathcal{A})$ are called **mutually singular** if there exists $A \in \mathcal{A}$ such that λ_1 is concentrated on A and λ_2 is concentrated on A^c. In this case we write $\lambda_1 \perp \lambda_2$.

Proposition U.30. Let $\lambda, \lambda_1, \lambda_2$ be measures on $(X, \mathcal{A})$ and μ be a positive measure.

(i) If λ is concentrated on A, so is $|\lambda|$.

(ii) If $\lambda_1 \perp \lambda_2$, then $|\lambda_1| \perp |\lambda_2|$.

(iii) If $\lambda_1 \perp \lambda$ and $\lambda_2 \perp \lambda$, then $\lambda_1 + \lambda_2 \perp \lambda$.

(iv) If $\lambda_1 \ll \mu$ and $\lambda_2 \ll \mu$, then $\lambda_1 + \lambda_2 \ll \mu$.

(v) If λ is signed, then $\lambda \ll \mu$ if and only if $\lambda^\pm \ll \mu$.

(vi) If $\lambda_1 \ll \mu$ and $\lambda_2 \perp \mu$, then $\lambda_1 \perp \lambda_2$.

(vii) If $\lambda \ll \mu$ and $\lambda \perp \mu$, then $\lambda = 0$.

Proof.

(i) For all measurable E, $|\lambda|(E) = \sup\limits_{\{E_n\}} \sum |\lambda(E \cap E_n)| = \sup\limits_{\{E_n\}} \sum |\lambda(E \cap E_n \cap A)| = |\lambda|(E \cap A)$

(ii) This follows from (i).

(iii) Say λ_i is concentrated on A_i. Then $\lambda_1 + \lambda_2$ is concentrated on $A := A_1 \cup A_2$ and λ is concentrated on A^c.

(iv) Obvious.

(v) Let (P, N) be a Hahn decomposition of λ. Let $A \in \mathcal{A}$ be such that $\mu(A) = 0$. Then $\mu(A \cap P) = 0 = \mu(A \cap N)$. If $\lambda \ll \mu$, then $\lambda^+(A) = \lambda(A \cap P) = 0 = \lambda(A \cap N) = \lambda^-(A)$, so $\lambda^\pm \ll \mu$. The converse follows from (iv).

(vi) Say λ_2 is concentrated on A. Then for all measurable $E \subseteq A$, since μ is concentrated on A^c, $\mu(E) = 0$, so that $\lambda_1(E) = 0$ by absolute continuity. Hence λ_1 is concentrated on A^c.

(vii) By (vi) we have $\lambda \perp \lambda$, so $\lambda = 0$ obviously.

□

Proposition U.31. Let μ be a positive measure on $(X, \mathcal{A})$. Let $F = \mathbb{R}$ or $\mathbb{C}$.

1. The subspace $M_{\mu,s} := \{\nu \in M(X, \mathcal{A}, F) \mid \nu \perp \mu\}$ is closed in $M(X, \mathcal{A}, F)$.
2. The subspace $M_{\mu,a} := \{\nu \in M(X, \mathcal{A}, F) \mid \nu \ll \mu\}$ is closed in $M(X, \mathcal{A}, F)$.

Proof.

1. Let $(\nu_n)_n \subseteq M_{\mu,s}$ be a sequence converging to ν. Say ν_n is concentrated on A_n and μ is concentrated on B_n with $A_n \cap B_n = \varnothing$. Put $A = \bigcup_{n=1}^{\infty} A_n$ and $B = \bigcap_{n=1}^{\infty} B_n$. Then μ is clearly concentrated on B. If E is measurable with $E \cap A = \varnothing$. Then in particular, $E \cap A_n = \varnothing$ for each n, so $\nu_n(E) = \nu_n(E \cap A_n) = 0$ for all n, whence $\nu(E) = \lim_{n\to\infty} \nu_n(E) = 0$. This shows ν is concentrated on A, so $\nu \perp \mu$.
2. Let $(\nu_n)_n \subseteq M_{\mu,a}$ be a sequence converging to ν. If $\mu(A) = 0$, then $\nu_n(A) = 0$ for all n, so $\nu(A) = \lim_{n\to\infty} \nu_n(A) = 0$. Thus $\nu \ll \mu$.

□

Theorem U.32. Let μ be a positive σ-finite measure on $(X, \mathcal{A})$, and λ be a positive σ-finite or complex measure. If $\lambda \ll \mu$, then there exists a unique function h which is

- finite nonnegative $\mathcal{A}$-measurable if λ is positive σ-finite, or
- complex μ-integrable if λ is complex

such that

$$\lambda(A) = \int_A h d\mu$$

for all $A \in \mathcal{A}$. The function h is called the **Radon-Nikodym derivative** of λ with respect to μ, and we write $h = \dfrac{d\lambda}{d\mu}$ and $d\lambda = h d\mu$.

Proof. If λ is complex, we can use Jordan decomposition for complex measures to write λ as a sum of positive finite measures which are absolutely continuous with respect to μ. If λ is positive σ-finite, say $X = \bigcup_{n=1}^{\infty} X_n$ with $\lambda(X_n) < \infty$ and $\mu(X_n) < \infty$ for each n, then by restricting to each X_n we can assume λ and μ are both finite. Hence we are reduced to show the following statement. If μ and λ are positive finite measures with $\lambda \ll \mu$, then there exists a real valued μ-integrable function h such that $\lambda(A) = \int_A h d\mu$ for all measurable A.

As said in the previous paragraph, we assume μ and λ are finite positive. Set $\tau = \mu + \lambda$. For every $\phi \in L^2(\tau)$, by Cauchy Schwarz we have

$$\left| \int_X \phi d\mu \right| \leqslant \int_X |\phi| d\mu \leqslant \int_X |\phi| d\tau \leqslant \left(\int_X |\phi|^2 d\tau \right)^{\frac{1}{2}} \tau(X)^{\frac{1}{2}}$$

Hence the map

$$\begin{aligned} L^2(\tau) &\longrightarrow \mathbb{C} \\ \phi &\longmapsto \int_X \phi d\mu \end{aligned}$$

is a bounded linear functional. The same is true for the map $\phi \mapsto \int_X \phi d\lambda$. Since $L^2(\tau)$ is a Hilbert space, by Riesz's Representation theorem there exist unique τ-square-integrable functions f and g such that

$$\int_X \phi d\lambda = \int_X f \phi d\tau, \qquad \int_X \phi d\mu = \int_X g \phi d\tau$$

for all $\phi \in L^2(\tau)$.

Let ϕ be a nonnegative $\mathcal{A}$-measurable function. Let $(\phi_n)_n$ be an increasing sequence of positive step functions such that $\phi_n \to \phi$ pointwise. By MCT applied to the identity

$$\int_X \phi_n d\lambda = \int_X f \phi_n d\tau$$

we see $\phi \in L^1(\lambda)$ if and only if $f\phi \in L^1(\tau)$. Similarly, $\phi \in L^1(\mu)$ if and only if $g\phi \in L^1(\tau)$.

Let $N = \{x \in X \mid g(x) = 0\}$. Then $\mu(N) = \int_X g \mathbf{1}_N d\tau = 0$, so $\lambda(N) = 0$ as well by absolute continuity $\lambda \ll \mu$. Define

$$h(x) = \begin{cases} \dfrac{f(x)}{g(x)} & \text{, if } x \in X \backslash N \\ 0 & \text{, if } x \in N \end{cases}$$

For every measurable $\phi : X \to \mathbb{C}$, $\phi \in L^1(\lambda)$ if and only if $f\phi = gh\phi \in L^1(\tau)$ (since $\lambda(N) = 0$), if and only if $h\phi \in L^1(\mu)$ (since $\mu(N) = 0$). Hence

$$\int_X \phi d\lambda = \int_X f\phi d\tau = \int_X gh\phi d\tau = \int_X h\phi d\mu$$

for all $\phi \in L^1(\lambda)$. In particular, by taking $\phi = g \in L^2(\tau) \subseteq L^1(\lambda)$ we obtain $h \in L^1(\mu)$.

It remains to show h is unique (up to a null function) in each case. Suppose $g, h : X \to [0, \infty)$ are $\mathcal{A}$-measurable and λ is finite such that

$$\lambda(A) = \int_A g d\mu = \int_A h d\mu$$

Then $\int_A (g - h) d\mu = 0$ for all measurable A. By taking $A = \{g \geqslant h\}$ and $A = \{g \leqslant h\}$, we see $(g - h)^+ = 0$ and $(g-h)^- = 0$ μ-almost everywhere, whence $g = h$ μ-almost everywhere. If λ is σ-finite, write $X = \bigcup_{n=1}^{\infty} X_n$ with $\lambda(X_n) < \infty$, then the preceding arguments show $g = h$ μ-almost everywhere on each X_n, whence on X. The case λ being complex can be dealt with in a similar fashion. □

Proposition U.33. Let μ be a complex measure on $(X, \mathcal{A})$. Then there exists a measurable function h such that $|h(x)| = 1$ for all $x \in X$ such that $d\mu = hd|\mu|$.

Proof. Since $\mu \ll |\mu|$, by Radon-Nikodym there exists $h \in L^1(|\mu|)$ such that $d\mu = hd|\mu|$. For each $r > 0$, put $A_r = \{x \in X \mid |h(x)| < r\}$, and let $\{E_n\}$ be a measurable partition of A_r. Then

$$\sum_n |\mu(E_n)| = \sum_n \left| \int_{E_n} hd|\mu| \right| \leqslant \sum_n r|\mu|(E_n) = r|\mu|(A_r)$$

so that $|\mu|(A_r) \leqslant r|\mu|(A_r)$. If $r < 1$, this forces $|\mu|(A_r) = 0$. Thus $|h| \geqslant 1$ a.e.

On the other hand, if $|\mu|(E) > 0$, we have

$$\left| \frac{1}{|\mu|(E)} \int_E hd|\mu| \right| = \frac{|\mu(E)|}{|\mu|(E)} \leqslant 1$$

Lemma U.34. Let $(X, \mathcal{A}, \mu)$ be a positive finite measure space and $f \in L^1(\mu)$. If $S \subseteq \mathbb{C}$ is a closed subset such that the average

$$A_E(f) = \frac{1}{\mu(E)} \int_E f d\mu$$

lie in S for every $E \in \mathcal{A}$ with $\mu(E) > 0$. Then $f(x) \in S$ for almost all $x \in X$.

Proof. The complement of S can be covered by countably closed ball, so it suffices to show $\mu(f^{-1}(D)) = 0$ for all closed ball $D \subseteq S^c$. If $\mu(f^{-1}(D)) > 0$,

$$\left| A_{f^{-1}(D)}(f) - \alpha \right| = \frac{1}{\mu(f^{-1}(D))} \left| \int_{f^{-1}(D)} (f - \alpha) d\mu \right| \leqslant \frac{1}{\mu(f^{-1}(D))} \int_{f^{-1}(D)} |f - \alpha| d\mu \leqslant r$$

where α is the center of D and r the radius of D. This is impossible for $A_{f^{-1}(D)}(f) \in S$ and $D \subseteq S^c$. □

By this lemma, we deduce that $|h| \leqslant 1$ a.e.

In sum, we have proved that $|h| = 1$ a.e. The proposition follows once we redefine h on $\{x \in X \mid |h(x)| \neq 1\}$ so that $h(x) = 1$ throughout the whole X. $\square$

Corollary U.34.1. Let μ be a positive measure on $(X, \mathcal{A})$ and $f \in L^1(\mu)$. Consider the finite measure μ_f given by

$$\mu_f(A) = \int_A f d\mu$$

Then $|\mu_f| = \mu_{|f|}$.

Proof. By the previous proposition, there exists a measurable function h of absolute value 1 such that $d\mu_f = hd|\mu_f|$. Then

$$hd|\mu_f| = d\mu_f = fd\mu$$

so that $d|\mu_f| = \overline{h}fd\mu$, where $\bar{\cdot}$ is complex conjugation. Since $|\mu_f| \geqslant 0$ and $\mu \geqslant 0$, it follows that $\overline{h}f \geqslant 0$ μ-almost everywhere, whence $\overline{h}f = |f|$ μ-almost everywhere, i.e., $|\mu_f| = |f|d\mu = \mu_{|f|}$. $\square$

Theorem U.35. Let μ be a positive measure. Let λ be either a real, complex or σ-finite positive measure. Then there are unique real, complex, or σ-finite measures λ_a and λ_s on $(X, \mathcal{A})$ such that

- $\lambda_a \ll \mu$ and $\lambda_s \perp \mu$, and
- $\lambda = \lambda_a + \lambda_s$.

The expression $\lambda = \lambda_a + \lambda_s$ is called the **Lebesgue decomposition** of λ. λ_a is called the **absolutely continuous part** of λ, and λ_s is called the **singular part** of λ.

Proof. First assume λ is positive finite. Define

$$\mathcal{N}_\mu := \{B \in \mathcal{A} \mid \mu(B) = 0\}$$

and let $\{B_n\}_n \subseteq \mathcal{N}_\mu$ be a sequence such that $\lambda(B_n) \to \sup\limits_{B \in \mathcal{N}_\mu} \lambda(B)$ as $n \to \infty$. Let $N = \bigcup\limits_{n \geqslant 1} B_n$ and define the measures λ_a, λ_s by the formulas

$$\lambda_a(A) = \lambda(A \cap N^c), \qquad \lambda_s = \lambda(A \cap N)$$

for all $A \in \mathcal{A}$. Then $\lambda = \lambda_a + \lambda_s$. Since $\mu(N) = 0$, we have $\lambda_s \perp \mu$. Since $\lambda(N) = \sup\limits_{B \in \mathcal{N}_\mu} \lambda(B)$, if $B \subseteq N^c$ is $\mathcal{A}$-measurable with $\mu(B) = 0$, then $\lambda(B) = 0$, for otherwise $N \cup B \in N_\mu$ and $\lambda(N \cup B) > \lambda(N)$. Hence $\lambda_a \ll \mu$.

Suppose λ is real or complex. Apply the preceding construction to the positive finite measure $|\lambda|$. Then we have a μ-null set N such that the Lebesgue decomposition $|\lambda| = |\lambda|_a + |\lambda|_s$ is given by

$$|\lambda|_a(A) := |\lambda|(A \cap N^c), \qquad |\lambda|_s(A) := |\lambda|(A \cap N)$$

Then the measures λ_a, λ_s defined by

$$\lambda_a(A) = \lambda(A \cap N^c), \qquad \lambda_s = \lambda(A \cap N)$$

clearly give the Lebesgue decomposition of λ.

Assume λ is σ-finite positive and let $\{X_n\}$ be a measurable partition of X such that $\lambda(X_n) < \infty$ for each n. Consider the restriction of λ and μ to each X_n; then we obtain a sequence $\{N_n\}$ of μ-null sets with $N_n \subseteq X_n$. Let $N = \bigcup_{n \geqslant 1} N_n$. Then the measures λ_a, λ_s defined by

$$\lambda_a(A) = \lambda(A \cap N^c), \qquad \lambda_s = \lambda(A \cap N)$$

give the Lebesgue decomposition of λ.

It remains to show the uniqueness. Let $\lambda = \lambda_a + \lambda_s = \lambda'_a + \lambda'_s$ be Lebesgue decompositions of λ. If λ is finite, then

$$\lambda_a - \lambda'_a = \lambda'_s - \lambda_s.$$

Since $\lambda_a - \lambda'_a \ll \mu$ and $\lambda'_s - \lambda_s \perp \mu$, it follows that $\lambda_a - \lambda'_a = \lambda'_s - \lambda_s = 0$, as wanted. The case λ being positive σ-finite is proved as usual by choosing a measurable partition of X with each piece finite under λ. $\square$

U.6 Bochner Integral

Let V be a Banach space. For a function $f : X \to V$ we aim to define an integral $\int_X f d\mu \in V$ such that for every continuous linear functional α on V we have

$$\alpha\left(\int_X f d\mu\right) = \int_X \alpha(f) d\mu$$

Let $(V, \|\cdot\|)$ be a Banach space and $(X, \mathcal{A}, \mu)$ a (positive) measure space. A **simple function** is a function $s : X \to V$ that can be written as

$$s = \sum_{j=1}^{n} \mathbf{1}_{A_j} b_j \tag{$\spadesuit$}$$

for some pairwise disjoint measurable sets $A_j \in \mathcal{A}$ of finite measure and some $b_j \in V$. Define

$$\int_X s d\mu := \sum_{j=1}^{n} \mu(A_j) b_j \in V$$

This is clearly independent of the expression ($\spadesuit$), and it satisfies

- $\left\| \int_X s d\mu \right\| \leqslant \int_X \|s\| \, d\mu$;

- for every linear functional $\alpha : V \to \mathbb{C}$, one has

$$\alpha\left(\int_X s d\mu\right) = \int_X \alpha(s) d\mu$$

Equip V with the Borel σ-algebra. A measurable function $f : X \to V$ is called **Bochner integrable** if there exists a sequence s_n of simple functions such that

$$\lim_{n\to\infty} \int_X \|f - s_n\| \, d\mu = 0$$

We call such (s_n) an **approximating sequence** of f.

Proposition U.36.

(a) If f is Bochner integrable and (s_n) is an approximating sequence, then the sequence of vectors $\int_X s_n d\mu$ converges. Its limit does not depend on the choice of the approximating sequence.

Define the **integral** of f to be this limit

$$\int_X f d\mu := \lim_{n\to\infty} \int_X s_n d\mu$$

(b) For every Bochner integrable function f one has

$$\left\| \int_X f d\mu \right\| \leqslant \int_X \|f\| \, d\mu < \infty$$

(c) Let f be Bochner integrable. For every continuous linear operator $T : V \to W$ to a Banach space W one has

$$T\left(\int_X f d\mu\right) = \int_X T(f) d\mu$$

(d) If $V = \mathbb{C}$, then the Bochner integral coincides with the usual integral.

Proof.

(a)

$$\begin{aligned}\left\| \int_X s_n d\mu - \int_X s_m d\mu \right\| &\leqslant \int_X \|s_n - s_m\| \, d\mu = \int_X \|s_n - f + f - s_m\| \, d\mu \\ &\leqslant \int_X \|s_n - f\| \, d\mu + \int_X \|f - s_m\| \, d\mu\end{aligned}$$

so this value tends to 0 as $n, m \to \infty$. Let (r_n) be another approximating sequence of f. Then

$$\begin{aligned}\left\| \int_X s_n d\mu - \int_X r_n d\mu \right\| &\leqslant \int_X \|s_n - r_n\| \, d\mu = \int_X \|s_n - f + f - r_n\| \, d\mu \\ &\leqslant \int_X \|s_n - f\| \, d\mu + \int_X \|f - r_n\| \, d\mu \to 0 \text{ as } n \to 0\end{aligned}$$

(b) Let (s_n) be an approximating sequence of f. Then $\|s_n\| \to \|f\|$ in $L^1(X)$, for

$$0 \leqslant \lim_{n\to\infty} \int_X |\,\|f\| - \|s_n\|\,| \leqslant \lim_{n\to\infty} \int_X \|f - s_n\| = 0$$

so that

$$\int_X \|f\| = \int_X \|f - s_n + s_n\| \leqslant \int_X \|f - s_n\| + \int_X \|s_n\| < \infty$$

Then

$$\left\| \int_X f \right\| = \lim_{n\to\infty} \left\| \int_X s_n \right\| \leqslant \lim_{n\to\infty} \int_X \|s_n\| = \int_X \|f\| < \infty$$

(c) By continuity, we have

$$T\left(\int_X f\right) = T\left(\lim_{n\to\infty} \int_X s_n\right) = \lim_{n\to\infty} \int_X T(s_n)$$

Since T is continuous, we can find $C > 0$ such that $\|T(v)\| \leqslant C\,\|v\|$ for all $v \in V$, and so $\|T(f)\| \leqslant C\,\|f\|$. In particular, $T(f)$ is integrable. Estimate

$$\left\| \int_X T(f) - \int_X T(s_n) \right\| \leqslant \int_X \|T(f) - T(s_n)\| = \int_X \|T(f - s_n)\| \leqslant C \int_X \|f - s_n\| \to 0$$

(d) It follows from the last estimate with $T = \mathrm{id}$.

□

Definition.

1. A function $f : X \to V$ is called **separable** if there exists a countable subset $C \subseteq V$ such that $f(X) \subseteq \overline{C}$, where $\overline{C}$ is the closure of C in V.

- If V is separable (as topological spaces), then any function is automatically separable in the sense above.

2. A function $f : X \to V$ is **essentially separable** if there exists a measurable zero set $N \subseteq X$ such that $f|_{X\setminus N}$ is separable.

Lemma U.37. Let X be a topological space and $f : X \to V$ a continuous function with σ-compact support. Then f is separable.

Proof. Being σ-compact, let $\operatorname{supp} f = \bigcup_{n=1}^{\infty} K_n$ with compact $K_n \subseteq X$. Then $f(X) \subseteq \bigcup_{n=1}^{\infty} f(K_n)$. Since f is continuous, $f(K_n)$ is compact; being a compact metric space, $f(K_n)$ is itself separable. □

Proposition U.38. For a measurable function $f : X \to V$ TFAE:

1. f is Bochner integrable.
2. f is essentially separable and $\displaystyle\int_X \|f\|\, d\mu < \infty$.

Proof. Suppose f is Bochner integrable. Then by Lemma U.36.(b), $\displaystyle\int_X \|f\|\, d\mu < \infty$. To show that f is essentially separable, let (s_n) be an approximating sequence of f. Since each s_n has finite image, the Banach subspace of V generated by the $s_n(X)$, $n \in \mathbb{N}$ is separable. The set $N := f^{-1}(V \backslash E)$ is a countable union $N = \bigcup_{n \geqslant 1} N_n$, where

$$N_n := \left\{ x \in X \mid \|f(x) - e\| \geqslant \frac{1}{n} \text{ for all } e \in E \right\}$$

Since $\displaystyle\lim_{n\to\infty} \int_X \|f - s_n\| = 0$, each N_n has zero measure, and so does N.

Conversely, suppose f is essentially separable and $\displaystyle\int_X \|f\| < \infty$. Up to a measure zero set we may assume f is separable. Let $C = \{c_n\}_{n \geqslant 1} \subseteq V$ be countable such that $f(X) \subseteq \overline{C}$. For each $(n, \delta) \in \mathbb{N} \times \mathbb{R}_{>0}$, put

$$A_n^\delta := \{x \in X \mid \|f(x)\| \geqslant \delta,\ \|f(x) - c_n\| < \delta\}$$

Since f is measurable, each A_n^δ is measurable. We make some modification:

$$D_n^\delta := A_n^\delta \backslash \bigcup_{k<n} A_k^\delta$$

Then for a fixed $\delta > 0$, the D_n^δ are pairwise disjoint, and

$$\bigcup_{n \geqslant 1} A_n^\delta = \bigsqcup D_n^\delta = f^{-1}(f(X) \backslash B_\delta(0))$$

for C is dense in $f(X)$. Since $\|f\|$ is integrable, $f^{-1}(f(X) \backslash B_\delta(0))$ has finite measure. Let

$$s_n = \sum_{j=1}^{n} \mathbf{1}_{D_j^{\frac{1}{n}}} c_j$$

Then s_n is a simple function. We content that $s_n \to f$ pointwise. Let $x \in X$.

- $f(x) = 0$. Then $s_n(x) = 0$ for every n. Great.
- $f(x) \neq 0$. Then $\|f(x)\| \geqslant \dfrac{1}{n}$ for some $n \in \mathbb{N}$, and for $m \geqslant n$ one has

  $$x \in \bigcup_{\nu \geqslant 1} D_\nu^{\frac{1}{m}}$$

  so that for each $m \geqslant n$ there exists a unique, by disjointness, ν_0 with $D_{\nu_0}^{\frac{1}{m}}$, and hence $s_m(x) = c_{\nu_0}$, and $\|f(x) - c_{\nu_0}\| < \dfrac{1}{m}$.

This shows $s_n \to f$. Also, by construction we have $\|s_n\| \leqslant 2\|f\|$. Since

- $\|f - s_n\| \to 0$ pointwise, and
- $\|f - s_n\| \leqslant \|f\| + \|s_n\| \leqslant 3\|f\|$,

by DCT we obtain $\int_X \|f - s_n\| \, d\mu \to 0$ as wanted. □

Corollary U.38.1. Let X be an LCH space and μ a Radon measure. Then every continuous function $f : X \to V$ with compact support is Bochner integrable.

Proof. Since $\|f\| \in C_c(X)$, it follows from Proposition and Lemma above that f is integrable.

□

Lemma U.39. Let G be an LCH group. If $f \in C_c(G)$ and $g \in L^1(G)$, then the Bochner integral

$$\int_G f(x) L_x g \, dx$$

exists in the Banach algebra $L^1(G)$, and equals the convolution product $f * g$.

Proof. Consider the function

$$\begin{aligned} \phi : G &\longrightarrow L^1(G) \\ x &\longmapsto f(x) L_x g \end{aligned}$$

This is continuous by Lemma 1.36, and since it has compact support (as f does), by Corollary U.38.1 it is Bochner integrable. Thus $\int_G \phi(x) \, dx \in L^1(G)$, and to see it coincides with $f * g$, it suffices to show

$$h * \int_G \phi(x) L_x g \, dx = h * f * g$$

for all $h \in C_c(G)$ by Lemma 1.38.

Let $h \in C_c(G)$. For $\phi \in L^1(G)$ and $y \in G$, the integral $h * \phi(y)$ exists, and

$$\begin{aligned} |h * \phi(y)| &\leqslant \int_G |h(z)| \phi(z^{-1} y) | dz \\ &\overset{1.11.4}{=} \int_G \Delta(z^{-1}) |h(yz^{-1})| |\phi(z)| dz \\ &\leqslant C \|\phi\|_1 \end{aligned}$$

with $C \geqslant 0$ for the function $z \mapsto \Delta(z^{-1}) |h(yz^{-1})|$ is continuous. This implies the linear functional

$$\begin{aligned} \alpha : L^1(G) &\longrightarrow \mathbb{C} \\ \phi &\longmapsto h * \phi(y) \end{aligned}$$

is continuous. Then for $\phi = \int_G f(x) L_x g \, dx$, it follows

$$\begin{aligned} h * \phi(y) = \alpha(\phi) &= \alpha\left(\int_G f(x) L_x g \, dx\right) \\ &= \int_G f(x)\alpha(L_x g) dx \\ &= \int_G \int_G g(x)h(z)g(x^{-1}z^{-1}y) dz dx \\ &= h * f * g(y) \end{aligned}$$

□

U.6.1 Cauchy's Integral Formula

Definition. Let $\Omega \subseteq \mathbb{C}$ be an open set, $f : \Omega \to V$ be holomorphic in the sense of Definition 2.1.1 and $\gamma : [0,1] \to \Omega$ be C^1. The **path integral** is defined as

$$\int_\gamma f(z) dz := \int_{[0,1]} f(\gamma(t))\gamma'(t) dt$$

Theorem U.40 (Cauchy's Integral Formula)**.** Let $\Omega \subseteq \mathbb{C}$ be an open set and $f : \Omega \to V$ be holomorphic. Suppose $D \subseteq \Omega$ is an open disc with $\overline{D} \subseteq \Omega$, then for every $z \in D$, we have

$$f(z) = \frac{1}{2\pi i} \int_{\partial D} \frac{f(\xi)}{\xi - z} d\xi$$

Proof. It follows from Hahn-Banach theorem, Proposition U.36.(c) and the usual Cauchy's Integral Formula in complex analysis. □

Corollary U.40.1. Let the situation be as in the theorem, and let D be an open disc around the origin such that $\overline{D} \subseteq \Omega$. Then there exist $v_n \in V$ such that

$$f(z) = \sum_{n=0}^{\infty} v_n z^n$$

holds for every $z \in D$, and the sum converges uniformly on every closed subset of D.

Proof. If $z \in D$ and $\xi \in \partial D$, then $\left|\frac{z}{\xi}\right| < 1$, which means that the geometric series

$$\sum_{n=0}^{\infty} \left(\frac{z}{\xi}\right)^n = \frac{1}{1 - z/\xi}$$

converges uniformly for (z, ξ) in $D \times \partial D$. Applying the Cauchy's formula, we obtain

$$\begin{aligned} f(z) &= \frac{1}{2\pi i}\int_{\partial D} \frac{f(\xi)}{\xi - z} d\xi = \frac{1}{2\pi i}\int_{\partial D} \frac{1}{\xi}\frac{f(\xi)}{1 - z/\xi} d\xi \\ &= \frac{1}{2\pi i}\int_{\partial D} \frac{f(\xi)}{\xi} \sum_{n=0}^{\infty} \left(\frac{z}{\xi}\right)^n d\xi \\ &= \frac{1}{2\pi i}\sum_{n=0}^{\infty} z^n \int_{\partial D} \frac{f(\xi)}{\xi^{n+1}} d\xi \end{aligned}$$

The last step is justifies by the uniform convergence. $\square$

Appendix V: Functional Analysis

V.1 Banach Space Basics

Definition.

1. A **(complex) topological vector space**, or **TVS** for short, is a complex vector space V together with a topology such that $(V, +)$ is a topological group and the scalar multiplication $\mathbb{C} \times V \to V$ is continuous.

2. For a TVS V, let $V^* := \mathrm{Hom}_{\mathbb{C}}(V, \mathbb{C})$ be the **algebraic dual** of V, and

$$V^\vee := \{\alpha \in V^* \mid T \text{ is continuous}\}$$

be the **continuous dual** of V.

Theorem V.1 (Hahn-Banach's)**.** Let M be s subspace of a Banach space V and let $\alpha : M \to \mathbb{C}$ be linear with $|\alpha(x)| \leqslant \|x\|$ for all $x \in M$. Then α extends to a linear function $V \to \mathbb{C}$ such that $|\alpha(x)| \leqslant \|x\|$ holds for all $x \in V$.

Corollary V.1.1. Let $E \subseteq V$ be a Banach subspace of V.

1. The restriction map $V^\vee \to E^\vee$ is surjective, and for $\alpha \in E^\vee$, its fibre contains $\alpha' \in V^\vee$ with $\|\alpha'\| = \|\alpha\|$.

2. $V^\vee$ separates points. In other words, if $v, w \in V$ such that $\alpha(v) = \alpha(w)$ for all $\alpha \in V^\vee$, then $v = w$.

Theorem V.2 (Uniform Boundedness Principle)**.** Let V be a Banach space and E a normed linear space. Let $\{\Lambda_\alpha\}_{\alpha \in A}$ be a collection of bounded linear maps from V to E. Then either there exists $M < \infty$ such that $\|\Lambda_\alpha\| \leqslant M$ for all $\alpha \in A$, or $\sup\limits_{\alpha \in A} \|\Lambda_\alpha v\| = \infty$ for all v in some dense G_δ set in X.

Theorem V.3 (Open Mapping Theorem)**.** Let $T : E \to F$ be a continuous surjective linear map between Banach spaces. Then T is an open map.

Theorem V.4 (Closed Graph Theorem)**.** Suppose that $T : V \to W$ is a linear map between Banach spaces such that the graph

$$\Gamma_T := \{(v, Tv) \mid v \in V\} \subseteq V \times W$$

is a closed in the product topology. Then T is bounded.

Lemma V.5. Let V be a Banach space and W a closed subspace. Then the quotient space V/W is also a Banach space equipped with the **quotient norm**

$$\|a + W\| := \inf\{\|a + w\| \mid w \in W\}$$

Proof. We must show it is V/W is complete with respect to the quotient norm. Let $(a_n+W)_n$ be a Cauchy sequence. Passing to a subsequence we may assume $\sum_{n=1}^{\infty} \|a_{n+1} - a_n + w_n\| < \infty$ for some $w_n \in W$. Since V is complete, $a := \sum_{n=1}^{\infty} (a_{n+1} - a_n + w_n)$ defines an element of V. Since the quotient map $v \mapsto v + W$ is norm-decreasing, we see $a_n + W \to a + W$ with respect to the quotient norm. □

Lemma V.6. Suppose F be a complete subspace of a normed vector space E. If F has finite codimension in E, then E is complete.

Proof. It at once reduces to the case E/F is one dimensional. Fix an $x \in E - F$. Suppose $(w_n)_n$ is Cauchy in E. Since the quotient map $E \to E/F$ is norm-decreasing, if we write $w_n = v_n + a_n x$ with $v_n \in F$ and $a_n \in \mathbb{C}$ for each n, then $(a_n x + F)_n$ is Cauchy in E/F. Recall that any finite dimensional normed space over $\mathbb{C}$ is complete. Thus $a_n x + F \to ax + F$ for some $a \in \mathbb{C}$ and by definition we can find $u_n \in F$ such that $\|a_n x + u_n - a\| \leqslant \|ax - a_n x + F\| + n^{-1}$ for each n. Therefore

$$\begin{aligned}\|(v_n - u_n) - (v_m - u_m)\| &\leqslant \|w_n - w_m\| + \|(a_n x + u_n) - (a_m x + u_m)\| \\ &\leqslant \|w_n - w_m\| + \|ax - a_n x + F\| + n^{-1} + \|ax - a_m x + F\| + m^{-1}\end{aligned}$$

and so $(v_n - u_n)_n$ is Cauchy. Since F is complete, $v_n - u_n \to v \in F$. Now define $w = v + a$.

$$\|w_n - w\| \leqslant \|(v_n - u_n) - v\| + \|(a_n x + u_n) - a\| \leqslant \|(v_n - u_n) - v\| + \|ax - a_n x + F\| + n^{-1}$$

The right hand side $\to 0$ as $n \to \infty$, from which we conclude $w_n \to w$. □

Lemma V.7. Let B be a Banach space and E be a proper closed subspace. For each $\varepsilon > 0$ there exists $x \in B$ with $\|x\| = 1$ such that $\inf_{y \in E} |x - y| > 1 - \varepsilon$.

Proof. If $\varepsilon \geqslant 1$, the assertion is obvious. Assume $0 < \varepsilon < 1$. Pick $x' \in B \backslash E$. Since E is closed, $d := \inf_{y \in E} \|x - y\| > 0$. Pick $y' \in E$ with $d \leqslant \|x' - y'\| \leqslant d + \eta$, where $\eta > 0$ is to be chosen. Put $x = \dfrac{x' - y'}{\|x' - y'\|}$. Then for $y \in E$, we have

$$\|x - y\| = \left\| \frac{x' - (y' + \|x' - y'\|\, y)}{\|x' - y'\|} \right\| \geqslant \frac{d}{\|x' - y'\|} \geqslant \frac{d}{d + \eta} = 1 - \varepsilon$$

once we choose $\eta = \dfrac{d\varepsilon}{1 - \varepsilon} > 0$. □

Proposition V.8. A Banach space is locally compact if and only if it is finite dimensional.

Proof. The if part is obvious. Suppose B is a locally compact Banach space. In particular, the unit ball B_1 is compact. Let $v_1 \in \partial B_1$. For $n \geqslant 2$, use the previous lemma to choose $v_1, \ldots, v_n \in \partial B_1$ inductively so that

$$\operatorname{disc}(v_i; \operatorname{span}\{v_1, \ldots, v_{i-1}\}) > \frac{1}{2}$$

If B is infinite dimensional, then we obtain an infinite sequence $(v_n)_n \subseteq \partial B_1$ with $\|v_i - v_j\| > \dfrac{1}{2}$ whenever $i \neq j$. But ∂B_1 is (sequentially) compact, this is a contradiction. □

Proposition V.9. A normed space is separable if and only if it admits a sequence of linearly independent vectors whose linear span is dense.

Proof. Let E be a separable normed space. Let C be a countable dense subset, and from C we choose a maximal linearly independent subset C'. By construction, $C \subseteq \operatorname{span} C'$, so $\operatorname{span} C'$ is dense. For the only if part, their linear span over $\mathbb{Q}$ is a countable dense subset. $\square$

V.2 Hilbert Space Basics

V.2.1 Riesz's Representation Theorem

In the subsection, let $(H, \langle , \rangle)$ be a Hilbert space, i.e., a complete inner product space over $\mathbb{C}$.

Definition.

1. Let S be a subset of H. The **orthogonal complement** of S is

$$S^\perp := \{v \in H \mid \langle v, w\rangle = 0 \text{ for all } w \in S\}$$

2. A projection $p : V \to V$ is called an **orthogonal projection** if it verifies the following equivalent conditions:

   (i) For all $v \in V$, $v - pv \in (\operatorname{Im} p)^\perp$.

   (ii) $(\operatorname{Im} p)^\perp = \ker p$ and $(\ker p)^\perp = (\operatorname{Im} p)$.

   (iii) p is self-adjoint.

We will show the equivalence between 2.(i),(ii),(iii) in the sequel. For this moment, by orthogonal projection we mean 2.(i).

Theorem V.10. If $W \subseteq H$ is a closed subspace, the orthogonal projection onto W exists.

Lemma V.11. If S is a subset of H, then $S^\perp$ is a closed subspace of H and $\overline{S} = (S^\perp)^\perp$.

Proof. It is clear that $S^\perp$ is a subspace of H. Let $(v_n)_n \subseteq S^\perp$ be a sequence converging to $v \in S$. Then for $w \in S$,

$$|\langle v, w\rangle| = |\langle v - v_n + v_n, w\rangle| = |\langle v - v_n, w\rangle| \leqslant \|v - v_n\| \, \|w\|$$

so that $\langle v, w\rangle = 0$ for all $w \in S$, i.e., $v \in S^\perp$. For the last assertion, note $S \subseteq (S^\perp)^\perp$ is clear, and since the latter is closed, it follows $\overline{S} \subseteq (S^\perp)^\perp$. For the reverse inclusion, we use Theorem V.10. First, note that $((S^\perp)^\perp)^\perp = S^\perp$. Indeed, for subsets $S \subseteq T$, one has $S^\perp \supseteq T^\perp$. Thus $S^\perp \supseteq ((S^\perp)^\perp)^\perp \supseteq S^\perp$ and we have equality throughout. Second, by Theorem for $v \in (S^\perp)^\perp$ we can write $v = x + y$ with $x \in \overline{S}$ and $y \in \overline{S}^\perp \subseteq S^\perp$. We must show $y = 0$, and proving $\langle y, w\rangle = 0$ for all $w \in H$ suffices.

- $w \in \overline{S}$. Then $\langle y, w\rangle = 0$ for $y \in S^\perp$ and w can be approximated by elements in S.

- $w \in \overline{S}^{\perp} \subseteq S^{\perp}$. Then

$$\langle y, w\rangle = \underbrace{\langle v, w\rangle}_{=0 \text{ since } v\in(S^{\perp})^{\perp}} - \underbrace{\langle x, w\rangle}_{=0 \text{ since } x\in\overline{S}} = 0$$

This proves $y = 0$, and thus $v \in \overline{S}$. □

Theorem V.12 (Riesz's Representation Theorem)**.** For $T \in H^{\vee}$, there exists a unique $y \in H$ such that $T(x) = \langle x, y\rangle$. In other words, the inner product $H \times H \to \mathbb{C}$ induces an isomorphism $H \cong H^{\vee}$.

Proof. The uniqueness is clear. Let $N := \ker T$. If $N = H$, then take $y = 0$. Otherwise, since N is closed, by Theorem V.10 we have $H = N \oplus N^{\perp}$, and T restricts to an isomorphism $T|_{N^{\perp}} : N^{\perp} \to \mathbb{C}$.

Let $v_0 \in N^{\perp}$ be of norm one and set $w = \overline{T(v_0)}v_0$. For each $v \in V$, we can write $v = x + \lambda v_0$ with $x \in N$, $\lambda \in \mathbb{C}$. Then

$$\langle v, w\rangle = \langle \lambda v_0, \overline{T(v_0)}v_0\rangle = \lambda T(v_0) = T(v)$$

□

Corollary V.12.1. For $T \in \mathcal{B}(H)$, there exists a unique $T^* \in \mathcal{B}(H)$, called the **adjoint** of T, such that

$$\langle Tv, w\rangle = \langle v, T^*w\rangle$$

for all $v, w \in H$.

V.2.2 Orthonormal Basis

Definition. Let H be a Hilbert space.

1. A subset $A \subseteq H$ is **orthonormal** if $\|u\| = 1$ for all $u \in A$ and $\langle u, v\rangle = 0$ for all $u \neq v \in A$.
2. A maximal (with respect to the inclusion) orthonormal subset A of H is called an **orthonormal basis** of H.

- A Zorn's lemma argument shows that every orthonormal subset of H is contained in an orthonormal basis of H. In particular, a nontrivial Hilbert space has an orthonormal basis.

Let S be an arbitrary set. For a function $\varphi : S \to [0, \infty]$, define

$$\sum_{s\in S} \varphi(s) := \sup\left\{\sum_{s\in S'} \varphi(s) \mid S' \subseteq S,\ \#S' < \infty\right\}$$

A moment consideration shows that this exactly means the integral of φ with respect to the counting measure $\#$ on S. In this situation, we write $\ell^p(S) = L^p(\#)$. Note that for $\varphi \in \ell^2(S)$, the set $\{x \in S \mid \varphi(x) \neq 0\}$ is at most countable.

Theorem V.13. Let A be an orthonormal set in H, and let $P = \operatorname{span}_{\mathbb{C}} A$.

1. We have the **Bessel's inequality**

$$\sum_{\alpha \in A} |\hat{x}(\alpha)|^2 \leqslant \|x\|^2$$

for all $x \in H$, where $\hat{x} : A \to \mathbb{C}$ is defined by $\hat{x}(\alpha) := \langle x, \alpha \rangle$.

2. The association

$$\begin{aligned} \Phi : H &\longrightarrow \ell^2(A) \\ x &\longmapsto \hat{x} \end{aligned}$$

is a surjective continuous linear map whose restriction to $\overline{P}$ is an isometry onto $\ell^2(A)$.

Proof. 1. holds with A replaced by any of its finite subset, so it holds for A as well by definition. In particular, this shows the map Φ defined in 2. is well-defined. It is clear that Φ is linear, and for continuity, one notes that for $x, y \in H$, by 1. we have

$$\|\Phi(y) - \Phi(x)\|_2 = \|\hat{y} - \hat{x}\|_2 \leqslant \|y - x\|$$

Then $\Phi|_P : P \to \ell^2(A)$ is an isometry whose image consists of those functions with finite support; clearly, $\Phi(P)$ is dense in $\ell^2(A)$. $\Phi|_P$ being isometric, we see that $\Phi|_{\overline{P}} : \overline{P} \to \ell^2(A)$ is then a surjective isometry. $\square$

Theorem V.14. Let A be an orthonormal set in H. TFAE:

(a) A is an orthonormal basis of H.

(b) The linear span of A is dense in H.

(c) The equality

$$\sum_{\alpha \in A} |\hat{x}(\alpha)|^2 = \|x\|^2$$

holds for all $x \in H$.

(d) The **Parseval's identity**

$$\sum_{\alpha \in A} \hat{x}(\alpha)\overline{\hat{y}(\alpha)} = \langle x, y \rangle$$

holds for all $x, y \in H$.

Proof. (a) $\Rightarrow$ (b) follows from Lemma V.10, and (b) $\Rightarrow$ (c) follows from Theorem V.13 by noting that LHS of (c) is $\|\hat{x}\|_2^2$. (c) $\Rightarrow$ (d) follows from the polarization identity

$$4\langle x, y \rangle = \|x + y\|^2 - \|x - y\|^2 + i\,\|x + iy\|^2 - i\,\|x - iy\|^2$$

Finally, if A is not maximal, then we can find $0 \neq u \in H \backslash A$ such that $\hat{u}(\alpha) = 0$ for all $\alpha \in A$. Then (d) leads to (with $x = y = u$) $0 = \|u\|^2 > 0$, a contradiction. $\square$

Proposition V.15. Let H be a Hilbert space and A, B be two orthonormal basis. Then $\#A = \#B$.

Proof. Let $A = \{e_i\}_{i\in I}$ and $B = \{f_j\}_{j\in J}$. By Parseval's identity, for every $f_j \in B$ we have

$$\sum_{i\in I} |\langle f_j, e_i\rangle|^2 = \|f_j\|^2 = 1$$

Let $I_j^n := \left\{ i \in I \mid \langle f_j, e_i\rangle \geqslant \frac{1}{n} \right\}$. The above identity implies $\#I_j^n < \infty$, so $I_j := \bigcup_{n\in\mathbb{N}} I_j^n$ is countable. By Parseval's identity again, we have

$$\sum_{j\in J} |\langle f_j, e_i\rangle|^2 = \|e_i\|^2 = 1$$

and this implies that $I = \bigcup_{j\in J} I_j$. Hence $\#I \leqslant \#(J \times \mathbb{N}) = \#J$. Symmetrically we have $\#J \leqslant \#I$, so $\#A = \#I = \#J = \#B$. □

Proposition V.16. Let $(x_n)_n$ be an orthogonal sequence in a Hilbert space H. Then the following are equivalent.

(i) $\sum_{n=1}^{\infty} x_n$ converges in norm topology of H.

(ii) $\sum_{n=1}^{\infty} \|x_n\|^2 < \infty$.

(iii) $\sum_{n=1}^{\infty} \langle x_n, y\rangle$ converges for all $y \in H$.

Proof. We have $\left\| \sum_{n=N}^{M} x_n \right\|^2 = \sum_{n=N}^{M} \|x_n\|^2$ by our assumption, so (ii) implies (i). By Cauchy-Schwarz, (i) implies (iii). Now assume (iii). For each $N \in \mathbb{N}$, define an operator Λ_N on H by

$$\Lambda_N y = \sum_{n=1}^{N} \langle x_n, y\rangle.$$

By (iii), the sequence $(\Lambda_N y)_N$ converges for all $y \in H$, so by uniformly boundedness principal, the sequence $(\|\Lambda_N\|)_N$ is bounded. But

$$\|\Lambda_N\| = \|\langle x_1 + \cdots + x_N, \cdot\rangle\| = \|x_1 + \cdots + x_N\| = \left(\|x_1\|^2 + \cdots + \|x_N\|^2\right)^{\frac{1}{2}}$$

we see (iii) implies (ii). □

V.2.3 Constructions

Proposition V.17. For a Hilbert space H, its continuous dual $H^\vee$ is also a Hilbert space, with the inner product

$$\langle f, g\rangle := \sum_{\alpha\in A} f(\alpha)\overline{g(\alpha)}$$

where A is an orthonormal basis of H. The sum is independent of the choice of orthonormal basis. In this way, the isomorphism described in Theorem V.12 is an antilinear norm-preserving isomorphism.

Proof. The sum always converges by Bessel's inequality and Rieze's Representation theorem, and it is clear that it defines an inner product on $H^\vee$. Let B be another orthonormal basis of H. Then we can find $a_{\alpha\beta} \in \mathbb{C}\,(\alpha \in A,\, \beta \in B)$ such that $\alpha = \sum\limits_{\beta\in B} a_{\alpha\beta}\beta.$, and they satisfy $\sum\limits_{\beta\in B} a_{\alpha\beta}\overline{a_{\alpha'\beta}} = \delta_{\alpha\alpha'}$ as well as $\sum\limits_{\alpha\in A} a_{\alpha\beta}\overline{a_{\alpha\beta'}} = \delta_{\beta\beta'}$; thus

$$\sum_{\alpha\in A} f(\alpha)\overline{g(\alpha)} = \sum_{\alpha\in A}\sum_{\beta,\beta'\in B} a_{\alpha\beta}\overline{a_{\alpha\beta'}} f(\beta)\overline{g(\beta')} = \sum_{\beta\in B} f(\beta)\overline{g(\beta)}$$

For the last assertion, we use the notation in the proof of Theorem V.12. We use an orthonormal basis of N together with v_0 to compute $\|T\|$; then

$$\|T\|^2 = T(v_0)\overline{T(v_0)} = T(w) = \langle w, w\rangle = \|w\|^2$$

In fact, the norm defined by this inner product is the same as the operator norm. Keep the notation as above; then

$$\|T\|_{\mathrm{op}}^2 = \left(\sup_{\|v\|=1} |T(v)|\right)^2 = |T(v_0)|^2 = T(v_0)\overline{T(v_0)} = \|T\|^2$$

as wanted. □

Definition. Let $\{H_i\}_{i\in I}$ be a family of Hilbert spaces. The algebraic direct sum has a natural inner product:

$$\langle v, w\rangle := \sum_{i\in I}\langle v_i, w_i\rangle$$

for $v = (v_i)$, $w = (w_i)$. The completion of $\bigoplus\limits_{i\in I} H_i$ is called the **Hilbert direct sum**, and is denoted by $\widehat{\bigoplus\limits_{i\in I}} H_i$. If there is no confusion, the hat is usually omitted.

- One can identity $\widehat{\bigoplus\limits_{i\in I}} H_i$ with the space $\left\{(v_i) \in \prod\limits_{i\in I} V_i \mid \sum\limits_{i\in I}\|v_i\|^2 < \infty\right\}$.

Definition. For two Hilbert spaces V, W, their algebraic tensor products (over the base field) $V\otimes W$ has a natural inner product:

$$\langle v\otimes w, v'\otimes w'\rangle = \langle v, v'\rangle\langle w, w'\rangle$$

The completion of $V\otimes W$ is called the **Hilbert tensor product**, and is denoted by $V\widehat{\otimes}W$.

Appendix W: Lie groups

W.1 Manifolds

W.1.1 I.F.T.

We start with some classical results in calculus.

Theorem W.1 (Implicit function theorem)**.** Let $\begin{array}{rcl} g : \mathbb{R}^n \times \mathbb{R}^m & \longrightarrow & \mathbb{R}^m \\ (x, y) & \longmapsto & g(x, y) \end{array}$ be C^1 and $\xi \in \mathbb{R}^n$, $\eta \in \mathbb{R}^m$ be given that $g(\eta, \xi) = 0$. Assume that the $\det\left(\dfrac{\partial g}{\partial y}(\xi, \eta)\right) \neq 0$. Then there exist positive numbers a, b such that there exists a unique C^1 function $\phi : B_a(\xi) \to B_b(\eta)$ with $\phi(\xi) = \eta$ such that

$$g^{-1}(0) \cap (B_a(\xi) \times B_b(\eta)) = \{(x, \phi(x)) \mid x \in B_a(\xi)\}$$

Furthermore, if g is C^p, so is ϕ (including $p = \infty$).

Theorem W.2 (Inverse function theorem)**.** Let $\theta : \mathbb{R}^m \to \mathbb{R}^m$ be C^1 with $\theta'(\eta)$ be invertible for some $\eta \in \mathbb{R}^m$. Then there exists an open neighborhood U of η such that

(i) $\theta(U)$ is open in $\mathbb{R}^m$

(ii) $\theta|_U : U \to \theta(U)$ is bijective, with inverse ϕ

(iii) $\phi : \theta(U) \to U$ is C^1

Theorem W.3 (Constant rank theorem)**.** Let $f : \mathbb{R}^m \to \mathbb{R}^n$ be C^1 and $p \in \mathbb{R}^m$. Suppose rank $f'(x) = r$ in a neighborhood U of p. Then under suitable (smooth) changes of coordinates near $p \in U$ and $f(p) \in \mathbb{R}^n$, the map f assumes the form

$$(x^1, \ldots, x^m) \mapsto (x^1, \ldots, x^r, 0, \ldots, 0)$$

Proof. Write $f = (f_1, \ldots, f_n)$. Since rank $f'|_U \equiv r$ and $p \in U$, we can replace U by a smaller neighborhood so that, WLOG, the $r \times r$ principal minor $\left(\dfrac{\partial f_i}{\partial x^j}\right)_{1 \leqslant i,j \leqslant r}$ is invertible. Now consider the map

$$\begin{array}{rcl} \Phi : U & \longrightarrow & \mathbb{R}^m \\ x = (x^1, \ldots, x^m) & \longmapsto & (f_1(x), \ldots, f_r(x), x^{r+1}, \ldots, x^m) \end{array}$$

By our choice, the Jacobian of this map is nonvanishing at p, so by restricting to a smaller neighborhood again, we may assume this map is a C^1-diffeomorphism. Now consider the commutative triangle

$$\begin{tikzcd} U \arrow[r, "f"] \arrow[d, "\wr"', "\Phi"] & \mathbb{R}^n \\ \Phi(U) \arrow[ur, "\overline{f}"'] & \end{tikzcd}$$

One easily sees that $\overline{f}(x^1, \ldots, x^m) = (x^1, \ldots, x^r, h(x^1, \ldots, x^m))$ for some C^1 map h. But rank $\overline{f}'$ is constant on $\Phi(U)$, by considering the Jacobian of $\overline{f}$ we see $\dfrac{\partial h}{\partial y} \equiv 0$, where $y = (x^{r+1}, \ldots, x^m)$. Hence h is independent of y. Finally, let $F : \mathbb{R}^n \to \mathbb{R}^n$ be the diffeomorphism defined by $F(u, v) = (u, v - h(u))$. Then $F \circ \overline{f}(x, y) = (x^1, \ldots, x^r, 0, \ldots, 0)$, as wanted. $\square$

W.1.2 Basic Definitions

Definition.

1. An n**-dimensional smooth manifold** M^n is a second countable Hausdorff space together with a (smooth) **atlas** $\{(U_\alpha, \varphi_\alpha : U_\alpha \to \varphi_\alpha(U_\alpha) \underset{\text{open}}{\subseteq} \mathbb{R}^n)\}$ such that the U_α covers M and the **transition map**
$$\varphi_{\alpha\beta} := \varphi_\beta \circ \varphi_\alpha^{-1} : \varphi_\alpha(U_\alpha \cap U_\beta) \to \varphi_\beta(U_\alpha \cap U_\beta)$$
is smooth (in the usual sense) for each α, β. An element in the atlas is called a **chart**. We also write $n = \dim_{\mathbb{R}} M$.

2. Let the notation be as above. For a chart $\{U, \varphi\}$, write $\varphi(p) = (x^1(p), \ldots, x^n(p))$ for each $p \in U$. The functions $x^1, \ldots, x^n$ are called the **local coordinates**.

3. Let M, N be two smooth manifolds and $f : M \to N$ a continuous map. f is said to be **smooth** if for each chart (U, φ) of M and (V, ψ) of N with $f(U) \subseteq V$, the composition
$$\psi \circ f \circ \varphi^{-1}\varphi(U) \to U \to V \to \psi(V)$$
is smooth.

Next we review the concepts of tangent vectors and tangent spaces.

Definition. Let M be a smooth manifold.

1. For $U \underset{\text{open}}{\subseteq} M$, denote by $C^\infty(U) = C^\infty(U, \mathbb{R})$ the space of smooth real-valued functions on U.

2. For $p \in M$, denote by $C_p^\infty(M)$ the **stalk** of smooth functions at p, i.e.
$$C_p^\infty(M) = \varinjlim_{p \in U \underset{\text{open}}{\subseteq} M} C^\infty(U)$$
where the $C^\infty(U)$ are directed in any obvious way. An element in $C_p^\infty(M)$ is called a **germ**.

3. For a smooth curve $\gamma : \mathbb{R} \to M$ with $\gamma(0) = p$, we define $D_\gamma : C_p^\infty(M) \to \mathbb{R}$ by
$$D_\gamma f = \left.\frac{d}{dt}\right|_{t=0} f(\gamma(t))$$
Each D_γ is called the **tangent vector** to γ at p. Denote by T_pM the space of all tangent vectors of p at M, and call it the **tangent space** of M at p.

4. For each $p \in M$, a **derivation at** p is an $\mathbb{R}$-linear map $D : C_p^\infty(M) \to \mathbb{R}$ satisfying the **Leibniz rule**:

$$D(fg) = f(p)D(g) + g(p)D(f)$$

Proposition W.4. Let M be an n-dimensional smooth manifold and $p \in M$.

1. T_pM is an n-dimensional real vector space. To be precise, let $x^1, \ldots, x^n$ be a local coordinate around p. Then $T_pM = \operatorname{span}_{\mathbb{R}} \left\{ \dfrac{\partial}{\partial x^i} \right\}_{i=1,\ldots,n}$.

2. Every derivation at p is a tangent vector at p.

Example W.5. $\mathbb{R}^n$ is an n-dimensional smooth manifold. For each $p \in \mathbb{R}^n$ and a vector $v \in \mathbb{R}^n$, define $\gamma_v = p + tv$. Then there is a linear isomorphism

$$\begin{aligned} \mathbb{R}^n &\longrightarrow T_p\mathbb{R}^n \\ v &\longmapsto D_{\gamma_v} \end{aligned}$$

Definition. Let M, N be smooth manifolds, $p \in M$ and $f : M \to N$ a smooth map. The **differential of** f **at** p is a linear map $f_* = f_{*p} : T_pM \to T_{f(p)}N$ defined by

$$f_* D_\gamma = D_{f \circ \gamma}$$

- If D is a derivation at p, we have $(f_*D)g = D(g \circ f)$.
- Let $\gamma : \mathbb{R} \to M$ be a smooth curve. Then $D_\gamma = \gamma_* \left(\left. \dfrac{d}{dt} \right|_{t=0} \right)$.

1. $f : M \to N$ is an **immersion** if f_* is injective at all points of M.

2. $f : M \to N$ is a **submersion** if f_* is surjective at all points of M.

3. If f is injective and immersive, then (M, f) is called an **immersed submanifold** of N.

4. $f : M \to N$ is an **embedding** if f is injective, immersive and *homeomorphic onto its image* $f(M)$, where $f(M)$ is equipped with the subspace topology from N. In this case, we say (M, f) is an **embedded/regular submanifold** of N.

For a future use, let us mention the famous Sard's theorem.

Definition. Let $f : M \to N$ be a smooth map between smooth manifolds.

1. A point $p \in M$ is called a **critical point of** f if $f_{*,p} : T_pM \to T_{f(p)}N$ is not surjective.

2. The image of a critical point under f is called a **critical value of** f.

3. A point $q \in N$ is called a **regular value of** f if it is not a critical value of f.

- Note that $q \in N$ is regular if and only if $q \notin f(M)$ or $q \in f(M)$ and f is submersive on the fibre $f^{-1}(q)$.

Theorem W.6 (Sard)**.** If $f : M \to N$ is smooth, then the set of critical values of f has measure zero.

Proposition W.7. Let M^m be a smooth manifold and $r \in \mathbb{N}_0$. If S is a subspace of M such that for each $p \in S$ we can find a chart (U, φ) of M near p such that $\varphi(U \cap S) = \{x^1 = \cdots = x^r = 0\} \subseteq \varphi(U)$, then S is itself a manifold of dimension $m - r$ (codimension r) and $(S, \iota : S \subseteq M)$ is a regular submanifold of M.

Proof. Let $\pi : \mathbb{R}^m \to \mathbb{R}^{m-r}$ be the projection to the last $m - r$ coordinates. Near each p we find a chart (U_p, φ_p) such that $\varphi_p(U \cap S) = \{x^1 = \cdots = x^r = 0\}$. Put $V_p := U_p \cap S$ and $\psi_p = \pi \circ \varphi_p : V_p \to \mathbb{R}^{m-r}$. We claim $\{(V_p, \psi_p)\}_{p \in M}$ forms a smooth atlas of S. It suffices to check the transition map is smooth. To this end, let $p \neq q \in M$ with $V_p \cap V_q \neq \varnothing$. If we write $\varphi_q \circ \varphi_p^{-1}(x^1, \ldots, x^m) = (h_1(x), \ldots, h_m(x))$, then $\psi_q \circ \psi_p^{-1}(x^{r+1}, \ldots, x^m) = (h_{r+1}(0, y), \ldots, h_m(0, y))$, where $y = (x^{r+1}, \ldots, x^m)$, which is clearly smooth. Finally, $\iota : S \subseteq M$ is clearly injective and immersive since in local coordinates it is an inclusion, and it is a topological embedding since S is with subspace topology. $\square$

Theorem W.8 (Constant rank theorem)**.** Let M^m, N^n be smooth manifolds and $p \in M$. Let $f : M \to N$ a smooth map such that $\operatorname{rank} f_{*,x} = r$ in an open neighborhood U of p. Then we can find a chart φ near p and a chart ψ near $f(p)$ such that f assumes the form

$$\begin{tikzcd}
M \arrow[r, "f"] & N \\
\mathbb{R}^m \arrow[u, "\varphi^{-1}"] \arrow[r, "f"] & \mathbb{R}^n \arrow[u, "\psi^{-1}"']
\end{tikzcd}$$

$$(x^1, \ldots, x^m) \longmapsto (x^1, \ldots, x^r, 0, \ldots, 0)$$

Corollary W.8.1. Let $f : M^m \to N^n$ be a smooth map and $q \in N$. Suppose f_* has constant rank r in a neighborhood of $f^{-1}(q)$, then $f^{-1}(q)$ is a regular submanifold of M of codimension r.

Proof. Let $p \in f^{-1}(q)$. By Constant rank theorem we can find a chart (U, φ) near p and a chart (V, ψ) near $f(p) = q$ with $f(U) \subseteq V$ and $\psi(q) = 0$ such that we have the commutative diagram

$$\begin{tikzcd}
U \arrow[r, "f"] & V \\
\varphi(U) \arrow[u, "\varphi^{-1}"] \arrow[r, "\overline{f}"] & \psi(V) \arrow[u, "\psi^{-1}"']
\end{tikzcd}$$

$$(x^1, \ldots, x^m) \longmapsto (x^1, \ldots, x^r, 0, \ldots, 0)$$

Then $\varphi(U \cap f^{-1}q) = \overline{f}^{-1}(0) = \{x^1 = \cdots = x^r = 0\}$, so that $f^{-1}(q)$ is a regular submanifold by Proposition W.7. $\square$

Corollary W.8.2. Let M^m, N^n be smooth manifolds and $f : M \to N$ a smooth map.

1. If f is immersive at $p \in M$, then f is locally an embedding near p, i.e., we can find an open neighborhood U of p and such that $f|_U : U \to N$ is an embedding.

2. If $q \in f(M)$ is a regular value of f, then $f^{-1}(q)$ is a regular submanifold of M.

3. If f is an embedding, then $(f(M), \iota : f(M) \subseteq N)$ is the regular submanifold of the type in Proposition W.7, i.e., it is locally the zero locus of some local coordinates.

Proof.

1. By Constant rank theorem, we can find a chart (U, φ) of M near p and a chart (V, ψ) near $f(p)$ with $f(U) \subseteq V$ such that $f : U \to V$ is of the form $(x^1, \ldots, x^m) \mapsto (x^1, \ldots, x^m, 0, \ldots, 0)$. This shows $f|_U \to N$ is an embedding.

2. This is just Corollary W.8.1 with $r = n$.

3. For each $p \in M$, let (U_p, φ_p) and (V_p, ψ_p) be the charts found as in 1. Since f is a *topological embedding*, we can find an open neighborhood W_p of $f(p)$ such that $f(U_p) = W_p \cap f(M)$. Replacing V_p by $V_p \cap W_p$, we are then in the situation as Proposition W.7.

$\square$

Lemma W.9. Let $M^m \subseteq \mathbb{R}^n$ be a embedded submanifold. Let $p \in M$ and assume there exists a smooth function $f : U \to \mathbb{R}$ defined on an open neighborhood U of p in $\mathbb{R}^n$ such that $f|_{U \cap M}$ is constant. Then the vector $\left(\dfrac{\partial f}{\partial x^1}, \ldots, \dfrac{\partial f}{\partial x^n}\right)\Big|_p$ is perpendicular to $T_pM \subseteq T_p\mathbb{R}^n = \mathbb{R}^n$.

Proof. Choose a local coordinate system $y^1, \ldots, y^n$ of $\mathbb{R}^n$ near p such that M is defined by $y^{m+1} = \cdots = y^n = 0$. Put $x^i = \theta_i(y^1, \ldots, y^n)$, so that $\dfrac{\partial}{\partial y^i} = \displaystyle\sum_{j=1}^n \frac{\partial \theta_j}{\partial y^i} \frac{\partial}{\partial x^j}$ near p. If we write $g(y^1, \ldots, y^n) = f(x^1, \ldots, x^n)$, then the constancy condition on f implies

$$0 = \frac{\partial g}{\partial y^i} = \sum_{j=1}^n \frac{\partial f}{\partial x^j} \frac{\partial \theta_j}{\partial y^i}$$

holds for any $1 \leqslant i \leqslant m$. These equations give exactly the result that we want. $\square$

Proposition W.10. Let $f : M \to N$ be a smooth map and $\varphi : S \to N$ be an immersed submanifold. Suppose $f(M) \subseteq \varphi(S)$ and let $f_0 : M \to S$ be the unique map such that $f = \varphi \circ f_0$.

(i) If f_0 is continuous, then f_0 is smooth.

(ii) If φ is an embedding, then f_0 is continuous.

Proof. (ii) is clear. For (i), assume f_0 is continuous. Let $p \in S$; by Corollary W.8.2.1. we can find an open neighborhood V of p in S such that $\varphi|_V : V \to N$ is an embedding. Then

$$f_0|_{f_0^{-1}(V)} = \left(\varphi|_V^{\varphi(N)}\right)^{-1} \circ f|_{f_0^{-1}(V)}^{\varphi(V)}$$

is smooth. □

W.1.3 Vector Fields, Local Flow and Integrability

Let M^m be a smooth manifold. We form the (set-theoretic) disjoint union

$$TM := \bigsqcup_{p \in M} T_pM = \{(p, v) \mid p \in M,\ v \in T_pM\}$$

This is called the **tangent bundle** of M. We are going to topologize TM and gives it a smooth structure.

We have the canonical projection $\pi : TM \to M$, sending (p, v) to p. Let (U, φ) be a chart on M and write $\varphi = (x^1, \ldots, x^m)$. Then for each $p \in U$, T_pM is spanned by the $\left.\frac{\partial}{\partial x^i}\right|_p$. By identifying the $\frac{\partial}{\partial x^i}$ with the standard basis of $\mathbb{R}^m$, we have a set-theoretic bijection:

$$\begin{aligned} \pi^{-1}(U) &\longrightarrow U \times \mathbb{R}^m \\ \left(p, \sum_{i=1}^m a_i \left.\frac{\partial}{\partial x^i}\right|_p\right) &\longmapsto (p, a_1, \ldots, a_m) \end{aligned}$$

We now use the cover $\{\pi^{-1}(U)\}$ to define a topology on TM. Say a subset $S \subseteq TM$ is open iff for every chart U on M, the image of $\pi^{-1}(U) \cap S$ in $U \times \mathbb{R}^n$ is open in the product topology. Then each $\pi^{-1}(U)$ is open in TM because if V is another chart, then $\pi^{-1}(U) \cap \pi^{-1}(V) = \pi^{-1}(U \cap V)$, and in $V \times \mathbb{R}^m$, it is $(U \cap V) \times \mathbb{R}^m$, an open subset. One checks easily that this topology is Hausdorff and second countable.

For a chart (U, φ) on M, let Φ be the composition $\pi^{-1}(U) \to U \times \mathbb{R}^m \to \varphi(U) \times \mathbb{R}^m$. Then $\{(U, \Phi)\}$ forms a smooth atlas on TM. We must show the transition map is smooth. Suppose $(V, \psi = (y^1, \ldots, y^m))$ is another chart that meets U. Let $f = \psi \circ \varphi^{-1}$ be the transition map. By definition, $f_{*,p} \left.\frac{\partial}{\partial x^i}\right|_p = \sum_j \left.\frac{\partial f_j}{\partial x^i}\right|_p \left.\frac{\partial}{\partial y^j}\right|_p$. If we write $J_p = \left(\left.\frac{\partial f_i}{\partial x^j}\right|_p\right)_{ij}$, then there is a smooth diffeomorphism

$$\begin{aligned} (U \cap V) \times \mathbb{R}^m &\longrightarrow (U \cap V) \times \mathbb{R}^m \\ (p, a_1, \ldots, a_m) &\longmapsto (p, (a_1 \ \cdots \ a_m) J_p^T) \end{aligned}$$

This is clearly a smooth map, for the transition map f is smooth so that the Jacobian J_p is smooth in p as well. This finishes our construction, and this atlas makes TM into a real $2m$-dimensional smooth manifold.

Definition. Let M be a smooth manifold. A **smooth vector bundle of rank** r **over** M is a pair $(E, \pi_E : E \to M)$ enjoying the following properties

(a) $\pi_E : E \to M$ is a surjective smooth map.

(b) Each fibre $\pi^{-1}(p)$ of $p \in M$ has a structure of real r-dimensional vector space.

(c) π_E is **locally trivial of rank** r, i.e., for each $p \in M$ there is an open neighborhood U of p and a smooth diffeomorphism $\phi : \pi_E^{-1}(U) \to U \times \mathbb{R}^r$ such that the triangle

$$\begin{tikzcd} \pi_E^{-1}(U) \arrow[rr, "\phi"] \arrow[dr, "\pi_E"'] & & U \times \mathbb{R}^r \arrow[dl, "\mathrm{pr}_1"] \\ & U & \end{tikzcd}$$

commutes, and for each $q \in U$, the restriction $\phi|_{\pi_E^{-1}(q)} : \pi_E^{-1}(q) \to \{q\} \times \mathbb{R}^r$ is a linear isomorphism. Such a (U, ϕ) is called a **local trivialization** of E.

- In this language, the tangent bundle TM is a smooth vector bundle of rank m over M^m.
- We say $M \times \mathbb{R}^d$ is the **trivial bundle** of rank d over M.

Definition. Let M be a smooth manifold and (E, π_E) a vector bundle over M.

1. A **section** of E is a continuous map $s : M \to E$ with $\pi \circ s = \mathrm{id}_M$.
2. A **vector field** of M is a section of the tangent bundle TM.

The set of all sections of E forms a real vector space. Denote by $\mathfrak{X}(M)$ the space of *smooth* vector fields of M.

3. If $f : M \to N$ is a smooth diffeomorphism and $X \in \mathfrak{X}(M)$, the **pushforward of** X **by** f is the vector field f_*X defined by $(f_*X)_q := f_{*,p}X_p$, where $q = f(p)$.

Definition. Let X be a smooth vector field on a smooth manifold M.

1. An **integral curve** of X is a smooth curve $\gamma : (a, b) \to M$ such that $\gamma_{*,s}\left(\left.\frac{d}{dt}\right|_{t=s}\right) = X_{\gamma(s)}$ for each $a < s < b$. We always assume $0 \in (a, b)$.
2. Let the notation be as in 1. If $p = \gamma(0)$, we say γ is an integral curve of X **starting from** p. γ is called **maximal** if its domain cannot be extended further.
3. A **local flow** is a smooth map $F : (a, b) \times M \to M$ (with $0 \in (a, b)$) such that $F(0, p) = p$ for all $p \in M$ and $F(s + t, p) = F(s, F(t, p))$ for all $p \in M$, $s, t \in (a, b)$ with $s + t \in (a, b)$.

4. A **local flow generated by** X is a local flow $F : (a, b) \times M \to M$ such that for each $p \in M$, $F(\cdot, p) : (a, b) \to M$ is an integral curve of X. The vector field X is called the **infinitesimal generator** of the flow F.

Theorem W.11. Let X be a smooth vector field on M, $p \in M$ and U an open neighborhood of p.

1. There exists a unique maximal integral curve of X starting from p.
2. There exists a local flow $F : (-\varepsilon, \varepsilon) \times W \to U$ generated by X for some small $\varepsilon > 0$ and $p \in W \underset{\text{open}}{\subseteq} U$.

Lemma W.12. Let M, N be smooth manifolds and $f : M \to N$ be a continuous map.

1. f is smooth if and only if $f^*g := g \circ f \in C^\infty(M)$ for all $g \in C^\infty(N)$.
2. A vector field X on M is smooth if and only if $Xf \in C^\infty(M)$ for all $f \in C^\infty(M)$.

Proof.

1. The only part is clear. For the if part, it suffices to note that every locally defined smooth function can be extended to a globally defined smooth function by using bump functions.
2. Locally, write $X = \sum a_i \dfrac{\partial}{\partial x^i}$. The rest is similar to 1.

□

For $X, Y \in \mathfrak{X}(M)$, $XY := X \circ Y : C^\infty(M) \to C^\infty(M)$ is an $\mathbb{R}$-linear map, but fails to be a derivation in general. Indeed, for $f, g \in C^\infty(M)$,

$$XY(fg) = X(Y(f)g + Y(g)f) = XY(f)g + XY(g)f + Y(f)X(g) + Y(g)X(f)$$

To cancel out the last to term, we subtract $YX(fg)$ so that

$$\begin{aligned} XY(fg) - YX(fg) &= XY(f)g + XY(g)f - (YX(f)g + YX(g)f) \\ &= (XY - YX)(f)g + (XY - YX)(g)f \end{aligned}$$

Then $XY - YX$ is a derivation on $C^\infty(M)$.

Definition. For $X, Y \in \mathfrak{X}(M)$, the **Lie bracket** $[X, Y]$ is a smooth vector field given by the derivation $XY - YX$. In other words, for $p \in M$ and $f \in C_p^\infty(M)$,

$$[X, Y]_p f := X_p(Yf) - Y_p(Xf)$$

Definition. A **Lie algebra over a field** k is a k-vector space L together with a alternating k-bilinear pairing $[,] : L \times L \to L$, called the **Lie bracket**, satisfying the **Jacobi's identity**:

$$[x, [y, z]] + [y, [z, x]] + [z, [x, y]] = 0 \qquad , \text{ for all } x, y, z \in L$$

For each $x \in L$ we associate it with an endomorphism $\mathrm{ad}_x : L \to L$ defined by $\mathrm{ad}_x\, y := [x, y]$. The map

$$\begin{aligned} \mathrm{ad} : L &\longrightarrow \mathrm{End}\, L \\ x &\longmapsto \mathrm{ad}_x \end{aligned}$$

is called the **adjoint representation** of the Lie algebra L. By Jacobi's identity, ad is a Lie algebra homomorphism.

Proposition W.13. For a smooth manifold M, the space $\mathfrak{X}(M)$ of smooth vector fields on M together with the Lie bracket defined above is a real Lie algebra.

Proposition W.14. Let $f : M \to N$ be a smooth diffeomorphism. For $X, Y \in \mathfrak{X}(M)$, we have

$$f_*[X, Y] = [f_*X, f_*Y]$$

Proof. This is a direct computation. Let $q = f(p)$, $p \in M$.

$$\begin{aligned} (f_*[X, Y])_q g &= f_{*,p}[X, Y]_p g = [X, Y]_p (g \circ f) \\ &= X_p(Y(g \circ f)) - Y_p(X(g \circ f)) \\ &= X_p((f_*Y)g \circ f) - Y_p((f_*X)g \circ f) \\ &= (f_*X)_q (f_*Y)g - (f_*Y)_q (f_*X)g = [f_*X, f_*X]_q g \end{aligned}$$

where we have used the equality $(f_*X)g \circ f = X(g \circ f)$ □

Definition. Let X be a vector field and $\varphi : (-\varepsilon, \varepsilon) \times W \to U$ be a local flow generated by X. For $t \in (-\varepsilon, \varepsilon)$, put $\varphi_t = \varphi(t, \cdot) : W \to U$. If Y is a vector field on X and $p \in W$, define the **Lie derivative of Y in X** by

$$(\mathcal{L}_X Y)_p = \left.\frac{d}{dt}\right|_{t=0} (\varphi_{-t})_{*,\varphi_t(p)} Y_{\varphi_t(p)} = \lim_{t \to 0} \frac{(\varphi_{-t})_{*,\varphi_t(p)} Y_{\varphi_t(p)} - Y_p}{t} \in T_p$$

Here note that $\varphi_0(p) = p$ for any $p \in W$, so $(\varphi_0)_{*,p} = \mathrm{id}$ on T_pM.

Proposition W.15. Let X be a vector field and φ the local flow generated by X.

(i) For $f \in C^\infty(M)$, we have $X_p(f) = \lim\limits_{t \to 0} \dfrac{f(\varphi_t(p)) - f(p)}{t}$.

(ii) For $Y \in \mathfrak{X}(M)$, we have $L_X Y = [X, Y]$.

Proof.

(i) The question is local, we may write $X = \sum\limits_{i=1}^n a_i \dfrac{\partial}{\partial x^i}$ and $\varphi_t = (\varphi_t^1, \ldots, \varphi_t^n)$ with $\varphi_0^i = x^i$ and $\dfrac{d}{dt}\varphi_t^i(p) = a_i(\varphi_t(p))$. Then

$$\left.\frac{d}{dt}\right|_{t=0} f(\varphi_t(p)) = \left.\frac{d}{dt}\right|_{t=0} f(\varphi_t^1(p), \ldots, \varphi_t^n(p)) = \sum_{i=1}^n \left.\frac{\partial f}{\partial x^i}\right|_p \left.\frac{d}{dt}\right|_{t=0} \varphi_t^i(p) = \sum_{i=1}^n a_i(p) \left.\frac{\partial f}{\partial x^i}\right|_p = X_p(f).$$

(ii) Let $f \in C^\infty(M)$. Then

$$(\mathcal{L}_X Y)_p(f) = \lim_{t\to 0} \frac{(\varphi_{-t})_{*,\varphi_t(p)} Y_{\varphi_t(p)}(f) - Y_p(f)}{t} = \lim_{t\to 0} \frac{Y_{\varphi_t(p)}(f\circ\varphi_{-t}) - Y_p(f)}{t}$$

where we use the identity $(f_*X)g\circ f = X(g\circ f)$ again. Write $f\circ\varphi_{-t} = f + th_t$ with $h_0 = -X(f)$. Applying Y, we obtain

$$Y_q(f\circ\varphi_{-t}) = Y_q(f) + Y_q(th_t) = Y_q(f) + tY_q(h_t).$$

Plugging into the expression above, we see

$$\begin{aligned}(\mathcal{L}_X Y)_p(f) &= \lim_{t\to 0} \frac{Y_{\varphi_t(p)}(f) + tY_{\varphi_t(p)}(h_t) - Y_p(f)}{t} \\ &= \lim_{t\to 0} \frac{Y_{\varphi_t(p)}(f) - Y_p(f)}{t} + Y_p(h_0) = X_p(Y(f)) - Y_p(X(f)).\end{aligned}$$

Here the last identity results from (i).

□

Corollary W.15.1. Let X, Y be two vector fields and φ_t, ψ_s the respective local flows defined on U. TFAE:

1. $[X, Y] = 0$ on U.
2. $(\varphi_t) * Y = Y$ for any small enough t.
3. For any $p \in U$, $\varphi_t \circ \psi_s(p) = \psi_s \circ \varphi_t(p)$ for any small t, s.

if and only if For any $p \in U$, $\varphi_t \circ \psi_s(p) = \psi_s \circ \varphi_t(p)$ for some $t, s > 0$

Definition. Let M^n be a smooth manifold and let $0 \leqslant k \leqslant n$.

1. A **rank k distribution of** TM is a subset $\mathcal{H} \subseteq TM$ satisfying
   (a) $\mathcal{H}_p := \mathcal{H} \cap T_pM$ is a k-dimensional subspace for any $p \in M$, and
   (b) $\mathcal{H}$ is smooth, in the sense that $\mathcal{H}_p = \operatorname{span}_{\mathbb{R}} \left\{ \sum_{j=1}^n a_{ij} \left.\frac{\partial}{\partial x^j}\right|_p \right\}_{i=1}^k$ holds in a small neighborhood p for any $p \in M$, where the a_{ij} are smooth.
2. A distribution $\mathcal{H}$ is **involutive** if for any $X, Y \in \mathfrak{X}(M)$ such that $X_p, Y_p \in \mathcal{H}$ for any $p \in M$, we have $[X, Y]_p \in \mathcal{H}$ for any $p \in M$.

Theorem W.16 (Frobenius')**.** Let M^n be a smooth manifold and $\mathcal{H}$ a rank k distribution on TM. If $\mathcal{H}$ is involutive, then for any $p \in M$, there exists a local chart $(U, x^1, \ldots, x^n)$ about p such that $\mathcal{H} = \operatorname{span}_{\mathbb{R}} \left\{ \frac{\partial}{\partial x^i} \right\}_{i=1}^k$ in U.

Proof. We prove this by induction on $k = \operatorname{rank} \mathcal{H}$; note that when $k = 1$, $\mathcal{H}$ is automatically involutive. Let $p \in M$ and choose a local chart $(U, x^1, \ldots, x^n)$ about p (where the image of p is the origin). Then $\mathcal{H} = \operatorname{span}_{\mathbb{R}} \left\{ \sum_{i=1}^n a_i \frac{\partial}{\partial x^i} \right\}$ in U. Let φ_t be the local flow in U generated by $X = \sum_{i=1}^n a_i \frac{\partial}{\partial x^i}$. Consider the function

$$F : (x^1, \ldots, x^{n-1}, t) \mapsto \varphi_t(x^1, \ldots, x^{n-1}, 0),$$

defined on a reasonable domain. We compute the Jacobian of F at the origin:

$$F'(0) = \begin{pmatrix} & a_1(0) \\ I_{n-1} & a_2(0) \\ & \vdots \\ O_{1,n-1} & a_n(0) \end{pmatrix}$$

Up to a rearrangement and shrinking U, we may assume a_n is nonvanishing in U, implying that $F'(0)$ is invertible. This means we can use $(x^1, \ldots, x^{n-1}, t)$ as a new coordinates. By construction, we have $F_{*,p} \left(\frac{\partial}{\partial t}\Big|_p \right) = X_{F(p)}$, or $F_* \frac{\partial}{\partial t} = X$, so that $(F^{-1})_* X = \frac{\partial}{\partial t}$. This means in the chart $(x^1, \ldots, x^{n-1}, t)$, we have $\mathcal{H} = \mathbb{R} \frac{\partial}{\partial t}$.

For general $k \geqslant 2$, take a small neighborhood around p so that $\mathcal{H} = \operatorname{span}_{\mathbb{R}}\{V_1, \ldots, V_k\}$, where the V_i are vector fields defined near p. By the $k = 1$ case applied to $\mathbb{R}V_k$, we can find a coordinate chart $(y^1, \ldots, y^n)$ such that $\mathbb{R}V_k = \mathbb{R} \frac{\partial}{\partial y^k}$. Write $[V_i, V_j] = V_{ij} + a_{ij} V_k \, (1 \leqslant i, j \leqslant k-1)$ for some $V_{ij} \in \operatorname{span}_{\mathbb{R}}\{V_1, \ldots, V_{k-1}\}$. By replacing V_i with $V_i - V_i(y^k) V_k$, we obtain

$$0 = V_{i,p}(V_j(y^k)) - V_{j,p}(V_i(y^k)) = [V_i, V_j]_p(y^k) = V_{ij,p}(y^k) + a_{ij}(p) V_{k,p}(y^k) = a_{ij}(p)$$

i.e., $a_{ij} = 0$. Thus $\operatorname{span}_{\mathbb{R}}\{V_1, \ldots, V_{k-1}\}$ is involutive, so by induction we can find $(z^1, \ldots, z^n)$ such that $\operatorname{span}_{\mathbb{R}}\{V_1, \ldots, V_{k-1}\} = \operatorname{span}_{\mathbb{R}} \left\{ \frac{\partial}{\partial z^i} \right\}_{i=1}^{k-1}$. So far we have arrived at

$$\mathcal{H} = \operatorname{span}_{\mathbb{R}} \left\{ \frac{\partial}{\partial z^1}, \ldots, \frac{\partial}{\partial z^{k-1}}, V_k \right\},$$

and we have found a function h such that $V_k(h) = 1$ while $\frac{\partial}{\partial z^i} h = 0$ for $1 \leqslant i \leqslant k-1$. In particular, we may assume $V_k \in \operatorname{span}_{\mathbb{R}} \left\{ \frac{\partial}{\partial z^i} \right\}_{i=k}^n$ Write $V_k = \sum_{i=k}^n b_i \frac{\partial}{\partial z^i}$. Since $\mathcal{H}$ is involutive,

$$\mathcal{H} \ni \sum_{j=1}^{k-1} a_j \frac{\partial}{\partial z^i} + c_i V_k = \left[\frac{\partial}{\partial z^i}, V_k \right] = \sum_{j=k}^n \frac{\partial b_j}{\partial z^i} \frac{\partial}{\partial z^j}.$$

Plugging h into the first expressing gives $c_i \equiv 0$, so the bracket actually lies in $\operatorname{span}_{\mathbb{R}}\{V_1, \ldots, V_{k-1}\}$, whence from the second expression we read off $\frac{\partial b_j}{\partial z^i} \equiv 0$ for $j \geqslant k$. It follows that the b_j only depend on the variables $z^k, \ldots, z^n$.

Using the coordinate $(z^1, \dots, z^{k-1}, z^k \dots, z^n)$, we identify $\mathbb{R}^n$ with $\mathbb{R}^{k-1} \times \mathbb{R}^{n-k+1}$. What we obtained above simply says that V_k is a vector field that entirely lies in $\{0\} \times \mathbb{R}^{n-k+1}$. By the case $k = 1$ applied the $\mathbb{R}V_k$ regarded as in the manifold $\mathbb{R}^{n-k+1}$, we can find $x^k, \dots, x^n$ such that $\mathbb{R}V_k = \mathbb{R}\dfrac{\partial}{\partial x^k}$. Now the coordinates $(z^1, \dots, z^{k-1}, x^k, \dots, x^n)$ about p in M does the job for us, and the induction step is completed. $\square$

Definition. Let M be a smooth manifold and $\mathcal{H} \subseteq TM$ a rank k distribution. An immersion $f : N \to M$ is called an **integral submanifold for $\mathcal{H}$** if for every $p \in N$, the image of $f_{*,p} : T_pN \to T_{f(p)}M$ is precisely $\mathfrak{H}_{f(p)}$.

Let $\mathcal{H} \subseteq TM$ be an involutive distribution of rank k. By Theorem W.16, for each $p \in M$ we can find a connected chart $(U_p, x_p^1, \dots, x_p^n)$ such that $\mathcal{H} = \operatorname{span}_{\mathbb{R}} \left\{ \dfrac{\partial}{\partial x_p^i} \right\}_{i=1}^k$ in U. Then the slices

$$S_p(\underline{c}) := \{x \in U_p \mid (x_p^{k+1}, \dots, x_p^n) = \underline{c}\},\ \underline{c} \in \mathbb{R}^{n-k}$$

are closed submanifolds of U_p such that $T_qS_p(\underline{c}) = \mathcal{H}_q$ for $q \in S_p(\underline{c})$. With our new language, each slice $S_p(\underline{c}) \to M$ is an integral submanifold for $\mathcal{H}$. In fact, if $f : N \to M$ is a *connected* integral submanifold for $\mathcal{H}$ with image $f(N) \subseteq U_p$, then $f(N) \subseteq S_p(\underline{c})$ for some $\underline{c} \in \mathbb{R}^{n-k}$. Indeed, if we denote by $\pi = (x_p^{k+1}, \dots, x_p^n) : \mathbb{R}^n \to \mathbb{R}^{n-k}$, then $(\pi \circ f)_* = 0$ identically, so $\pi \circ f$ is a constant, as N is connected.

We are going to construct a maximal integral submanifold for $\mathcal{H}$ passing through a fixed point $p \in M$. Let $p \in M$. Define

$$K = \left\{ x \in M \mid \begin{array}{l} \text{there exists a piecewise smooth } \gamma : (-\varepsilon, 1+\varepsilon) \to M \text{ such that } \gamma(0) = p,\ \gamma(1) = p, \\ \text{and all of its smooth pieces are integral curves for } \mathcal{H} \end{array} \right\}.$$

We topologize K as follows. By second countability of M, we can find a countable subset $F \subseteq M$ such that $\{U_q \mid q \in F\}$ covers M; we assume $p \in F$. For each $x \in K$, there exists $q_x \in F$ such that $x \in U_{q_x}$, so $x \in S_{q_x} := S_{q_x}(\underline{c})$ for some $\underline{c} \in \mathbb{R}^{n-k}$. Note that $S_{q_x} \subseteq K$. We use the cover $\{(S_{q_x}, x_{q_x}^1|_{S_{q_x}}, \dots, x_{q_x}^k|_{S_{q_x}}) \mid x \in K\}$ to give K a smooth structure. If we can show this topology is second countable, then K is then a (path-)connected k-dimensional smooth manifold. For this, fix some $q \in F$, and we only need to show the set

$$\{\underline{c} \in \mathbb{R}^{n-k} \mid S_q(\underline{c}) \subseteq K\}$$

is at most countable. Each point $x \in K$ is joined to p by a piecewise smooth curve γ. Take $(q_i)_{i=0}^n \subseteq F$ with $q_0 = p$ so that $\gamma|_{[0,1]}$ passes through U_{q_i} in order. Since each smooth part of γ is an integral curve, it starts with p in U_0, and passes through U_{q_1} in some slices, and then passes through U_{q_2} in some slices, and so on. To show the countability, it suffices to show that for $q, r \in F$ and $\underline{c} \in \mathbb{R}^{n-k}$, the set

$$\{\underline{d} \in \mathbb{R}^{n-k} \mid S_q(\underline{c}) \cap S_r(\underline{d}) \neq \varnothing\}$$

is at most countable. For this, notice that $S_q(\underline{c}) \cap U_r$ is an open submanifold of U_r, so it has (at most) countably many connected components, each being an integral submanifold for $\mathcal{H}$ in U_r. Hence $S_q(\underline{c}) \cap U_r$ is contained only in a countably union of the slices $S_r(\underline{d})$ in U_r.

Hence, the inclusion $f : K \to M$ is then a connected integral submanifold for $\mathcal{H}$ passing through p. We claim it is maximal. If $g : N \to M$ is a connected integral submanifold passing through p, then any point in N can be joined to $g^{-1}(p)$ by a piecewise smooth curve, as N is path-connected. Hence every point in $g(N)$ is connected to p by a piecewise smooth curve in $g(N)$, whence by a piecewise integral curve for $\mathcal{H}$. In conclusion, $g(N) \subseteq K$.

Corollary W.16.1. Let M^n be a smooth manifold and $\mathcal{H}$ a rank k involutive distribution on TM. Then for each $p \in M$, there exists a maximal connected integral submanifold $K \to M$ for $\mathcal{H}$ passing through p.

Integral submanifolds behave well categorically, in the following sense.

Lemma W.17. Let $f : N \to M$ be a smooth map and $\varphi : P \to M$ an integral submanifold for an involutive distribution $\mathcal{H} \subseteq TM$ of rank k. Suppose $f(N) \subseteq \varphi(P)$, and let $f_0 : N \to P$ be the unique map such that $f = \varphi \circ f_0$. Then f_0 is continuous. In particular, f_0 is smooth by Corollary W.10.

Proof. Let U be an open set in P, $p \in U$ and $x \in f_0^{-1}(p)$. Take a chart $(V, y^1, \ldots, y^m)$ centred at $\varphi(p)$ such that $S(\underline{c}) := \{y \in V \mid (y^{k+1}, \ldots, y^m) = \underline{c}\}$ are integral submanifolds for $\mathcal{H}$ in V, and pick an neighborhood $p \in U'$ in U such that $\varphi(U') = S(\underline{0})$. Take W to be the connected component of $f^{-1}(V)$ containing x; then W is open in N. To show the continuity it suffices to show $f_0(W) \subseteq U'$. Since φ is injective, it suffices to show $f(W) \subseteq S(\underline{0})$. Furthermore, since $f(x) \in f(W) \cap S(\underline{0})$ and $f(W)$ lies in a component of $V \cap \varphi(P)$, it suffices to show each component C of $V \cap \varphi(P)$ is contained in some $S(\underline{c})$.

Let $\pi = (y^{k+1}, \ldots, y^m) : V \to \mathbb{R}^{m-k}$. Since P is separable and $V \cap \varphi(P)$ is a disjoint union of the slices $S(\underline{c})$, the image $\pi(V \cap \varphi(P))$ is at most countable. Since C is connected, $\pi(C)$ is a connected countable set in $\mathbb{R}^{m-k}$, whence $\#\pi(C) = 1$. $\square$

W.2 Lie Groups and Examples

Definition. A **Lie group** G is a smooth manifold G which is also an abstract group such that the multiplication $\mu : G \times G \to G$ and the inversion $\mathrm{inv} : G \to G$ are smooth.

- For $g \in G$, let $\ell_g : G \to G$ be defined by $\ell_g(x) = gx$. Similarly, define $r_g : G \to G$ by $r_g(x) = xg$.

A **homomorphism of two Lie groups** $G \to H$ is a smooth map that is also a group homomorphism.

Lemma W.18. Let G be a Lie group and e be the identity element of G.

1. $\mu_{*,(e,e)} : T_eG \times T_eG \to T_eG$ is just addition, and $\mathrm{inv}_{*,e} : T_eG \to T_eG$ is multiplication by -1.
2. In the definition of a Lie group, we do not need to assume the smoothness of inv.

Proof.

1. Let $\gamma : (-\varepsilon, \varepsilon) \to G$ be a curve with $\gamma(0) = e$ and put $X := \gamma_{*,e}\left(\left.\frac{d}{dt}\right|_{t=0}\right) \in T_eG$. We compute $\mu_*(X, 0)$, and the result will ensue by linearity. Define $\tilde{\gamma}(t) := (\gamma(t), e)$. We have

$$\mu_*(X,0)f = \mu_*\tilde{\gamma}_*\left(\left.\frac{d}{dt}\right|_{t=0}\right)f = \left.\frac{d}{dt}\right|_{t=0} f((\mu \circ \gamma)(t)) = \left.\frac{d}{dt}\right|_{t=0} f(\gamma(t)) = \gamma_*\left(\left.\frac{d}{dt}\right|_{t=0}\right)f = Xf$$

so that $\mu_*(X, 0) = X$.

The second statement follows from chain rules. We have $\mu(\gamma(t), \mathrm{inv}(\gamma(t))) = e$, so taking differential, we have

$$0 = \left.\frac{d}{dt}\right|_{t=0} \mu(\gamma(t), \mathrm{inv}(\gamma(t))) = \mu_*(\mathrm{id}, \mathrm{inv})_*\gamma_*\left(\left.\frac{d}{dt}\right|_{t=0}\right) = \mu_*(\mathrm{id}, \mathrm{inv}_*)X = X + \mathrm{inv}_*X$$

so that $\mathrm{inv}_*X = -X$.

2. It suffices to show there exists a unit-neighborhood U of G such that $\mathrm{inv}|_U$ is smooth. Indeed, for $g \neq 1 \in G$, the diagram

$$\begin{tikzcd} gU \arrow[r, "\mathrm{inv}|_{gU}"] \arrow[d, "\ell_{g^{-1}}"'] & G \arrow[d, "r_g"] \\ U \arrow[r, "\mathrm{inv}|_U"] & G \end{tikzcd}$$

commutes, and since ℓ_g and r_g are diffeomorphisms, it follows that $\mathrm{inv}|_{gU}$ is smooth as well. Letting g run over G proves the global smoothness of inv.

To find U, we use Implicit function theorem. By 1, μ_* is an isomorphism in the second argument, so we can find open unit-neighborhoods U, V and a smooth map $\phi : U \to V$ such that $\mu(g, \phi(g)) = e$ for $g \in U$. But then ϕ coincides with inv on U, so inv is smooth as well.

□

Example W.19 (0-dimensional Lie groups)**.** All finite groups with discrete topology are zero-dimensional Lie groups. Some important examples are the symmetric groups S_n, the alternating groups A_n and the cyclic groups $C_n = \mathbb{Z}/n$.

Example W.20 (Direct product)**.** If G, H are Lie groups, so is their direct product $G \times H$ with usual group structure and smooth structure. For instance, the n-dimensional torus $\mathbb{R}^n/\mathbb{Z}^n \cong (\mathbb{R}/\mathbb{Z})^n \cong (S^1)^n$ is a Lie group.

Example W.21 (Identity component)**.** Let G be a Lie group and G^0 be the **identity component** of G, i.e., the connected component containing the identity element e. Then

- G^0 is a normal subgroup of G. Since $\mathrm{inv} : G \to G$ is a homeomorphism, $\mathrm{inv}\, G^0$ is a connected subset of G containing e, so $\mathrm{inv}\, G^0 \subseteq G^0$. Now for $g \in G^0$, gG^0 is connected and contains e as well, so $gG^0 \subseteq G^0$. The normality is shown in the same fashion.

- G^0 is open. Indeed, this follows from the fact that G^0 is locally connected.

Hence G^0 is an open (and closed) subgroup of G, so it is itself a Lie group.

Example W.22 (Matrix groups)**.** Let V be a finite dimensional real/complex vector space. The group of linear isomorphisms $\operatorname{Aut} V$ is an open subspace of $\operatorname{End} V$, and hence it has a smooth structure. Taking a basis for V, we see that the multiplication on $\operatorname{Aut} V$ is just a polynomial map, so $\operatorname{Aut} V$ is a Lie group. Hence $\mathrm{GL}_n(\mathbb{R})$ and $\mathrm{GL}_n(\mathbb{C})$ are Lie groups.

- $\mathrm{GL}_n(\mathbb{R})$ has two connected components. The identity component is

$$\mathrm{GL}_n(\mathbb{R})^+ := \{g \in \mathrm{GL}_n(\mathbb{R}) \mid \det g > 0\}$$

  and the other component is $\{g \in \mathrm{GL}_n(\mathbb{R}) \mid \det g < 0\}$. To see $\mathrm{GL}_n(\mathbb{R})^+$ is connected, note that it is generated (as groups) by elementary matrices with positive determinant.

- $\mathrm{GL}_n(\mathbb{C})$ is connected.

- The tangent space of $\mathrm{GL}_n(k)$ at the identity for $k = \mathbb{R}$, $\mathbb{C}$ can be regarded as the matrix ring $M_n(k)$.

Consider the special linear groups $\mathrm{SL}_n(\mathbb{R})$ and $\mathrm{SL}_n(\mathbb{C})$. We show they are Lie groups. Let $k = \mathbb{R}$, $\mathbb{C}$.

- 1 is a regular value of $\det : \mathrm{GL}_n(k) \to k^\times$. To show this, we compute the differential of det. For $A = (a_{ij}) \in \mathrm{GL}_n(k)$, by definition we have

$$\det A = (-1)^i a_{i1} m_{i1} + \cdots + (-1)^{ij} a_{ij} m_{ij} + \cdots + (-1)^{in} a_{in} m_{in}$$

  where the m_{ij} are determinant of the matrix obtained by deleting the i-th row and the j-th column of A. Hence $\det' A = ((-1)^{ij} m_{ij})_{1 \leqslant i,j \leqslant n}$, so a matrix $A \in \mathrm{GL}_n(k)$ is a critical point of det if and only all $m_{ij} = 0$. For $A \in \mathrm{SL}_n(k)$, since $\det A = 1$, it follows every matrix in $\mathrm{SL}_n(k)$ is a regular point of det. By regular level set theorem, $\mathrm{SL}_n(k)$ is then a regular submanifold of $\mathrm{GL}_n(k)$ of codimension 1.

- $\mathrm{SL}_n(k)$ is a Lie group. It remains to show the multiplication is smooth, and this follows easily from Proposition W.10.

Example W.23 (Orthogonal groups)**.** Let k be a field and B a k-bilinear form on a finite dimensional k-vector space V. Then we have an associated group

$$\{T \in \operatorname{Aut}_k V \mid B(Tv, Tw) = B(v, w) \text{ for all } v, w \in V\}$$

- $k = \mathbb{R}$, B the standard inner product on $\mathbb{R}^n$. Then we have the **orthogonal group**

$$\mathrm{O}(n) := \{A \in \mathrm{GL}_n(\mathbb{R}) \mid A^T A = I_n\}$$

  As $\mathrm{GL}_n(\mathbb{R})$, $\mathrm{O}_n(\mathbb{R})$ also splits into two connected components, and the identity component is the **special orthogonal groups**

$$\mathrm{SO}(n) := \{A \in \mathrm{O}_n(\mathbb{R}) \mid \det A = 1\}$$

- $k = \mathbb{C}$, B the standard Hermitian product on $\mathbb{C}^n$. Then we have the **unitary groups**

$$\mathrm{U}(n) := \{A \in \mathrm{GL}_n(\mathbb{C}) \mid A^*A = I_n\}$$

We can similar define the **special unitary group**

$$\mathrm{SU}(n) := \{A \in \mathrm{U}_n(\mathbb{C}) \mid \det A = 1\}$$

These groups are compact, for they are closed and bounded in the affine space k^{n^2}. We show $\mathrm{O}(n)$ is a regular submanifold of $\mathrm{GL}_n(\mathbb{R})$.

- Define $f : \mathrm{GL}_n(\mathbb{R}) \to \mathrm{GL}_n(\mathbb{R})$ by $f(A) = A^TA$. We prove f has constant rank on $\mathrm{GL}_n(\mathbb{R})$, so that $\mathrm{O}(n) = f^{-1}(I_n)$ is a regular submanifold by constant level set theorem.

- For $A, B \in \mathrm{GL}_n(\mathbb{R})$, we have $f(AB) = (AB)^TAB = B^Tf(A)B$, so that

$$f \circ r_B = \ell_{(B^T)^{-1}} \circ r_B \circ f$$

Taking differential at A, we have

$$f_{*,AB} \circ (r_B)_{*,A} = (\ell_{(B^T)^{-1}})_{*,f(A)B} \circ (r_B)_{*,f(A)} \circ f_{*,A}$$

Since the left and right translation are diffeomorphisms, we see $\operatorname{rank} f_{*,AB} = \operatorname{rank} f_{*,A}$ for all $A, B \in \mathrm{GL}_n(\mathbb{R})$. Since B is arbitrary, this proves f has constant rank.

- We determine $\operatorname{rank} f_{*,I_n}$. Let $X \in M_n(\mathbb{R})$ and $\gamma : \mathbb{R} \to \mathrm{GL}_n(\mathbb{R})$ a curve such that $\gamma(0) = I_n$ and $\gamma_{*,0} \left.\dfrac{d}{dt}\right|_{t=0} = X$. Then

$$f_{*,I_n}X = f_{*,I_n}\gamma_{*,0} \left.\frac{d}{dt}\right|_{t=0} = \left.\frac{d}{dt}\right|_{t=0} \gamma(t)^T\gamma(t) = X^T + X$$

Hence $\operatorname{rank} f_{*,I_n} = \dfrac{n^2+n}{2}$, and thus $\dim_{\mathbb{R}} \mathrm{O}(n) = n^2 - \dfrac{n^2-n}{2} = \dfrac{n^2+n}{2}$.

W.2.1 Symplectic groups

Recall the **real quaternion algebra** is a 4-dimensional real algebra

$$\mathbb{H} = \mathbb{R} + \mathbb{R}i + \mathbb{R}j + \mathbb{R}ij$$

with $i^2 = j^2$, $ij = -ji$. Concretely, it is the $\mathbb{R}$-algebra of 2×2 complex matrices of the form

$$\begin{pmatrix} a & b \\ -\bar{b} & \bar{a} \end{pmatrix} = \begin{pmatrix} a & \\ & \bar{a} \end{pmatrix} + \begin{pmatrix} & b \\ -\bar{b} & \end{pmatrix} = a + bj$$

with matrix multiplication and addition. Here we identify $z \in \mathbb{C}$ with $\begin{pmatrix} z & \\ & \overline{z} \end{pmatrix} \in \mathbb{H}$, and $j := \begin{pmatrix} & 1 \\ -1 & \end{pmatrix}$. The multiplication rule is given by

$$zj = j\overline{z} \text{ for } z \in \mathbb{C}, \qquad j^2 = -1$$

If $a + bj \neq 0$, then its determinant $|a|^2 + |b|^2$ is nonzero, and its multiplicative inverse is $\dfrac{\overline{a} - bj}{|a|^2 + |b|^2}$. This show $\mathbb{H}$ is a division ring, and we can think $\mathbb{C}$ as a subfield of $\mathbb{H}$.

- The center Z of $\mathbb{H}$ is $\mathbb{R}$. Clearly, $\mathbb{R} \subseteq Z$, and Z is a finite field extension of $\mathbb{R}$. If $\mathbb{R} \subsetneq Z$, then $Z = \mathbb{C}$. However, if we take $z \in \mathbb{H} \backslash \mathbb{C}$, we have $Z(x) = \mathbb{C}(x) = \mathbb{C}$ which is absurd. Hence $\mathbb{R} = Z$.

- $\mathbb{H}$ is a $\mathbb{C}$-vector space with $\mathbb{C}$ acting by the left multiplication. $\{1, j\}$ forms a $\mathbb{C}$-basis for $\mathbb{H}$.

- There is an involution, called **conjugation**, on $\mathbb{H}$ given by

$$\begin{aligned} \mathbb{H} &\longrightarrow \mathbb{H} \\ h = a + bj &\longmapsto \overline{h} := \overline{a} - bj \end{aligned}$$

  This is the restriction of the conjugate transpose $g \mapsto g^*$ on $\mathrm{GL}(2, \mathbb{C})$ to $\mathbb{H}$. Hence, it is an anti-$\mathbb{R}$-automorphism, which coincides with complex conjugation on $\mathbb{C}$. The **norm** for h is defined as $N(h) := h\overline{h} = \det h$, so if $h \neq 0$, the multiplicative inverse of h is $N(h)^{-1}\overline{h}$.

- As a real vector space, $\mathbb{H}$ has a basis consisting of

$$1 = \begin{pmatrix} 1 & \\ & 1 \end{pmatrix}, \qquad i = \begin{pmatrix} i & \\ & -i \end{pmatrix}, \qquad j = \begin{pmatrix} & 1 \\ -1 & \end{pmatrix}, \qquad k = \begin{pmatrix} & i \\ i & \end{pmatrix} = ij$$

  The quaternion of the form $ai + bj + ck$ with $a, b, c \in \mathbb{R}$ is called a **pure quaternion**. Every $h \in \mathbb{H}$ has a unique decomposition $h = r + q$ with r real and q pure. We have $\overline{r + q} = r - q$, and therefore $N(r + q) = r^2 - q^2$. Hence if q is pure, $N(q) = -q^2$ so that $q^2 \leqslant 0$.

  - The subspace of pure quaternions can be characterized only using the ring structure on $\mathbb{H}$: If $h = r + q$ with r real and q pure, then $h^2 = r^2 + q^2 + 2rq$ is real if and only if $r = 0$ or $q = 0$, and is non-positive if and only if $r = 0$.

- The standard isomorphisms $\mathbb{H} \cong \mathbb{R}^4$ and $\mathbb{H} \cong \mathbb{C}^2$ are norm-preserving. The group

$$\mathrm{Sp}(1) := \{h \in \mathbb{H} \mid N(h) = 1\}$$

  is called the **quaternion group**, or group of unit quaternions. In terms of matrices, we see

$$\mathrm{Sp}(1) = \left\{ \begin{pmatrix} a & b \\ -\overline{b} & \overline{a} \end{pmatrix} \mid a, b \in \mathbb{C},\ |a|^2 + |b|^2 = 1 \right\} = \mathrm{SU}(2)$$

  and the isomorphism $\mathbb{H} \cong \mathbb{R}^4$ identifies $\mathrm{Sp}(1)$ with the unit sphere S^3.

The basic statements of linear algebra may be formulated for modules over division rings. For example,

$$\mathrm{End}_{\mathbb{H}}(\mathbb{H}^n) = \{f \in \mathrm{End}_{\mathbb{R}}(\mathbb{H}^n) \mid f(zx) = zf(x) \text{ for all } z \in \mathbb{H},\, x \in \mathbb{H}^n\}$$

can be identified with the matrix ring $M_n(\mathbb{H})$. Under the identification, we see

$$\mathrm{GL}_n(\mathbb{H}) = \mathrm{Aut}_{\mathbb{H}}(\mathbb{H}^n) = \mathrm{End}_{\mathbb{H}}(\mathbb{H}^n) \cap \mathrm{Aut}_{\mathbb{R}}(\mathbb{H}^n).$$

We topologize $\mathbb{H}$ by the norm defined above; equivalently, since $\mathbb{H}$ is a finite dimensional real vector space, it is automatically a real Banach space. Then $\mathrm{GL}_n(\mathbb{H})$ is an open subspace of $M_n(\mathbb{H})$, and is a $4n^2$-dimensional Lie groups.

- The standard isomorphism $\mathbb{H} = \mathbb{C} + \mathbb{C}j \cong \mathbb{C}^2$ induces a standard $\mathbb{C}$-isomorphism $\mathbb{H}^n = \mathbb{C}^n + \mathbb{C}^n j \cong \mathbb{C}^{2n}$. Left multiplication by j induces an $\mathbb{R}$-endomorphism

$$\begin{aligned} j : \mathbb{C}^{2n} \cong \mathbb{H}^n &\longrightarrow \mathbb{H}^n \cong \mathbb{C}^{2n} \\ (u, v) = u + vj &\longmapsto j(u + vj) = -\overline{v} + \overline{u}j = (-\overline{v}, \overline{u}) \end{aligned}$$

  An $\mathbb{H}$-endomorphism $\varphi : \mathbb{H}^n \to \mathbb{H}^n$ is the same of a $\mathbb{C}$-endomorphism $\varphi : \mathbb{C}^{2n} \to \mathbb{C}^{2n}$ that commutes with j. Thus in matrix, φ assumes the form

$$\begin{pmatrix} A & -\overline{B} \\ B & \overline{A} \end{pmatrix} \in M_{2n}(\mathbb{C})$$

  with $A, B \in M_n(\mathbb{C})$.

- There is a inner product on $\mathbb{H}^n$, the **standard symplectic product**:

$$\langle h, k \rangle := \sum_{i=1}^{n} h_i \overline{k_i}$$

  for $h = (h_1, \ldots, h_n)$, $k = (k_1, \ldots, k_n) \in \mathbb{H}^n$. The corresponding norm is $\langle h, h \rangle = \sum\limits_{i=1}^{n} N(h_i) \geqslant 0$. The **(compact) symplectic group** is defined to be the norm-preserving $\mathbb{H}$-automorphisms:

$$\begin{aligned} \mathrm{Sp}(n) :=& \{\varphi \in \mathrm{GL}_n(\mathbb{H}) \mid N(\varphi(h)) = N(h) \text{ for all } h \in \mathbb{H}\} \\ =& \{\varphi \in \mathrm{GL}_n(\mathbb{H}) \mid \langle \varphi h, \varphi k \rangle = \langle h, k \rangle \text{ for all } h, k \in \mathbb{H}\} \end{aligned}$$

  Under the standard (norm-preserving) isomorphism $\mathbb{H}^n \cong \mathbb{C}^{2n}$, we have

$$\mathrm{Sp}(n) = \left\{ \begin{pmatrix} A & -\overline{B} \\ B & \overline{A} \end{pmatrix} \in \mathrm{U}(2n) \mid A, B \in M_n(\mathbb{C}) \right\}$$

  An element in $\mathrm{Sp}(n)$ is called a **symplectic matrix**.

- Left multiplication by j is not $\mathbb{C}$-linear. However, right multiplication by j is $\mathbb{C}$-linear:

$$\begin{aligned} J : \mathbb{C}^{2n} \cong \mathbb{H}^n &\longrightarrow \mathbb{H}^n \cong \mathbb{C}^{2n} \\ (u, v) = u + vj &\longmapsto (u + vj)j = -v + uj = (-v, u) \end{aligned}$$

  In matrix form, $J = \begin{pmatrix} 0 & -I_n \\ I_n & 0 \end{pmatrix} \in \mathrm{GL}_n(\mathbb{C})$. Denote by $c : \mathbb{C}^{2n} \to \mathbb{C}^{2n}$ the complex conjugation. Then $j = c \circ J$, and a unitary matrix A is symplectic if and only if $AcJ = cJA$. Sine $Ac = c\overline{A}$, it becomes $c\overline{A}J = cJA$, i.e., $JA = \overline{A}J = (A^T)^{-1}J$, or $A^TJA = J$. Dropping the condition that A being unitary, we obtain the **complex symplectic group**

$$\mathrm{Sp}(2n, \mathbb{C}) = \{A \in \mathrm{GL}_{2n}(\mathbb{C}) \mid A^TJA = J\}$$

  Then $\mathrm{Sp}(n) = \mathrm{Sp}(2n, \mathbb{C}) \cap U(2n)$.

- That $\mathrm{Sp}(2n, \mathbb{C})$ and $\mathrm{Sp}(n)$ are Lie groups can be shown in the same way as for $\mathrm{O}(n)$. We have the following lattices

$$\begin{tikzcd}
 & & \mathrm{GL}_{2n}(\mathbb{R})^+ & & \\
\mathrm{GL}_n(\mathbb{R})^+ \arrow[r] & \mathrm{GL}_n(\mathbb{R}) \arrow[r] & \mathrm{GL}_n(\mathbb{C}) \arrow[u] \arrow[r] & \mathrm{GL}_n(\mathbb{H}) \arrow[r] & \mathrm{GL}_{2n}(\mathbb{C}) \\
\mathrm{SO}(n) \arrow[u] \arrow[r] & \mathrm{O}(n) \arrow[u] \arrow[r] & \mathrm{U}(n) \arrow[u] \arrow[r] \arrow[d] & \mathrm{Sp}(n) \arrow[u] \arrow[r] \arrow[d] & \mathrm{Sp}(2n, \mathbb{C}) \arrow[u] \\
 & & \mathrm{SO}(2n) & \mathrm{U}(2n) &
\end{tikzcd}$$

- Consider the "adjoint" action of $\mathbb{H}^\times$ on $\mathbb{H}$:

$$\begin{aligned} \mathrm{Ad} : \mathbb{H}^\times \times \mathbb{H} &\longrightarrow \mathbb{H} \\ (q, x) &\longmapsto \mathrm{Ad}(q)(x) := qxq^{-1}. \end{aligned}$$

  Note that $N(qxq^{-1}) = N(q)N(x)N(q)^{-1} = N(x)$. We claim if x is pure quaternion, then so is qxq^{-1}. Indeed, if we write $qxq^{-1} = r + s$ with r real and s pure, then

$$0 \geqslant -N(x) = -qN(x)q^{-1} = qx^2q^{-1} = (qxq^{-1})^2 = r^2 + s^2 + 2rs.$$

  Since rs is nonreal unless $r = 0$ or $s = 0$ and $s^2 \leqslant 0$, we must have $qxq^{-1} = s$ is pure. Hence the adjoint action of $\mathbb{H}^\times$ on $\mathbb{H}$ stabilizes the subspace of pure quaternions $\operatorname{Im}\mathbb{H}$.

  The subspace $\operatorname{Im}\mathbb{H}$ is clearly isomorphic to $\mathbb{R}^3$ as real vector spaces:

$$\begin{aligned} \psi : \operatorname{Im}\mathbb{H} &\longrightarrow \mathbb{R}^3 \\ ai + bj + ck &\longmapsto (a, b, c). \end{aligned}$$

Actually it gives more: it is norm-preserving, and for $v, u \in \operatorname{Im}\mathbb{H}$, we have

$$vu = -\psi(v) \cdot \psi(u) + \psi^{-1}(\psi(v) \times \psi(u)). \tag{$\spadesuit$}$$

In view of this formula, it is natural to define an inner product $(\ ,\ )$ on $\operatorname{Im}\mathbb{H}$ by $(u, v) := \operatorname{Re}(u\overline{v})$; this turns ψ into a linear isometry. Now consider the restriction of Ad to $\operatorname{Sp}(1) \times \operatorname{Im}\mathbb{H}$; since $N(\operatorname{Ad}(q)x) = N(x)$, it induces a map

$$\begin{aligned} \operatorname{Ad} : \operatorname{Sp}(1) &\longrightarrow \operatorname{O}(\operatorname{Im}\mathbb{H}) \\ q &\longmapsto [\operatorname{Ad}(q) : x \mapsto qxq^{-1} = qx\overline{q}]. \end{aligned}$$

Identifying $\operatorname{O}(\operatorname{Im}\mathbb{H})$ with $\operatorname{O}(3)$, this yields a continuous homomorphism $\operatorname{Ad} : \operatorname{Sp}(1) \to \operatorname{O}(3)$. Since $\operatorname{Sp}(1) \cong S^3$ is connected and $\operatorname{Ad}(1) = \operatorname{id} \in \operatorname{SO}(3)$, we see $\operatorname{Ad}(\operatorname{Sp}(1)) \subseteq \operatorname{SO}(3)$. The kernel $\ker \operatorname{Ad} \leqslant \operatorname{Sp}(1)$ is $\{\pm 1\}$, which can be easily seen from ($\spadesuit$). We contend $\operatorname{Ad} : \operatorname{Sp}(1) \to \operatorname{SO}(3)$ is surjective by showing its image contains all rotations.

Restricting ψ to the norm one elements, we obtain $\psi : \operatorname{Im}\mathbb{H} \cap \operatorname{Sp}(1) \cong S^2$. From the formula ($\spadesuit$) again, we see each element $v \in \operatorname{Im}\mathbb{H} \cap \operatorname{Sp}(1)$ can be completed to a basis $\{v, w, u\}$ such that $v, w, u \in \operatorname{Im}\mathbb{H} \cap \operatorname{Sp}(1)$ with $vw = u$, $wu = v$, $uv = w$ and $v^2 = w^2 = u^2 = -1$.

Pick $v \in \operatorname{Im}\mathbb{H} \cap \operatorname{Sp}(1)$ and w, u as above. We claim that

$$\operatorname{Ad}(\cos\theta + v\sin\theta)$$

fixes v and acts as rotation by 2θ on $\operatorname{span}\{w, u\} \subseteq \operatorname{Im}\mathbb{H}$. To begin with, we observe that

$$\overline{\cos\theta + v\sin\theta} = \cos\theta - v\sin\theta.$$

This can be seen as follows: the natural identification $\mathbb{H} \cong \mathbb{R}^4$ preserves norm, so

$$\|\cos\theta + v\sin\theta\|^2 = \cos^2\theta + \sin^2\theta - v\cos\theta\sin\theta + v\sin\theta\cos\theta = 1$$

while $(\cos\theta + v\sin\theta)(\cos\theta - v\sin\theta) = 1$. This verifies our observation. Now compute

$$\begin{aligned} \operatorname{Ad}(\cos\theta + v\sin\theta)v &= (\cos\theta + v\sin\theta)v(\cos\theta - v\sin\theta) \\ &= (-\sin\theta + v\cos\theta)(\cos\theta - v\sin\theta) = v \\ \operatorname{Ad}(\cos\theta + v\sin\theta)u &= (\cos\theta + v\sin\theta)u(\cos\theta - v\sin\theta) \\ &= (u\cos\theta + w\sin\theta)(\cos\theta - v\sin\theta) = u\cos 2\theta + w\sin 2\theta \\ \operatorname{Ad}(\cos\theta + v\sin\theta)w &= (\cos\theta + v\sin\theta)w(\cos\theta - v\sin\theta) \\ &= (w\cos\theta - u\sin\theta)(\cos\theta - v\sin\theta) = w\cos 2\theta - u\sin 2\theta \end{aligned}$$

This proves the claim.

In conclusion, we obtain a short exact sequence of Lie groups:

$$1 \longrightarrow \{\pm 1\} \longrightarrow \operatorname{Sp}(1) \xrightarrow{\operatorname{Ad}} \operatorname{SO}(3) \longrightarrow 1$$

so that $\operatorname{Sp}(1)$ is a double cover of $\operatorname{SO}(3)$.

W.2.2 Spin

Definition. Let (V, Q) be a finite dimensional real quadratic space. The **Clifford algebra** $C(Q)$ is a unital $\mathbb{R}$-algebra together with an $\mathbb{R}$-linear map $i = i_Q : V \to C(Q)$ such that

(i) $i(x)^2 = -Q(x).1$ in $C(Q)$ for all $x \in V$, and

(ii) if A is any unital $\mathbb{R}$-algebra with an $\mathbb{R}$-linear map $j : V \to A$ such that $j(x)^2 = -Q(x).1$, then there exists a unique unital $\mathbb{R}$-algebra homomorphism $\phi_j : C(Q) \to A$ fitting in the commutative triangle:

$$\begin{tikzcd} C(Q) \arrow[rr] & & A \\ & V \arrow[ul, "i"] \arrow[ur, "j"'] & \end{tikzcd}$$

By universal property nonsense, a Clifford algebra is unique up to a unique isomorphism. It is easy to construct a Clifford algebra $C(Q)$ as a quotient of the tensor algebra: define I to be the two-sided ideal of IV generated by $\{x \otimes x + Q(x).1 \mid x \in V\}$, and define

$$C(Q) = TV/I.$$

An obvious candidate for $i : V \to C(Q)$ is the composition $V \to TV \to TV/I = C(Q)$.

Example W.24. Let $Q : \mathbb{R}^n \to \mathbb{R}$ be the usual norm function: $Q(x) = \|x\|^2$. Denote by $C_n = C_n(\mathbb{R}) = C(\|\cdot\|^2)$ the corresponding Clifford algebra. Let $\{e_i\}_{i=1}^n$ be the standard basis for $\mathbb{R}^n$. Then their image in C_n satisfies

$$e_i^2 = -1, \qquad e_i e_j = -e_j e_i \text{ if } i \neq j.$$

If $n = 0$, by construction we have $C_0 = \mathbb{R}$. If $n = 1$, we have an algebra isomorphism

$$\begin{aligned} C_1 &\longrightarrow \mathbb{C} \\ a + be_1 &\longmapsto a + bi \end{aligned}$$

and if $n = 2$, we have

$$\begin{aligned} C_2 &\longrightarrow \mathbb{H} \\ a + be_1 + ce_2 + de_1e_2 &\longmapsto a + bi + cj + dk \end{aligned}$$

We introduce two canonical maps on $C(Q)$. The opposite algebra $C(Q)^{\mathrm{op}}$ satisfies the universal properties (ii) for any opposite unital $\mathbb{R}$-algebras, and hence for all unital $\mathbb{R}$-algebras. Hence there exists a unique unital $\mathbb{R}$-algebra anti-isomorphism

$$(\cdot)^t : C(Q) \longrightarrow C(Q)$$

satisfying $(xy)^t = y^t x^t$ and $(x^t)^t = x$ for any $x, y \in C(Q)$; it is uniquely determined by $x^t = x$ for $x \in i(V)$.

By the universal property, the $\mathbb{R}$ isomorphism $V \to C(Q)$ given by $x \mapsto -x$ extends uniquely to a unique $\mathbb{R}$-algebra automorphism

$$\alpha : C(Q) \longrightarrow C(Q)$$

satisfying $\alpha \circ \alpha = \mathrm{id}$ and $\alpha(x) = -x$ for $x \in i(V)$. For $\nu \in \mathbb{N}_{\geqslant 0}$

$$C(Q)^\nu = \{x \in C(Q) \mid \alpha(x) = (-1)^\nu x\}$$

to be the eigenspace of α with eigenvalue $(-1)^\nu$; we have $C(Q) = C(Q)^0 \oplus C(Q)^1$. This is a $\mathbb{Z}/2\mathbb{Z}$-grading on $C(Q)$ in the sense that $xy \in C(Q)^{\nu+\mu}$ whenever $x \in C(Q)^\nu$ and $y \in C(Q)^\mu$.

In general, if $A = A^0 \oplus A^1$ and $B = B^0 \oplus B^1$ are two $\mathbb{Z}/2\mathbb{Z}$-graded algebras, their (vector space) tensor product $A \otimes B$ naturally inherits a $\mathbb{Z}/2\mathbb{Z}$-grading given by

$$\begin{aligned}(A \otimes B)^0 &= (A^0 \otimes B^0) \oplus (A^1 \otimes B^1)\\ (A \otimes B)^1 &= (A^0 \otimes B^1) \oplus (A^1 \otimes B^0)\end{aligned}$$

with multiplication

$$(a' \otimes b)(a \otimes b') = (-1)^{\nu\mu}(a'a) \otimes (bb')$$

where $a \in A^\mu$ and $b \in B^\nu$. To stress this extra structure, we denote such tensor product as $A \otimes^2 B$.

If (V, P) and (W, Q) are real quadratic spaces, we can form the orthogonal direct sum $(V \oplus W, P \oplus Q)$ by $(P \oplus Q)(v, w) = P(v) + Q(w)$. Define

$$\begin{aligned} f : V \oplus W &\longrightarrow C(P) \otimes^2 C(Q)\\ (v, w) &\longmapsto i_P v \otimes 1 + 1 \otimes i_Q w. \end{aligned}$$

We check $f(v, w)^2 = -(P(v) + Q(w))(1 \otimes 1)$, so that by universal property this extends to a map $f : C(P \oplus Q) \to C(P) \otimes^2 C(Q)$ on Clifford algebras. Indeed,

$$\begin{aligned} f(v, w)^2 &= (i_P v \otimes 1 + 1 \otimes i_Q w)^2\\ &= (i_P v \otimes 1)^2 + (i_P v \otimes 1)(1 \otimes i_Q w) + (1 \otimes i_Q w)(i_P v \otimes 1) + (1 \otimes i_Q w)^2\\ &= -P(v).1 \otimes 1 + i_P v \otimes i_Q w - i_P v \otimes i_Q w - 1 \otimes Q(w).1\\ &= -(P(v) + Q(w))(1 \otimes 1). \end{aligned}$$

We claim f is an isomorphism by constructing its inverse. This is easy. The inclusions $V \to V \oplus W$, $W \to V \oplus W$ induces homomorphisms $\varphi : C(P) \to C(P \oplus Q)$, $\psi : C(Q) \to C(P \oplus Q)$. Then

$$\begin{aligned} C(P) \otimes^2 C(Q) &\longrightarrow C(P) \oplus C(Q)\\ v \otimes w &\longmapsto \varphi(v)\psi(w) \end{aligned}$$

is the inverse of f.

Recall that every real quadratic space (V, Q) admits an orthogonal decomposition $(V, Q) \cong \bigoplus_{i=1}^{n}(V_i, Q_i)$ with $\dim V_i = 1$ (so that $\dim V = n$). The above isomorphism implies that

$$C(Q) = C(Q_1) \otimes^2 C(Q_2) \otimes^2 \cdots \otimes^2 C(Q_n)$$

We compute $C(Q)$ when $\dim V = 1$. By construction this is simply

$$C(Q) \cong \frac{\mathbb{R}[X]}{X^2 + Q(e_1)}$$

where e_1 is any nonzero vector in V. In particular, $\dim C(Q) = 2$. For general V, we conclude from the above decomposition that

$$\dim_{\mathbb{R}} C(Q) = 2^n.$$

Moreover, if $e_1, \ldots, e_n$ are a basis for V, then

$$C(Q) = \operatorname{span}_{\mathbb{R}}\{e_{i_1} \cdots e_{i_k} \mid 1 \leqslant i_1 < \cdots < i_k \leqslant n\}.$$

In particular, the canonical map $i : V \to C(Q)$ is injective, so we may view V as a subspace of $C(Q)$ without loss of generality.

We introduce more terms. Composing α and $(\cdot)^t$, we obtain an algebra anti-automorphism

$$\begin{aligned} (\cdot)^* : C(Q) &\longrightarrow C(Q) \\ x &\longmapsto \alpha(x)^t = \alpha(x^t). \end{aligned}$$

Define the **norm** $N : C(Q) \to C(Q)$ by $N(x) = xx^*$. For $x \in V$, we have $N(x) = x(-x) = Q(x).1$. Also, put

$$\Gamma(Q) = \{x \in C(Q)^\times \mid \alpha(x).v.x^{-1} \in V \text{ for all } v \in V\}.$$

This is a subgroup of the unit group $C(Q)^\times$, called the **Clifford group** of Q.

Lemma W.25. $\alpha : C(Q) \to C(Q)$ and $(\cdot)^t : C(Q) \to C(Q)$ induces an automorphism and anti-automorphism of the group $\Gamma(Q)$ respectively.

Let us turn to the case the Clifford algebra $C_n = C(\|\cdot\|^2)$ of the euclidean space $(\mathbb{R}^n, \|\cdot\|^2)$. Then

$$C_n = \bigotimes_{i=1}^{n}(\mathbb{R} + \mathbb{R}e_i) \supseteq \mathbb{R}^n$$

with a basis $e_{i_1} \cdots e_{i_k}$ $(1 \leqslant i_1 < \cdots < i_k \leqslant n)$ and relations $e_i^2 = -1$, $e_i e_j = -e_j e_i$ $(i \neq j)$. Put $\Gamma_n = \Gamma(\|\cdot\|^2)$. From the definition of Γ_n, it admits a representation

$$\rho : \Gamma_n \longrightarrow \operatorname{Aut} \mathbb{R}^n$$

defined as $\rho(x)v = \alpha(x)vx^{-1}$ $(x \in \Gamma_n,\ v \in \mathbb{R}^n)$.

Lemma W.26. $\ker \rho = \mathbb{R}^\times.1 \in C_n$.

Proof. Let $x \in \ker \rho$; then $\alpha(x)v = vx$ for all $v \in \mathbb{R}^n$. Write $x = x^0 + x^1$ with $x^i \in C_n^i$. Then

$$vx^0 + vx^1 = vx = \alpha(x)v = (x^0 - x^1)v$$

so that $x^0 v = vx^0$ and $-x^1 v = vx^1$ for all $v \in \mathbb{R}^n$. Write

$$x^0 = a^0 + e_1 b^1$$

with $a^0 \in C_n^0$, $b^1 \in C_n^1$ such that they do not contain any e_1 in their basis expression. Taking $v = e_1$ yields

$$a^0 + e_1 b^1 = x^0 = e_1^{-1} x^0 e_1 = -e_1(a^0 + e_1 b^1)e_1 = a^0 + b^1 e_1 = a^0 - e_1 b^1$$

so that $e_1 b^1 = 0$. Hence x^0 contains no monomial with a factor e_1. Repeating this argument shows that $x^0 \in \mathbb{R}.1$. Next write $x^1 = a^1 + e_1 b^0$ with $a^1 \in C_n^1$, $b^0 \in C_n^0$ such that they do not contain any e_1 in their basis expression. Then

$$a^1 + e_1 b^0 = x^1 = -e_1^{-1} x^1 e_1 = e_1(a^1 + e_1 b^0)e_1 = a^1 - e_1 b^0$$

so that $e_1 b^0 = 0$. The same argument implies $x^1 \in \mathbb{R}.1 \subseteq C_n^0$, whence $x^1 \in C_n^0 \cap C_n^1 = \{0\}$, i.e., $x^1 = 0$. Hence $x = x^0 \in \mathbb{R} \cap \Gamma_n = \mathbb{R}^\times$. □

Lemma W.27. $N(\Gamma_n) \subseteq \mathbb{R}^\times$, and $N|_{\Gamma_n} : \Gamma_n \to \mathbb{R}^\times$ is a homomorphism with $N(\alpha(x)) = N(x)$.

Proof. Let $x \in \Gamma_n$. We claim $N(x) \in \ker \rho$ so that $N(x) \in \mathbb{R}^\times.1$. For $v \in \mathbb{R}^n$, we compute

$$\rho(N(x))v = \alpha(N(x))vN(x)^{-1} = \alpha(x)x^t v(x^*)^{-1}x^{-1} = \alpha(x)(\alpha(x^{-1})vx)^t x^{-1} = \alpha(x)\alpha(x)^{-1}vxx^{-1} = v.$$

Hence $N(x) \in \ker \rho$. For $x, y \in \Gamma_n$, since $N(y) \in \mathbb{R}^\times$, we have

$$N(xy) = xy(xy)^* = xyy^*x^* = xN(y)x^* = xx^*N(y) = N(x)N(y).$$

Last,

$$N(\alpha(x)) = \alpha(x)\alpha(x)^* = \alpha(x)x^t = \alpha(xx^*) = \alpha(N(x)) = N(x)$$

as $N(x) \in \mathbb{R}$. □

Lemma W.28. $\mathbb{R}^n \backslash \{0\} \subseteq \Gamma_n$, and if $x \in \mathbb{R}^n \backslash \{0\}$, then $\rho(x)$ is the reflection in the hyperplane orthogonal to x. Also, $\rho(\Gamma_n) \subseteq \mathrm{O}(n)$.

Proof. Let $x \in \mathbb{R}^n \backslash \{0\}$. Choose a basis $\{e_i\}_{i=1}^n$ such that $x = \|x\| e_1$. Since $\mathbb{R}^\times.1 \subseteq \ker \rho$, we have $\rho(x) = \rho(\|x\| e_1) = \rho(e_1)$; we may assume $e_1 = x$. Then we have

$$\begin{aligned}
\rho(e_1)e_1 &= \alpha(e_1)e_1 e_1^{-1} = -e_1 \\
\rho(e_1)e_j &= \alpha(e_1)e_j e_1^{-1} = e_1 e_j e_1 = e_j \ (j \neq 1).
\end{aligned}$$

This proves the first part. For $x \in \Gamma_n$, $v \in \mathbb{R}^n \backslash \{0\}$,

$$N(\rho(x)v) = N(\alpha(x)vx^{-1}) = N(\alpha(x))N(v)N(x)^{-1} = N(v),$$

so that $\rho(x) \in \mathrm{O}(n)$ (as $N(v) = \|v\|^2 .1$). $\square$

Definition. For $n \geqslant 1$, define

$$\mathrm{Pin}(n) = \ker N|_{\Gamma_n} = \{x \in C(\|\cdot\|^2)^\times \mid xx^* = 1,\ \alpha(x)vx^{-1} \in \mathbb{R}^n \text{ for all } v \in \mathbb{R}^n\}$$

Theorem W.29. The sequence

$$\{1\} \longrightarrow \{\pm 1\} \longrightarrow \mathrm{Pin}(n) \xrightarrow{\ \rho\ } \mathrm{O}(n) \longrightarrow \{1\}$$

is a short exact sequence of groups.

Proof. By Cartan-Dieudonne theorem and Lemma W.28, $\rho : \mathrm{Pin}(n) \to \mathrm{O}(n)$ is surjective. Finally,

$$\ker(\rho|_{\mathrm{Pin}(n)}) = \ker \rho \cap \ker N = (\mathbb{R}^\times .1) \cap \ker N = \{\pm 1\}.$$

$\square$

Via this exact sequence, we can topologize the set $\mathrm{Pin}(n)$, making it a topological group. With this topology, $\rho : \mathrm{Pin}(n) \to \mathrm{O}(n)$ becomes a double cover. Moreover, since $\mathrm{O}(n)$ is a Lie group, $\mathrm{Pin}(n)$ then admits a unique smooth structure so that $\mathrm{Pin}(n)$ is a Lie group and ρ is a smooth homomorphism.

Define the **spin group**

$$\mathrm{Spin}(n) := \rho^{-1}(\mathrm{SO}(n))$$

Then we have a short exact sequence of Lie groups:

$$\{1\} \longrightarrow \{\pm 1\} \longrightarrow \mathrm{Spin}(n) \xrightarrow{\ \rho\ } \mathrm{SO}(n) \longrightarrow \{1\}$$

When $n = 1$, we have $C_1 = \mathbb{C}$ and

$$\mathrm{Pin}(1) = \{z \in S^1 \mid \overline{z}(i\mathbb{R})z^{-1} \subseteq i\mathbb{R}\} = \langle i \rangle = \mathbb{Z}/4\mathbb{Z},$$

so that $\mathrm{Spin}(1) = \{\pm 1\} = \mathbb{Z}/2\mathbb{Z}$. For $n \geqslant 2$, we claim

$$\rho : \mathrm{Spin}(n) \to \mathrm{SO}(n)$$

is a nontrivial double cover. To see this we claim $\mathrm{Spin}(n)$ is path-connected. To show this it suffices to show there is a path connecting 1 and -1. Define $\gamma : [0, \pi] \to C_n$ by

$$\gamma(t) = \cos(t) + \sin(t)e_1e_2$$

The two endpoints are exactly ± 1, and this is so constructed that

$$\gamma(t)^{-1} = \cos(t) - \sin(t)e_1e_2 = \gamma(t)^*$$

An easy computation shows that $\gamma(t) \in \mathrm{Pin}(n)$. It follows from an argument that $\gamma(t) \in \mathrm{Spin}(n)$ for all t. Since $\mathrm{SO}(n)$ is connected, the preceding result also implies that $\mathrm{Spin}(n)$ is connected for $n \geqslant 2$. Since $\mathrm{SO}(n)$ is the identity component of $\mathrm{O}(n)$, we conclude that $\mathrm{Spin}(n)$ is the identity component of $\mathrm{Pin}(n)$.

We consider the case $n = 3$. We have an injective algebra homomorphism

$$\begin{aligned} \kappa : \mathbb{H} &\longrightarrow C_3 \\ a + bi + cj + dk &\longmapsto a + be_2e_3 + ce_3e_1 + de_1e_2 \end{aligned}$$

It follows from the definition of $(\cdot)^t$ that this map is *-equivariant and hence norm-preserving, and it restricts to an injective homomorphism $\kappa : \mathbb{H}^\times \to \Gamma_3$. There is a commutative square:

$$\begin{tikzcd} \mathbb{H}^\times \times \operatorname{Im}\mathbb{H} \arrow[r, "\mathrm{Ad}"] \arrow[d, "\kappa\times\psi"] & \operatorname{Im}\mathbb{H} \arrow[d, "\psi"] \\ \Gamma_3 \times \mathbb{R}^3 \arrow[r, "\rho"] & \mathbb{R}^3 \end{tikzcd}$$

which can be easily checked. Restricting κ to norm one elements, we obtain $\kappa : \mathrm{Sp}(1) \to \mathrm{Spin}(3)$. Since both sides have the same dimension and are connected, this is an isomorphism.

There is another description of $\mathrm{Spin}(n)$:

Lemma W.30. $\mathrm{Spin}(n) = \mathrm{Pin}(n) \cap C_n^0$.

W.3 Left Invariant Vector Fields and Exponential Maps

Proposition W.31. Let G be a Lie group. Then G is parallelizable, i.e., the tangent bundle TG is isomorphic to the trivial bundle $G \times T_eG$.

Proof. The isomorphism is given by

$$\begin{tikzcd} (g, X) \arrow[r, mapsto] & (g, (\ell_g)_{*,e}X) \\ G \times T_eG \arrow[rr] \arrow[dr, "\mathrm{pr}_1"'] & & TG \arrow[dl, "\pi"] \\ & G & \end{tikzcd}$$

□

Definition. For a Lie group G, the vector space $\mathrm{Lie}(G) := T_eG$ is called the **Lie algebra** of G.

1. A smooth vector field $X \in \mathfrak{X}(G)$ is **left invariant** if for all $g \in G$, the diagram

$$\begin{tikzcd} TG \arrow[r, "(\ell_g)_*"] & TG \\ G \arrow[u, "X"] \arrow[r, "\ell_g"] & G \arrow[u, "X"'] \end{tikzcd}$$

commutes, i.e, $(\ell_g)_{*,h}X_h = X_{gh}$, or simply, $(\ell_g)_*X = X$, i.e, X is invariant under pushforward by ℓ_g. Denote by $L_G \subseteq \mathfrak{X}(G)$ the space of left-invariant smooth vector field on X.

- For $X \in \mathrm{Lie}(G)$ via the trivialization $G \times T_eG \cong TG$ we associate it with a left-invariant vector field $g \mapsto (\ell_g)_{*,e}X$. This establishes a one-one correspondence between $\mathrm{Lie}(G)$ and L_G.

- To justify the name of $\mathrm{Lie}(G)$, recall in Proposition W.13 we saw that $\mathfrak{X}(G)$ is a Lie algebra. We restrict the bracket to L_G and transfer it to $\mathrm{Lie}(G)$. To show it is well-defined, we must show if $X, Y \in \mathfrak{X}(G)$ is left-invariant, then so is $[X, Y]$; but this follows from Proposition W.14.

- Explicitly, let $X, Y \in T_eG$ and put $X', Y' \in \mathfrak{X}(G)$ be the associated left-invariant vector fields. Then $[X, Y] \in T_eG$ is defined by $[X', Y']_e$.

Proposition W.32. For $f : G \to H$ a homomorphism of Lie groups, the differential $f_{*,e} : T_eG \to T_eH$ is a Lie algebra homomorphism.

Proof. For $X \in \mathrm{Lie}(G)$, let $X' \in L_G$ be its associated left-invariant vector field. Then we must show the identity in $\mathrm{Lie}(H)$

$$f_{*,e}[X', Y']_e = [(f_{*,e}X)', (f_{*,e}Y)']_e$$

For $h \in C^\infty(H)$, compute

$$\begin{aligned} [(f_{*,e}X)', (f_{*,e}Y)']_e h &= (f_{*,e}X)'_e((f_{*,e}Y)'h) - (f_{*,e}Y)'_e((f_{*,e}X)'h) \\ &= f_{*,e}X((f_{*,e}Y)'h) - f_{*,e}Y((f_{*,e}X)'h) \\ &= X((f_{*,e}Y)'h \circ f) - Y((f_{*,e}X)'h \circ f) \end{aligned}$$

On the other hand,

$$f_{*,e}[X', Y']_e h = [X', Y']_e(h \circ f) = X(Y'(h \circ f)) - Y(X'(h \circ f))$$

Hence, it suffices to show $Y'(h \circ f) = (f_{*,e}Y)'h \circ f$. For $g \in G$,

$$\begin{aligned} (f_{*,e}Y)'h \circ f|_g &= (f_{*,e}Y)'_{f(g)}h \\ &= (\ell_{f(g)})_{*,e}f_{*,e}Yh \\ &= Y(h \circ \ell_{f(g)} \circ f) = Y(h \circ f \circ \ell_g) \\ &= (\ell_g)_{*,e}Y(h \circ f) \\ &= Y'_g(h \circ f) \end{aligned}$$

Here the identity $\ell_{f(g)} \circ f = f \circ \ell_g$ holds due to the fact that f is a group homomorphism. $\square$

W.3.1 Exponential Maps

Suppose X is a left-invariant vector field on a Lie group G and α is an integral curve of X starting from e. Then $g\alpha = \ell_{g^{-1}}\alpha$ is an integral curve of X starting from g for every $g \in G$; indeed,

$$(\ell_{g^{-1}}\alpha)_{*,s}\left(\left.\frac{d}{dt}\right|_{t=s}\right) = (\ell_{g^{-1}})_{*,\alpha(s)}X_{\alpha(s)} = X_{g\alpha(s)}$$

This shows $\alpha(s+t) = \alpha(s)\alpha(t)$ whenever $s, t, s+t$ lie in the domain of α, for $t \mapsto \alpha(s+t)$ and $t \mapsto \alpha(s)\alpha(t)$ are both integral curves for X starting from $\alpha(s)$. Hence the maximal integral curve α for X starting from e is defined over $\mathbb{R}$, and the local flow generated by X is actually global, and is given by $F(t, g) := \ell_{g^{-1}}\alpha(t) = g\alpha(t)$.

A homomorphism $\mathbb{R} \to G$ of Lie groups is called a **one-parameter subgroup** of G. The above discussion shows that the integral curve of X starting from e is a one-parameter subgroup. In fact, we have a correspondence

$$\begin{aligned} \mathrm{Hom}_{\mathrm{LieGp}}(\mathbb{R}, G) &\longrightarrow \mathrm{Lie}(G) \\ \alpha &\longmapsto \alpha_{*,0}\left(\left.\frac{d}{dt}\right|_{t=0}\right) \\ \alpha^X &\longleftarrow\!\shortmid X \end{aligned}$$

where in the second map we identify $\mathrm{Lie}(G)$ with L_G, and α^X is the integral curve for X starting from e.

Proposition W.33. The **exponential map** defined by

$$\begin{aligned} \exp : \mathrm{Lie}(G) &\longrightarrow G \\ X &\longmapsto \alpha^X(1) \end{aligned}$$

is smooth, and its differential at 0 is the identity map.

Proof. The map

$$\begin{aligned} \mathbb{R} \times G \times \mathrm{Lie}(G) &\longrightarrow G \times \mathrm{Lie}(G) \\ (t, g, X) &\longmapsto (g\alpha^X(t), X) \end{aligned}$$

is a flow of the smooth vector field $(g, X) \mapsto (X_g, 0) \in T_{(g,X)}(G \times \mathrm{Lie}(G)) \cong T_gG \times \mathrm{Lie}(G)$, so it is smooth. Thus its restriction to $\{1\} \times \{e\} \times \mathrm{Lie}(G)$, which is $X \mapsto (\alpha^X(1), X)$, is smooth, and so is the map $\exp : X \mapsto \alpha^X(1)$.

For the second statement, note that both $s \mapsto \alpha^{tX}(s)$ and $s \mapsto \alpha^X(ts)$ are integral curves of tX starting from e, so they are the same map. In particular,

$$\exp(tX) = \alpha^{tX}(1) = \alpha^X(t)$$

so that $\left.\frac{d}{dt}\right|_{t=0} \exp(tX) = \left.\frac{d}{dt}\right|_{t=0} \alpha^X(t) = X$. Hence $\exp_{*,0} = \mathrm{id}_{\mathrm{Lie}(G)}$. $\square$

Proposition W.34 (Naturality)**.** A homomorphism $f : G \to H$ of Lie groups induces a commutative diagram

$$\begin{tikzcd} \mathrm{Lie}(G) \arrow[r, "f_{*,e}"] \arrow[d, "\exp_G"] & \mathrm{Lie}(H) \arrow[d, "\exp_H"] \\ G \arrow[r, "f"'] & H \end{tikzcd}$$

Proof. Let $X \in \mathrm{Lie}(G)$. The one parameter subgroup $t \mapsto f(\alpha^X(t))$ has differential

$$(f \circ \alpha^X)_{*,0} \left(\left.\frac{d}{dt}\right|_{t=0} \right) = f_{*,e} \alpha^X_{*,0} \left(\left.\frac{d}{dt}\right|_{t=0} \right) = f_{*,e} X$$

at the origin, so

$$\exp(f_{*,e} X) = f(\alpha^X(1)) = f(\exp(X))$$

$\square$

W.3.2 Adjoint Representations

With the group structure of G, we have an alternative description of the Lie bracket on $\mathrm{Lie}(G)$.

Lemma W.35. For each $g \in G$, let $c(g) : G \to G$ be defined by $c(g)x := gxg^{-1}$. The homomorphism

$$\begin{aligned} \mathrm{Ad} : G &\longrightarrow \mathrm{Aut}\,\mathrm{Lie}(G) \\ g &\longmapsto c(g)_{*,e} \end{aligned}$$

is smooth, called the **adjoint representation** of G.

Proof. Locally, Ad sends g to the Jacobian matrix of $c(g)$. Since $c(g)$ is smooth, so is each entry of its Jacobian. $\square$

Proposition W.36. The differential $\mathrm{Ad}_{*,e} : \mathrm{Lie}(G) \to \mathrm{End}\,\mathrm{Lie}(G)$ coincides with the adjoint representation of the Lie algebra $\mathrm{Lie}(G)$.

Proof. For each $X, Y \in \mathrm{Lie}(G)$, we must show $\mathrm{Ad}_*(X)Y = \mathrm{ad}_X Y = [X, Y]$. For $f \in C^\infty(G)$, we have

$$\begin{aligned}
\mathrm{Ad}_*(X)Yf &= \left.\frac{d}{ds}\right|_{s=0} \mathrm{Ad}(\alpha^X(s))Yf = \left.\frac{d}{ds}\right|_{s=0} c(\alpha^X(s))_* Yf \\
&= \left.\frac{d}{ds}\right|_{s=0} Y(f \circ c(\alpha^X(s))) \\
&= \left.\frac{d}{ds}\right|_{s=0} \left(\alpha^Y(t)_* \left.\frac{d}{dt}\right|_{t=0}\right)(f \circ c(\alpha^X(s))) \\
&= \left.\frac{d}{ds}\right|_{s=0} \left.\frac{d}{dt}\right|_{t=0} f(\alpha^X(s)\alpha^Y(t)\alpha^X(-s)) \\
&= \left.\frac{d}{dt}\right|_{t=0} \left.\frac{d}{ds}\right|_{s=0} f(\alpha^X(s)\alpha^Y(t)\alpha^X(-s)) \\
&= \left.\frac{d}{dt}\right|_{t=0} \left(\left.\frac{d}{ds}\right|_{s=0} f(\alpha^X(s)\alpha^Y(t)) + \left.\frac{d}{ds}\right|_{s=0} f(\alpha^Y(t)\alpha^X(-s))\right) \\
&= XYf - YXf = [X, Y]f
\end{aligned}$$

Here we used chain rule: for real-valued $f(x, y)$, we have $\left.\frac{d}{dt}\right|_{t=0} f(t, t) = f_x(0, 0) + f_y(0, 0)$. □

Example W.37. A finite dimensional real vector space V, as a Lie group, coincides with its Lie algebra T_0V, and for $v \in T_0V = V$, $\alpha^v(t) := tv$ is the integral curve (one parameter subgroup) of v. T_0V is an abelian Lie algebra, for its adjoint representation is a trivial map. Generally, an abelian Lie group has abelian Lie algebra.

The torus $\mathbb{R}^n/\mathbb{Z}^n$ has Lie algebra $\mathbb{R}^n$, and the one parameter subgroup for $v \in \mathbb{R}^n$ is $\alpha^v(t) = tv \bmod \mathbb{Z}^n$.

Example W.38 (General linear groups)**.** Let V be a finite dimensional real/complex/quaternionic vector space. The group $\mathrm{Aut}\, V$ of automorphisms on V has Lie algebra $\mathrm{End}\, V$. The one parameter subgroup for $X \in \mathrm{End}\, V$ is

$$\alpha^X(t) := e^{tX} = \sum_{n=0}^{\infty} \frac{(tX)^n}{n!}$$

In the proof of Proposition W.36, we see

$$\mathrm{ad}_X Y = \left.\frac{\partial}{\partial s}\right|_{=0} \left.\frac{\partial}{\partial t}\right|_{t=0} \alpha^X(s)\alpha^Y(t)\alpha^X(-s)$$

We use this formula to compute the Lie bracket on $\mathrm{End}\, V$. We have

$$\alpha^X(s)\alpha^Y(t)\alpha^X(-s) \equiv (1 + sX)(1 + tY)(1 - sX) \equiv 1 + tY + st(XY - YX) \pmod{(s^2, t^2)}$$

Taking derivatives yields $\mathrm{ad}_X Y = XY - YX$.

Also, the adjoint representation $\mathrm{Ad}(g) : \mathrm{End}\, V \to \mathrm{End}\, V$ is given by $\mathrm{Ad}(g)Y = gYg^{-1}$. To see this,

$$c(g)\alpha^Y(t) = ge^{tY}g^{-1} = e^{tgYg^{-1}} = \alpha^{gYg^{-1}}(t)$$

Differentiating at $t = 0$, we obtain $\mathrm{Ad}(g)Y = gYg^{-1}$.

Example W.39 (Matrix groups)**.** By Proposition W.34, if $G \subseteq \operatorname{Aut} V$ is a closed Lie subgroup, then the exponential map on G is the same as that of on $\operatorname{Aut} V$. Hence for $X \in \operatorname{Lie}(G) \leqslant \operatorname{End} V$, $\exp_G X = \sum_{n=0}^{\infty} \frac{(tX)^n}{n!}$

- $\operatorname{Lie}(\mathrm{SL}(n)) = \mathfrak{sl}(n) := \{A \in \mathfrak{gl}(n) \mid \operatorname{tr} A = 0\}$.
- $\operatorname{Lie}(\mathrm{SO}(n)) = \mathfrak{so}(n) := \{A \in \mathfrak{gl}_n(\mathbb{R}) \mid A + A^T = 0\}$.
- $\operatorname{Lie}(\mathrm{U}(n)) = \mathfrak{u}(n) := \{A \in \mathfrak{gl}_n(\mathbb{C}) \mid A + A^* = 0\}$.
- $\operatorname{Lie}(\mathrm{SU}(n)) = \mathfrak{su}(n) := \{A \in \mathfrak{u}(n) \mid \operatorname{tr} A = 0\}$.
- $\operatorname{Lie}(\mathrm{Sp}(n)) = \mathfrak{sp}(n) := \{A \in \mathfrak{gl}_n(\mathbb{H}) \subseteq \mathfrak{gl}_{2n}(\mathbb{C}) \mid A + A^* = 0\} = \left\{ \begin{pmatrix} A & -\overline{B} \\ B & \overline{A} \end{pmatrix} \in \mathfrak{u}(2n) \mid A, B \in M_n(\mathbb{C}) \right\}$

Note that $\exp_{\mathrm{SO(n)}}$ and $\exp_{\mathrm{U(n)}}$ are surjective, while $\exp_{\mathrm{SL_2(\mathbb{R})}}$ is not.

W.3.3 Applications

Proposition W.40 (No small subgroup)**.** Let G be a Lie group. Then G admits a unit-neighborhood U such that U contains no nontrivial subgroup of G.

Proof. Let V, U be unit neighborhoods of $\operatorname{Lie}(G)$ and G, respectively, such that $\exp_G : V \to U$ is a homeomorphism. If $x \in U$ is such that $x^n \in U$ for all $n \in \mathbb{Z}$, say $x = \exp a$ for $a \in V$, then $x^n = (\exp a)^n = \exp(na)$, and thus $na \in V$ for all $n \in \mathbb{Z}$. If we pick V to a bounded set, then this forces $a = 0$, and thus $x = e$. □

Lemma W.41. A connected topological group G is generated (as groups) by any unit-neighborhood.

Proof. Let U be an open unit-neighborhood of G. Then the subgroup $\langle U \rangle = \bigcup_{n=1}^{\infty} U^n$ generated by U is open, and hence closed. G being connected, it follows that $G = \langle U \rangle$. □

Proposition W.42. A homomorphism of connected Lie groups is determined by its differential at the identity.

Proof. Let $f : G \to H$ be a homomorphism of Lie groups. Consider the diagram

$$\begin{tikzcd} \operatorname{Lie}(G) \arrow[r, "f_{*,e}"] \arrow[d, "\exp_G"] & \operatorname{Lie}(H) \arrow[d, "\exp_H"] \\ G \arrow[r, "f"'] & H \end{tikzcd}$$

Since $\exp_G$ is a local diffeomorphism, we can find an open unit-neighborhood U of $\operatorname{Lie}(G)$ such that $\exp_G|_U$ is bijective and $\exp_G(U) \subseteq G$ is open. Since G is connected, $G = \langle \exp_G(U) \rangle$, and the result follows. □

Corollary W.42.1. Let G be a connected Lie group. TFAE:

1. G is abelian.
2. $\mathrm{Ad} : G \to \mathrm{Aut}\,\mathrm{Lie}(G)$ is trivial.
3. $\mathrm{ad} : \mathrm{Lie}(G) \to \mathrm{End}\,\mathrm{Lie}(G)$ is trivial.

Corollary W.42.2. Every compact connected holomorphic Lie group G is abelian.

Proof. The adjoint representation $\mathrm{Ad} : G \to \mathrm{Aut}\,\mathrm{Lie}(G)$ is constant by the Liouville's theorem. □

Proposition W.43. For a Lie group G, its identity component G^0 is abelian if and only if $\exp : \mathrm{Lie}(G) \to G$ is a homomorphism.

Proof. WLOG we assume $G = G^0$ is connected. If $\exp : \mathrm{Lie}(G) \to G$ is a homomorphism, then since $\mathrm{Lie}(G)$ is an abelian group, its image is abelian, and since $G = \langle \exp_G \mathrm{Lie}(G) \rangle$, G is abelian. Suppose G is abelian. Then the multiplication $\mu : G \times G \to G$ is a Lie group homomorphism, and the result follows from the naturality. □

Corollary W.43.1. If G^0 is abelian, then $\exp : \mathrm{Lie}(G^0) = \mathrm{Lie}(G) \to G^0$ is surjective.

Lemma W.44. Let G, H be Lie groups and $f : G \to H$ a bijective (abstract) group homomorphism. If f is a local diffeomorphism, then $f : G \to H$ is a Lie group isomorphism.

Theorem W.45. A connected abelian Lie group G is isomorphic to $\mathbb{T}^t \times \mathbb{R}^r$.

Proof. The exponential map $\exp_G : \mathrm{Lie}(G) \to G$ is a surjective homomorphism, so we have an abstract group isomorphism $G \cong \mathrm{Lie}(G)/\ker \exp_G$. Since $\exp_G$ is a local diffeomorphism, $\ker \exp_G \subseteq \mathrm{Lie}(G)$ is discrete; in particular, this shows $G \cong \mathrm{Lie}(G)/\ker \exp_G$ is a local diffeomorphism, if we view $\mathrm{Lie}(G)/\ker \exp_G$ as a product of a torus and a vector space. Hence $G \cong \mathrm{Lie}(G)/\ker \exp_G$ is a Lie group isomorphism.

Here we use the fact that every discrete subgroup of $\mathbb{R}^n$ is a finite rank abelian subgroup. □

Corollary W.45.1. A compact abelian Lie group G is isomorphism to $\mathbb{T}^t \times \mathbb{R}^r \times F$, where F is a finite abelian group.

Proof. The identity component G^0 of G is connected abelian, so $G^0 \cong \mathbb{T}^t \times \mathbb{R}^r$, and since G is compact, $G/G^0 =: F$ is finite abelian. So far we have an short exact sequence of abelian groups

$$0 \longrightarrow G^0 \longrightarrow G \xrightarrow{p} F \longrightarrow 0$$

Since G^0 is divisible, this short exact sequence splits, so we can find a section $s : F \to G$ of p. Consider the maps

$$\begin{aligned} G^0 \times F &\longrightarrow G \\ (g, f) &\longmapsto g + s(f) \\ (g - (s \circ p)(g), p(g)) &\longleftarrow\!\shortmid\; g \end{aligned}$$

They are mutually inverses, so we have an abstract group isomorphism. Again, this is a local diffeomorphism, so $G^0 \times F \cong G$ as Lie groups. $\square$

W.3.4 Universal enveloping algebras

In this subsubsection, let k be any field and L an (possibly infinite dimensional) Lie algebra over k. Notice that a associative algebra A is naturally a Lie algebra: $(a, b) \times A \times A \mapsto ab - ba$ is clearly a Lie bracket.

Definition. The universal enveloping algebra of L is an associative unital algebra $U(L)$ over k together with a k-linear map $\iota : L \to U(L)$ representing the functor $\mathbf{Alg}_k \ni A \mapsto \mathrm{Hom}_{\mathbf{LieAlg}_k}(L, A)$. In other words, the map

$$\begin{array}{ccc} \mathrm{Hom}_{\mathbf{Alg}_k}(U(L), A) & \longrightarrow & \mathrm{Hom}_{\mathbf{LieAlg}_k}(L, A) \\ \phi & \longmapsto & \phi \circ \iota \end{array}$$

is a bijection functorial in A. Here $\mathbf{Alg}_k$ denotes the category of unital associative k-algebras and $\mathbf{LieAlg}_k$ stands for the obvious category.

The existence can be easily established. Indeed, let J be the two sided ideal of the tensor algebra TL generated by the elements $x \otimes y - y \otimes x - [x, y]\,(x, y \in L)$. Put

$$U(L) = TL/J$$

and denote by $\iota : L \to U(L)$ the composition of natural maps $L \to TL \to U(L)$. One readily checks that $(U(L), \iota)$ really represents the functor described above.

Let $\pi : TL \to U(L)$ be the natural projection. Put $T_m L = \bigoplus_{0 \leqslant n \leqslant m} T^n L$ and $U_m = \pi(T_m)$; for convenience, put $U_{-1} := 0$. In this way we obtain a filtration

$$0 = U_{-1} \subseteq k = U_0 \subseteq U_1 \subseteq \cdots \subseteq U_m \subseteq U_{m+1} \subseteq \cdots \subseteq U(L)$$

of the universal enveloping algebra $U(L)$. The filtration respects the multiplication: $U_p U_q \subseteq U_{p+q}$, which makes the associated $\mathbb{Z}_{\geqslant 0}$-graded abelian group

$$G = \bigoplus_{m \geqslant 0} G^m := \bigoplus_{m \geqslant 0} U_m / U_{m-1}$$

a unital associative k-algebra.

Since $\pi(T^m L) \subseteq \pi(T_m) = U_m$, the composition $\phi_m : T^m L \to U_m \to G^m$ of natural maps is well-defined. As $\pi(T_m \backslash T_{m-1}) = U_m \backslash U_{m-1}$, the map ϕ_m is surjective. The maps $\phi_m\,(m \geqslant 0)$ then together define a surjective linear map $\phi : TL \to G$ that sends 1 to 1. It is easy to see that ϕ is an algebra homomorphism. Moreover,

Lemma W.46. $\phi(x \otimes y - y \otimes x) = 0 \in G$ for any $x, y \in L$.

Proof. By definition, $\pi(x \otimes y - y \otimes x) \in U_2$. On the other hand,

$$\pi(x \otimes y - y \otimes x) = \pi(x)\pi(y) - \pi(y)\pi(x) = [\pi(x), \pi(y)] = \pi([x, y]) \in U_1,$$

so that $\phi(x \otimes y - y \otimes x) \in U_1/U_1 = 0$. □

By the lemma, ϕ factors through the projection $TL \to \text{Sym } L$, inducing a unital k-algebra homomorphism $\omega : \text{Sym } L \to G$.

Theorem W.47 (Poincaré-Birkhoff-Witt; PBW)**.** The homomorphism

$$\omega : \text{Sym } L \to G$$

is an isomorphism of (graded) k-algebras.

W.4 Lie Subgroups

Definition. A **Lie subgroup** of a Lie group G is an injective homomorphism of Lie groups $f : H \to G$.

- H needs not be a regular submanifold of G. Nevertheless, H is always an immersed submanifold of G since such an f must have constant rank. One way to see this is using naturality of exps.

Lemma W.48. Let G be a Lie subgroup and $f : H \to G$ a Lie subgroup. Then

$$f_{*,e}(\text{Lie}(H)) = \{X \in \text{Lie}(G) \mid \exp_G(tX) \in f(H) \text{ for all } t \in \mathbb{R}\}.$$

Proof. Let $X \in \text{Lie}(H)$. By naturality we have $f(\exp_H(tX)) = \exp_G(f_{*,e}(tX))$ for all $t \in \mathbb{R}$, so $f_{*,e}(X)$ lies in RHS. Conversely, let $X \in \text{Lie}(G)$ such that $\exp_G(tX) \in f(H)$ for all $t \in \mathbb{R}$. Define $\gamma : \mathbb{R} \to H$ by $\gamma(t) := f^{-1}(\exp_G(tX))$, and put $Y = \gamma_{*,0}\left(\left.\frac{d}{dt}\right|_{t=0}\right) \in \text{Lie}(H)$. Then $\gamma(t) = \exp_H(tY)$, and by naturality

$$\exp_G(tf_{*,e}(Y)) = f(\exp_H(tY)) = f(\gamma(t)) = \exp_G(tX).$$

Since $\exp_G$ is a local diffeomorphism, this implies $X = f_{*,e}(Y)$. □

Theorem W.49 (Cartan)**.** An abstract subgroup H of a Lie group G is a regular submanifold of G if and only if H is closed in G.

Proof. Note that a regular submanifold is locally closed. Thus we can find an open unit-neighborhood U of G such that $H \cap U$ is closed in U. We claim $H = \overline{H}$. For $h \in \overline{H}$, by definition we have $g \in hU \cap H$, so $h^{-1}g \in U \cap \overline{H} = U \cap H$, hence $h \in H$. Hence H is closed.

Conversely, we assume H is a closed abstract subgroup of G. By translation, it suffices to find an open unit-neighborhood of G on which H is a submanifold.

1° Let V, U be open unit-neighborhoods of $\text{Lie}(G)$ and G, respectively, such that $\exp_G : V \to U$ is a smooth diffeomorphism. Put $\log : U \to V$ to be its inverse.

2° Consider the set $B = \{X \in \mathrm{Lie}(G) \mid \exists (h_n) \subseteq \log(H \cap U) \text{ with } h_n \to 0 \text{ such that } \frac{h_n}{|h_n|} \to X\}$. Then $\exp_G tX \in H$ for all $t \in \mathbb{R}$ and $X \in B$.

Since $|h_n| \to 0$, if we take $m_n = \left\lfloor \frac{t}{h_n} \right\rfloor \in \mathbb{Z}$, we have $m_n|h_n| \to t$, so that

$$\exp m_n h_n = \exp\left(m_n|h_n|\frac{h_n}{|h_n|}\right) \to \exp tX$$

On the other hand, $\exp m_n h_n = (\exp h_n)^{m_n} \in H$. Since H is closed, $\exp tX \in H$.

3° The set $\mathfrak{h} = \{tX \mid t \in \mathbb{R},\ X \in B\}$ is a linear subspace of $\mathrm{Lie}(G)$.

Let $X, Y \in \mathfrak{h}$ and consider $h(t) := \log(\exp(tX)\exp(tY))$. Then $\lim_{t\to 0} \frac{h(t)}{t} = h_{*,0}\left(\left.\frac{d}{dt}\right|_{t=0}\right) = X + Y$, and $\frac{h(t)}{|h(t)|} = \frac{h(t)}{t}\frac{t}{|h(t)|} \to \frac{X+Y}{|X+Y|}$. Thus $\frac{X+Y}{|X+Y|} \in B$ and $X + Y \in \mathfrak{h}$.

4° $\exp \mathfrak{h}$ is a unit-neighborhood of H.

Let D be any linear complement of $\mathfrak{h}$ in $\mathrm{Lie}(G)$. Consider the map

$$\begin{aligned} \psi : D \oplus \mathfrak{h} &\longrightarrow G \\ (X, Y) &\longmapsto \exp X \exp Y \end{aligned}$$

Since $\exp_* = \mathrm{id}_{\mathrm{Lie}(G)}$, ψ is a local diffeomorphism. We prove 4° by contradiction. Choose $(X_n, Y_n) \in D \oplus \mathfrak{h}$ with $\psi(X_n, Y_n) \in H$, $X_n \neq 0$ and $(X_n, Y_n) \to 0$. Since D is closed, we can find $X \in D$ such that, by passing to subsequences, $\frac{X_n}{|X_n|} \to X$; since $|X| = 1$, $X \neq 0$. Because $\exp Y_n \in H$ by 2°, $\exp X_n \in H$ as well, so $X \in B \subseteq \mathfrak{h}$ by 2°, a contradiction.

This completes the proof because by 4° we can find an open unit-neighborhood W of G such that $W \cap H = U \cap \exp \mathfrak{h}$ and on it exp is invertible, so we obtain a chart of H around the identity. In addition, by 4°and Lemma W.48,

5° The equality holds:

$$\mathrm{Lie}(H) = \mathfrak{h} = \{X \in \mathrm{Lie}(G) \mid \exp_G(tX) \in H \text{ for all } t \in \mathbb{R}\}$$

□

Corollary W.49.1. Let G, H be Lie groups and $f : G \to H$ be a continuous abstract group homomorphisms. Then f is automatically smooth, so f is a Lie group homomorphism.

Proof. Consider the graph of f:

$$\Gamma_f := \{(g, f(g)) \mid g \in G\} \subseteq G \times H$$

By assumption Γ_f is a closed subgroup of $G \times H$, so Γ_f is a closed Lie subgroup of $G \times H$. Now consider the first projection $\pi = \mathrm{pr}_1|_{\Gamma_f} : \Gamma_f \to G$; it is clear that π is a smooth homeomorphism. We claim $\pi_{*,e}$ is an isomorphism. Assuming this, we see π^{-1} is a local diffeomorphism, and hence π^{-1} is smooth everywhere. Hence $f = \mathrm{pr}_2 \circ \pi^{-1}$ is smooth.

View $\mathrm{Lie}(\Gamma_f) \subseteq \mathrm{Lie}(G) \times \mathrm{Lie}(H)$. Then $\mathrm{Lie}(\Gamma_f) \cap \mathrm{Lie}(H) = 0$. For if $X \in \mathrm{Lie}(\Gamma_h) \cap \mathrm{Lie}(H)$, then $\exp tX \in \Gamma_f \cap H = \{e\}$, i.e., $X = 0$. This shows $\pi_* : \mathrm{Lie}(\Gamma_f) \to \mathrm{Lie}(G)$ is injective. Hence $\dim \mathrm{Lie}(\Gamma_f) \leqslant \dim \mathrm{Lie}(G)$ with equality if and only if π_* is an isomorphism. But π is a homeomorphism, by invariance of dimension, we must have $\dim \Gamma_f = \dim G$ as smooth manifolds, and hence equality holds. $\square$

Remark. Alternatively, we can show $\dim \mathrm{Lie}(\Gamma_f) = \dim \mathrm{Lie}(G)$ by measure theory. If $\dim \mathrm{Lie}(\Gamma_f) < \dim \mathrm{Lie}(G)$, then the (smooth) image of Γ_f in G is of measure zero, a contradiction to the bijectivity of π. Note that here the second countability of Γ_f and G are used. However, if we do not impose the second countability on the definition of manifolds, the proof still works with slight modification. To show injectivity of π_* we may replace Γ_f by its identity component, and thus we may assume G and Γ_f are connected; but this automatically forces G and Γ_f to be second countable, thanks to the fact euclidean spaces are second countable.

W.5 Correspondence between Lie Groups and Lie Algebras

Definition. Let $p : \tilde{X} \to X$ be a continuous map between topological space. p is called a **covering map** if each $x \in X$ admits an open neighborhood U such that $f^{-1}(U)$ is a union of disjoint open sets in $\tilde{X}$ each of which being mapped homeomorphically onto U by p.

Lemma W.50. Let $f : G \to H$ be a continuous group homomorphism between topological groups. If H is connected and there exists a unit-neighborhood U of G such that $f(U)$ is open and $f|_U^{f(U)} : U \to f(U)$ is a homeomorphism, then f is a covering map.

Proof. We have $f^{-1}(f(U)) = \bigsqcup_{x \in \ker f} Ux$. Hence the covering property is verified at the identity of H. Since H is connected and $f(U)$ is open, f is surjective. Then for $h \in H$ and any $g \in f^{-1}(h)$, we have $f^{-1}(f(gU)) = \bigsqcup_{x \in \ker f} gUx$.

$\square$

Theorem W.51. Let $f : G \to H$ be a homomorphism of Lie groups.

(i) The differential $f_* : \mathrm{Lie}(G) \to \mathrm{Lie}(H)$ is a Lie algebra homomorphism.

(ii) $\ker f_* = \mathrm{Lie}(\ker f)$. Thus f_* is injective if and only if $\ker f$ is discrete.

(iii) If H is connected, f_* is surjective if and only if f is surjective, and f_* is an isomorphism if and only if f is a covering map.

Proof.

1. This is Proposition W.32.

2. For $X \in \mathrm{Lie}(G)$, by naturality $f_* X = 0$ implies $f(\exp(tX)) = 0$ for all $t \in \mathbb{R}$, and vice versa. By Lemma W.48, the last condition is equivalent to $X \in \mathrm{Lie}(\ker f)$. For the second assertion, note that for a Lie group G, $\mathrm{Lie}(G) = 0$ if and only if G is discrete.

3. If f_* is surjective, then by Constant rank theorem the image of f contains a unit neighborhood U of H. Since H is connected, $H = \langle U \rangle \subseteq f(G) \subseteq H$. If f is surjective and f is nowhere submersive, then by Sard's theorem, $f(G) \subseteq H$ is of measure zero, a contradiction. Hence f is submersive at some point of G, and by translation f_* is surjective.

   If f_* is an isomorphism, f is a local diffeomorphism, and hence by Lemma f is a covering map. Conversely, if f is a covering map, then $\ker f$ is discrete and f is surjective, so f_* is bijective by (ii) and the first statement of (iii).

□

Proposition W.52. Let G be a Lie group. For $X, Y \in \mathrm{Lie}(G)$, $[X, Y] = 0$ if and only if $e^{tX} e^{sY} = e^{sY} e^{tX}$.

Proof.

$$\mathrm{ad}_X Y = \mathrm{Ad}_{*,e}(X) Y = \frac{d}{dt}\bigg|_{t=0} \mathrm{Ad}(e^{tX}) Y = \frac{d}{dt}\bigg|_{t=0} c(e^{tX})_* Y = \frac{d}{dt}\bigg|_{t=0} (c(e^{tX}) e^{sY})_*$$

Hence $\mathrm{ad}_X Y = 0$ if and only if $(c(e^{tX}) e^{sY})_*$ is constant in t. But it is

$$(c(e^{tX}) e^{sY})_* = (e^{tX} e^{sY} e^{-tX})_* = e^{sY}_* = Y$$

so $c(e^{tX}) e^{sY}$ is an integral curve of Y starting from e, and thus $c(e^{tX}) e^{sY} = e^{sY}$. □

Corollary W.52.1. For a connected Lie group, we have $Z(\mathrm{Lie}(G)) = \mathrm{Lie}\, Z(G)$.

Proof. Note that $Z(G)$ is a closed subgroup of G, so it is a Lie group by Theorem W.49. For $X \in \mathrm{Lie}(G)$, by the last proposition we have $X \in Z(\mathrm{Lie}(G))$ if and only if e^{tX} commutes with elements of the form e^Y. But such elements generate G, so e^{tX} commutes with G, i.e., $e^{tX} \in Z(G)$. Taking differential yields $X \in \mathrm{Lie}(Z(G))$. □

Theorem W.53. Let G be a connected Lie group. Then the map

$$\begin{aligned} \{H \leqslant G : \text{Lie subgroup}\} &\longrightarrow \{\mathfrak{h} \subseteq \mathrm{Lie}(G) : \text{Lie subalgebra}\} \\ H &\longmapsto \mathrm{Lie}(H) \end{aligned}$$

is a bijection with inverse $\mathfrak{h} \mapsto \langle \exp_G \mathfrak{h} \rangle \leqslant G$.

Proof. The injectivity is clear. Let $\mathfrak{h} \subseteq \mathrm{Lie}(G)$ be a Lie subalgebra. By regarding $\mathrm{Lie}(G)$ as the space of left-invariant vector fields L_G of G, the subalgebra $\mathfrak{h}$ corresponds to an involutive distribution of TG of rank $\dim_{\mathbb{R}} \mathfrak{h}$. By Frobenius integrability theorem, there exists a unique connected (immersed) submanifold $f : H \to G$ such that $f_{*,h}(T_h H) = (\ell_{f(h)})_{*,e}\mathfrak{h}$ for all $h \in H$. Define $\mathrm{mult}_H : H \times H \to G$ and $\mathrm{inv}_H : H \to G$ by $\mathrm{mult}_H = \mathrm{mult}_G \circ (f \times f)$ and $\mathrm{inv}_H = \mathrm{inv}_G \circ f$. We claim their image lie in H, and by Lemma W.17 they are automatically smooth with respect to the smooth structure of H. It suffices to show (we use usual notations for groups) $hk^{-1}H = H$ for $h, k \in H$. Note that $e \in hk^{-1}H$ and it is connected, so we only need to show it is an integral submanifold for $\mathfrak{h}$. This follows from the uniqueness part of integrability theorem, as $T_{hk^{-1}p}hk^{-1}H = (\ell_{hk^{-1}p})_{*,e}\mathfrak{h} = T_{hk^{-1}p}H$. Hence H is a Lie subgroup of G with Lie algebra $\mathfrak{k}$. This shows the surjectivity.

□

Theorem W.54. Let G, H be two connected Lie groups with G simply connected. If $\varphi : \mathrm{Lie}(G) \to \mathrm{Lie}(H)$ is a Lie algebra homomorphism, then there exists a Lie group homomorphism $f : G \to H$ such that $f_{*,e} = \varphi$.

Proof. Define

$$\mathfrak{a} = \{(X, \varphi(X)) \in \mathrm{Lie}(G) \oplus \mathrm{Lie}(H) \mid X \in \mathrm{Lie}(G)\} \subseteq \mathrm{Lie}(G \times H).$$

Note the Lie bracket on $\mathrm{Lie}(G \times H)$ is the same as that on the Lie algebra direct sum $\mathrm{Lie}(G) \oplus \mathrm{Lie}(H)$. Since φ preserves Lie bracket, $\mathfrak{a}$ is a Lie subalgebra of $\mathrm{Lie}(G \times H)$. By Theorem W.53, there exists a connected Lie subgroup $D \leqslant G \times H$ with Lie algebra $\mathfrak{a}$. The composition $g : D \subseteq G \times H \xrightarrow{\mathrm{pr}_1} G$ is a Lie group homomorphism whose differential is $\mathfrak{a} \xrightarrow{\mathrm{pr}_1} \mathrm{Lie}(G)$, an isomorphism. By Theorem W.51.(iii), g is a covering map. Since we assume G is simply connected, g is actually an isomorphism. Then the homomorphism $f := \mathrm{pr}_2 \circ g^{-1} : G \to H$ does the job.

□

W.6 Lie Group Actions

W.6.1 Topological Group Actions

For G an abstract group, a **(left) G-set** is a nonempty set X together with a group homomorphism $G \to S_X$, where S_X denotes the group of all bijections on X. For two G-sets X, Y, a set-theoretic map $f : X \to Y$ is called G-equivariant if $f(gx) = gf(x)$ for all $x \in X$ and $g \in G$. In this way we obtain the category of G-sets.

Let G be a topological group. A **(left) G-space** is a nonempty topological space X on which G acts on the left such that the action map $G \times X \to X$ is continuous. For two G-spaces X, Y, a continuous map $f : X \to Y$ is called **G-equivariant** if $f(gx) = gf(x)$ for all $g \in G$ and $x \in X$.

- For $x \in X$, the subset $Gx := \{gx \mid g \in G\} \subseteq X$ is called the **G-orbit** of x, and the subgroup $G_x := \{g \in G \mid gx = x\}$ is called the **isotropy group** of x, or the **stabilizer** of x.

- The set of all G-orbits is denoted by ${}_G\backslash^X$, and is called the **orbit space**. We equip ${}_G\backslash^X$ with the final topology with respect to the projection $X \to {}_G\backslash^X$.
- With the above topology, ${}_G\backslash^X$ is a G-space and the projection $X \to {}_G\backslash^X$ is open and G-equivariant.
- If G is a Lie group, a **smooth G-space** is required to be a smooth manifold on which G acts smoothly.

Definition. Let G be a group and X a G-set.

1. The action is **faithful/effective** if the associated map $G \to S_X$ is injective, where S_X is the group of all bijections on X.
2. The action is **free** if all stabilizers are trivial.
3. The action is **transitive** if $\#{}_G\backslash^X = 1$. In this case, X is called a **homogeneous** G-set.
4. If the action is free and transitive, then X is called a **principal homogeneous G-set**, or a **G-torsor**.

- For a subgroup $H \leqslant G$, the left coset space G/H is a homogeneous G-set. Conversely, if X is a homogeneous G-set, any point $x \in X$ gives a set-theoretic G-equivariant bijection $G/G_x \cong X$.
- For X a homogeneous space and $x \in X$, the map $G/G_x \to X$ is always continuous, but it may not be a homeomorphism.
- For a subgroup H, the G-action on G/H is faithful if and only if H contains no trivial normal subgroup.

Example W.55. Consider the real line $(\mathbb{R}, +)$ with usual euclidean topology and the torus $(\mathbb{R}^2/\mathbb{Z}^2, +)$ with usual topology. Let $\alpha \in \mathbb{R} \backslash \mathbb{Q}$ and consider the topological subgroup $F := \{(x, \alpha x) \mid x \in \mathbb{R}\} \leqslant \mathbb{R}^2/\mathbb{Z}^2$. This is isomorphic to $\mathbb{R}$ as abstract groups, and the identity map $\mathbb{R} \to F$ is continuous, but it is not open.

Proposition W.56. Let G a topological group and $H \leqslant G$ a subgroup. Equip the left coset space G/H with the quotient topology.

1. The projection $\pi : G \to G/H$ is an open map. If H is compact, then π is also a closed map.
2. If H is normal, G/H is a topological group.
3. The natural G-action on G/H is continuous, and G/H is a homogeneous G-space.
4. If G is Hausdorff, then G/H is Hausdorff if and only if H is closed.
5. If H and G/H are compact, then so is G.
6. If H and G/H are connected, then so is G.

Proof.

1. For $U \subseteq G$ open, $\pi^{-1}\pi(U) = \bigcup_{g \in H} gU$ is open. If H is compact, for $C \subseteq G$ closed, $\pi^{-1}\pi(C) = CH$ is closed.

4. If G/H is Hausdorff, then $H = \pi^{-1}(eH)$ is closed. Suppose H is closed. We must show the diagonal $\Delta = \Delta_{G/H}$ is closed in $G/H \times G/H$. It suffices to show $(\pi \times \pi)^{-1}(\Delta)$ is closed in $G \times G$. Note that $(g, g') \in (\pi \times \pi)^{-1}(\Delta)$ if and only if $gH = g'H$, i.e, $g^{-1}g' \in H$. Hence $(\pi \times \pi)^{-1}(\Delta) = q^{-1}(H)$ is a closed set, where $q : G \times G \to G$ is defined by $q(gg') = g^{-1}g'$.

5. $\pi : G \to G/H$ is closed (by 1.) with compact fibre (each of which homeomorphic to H), so it is proper by Proposition T.6. Since G/H is compact by assumption, $G = \pi^{-1}(G/H)$ is compact.

6. We prove if G/H is connected and G is not connected, then H is not connected. Let U, V be disjoint nonempty open sets in G with $U \cup V = G$. Since π is open, by connectivity of G/H we have $\pi(U) \cap \pi(V) \neq \varnothing$, i.e., we can find a coset gH with $U \cap gH \neq \varnothing \neq V \cap gH$. But then $gH = (U \cup V) \cap gH = (U \cap gH) \cup (V \cap gH)$ so $gH \cong H$ is not connected.

□

Lemma W.57. Let X be a homogeneous G-space. Suppose there exists $x_0 \in X$ such that every open unit-neighborhood U of G the subspace $Ux_0 \subseteq X$ contains x_0 in its interior. Then the natural map $G/G_x \to X$ is open for all $x \in X$, and hence a homeomorphism.

Proof. By translation, it suffice to show $G/G_{x_0} \to X$ is open. We must show for every open subset V of G, $Vx_0 \subseteq X$ is open. Let $g \in V$; then $g^{-1}V$ is an open unit neighborhood of G, so x_0 lies in the interior of $g^{-1}Vx_0$. Hence gx_0 lies in the interior of Vx_0 for all $g \in V$, and the proof is done. □

Theorem W.58. Let X be a homogeneous G-space. Suppose

- G is LCH and separable, and
- X is Hausdorff and a Baire space.

Then $G/G_x \to X$ are homeomorphisms for all $x \in X$.

Proof. Let $x_0 \in X$ be any point and U an open unit-neighborhood of G. By the last lemma, it suffices to show x_0 lies in the interior of Ux_0. Let $W \subseteq U$ be a compact symmetric unit-neighborhood such that $W^2 \subseteq U$. Let $S \subseteq G$ be a countable dense subset. Then it is easy to see $G = \bigcup_{s \in S} sW$, so $X = \bigcup_{s \in S} sWx_0$ is a countable union of compact sets which are closed since X is Hausdorff. Since X is Baire, $s'Wx_0$ has nonempty interior for some $s' \in S$, i.e., $s'gx_0$ lies in the interior of $s'Wx_0$ for some $g \in W$. Hence x_0 lies in the interior of $g^{-1}Wx_0 = Wx_0 \subseteq Ux_0$, as wanted. □

W.6.2 Principal Bundles

Definition. A **fibre bundle** (E, B, π, F) consisting of three topological spaces E, B, F together with a continuous surjection $\pi : E \to B$ such that each point $x \in B$ has an open neighborhood U for which there is a homeomorphism $\phi : \pi^{-1}(U) \to U \times F$ making the triangle

$$\begin{tikzcd} \pi^{-1}(U) \arrow[rr, "\phi"] \arrow[dr, "\pi"'] & & U \times F \arrow[dl, "\mathrm{pr}_1"] \\ & U & \end{tikzcd}$$

commute, where $\pi^{-1}(U)$ is given the subspace topology from E.

- E is called the **total space**, B the **base space**, F the **fibre space** and π the **bundle projection**. The homeomorphism ϕ in the definition is called a **local trivialization**. A fibre bundle (E, B, π, F) is sometimes written as $F \to E \xrightarrow{\pi} B$, or simply $E \xrightarrow{\pi} B$.

- ϕ induces a homeomorphism on the fibre $\pi^{-1}(b) =: E_b \to \{b\} \times F$ for each $b \in U$. Also, since pr_1 is open, the bundle projection π is an open map.

- A cover map $p : Y \to X$ with X connected is a fibre bundle $(Y, X, p, p^{-1}(x))$, where $x \in X$ is an arbitrary point and $p^{-1}(x)$ is a discrete subspace of Y. Moreover, a fibre bundle (E, B, π, F) is a cover if and only if F is discrete.

A **morphism between fibre bundles** (E, B, π, F) and (E', B', π', F') consists of two continuous maps $\phi : E \to E'$ and $\psi : B \to B'$ making the diagram

$$\begin{tikzcd} E \arrow[r, "\phi"] \arrow[d, "\pi"] & E' \arrow[d, "\pi'"] \\ B \arrow[r, "\psi"] & B' \end{tikzcd}$$

commute.

- Since π is surjective, ψ is determined by ϕ. We say ϕ is **the bundle map covering ψ.**

Definition. Let G be a topological group. A **principal G-bundle** is a fibre bundle $G \to P \xrightarrow{\pi} X$ together with a continuous right G-action $P \times G \to P$ such that each local trivialization $\pi^{-1}(U) \to U \times G$ is G-equivariant, where G acts on $U \times G$ by right translation on G.

- G is called the **structure group** of the principal bundle. Each local trivialization induces a homeomorphism $P_x \to G$ with inverse $G \to P_x$ given by $g \mapsto xg$.

- By definition, π induces a continuous bijection $P/G \to X$, and each fibre is a G-torsor.
- For a Lie group G, a principal G-bundle $\pi : P \to X$ is **smooth** if all maps and actions involved are smooth. Then the bundle map π is submersive since locally it is $\pi^{-1}(U) \to U \times G \xrightarrow{\mathrm{pr}_1} U$.

A **morphism from a principal G-bundle $P \xrightarrow{\pi} X$ to a principal H-bundle $Q \xrightarrow{p} Y$** is a morphism (ϕ, ψ) of fibre bundles $P \xrightarrow{\pi} X$ to $Q \xrightarrow{p} Y$ together with a continuous group homomorphism $\theta : G \to H$ such that $\phi(yg) = \phi(p)\theta(g)$ for all $y \in P$ and $g \in G$.

Let $P \xrightarrow{\pi} X$ be a principal G-bundle, and let $\phi : \pi^{-1}(U) \to U \times G$ be a local trivialization. Consider the continuous map $\sigma : U = U \times \{1\} \xrightarrow{\phi^{-1}} \pi^{-1}(U) \subseteq P$; clearly, $\pi \circ \sigma = \mathrm{id}_U$, so $\sigma : U \to P$ is a section to the bundle projection $\pi : P \to X$. Put $\psi := \mathrm{pr}_2 \circ \phi$. Then for $g \in G$, we have

$$\psi(\sigma(x)g) = \psi(\sigma(x))g = g$$

This means $\sigma(x)g$ covers the whole fibre P_x when g runs over G, and thus ψ is determined by σ. Therefore ϕ is completely determined by σ; explicitly

$$\phi(\sigma(x)g) = (x, \psi(\sigma(x)g)) = (x, g)$$

Hence to find a local trivialization is the same as to find a local section of π.

Let $\phi : \pi^{-1}(U) \to U \times G$ and $\psi : \pi^{-1}(V) \to V \times G$ be two local trivializations with $U \cap V \neq \varnothing$. On the intersection we have the transition map:

$$\psi \circ \phi^{-1} : (U \cap V) \times G \to (U \cap V) \times G$$

This induces a map $\theta : U \cap V \to G$ satisfying $(\psi \circ \phi^{-1})(x, 1) = (x, \theta(x))$. Since ψ and ϕ are G-equivariant, we have

$$(\psi \circ \phi^{-1})(x, g) = (\psi \circ \phi^{-1})(x, 1)g = (x, \theta(x))g = (x, \theta(x)g)$$

On the other hands, $(\psi \circ \phi^{-1})(x, 1) = (x, \theta(x)) = (x, 1)\theta(x)$ implies $\sigma_\phi(x) = \sigma_\psi(x)\theta(x)$, where σ_ϕ and σ_ψ are their associated local sections.

Let $\phi_i : \pi^{-1}(U_i) \to U_i \times G$ be three local trivializations with $U_1 \cap U_2 \cap U_3 \neq \varnothing$. Then we have three transition maps $\theta_{ij} := \phi_j \circ \phi_i^{-1}$ which satisfy

$$(x, \theta_{13}(x)) = \phi_3\phi_1^{-1}(x, 1) = \phi_3\phi_2^{-1}\phi_2\phi_1^{-1}(x, 1) = \phi_3\phi_2^{-1}(x, \theta_{12}(x)) = (x, \theta_{23}(x)\theta_{12}(x))$$

i.e., the **cocycle condition**

$$\theta_{13}(x) = \theta_{23}(x)\theta_{12}(x)$$

Conversely,

Theorem W.59. Let X be a space with an open cover $\{U_\alpha\}$ and G a topological group. Suppose for each pair (α, β), there exists $\theta_{\alpha,\beta} : U_{\alpha\beta} \to G$ such that each triple (α, β, γ) satisfies the cocycle condition. Then there exists a unique (up to isomorphism) principal G-bundle $P \xrightarrow{\pi} X$ with local trivializations ϕ_α such that the transition maps are given by the $\theta_{\alpha\beta}$.

Definition. Let G be a topological group, $P \xrightarrow{\pi} X$ a principal G-bundle and F a left G-space. Let G act on the product space $P \times F$ diagonally, i.e., $(p, v)g := (pg, g^{-1}v)$. The (right) coset space $(P \times F)/G$ is called the **associated bundle of P with typical fibre F and base X**, and is denoted by $E = P \times^G F$.

- We topologize $P \times_G F$ as usual so that $P \times G \to P \times_G F$ is an open continuous map.
- The bundle projection $\pi : P \to X$ gives rise to a continuous surjection $P \times F \xrightarrow{\mathrm{pr}_1} P \xrightarrow{\pi} X$. Since the G-action on P preserves fibres, the map induces a continuous surjection $\pi_E : P \times^G F \to X$.
- For $(p, v) \in P \times^G F$, denote by $[p, v]$ its equivalence class in $P \times_G F$. For $w, v \in F$, $[p, v] = [p, w]$ if and only if $p = pg$, $v = g^{-1}w$ for some $g \in G$. Since G acts freely on P, we have $g = 1$, so $v = w$. Hence for each $p \in P$ (say $x = \pi(p)$), the map

$$\begin{aligned} F &\longrightarrow E_x \\ v &\longmapsto [p, v] \end{aligned}$$

  is a homeomorphism into. This is also surjective: for $[q, v] \in E_x$, we have $q = pg$ for some $g \in G$, so $[q, v] = [pg, v] = [p, gv]$. Hence *each $p \in P$ parametrizes the fibre E_x ($x = \pi(p)$) by the typical fibre F.*
- Let $\sigma : U \to P$ be a local section of π. The map

$$\begin{aligned} \psi_\sigma : U \times F &\longrightarrow \pi_E^{-1}(U) \\ (x, v) &\longmapsto [\sigma(x), v] \end{aligned}$$

  is a homeomorphism (note that the local section σ is open). Thus the associated bundle $E \xrightarrow{\pi_E} X$ is a fibre bundle.
- Suppose $\sigma : U \to P$ and $\tau : V \to P$ are two local sections, and $\theta : U \cap V \to G$ be the transition map so that $\tau(x) = \sigma(x)\theta(x)$ for $x \in U \cap V$. If we denote by ψ_σ, ψ_τ the associated trivializations, then

$$(\psi_\sigma \circ \psi_\tau^{-1})(x, v) = \psi_\sigma^{-1}([\tau(x), v]) = \psi_\sigma^{-1}([\sigma(x), \theta(x)v]) = (x, \theta(x)v)$$

Proposition W.60. Let G be a Lie group, $P \xrightarrow{\pi} X$ be a smooth principal G-bundle and F a smooth left G-space. Then the associated bundle $E = P \times^G F$ is a smooth manifold with the bundle projection $\pi_E : E \to X$ submersive. In particular, the fibres E_x are closed regular submanifolds of E, and the parametrizations $F \ni v \mapsto [p, v] \in E_{\pi(p)}$ are smooth for all $p \in P$.

W.6.3 Homogeneous Spaces

Proposition W.61. Let G be a Lie group and $H \leqslant G$ a closed subgroup. Then G/H has a unique smooth structure that makes the projection $G \to G/H$ a smooth principal bundle with the structure group H (with H acting on G by right translation).

Proof. Choose a Euclidean metric on $\mathrm{Lie}(G)$, and let $\mathrm{Lie}(G) = V \oplus \mathrm{Lie}(H)$ be an orthogonal decomposition. Let $V_\varepsilon := \{X \in V \mid |X| < \varepsilon\}$ and $D_\varepsilon := \exp V_\varepsilon$. Take $\varepsilon > 0$ so small that $\exp : V_\varepsilon \to D_\varepsilon$ is a diffeomorphism; this makes D_ε is smooth manifold with $T_e D_\varepsilon = V$.

1) <u>For $\varepsilon > 0$ small enough, the map</u>

$$\begin{aligned} \mu : D_\varepsilon \times H &\longrightarrow G \\ (g, h) &\longmapsto gh \end{aligned}$$

<u>is an open embedding</u>. The differential of μ at (e, e) is addition $V \oplus \mathrm{Lie}(H) \to \mathrm{Lie}(G)$, so we can find $\varepsilon > 0$ small enough and some open unit-neighborhood U of H such that $\mu : D_\varepsilon \times U \to D_\varepsilon U$ is diffeomorphism. Translating shows $\mu : D_\varepsilon \times H \to G$ is a local diffeomorphism everywhere.

It remains to show we can make μ injective by choosing ε small enough. Let $d_1, d_2 \in D_\varepsilon$ and $h_1, h_2 \in H$ such that $d_1 h_1 = d_2 h_2$. Then $h_1 h_2^{-1} = d_1^{-1} d_2$. Let $V \underset{\text{open}}{\subseteq} G$ such that $U = V \cap H$ and let $\varepsilon > 0$ so small that $D_\varepsilon^{-1} D_\varepsilon \subseteq V$. Then $h := h_1 h_2^{-1} = d_1^{-1} d_2 \in V \cap H = U$. Since μ is injective on $D_\varepsilon \times U$, that $\mu(d_1, h) = \mu(d_2, e)$ implies $d_1 = d_2$ and $h = e$, or $h_1 = h_2$, showing the global injectivity.

2) The sets $U_g := g D_\varepsilon H$, $g \in G$ are invariant under right H-action. For $g \in G$, define $h_g : U_g / H \to D_\varepsilon$ by

$$h_g^{-1} : D_\varepsilon \xrightarrow[\sim]{} D_\varepsilon \times \{e\} \hookrightarrow D_\varepsilon \times H \xrightarrow[\sim]{\mu} D_\varepsilon H \xrightarrow[\sim]{\ell_g} g D_\varepsilon H =: U_g \xrightarrow{\pi} U_g / H$$

h_g^{-1} is a continuous bijection, and is open since it is a composition of open maps, so it is a homeomorphism. Let $g, g' \in G$ such that we can find $x, y \in D_\varepsilon$ with $h_g^{-1}(x) = h_{g'}^{-1}(y)$. This means $gx = g'yh$ for some $h \in H$, so that $yh = g'^{-1} g x$, which implies $y = \mathrm{pr}_1(y, h) = \mathrm{pr}_1 \mu^{-1}(g'^{-1} g x)$, which is smooth in x. Hence G/H is made into a smooth manifold with the atlas h_g, $g \in G$.

For $g \in G$, define $\varphi_g : U_g \to U_g / H \times H$ by

$$\varphi_g^{-1} : U_g / H \times H \xrightarrow{h_g \times \mathrm{id}} D_\varepsilon \times H \xrightarrow{\mu} D_\varepsilon H \xrightarrow{\ell_g} g D_\varepsilon H = U_g$$

In the course of the proof we also see that $\dim G/H = \dim G - \dim H$ and $T_e(G/H) \cong \mathrm{Lie}(G)/\mathrm{Lie}(H)$ given by the splitting $\mathrm{Lie}(G) = V \oplus \mathrm{Lie}(H)$.

3) It remains to show G/H is Hausdorff and second countable. The second follows from the openness of $G \to G/H$, and the first is Proposition W.56.4.

4) The uniqueness follows from the following lemma.

Lemma W.62. Let X, Y, Z be smooth manifolds with a commutative diagram with each map continuous

$$\begin{tikzcd} X \arrow[rr, "f"] \arrow[dr, "\pi"'] & & Y \\ & Z \arrow[ur, "g"'] & \end{tikzcd}$$

If f is smooth and π is submersive, then g is smooth on $\pi(X)$.

Proof. Since π is submersive, $\pi(X)$ is open in Z. Let $z \in \pi(X)$ and $x \in \pi^{-1}(z)$. By Constant rank theorem we can find a chart U about x such that $\pi : U \to \pi(U)$ takes the form $(x^1, \dots, x^n) \mapsto (x^1, \dots, x^s)$, where $n = \dim X$, $s = \dim Z$. Define $\psi : \pi(U) \to U$ by $(x^1, \dots, x^s) \mapsto (x^1, \dots, x^s, 0, \dots, 0)$. Then ψ is smooth on $\pi(U)$, and on $\pi(U)$ we have $g = g \circ \pi \circ \psi = f \circ \psi$ is smooth. □

To conclude the proof, it suffices to take $X = G$, $Y = Z = G/H$, $f = \pi$ the canonical projection $G \to G/H$ and $g = \mathrm{id}_{G/H}$.

□

Corollary W.62.1. If $N \trianglelefteq G$ is a closed normal subgroup of a Lie group G, then G/N with canonical smooth and group structures is a Lie group.

Corollary W.62.2. Let $f : G \to H$ be a Lie group homomorphism and $N \trianglelefteq G$ be a normal closed subgroup contained in $\ker f$. Then the induced abstract group homomorphism $G/N \to H$ is a Lie group homomorphism.

Proposition W.63. Let G be a Lie group and M be a smooth homogeneous left G-space. Then the map

$$\begin{aligned} f : G/G_x &\longrightarrow M \\ gG_x &\longmapsto gx \end{aligned}$$

is a diffeomorphism. (See also Theorem W.58.) In particular, $G_x \to G \to M$ is a smooth principal bundle.

Proof. It is clear that f is a continuous bijection. Also, f is smooth, since locally it is $U \to U \times G_x \cong \pi^{-1}(U) \to M$ where $\pi : G \to G/G_x$ is the projection. It remains to show f is immersive (then f is automatically a diffeomorphism). Since f is G-equivariant, it has constant rank. Now since f is injective, it must be immersive by Constant rank theorem. □

Example W.64.

1. $\mathrm{SO}(n)$ acts smoothly on $\mathbb{R}^n$ on the left, and when $n \geqslant 2$ the orbits are the spheres $\{x \in \mathbb{R}^n \mid |x| = r\}$, $r \geqslant 0$. To see this, let $x \neq y \in \mathbb{R}^n$ such that $|x| = |y| = r > 0$. Consider the plane spanned by x and y and let y' be a unit vector lying on the plane that is orthonormal to x. Extending x/r and y' to be orthonormal basis of $\mathbb{R}^n$, we may assume $x = (r, 0, 0, \dots, 0)$ and $y = (a, b, 0, \dots, 0)$ with $a^2 + b^2 = r^2$. Now we can easily find $A \in \mathrm{SO}(2)$ such that $A := A \oplus I_{n-2} \in \mathrm{SO}(n)$ takes x to y.

   Particularly, $\mathrm{SO}(n)$ acts transitively on the $n-1$-sphere S^{n-1}. The unit vector $e_1 = (1, 0, \dots, 0)$ has stabilizer consisting of matrices of the form

$$\begin{pmatrix} 1 & 0 \\ 0 & C \end{pmatrix}$$

with $C \in \mathrm{SO}(n-1)$, so we have a diffeomorphism $\mathrm{SO}(n)/\mathrm{SO}(n-1) \cong S^{n-1}$, and the $\mathrm{SO}(n-1)$-principal bundle

$$\mathrm{SO}(n-1) \to \mathrm{SO}(n) \to S^{n-1}$$

Similarly, the $\mathrm{O}(n)$-action on S^{n-1} gives rise to the $\mathrm{O}(n-1)$-principal bundle

$$\mathrm{O}(n-1) \to \mathrm{O}(n) \to S^{n-1}$$

2. Similarly, the natural $\mathrm{SU}(n)$-action on $\mathbb{C}^n$ has orbits $\{z \in \mathbb{C}^n \mid |z| = r\}$, $r \geqslant 0$. In particular, $\mathrm{SU}(n)$ acts transitively on S^{2n-1}, and the stabilizer of $e_1 \in \mathbb{C}^n$ is $\mathrm{SU}(n-1) \subseteq \mathrm{SU}(n)$. This gives a diffeomorphism $\mathrm{SU}(n)/\mathrm{SU}(n-1) \cong S^{2n-1}$ and an $\mathrm{SU}(n-1)$-principal bundle

$$\mathrm{SU}(n-1) \to \mathrm{SU}(n) \to S^{2n-1}$$

Similarly, there is a $\mathrm{U}(n-1)$-principal bundle

$$\mathrm{U}(n-1) \to \mathrm{U}(n) \to S^{2n-1}$$

A similar argument shows that letting $\mathrm{Sp}(n)$ act on $\mathbb{H}^n$ gives rise to a principal bundle

$$\mathrm{Sp}(n-1) \to \mathrm{Sp}(n) \to S^{4n-1}$$

3. The group $\mathrm{SU}(n)$ acts on $\mathbb{C}^n$, and the induced action on $\mathbb{C}\mathbf{P}^{n-1} = \mathbb{P}^{n-1}_{\mathbb{C}} = \mathbf{P}(\mathbb{C}^n)$ is transitive. The stabilizer of the line $\mathbb{C}e_1$ consists of matrices of the form

$$\begin{pmatrix} a & 0 \\ 0 & C \end{pmatrix} \in \mathrm{SU}(n)$$

and this is the image of the embedding

$$\begin{aligned} \mathrm{U}(n-1) &\longrightarrow \mathrm{SU}(n) \\ C &\longmapsto \begin{pmatrix} \det C^{-1} & 0 \\ 0 & C \end{pmatrix} \end{aligned}$$

In this way we get a principal bundle

$$\mathrm{U}(n-1) \to \mathrm{SU}(n) \to \mathbf{P}(\mathbb{C}^n) \qquad (\spadesuit)$$

Similarly we have an embedding $\mathrm{O}(n-1) \to \mathrm{SU}(n)$ and a principal bundle

$$\mathrm{O}(n-1) \to \mathrm{SO}(n) \to \mathbf{P}(\mathbb{R}^n)$$

4. In the case (♠)($n = 2$), we have $\mathrm{U}(1) = S^1$, $\mathrm{SU}(2) = S^3$ and $\mathbf{P}(\mathbb{C}^2) = S^2$ the Riemann sphere. Hence we have a principal bundle

$$S^1 \to S^3 \to S^2$$

This is called the **Hopf fibration**. Explicitly, the bundle projection is

$$\begin{aligned} S^3 &\longrightarrow S^2 \\ (a, b) &\longmapsto [a : b] = \frac{a}{b} \end{aligned}$$

In polar coordinates, it is

$$(r_0 e^{i\theta_0}, r_1 e^{i\theta_1}) \mapsto \frac{r_0}{r_1} e^{i(\theta_0 - \theta 1)}, \ r_0^2 + r_1^2 = 1$$

5. Let U an n-dimensional real/complex/quaternionic inner product space. An **orthonormal k-frame** is an ordered orthonormal set in U. The **(compact) Stiefel manifold** $V_k(U)$ is the set of all orthonormal k-frames of U. Topologize $V_k(U)$ as a (closed) subspace of the product of k-copies of the unit sphere in U. The orthogonal group $\mathrm{O}(U)$ acts $V_k(U)$ on the left continuously and transitively.

   Let us consider the case $U = \mathbb{R}^n$. Then the stabilizer of $(e_1, \ldots, e_k)$ is $\mathrm{O}(n-k)$, and thus by Theorem W.58 we have a homeomorphism $\mathrm{O}(n)/\mathrm{O}(n-k) \cong V_k(\mathbb{R}^n)$ (of course one can use this to topologize $V_k(\mathbb{R}^n)$). We use this bijection to give a smooth structure on $V_k(\mathbb{R}^n)$, so we have a principal bundle

$$\mathrm{O}(n-k) \to \mathrm{O}(n) \to V_k(\mathbb{R}^n)$$

   Similarly, we have

$$\mathrm{U}(n-k) \to \mathrm{U}(n) \to V_k(\mathbb{C}^n)$$

$$\mathrm{Sp}(n-k) \to \mathrm{Sp}(n) \to V_k(\mathbb{H}^n)$$

   In fact, for $k < m \leqslant n$, we have a fibre bundle

$$V_{m-k}(\mathbb{R}^{n-k}) \to V_m(\mathbb{R}^n) \to V_k(\mathbb{R}^n)$$

   where the last projection sends an m-frame onto the k-frame formed by its first k-vectors.

6. Let U an n-dimensional real/complex/quaternionic vector space. The **Grassmannian** $\mathrm{Gr}_k(U)$ is the set of all k-planes in U. There is a natural surjection $\pi : V_k(U) \to \mathrm{Gr}_k(U)$ sending an k-frame to the plane it spans, and we use this to topologize $\mathrm{Gr}_k(U)$; this coincides with the usual topology imposed on $\mathrm{Gr}_k(U)$. Consider the case $U = \mathbb{R}^n$. Then the natural surjection is a principal $\mathrm{O}(k)$-bundle:

$$\mathrm{O}(k) \to V_k(\mathbb{R}^n) \to \mathrm{Gr}_k(\mathbb{R}^n)$$

The orthogonal group $\mathrm{O}(U)$ also acts on $\mathrm{Gr}_k(U)$ transitively. The stabilizer of the k-plane $\mathbb{R}^k = \mathbb{R}^k \times \{0\} \subseteq \mathbb{R}^k \times \mathbb{R}^{n-k}$ is $\mathrm{O}(k) \times \mathrm{O}(n-k) \subseteq \mathrm{O}(n)$ via the embedding

$$(A, B) \mapsto \begin{pmatrix} A & \\ & B \end{pmatrix}$$

Hence we have a homeomorphism $\mathrm{Gr}_k(\mathbb{R}^n) \cong \dfrac{\mathrm{O}(n)}{\mathrm{O}(k) \times \mathrm{O}(n-k)}$, and we can use this map to define a smooth structure on $\mathrm{Gr}_k(\mathbb{R}^n)$, and hence we obtain a principal bundle

$$\mathrm{O}(k) \times \mathrm{O}(n-k) \to \mathrm{O}(n) \to \mathrm{Gr}_k(\mathbb{R}^n)$$

Of course, we can reverse the process: use the usual smooth structure on $\mathrm{Gr}_k(\mathbb{R}^n)$ and show the $\mathrm{O}(n)$-action on is smooth. Similarly, we have

$$\mathrm{U}(k) \times \mathrm{U}(n-k) \to \mathrm{U}(n) \to \mathrm{Gr}_k(\mathbb{C}^n)$$

$$\mathrm{Sp}(k) \times \mathrm{Sp}(n-k) \to \mathrm{Sp}(n) \to \mathrm{Gr}_k(\mathbb{H}^n)$$

On the other hand, $\mathrm{GL}_n(\mathbb{R})$ acts transitively on $\mathrm{Gr}_k(\mathbb{R}^n)$, and the stabilizer of $\mathbb{R}^k$ is the subgroup P consisting matrices of the form

$$\begin{pmatrix} A & * \\ & C \end{pmatrix}, \ A \in \mathrm{GL}_k(\mathbb{R}), \ C \in \mathrm{GL}_{n-k}(\mathbb{R})$$

so we have diffeomorphisms

$$\mathrm{GL}_n(\mathbb{R})/P \cong \mathrm{Gr}_k(\mathbb{R}^n) \cong \frac{\mathrm{O}(n)}{\mathrm{O}(k) \times \mathrm{O}(n-k)}$$

7. We have a natural action of $\mathrm{SL}_2(\mathbb{R})$ on the upper half plane $\mathbb{H}$ (not confused with the quaternion). It is smooth and transitive, and the stabilizer of i is $\mathrm{SO}(n)$, so we have a diffeomorphism $\mathrm{SL}_2(\mathbb{R})/\mathrm{SO}(2) \cong \mathbb{H}$.

   Also, $\mathrm{SL}_2(\mathbb{C})$ acts on $\mathbb{P}^1_{\mathbb{C}}$ smoothly and transitively, and the stabilizer of ∞ is the subgroup P of upper triangular matrices with determinant 1. Hence $\mathbb{P}^1_{\mathbb{C}} \cong \mathrm{SL}_2(\mathbb{C})/P$.

Definition. An element x of a topological group G is called a **topological generator** if $G = \overline{\langle x \rangle}$.

Theorem W.65 (Kronecker)**.** A vector $v = (v_1, \ldots, v_n) \in \mathbb{R}^n$ represents a topological generator of $\mathbb{T}^n := \mathbb{R}^n/\mathbb{Z}^n$ if and only if $1, v_1, \ldots, v_n$ are $\mathbb{Q}$-linearly independent.

Proof. The exact sequence

$$0 \longrightarrow \mathbb{Z}^n \longrightarrow \mathbb{R}^n \longrightarrow \mathbb{T}^n \longrightarrow 0$$

defining the torus canonically identifies $\mathbb{R}^n = \operatorname{Lie}\mathbb{R}^n = \operatorname{Lie}\mathbb{T}^n$, so the projection $\mathbb{R}^n \to \mathbb{T}^n$ is the same as the exponential map by naturality, so a Lie group homomorphism $f : \mathbb{T}^n \to S^1$ induces a commutative diagram

$$\begin{tikzcd}
0 \arrow[r] & \mathbb{Z}^n \arrow[r] \arrow[d, "f_*|_{\mathbb{Z}^n}"] & \mathbb{R}^n \arrow[r] \arrow[d, "f_*"] & \mathbb{T}^n \arrow[d, "f"] \\
0 \arrow[r] & \mathbb{Z} \arrow[r] & \mathbb{R} \arrow[r] & S^1
\end{tikzcd}$$

so that $f_*(v_1, \ldots, v_n) = \alpha_1 v_1 + \cdots + \alpha_n v_n$ for some $\alpha_i \in \mathbb{Z}$. Now the following statements are equivalent:

(i) $1, v_1, \ldots, v_n$ are linearly dependent over $\mathbb{Q}$.

(ii) $\sum_{i=1}^{n} a_i v_i \in \mathbb{Q}$ for some non-all-zero $a_i \in \mathbb{Q}$.

(iii) $\sum_{i=1}^{n} a_i v_i \in \mathbb{Z}$ for some non-all-zero $a_i \in \mathbb{Z}$.

(iv) $v \bmod \mathbb{Z}^n$ is in the kernel of some nontrivial homomorphism $f : \mathbb{T}^n \to S^1$.

(v) $v \bmod \mathbb{Z}^n$ is not a topological generator.

(i) $\Leftrightarrow$(ii) $\Leftrightarrow$(iii) are clear. (iii) $\Leftrightarrow$(iv) follows from the first paragraph. (iv) $\Leftrightarrow$(v) is obvious, so it remains to show (v) $\Leftrightarrow$(iv). A nongenerator $[v] := v \mod \mathbb{Z}^n$ is contained is a proper closed subgroup $H \leqslant \mathbb{T}^n$, and the quotient $\mathbb{T}^n/H$ is a nontrivial compact connected abelian Lie group, so $\mathbb{T}^n/H \cong \mathbb{T}^k$ for some $k \in \mathbb{N}$, and $[v]$ lies in the kernel of the nontrivial homomorphism

$$\mathbb{T}^n \to \mathbb{T}^n/H \cong \mathbb{T}^k = S^1 \times \cdots \times S^1 \xrightarrow{\mathrm{pr}_1} S^1$$

□

Corollary W.65.1. A compact Lie group contains a dense cyclic subgroup if and only if the group is isomorphic to $\mathbb{T}^n \cong \mathbb{Z}/\ell\mathbb{Z}$ for some $k \in \mathbb{N}_0$, $\ell \in \mathbb{N}$.

Proof. Take a topological generator $t \in \mathbb{T}^n$ and pick $x \in \mathbb{T}^n$ such that $\ell x = t$. Then the subgroup generated by $(x, 1)$ is dense in $\mathbb{T}^n \cong \mathbb{Z}/\ell\mathbb{Z}$. Conversely, if a compact Lie group G contains a dense cyclic subgroup, say $\langle a \rangle$, then G is abelian, so $G \cong \mathbb{T}^n \times F$ for some finite F. But $\mathrm{pr}_2(a)$ generates F, so F is cyclic.

□

W.6.4 Proper Free Actions

Definition. Let G be a topological group and M a right G-space. We say the G-action on M is **proper** if the continuous map

$$\begin{aligned} M \times G &\longrightarrow M \times M \\ (m, g) &\longmapsto (m, mg) \end{aligned}$$

is proper.

Theorem W.66. Let G be a Lie group and M a smooth right G-space on which the action is proper and free. Then the orbit space M/G has a unique smooth structure of dimension $\dim M - \dim G$ such that the canonical projection $M \to M/G$ is a smooth principal G-bundle.

W.7 Integration on Lie groups

In this subsection, for a topological space X, we use $C_c(X)$ to denote the space of real-valued continuous functions with compact support (rather than the complex-valued ones).

W.7.1 Smooth partition of unity

Let M be a smooth manifold of dimension n. By Theorem T.16, M is paracompact (the second countability is imposed to a manifold in our definition). In this subsubsection we construct a "smooth" partition of unity subordinate to any given open cover of M, by modifying the proof of Theorem T.15.

Define $f : \mathbb{R} \to \mathbb{R}$ by $f(t) = \mathbf{1}_{\geqslant 0}(t)e^{-\frac{1}{t}}$. A direct computation of derivatives of f at $t = 0$ shows that $f \in C^\infty(\mathbb{R})$. For $0 < a < b$, the function $f_1(x) := f((b-a)(x-a))$ is then a smooth function with $f_1(x) = 0$ if and only if $x \leqslant a$ or $x \geqslant b$. Define $f_2 : \mathbb{R} \to \mathbb{R}$ by

$$f_2(x) := \int_x^b f_1(t)dt \Big/ \int_a^b f_1(t)dt$$

Then $f_2 \in C^\infty(\mathbb{R})^+$ and $f_2(x) = \begin{cases} 1 & \text{, if } x \leqslant a \\ 0 & \text{, if } x \geqslant b \end{cases}$ Define $\phi : \mathbb{R}^n \to \mathbb{R}$ by

$$\phi(x) = f_2(\|x\|^2)$$

We have $\phi \in C^\infty(\mathbb{R}^n)^+$ and $\phi(x) = \begin{cases} 1 & \text{, if } \|x\|^2 \leqslant a \\ 0 & \text{, if } \|x\|^2 \geqslant b \end{cases}$. Using this function, we first prove a smooth version of Urysohn's lemma:

Lemma W.67. Let M^n be a smooth manifold, K a compact subset and U an open neighborhood of K in M. Then we can find $\psi \in C^\infty(M)^+$ such that $\psi|_K = 1$ and $\psi|_{M\setminus U} = 0$.

Proof. Let $p \in K$ and choose an open chart $\varphi_p : U_p \to \mathbb{R}^n$ of p in U with $\varphi_p(p) = 0$. The preceding discussion allows us to find $\phi_p \in C_c^\infty(U_p)^+$ and $0 < a < b$ with $\overline{B_b(0)} \subseteq \varphi_p(U_p)$ with $\phi_p(x) = \begin{cases} 1 & \text{, if } \|\varphi_p(x)\|^2 \leqslant a \\ 0 & \text{, if } \|\varphi_p(x)\|^2 \geqslant b \end{cases}$ Let us put $V_p := \varphi_p^{-1}(\overline{B_a(0)})$. By compactness of K, we can find $p_1, \ldots, p_n \in K$ such that

$$K \subseteq \bigcup_{i=1}^n V_i \subseteq \bigcup_{i=1}^n \operatorname{supp} \phi_i \subseteq U$$

where $V_i := V_{p_i}$ and $\phi_i := \phi_{p_i}$. Since $\operatorname{supp}\phi_i \subseteq U_{p_i}$, we can extend ϕ_i to the whole M by setting $\phi_i|_{M\setminus U_{p_i}} = 0$, so that we can view $\phi_i \in C_c^\infty(M)^+$. Finally define

$$\psi = 1 - (1-\phi_1)(1-\phi_2)\cdots(1-\phi_n) \in C^\infty(M).$$

As $K \subseteq \bigcup_{i=1}^n V_i$, ψ is identically 1 on K, and as $\bigcup_{i=1}^n \operatorname{supp}\phi_i \subseteq U$, ψ vanishes outside U. In addition, for $x \in U\setminus K$, by construction (of ϕ_p) we have $\psi(x) \in [0,1]$, so in fact $\psi \in C^\infty(M,[0,1])$. □

We proceed to construct a smooth partition of unity. Let $p \in M$ and let U be an open neighborhood of p. By the preceding discussion, we can find $g_p \in C_c^\infty(U)^+$ with $g_p(p) > 0$ and $\operatorname{supp} g_p$ having nonempty interior. That $\operatorname{supp} g_p \subseteq U$ implies that g_p extends (by zero) to a global smooth function, which we still denote by $g_p \in C_c^\infty(M)^+$.

We start our construction of a smooth partition of unity subordinate to the given open cover $(U_\alpha)_\alpha$ of M. By Theorem T.16, M is σ-compact, so there exists an increasing sequence $(E_n)_n$ of relatively compact open sets of M with $\overline{E_n} \subseteq E_{n+1}$. For each $n \in \mathbb{N}$, the annulus $K_n := \overline{E_n}\setminus E_{n-1}$ is compact, and it contained in the open annulus $A_n := E_{n+1}\setminus\overline{E_{n-2}}$. For each $p \in K_n$, let W_p be an chart about p contained in $U_\alpha \cap A_n$, and let $g_p \in C_c^\infty(M)^+$ be such that $g_p(p) > 0$ and $\operatorname{supp} g_p$ has nonempty interior V_p and is contained in W_p. We can find a finite subcollection of $\{V_p\}$ that covers K_n. Collecting these V_p, we obtain a locally finite open refinement $(V_n)_{n\in\mathbb{N}}$ of $(U_\alpha)_\alpha$.

Let g_n be the corresponding function to V_n. For each $p \in M$, we see $\{n \in \mathbb{N} \mid g_n(p) \neq 0\}$ is nonempty and finite. Thus it is legal to define

$$\psi_n(p) := \frac{g_n(p)}{\sum_m g_m(p)} \in [0,1]$$

for each $p \in M$ and $n \in \mathbb{N}$. Clearly, $\psi_n \in C^\infty(M)$, $\operatorname{supp}\psi_n = \operatorname{supp} g_n = \overline{V_n}$ is compact, and $\sum_n \psi_n \equiv 1$. This finishes our construction.

In addition, if we define $\psi_\alpha := \sum_{V_n \subseteq U_\alpha} \psi_n$ ($\psi_\alpha = 0$ if the range of summation is empty), then $(\psi_\alpha)_\alpha$ is a partition of unity with the same index set as U_α, but each $\operatorname{supp}\psi_\alpha$ fails to be compact.

Let $\ell \in \mathbb{N}$. If we define $\phi_n := \dfrac{g_n^\ell}{\sum_m g_m^\ell}$, then $(\phi_n)_n$ is a smooth partition of unity subordinate to $(U_\alpha)_\alpha$ with smooth ℓ-th root.

W.7.2 Tangent frame bundles, density bundles and integration

Let V be an n-dimensional real vector space. A **frame** on V is an ordered basis $\beta = (\beta_1 \cdots \beta_n)$ for V. Denote by $F(V)$ the set of all frames on V. There is a canonical bijection $F(V) \cong \operatorname{Isom}_{\mathbb{R}}(\mathbb{R}^n, V)$, by sending β to a linear map T_β defined by $e_i \mapsto \beta_i$. The general linear group $\mathrm{GL}_n(\mathbb{R})$ acts on the set $F(V)$ on the right, and the action is free and transitive.

Let M be an n-dimensional smooth manifold. Define the **(tangent) frame bundle** of M

$$FM = F(TM) := \bigsqcup_{p\in M} F(T_pM).$$

This is a smooth principal $\mathrm{GL}_n(\mathbb{R})$-bundle, which we show now. Let us put $\pi : FM \to M$ to be the projection. Let (U, φ) be a local coordinates of M; then $\varphi_{*,p} : T_pU \to \mathbb{R}^n$ is a linear isomorphism. The fibrewise bijections $F(T_pM) \to \mathrm{Isom}(\mathbb{R}^n, V) \cong \mathrm{GL}_n(\mathbb{R})$ glues to a bijection

$$\begin{aligned} \pi^{-1}(U) &\longrightarrow U \times \mathrm{GL}_n(\mathbb{R}) \\ (p, \beta) &\longmapsto (p, \varphi_{*,p} \circ T_\beta) \end{aligned}$$

The general linear group $\mathrm{GL}_n(\mathbb{R})$ acts on the bundle FM from the right in a natural way, making this bijection $\mathrm{GL}_n(\mathbb{R})$-equivariant. We now equip FM with the final topology induced by the "inclusions" $U \times \mathrm{GL}_n(\mathbb{R}) \to \pi^{-1}(U) \subseteq FM$. It is clear that this topology and the above local trivialization make FM into a smooth principal $\mathrm{GL}_n(\mathbb{R})$-bundle as we promised. For convenience, we denote the section $U \to U \times \{\mathrm{id}\} \subseteq U \times \mathrm{GL}_n(\mathbb{R}) \cong \pi^{-1}(U)$ by $\sigma_\varphi = \sigma_{(U,\varphi)}$; explicitly, $\sigma(p) = (p, \varphi_{*,p}^{-1}(e_1), \ldots, \varphi_{*,p}^{-1}(e_n))$.

Let $s \in \mathbb{R}$ and define the one-dimensional representation $\rho_s : \mathrm{GL}_n(\mathbb{R}) \to \mathbb{R}^\times = \mathrm{GL}(\mathbb{R})$ by $\rho_s(A) = |\det A|^{-s}$. The associated bundle $FM \times_{\mathrm{GL}_n(\mathbb{R})} (\mathbb{R}, \rho_s)$ of $FM \to M$ intertwining ρ_s is called the **s-density bundle** of X, and is denoted by $\mathrm{Vol}_s M$. A **continuous** (resp. **smooth**) **s-density** is then a continuous (resp. smooth) global section of the s-density bundle $\mathrm{Vol}_s M$. Particularly, a **density** is referred to as a 1-density, and we simply write $\mathrm{Vol}(M) := \mathrm{Vol}_1 M$. Also, we put $\Gamma(U, \mathrm{Vol}_s M)$ (resp. $\Gamma^\infty(U, \mathrm{Vol}_s M)$) be the set of all continuous (resp. smooth) s-densities defined on $U \underset{\text{open}}{\subseteq} M$.

Let (U, φ) be a local chart of M, and let $\sigma = \sigma_\varphi : U \to \pi^{-1}(U) \subseteq FM$ be the associated local section. Then a local trivialization of $\mathrm{Vol}_s M$ is given by

$$\begin{aligned} \varphi(U) \times \mathbb{R} \longrightarrow U \times \mathbb{R} &\longrightarrow \pi_d^{-1}(U) \\ (p, v) &\longmapsto [\sigma(p), v] \end{aligned}$$

where $\pi_d : \mathrm{Vol}_s M \to M$ denotes the projection, and $[\sigma(p), v]$ denotes the class of $(\sigma(p), v)$ in $\mathrm{Vol}_s M$. We describe its inverse: if $[(p, \beta), w] \in \pi_d^{-1}(U)$, pick $A \in \mathrm{GL}_n(\mathbb{R})$ such that $\beta = (\varphi_{*,p}^{-1}(e_i))_{1 \leqslant i \leqslant n} A$, and send $[(p, \beta), w]$ to $(p, |\det A|^{-s} w)$. If (V, ψ) is another local chart with $U \cap V \neq \varnothing$, then the corresponding transition map on $\mathrm{Vol}_s M$ is then

$$\begin{aligned} \varphi(U \cap V) \times \mathbb{R} &\longrightarrow \psi(U \cap V) \times \mathbb{R} \\ (\varphi(p), v) &\longmapsto (\psi(p), |\det(\theta_{UV})_{*,\varphi(p)}|^{-s} v) \end{aligned} \tag{$\clubsuit$}$$

where $\theta_{UV} = \psi \circ \varphi^{-1} : \varphi(U \cap V) \to \psi(U \cap V)$ is the transition map on M.

The representation ρ_s can be realized in terms of functions. Consider the set $\mathrm{Vol}_s(\mathbb{R}^n)$ of functions $\omega : \mathbb{R}^n \times \cdots \times \mathbb{R}^n \to \mathbb{R}$ satisfying $\omega(Av_1, \ldots, Av_n) = |\det A|^s \omega(v_1, \ldots, v_n)$. It is clear that $\mathrm{Vol}_s(\mathbb{R}^n)$ is one-dimensional, and $\mathrm{GL}_n(\mathbb{R})$ acts on $\mathrm{Vol}_s(\mathbb{R}^n)$ on the left by

$$(A^*\omega)(v_1, \ldots, v_n) := \omega(A^{-1}v_1, \ldots, A^{-1}v_n) = |\det A|^{-s} \omega(v_1, \ldots, v_n).$$

Thus, via the local chart (U, φ), we can understand each element $v \in \mathbb{R}$ in the fibre of $\mathrm{Vol}_s M$ at p as a function $\omega_p : T_pM \times \cdots \times T_pM \to \mathbb{R}$ satisfying $\omega_p(\varphi_{*,p}^{-1}(e_1), \ldots, \varphi_{*,p}^{-1}(e_n)) = v$ and

$$\omega_p(AX_1, \ldots, AX_n) = |\det A|\omega_p(X_1, \ldots, X_n).$$

where $A \in \mathrm{GL}(T_pM)$ and $\det A := \det(\varphi_{x,p}^{-1} \circ A \circ \varphi_{x,p})$. Such a function is called an **s-density** on the vector space T_pM. For a section ω of $\mathrm{Vol}_s M$, we shall always think of ω_p as an s-density on T_pM. We say an s-density ω is **positive** if ω_p is a non-negative function on $T_pM \times \cdots \times T_pM$; this is a well-defined notion independent of the choice of φ due to the transition rule (♣) of $\mathrm{Vol}_s M$.

Let $f : M \to N$ be a smooth map of smooth manifolds. For a section ω of $\mathrm{Vol}_s N$ defined on V, we define the **pullback density** $f^*\omega : f^{-1}(V) \to \mathrm{Vol}_s M$ by means of

$$(f^*\omega)_p(X_1, \ldots, X_n) = \omega_{f(p)}(f_{*,p}X_1, \ldots, f_{*,p}X_n).$$

It is easy to see $f^*\omega$ defines a section of $\mathrm{Vol}_s M$ on $f^{-1}(V)$, and it is smooth as long as ω is.

Let $U \subseteq \mathbb{R}^n$ be an open set. On $\mathbb{R}^n$ there is a natural s-density λ determined by $\lambda(e_1, \ldots, e_n) = 1$. If $\omega : U \to \mathrm{Vol}_s(U)$ is a continuous (resp. smooth) section of compact support, there exists a continuous (resp. smooth) function $c : U \to \mathbb{R}$ such that $\omega_p = c(p)\lambda$. Indeed, $c(p)$ is the image of p under the composition $U \to \mathrm{Vol}_s(U) \cong U \times \mathbb{R} \xrightarrow{\mathrm{pr}_2} \mathbb{R}$, where we use id_U as the chart on U. Now define

$$\int_U \omega := \int_{\mathbb{R}^n} c(p) d\lambda(p)$$

where $d\lambda$ is the usual Lebesgue measure on $\mathbb{R}^n$.

Now let (U, φ) be a local chart of M, $\omega \in \mathrm{Vol}(M)$ with $\operatorname{supp} \omega \subseteq U$. Define

$$\int_M \omega := \int_{\varphi(U)} (\varphi^{-1})^*\omega.$$

To see this is well-defined let (V, ψ) be another local chart with $\operatorname{supp} \omega \subseteq V$. We must show

$$\int_{\varphi(U)} (\varphi^{-1})^*\omega = \int_{\psi(V)} (\psi^{-1})^*\omega.$$

Put $\theta = \psi \circ \varphi^{-1}$ to be the transition map. By definition we have $\theta^*(\psi^{-1})^*\omega = (\varphi^{-1})^*\omega$. Thus it suffices to show

$$\int_W \theta^*\eta = \int_{\theta(W)} g\eta$$

where $W, \theta(W)$ are opens in $\mathbb{R}^n$, $\theta : W \to \theta(W)$ is a diffeomorphism, $g \in C_c(\theta(W))$ and η is a density on $\theta(W)$. Write $\eta_p = c(p)\lambda$. Then

$$(\theta^*\eta)_q(X_1, \ldots, X_n) = c(\theta(q))\eta_{\theta(q)}(\theta_{*,q}X_1, \ldots, \theta_{*,q}X_n) = c(\theta(q))|\det \theta_{*,q}|\eta_{\theta(q)}(X_1, \ldots, X_n)$$

so that the desired identity now follows from from the change of variables rule on $\mathbb{R}^n$.

Denote by $\Gamma_c(M, \mathrm{Vol}_s M) \subseteq \Gamma(M, \mathrm{Vol}_s M)$ the set of s-densities of compact supports. By a partition of unity argument, we obtain a unique linear functional

$$\int_M : \Gamma_c(M, \mathrm{Vol}(M)) \longrightarrow \mathbb{R}$$

satisfying $\displaystyle\int_M \omega = \int_{\mathbb{R}^n} c d\lambda$ as long as $\operatorname{supp} \omega$ lies in a local chart (U, φ) of M and $(\varphi^{-1})^* \omega = c\lambda$. In addition, if ω is a positive density, we see from the construction that $\displaystyle\int_M \omega \geqslant 0$.

If ω is a positive density, the composition

$$\begin{aligned} C_c(M) &\longrightarrow \Gamma_c(M, \mathrm{Vol}(M)) \xrightarrow{\int_M} \mathbb{R} \\ f &\longmapsto f\omega \end{aligned}$$

defines a positive linear functional on $C_c(M)$, so by Riesz's representation theorem, there exists a outer Radon measure $\mu = \mu_\omega$ on M such that

$$\int_M f d\mu = \int_M f\omega$$

for any $f \in C_c(M)$. Generally, if ω is an arbitrary density, we can define a positive density $|\omega|$ by setting $|\omega|_p := |\omega_p|$ for any $p \in M$, which is well-defined because of (♣), so it gives rise to a outer Radon measure $\mu_{|\omega|}$. If $|\omega|$ is smooth (e.g. ω is smooth and positive), then $|\omega|$ is a smooth measure in the sense of 10.5. Conversely, if μ is a smooth measure on M, then the Radon Nikodym derivatives of μ (with respect to the Lebesgue measure on any chart) glue to a smooth positive density on M.

W.7.3 Differential forms

Let V an n-dimensional vector space over a field k. Consider the space $A^p(V)$ of all alternating p-linear map on V. Clearly, $A^p(V) \cong \mathrm{Hom}_k(\bigwedge^p V, F) \cong \bigwedge^k V^\vee$. To put it explicitly, let $x^1, \ldots, x^n$ be a k-basis for V and put $dx^1, \ldots, dx^n$ to be the dual basis in $V^\vee$. For $I = \{i_1, \ldots, i_p\} \subseteq [n]$ with $1 \leqslant i_1 < i_2 < \cdots < i_p \leqslant n$, define $dx^I \in A^p(V)$ by $dx^I(x_{j_1} \wedge \cdots \wedge x_{j_p}) = \delta_{IJ}$ for all $1 \leqslant j_1 < \cdots < j_p \leqslant n$ and extending linearly. Then for $f \in A^p(V)$, we have $f = \displaystyle\sum_{I: i_1 < \cdots < i_p} f(x^{i_1}, \ldots, x^{i_p}) dx^I$. Thus

$$A^p(V) = \mathrm{span}_F\{dx^I \mid I : i_1 < \cdots < i_k\} \cong \bigwedge^p V^\vee$$

We then transfer the wedge product on $\bigwedge^p V^\vee$ to $A^p(V)$ via this bijection. Under this bijection, we have $dx^I = dx^{i_1} \wedge \cdots \wedge dx^{i_p}$.

Let M be an n-dimensional smooth manifold. We have constructed its tangent bundle TM. Define the **cotangent bundle** $(TM)^\vee$ to be the dual bundle of TM. Set-theoretically, $(TM)^\vee = \displaystyle\bigsqcup_{p \in M} (T_p M)^\vee$ is the disjoint union of the **cotangent spaces**, and we have a natural projection $\pi : (TM)^\vee \to M$. Let

U be a chart on M and $x^1, \ldots, x^n$ its local coordinates. Then for $p \in U$, $dx^1, \ldots, dx^n$ is a basis for $(T_pM)^\vee$ and we have a bijection $\pi^{-1}(U) \cong U \times \mathbb{R}^n$. In this way we turn $(TM)^\vee$ into a $2n$-dimensional smooth manifold. In a similar fashion we construct the **k-th exterior power** of the cotangent bundle $\bigwedge^k (TM)^\vee := \bigsqcup_{p \in M} \bigwedge^k (T_pM)^\vee$.

Denote by $\Omega^k M$ the continuous global sections of the bundle $\bigwedge^k (TM)^\vee \to M$. An element in $\Omega^k M$ is called a **k-form**. Locally, a k-form ω has the form $\sum a_I dx^I$, where the a_I are continuous functions. From this one sees $\Omega^k M$ is a module over the ring $C^\infty(M)$ of real-valued smooth functions on M. An element in $\Omega^n M$, where $n = \dim M$, is called a **top form** on M.

For $\omega \in \Omega^k M$ and $X_1, \ldots, X_k$ vector fields on M, define $\omega(X_1, \ldots, X_k)$ to be a (continuous) function $M \to \mathbb{R}$ by

$$\omega(X_1, \ldots, X_k)(p) := \omega_p(X_{1,p}, \ldots, X_{k,p})$$

The wedge product defined fibrewise gives a globally defined wedge product:

$$\wedge : \Omega^k M \times \Omega^\ell M \longrightarrow \Omega^{k+\ell} M$$

For a smooth map $\varphi : M \to N$, define the **pullback map** $\varphi^* : \Omega^k N \to \Omega^k M$ by the formula

$$(\varphi^* \omega)_p (X_1, \ldots, X_k) = \omega_{\varphi(p)}(\varphi_{*,p} X_1, \ldots, \varphi_{*,p} X_k).$$

When $k = 0$ it is just the pullback map $C(N) \to C(M)$ given by $f \mapsto f \circ \varphi$. Let U, V be open sets in $\mathbb{R}^n$ and $\varphi : U \to V$ a smooth map. Then for $\omega = f dx^1 \wedge \cdots \wedge dx^n$, we have

$$(\varphi^* \omega)_p = f(\varphi(p)) \det \varphi_{*,p} \; dx^1 \wedge \cdots \wedge dx^n$$

Concisely, $\varphi^* \omega = (f \circ \varphi) \det \varphi_* \; dx^1 \wedge \cdots \wedge dx^n$

Let ω be a top form on M. For each $p \in M$, ω_p is a function $T_pM \times \cdots \times T_pM \to \mathbb{R}$ satisfying $\omega_p(AX_1, \ldots, AX_n) = (\det A)\omega_p(X_1, \ldots, X_n)$ for any $X_i \in T_pM$ and $A \in \mathrm{GL}(T_pM)$, where $\det A := \det(\varphi_{*,p} \circ A \circ \varphi_{*,p}^{-1})$ for any local chart φ about p. Clearly, the map $|\omega| : M \to \bigsqcup_{p \in M} \{T_pM \times \cdots \times T_pM \to \mathbb{R}\}$ defined by

$$|\omega_p|(X_1, \ldots, X_n) := |\omega_p(X_1, \ldots, X_n)|$$

is a continuous section of $\mathrm{Vol}(M)$, and it defines a positive density $|\omega| \in \Gamma(M, \mathrm{Vol}(M))$.

W.7.4 Orientations and integration

Let V be an n-dimensional real vector space. There is a natural map $F(V) \to \bigwedge^n V^\vee \backslash \{0\}$ given by $\beta \mapsto d\beta := d\beta_1 \wedge \cdots \wedge d\beta_n$. We say two frames β and γ are equivalent, and write $\beta \sim \gamma$, if $d\beta$ and $d\gamma$ lies in the same connected component in $\bigwedge^n V^\vee \backslash \{0\}$. An **orientation** is then an equivalence class of $\sim$ on $F(V)$, or equivalently, a connected component of $\bigwedge^n V^\vee \backslash \{0\}$. It follows that V admits precisely two orientations.

Let M be an n-dimensional smooth manifold. A continuous local section of the tangent frame bundle FM defined over U is called a **(continuous) frame** on U. For an open set $U \subseteq M$, we say two frames X and Y are equivalent, and write $X \sim Y$, if $X_p \sim Y_p$ for any $p \in M$. Note that if U is a local chart on M, then $X = (X_1, \dots, X_n)$ for some continuous vector fields X_i on U. An **orientation** on M is a collection $\mu = (\mu_p)_{p \in M}$ such that each μ_p is an orientation on T_pM and for each $p \in M$, there are an open neighborhood U and a frame X on U on U satisfying $X_p \in \mu_p$. In other words, we require an orientation on M is locally represented by a continuous frame.

Definition.

1. A smooth manifold that admits an orientation is called **orientable.**

2. A smooth manifold with an orientation is called **oriented**.

Suppose M is connected, and let μ, ν be two orientations on M. Define $f : M \to \{\pm 1\}$ by $f(p) = \begin{cases} 1 & \text{, if } \mu_p = \nu_p \\ -1 & \text{, if } \mu_p \neq \nu_p \end{cases}$. Let $(U, x^1, \dots, x^n)$ be a connected local chart of M on which, say, μ and ν are represented by frames X and Y, respectively. Then on U, we can write $d(Y_p) = \alpha_Y(p) dx^1 \wedge \cdots \wedge dx^n$ and $d(X_p) = \alpha_X(p) dx^1 \wedge \cdots \wedge dx^n$ for some non-vanishing continuous functions α_X, α_Y on U. Hence $d(Y_p) = (\alpha_Y \alpha_X^{-1})(p) d(X_p)$ with $\alpha_Y \alpha_X^{-1}$ continuous and non-vanishing. Hence $p \mapsto \operatorname{sgn}(\alpha_Y \alpha_X^{-1})(p)$ is constant on U, which implies that f is locally constant on M. As we assume M is connected, f is constant on the whole M. Hence either $\mu = \nu$ or $\mu = -\nu$. This proves

Lemma W.68. There are precisely two orientations on a connected orientable smooth manifold.

Suppose M is oriented by μ. Let $(U, x^1, \dots, x^n)$ be a connected local coordinates of M, and suppose μ is represented by $(X_1, \dots, X_n)$ on μ. Write $X_i = \sum_{j=1}^{n} a_{ij} \frac{\partial}{\partial x^j}$ so that $dx^1 \wedge \cdots \wedge dx^n(X_1, \dots, X_n) = \det(a_{ij})$. Replacing x^1 by $-x^1$ if necessary, we may assume $\det(a_{ij}) > 0$. Let $\{(U_\alpha, x_\alpha^1, \dots, x_\alpha^n)\}_\alpha$ be such a collection of local charts with $\{U_\alpha\}$ being a cover of M. Let $(\psi_\alpha)_\alpha$ be a (smooth) partition of unity subordinate to $\{U_\alpha\}$ as in W.7.1, and define

$$\omega = \sum_\alpha \psi_\alpha dx_\alpha^1 \wedge \cdots \wedge dx_\alpha^n.$$

This is a well-defined smooth top form on M. For $p \in M$, we have $\psi_\alpha(p) \geqslant 0$ for any α and > 0 for at least one α, and thus

$$\omega_p(X_{1,p}, \dots, X_{n,p}) = \sum_\alpha \psi_\alpha(p)(dx_\alpha^1 \wedge \cdots \wedge dx_\alpha^n)_p(X_{1,p}, \dots, X_{n,p}) > 0$$

i.e., ω is a (smooth) volume form on M.

Conversely, let ω be a (smooth) volume form on M. At each point $p \in M$ choose an ordered basis $(X_{1,p}, \dots, X_{n,p})$ of T_pM such that $\omega_p(X_{1,p}, \dots, X_{n,p}) > 0$. Let $(U, \varphi, x^1, \dots, x^n)$ be a connected local coordinates of M, and write $(\varphi^{-1})^*\omega = f dx^1 \wedge \cdots \wedge dx^n$ for a smooth nonvanishing function f;

by suitable replacement we may assume $f > 0$ on U. Write $X_i = \sum_{j=1}^{n} a_{ij} \frac{\partial}{\partial x^j}$ on U; then by our choice, we have $f(\varphi(p)) \det(a_{ij}(\varphi(p))) > 0$ for every $p \in U$, or $\det(a_{ij}(\varphi(p))) > 0$, so $(X_{1,p}, \ldots, X_{n,p}) \sim \left(\left.\frac{\partial}{\partial x^1}\right|_p, \ldots, \left.\frac{\partial}{\partial x^n}\right|_p \right)$ for $p \in U$. Hence $p \mapsto (X_{1,p}, \ldots, X_{n,p})$ defines a orientation on M. Thus

Lemma W.69. A smooth manifold is orientable if and only if it admits a (smooth) volume form.

An easy argument shows that if ω and ω' are two smooth volume forms on M, then $\omega = f\omega'$ for a unique nowhere vanishing smooth function f on M. Say $\omega \sim \omega'$ if $\omega = f\omega'$ for some smooth $f > 0$ on M. This sets up an equivalence relations on the set of smooth volume forms on M, and it is bijection with the orientations on M. Thus an oriented smooth manifold can be described as a pair $(M, [\omega])$, where $[\omega] = \{f\omega \mid f \in C^\infty(M),\ f > 0\}$ is a equivalence class containing the volume form ω.

Definition. A diffeomorphism $\varphi : (M, [\omega]) \to (N, [\eta])$ of smooth manifolds is **orientation-preserving** (resp. **orientation-reversing**) if $[\varphi^*\omega] = [\eta]$ (resp. $[\varphi^*\omega] = [-\eta]$).

Let $(M, [\omega])$ be an oriented smooth manifold. Let $(U, x^1, \ldots, x^n)$ and $(V, y^1, \ldots, y^n)$ be two local charts on M intersecting nontrivially. Assume $[dx^1 \wedge \cdots \wedge dx^n] = [\omega|_U]$ and $[dy^1 \wedge \cdots \wedge dy^n] = [\omega|_V]$. If we put θ to be the transition map from U to V, then $dx^1 \wedge \cdots \wedge dx^n = \det(\theta_*) dy^1 \wedge \cdots \wedge dy^n$ on $U \cap V$. A moment consideration shows that we must have $\det(\theta_*) > 0$ on $U \cap V$. Conversely, if M is a smooth manifold that admits an **oriented atlas**, i.e., a smooth atlas such that the Jacobian of every transition map is positive, one may construct (smooth) volume form ω on M by a (smooth) partition of unity argument (we always orient $\mathbb{R}^n$ by $dx^1 \wedge \cdots \wedge dx^n$) in a way that $[\omega|_U] = [dx^1 \wedge \cdots \wedge dx^n]$ for any oriented chart $(U, x^1, \ldots, x^n)$. Given an oriented atlas on M, there is a unique maximal oriented atlas (with respect to inclusion) containing it, and the set of all maximal oriented atlas is in bijection with the equivalence classes of smooth volume forms. Hence we can *specify* an orientation of M by a maximal oriented atlas.

For an oriented manifold M, we usually denote by $-M$ the same manifold but with *opposite orientation*. Precisely, if $(U, x^1, \ldots, x^n)$ is an oriented chart on M, then $(U, -x^1, x^2 \ldots, x^n)$ is set to be an oriented chart on $-M$.

Finally, we discuss the theory of integration via forms. Assume M is an oriented manifold, and we only use oriented charts on M. Denote by $\Omega_c^n(M)$ the space of continuous top forms with compact support. Similarly to the argument for densities with compact support, we obtain

Proposition W.70. Let M^n be an oriented manifold. Then there exists a unique linear map

$$\begin{aligned} \int_M : \Omega_c^n M &\longrightarrow \mathbb{R} \\ \alpha &\longmapsto \int_M \alpha \end{aligned}$$

such that if $\varphi : U \to \varphi(U) \subseteq \mathbb{R}^n$ is an oriented chart of M and if $\alpha \in \Omega_c^n M$ with $\operatorname{supp} \alpha \subseteq U$, then

$$\int_M \alpha = \int_{\mathbb{R}^n} (\varphi^{-1})^* \alpha := \int_{\mathbb{R}^n} a(x) dx^1 \cdots dx^n$$

where $(\varphi^{-1})^* \alpha = a(x) dx^1 \wedge \cdots \wedge dx^n$ and the last expression is the usual integration on $\mathbb{R}^n$.

Hence, for any top form ω on M, there is a linear functional

$$\begin{aligned} I_\omega : C_c(M) &\longrightarrow \Omega_c^M \xrightarrow{\int_M} \mathbb{R} \\ f &\longmapsto f\omega \end{aligned}$$

On the other hand, recall that we can construct a positive density $|\omega|$ for any top form ω, which induces an outer Radon measure $\mu_{|\omega|}$ on M. So we have two theories for integration on M:

$$\int_M f\omega =: I_\omega(f) \qquad \text{and} \qquad \int_M f|\omega| =: \int_M f d\mu_{|\omega|}.$$

In the oriented case, there is another way to define $|\omega|$ in terms of forms but not densities. If $(U, \varphi, x^1, \ldots, x^n)$ is an oriented local chart, then $(\varphi^{-1})^* \omega = f dx^1 \wedge \cdots \wedge dx^n$ for some continuous function f defined on U. Define a form $|\omega|$ by requiring $(\varphi^{-1})^* |\omega| := |f| dx^1 \wedge \cdots \wedge dx^n$. This is well-defined since we only use oriented charts on M. Although two defined $|\omega|$ are totally different, it is clear that $I_{|\omega|} = \int_M d\mu_{|\omega|}$. Particularly, if ω is a volume form representing the orientation of M, then $|\omega| = \omega$ (this holds as we only use oriented charts), and the two ways to integrating functions coincide.

W.7.5 Left Invariant densities and forms

Let M be a smooth manifold with a smooth Lie group action G. For $g \in G$, we write ℓ_g for the diffeomorphism $p \mapsto g.p$ on M.

Definition. An s-density (resp. form) ω on M is called G**-invariant** if $\ell_g^* \omega = \omega$ for any $g \in G$.

- Denote by $\operatorname{Vol}_s(M)^G$ (resp. $(\Omega^k M)^G$) the set of G-invariant s-densities (resp. k-forms) on M.
- As the G-action is smooth, we easily see that any G-invariant s-density (resp. k-form) is automatically smooth.

Assume either ω is a G-invariant density on M, or M is oriented, ω is G-invariant top form and G acts on M in an orientation-preserving way. Then for any $f \in C_c(M)$,

$$\int_M (\ell_g^* f)\omega = \int_M (\ell_g^* f)(\ell_g^* \omega) = \int_M \ell_g^*(f\omega) = \int_M f\omega. \qquad (\spadesuit)$$

The last equality holds for the first case since ω is a density, and holds for the second case as ℓ_g is orientation-preserving.

Now consider the Lie group G of dimension n acting on itself. There are two bijections, one being canonical:

$$\begin{aligned} (\Omega^n G)^G &\longrightarrow \bigwedge^n \operatorname{Lie}(G) \\ \omega &\longmapsto \omega_e \end{aligned}$$

and the other canonical up to a choice of local charts about the identity $e \in G$:

$$\begin{aligned} \operatorname{Vol}(G)^G &\longrightarrow \operatorname{Vol}(\operatorname{Lie}(G)) \\ \eta &\longmapsto \eta_e \end{aligned}$$

Since $\dim \operatorname{Vol}(\operatorname{Lie}(G)) = 1$, it follows that G admits a left-invariant smooth positive density η, and such η is unique up to a positive scalar. Also, as $\dim_{\mathbb{R}} \bigwedge^n \operatorname{Lie}(G) = 1$, G admits a left-invariant smooth volume form ω such that ω_e represents a to-be-given orientation of $\operatorname{Lie}(G)$, and such ω is unique up to a positive scalar as well. Clearly, $|\omega|$ is a left-invariant smooth positive density as long as ω is a left-invariant smooth volume form.

Let η be a left-invariant positive density on G. By (♠), we see the outer Radon measure μ_η is left-invariant. In other words μ_η is a left Haar measure on G.

Example W.71 (Invariant top forms on $\mathrm{GL}_n(\mathbb{R})$)**.** Let us begin with finding a left invariant Haar measure on $\mathrm{GL}_n(\mathbb{R})$. It is easy to guess one, but here we use a quite general method to work it out. First we identify $T_p \mathrm{GL}_n(\mathbb{R}) \cong \mathbb{R}^{n\times n} \cong \mathfrak{gl}_n(\mathbb{R})$ as usual, and write the e_{ij} for the standard basis of $\mathfrak{gl}_n(\mathbb{R})$. Next, let $X_{ij} : \mathrm{GL}_n(\mathbb{R}) \to \mathfrak{gl}_n(\mathbb{R})$ be defined by $X_{ij}(g) = ge_{ij}$; these are left-invariant vector fields on $\mathrm{GL}_n(\mathbb{R})$ with $X_{ij}(e) = e_{ij}$. In fact, $(X_{ij})_{1\leqslant i,j\leqslant n}$ forms a global frame of $\mathrm{GL}_n(\mathbb{R})$. Define $(\omega_{ij})_{1\leqslant i,j\leqslant n}$ to be the differential one forms dual to $(X_{ij})_{1\leqslant i,j\leqslant n}$, in the sense that $\omega_{ij,g}(X_{k\ell,g}) = \delta_{ij}\delta_{k\ell}$ for any $g \in \mathrm{GL}_n(\mathbb{R})$. We claim that the top form $\omega := \omega_{11} \wedge \omega_{12} \wedge \cdots \wedge \omega_{n,n-1} \wedge \omega_{nn}$ is left-invariant. Let $g \in \mathrm{GL}_n(\mathbb{R})$. We need to show $(\ell_g^* \omega)_h = \omega_h$ for any $h \in \mathrm{GL}_n(\mathbb{R})$. Since $(X_{ij,h})$ is a basis for $T_h \mathrm{GL}_n(\mathbb{R})$, it suffices to check $\omega_{gh}((\ell_g)_{*,h} X_{ij,h}) = \omega_h(X_{ij,h})$. This is clear, as

$$\omega_{gh}((\ell_g)_{*,h} X_{ij,h}) = \omega_{gh}(X_{ij,gh}) = \omega_{gh}(X_{ij,gh}) = 1 = \omega_h(X_{ij,h}).$$

Finally, write $\omega = f dx^{11} \wedge dx^{12} \wedge \cdots \wedge dx^{nn}$ for some $f \in C(\mathrm{GL}_n(\mathbb{R}))$. We have

$$1 = \omega_g(X_{ij,g}) = f(g) \prod_{i=1}^n dx^{i1} \wedge \cdots \wedge dx^{in} \left(\sum_{k=1}^n g_{k1} e_{k1}, \ldots, \sum_{k=1}^n g_{kn} e_{kn} \right) = f(g)(\det g)^n.$$

Therefore, we obtain

$$\omega = \frac{dx^{11} \wedge dx^{12} \wedge \cdots \wedge dx^{nn}}{(\det g)^n}$$

We compute the modular function Δ_G of G. Let $f \in C_c(G)$ and $g \in G$. Then

$$\int_G r_g^* f \eta = \int_G (r_g)^* (f r_{g^{-1}}^* \eta) = \int_G f r_{g^{-1}}^* \eta = \int_G f c_g^* \omega = \int_G f \, |\det \operatorname{Ad}(g)| \eta.$$

By Theorem 1.11.3., we deduce that

$$\Delta_G(g) = |\det \mathrm{Ad}(g^{-1})|$$

for every $g \in G$. Similarly, for a left-invariant volume form ω on G, we have

$$r_g^*\omega = (\det \mathrm{Ad}(g))\omega.$$

Hence, a left-invariant volume form on G is right invariant if and only if $\det \mathrm{Ad}(g) : G \to \mathbb{R}^\times$ is trivial.

Next, let H be a closed subgroup of G. By Theorem W.49, H is a regular submanifold of G, so it is a Lie group as well. We discuss the integration on the homogeneous space G/H. We already see in Theorem 1.17 that a G-invariant exists on G/H if and only if $\Delta_G|_H = \Delta_H$. By the result above, this is equivalent to saying $|\det \mathrm{Ad}_G(h)| = |\det \mathrm{Ad}_H(h)|$ for any $h \in H$.

For $g \in G$, denote by $\ell_g : G/H \to G/H$ the map given by $g'H \mapsto gg'H$. For $h \in H$, the conjugation $c(h) : G \to G$ defined by $c(h)g := hgh^{-1}$ descends to a map $c(h) : G/H \to G/H$, which coincides with ℓ_h. Indeed $c(h)(gH) = hgh^{-1}H = hgH = \ell_h(gH)$. Define

$$\begin{aligned} \mathrm{Ad}_{G/H} : H &\longrightarrow \mathrm{Aut}\, T_{eH}G/H \\ h &\longmapsto (\ell_h)_{*,eH} \end{aligned}$$

It is easy to see that $\det \mathrm{Ad}_{G/H}(h) = \dfrac{\det \mathrm{Ad}_G(h)}{\det \mathrm{Ad}_H(h)}$ (c.f. Proposition W.61).

A density η on G/H is G-invariant if and only if $\eta = \ell_g^*\eta$. In particular, we have $\eta = \ell_h^*\eta$ for any $h \in H$. Evaluating at $H = eH$, we must have

$$\eta_H(X_1, \ldots, X_n) = \eta_{hH}((\ell_h)_{*,hH}X_1, \ldots, (\ell_h)_{*,hH}X_n) = |\det(\ell_h)_{*,eH}|\eta_H(X_1, \ldots, X_n)$$

for any $X_i \in T_H(G/H)$, so that $1 = |\det(\ell_h)_{*,eH}| = |\det \mathrm{Ad}_{G/H}(h)|$ for any $h \in H$. Conversely, if $|\det \mathrm{Ad}_{G/H}(h)| = 1$ for any $h \in H$, it is easy to show G/H admits a nontrivial G-invariant positive density on G/H. By the same computation one shows that G/H admits a G-invariant volume form if and only if $\det \mathrm{Ad}_{G/H}(h) = 1$ for any $h \in H$.

In summary:

Proposition W.72. Let G be a Lie group and H a closed subgroup of G.

1. Any left Haar measure on G is a smooth measure in the sense of 10.5.

2. $\Delta_G(g) = |\det \mathrm{Ad}_G(g^{-1})|$ for any $g \in G$.

3. G/H admits a nontrivial G-invariant positive density if and only if

$$|\det \mathrm{Ad}_G(h)| = |\det \mathrm{Ad}_H(h)|$$

for any $h \in H$. This is the case if H is compact.

4. G/H admits a G-invariant volume form if and only if

$$\det \mathrm{Ad}_G(h) = \det \mathrm{Ad}_H(h)$$

for any $h \in H$.

5. If H is connected, then $\det \mathrm{Ad}_{G/H}(h) = |\det \mathrm{Ad}_{G/H}(h)|$ for any $h \in H$.

Proof. The function $h \mapsto |\det \mathrm{Ad}_{G/H}(h)|$ is a continuous homomorphism $H \to \mathbb{R}_{>0}$, so if H is compact, it must be trivial as $\{1\}$ is the only compact subgroup of $\mathbb{R}_{>0}$. Similarly, the function $h \mapsto \det \mathrm{Ad}_{G/H}(h)$ is a continuous homomorphism $H \to \mathbb{R}^\times$, so if H is connected, its image must lie entirely in $\mathbb{R}_{>0}$, hence 4. $\square$

Let us conclude this subsubsection by a discussion on integration on principal G-bundles, where G is a LCH group or a Lie group. Let $\pi : M \to X$ be a principal G-bundle with M, X LCH; in fact, it suffices to assume X is Hausdorff, as $M/G \cong X$ and M/G is always locally compact. Let dg be a left Haar measure on G. For $f \in C_c(M)$, define a function $\int_G f : M \to \mathbb{R}$ by

$$\left(\int_G f\right)(x) := \int_G f(xg)dg$$

The integral exists as the domain of integration is actually $\pi^{-1}(xG) \cap \operatorname{supp} f$, which is compact. Since dg is left-invariant, the integral is G-invariant, so $\int_G f \in C_c(X)$. By the way, this shows that $\int_G$ defines a map $C_c(M) \to C_c(X)$, and this is usually called the **integration along the fibre**, and is denoted by $f \mapsto \pi_* f$ (pushforward f along the bundle projection π).

Now take any outer Radon measure μ on X, and form the linear functional $C_c(M) \to \mathbb{R}$ by

$$f \mapsto \int_X \left(\int_G f\right) d\mu$$

This is positive linear functional on $C_c(M)$, so by Riesz's representation theorem it is uniquely determined by a outer Radon measure on M, which we denote by $d\mu \otimes dg$. Hence

$$\int_M f d\mu \otimes dg = \int_X \left(\int_G f\right) d\mu = \int_X \left(\int_G f(xg)dg\right) d\mu(x)$$

The measure $d\mu \otimes dg$ is G-invariant if and only if

$$\int_X \left(\int_G f(xg)dg\right) d\mu(x) = \int_X \left(\int_G f(xgg')dg\right) d\mu(x) = \Delta_G((g')^{-1}) \int_X \left(\int_G f(xg)dg\right) d\mu(x)$$

for any $g' \in G$, i.e., G is unimodular.

Conversely, suppose we are given an outer Radon measure ν on M. A similar argument in Lemma 1.15, together with a version of Lemma 1.14, shows that $\pi_* : C_c(M) \to C_c(X)$ is surjective. Define a linear functional $I : C_c(X) \to I$ as follows. For $g \in C_c(X)$, take any $f \in C_c(M)$ with $\pi_* f = g$ and put

$I(g) := \int_M f d\nu$. The same argument in Theorem 1.17.(i) shows I is positive and well-defined as long as we assume ν satisfies

$$\int_M f(xg)d\nu(x) = \Delta_G(g^{-1}) \int_M f(x)d\nu(x) \ (g \in G).$$

By Riesz's representation theorem again I is uniquely determined by an outer Radon measure, which we usually denote by $d\mu = \dfrac{d\nu}{dg}$ (not confused with Radon Nikodym derivative!). Again, it satisfies

$$\int_M f d\nu = \int_X \left(\int_G f(xg)dg \right) d\mu(x)$$

Let us go to the smooth world. Let G be a Lie group and assume $\pi : M \to X$ is a smooth principal G-bundle. We claim if the measure μ on X is from some positive density, then the measure $d\mu \otimes dg$ on M is also from a positive density. Since G is a Lie group, dg is a smooth measure, and we identify it with a left-invariant positive density on G. Define $d\mu \otimes dg$ as a density on G as follows. For $p \in M$, take a local trivialization $U \times G$ around p, and say p corresponds to $(x, g) \in U \times G$. Then $T_pM \cong T_xU \times T_gG$, and define

$$(d\mu \otimes dg)_p(X_1, \dots, X_n, Y_1, \dots, Y_m) = (d\mu)_p(X_1, \dots, X_n)(dg)_g(Y_1, \dots, Y_m).$$

To show this is well-defined, if $V \times G$ is another trivialization about p, then there exists $\theta : U \cap V \to G$ such that the transition map from $U \times G$ to $V \times G$ is given by $(x', g') \mapsto (x', \theta(x')g')$. Using the fact that dg is left-invariant, we see $d\mu \otimes dg$ is a well-defined density on M. It follows from Fubini's the integration against the density $d\mu \otimes dg$ is really the same as what we define above.

W.7.6 Invariant de Rham cohomology

In this subsubsection, for a smooth manifold M, we use $\Omega^k M$ denote the smooth global sections of $\bigwedge^k (TM)^\vee \to M$ instead. There is a **de Rham complex**

$$0 \longrightarrow \Omega^0 M \xrightarrow{d} \Omega^1 M \xrightarrow{d} \Omega^2 M \longrightarrow \cdots \longrightarrow \Omega^m M \xrightarrow{d} 0$$

where $m = \dim M$, and $d : \Omega^k M \to \Omega^{k+1} M$ is the **exterior derivative**. The cohomology of the de Rham complex is called the **de Rham cohomology** of M, and is denoted by $H^\bullet_{\mathrm{dR}}(M, \mathbb{R})$.

We briefly recall the definition of d. For a smooth function $f : \mathbb{R}^n \to \mathbb{R}$, define $df = \sum_{i=1}^n \dfrac{\partial f}{\partial x^i} dx^i$ and $d(f dx_1 \wedge \cdots \wedge dx_k) = df \wedge dx_1 \wedge \cdots \wedge dx_k$, where $1 \leqslant k \leqslant n$. For a smooth k-form ω on M which has the form $\omega = \sum f_I dx^I$ on the chart $(U, x_1, \dots, x_n)$, define

$$(d\omega)_p = \sum (df_I)_p \wedge dx_1 \wedge \cdots \wedge dx_n$$

for $p \in U$. Such defined d is independent of the local chart, and gives a well-defined global map $d : \Omega^k M \to \Omega^{k+1} M$ satisfying $d^2 = 0$. Note that $d_M \varphi^* \omega = \varphi^* d_N \omega$ for any smooth form ω on N and for any smooth function $\varphi : M \to N$.

Lemma W.73. For a smooth k-form ω and smooth vector fields $X_0, \dots, X_k$ on M, we have

$$d\omega(X_0, \dots, X_k) = \sum_{i=0}^{k} (-1)^i X_i(\omega(X_0, \dots, \widehat{X_i}, \dots, X_k)) + \sum_{0 \leqslant i < j \leqslant k} (-1)^{i+j} \omega([X_i, X_j], X_0, \dots, \widehat{X_i}, \dots, \widehat{X_j}, \dots, X_k),$$

where $[X_i, X_j]$ is the Lie bracket.

Suppose G is a Lie group acting on M smoothly; denote by $\alpha : G \to \mathrm{Diffeo}(M)$ the action map. A smooth k-form $\omega \in \Omega^k M$ is called **(G-)invariant** if $\alpha(g)^*\omega = \omega$ for any $g \in G$. Put $(\Omega^k M)^G$ to be the subspace of invariant k-forms. Since $d\alpha(g)^* = \alpha(g)^* d$, the sequence $((\Omega^\bullet M)^G, d)$ forms a subcomplex of the de Rham complex $(\Omega^\bullet M, d)$. The inclusion then induces a map on cohomology groups $H^k((\Omega^\bullet M)^G, d) \to H^k_{\mathrm{dR}}(M, \mathbb{R})$. Again, as $d\alpha(g)^* = \alpha(g)^* d$, the group G acts on $H^k_{\mathrm{dR}}(M, \mathbb{R})$ naturally, and the above cohomology map goes into $H^k_{\mathrm{dR}}(M, \mathbb{R})^G$. Hence we have a map

$$H^k((\Omega^\bullet M)^G, d) \to H^k_{\mathrm{dR}}(M, \mathbb{R})^G.$$

We are going to show that this is an isomorphism when G is a compact Lie group.

Assume G is compact; we normalize the Haar measure so that the total volume of G is 1. Define $I : \Omega^k M \to \Omega^k M$ by

$$I(\omega)_p(X_1, \dots, X_k) = \int_G (\alpha(g)^*\omega)_p(X_1, \dots, X_k) dg = \int_G \omega_{\alpha(g)p}(\alpha(g)_{*,p} X_1, \dots, \alpha(g)_{*,p} X_k) dg.$$

The integral is well-defined as the integrand is continuous in g and G is compact. Taking a chart about p, we can see $I(\omega)$ is really a smooth k-form, so I is well-defined. In fact, $I(\omega) \in (\Omega^k M)^G$ because dg is a left-invariant Haar measure. Thus I actually defines a map $I : \Omega^k M \to (\Omega^k M)^G$.

Lemma W.74. $I : \Omega^k M \to (\Omega^k M)^G$ restricts to the identity on $(\Omega^k M)^G$, and is a chain map.

Proof. The first is clear. For the second, we apply Lemma W.73 to compute

$$\begin{aligned} d(I\omega)(X_0, \dots, X_k)(p) &= \sum_{i=0}^{k} (-1)^i X_i((I\omega)(X_0, \dots, \widehat{X_i}, \dots, X_k))(p) \\ &\quad + \sum_{0 \leqslant i < j \leqslant k} (-1)^{i+j} (I\omega)_p([X_i, X_j], X_0, \dots, \widehat{X_i}, \dots, \widehat{X_j}, \dots, X_k) \\ &= \sum_{i=0}^{k} (-1)^i X_{i,p} \int_G \omega_p(\alpha(g)_{*,p} X_0, \dots, \widehat{\alpha(g)_{*,p} X_{i,p}}, \dots, \alpha(g)_{*,p} X_{k,p})) dg \\ &\quad + \sum_{0 \leqslant i < j \leqslant k} (-1)^{i+j} \int_G \omega_p(\alpha_{*,p}[X_i, X_j]_p, X_{0,p}, \dots, \widehat{\alpha(g)_{*,p} X_{i,p}}, \dots, \widehat{\alpha(g)_{*,p} X_{j,p}}, \dots, \alpha(g)_{*,p} X_{k,p}) dg \end{aligned}$$

Since G is compact, $X_{i,p}$ and $\int$ interchange, giving

$$d(I\omega)(X_0, \dots, X_k)(p) = \int_G (d\omega)_p(\alpha(g)_{*,p} X_{0,k}, \dots, \alpha(g)_{*,p} X_{k,p}) dg = I(d\omega)(X_0, \dots, X_k)(p)$$

□

Theorem W.75. The inclusion $((\Omega^\bullet M)^G, d) \to (\Omega^\bullet M, d)$ induces an isomorphism

$$H^k((\Omega^\bullet M)^G, d) \cong H^k_{\mathrm{dR}}(M, \mathbb{R})^G.$$

Proof. Denote by J the inclusion. We saw right before Lemma W.74 that $I \circ J = \mathrm{id}$. In particular, this shows $J^* : H^k((\Omega^\bullet M)^G, d) \to H^k_{\mathrm{dR}}(M, \mathbb{R})$ is injective and $I^* : H^k_{\mathrm{dR}}(M, \mathbb{R}) \to H^k((\Omega^\bullet M)^G, d)$ is surjective. We need to show J^* is surjective onto $H^k_{\mathrm{dR}}(M, \mathbb{R})^G$.

Let $\omega \in \Omega^k M$ be a closed form represent a class in $H^k_{\mathrm{dR}}(M, \mathbb{R})^G$. Let $g \in G$. Since ω represents a class invariant under G, we have $\omega - \alpha(g)^*\omega = d\eta$ for some $\eta = \eta_g \in \Omega^{k-1}M$. Therefore, for any smooth p-cycle $\gamma \in \Delta_p(M)$, by Stokes' theorem we have

$$\int_\gamma \omega - \int_\gamma \alpha(g)^*\omega = \int_\gamma d\eta = \int_{\partial\gamma} \eta = 0,$$

and hence by Fubini,

$$\int_\gamma I(\omega) = \int_\gamma \left(\int_G \alpha(g)^*\omega dg \right) = \int_G (\alpha(g)^*\omega)\, dg = \int_G \left(\int_\gamma \omega \right) dg = \int_\gamma \omega.$$

By de Rham theorem, $I(\omega)$ and ω represent the same cohomology class, and this finishes the proof. □

Assume further G is connected. Then $\alpha(g)$ is homotopic to $\alpha(e) = \mathrm{id}_M$, implying the G-action on $H^\bullet_{\mathrm{dR}}(M, \mathbb{R})$ is trivial. Hence taking $M = G$ in Theorem W.75 yields

Corollary W.75.1. Let G be a compact connected Lie group. Then $H^\bullet_{\mathrm{dR}}(G, \mathbb{R})$ is naturally isomorphic to $H^\bullet(\bigwedge^\bullet \mathrm{Hom}_{\mathbb{R}}(\mathrm{Lie}(G), \mathbb{R}), d)$, where d is given by

$$d\omega(X_0, \ldots, X_k) = \sum_{0 \leqslant i < j \leqslant k} (-1)^{i+j} \omega([X_i, X_j], X_0, \ldots, \widehat{X_i}, \ldots, \widehat{X_j}, \ldots, X_k).$$

W.8 Riemannian structures and Lie groups

W.8.1 Tensors

Let M^n be a smooth manifold. For $p, q \in \mathbb{Z}_{\geqslant 0}$, we can form the bundle

$$T^{p,q}M = \underbrace{TM \otimes \cdots \otimes TM}_{p \text{ times}} \otimes \underbrace{TM^\vee \otimes \cdots \otimes TM^\vee}_{q \text{ times}}$$

A **(smooth)** (p, q)**-tensor** defined on an open subspace $U \subseteq M$ is then a (smooth) section $t : U \to T^{p,q}M$ of the bundle projection $\pi^{p,q} : T^{p,q}M \to M$.

- A $(0, 0)$-tensor is simply a smooth function on M.
- We call a $(0, k)$-tensor a **covariant k-tensor**.
- We call a $(k, 0)$-tensor a **contravariant k-tensor**.

Let ω be a covariant k-tensor with $k > 0$. For each $p \in M$, $\omega_p \in T_pM^\vee \otimes \cdots \otimes T_pM^\vee$ can be viewed canonically as a linear functional

$$\omega_p : T_pM \otimes \cdots \otimes T_pM \to \mathbb{R}$$

We say ω is a **symmetric** (resp. **alternating**) tensor if ω_p is symmetric (resp. alternating) for each $p \in M$. For each covariant k-tensor ω, we define

$$\mathrm{sym}^k\omega(v_1, \ldots, v_k) := \frac{1}{k!} \sum_{\sigma \in S_k} \omega(v_{\sigma(1)}, \ldots, v_{\sigma(k)})$$

which is a symmetric k-tensor. Likewise, we define

$$\mathrm{alt}^k\omega(v_1, \ldots, v_k) := \frac{1}{k!} \sum_{\sigma \in S_k} \mathrm{sgn}(\sigma)\omega(v_{\sigma(1)}, \ldots, v_{\sigma(k)})$$

W.8.2 Affine connections

Recall that for a smooth manifold M, the symbol $\mathfrak{X}(M)$ denotes the set of all smooth vector fields on M. For $f \in C^\infty(M)$ and $X \in \mathfrak{X}(M)$, we can define $fX \in \mathfrak{X}(M)$ by $(fX)_p(g) = f(p)X_p(g)$ for any $g \in C^\infty(M)$ and $p \in M$. In this way $\mathfrak{X}(M)$ becomes a $C^\infty(M)$-module.

Definition. Let M be a smooth manifold. An **affine connection** is a $C^\infty(M)$-module homomorphism $\nabla : \mathfrak{X}(M) \to \mathrm{End}_{\mathbb{R}}\, \mathfrak{X}(M)$ satisfying the **Leibniz's rule**:

$$\nabla_X(fY) = f\nabla_X(Y) + (Xf)Y$$

for $X, Y \in \mathfrak{X}(M)$ and $f \in C^\infty(M)$, where $\nabla_X \in \mathrm{End}_{\mathbb{R}}\, \mathfrak{X}(M)$ usually denotes the image of X under ∇.

Let U be any open subspace of M. An affine connection ∇ on M induces an affine connection ∇_U on U: let $X, Y \in \mathfrak{X}(U)$ and $p \in U$. Pick any $X', Y' \in \mathfrak{X}(M)$ such that X', Y' agree with X, Y on a neighborhood V of p in U. For $q \in V$, define

$$((\nabla_U)_X(Y))_q := (\nabla_{X'}(Y'))_q$$

The following lemma guarantees that the right hand side is independent of the choice of X', Y'.

Lemma W.76. Let ∇ be an affine connection on M, U an open subspace and $X, Y \in \mathfrak{X}(M)$. If $X|_U = 0$ or $Y|_U = 0$, then $\nabla_X(Y)|_U = 0$.

Proof. Let $p \in U$ and $g \in C^\infty(M)$. By Urysohn's lemma, we can find $f \in C^\infty(M)$ such that $f(p) = 0$ and $f|_{M \setminus U} = 1$. If $X|_U = 0$, then $fX = X$, and

$$\nabla_X(Y)g(p) = \nabla_{fX}(Y)g(p) = f(p)\nabla_X(Y)g(p) = 0.$$

Similarly, if $Y|_U = 0$, then $fY = Y$, and

$$\nabla_X(Y)g(p) = \nabla_X(fY)g(p) = f(p)\nabla_X(Y)g(p) + (X_pf)(Y_pg) = 0$$

□

Take any local chart $(U, x^1, \ldots, x^n)$ of M. Then on U,

$$(\nabla_U)_{\frac{\partial}{\partial x^i}} \frac{\partial}{\partial x^j} = \sum_{k=1}^{n} \Gamma_{ij}^{\ k} \frac{\partial}{\partial x^k} \tag{$\clubsuit$}$$

for some (uniquely defined) smooth functions $\Gamma_{ij}^{\ k} \in C^\infty(U)$. If $y^1, \ldots, y^n$ is another local coordinates of U, we obtain another collection of smooth functions $\{\Gamma'^{\ k}_{ij}\}$ by

$$(\nabla_U)_{\frac{\partial}{\partial y^i}} \frac{\partial}{\partial y^j} = \sum_{k=1}^{n} \Gamma'^{\ k}_{ij} \frac{\partial}{\partial y^k}.$$

Using Leibniz's rule and chain rule, a tedious computation shows that

$$\Gamma'^{\ \gamma}_{\alpha\beta} = \sum_{i,j,k} \frac{\partial x^i}{\partial y^\alpha} \frac{\partial x^j}{\partial y^\beta} \frac{\partial y^\gamma}{\partial x^k} \Gamma_{ij}^{\ k} + \sum_j \frac{\partial^2 x^j}{\partial y^\alpha \partial y^\beta} \frac{\partial y^\gamma}{\partial x^j}. \tag{$\spadesuit$}$$

On the other hand, if $\mathcal{U}$ is an open cover of M and on each $U \in \mathcal{U}$ is a collection of function $\Gamma_{ij}^{\ k}$ satisfying ($\spadesuit$) whenever two open sets in $\mathcal{U}$ overlap, we can define an affine connection ∇_U on each $U \in \mathcal{U}$ by ($\clubsuit$), and hence an affine connection ∇ on M by

$$\nabla_X(Y)_p = (\nabla_U)_{X'}(Y')_p$$

where $p \in M$, $U \in \mathcal{U}$ is any open set containing p, and X', Y' are vector fields on U obtained from restriction of X, Y to U.

Lemma W.77. Let $X, Y \in \mathfrak{X}(M)$ and suppose $X_p = 0$ for some $p \in M$. Then $\nabla_X(Y)_p = 0$ as well.

Proof. Let $(U, x^1, \ldots, x^n)$ be a local chart near p, and write $X = \sum_{i=1}^{n} f_i \frac{\partial}{\partial x^i}$. Then $f_i(p) = 0$, and

$$\nabla_X(Y)_p = \sum_{i=1}^{n} f_i(p) \nabla_{\frac{\partial}{\partial x^i}}(Y)_p = 0.$$

□

In particular, if $v \in T_pM$ and $Y \in \mathfrak{X}(M)$, we may define

$$\nabla_v(Y) := \nabla_X(Y)_p \in T_pM$$

where X is any smooth vector field on X with $X_p = v$ (this always exists by, for example Urysohn's lemma). In other words, for each p, an affine connection ∇ on M induces a well-defined $\mathbb{R}$-linear map

$$\nabla : T_pM \longrightarrow \mathrm{Hom}_{\mathbb{R}}(\mathfrak{X}(M), T_p(M))$$

Let M be a smooth manifold and ∇ an affine connection on M. We can regard ∇ as a $\mathfrak{X}(M)$-valued $\mathbb{R}$-linear pairing:

$$\nabla : \mathfrak{X}(M) \times \mathfrak{X}(M) \longrightarrow \mathfrak{X}(M)$$

that is $C^\infty(M)$-linear in the first argument and satisfies the Leibniz's rule in the second argument. In other word, ∇ is an $\mathbb{R}$-linear map

$$\nabla : \Gamma(TM) \longrightarrow \Gamma((TM)^\vee \otimes TM)$$

that satisfies the Leibniz's rule in the sense that

$$\nabla(fX) = df \otimes X + f\nabla(X)$$

for any $f \in C^\infty(M)$ and $X \in \Gamma(TM) = \mathfrak{X}(M)$. Here for a smooth vector bundle $E \to M$, we denote by $\Gamma(E)$ its smooth global sections. Generally,

Definition. Let $E \to M$ be a smooth vector bundle. A **bundle connection** is an $\mathbb{R}$-linear map

$$\nabla : \Gamma(E) \longrightarrow \Gamma((TM)^\vee \otimes E)$$

satisfying

$$\nabla(fs) = df \otimes s + f\nabla(s)$$

for any $f \in C^\infty(M)$ and $s \in \Gamma(E)$.

In this language, an affine connection on M is simply a bundle connection on the tangent bundle TM of M. Similar to the situation above, a bundle connection can also be seen as a map $\nabla : \Gamma(TM) \times \Gamma(E) \to \Gamma(E)$, or $\nabla : \Gamma(TM) \to \operatorname{End}_{\mathbb{R}} \Gamma(E)$.

Suppose $f : M \to N$ is a diffeomorphism and ∇ is an affine connection on N. We can pullback ∇ to an affine connection $f^*\nabla$ on M by

$$(f^*\nabla)_X(Y) := (f^{-1})_*(\nabla_{f_*X}(f_*Y)).$$

If $M = N$, i.e., f is a diffeomorphism on M, we say f is an **affine transformation** if $f^*\nabla = \nabla$, i.e., ∇ is **f-invariant**.

Definition. Let G be a Lie group. An affine connection ∇ is called **left-invariant** if for each $g \in G$, the left translation $\ell_g : G \to G$ by g is an affine transformation.

- Let $X_1, \ldots, X_n$ be a basis of L_G, the space of left-invariant vector fields on G. Then if ∇ is left-invariant, each $\nabla_{X_i} X_j$ is left-invariant as well. In fact, for each $g \in G$,

$$(\ell_g)_* \nabla_{X_i} X_j = (\ell_g)_*(\nabla_{(\ell_{g^{-1}})_* X_i}(\ell_{g^{-1}})_* X_j) = ((\ell_g)^*\nabla)_{X_i}(X_j) = \nabla_{X_i}(X_j)$$

- We can define an affine connection ∇ on G by requiring $\nabla_{X_i}(X_j)$ to be any left invariant vector fields. For any vector fields X, Y, write $X = \sum_{i=1}^n f_i X_i$ and $Y = \sum_{i=1}^n g_i X_i$ for some smooth functions $f_i, g_i \in C^\infty(G)$. Then

$$\nabla_X Y := \sum_{i,j} f_i \nabla_{X_i}(g_j X_j) := \sum_{i,j} (f_i X_i(g_j) X_j + g_j \nabla_{X_i} X_j)$$

so for any $g \in G$,

$$\nabla_{(\ell_g)_* X}((\ell_g)_* Y) = (\ell_g)_*(\nabla_X Y)$$

Hence, such defined affine connection ∇ is left-invariant. In particular, if we recall $\mathrm{Lie}(G) \cong L_G$, it follows that there is a bijection:

$$\begin{aligned} \{\text{left-invariant affine connections on } G\} &\longrightarrow \mathrm{Hom}_{\mathbb{R}}(\mathrm{Lie}(G) \otimes_{\mathbb{R}} \mathrm{Lie}(G), \mathrm{Lie}(G)) \\ \nabla &\longmapsto [(X, Y) \mapsto \nabla_{X'}(Y')_e] \end{aligned}$$

where $X', Y' \in L_G$ are the left-invariant vector fields with $X'_e = X$, $Y'_e = Y$.

W.8.3 Parallelism

Definition. Let M be a smooth manifold with an affine connection ∇. A vector field $X \in \mathfrak{X}(M)$ is said to be **parallel along a curve** $\gamma : I \to M$ if

$$\nabla_{\gamma'(t)} X = 0 \in T_{\gamma(t)} M$$

for any $t \in I$.

Let X, Y be two vector fields and $(U, x^1, \ldots, x^n)$ a local chart. Write $X = \sum_{i=1}^n X^i \frac{\partial}{\partial x^i}$ and $Y = \sum_{i=1}^n Y^i \frac{\partial}{\partial x^i}$. Then

$$\begin{aligned} \nabla_X Y &= \sum_{j=1}^n \nabla_X Y^j \frac{\partial}{\partial x^j} = \sum_{j=1}^n X(Y^j) \frac{\partial}{\partial x^j} + \sum_{j=1}^n Y^j \nabla_X \frac{\partial}{\partial x^j} \\ &= \sum_{j=1}^n X(Y^j) \frac{\partial}{\partial x^j} + \sum_{i,j,k=1}^n X^i Y^j \Gamma_{ij}{}^k \frac{\partial}{\partial x^k} \\ &= \sum_{k=1}^n \left(X(Y^k) + \sum_{i,j=1}^n X^i Y^j \Gamma_{ij}{}^k \right) \frac{\partial}{\partial x^k} \end{aligned}$$

In particular, for a fixed vector field X and $p \in U$, the vector $(\nabla_X Y)_p$ only depends on the $X_p(Y^j)$ and $Y^j(p)$.

Let $\gamma : I \to M$ be a curve, and write $\gamma'(t) = \sum_{i=1}^n a_i(t) \left. \frac{\partial}{\partial x^i} \right|_{\gamma(t)}$ in U. Fix $s \in I$. We can always find smooth functions $f_1, \ldots, f_n$ defined near $p = \gamma(s)$ satisfying

$$(x^i \circ \gamma)(s) = f_i(s) \text{ and } X_p f_i = a_i(s) \text{ for } 1 \leqslant i \leqslant n.$$

Form a vector field $Y = \sum_{i=1}^{n} f_i \frac{\partial}{\partial x^i}$. Then

$$\begin{aligned}(\nabla_X Y)_p &= \sum_{k=1}^{n} \left(X_p(f_k) + \sum_{i,j=1}^{n} X^i(p) f_j(p) \Gamma_{ij}{}^k(p) \right) \frac{\partial}{\partial x^k}\bigg|_p \\ &= \sum_{k=1}^{n} \left(a_i(s) + \sum_{i,j=1}^{n} X^i(p)(x^j \circ \gamma)(s) \Gamma_{ij}{}^k(p) \right) \frac{\partial}{\partial x^k}\bigg|_p .\end{aligned}$$

The right hand side is independent of the choice of the auxiliary vector field Y. By abuse of notation, denote this value by $(\nabla_X \gamma')_{\gamma(s)}$.

Definition. A curve $\gamma : I \to M$ is called a **geodesic** if

$$(\nabla_{\gamma'} \gamma')_{\gamma(t)} = 0 \in T_{\gamma(t)} M$$

for any $t \in I$.

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